



US012406749B2

(12) **United States Patent**
Shen et al.

(10) **Patent No.:** **US 12,406,749 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SYSTEMS AND METHODS FOR
PREDICTING REPAIR OUTCOMES IN
GENETIC ENGINEERING**

(71) Applicants: **The Broad Institute, Inc.**, Cambridge,
MA (US); **Massachusetts Institute of
Technology**, Cambridge, MA (US); **The
Brigham and Women's Hospital, Inc.**,
Boston, MA (US); **President and
Fellows of Harvard College**,
Cambridge, MA (US)

(72) Inventors: **Max Walt Shen**, Cambridge, MA (US);
Jonathan Yee-Ting Hsu, Cambridge,
MA (US); **Mandana Arbab**,
Cambridge, MA (US); **David K.
Gifford**, Cambridge, MA (US); **David
R. Liu**, Cambridge, MA (US); **Richard
Irving Sherwood**, Cambridge, MA
(US)

(73) Assignees: **The Broad Institute, Inc.**, Cambridge,
MA (US); **Massachusetts Institute of
Technology**, Cambridge, MA (US); **The
Brigham and Women's Hospital, Inc.**,
Boston, MA (US); **President and
Fellows of Harvard College**,
Cambridge, MA (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 351 days.

(21) Appl. No.: **16/772,747**

(22) PCT Filed: **Dec. 15, 2018**

(86) PCT No.: **PCT/US2018/065886**

§ 371 (c)(1),

(2) Date: **Jun. 12, 2020**

(87) PCT Pub. No.: **WO2019/118949**

PCT Pub. Date: **Jun. 20, 2019**

(65) **Prior Publication Data**

US 2022/0238182 A1 Jul. 28, 2022

Related U.S. Application Data

(60) Provisional application No. 62/669,771, filed on May
10, 2018, provisional application No. 62/599,623,
filed on Dec. 15, 2017.

(51) **Int. Cl.**

G16B 20/30 (2019.01)

A61K 31/7088 (2006.01)

A61K 38/46 (2006.01)

C12N 9/22 (2006.01)

C12N 15/10 (2006.01)

C12N 15/11 (2006.01)

G16B 40/20 (2019.01)

(52) **U.S. Cl.**

CPC **G16B 20/30** (2019.02); **A61K 31/7088**
(2013.01); **A61K 38/465** (2013.01); **C12N 9/22**
(2013.01); **C12N 15/1089** (2013.01); **C12N**
15/11 (2013.01); **G16B 40/20** (2019.02); **C12N**
2310/20 (2017.05); **C12N 2800/80** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,182,449 A	1/1980	Kozlow
4,186,183 A	1/1980	Steck et al.
4,217,344 A	8/1980	Vanlerberghe et al.
4,235,871 A	11/1980	Papahadjopoulos et al.
4,261,975 A	4/1981	Fullerton et al.
4,485,054 A	11/1984	Mezei et al.
4,501,728 A	2/1985	Geho et al.
4,663,290 A	5/1987	Weis et al.
4,737,323 A	4/1988	Martin et al.
4,774,085 A	9/1988	Fidler
4,797,368 A	1/1989	Carter et al.

(Continued)

FOREIGN PATENT DOCUMENTS

AU	2012244264 A1	11/2012
AU	2015252023 A1	11/2015

(Continued)

OTHER PUBLICATIONS

Kuan, P.F., Powers, S., He, S. et al. A systematic evaluation of
nucleotide properties for CRISPR sgRNA design. BMC Bioinform-
atics 18, 297 (2017) (Year: 2017).*

Ousterout et al. Multiplex CRISPR/Cas9-based genome editing for
correction of dystrophin mutations that cause Duchenne muscular
dystrophy. Nat Commun 6, 6244 (2015) (Year: 2015).*

Seonwoo Min, Byunghan Lee, Sungroh Yoon, Deep learning in
bioinformatics, Briefings in Bioinformatics, vol. 18, Issue 5, Sep.
2017, pp. 851-869 (Year: 2017).*

Qiuyu Zhu et al. Novel Thrombotic Function of a Human SNP in
STXBP5 Revealed by CRISPR/Cas9 Gene Editing in Mice. Arte-
riosclerosis, Thrombosis, and Vascular Biology. vol. 37, No. 2, 2017
pp. 264-270 (Year: 2017).*

(Continued)

Primary Examiner — Olivia M. Wise

Assistant Examiner — Jonathan Edward Hayes

(74) *Attorney, Agent, or Firm* — Wolf, Greenfield &
Sacks, P.C.

(57)

ABSTRACT

The specification provides methods for introducing a desired
genetic change in a nucleotide sequence using a double-
strand break (DSB)-inducing genome editing system, the
method comprising: identifying one or more available cut
sites in a nucleotide sequence; analyzing the nucleotide
sequence and available cut sites with a computational model
to identify the optimal cut site for introducing the desired
genetic change into the nucleotide sequence; and contacting
the nucleotide sequence with a DSB-inducing genome edit-
ing system, thereby introducing the desired genetic change
in the nucleotide sequence at the cut site.

13 Claims, 72 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited

U.S. PATENT DOCUMENTS

4,837,028	A	6/1989	Allen	7,223,545	B2	5/2007	Liu et al.
4,873,316	A	10/1989	Meade et al.	7,354,761	B2	4/2008	Schultz et al.
4,880,635	A	11/1989	Janoff et al.	7,368,275	B2	5/2008	Schultz et al.
4,889,818	A	12/1989	Gelfand et al.	7,419,669	B2	9/2008	Kosmatopoulos et al.
4,897,355	A	1/1990	Eppstein et al.	7,442,160	B2	10/2008	Liu et al.
4,906,477	A	3/1990	Kurono et al.	7,476,500	B1	1/2009	Liu et al.
4,911,928	A	3/1990	Wallach	7,476,734	B2	1/2009	Liu
4,917,951	A	4/1990	Wallach	7,479,573	B2	1/2009	Chu et al.
4,920,016	A	4/1990	Allen et al.	7,488,718	B2	2/2009	Scheinberg et al.
4,921,757	A	5/1990	Wheatley et al.	7,491,494	B2	2/2009	Liu et al.
4,946,787	A	8/1990	Eppstein et al.	7,541,450	B2	6/2009	Liu et al.
4,965,185	A	10/1990	Grischenko et al.	7,557,068	B2	7/2009	Liu et al.
5,017,492	A	5/1991	Kotewicz et al.	7,595,179	B2	9/2009	Chen et al.
5,047,342	A	9/1991	Chatterjee	7,638,300	B2	12/2009	Schultz et al.
5,049,386	A	9/1991	Eppstein et al.	7,670,807	B2	3/2010	Lampson et al.
5,079,352	A	1/1992	Gelfand et al.	7,678,554	B2	3/2010	Liu et al.
5,139,941	A	8/1992	Muzyczka et al.	7,713,721	B2	5/2010	Schultz et al.
5,173,414	A	12/1992	Lebkowski et al.	7,771,935	B2	8/2010	Liu et al.
5,223,409	A	6/1993	Ladner et al.	7,794,931	B2	9/2010	Breaker et al.
5,244,797	A	9/1993	Kotewicz et al.	7,807,408	B2	10/2010	Liu et al.
5,270,179	A	12/1993	Chatterjee	7,851,658	B2	12/2010	Liu et al.
5,374,553	A	12/1994	Gelfand et al.	7,915,025	B2	3/2011	Schultz et al.
5,405,776	A	4/1995	Kotewicz et al.	7,919,277	B2	4/2011	Russell et al.
5,436,149	A	7/1995	Barnes	7,993,672	B2	8/2011	Huang et al.
5,449,639	A	9/1995	Wei et al.	7,998,904	B2	8/2011	Liu et al.
5,496,714	A	3/1996	Comb et al.	7,999,071	B2	8/2011	Schlom et al.
5,512,462	A	4/1996	Cheng	8,012,739	B2	9/2011	Schultz et al.
5,580,737	A	12/1996	Polisky et al.	8,017,323	B2	9/2011	Liu et al.
5,614,365	A	3/1997	Tabor et al.	8,017,755	B2	9/2011	Liu et al.
5,652,094	A	7/1997	Usman et al.	8,030,074	B2	10/2011	Schultz et al.
5,658,727	A	8/1997	Barbas et al.	8,067,556	B2	11/2011	Hogrefe et al.
5,668,005	A	9/1997	Kotewicz et al.	8,114,648	B2	2/2012	Schultz et al.
5,677,152	A	10/1997	Birch et al.	8,173,364	B2	5/2012	Schultz et al.
5,767,099	A	6/1998	Harris et al.	8,173,392	B2	5/2012	Schultz et al.
5,780,053	A	7/1998	Ashley et al.	8,183,012	B2	5/2012	Schultz et al.
5,830,430	A	11/1998	Unger et al.	8,183,178	B2	5/2012	Liu et al.
5,834,247	A	11/1998	Comb et al.	8,206,914	B2	6/2012	Liu et al.
5,835,699	A	11/1998	Kimura	8,354,380	B2	1/2013	Liu et al.
5,844,075	A	12/1998	Kawakami et al.	8,361,725	B2	1/2013	Russell et al.
5,849,548	A	12/1998	Haseloff et al.	8,394,604	B2	3/2013	Liu et al.
5,851,548	A	12/1998	Dattagupta et al.	8,440,431	B2	5/2013	Voytas et al.
5,855,910	A	1/1999	Ashley et al.	8,440,432	B2	5/2013	Voytas et al.
5,856,463	A	1/1999	Blankenborg et al.	8,450,471	B2	5/2013	Voytas et al.
5,962,313	A	10/1999	Podsakoff et al.	8,492,082	B2	7/2013	De Francisicis et al.
5,981,182	A	11/1999	Jacobs, Jr. et al.	8,546,553	B2	10/2013	Terns et al.
6,015,794	A	1/2000	Haseloff et al.	8,569,256	B2	10/2013	Heyes et al.
6,057,153	A	5/2000	George et al.	8,586,363	B2	11/2013	Voytas et al.
6,063,608	A	5/2000	Kotewicz et al.	8,680,069	B2	3/2014	de Fougerolles et al.
6,077,705	A	6/2000	Duan et al.	8,691,729	B2	4/2014	Liu et al.
6,156,509	A	12/2000	Schellenberger	8,691,750	B2	4/2014	Constien et al.
6,183,998	B1	2/2001	Ivanov et al.	8,697,359	B1	4/2014	Zhang
6,355,415	B1	3/2002	Wagner et al.	8,697,439	B2	4/2014	Mangeot et al.
6,429,298	B1	8/2002	Ellington et al.	8,697,853	B2	4/2014	Voytas et al.
6,453,242	B1	9/2002	Eisenberg et al.	8,709,466	B2	4/2014	Coady et al.
6,479,264	B1	11/2002	Louwrier	8,728,526	B2	5/2014	Heller
6,503,717	B2	1/2003	Case et al.	8,748,667	B2	6/2014	Budzik et al.
6,534,261	B1	3/2003	Cox, III et al.	8,758,810	B2	6/2014	Okada et al.
6,558,671	B1	5/2003	Slingluff et al.	8,759,103	B2	6/2014	Kim et al.
6,589,768	B1	7/2003	Kotewicz et al.	8,759,104	B2	6/2014	Unciti-Broceta et al.
6,599,692	B1	7/2003	Case et al.	8,771,728	B2	7/2014	Huang et al.
6,607,882	B1	8/2003	Cox, III et al.	8,790,664	B2	7/2014	Pitard et al.
6,610,522	B1	8/2003	Kotewicz et al.	8,795,965	B2	8/2014	Zhang
6,689,558	B2	2/2004	Case et al.	8,822,663	B2	9/2014	Schrum et al.
6,716,973	B2	4/2004	Baskerville et al.	8,835,148	B2	9/2014	Janulaitis et al.
6,824,978	B1	11/2004	Cox, III et al.	8,846,578	B2	9/2014	McCray et al.
6,933,113	B2	8/2005	Case et al.	8,871,445	B2	10/2014	Cong et al.
6,979,539	B2	12/2005	Cox, III et al.	8,889,418	B2	11/2014	Zhang et al.
7,013,219	B2	3/2006	Case et al.	8,900,814	B2	12/2014	Yasukawa et al.
7,045,337	B2	5/2006	Schultz et al.	8,945,839	B2	2/2015	Zhang
7,067,650	B1	6/2006	Tanaka	8,975,232	B2	3/2015	Liu et al.
7,070,928	B2	7/2006	Liu et al.	8,993,233	B2	3/2015	Zhang et al.
7,078,208	B2	7/2006	Smith et al.	8,999,641	B2	4/2015	Zhang et al.
7,083,970	B2	8/2006	Schultz et al.	9,023,594	B2	5/2015	Liu et al.
7,163,824	B2	1/2007	Cox, III et al.	9,023,649	B2	5/2015	Mali et al.
7,192,739	B2	3/2007	Liu et al.	9,034,650	B2	5/2015	Padidam
				9,068,179	B1	6/2015	Liu et al.
				9,150,626	B2	10/2015	Liu et al.
				9,163,271	B2	10/2015	Schultz et al.
				9,163,284	B2	10/2015	Liu et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

9,181,535 B2	11/2015	Liu et al.	10,954,548 B2	3/2021	Liu et al.
9,200,045 B2	12/2015	Liu et al.	11,046,948 B2	6/2021	Liu et al.
9,221,886 B2	12/2015	Liu et al.	11,053,481 B2	7/2021	Liu et al.
9,228,207 B2	1/2016	Liu et al.	11,124,782 B2	9/2021	Liu et al.
9,234,213 B2	1/2016	Wu	11,214,780 B2	1/2022	Liu et al.
9,243,038 B2	1/2016	Liu et al.	11,268,082 B2	3/2022	Liu et al.
9,267,127 B2	2/2016	Liu et al.	11,299,755 B2	4/2022	Liu et al.
9,290,773 B2	3/2016	Edgerton	11,306,324 B2	4/2022	Liu et al.
9,322,006 B2	4/2016	Liu et al.	11,319,532 B2	5/2022	Liu et al.
9,322,037 B2	4/2016	Liu et al.	11,447,770 B1	9/2022	Liu et al.
9,340,799 B2	5/2016	Liu et al.	11,542,496 B2	1/2023	Liu et al.
9,340,800 B2	5/2016	Liu et al.	11,542,509 B2	1/2023	Maianti et al.
9,359,599 B2	6/2016	Liu et al.	11,560,566 B2	1/2023	Liu et al.
9,388,430 B2	7/2016	Liu et al.	11,578,343 B2	2/2023	Liu et al.
9,394,537 B2	7/2016	Liu et al.	11,643,652 B2	5/2023	Liu et al.
9,434,774 B2	9/2016	Liu et al.	11,661,590 B2	5/2023	Liu et al.
9,458,484 B2	10/2016	Ma et al.	11,702,651 B2	7/2023	Liu et al.
9,512,446 B1	12/2016	Joung et al.	11,732,274 B2	8/2023	Liu et al.
9,526,784 B2	12/2016	Liu et al.	11,795,443 B2	10/2023	Liu et al.
9,534,210 B2	1/2017	Park et al.	11,795,452 B2	10/2023	Liu et al.
9,580,698 B1	2/2017	Xu et al.	11,820,969 B2	11/2023	Maianti et al.
9,610,322 B2	4/2017	Liu et al.	11,898,179 B2	2/2024	Maianti et al.
9,637,739 B2	5/2017	Siksnyis et al.	11,912,985 B2	2/2024	Liu et al.
9,663,770 B2	5/2017	Rogers et al.	11,920,181 B2	3/2024	Liu et al.
9,695,446 B2	7/2017	Mangeot et al.	11,932,884 B2	3/2024	Liu et al.
9,737,604 B2	8/2017	Liu et al.	11,999,947 B2	6/2024	Liu et al.
9,738,693 B2	8/2017	Telford et al.	12,006,520 B2	6/2024	Liu et al.
9,753,340 B2	9/2017	Saitou	12,031,126 B2	7/2024	Liu et al.
9,771,574 B2	9/2017	Liu et al.	12,043,852 B2	7/2024	Liu et al.
9,783,791 B2	10/2017	Hogrefe et al.	12,084,663 B2	9/2024	Maianti et al.
9,816,093 B1	11/2017	Donohoue et al.	12,157,760 B2	12/2024	Liu et al.
9,840,538 B2	12/2017	Telford et al.	2003/0082575 A1	5/2003	Schultz et al.
9,840,690 B2	12/2017	Karli et al.	2003/0087817 A1	5/2003	Cox et al.
9,840,699 B2	12/2017	Liu et al.	2003/0096337 A1	5/2003	Hillman et al.
9,840,702 B2	12/2017	Collingwood et al.	2003/0108885 A1	6/2003	Schultz et al.
9,850,521 B2	12/2017	Braman et al.	2003/0119764 A1	6/2003	Loeb et al.
9,873,907 B2	1/2018	Zeiner et al.	2003/0167533 A1	9/2003	Yadav et al.
9,879,270 B2	1/2018	Hittinger et al.	2003/0203480 A1	10/2003	Kovesdi et al.
9,914,939 B2	3/2018	Church et al.	2004/0003420 A1	1/2004	Kuhn et al.
9,932,567 B1	4/2018	Xu et al.	2004/0115184 A1	6/2004	Smith et al.
9,938,288 B1	4/2018	Kishi et al.	2004/0156861 A1	8/2004	Figdor et al.
9,944,933 B2	4/2018	Storici et al.	2004/0197892 A1	10/2004	Moore et al.
9,982,279 B1	5/2018	Gill et al.	2004/0203109 A1	10/2004	Lal et al.
9,999,671 B2	6/2018	Liu et al.	2005/0136429 A1	6/2005	Guarente et al.
10,011,868 B2	7/2018	Liu et al.	2005/0222030 A1	10/2005	Allison
10,053,725 B2	8/2018	Liu et al.	2005/0260626 A1	11/2005	Lorens et al.
10,059,940 B2	8/2018	Zhong	2006/0088864 A1	4/2006	Smolke et al.
10,077,453 B2	9/2018	Liu et al.	2006/0104984 A1	5/2006	Littlefield et al.
10,113,163 B2	10/2018	Liu et al.	2006/0246568 A1	11/2006	Honjo et al.
10,150,955 B2	12/2018	Lambowitz et al.	2007/0015238 A1	1/2007	Snyder et al.
10,167,457 B2	1/2019	Liu et al.	2007/0049533 A1	3/2007	Liu et al.
10,179,911 B2	1/2019	Liu et al.	2007/0264692 A1	11/2007	Liu et al.
10,189,831 B2	1/2019	Arrington et al.	2007/0269817 A1	11/2007	Shapero
10,202,593 B2	2/2019	Liu et al.	2008/0008697 A1	1/2008	Mintier et al.
10,202,658 B2	2/2019	Parkin et al.	2008/0051317 A1	2/2008	Church et al.
10,227,581 B2	3/2019	Liu et al.	2008/0124725 A1	5/2008	Barrangou et al.
10,323,236 B2	6/2019	Liu et al.	2008/0182254 A1	7/2008	Hall et al.
10,336,997 B2	7/2019	Liu et al.	2008/0220502 A1	9/2008	Schellenberger et al.
10,358,670 B2	7/2019	Janulaitis et al.	2008/0241917 A1	10/2008	Akita et al.
10,392,674 B2	8/2019	Liu et al.	2008/0268516 A1	10/2008	Perreault et al.
10,407,474 B2	9/2019	Liu et al.	2009/0111119 A1	4/2009	Doyon et al.
10,407,697 B2	9/2019	Doudna et al.	2009/0130718 A1	5/2009	Short
10,465,176 B2	11/2019	Liu et al.	2009/0215878 A1	8/2009	Tan et al.
10,508,298 B2	12/2019	Liu et al.	2009/0234109 A1	9/2009	Han et al.
10,583,201 B2	3/2020	Chen et al.	2010/0076057 A1	3/2010	Sontheimer et al.
10,597,679 B2	3/2020	Liu et al.	2010/0093617 A1	4/2010	Barrangou et al.
10,612,011 B2	4/2020	Liu et al.	2010/0104690 A1	4/2010	Barrangou et al.
10,640,767 B2	5/2020	Maianti et al.	2010/0273857 A1	10/2010	Thakker et al.
10,682,410 B2	6/2020	Liu et al.	2010/0305197 A1	12/2010	Che
10,704,062 B2	7/2020	Liu et al.	2010/0316643 A1	12/2010	Eckert et al.
10,745,677 B2	8/2020	Maianti et al.	2011/0016540 A1	1/2011	Weinstein et al.
10,858,639 B2	12/2020	Liu et al.	2011/0059160 A1	3/2011	Essner et al.
10,912,833 B2	2/2021	Liu et al.	2011/0059502 A1	3/2011	Chalasani
10,930,367 B2	2/2021	Zhang et al.	2011/0104787 A1	5/2011	Church et al.
10,947,530 B2	3/2021	Liu et al.	2011/0177495 A1	7/2011	Liu et al.
			2011/0189775 A1	8/2011	Ainley et al.
			2011/0189776 A1	8/2011	Terns et al.
			2011/0217739 A1	9/2011	Terns et al.
			2011/0301073 A1	12/2011	Gregory et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2012/0129759 A1	5/2012	Liu et al.	2015/0132269 A1	5/2015	Orkin et al.
2012/0141523 A1	6/2012	Castado et al.	2015/0140664 A1	5/2015	Byrne et al.
2012/0244601 A1	9/2012	Bertozzi et al.	2015/0159172 A1	6/2015	Miller et al.
2012/0270273 A1	10/2012	Zhang et al.	2015/0165054 A1	6/2015	Liu et al.
2012/0322861 A1	12/2012	Byrne et al.	2015/0166980 A1	6/2015	Liu et al.
2013/0022980 A1	1/2013	Nelson et al.	2015/0166981 A1	6/2015	Liu et al.
2013/0059931 A1	3/2013	Petersen-Mahrt et al.	2015/0166982 A1	6/2015	Liu et al.
2013/0108657 A1	5/2013	Yee et al.	2015/0166983 A1	6/2015	Liu et al.
2013/0117869 A1	5/2013	Duchateau et al.	2015/0166984 A1	6/2015	Liu et al.
2013/0130248 A1	5/2013	Haurwitz et al.	2015/0166985 A1	6/2015	Liu et al.
2013/0158245 A1	6/2013	Russell et al.	2015/0191744 A1	7/2015	Wolfe et al.
2013/0165389 A1	6/2013	Schellenberger et al.	2015/0197759 A1	7/2015	Xu et al.
2013/0212725 A1	8/2013	Kuhn et al.	2015/0211058 A1	7/2015	Carstens
2013/0309720 A1	11/2013	Schultz et al.	2015/0218573 A1	8/2015	Loque et al.
2013/0344117 A1	12/2013	Mirosevich et al.	2015/0225773 A1	8/2015	Farmer et al.
2013/0345064 A1	12/2013	Liu et al.	2015/0241440 A1	8/2015	Fasan et al.
2014/0004280 A1	1/2014	Loomis	2015/0252358 A1	9/2015	Maeder et al.
2014/0005269 A1	1/2014	Ngwuluka et al.	2015/0275202 A1	10/2015	Liu et al.
2014/0017214 A1	1/2014	Cost	2015/0291965 A1	10/2015	Zhang et al.
2014/0018404 A1	1/2014	Chen et al.	2015/0307889 A1	10/2015	Petolino et al.
2014/0044793 A1	2/2014	Goll et al.	2015/0315252 A1	11/2015	Haugwitz et al.
2014/0065711 A1	3/2014	Liu et al.	2015/0344549 A1	12/2015	Muir et al.
2014/0068797 A1	3/2014	Doudna et al.	2016/0017393 A1	1/2016	Jacobson et al.
2014/0127752 A1	5/2014	Zhou et al.	2016/0017396 A1	1/2016	Cann et al.
2014/0128449 A1	5/2014	Liu et al.	2016/0032292 A1	2/2016	Storici et al.
2014/0141094 A1	5/2014	Smyth et al.	2016/0032353 A1	2/2016	Braman et al.
2014/0141487 A1	5/2014	Feldman et al.	2016/0040155 A1	2/2016	Maizels et al.
2014/0179770 A1	6/2014	Zhang et al.	2016/0046952 A1	2/2016	Hittinger et al.
2014/0186843 A1	7/2014	Zhang et al.	2016/0046961 A1	2/2016	Jinek et al.
2014/0186919 A1	7/2014	Zhang et al.	2016/0046962 A1	2/2016	May et al.
2014/0186958 A1	7/2014	Zhang et al.	2016/0053272 A1	2/2016	Wurtzel et al.
2014/0201858 A1	7/2014	Ostertag et al.	2016/0053304 A1	2/2016	Wurtzel et al.
2014/0234289 A1	8/2014	Liu et al.	2016/0074535 A1	3/2016	Ranganathan et al.
2014/0248702 A1	9/2014	Zhang et al.	2016/0076093 A1	3/2016	Shendure et al.
2014/0273037 A1	9/2014	Wu	2016/0090603 A1	3/2016	Carnes et al.
2014/0273226 A1	9/2014	Wu	2016/0090622 A1	3/2016	Liu et al.
2014/0273230 A1	9/2014	Chen et al.	2016/0115488 A1	4/2016	Zhang et al.
2014/0273234 A1	9/2014	Zhang et al.	2016/0138046 A1	5/2016	Wu
2014/0283156 A1	9/2014	Zador et al.	2016/0153003 A1	6/2016	Joung et al.
2014/0295556 A1	10/2014	Joung et al.	2016/0186214 A1	6/2016	Brouns et al.
2014/0295557 A1	10/2014	Joung et al.	2016/0200779 A1	7/2016	Liu et al.
2014/0342456 A1	11/2014	Mali et al.	2016/0201040 A1	7/2016	Liu et al.
2014/0342457 A1	11/2014	Mali et al.	2016/0201089 A1	7/2016	Gersbach et al.
2014/0342458 A1	11/2014	Mali et al.	2016/0206566 A1	7/2016	Lu et al.
2014/0349400 A1	11/2014	Jakimo et al.	2016/0208243 A1	7/2016	Zhang et al.
2014/0356867 A1	12/2014	Peter et al.	2016/0208288 A1	7/2016	Liu et al.
2014/0356956 A1	12/2014	Church et al.	2016/0215275 A1	7/2016	Zhong
2014/0356958 A1	12/2014	Mali et al.	2016/0215276 A1	7/2016	Liu et al.
2014/0356959 A1	12/2014	Church et al.	2016/0215300 A1	7/2016	May et al.
2014/0357523 A1	12/2014	Zeiner et al.	2016/0244784 A1	8/2016	Jacobson et al.
2014/0377868 A1	12/2014	Joung et al.	2016/0244829 A1	8/2016	Bang et al.
2015/0010526 A1	1/2015	Liu et al.	2016/0264934 A1	9/2016	Giallourakis et al.
2015/0031089 A1	1/2015	Lindstrom	2016/0272593 A1	9/2016	Ritter et al.
2015/0031132 A1	1/2015	Church et al.	2016/0272965 A1	9/2016	Zhang et al.
2015/0031133 A1	1/2015	Church et al.	2016/0281072 A1	9/2016	Zhang
2015/0044191 A1	2/2015	Liu et al.	2016/0298136 A1	10/2016	Chen et al.
2015/0044192 A1	2/2015	Liu et al.	2016/0304846 A1	10/2016	Liu et al.
2015/0044772 A1	2/2015	Zhao	2016/0304855 A1	10/2016	Stark et al.
2015/0050699 A1	2/2015	Siksnys et al.	2016/0312304 A1	10/2016	Sorrentino et al.
2015/0056177 A1	2/2015	Liu et al.	2016/0319262 A1	11/2016	Doudna et al.
2015/0056629 A1	2/2015	Guthrie-Honea	2016/0333389 A1	11/2016	Liu et al.
2015/0064138 A1	3/2015	Lu et al.	2016/0340622 A1	11/2016	Abdou
2015/0064789 A1	3/2015	Paschon et al.	2016/0340661 A1	11/2016	Cong et al.
2015/0071898 A1	3/2015	Liu et al.	2016/0340662 A1	11/2016	Zhang et al.
2015/0071899 A1	3/2015	Liu et al.	2016/0345578 A1	12/2016	Barrangou et al.
2015/0071900 A1	3/2015	Liu et al.	2016/0346360 A1	12/2016	Quake et al.
2015/0071901 A1	3/2015	Liu et al.	2016/0346361 A1	12/2016	Quake et al.
2015/0071902 A1	3/2015	Liu et al.	2016/0346362 A1	12/2016	Quake et al.
2015/0071903 A1	3/2015	Liu et al.	2016/0348074 A1	12/2016	Quake et al.
2015/0071906 A1	3/2015	Liu et al.	2016/0348096 A1	12/2016	Liu et al.
2015/0079680 A1	3/2015	Bradley et al.	2016/0350476 A1	12/2016	Quake et al.
2015/0079681 A1	3/2015	Zhang	2016/0355796 A1	12/2016	Davidson et al.
2015/0098954 A1	4/2015	Hyde et al.	2016/0369262 A1	12/2016	Reik et al.
2015/0118216 A1	4/2015	Liu et al.	2017/0009224 A1	1/2017	Liu et al.
2015/0128300 A1	5/2015	Warming et al.	2017/0009242 A1	1/2017	McKinley et al.
			2017/0014449 A1	1/2017	Bangera et al.
			2017/0020922 A1	1/2017	Wagner et al.
			2017/0022251 A1	1/2017	Rammensee et al.
			2017/0037432 A1	2/2017	Donohoue et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2017/0044520	A1	2/2017	Liu et al.	2018/0273939	A1	9/2018	Yu et al.
2017/0044592	A1	2/2017	Peter et al.	2018/0273976	A1	9/2018	Ümit et al.
2017/0053729	A1	2/2017	Kotani et al.	2018/0282722	A1	10/2018	Jakimo et al.
2017/0058271	A1	3/2017	Joung et al.	2018/0298391	A1	10/2018	Jakimo et al.
2017/0058272	A1	3/2017	Carter et al.	2018/0305688	A1	10/2018	Zhong
2017/0058298	A1	3/2017	Kennedy et al.	2018/0305704	A1	10/2018	Zhang
2017/0073663	A1	3/2017	Wang et al.	2018/0312822	A1	11/2018	Lee et al.
2017/0073670	A1	3/2017	Nishida et al.	2018/0312825	A1	11/2018	Liu et al.
2017/0087224	A1	3/2017	Quake	2018/0312828	A1	11/2018	Liu et al.
2017/0087225	A1	3/2017	Quake	2018/0312835	A1	11/2018	Yao et al.
2017/0088587	A1	3/2017	Quake	2018/0327756	A1	11/2018	Zhang et al.
2017/0088828	A1	3/2017	Quake	2018/0346927	A1	12/2018	Doudna et al.
2017/0107536	A1	4/2017	Zhang et al.	2018/0371497	A1	12/2018	Gill et al.
2017/0107560	A1	4/2017	Peter et al.	2019/0010481	A1	1/2019	Joung et al.
2017/0114367	A1	4/2017	Hu et al.	2019/0032053	A1	1/2019	Ji et al.
2017/0121693	A1	5/2017	Liu et al.	2019/0055543	A1	2/2019	Tran et al.
2017/0145394	A1	5/2017	Yeo et al.	2019/0055549	A1*	2/2019	Capurso C12N 15/102
2017/0145405	A1	5/2017	Tang et al.	2019/0062734	A1	2/2019	Cotta-Ramusino et al.
2017/0145438	A1	5/2017	Kantor	2019/0093099	A1	3/2019	Liu et al.
2017/0152528	A1	6/2017	Zhang	2019/0185883	A1	6/2019	Liu et al.
2017/0152787	A1	6/2017	Kubo et al.	2019/0218547	A1	7/2019	Lee et al.
2017/0159033	A1	6/2017	Kamtekar et al.	2019/0225955	A1	7/2019	Liu et al.
2017/0166928	A1	6/2017	Vyas et al.	2019/0233847	A1	8/2019	Savage et al.
2017/0175104	A1	6/2017	Doudna et al.	2019/0241633	A1	8/2019	Fotin-Mleczek et al.
2017/0175142	A1	6/2017	Zhang et al.	2019/0256842	A1	8/2019	Liu et al.
2017/0191047	A1	7/2017	Terns et al.	2019/0264202	A1	8/2019	Church et al.
2017/0191078	A1	7/2017	Zhang et al.	2019/0276816	A1	9/2019	Liu et al.
2017/0198269	A1	7/2017	Zhang et al.	2019/0309290	A1	10/2019	Neuteboom et al.
2017/0198277	A1	7/2017	Kmiec et al.	2019/0322992	A1	10/2019	Liu et al.
2017/0198302	A1	7/2017	Feng et al.	2019/0330619	A1	10/2019	Smith et al.
2017/0211061	A1	7/2017	Weiss et al.	2019/0352632	A1	11/2019	Liu et al.
2017/0224843	A1	8/2017	Deglon et al.	2019/0367891	A1	12/2019	Liu et al.
2017/0226522	A1	8/2017	Hu et al.	2020/0010818	A1	1/2020	Liu et al.
2017/0233703	A1	8/2017	Xie et al.	2020/0010835	A1	1/2020	Maianti et al.
2017/0233708	A1	8/2017	Liu et al.	2020/0056206	A1	2/2020	Tremblay et al.
2017/0233756	A1	8/2017	Begemann et al.	2020/0063127	A1	2/2020	Lu et al.
2017/0247671	A1	8/2017	Yung et al.	2020/0071722	A1	3/2020	Liu et al.
2017/0247703	A1	8/2017	Sloan et al.	2020/0109398	A1	4/2020	Rubens et al.
2017/0268022	A1	9/2017	Liu et al.	2020/0157570	A1	5/2020	Loiler
2017/0275648	A1	9/2017	Barrangou et al.	2020/0172931	A1	6/2020	Liu et al.
2017/0275665	A1	9/2017	Silas et al.	2020/0181619	A1	6/2020	Tang et al.
2017/0283797	A1	10/2017	Robb et al.	2020/0190493	A1	6/2020	Liu et al.
2017/0283831	A1	10/2017	Zhang et al.	2020/0216833	A1	7/2020	Liu et al.
2017/0306306	A1	10/2017	Potter et al.	2020/0255868	A1	8/2020	Liu et al.
2017/0314016	A1	11/2017	Kim et al.	2020/0277587	A1	9/2020	Liu et al.
2017/0362635	A1	12/2017	Chamberlain et al.	2020/0318116	A1	10/2020	Freier
2018/0023062	A1	1/2018	Lamb et al.	2020/0323984	A1	10/2020	Liu et al.
2018/0037877	A1	2/2018	Gao et al.	2020/0399619	A1	12/2020	Maianti et al.
2018/0064077	A1	3/2018	Dunham et al.	2020/0399626	A1	12/2020	Liu et al.
2018/0066258	A1	3/2018	Powell	2021/0054416	A1	2/2021	Liu et al.
2018/0068062	A1	3/2018	Zhang et al.	2021/0115428	A1	4/2021	Maianti et al.
2018/0073012	A1	3/2018	Liu et al.	2021/0196809	A1	7/2021	Maianti et al.
2018/0080051	A1	3/2018	Sheikh et al.	2021/0198330	A1	7/2021	Liu et al.
2018/0087046	A1	3/2018	Badran et al.	2021/0214698	A1	7/2021	Liu et al.
2018/0100147	A1	4/2018	Yates et al.	2021/0230577	A1	7/2021	Liu et al.
2018/0105867	A1	4/2018	Xiao et al.	2021/0254127	A1	8/2021	Liu et al.
2018/0119118	A1	5/2018	Lu et al.	2021/0315994	A1	10/2021	Liu et al.
2018/0127759	A1	5/2018	Lu et al.	2021/0317440	A1	10/2021	Liu et al.
2018/0127780	A1	5/2018	Liu et al.	2022/0033785	A1	2/2022	Liu et al.
2018/0155708	A1	6/2018	Church et al.	2022/0119785	A1	4/2022	Liu et al.
2018/0155720	A1	6/2018	Donohoue et al.	2022/0170013	A1	6/2022	Liu et al.
2018/0163213	A1	6/2018	Aneja et al.	2022/0177877	A1	6/2022	Church et al.
2018/0170984	A1	6/2018	Harris et al.	2022/0204975	A1	6/2022	Liu et al.
2018/0179503	A1	6/2018	Maianti et al.	2022/0213507	A1	7/2022	Liu et al.
2018/0179547	A1	6/2018	Zhang et al.	2022/0220462	A1	7/2022	Liu et al.
2018/0201921	A1	7/2018	Malcolm	2022/0249697	A1	8/2022	Liu et al.
2018/0230464	A1	8/2018	Zhong	2022/0282275	A1	9/2022	Liu et al.
2018/0230471	A1	8/2018	Storici et al.	2022/0290115	A1	9/2022	Liu et al.
2018/0236081	A1	8/2018	Liu et al.	2022/0307001	A1	9/2022	Liu et al.
2018/0237787	A1	8/2018	Maianti et al.	2022/0307003	A1	9/2022	Liu et al.
2018/0245066	A1	8/2018	Yao et al.	2022/0315906	A1	10/2022	Liu et al.
2018/0245075	A1	8/2018	Khalil et al.	2022/0356469	A1	11/2022	Liu et al.
2018/0258418	A1	9/2018	Kim	2022/0380740	A1	12/2022	Liu et al.
2018/0265864	A1	9/2018	Li et al.	2022/0389395	A1	12/2022	Liu et al.
2018/0273935	A1	9/2018	Lane et al.	2023/0002745	A1	1/2023	Liu et al.
				2023/0021641	A1	1/2023	Liu et al.
				2023/0056852	A1	2/2023	Liu et al.
				2023/0058176	A1	2/2023	Liu et al.
				2023/0078265	A1	3/2023	Liu et al.

(56)	References Cited			CN	104651398	A	5/2015
	U.S. PATENT DOCUMENTS			CN	104651399	A	5/2015
				CN	104651401	A	5/2015
				CN	104673816	A	6/2015
2023/0086199	A1	3/2023	Liu et al.	CN	104725626	A	6/2015
2023/0090221	A1	3/2023	Liu et al.	CN	104726449	A	6/2015
2023/0108687	A1	4/2023	Liu et al.	CN	104726494	A	6/2015
2023/0123669	A1	4/2023	Liu et al.	CN	104745626	A	7/2015
2023/0127008	A1	4/2023	Liu et al.	CN	104762321	A	7/2015
2023/0159913	A1	5/2023	Liu et al.	CN	104805118	A	7/2015
2023/0193295	A1	6/2023	Maianti et al.	CN	104846010	A	8/2015
2023/0220374	A1	7/2023	Liu et al.	CN	104805078	A	9/2015
2023/0272425	A1	8/2023	Liu et al.	CN	104805099	A	9/2015
2023/0279443	A1	9/2023	Liu et al.	CN	104894068	A	9/2015
2023/0332144	A1	10/2023	Liu et al.	CN	104894075	A	9/2015
2023/0340465	A1	10/2023	Liu et al.	CN	104928321	A	9/2015
2023/0340466	A1	10/2023	Liu et al.	CN	105039339	A	11/2015
2023/0340467	A1	10/2023	Liu et al.	CN	105039399	A	11/2015
2023/0348883	A1	11/2023	Liu et al.	CN	105063061	A	11/2015
2023/0357766	A1	11/2023	Liu et al.	CN	105087620	A	11/2015
2023/0383289	A1	11/2023	Liu et al.	CN	105112422	A	12/2015
2024/0035017	A1	2/2024	Liu et al.	CN	105112445	A	12/2015
2024/0076652	A1	3/2024	Liu et al.	CN	105112519	A	12/2015
2024/0110166	A1	4/2024	Maianti et al.	CN	105121648	A	12/2015
2024/0124866	A1	4/2024	Liu et al.	CN	105132427	A	12/2015
2024/0173430	A1	5/2024	Liu et al.	CN	105132451	A	12/2015
2024/0209329	A1	6/2024	Liu et al.	CN	105177038	A	12/2015
2024/0229077	A1	7/2024	Liu et al.	CN	105177126	A	12/2015
2024/0271116	A1	8/2024	Maianti et al.	CN	105210981	A	1/2016
2024/0287487	A1	8/2024	Liu et al.	CN	105219799	A	1/2016
2024/0327872	A1	10/2024	Liu et al.	CN	105238806	A	1/2016
				CN	105255937	A	1/2016
				CN	105274144	A	1/2016
				CN	105296518	A	2/2016
				CN	105296537	A	2/2016
AU	2015101792	A4	1/2016	CN	105316324	A	2/2016
AU	2012354062	B2	9/2017	CN	105316327	A	2/2016
BR	112015013786	A2	7/2017	CN	105316337	A	2/2016
CA	2480696	A1	10/2003	CN	105331607	A	2/2016
CA	2894668	A1	6/2014	CN	105331608	A	2/2016
CA	2894681	A1	6/2014	CN	105331609	A	2/2016
CA	2894684	A1	6/2014	CN	105331627	A	2/2016
CA	2852593	A1	11/2015	CN	105400773	A	3/2016
CN	1069962	A	3/1993	CN	105400779	A	3/2016
CN	101460619	A	6/2009	CN	105400810	A	3/2016
CN	101873862	A	10/2010	CN	105441451	A	3/2016
CN	102057039	A	5/2011	CN	105462968	A	4/2016
CN	102892777	A	1/2013	CN	105463003	A	4/2016
CN	103224947	A	7/2013	CN	105463027	A	4/2016
CN	103233028	A	8/2013	CN	105492608	A	4/2016
CN	103388006	A	11/2013	CN	105492609	A	4/2016
CN	103614415	A	3/2014	CN	105505976	A	4/2016
CN	103642836	A	3/2014	CN	105505979	A	4/2016
CN	103668472	A	3/2014	CN	105518134	A	4/2016
CN	103820441	A	5/2014	CN	105518135	A	4/2016
CN	103820454	A	5/2014	CN	105518137	A	4/2016
CN	103911376	A	7/2014	CN	105518138	A	4/2016
CN	103923911	A	7/2014	CN	105518139	A	4/2016
CN	103088008	A	8/2014	CN	105518140	A	4/2016
CN	103981211	A	8/2014	CN	105543228	A	5/2016
CN	103981212	A	8/2014	CN	105543266	A	5/2016
CN	104004778	A	8/2014	CN	105543270	A	5/2016
CN	104004782	A	8/2014	CN	105567688	A	5/2016
CN	104017821	A	9/2014	CN	105567689	A	5/2016
CN	104109687	A	9/2014	CN	105567734	A	5/2016
CN	104178461	A	12/2014	CN	105567735	A	5/2016
CN	104342457	A	2/2015	CN	105567738	A	5/2016
CN	104404036	A	3/2015	CN	105593367	A	5/2016
CN	104450774	A	3/2015	CN	105594664	A	5/2016
CN	104480144	A	4/2015	CN	105602987	A	5/2016
CN	104498493	A	4/2015	CN	105624146	A	6/2016
CN	104504304	A	4/2015	CN	105624187	A	6/2016
CN	104531704	A	4/2015	CN	105646719	A	6/2016
CN	104531705	A	4/2015	CN	105647922	A	6/2016
CN	104560864	A	4/2015	CN	105647962	A	6/2016
CN	104561095	A	4/2015	CN	105647968	A	6/2016
CN	104593418	A	5/2015	CN	105647969	A	6/2016
CN	104593422	A	5/2015	CN	105671070	A	6/2016
CN	104611370	A	5/2015	CN	105671083	A	6/2016
CN	104651392	A	5/2015	CN			

(56)	References Cited			CN	106399306	A	2/2017
	FOREIGN PATENT DOCUMENTS			CN	106399311	A	2/2017
				CN	106399360	A	2/2017
				CN	106399367	A	2/2017
				CN	106399375	A	2/2017
CN	105695485	A	6/2016	CN	106399377	A	2/2017
CN	105779448	A	7/2016	CN	106434651	A	2/2017
CN	105779449	A	7/2016	CN	106434663	A	2/2017
CN	105802980	A	7/2016	CN	106434688	A	2/2017
CN	105821039	A	8/2016	CN	106434737	A	2/2017
CN	105821040	A	8/2016	CN	106434748	A	2/2017
CN	105821049	A	8/2016	CN	106434752	A	2/2017
CN	105821072	A	8/2016	CN	106434782	A	2/2017
CN	105821075	A	8/2016	CN	106446600	A	2/2017
CN	105821116	A	8/2016	CN	106479985	A	3/2017
CN	105838733	A	8/2016	CN	106480027	A	3/2017
CN	105861547	A	8/2016	CN	106480036	A	3/2017
CN	105861552	A	8/2016	CN	106480067	A	3/2017
CN	105861554	A	8/2016	CN	106480080	A	3/2017
CN	105886498	A	8/2016	CN	106480083	A	3/2017
CN	105886534	A	8/2016	CN	106480097	A	3/2017
CN	105886616	A	8/2016	CN	106544351	A	3/2017
CN	105907758	A	8/2016	CN	106544353	A	3/2017
CN	105907785	A	8/2016	CN	106544357	A	3/2017
CN	105925608	A	9/2016	CN	106554969	A	4/2017
CN	105934516	A	9/2016	CN	106566838	A	4/2017
CN	105950560	A	9/2016	CN	106701763	A	5/2017
CN	105950626	A	9/2016	CN	106701808	A	5/2017
CN	105950633	A	9/2016	CN	106701818	A	5/2017
CN	105950639	A	9/2016	CN	106701823	A	5/2017
CN	105985985	A	10/2016	CN	106701830	A	5/2017
CN	106011104	A	10/2016	CN	106754912	A	5/2017
CN	106011150	A	10/2016	CN	106755026	A	5/2017
CN	106011167	A	10/2016	CN	106755077	A	5/2017
CN	106011171	A	10/2016	CN	106755088	A	5/2017
CN	106032540	A	10/2016	CN	106755091	A	5/2017
CN	106047803	A	10/2016	CN	106755097	A	5/2017
CN	106047877	A	10/2016	CN	106755424	A	5/2017
CN	106047930	A	10/2016	CN	106801056	A	6/2017
CN	106086008	A	11/2016	CN	106834323	A	6/2017
CN	106086028	A	11/2016	CN	106834341	A	6/2017
CN	106086061	A	11/2016	CN	106834347	A	6/2017
CN	106086062	A	11/2016	CN	106845151	A	6/2017
CN	106103475	A	11/2016	CN	106868008	A	6/2017
CN	106109417	A	11/2016	CN	106868031	A	6/2017
CN	106119275	A	11/2016	CN	106906240	A	6/2017
CN	106119283	A	11/2016	CN	106906242	A	6/2017
CN	106148286	A	11/2016	CN	106916820	A	7/2017
CN	106148370	A	11/2016	CN	106916852	A	7/2017
CN	106148416	A	11/2016	CN	106939303	A	7/2017
CN	106167525	A	11/2016	CN	106947750	A	7/2017
CN	106167808	A	11/2016	CN	106947780	A	7/2017
CN	106167810	A	11/2016	CN	106957830	A	7/2017
CN	106167821	A	11/2016	CN	106957831	A	7/2017
CN	106172238	A	12/2016	CN	106957844	A	7/2017
CN	106190903	A	12/2016	CN	106957855	A	7/2017
CN	106191057	A	12/2016	CN	106957858	A	7/2017
CN	106191061	A	12/2016	CN	106967697	A	7/2017
CN	106191062	A	12/2016	CN	106967726	A	7/2017
CN	106191064	A	12/2016	CN	106978428	A	7/2017
CN	106191071	A	12/2016	CN	106987570	A	7/2017
CN	106191099	A	12/2016	CN	106987757	A	7/2017
CN	106191107	A	12/2016	CN	107012164	A	8/2017
CN	106191113	A	12/2016	CN	107012174	A	8/2017
CN	106191114	A	12/2016	CN	107012213	A	8/2017
CN	106191116	A	12/2016	CN	107012250	A	8/2017
CN	106191124	A	12/2016	CN	107022562	A	8/2017
CN	106222177	A	12/2016	CN	107034188	A	8/2017
CN	106222193	A	12/2016	CN	107034218	A	8/2017
CN	106222203	A	12/2016	CN	107034229	A	8/2017
CN	106232823	A	12/2016	CN	107043775	A	8/2017
CN	106244555	A	12/2016	CN	107043779	A	8/2017
CN	106244557	A	12/2016	CN	107043787	A	8/2017
CN	106244591	A	12/2016	CN	107058320	A	8/2017
CN	106244609	A	12/2016	CN	107058328	A	8/2017
CN	106282241	A	1/2017	CN	107058358	A	8/2017
CN	106318934	A	1/2017	CN	107058372	A	8/2017
CN	106318973	A	1/2017	CN	107083392	A	8/2017
CN	106350540	A	1/2017	CN	107099533	A	8/2017
CN	106367435	A	2/2017				

(56)	References Cited			CN	107619829	A	1/2018
	FOREIGN PATENT DOCUMENTS			CN	107619837	A	1/2018
				CN	107630006	A	1/2018
				CN	107630041	A	1/2018
CN	107099850	A	8/2017	CN	107630042	A	1/2018
CN	107119053	A	9/2017	CN	107630043	A	1/2018
CN	107119071	A	9/2017	CN	107641631	A	1/2018
CN	107129999	A	9/2017	CN	107653256	A	2/2018
CN	107130000	A	9/2017	CN	107686848	A	2/2018
CN	107142272	A	9/2017	CN	206970581	U	2/2018
CN	107142282	A	9/2017	CN	107760652	A	3/2018
CN	107177591	A	9/2017	CN	107760663	A	3/2018
CN	107177595	A	9/2017	CN	107760684	A	3/2018
CN	107177625	A	9/2017	CN	107760715	A	3/2018
CN	107177631	A	9/2017	CN	107784200	A	3/2018
CN	107190006	A	9/2017	CN	107794272	A	3/2018
CN	107190008	A	9/2017	CN	107794276	A	3/2018
CN	107217042	A	9/2017	CN	107815463	A	3/2018
CN	107217075	A	9/2017	CN	107828738	A	3/2018
CN	107227307	A	10/2017	CN	107828794	A	3/2018
CN	107227352	A	10/2017	CN	107828826	A	3/2018
CN	107236737	A	10/2017	CN	107828874	A	3/2018
CN	107236739	A	10/2017	CN	107858346	A	3/2018
CN	107236741	A	10/2017	CN	107858373	A	3/2018
CN	107245502	A	10/2017	CN	107880132	A	4/2018
CN	107254485	A	10/2017	CN	107881184	A	4/2018
CN	107266541	A	10/2017	CN	107893074	A	4/2018
CN	107267515	A	10/2017	CN	107893075	A	4/2018
CN	107287245	A	10/2017	CN	107893076	A	4/2018
CN	107298701	A	10/2017	CN	107893080	A	4/2018
CN	107299114	A	10/2017	CN	107893086	A	4/2018
CN	107304435	A	10/2017	CN	107904261	A	4/2018
CN	107312785	A	11/2017	CN	107937427	A	4/2018
CN	107312793	A	11/2017	CN	107937432	A	4/2018
CN	107312795	A	11/2017	CN	107937501	A	4/2018
CN	107312798	A	11/2017	CN	107974466	A	5/2018
CN	107326042	A	11/2017	CN	107988229	A	5/2018
CN	107326046	A	11/2017	CN	107988246	A	5/2018
CN	107354156	A	11/2017	CN	107988256	A	5/2018
CN	107354173	A	11/2017	CN	107988268	A	5/2018
CN	107356793	A	11/2017	CN	108018316	A	5/2018
CN	107362372	A	11/2017	CN	108034656	A	5/2018
CN	107365786	A	11/2017	CN	108048466	A	5/2018
CN	107365804	A	11/2017	CN	108102940	A	6/2018
CN	107384894	A	11/2017	CN	108103090	A	6/2018
CN	107384922	A	11/2017	CN	108103092	A	6/2018
CN	107384926	A	11/2017	CN	108103098	A	6/2018
CN	107400677	A	11/2017	CN	108103586	A	6/2018
CN	107418974	A	12/2017	CN	108148835	A	6/2018
CN	107435051	A	12/2017	CN	108148837	A	6/2018
CN	107435069	A	12/2017	CN	108148873	A	6/2018
CN	107446922	A	12/2017	CN	108192956	A	6/2018
CN	107446923	A	12/2017	CN	108243575	A	7/2018
CN	107446924	A	12/2017	CN	108251423	A	7/2018
CN	107446932	A	12/2017	CN	108251451	A	7/2018
CN	107446951	A	12/2017	CN	108251452	A	7/2018
CN	107446954	A	12/2017	CN	108342480	A	7/2018
CN	107460196	A	12/2017	CN	108359691	A	8/2018
CN	107474129	A	12/2017	CN	108359712	A	8/2018
CN	107475300	A	12/2017	CN	108384784	A	8/2018
CN	107488649	A	12/2017	CN	108396027	A	8/2018
CN	107502608	A	12/2017	CN	108410877	A	8/2018
CN	107502618	A	12/2017	CN	108410906	A	8/2018
CN	107513531	A	12/2017	CN	108410907	A	8/2018
CN	107519492	A	12/2017	CN	108410911	A	8/2018
CN	107523567	A	12/2017	CN	108424931	A	8/2018
CN	107523583	A	12/2017	CN	108441519	A	8/2018
CN	107541525	A	1/2018	CN	108441520	A	8/2018
CN	107557373	A	1/2018	CN	108472314	A	8/2018
CN	107557378	A	1/2018	CN	108486108	A	9/2018
CN	107557381	A	1/2018	CN	108486111	A	9/2018
CN	107557390	A	1/2018	CN	108486145	A	9/2018
CN	107557393	A	1/2018	CN	108486146	A	9/2018
CN	107557394	A	1/2018	CN	108486154	A	9/2018
CN	107557455	A	1/2018	CN	108486159	A	9/2018
CN	107574179	A	1/2018	CN	108486234	A	9/2018
CN	107586777	A	1/2018	CN	108504657	A	9/2018
CN	107586779	A	1/2018	CN	108504685	A	9/2018
CN	107604003	A	1/2018	CN	108504693	A	9/2018

(56)	References Cited			GB	2531454	A	4/2016
	FOREIGN PATENT DOCUMENTS			GB	2542653	A	3/2017
				HK	1208045	A1	2/2016
				JP	2007-501626	A	2/2007
CN	108513575	A	9/2018	JP	2008-515405	A	5/2008
CN	108546712	A	9/2018	JP	2010-033344	A	2/2010
CN	108546717	A	9/2018	JP	2010-535744	A	11/2010
CN	108546718	A	9/2018	JP	2010-539929	A	12/2010
CN	108559730	A	9/2018	JP	2011-081011	A	4/2011
CN	108559732	A	9/2018	JP	2011-523353	A	8/2011
CN	108559745	A	9/2018	JP	2012-525146	A	10/2012
CN	108559760	A	9/2018	JP	2012-210172	A	11/2012
CN	108570479	A	9/2018	JP	2012-531909	A	12/2012
CN	108588071	A	9/2018	JP	2015-523856	A	8/2015
CN	108588123	A	9/2018	JP	2015-532654	A	11/2015
CN	108588128	A	9/2018	JP	2016-525888	A	9/2016
CN	108588182	A	9/2018	JP	2016-534132	A	11/2016
CN	108610399	A	10/2018	JP	2017-500035	A	1/2017
CN	108611364	A	10/2018	JP	2018-521045	A	8/2018
CN	108624622	A	10/2018	JP	2019-506123	A	2/2019
CN	108642053	A	10/2018	JP	6629734	B2	1/2020
CN	108642055	A	10/2018	JP	6633524	B2	1/2020
CN	108642077	A	10/2018	JP	6830517	B2	2/2021
CN	108642078	A	10/2018	JP	7324523	B2	8/2023
CN	108642090	A	10/2018	KR	101584933	B1	1/2016
CN	108690844	A	10/2018	KR	2016-0050069	A	5/2016
CN	108699542	A	10/2018	KR	20160133380	A	11/2016
CN	108707604	A	10/2018	KR	20170037025	A	4/2017
CN	108707620	A	10/2018	KR	20170037028	A	4/2017
CN	108707621	A	10/2018	KR	101748575	B1	6/2017
CN	108707628	A	10/2018	KR	20170128137	A	11/2017
CN	108707629	A	10/2018	KR	2018-0022465	A	3/2018
CN	108715850	A	10/2018	RU	2016104674	A	8/2017
CN	108728476	A	11/2018	RU	2634395	C1	10/2017
CN	108728486	A	11/2018	RU	2652899	C1	5/2018
CN	108753772	A	11/2018	RU	2015128057	A	3/2019
CN	108753783	A	11/2018	RU	2015128098	A	3/2019
CN	108753813	A	11/2018	RU	2687451	C1	5/2019
CN	108753817	A	11/2018	RU	2019112514	A	6/2019
CN	108753832	A	11/2018	RU	2019127300	A	9/2019
CN	108753835	A	11/2018	RU	2701850	C2	10/2019
CN	108753836	A	11/2018	SG	10201707569	Y	10/2017
CN	108795902	A	11/2018	SG	10201710486X		1/2018
CN	108822217	A	11/2018	SG	10201710487		1/2018
CN	108823248	A	11/2018	SG	10201710488	T	1/2018
CN	108823249	A	11/2018	TW	1608100	B	12/2017
CN	108823291	A	11/2018	TW	201809272	A	3/2018
CN	108841845	A	11/2018	TW	2018-29773	A	8/2018
CN	108853133	A	11/2018	WO	WO 1990/002809		3/1990
CN	108866093	A	11/2018	WO	WO 1991/003162	A1	3/1991
CN	108893529	A	11/2018	WO	WO 1991/016024	A1	10/1991
CN	108913664	A	11/2018	WO	WO 1991/017271	A1	11/1991
CN	108913691	A	11/2018	WO	WO 1991/017424	A1	11/1991
CN	108913714	A	11/2018	WO	WO 1992/006188	A2	4/1992
CN	108913717	A	11/2018	WO	WO 1992/006200	A1	4/1992
CN	208034188	U	11/2018	WO	WO 1992/007065	A1	4/1992
CN	109517841	A	3/2019	WO	WO 1993/015187	A1	8/1993
EP	0264166	A1	4/1988	WO	WO 1993/024641	A2	12/1993
EP	0321201	B2	6/1989	WO	WO 1994/018316	A2	8/1994
EP	0519463	A1	12/1992	WO	WO 1994/026877	A1	11/1994
EP	1085892	A2	3/2001	WO	WO 1996/004403	A1	2/1996
EP	1092770	A2	4/2001	WO	WO 1996/010640	A1	4/1996
EP	2604255	A1	6/2013	WO	WO 1998/032845	A1	7/1998
EP	2840140	A1	2/2015	WO	WO 2001/036452	A2	5/2001
EP	2877490	A2	6/2015	WO	WO 2001/038547	A2	5/2001
EP	2966170	A1	1/2016	WO	WO 2002/059296	A2	8/2002
EP	3009511	A2	4/2016	WO	WO 2002/068676	A2	9/2002
EP	3031921	A1	6/2016	WO	WO 2002/103028	A2	12/2002
EP	3045537	A1	7/2016	WO	WO 2003/004608	A2	1/2003
EP	3115457	A	1/2017	WO	WO 2004/007684	A2	1/2004
EP	3144390	A1	3/2017	WO	WO 2005/014791	A2	2/2005
EP	3199632	A1	8/2017	WO	WO 2005/019415	A2	3/2005
EP	3216867	A1	9/2017	WO	WO 2006/002547	A1	1/2006
EP	3252160	A1	12/2017	WO	WO 2006/042112	A2	4/2006
EP	2498823	B1	8/2018	WO	WO 2007/025097	A2	3/2007
EP	3365437	A1	8/2018	WO	WO 2007/037444	A1	4/2007
EP	3450553	B1	12/2019	WO	WO 2007/066923	A1	6/2007
ES	2740248	T3	2/2020	WO	WO 2007/136815	A2	11/2007
GB	2528177	A	1/2016	WO	WO 2007/143574	A1	12/2007

(56) References Cited

FOREIGN PATENT DOCUMENTS

WO	WO 2008/005529	A2	1/2008	WO	WO 2013/192278	A1	12/2013
WO	WO 2008/108989	A2	9/2008	WO	WO 2013/142378	A9	1/2014
WO	WO 2009/002418	A2	12/2008	WO	WO 2014/004336	A2	1/2014
WO	WO 2009/019317	A1	2/2009	WO	WO 2014/005042	A2	1/2014
WO	WO 2009/098290	A1	8/2009	WO	WO 2014/011237	A1	1/2014
WO	WO 2009/134808	A2	11/2009	WO	WO 2014/011901	A2	1/2014
WO	WO 2010/011961	A2	1/2010	WO	WO 2014/018423	A2	1/2014
WO	WO 2010/012902	A1	2/2010	WO	WO 2014/020608	A1	2/2014
WO	WO 2010/028347	A2	3/2010	WO	WO 2014/022120	A1	2/2014
WO	WO 2010/054108	A2	5/2010	WO	WO 2014/022702	A2	2/2014
WO	WO 2010/054154	A2	5/2010	WO	WO 2014/036219	A2	3/2014
WO	WO 2010/068289	A2	6/2010	WO	WO 2014/039513	A2	3/2014
WO	WO 2010/075424	A2	7/2010	WO	WO 2014/039523	A1	3/2014
WO	WO 2010/091122	A1	8/2010	WO	WO 2014/039585	A2	3/2014
WO	WO 2010/102257	A2	9/2010	WO	WO 2014/039684	A1	3/2014
WO	WO 2010/104749	A2	9/2010	WO	WO 2014/039692	A2	3/2014
WO	WO 2010/129019	A2	11/2010	WO	WO 2014/039702	A2	3/2014
WO	WO 2010/129023	A2	11/2010	WO	WO 2014/039872	A1	3/2014
WO	WO 2010/132092	A2	11/2010	WO	WO 2014/039970	A1	3/2014
WO	WO 2010/144150	A2	12/2010	WO	WO 2014/041327	A1	3/2014
WO	WO 2011/002503	A1	1/2011	WO	WO 2014/043143	A1	3/2014
WO	WO 2011/017293	A2	2/2011	WO	WO 2014/047103	A2	3/2014
WO	WO 2011/053868	A1	5/2011	WO	WO 2014/055782	A1	4/2014
WO	WO 2011/053982	A2	5/2011	WO	WO 2014/059173	A2	4/2014
WO	WO 2011/068810	A1	6/2011	WO	WO 2014/059255	A1	4/2014
WO	WO 2011/075627	A1	6/2011	WO	WO 2014/065596	A1	5/2014
WO	WO 2011/091311	A2	7/2011	WO	WO 2014/066505	A1	5/2014
WO	WO 2011/091396	A1	7/2011	WO	WO 2014/068346	A2	5/2014
WO	WO 2011/109031	A1	9/2011	WO	WO 2014/070887	A1	5/2014
WO	WO 2011/143124	A2	11/2011	WO	WO 2014/071006	A1	5/2014
WO	WO 2011/147590	A2	12/2011	WO	WO 2014/071219	A1	5/2014
WO	WO 2011/159369	A1	12/2011	WO	WO 2014/071235	A1	5/2014
WO	WO 2012/054726	A1	4/2012	WO	WO 2014/072941	A1	5/2014
WO	WO 2012/061815	A2	5/2012	WO	WO 2014/081729	A1	5/2014
WO	WO 2012/065043	A2	5/2012	WO	WO 2014/081730	A1	5/2014
WO	WO 2012/088381	A2	6/2012	WO	WO 2014/081855	A1	5/2014
WO	WO 2012/125445	A2	9/2012	WO	WO 2014/082644	A1	6/2014
WO	WO 2012/138927	A2	10/2012	WO	WO 2014/085261	A1	6/2014
WO	WO 2012/149470	A1	11/2012	WO	WO 2014/085593	A1	6/2014
WO	WO 2012/158985	A2	11/2012	WO	WO 2014/085830	A2	6/2014
WO	WO 2012/158986	A2	11/2012	WO	WO 2014/089212	A1	6/2014
WO	WO 2012/164565	A1	12/2012	WO	WO 2014/089290	A1	6/2014
WO	WO 2012/170930	A1	12/2012	WO	WO 2014/089348	A1	6/2014
WO	WO 2013/012674	A1	1/2013	WO	WO 2014/089513	A1	6/2014
WO	WO 2013/013105	A1	1/2013	WO	WO 2014/089533	A2	6/2014
WO	WO 2013/039857	A1	3/2013	WO	WO 2014/089541	A2	6/2014
WO	WO 2013/039861	A2	3/2013	WO	WO 2014/093479	A1	6/2014
WO	WO 2013/040093	A2	3/2013	WO	WO 2014/093595	A1	6/2014
WO	WO 2013/045632	A1	4/2013	WO	WO 2014/093622	A2	6/2014
WO	WO 2013/047844	A1	4/2013	WO	WO 2014/093635	A1	6/2014
WO	WO 2013/066438	A2	5/2013	WO	WO 2014/093655	A2	6/2014
WO	WO 2013/086441	A2	6/2013	WO	WO 2014/093661	A2	6/2014
WO	WO 2013/086444	A2	6/2013	WO	WO 2014/093694	A1	6/2014
WO	WO 2013/098244	A1	7/2013	WO	WO 2014/093701	A1	6/2014
WO	WO 2013/119602	A1	8/2013	WO	WO 2014/093709	A1	6/2014
WO	WO 2013/120022	A2	8/2013	WO	WO 2014/093712	A1	6/2014
WO	WO 2013/122617	A1	8/2013	WO	WO 2014/093718	A1	6/2014
WO	WO 2013/126794	A1	8/2013	WO	WO 2014/093736	A1	6/2014
WO	WO 2013/130683	A2	9/2013	WO	WO 2014/093768	A1	6/2014
WO	WO 2013/130824	A1	9/2013	WO	WO 2014/093852	A1	6/2014
WO	WO 2013/141680	A1	9/2013	WO	WO 2014/096972	A2	6/2014
WO	WO 2013/142578	A1	9/2013	WO	WO 2014/099744	A1	6/2014
WO	WO 2013/152359	A1	10/2013	WO	WO 2014/099750	A2	6/2014
WO	WO 2013/160230	A1	10/2013	WO	WO 2014/104878	A1	7/2014
WO	WO 2013/166315	A1	11/2013	WO	WO 2014/110006	A1	7/2014
WO	WO 2013/169398	A2	11/2013	WO	WO 2014/110552	A1	7/2014
WO	WO 2013/169802	A1	11/2013	WO	WO 2014/113493	A1	7/2014
WO	WO 2013/176772	A1	11/2013	WO	WO 2014/123967	A2	8/2014
WO	WO 2013/176915	A1	11/2013	WO	WO 2014/124226	A1	8/2014
WO	WO 2013/176916	A1	11/2013	WO	WO 2014/125668	A1	8/2014
WO	WO 2013/181440	A1	12/2013	WO	WO 2014/127287	A1	8/2014
WO	WO 2013/186754	A2	12/2013	WO	WO 2014/128324	A1	8/2014
WO	WO 2013/188037	A2	12/2013	WO	WO 2014/128659	A1	8/2014
WO	WO 2013/188522	A2	12/2013	WO	WO 2014/130706	A1	8/2014
WO	WO 2013/188638	A2	12/2013	WO	WO 2014/130955	A1	8/2014
				WO	WO 2014/131833	A1	9/2014
				WO	WO 2014/138379	A1	9/2014
				WO	WO 2014/143381	A1	9/2014
				WO	WO 2014/144094	A1	9/2014

(56) References Cited

FOREIGN PATENT DOCUMENTS

WO	WO 2014/144155	A1	9/2014	WO	WO 2015/026887	A1	2/2015
WO	WO 2014/144288	A1	9/2014	WO	WO 2015/027134	A1	2/2015
WO	WO 2014/144592	A2	9/2014	WO	WO 2015/028969	A2	3/2015
WO	WO 2014/144761	A2	9/2014	WO	WO 2015/030881	A1	3/2015
WO	WO 2014/144951	A1	9/2014	WO	WO 2015/031619	A1	3/2015
WO	WO 2014/145599	A2	9/2014	WO	WO 2015/031775	A1	3/2015
WO	WO 2014/145736	A2	9/2014	WO	WO 2015/032494	A2	3/2015
WO	WO 2014/150624	A1	9/2014	WO	WO 2015/033293	A1	3/2015
WO	WO 2014/152432	A2	9/2014	WO	WO 2015/034872	A2	3/2015
WO	WO 2014/152940	A1	9/2014	WO	WO 2015/034885	A1	3/2015
WO	WO 2014/153118	A1	9/2014	WO	WO 2015/035136	A2	3/2015
WO	WO 2014/153470	A2	9/2014	WO	WO 2015/035139	A2	3/2015
WO	WO 2014/158593	A1	10/2014	WO	WO 2015/035162	A2	3/2015
WO	WO 2014/161821	A1	10/2014	WO	WO 2015/040075	A1	3/2015
WO	WO 2014/164466	A1	10/2014	WO	WO 2015/040402	A1	3/2015
WO	WO 2014/165177	A1	10/2014	WO	WO 2015/042393	A2	3/2015
WO	WO 2014/165349	A1	10/2014	WO	WO 2015/042585	A1	3/2015
WO	WO 2014/165612	A2	10/2014	WO	WO 2015/048577	A2	4/2015
WO	WO 2014/165707	A2	10/2014	WO	WO 2015/048690	A1	4/2015
WO	WO 2014/165825	A2	10/2014	WO	WO 2015/048707	A2	4/2015
WO	WO 2014/172458	A1	10/2014	WO	WO 2015/048801	A2	4/2015
WO	WO 2014/172470	A2	10/2014	WO	WO 2015/049897	A1	4/2015
WO	WO 2014/172489	A2	10/2014	WO	WO 2015/051191	A1	4/2015
WO	WO 2014/173955	A1	10/2014	WO	WO 2015/052133	A1	4/2015
WO	WO 2014/182700	A1	11/2014	WO	WO 2015/052231	A2	4/2015
WO	WO 2014/183071	A2	11/2014	WO	WO 2015/052335	A1	4/2015
WO	WO 2014/184143	A1	11/2014	WO	WO 2015/053995	A1	4/2015
WO	WO 2014/184741	A1	11/2014	WO	WO 2015/054253	A1	4/2015
WO	WO 2014/184744	A1	11/2014	WO	WO 2015/054315	A1	4/2015
WO	WO 2014/186585	A2	11/2014	WO	WO 2015/057671	A1	4/2015
WO	WO 2014/186686	A2	11/2014	WO	WO 2015/057834	A1	4/2015
WO	WO 2014/190181	A1	11/2014	WO	WO 2015/057852	A1	4/2015
WO	WO 2014/191128	A1	12/2014	WO	WO 2015/057976	A1	4/2015
WO	WO 2014/191518	A1	12/2014	WO	WO 2015/057980	A1	4/2015
WO	WO 2014/191521	A2	12/2014	WO	WO 2015/059265	A1	4/2015
WO	WO 2014/191525	A1	12/2014	WO	WO 2015/065964	A1	5/2015
WO	WO 2014/191527	A1	12/2014	WO	WO 2015/066119	A1	5/2015
WO	WO 2014/193583	A2	12/2014	WO	WO 2015/066634	A2	5/2015
WO	WO 2014/194190	A1	12/2014	WO	WO 2015/066636	A2	5/2015
WO	WO 2014/197568	A2	12/2014	WO	WO 2015/066637	A1	5/2015
WO	WO 2014/197748	A2	12/2014	WO	WO 2015/066638	A2	5/2015
WO	WO 2014/199358	A1	12/2014	WO	WO 2015/066643	A1	5/2015
WO	WO 2014/200659	A1	12/2014	WO	WO 2015/069682	A2	5/2015
WO	WO 2014/201015	A2	12/2014	WO	WO 2015/070083	A1	5/2015
WO	WO 2014/204578	A1	12/2014	WO	WO 2015/070193	A1	5/2015
WO	WO 2014/204723	A1	12/2014	WO	WO 2015/070212	A1	5/2015
WO	WO 2014/204724	A1	12/2014	WO	WO 2015/071474	A2	5/2015
WO	WO 2014/204725	A1	12/2014	WO	WO 2015/073683	A2	5/2015
WO	WO 2014/204726	A1	12/2014	WO	WO 2015/073867	A1	5/2015
WO	WO 2014/204727	A1	12/2014	WO	WO 2015/073990	A1	5/2015
WO	WO 2014/204728	A1	12/2014	WO	WO 2015/075056	A1	5/2015
WO	WO 2014/204729	A1	12/2014	WO	WO 2015/075154	A2	5/2015
WO	WO 2014/205192	A2	12/2014	WO	WO 2015/075175	A1	5/2015
WO	WO 2014/207043	A1	12/2014	WO	WO 2015/075195	A1	5/2015
WO	WO 2015/002780	A1	1/2015	WO	WO 2015/075557	A2	5/2015
WO	WO 2015/004241	A2	1/2015	WO	WO 2015/077058	A2	5/2015
WO	WO 2015/006290	A1	1/2015	WO	WO 2015/077290	A2	5/2015
WO	WO 2015/006294	A2	1/2015	WO	WO 2015/077318	A1	5/2015
WO	WO 2015/006437	A1	1/2015	WO	WO 2015/079056	A1	6/2015
WO	WO 2015/006498	A2	1/2015	WO	WO 2015/079057	A2	6/2015
WO	WO 2015/006747	A2	1/2015	WO	WO 2015/086795	A1	6/2015
WO	WO 2015/007194	A1	1/2015	WO	WO 2015/086798	A2	6/2015
WO	WO 2015/010114	A1	1/2015	WO	WO 2015/088643	A1	6/2015
WO	WO 2015/011483	A1	1/2015	WO	WO 2015/089046	A1	6/2015
WO	WO 2015/013583	A2	1/2015	WO	WO 2015/089077	A2	6/2015
WO	WO 2015/021990	A1	2/2015	WO	WO 2015/089277	A1	6/2015
WO	WO 2015/017866	A1	2/2015	WO	WO 2015/089351	A1	6/2015
WO	WO 2015/018503	A1	2/2015	WO	WO 2015/089354	A1	6/2015
WO	WO 2015/021353	A1	2/2015	WO	WO 2015/089364	A1	6/2015
WO	WO 2015/021426	A1	2/2015	WO	WO 2015/089406	A1	6/2015
WO	WO 2015/024017	A2	2/2015	WO	WO 2015/089419	A2	6/2015
WO	WO 2015/024986	A1	2/2015	WO	WO 2015/089427	A1	6/2015
WO	WO 2015/026883	A1	2/2015	WO	WO 2015/089462	A1	6/2015
WO	WO 2015/026885	A1	2/2015	WO	WO 2015/089465	A1	6/2015
WO	WO 2015/026886	A1	2/2015	WO	WO 2015/089473	A1	6/2015
				WO	WO 2015/089486	A1	6/2015
				WO	WO 2015/095804	A1	6/2015
				WO	WO 2015/099850	A1	7/2015
				WO	WO 2015/100929	A1	7/2015

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	WO 2015/103057	A1	7/2015	WO	WO 2015/168547	A2	11/2015
WO	WO 2015/103153	A1	7/2015	WO	WO 2015/168800	A1	11/2015
WO	WO 2015/105928	A1	7/2015	WO	WO 2015/171603	A1	11/2015
WO	WO 2015/108993	A1	7/2015	WO	WO 2015/171894	A1	11/2015
WO	WO 2015/109752	A1	7/2015	WO	WO 2015/171932	A1	11/2015
WO	WO 2015/110474	A1	7/2015	WO	WO 2015/172128	A1	11/2015
WO	WO 2015/112790	A2	7/2015	WO	WO 2015/173436	A1	11/2015
WO	WO 2015/112896	A2	7/2015	WO	WO 2015/175642	A2	11/2015
WO	WO 2015/113063	A1	7/2015	WO	WO 2015/179540	A1	11/2015
WO	WO 2015/114365	A1	8/2015	WO	WO 2015/183025	A1	12/2015
WO	WO 2015/115903	A1	8/2015	WO	WO 2015/183026	A1	12/2015
WO	WO 2015/116686	A1	8/2015	WO	WO 2015/183885	A1	12/2015
WO	WO 2015/116969	A2	8/2015	WO	WO 2015/184259	A1	12/2015
WO	WO 2015/117021	A1	8/2015	WO	WO 2015/184262	A1	12/2015
WO	WO 2015/117041	A1	8/2015	WO	WO 2015/184268	A1	12/2015
WO	WO 2015/117081	A2	8/2015	WO	WO 2015/188056	A1	12/2015
WO	WO 2015/118156	A1	8/2015	WO	WO 2015/188065	A1	12/2015
WO	WO 2015/119941	A2	8/2015	WO	WO 2015/188094	A1	12/2015
WO	WO 2015/121454	A1	8/2015	WO	WO 2015/188109	A1	12/2015
WO	WO 2015/122967	A1	8/2015	WO	WO 2015/188132	A1	12/2015
WO	WO 2015/123339	A1	8/2015	WO	WO 2015/188135	A1	12/2015
WO	WO 2015/124715	A1	8/2015	WO	WO 2015/188191	A1	12/2015
WO	WO 2015/124718	A1	8/2015	WO	WO 2015/189693	A1	12/2015
WO	WO 2015/126927	A2	8/2015	WO	WO 2015/191693	A2	12/2015
WO	WO 2015/127428	A1	8/2015	WO	WO 2015/191899	A1	12/2015
WO	WO 2015/127439	A1	8/2015	WO	WO 2015/191911	A2	12/2015
WO	WO 2015/129686	A1	9/2015	WO	WO 2015/193858	A1	12/2015
WO	WO 2015/131101	A1	9/2015	WO	WO 2015/195547	A1	12/2015
WO	WO 2015/133554	A1	9/2015	WO	WO 2015/195621	A1	12/2015
WO	WO 2015/134121	A2	9/2015	WO	WO 2015/195798	A1	12/2015
WO	WO 2015/134812	A1	9/2015	WO	WO 2015/198020	A1	12/2015
WO	WO 2015/136001	A1	9/2015	WO	WO 2015/200334	A1	12/2015
WO	WO 2015/138510	A1	9/2015	WO	WO 2015/200378	A1	12/2015
WO	WO 2015/138739	A2	9/2015	WO	WO 2015/200555	A2	12/2015
WO	WO 2015/138855	A1	9/2015	WO	WO 2015/200805	A2	12/2015
WO	WO 2015/138870	A2	9/2015	WO	WO 2016/001978	A1	1/2016
WO	WO 2015/139008	A1	9/2015	WO	WO 2016/004010	A1	1/2016
WO	WO 2015/139139	A1	9/2015	WO	WO 2016/004318	A1	1/2016
WO	WO 2015/143046	A2	9/2015	WO	WO 2016/007347	A1	1/2016
WO	WO 2015/143177	A1	9/2015	WO	WO 2016/007604	A1	1/2016
WO	WO 2015/145417	A1	10/2015	WO	WO 2016/007948	A1	1/2016
WO	WO 2015/148431	A1	10/2015	WO	WO 2016/011080	A2	1/2016
WO	WO 2015/148670	A1	10/2015	WO	WO 2016/011210	A2	1/2016
WO	WO 2015/148680	A1	10/2015	WO	WO 2016/011428	A1	1/2016
WO	WO 2015/148760	A1	10/2015	WO	WO 2016/012544	A2	1/2016
WO	WO 2015/148761	A1	10/2015	WO	WO 2016/012552	A1	1/2016
WO	WO 2015/148860	A1	10/2015	WO	WO 2016/014409	A1	1/2016
WO	WO 2015/148863	A2	10/2015	WO	WO 2016/014565	A2	1/2016
WO	WO 2015/153760	A2	10/2015	WO	WO 2016/014794	A1	1/2016
WO	WO 2015/153780	A1	10/2015	WO	WO 2016/014837	A1	1/2016
WO	WO 2015/153789	A1	10/2015	WO	WO 2016/016119	A1	2/2016
WO	WO 2015/153791	A1	10/2015	WO	WO 2016/016358	A1	2/2016
WO	WO 2015/153889	A2	10/2015	WO	WO 2016/019144	A2	2/2016
WO	WO 2015/153940	A1	10/2015	WO	WO 2016/020399	A1	2/2016
WO	WO 2015/155341	A1	10/2015	WO	WO 2016/021972	A1	2/2016
WO	WO 2015/155686	A2	10/2015	WO	WO 2016/021973	A1	2/2016
WO	WO 2015/157070	A2	10/2015	WO	WO 2016/022363	A2	2/2016
WO	WO 2015/157534	A1	10/2015	WO	WO 2016/022866	A1	2/2016
WO	WO 2015/159068	A1	10/2015	WO	WO 2016/022931	A1	2/2016
WO	WO 2015/159086	A1	10/2015	WO	WO 2016/025131	A1	2/2016
WO	WO 2015/159087	A1	10/2015	WO	WO 2016/025469	A1	2/2016
WO	WO 2015/160683	A1	10/2015	WO	WO 2016/025759	A1	2/2016
WO	WO 2015/161276	A2	10/2015	WO	WO 2016/026444	A1	2/2016
WO	WO 2015/163733	A1	10/2015	WO	WO 2016/028682	A1	2/2016
WO	WO 2015/164740	A1	10/2015	WO	WO 2016/028843	A1	2/2016
WO	WO 2015/164748	A1	10/2015	WO	WO 2016/028887	A1	2/2016
WO	WO 2015/165274	A1	11/2015	WO	WO 2016/033088	A1	3/2016
WO	WO 2015/165275	A1	11/2015	WO	WO 2016/033230	A1	3/2016
WO	WO 2015/165276	A1	11/2015	WO	WO 2016/033246	A1	3/2016
WO	WO 2015/166272	A2	11/2015	WO	WO 2016/033298	A1	3/2016
WO	WO 2015/167766	A1	11/2015	WO	WO 2016/035044	A1	3/2016
WO	WO 2015/167956	A1	11/2015	WO	WO 2016/035918	A1	3/2016
WO	WO 2015/168125	A1	11/2015	WO	WO 2016/036754	A1	3/2016
WO	WO 2015/168158	A1	11/2015	WO	WO 2016/037157	A2	3/2016
WO	WO 2015/168404	A1	11/2015	WO	WO 2016/040030	A1	3/2016
				WO	WO 2016/040594	A1	3/2016
				WO	WO 2016/044182	A1	3/2016
				WO	WO 2016/044416	A1	3/2016
				WO	WO 2016/046635	A1	3/2016

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	WO 2016/049024	A2	3/2016	WO	WO 2016/109255	A1	7/2016
WO	WO 2016/049163	A2	3/2016	WO	WO 2016/109840	A2	7/2016
WO	WO 2016/049230	A1	3/2016	WO	WO 2016/110214	A1	7/2016
WO	WO 2016/049251	A1	3/2016	WO	WO 2016/110453	A1	7/2016
WO	WO 2016/049258	A2	3/2016	WO	WO 2016/110511	A1	7/2016
WO	WO 2016/053397	A2	4/2016	WO	WO 2016/110512	A1	7/2016
WO	WO 2016/054326	A1	4/2016	WO	WO 2016/111546	A2	7/2016
WO	WO 2016/057061	A2	4/2016	WO	WO 2016/112242	A1	7/2016
WO	WO 2016/057821	A2	4/2016	WO	WO 2016/112351	A1	7/2016
WO	WO 2016/057835	A2	4/2016	WO	WO 2016/112963	A1	7/2016
WO	WO 2016/057850	A1	4/2016	WO	WO 2016/113357	A1	7/2016
WO	WO 2016/057951	A2	4/2016	WO	WO 2016/114972	A1	7/2016
WO	WO 2016/057961	A1	4/2016	WO	WO 2016/115179	A1	7/2016
WO	WO 2016/061073	A1	4/2016	WO	WO 2016/115326	A1	7/2016
WO	WO 2016/061374	A1	4/2016	WO	WO 2016/115355	A1	7/2016
WO	WO 2016/061481	A1	4/2016	WO	WO 2016/116032	A1	7/2016
WO	WO 2016/061523	A1	4/2016	WO	WO 2016/120480	A1	8/2016
WO	WO 2016/064894	A2	4/2016	WO	WO 2016/123071	A1	8/2016
WO	WO 2016/065364	A1	4/2016	WO	WO 2016/123230	A1	8/2016
WO	WO 2016/069282	A1	5/2016	WO	WO 2016/123243	A1	8/2016
WO	WO 2016/069283	A1	5/2016	WO	WO 2016/123578	A1	8/2016
WO	WO 2016/069591	A2	5/2016	WO	WO 2016/126747	A1	8/2016
WO	WO 2016/069774	A1	5/2016	WO	WO 2016/130600	A2	8/2016
WO	WO 2016/069910	A1	5/2016	WO	WO 2016/130697	A1	8/2016
WO	WO 2016/069912	A1	5/2016	WO	WO 2016/131009	A1	8/2016
WO	WO 2016/070037	A2	5/2016	WO	WO 2016/132122	A1	8/2016
WO	WO 2016/070070	A1	5/2016	WO	WO 2016/133165	A1	8/2016
WO	WO 2016/070129	A1	5/2016	WO	WO 2016/135507	A1	9/2016
WO	WO 2016/072399	A1	5/2016	WO	WO 2016/135557	A2	9/2016
WO	WO 2016/072936	A1	5/2016	WO	WO 2016/135558	A2	9/2016
WO	WO 2016/073433	A1	5/2016	WO	WO 2016/135559	A2	9/2016
WO	WO 2016/073559	A1	5/2016	WO	WO 2016/137774	A1	9/2016
WO	WO 2016/073990	A2	5/2016	WO	WO 2016/137949	A1	9/2016
WO	WO 2016/075662	A2	5/2016	WO	WO 2016/141224	A1	9/2016
WO	WO 2016/076672	A1	5/2016	WO	WO 2016/141893	A1	9/2016
WO	WO 2016/077273	A1	5/2016	WO	WO 2016/142719	A1	9/2016
WO	WO 2016/077350	A1	5/2016	WO	WO 2016/145150	A2	9/2016
WO	WO 2016/080097	A1	5/2016	WO	WO 2016/148994	A1	9/2016
WO	WO 2016/080795	A1	5/2016	WO	WO 2016/149484	A2	9/2016
WO	WO 2016/081923	A2	5/2016	WO	WO 2016/149547	A1	9/2016
WO	WO 2016/081924	A1	5/2016	WO	WO 2016/150336	A1	9/2016
WO	WO 2016/082135	A1	6/2016	WO	WO 2016/150855	A1	9/2016
WO	WO 2016/083811	A1	6/2016	WO	WO 2016/154016	A2	9/2016
WO	WO 2016/084084	A1	6/2016	WO	WO 2016/154579	A2	9/2016
WO	WO 2016/084088	A1	6/2016	WO	WO 2016/154596	A1	9/2016
WO	WO 2016/086177	A2	6/2016	WO	WO 2016/155482	A1	10/2016
WO	WO 2016/089433	A1	6/2016	WO	WO 2016/161004	A1	10/2016
WO	WO 2016/089866	A1	6/2016	WO	WO 2016/161207	A1	10/2016
WO	WO 2016/089883	A1	6/2016	WO	WO 2016/161260	A1	10/2016
WO	WO 2016/090385	A1	6/2016	WO	WO 2016/161380	A1	10/2016
WO	WO 2016/094679	A1	6/2016	WO	WO 2016/161446	A1	10/2016
WO	WO 2016/094845	A2	6/2016	WO	WO 2016/164305	A1	10/2016
WO	WO 2016/094867	A1	6/2016	WO	WO 2016/164356	A1	10/2016
WO	WO 2016/094872	A1	6/2016	WO	WO 2016/164797	A1	10/2016
WO	WO 2016/094874	A1	6/2016	WO	WO 2016/166340	A1	10/2016
WO	WO 2016/094880	A1	6/2016	WO	WO 2016/167300	A1	10/2016
WO	WO 2016/094888	A1	6/2016	WO	WO 2016/168631	A1	10/2016
WO	WO 2016/097212	A1	6/2016	WO	WO 2016/170484	A1	10/2016
WO	WO 2016/097231	A2	6/2016	WO	WO 2016/172359	A2	10/2016
WO	WO 2016/097751	A1	6/2016	WO	WO 2016/172727	A1	10/2016
WO	WO 2016/099887	A1	6/2016	WO	WO 2016/174056	A1	11/2016
WO	WO 2016/100272	A1	6/2016	WO	WO 2016/174151	A1	11/2016
WO	WO 2016/100389	A1	6/2016	WO	WO 2016/174250	A1	11/2016
WO	WO 2016/100568	A1	6/2016	WO	WO 2016/176191	A1	11/2016
WO	WO 2016/100571	A1	6/2016	WO	WO 2016/176404	A1	11/2016
WO	WO 2016/100951	A2	6/2016	WO	WO 2016/176690	A2	11/2016
WO	WO 2016/100955	A2	6/2016	WO	WO 2016/177682	A1	11/2016
WO	WO 2016/100974	A1	6/2016	WO	WO 2016/178207	A1	11/2016
WO	WO 2016/103233	A2	6/2016	WO	WO 2016/179038	A1	11/2016
WO	WO 2016/104716	A1	6/2016	WO	WO 2016/179112	A1	11/2016
WO	WO 2016/106236	A1	6/2016	WO	WO 2016/181357	A1	11/2016
WO	WO 2016/106239	A1	6/2016	WO	WO 2016/182893	A1	11/2016
WO	WO 2016/106244	A1	6/2016	WO	WO 2016/182917	A1	11/2016
WO	WO 2016/106338	A2	6/2016	WO	WO 2016/182959	A1	11/2016
WO	WO 2016/108926	A1	7/2016	WO	WO 2016/183236	A1	11/2016
				WO	WO 2016/183298	A2	11/2016
				WO	WO 2016/183345	A1	11/2016
				WO	WO 2016/183402	A2	11/2016
				WO	WO 2016/183438	A1	11/2016

(56) References Cited

FOREIGN PATENT DOCUMENTS

WO	WO 2016/183448	A1	11/2016	WO	WO 2017/024343	A1	2/2017
WO	WO 2016/184955	A2	11/2016	WO	WO 2017/024602	A1	2/2017
WO	WO 2016/184989	A1	11/2016	WO	WO 2017/025323	A1	2/2017
WO	WO 2016/185411	A1	11/2016	WO	WO 2017/027423	A1	2/2017
WO	WO 2016/186745	A1	11/2016	WO	WO 2017/028768	A1	2/2017
WO	WO 2016/186772	A2	11/2016	WO	WO 2017/029664	A1	2/2017
WO	WO 2016/186946	A1	11/2016	WO	WO 2017/031360	A1	2/2017
WO	WO 2016/186953	A1	11/2016	WO	WO 2017/031483	A1	2/2017
WO	WO 2016/187717	A1	12/2016	WO	WO 2017/035416	A2	3/2017
WO	WO 2016/187904	A1	12/2016	WO	WO 2017/040348	A1	3/2017
WO	WO 2016/191684	A1	12/2016	WO	WO 2017/040511	A1	3/2017
WO	WO 2016/191869	A1	12/2016	WO	WO 2017/040709	A1	3/2017
WO	WO 2016/196273	A1	12/2016	WO	WO 2017/040786	A1	3/2017
WO	WO 2016/196282	A1	12/2016	WO	WO 2017/040793	A1	3/2017
WO	WO 2016/196308	A1	12/2016	WO	WO 2017/040813	A2	3/2017
WO	WO 2016/196361	A1	12/2016	WO	WO 2017/043573	A1	3/2017
WO	WO 2016/196499	A1	12/2016	WO	WO 2017/043656	A1	3/2017
WO	WO 2016/196539	A2	12/2016	WO	WO 2017/044419	A1	3/2017
WO	WO 2016/196655	A1	12/2016	WO	WO 2017/044776	A1	3/2017
WO	WO 2016/196805	A1	12/2016	WO	WO 2017/044857	A2	3/2017
WO	WO 2016/196887	A1	12/2016	WO	WO 2017/048390	A1	3/2017
WO	WO 2016/197132	A1	12/2016	WO	WO 2017/049129	A2	3/2017
WO	WO 2016/197133	A1	12/2016	WO	WO 2017/050963	A1	3/2017
WO	WO 2016/197354	A1	12/2016	WO	WO 2017/053312	A1	3/2017
WO	WO 2016/197355	A1	12/2016	WO	WO 2017/053431	A2	3/2017
WO	WO 2016/197356	A1	12/2016	WO	WO 2017/053713	A1	3/2017
WO	WO 2016/197357	A1	12/2016	WO	WO 2017/053729	A1	3/2017
WO	WO 2016/197358	A1	12/2016	WO	WO 2017/053753	A1	3/2017
WO	WO 2016/197359	A1	12/2016	WO	WO 2017/053762	A1	3/2017
WO	WO 2016/197360	A1	12/2016	WO	WO 2017/053879	A1	3/2017
WO	WO 2016/197361	A1	12/2016	WO	WO 2017/058658	A2	4/2017
WO	WO 2016/197362	A1	12/2016	WO	WO 2017/059241	A1	4/2017
WO	WO 2016/198361	A1	12/2016	WO	WO 2017/062605	A1	4/2017
WO	WO 2016/198500	A1	12/2016	WO	WO 2017/062723	A1	4/2017
WO	WO 2016/200263	A1	12/2016	WO	WO 2017/062754	A1	4/2017
WO	WO 2016/201047	A1	12/2016	WO	WO 2017/062855	A1	4/2017
WO	WO 2016/201138	A1	12/2016	WO	WO 2017/062886	A1	4/2017
WO	WO 2016/201152	A1	12/2016	WO	WO 2017/062983	A1	4/2017
WO	WO 2016/201153	A1	12/2016	WO	WO 2017/064439	A1	4/2017
WO	WO 2016/201155	A1	12/2016	WO	WO 2017/064546	A1	4/2017
WO	WO 2016/205276	A1	12/2016	WO	WO 2017/064566	A2	4/2017
WO	WO 2016/205613	A1	12/2016	WO	WO 2017/066175	A1	4/2017
WO	WO 2016/205623	A1	12/2016	WO	WO 2017/066497	A2	4/2017
WO	WO 2016/205680	A1	12/2016	WO	WO 2017/066588	A2	4/2017
WO	WO 2016/205688	A2	12/2016	WO	WO 2017/066707	A1	4/2017
WO	WO 2016/205703	A1	12/2016	WO	WO 2017/066781	A1	4/2017
WO	WO 2016/205711	A1	12/2016	WO	WO 2017/068077	A1	4/2017
WO	WO 2016/205728	A1	12/2016	WO	WO 2017/068377	A1	4/2017
WO	WO 2016/205745	A2	12/2016	WO	WO 2017/069829	A2	4/2017
WO	WO 2016/205749	A1	12/2016	WO	WO 2017/070029	A1	4/2017
WO	WO 2016/205759	A1	12/2016	WO	WO 2017/070032	A1	4/2017
WO	WO 2016/205764	A1	12/2016	WO	WO 2017/070169	A1	4/2017
WO	WO 2017/001572	A1	1/2017	WO	WO 2017/070284	A1	4/2017
WO	WO 2017/001988	A1	1/2017	WO	WO 2017/070598	A1	4/2017
WO	WO 2017/004261	A1	1/2017	WO	WO 2017/070605	A1	4/2017
WO	WO 2017/004279	A2	1/2017	WO	WO 2017/070632	A2	4/2017
WO	WO 2017/004616	A1	1/2017	WO	WO 2017/070633	A2	4/2017
WO	WO 2017/005807	A1	1/2017	WO	WO 2017054721	A1	4/2017
WO	WO 2017/009399	A1	1/2017	WO	WO 2017/072590	A1	5/2017
WO	WO 2017/010556	A1	1/2017	WO	WO 2017/074526	A1	5/2017
WO	WO 2017/011519	A1	1/2017	WO	WO 2017/074962	A1	5/2017
WO	WO 2017/011721	A1	1/2017	WO	WO 2017/075261	A1	5/2017
WO	WO 2017/011804	A1	1/2017	WO	WO 2017/075335	A1	5/2017
WO	WO 2017/015015	A1	1/2017	WO	WO 2017/075475	A1	5/2017
WO	WO 2017/015101	A1	1/2017	WO	WO 2017/077135	A1	5/2017
WO	WO 2017/015545	A1	1/2017	WO	WO 2017/077329	A2	5/2017
WO	WO 2017/015567	A1	1/2017	WO	WO 2017/078751	A1	5/2017
WO	WO 2017/015637	A1	1/2017	WO	WO 2017/079400	A1	5/2017
WO	WO 2017/017016	A1	2/2017	WO	WO 2017/079428	A1	5/2017
WO	WO 2017/019867	A1	2/2017	WO	WO 2017/079673	A1	5/2017
WO	WO 2017/019895	A1	2/2017	WO	WO 2017/079724	A1	5/2017
WO	WO 2017/023803	A1	2/2017	WO	WO 2017/081097	A1	5/2017
WO	WO 2017/023974	A1	2/2017	WO	WO 2017/081288	A1	5/2017
WO	WO 2017/024047	A1	2/2017	WO	WO 2017/083368	A1	5/2017
WO	WO 2017/024319	A1	2/2017	WO	WO 2017/083722	A1	5/2017
				WO	WO 2017/083766	A1	5/2017
				WO	WO 2017/087395	A1	5/2017
				WO	WO 2017/090724	A1	6/2017
				WO	WO 2017/091510	A1	6/2017

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	WO 2017/091630	A1	6/2017	WO	WO 2017/178590	A1	10/2017
WO	WO 2017/092201	A1	6/2017	WO	WO 2017/180694	A1	10/2017
WO	WO 2017/093370	A1	6/2017	WO	WO 2017/180711	A1	10/2017
WO	WO 2017/093969	A1	6/2017	WO	WO 2017/180915	A2	10/2017
WO	WO 2017/095111	A1	6/2017	WO	WO 2017/180926	A1	10/2017
WO	WO 2017/096041	A1	6/2017	WO	WO 2017/181107	A2	10/2017
WO	WO 2017/096237	A1	6/2017	WO	WO 2017/181735	A2	10/2017
WO	WO 2017/100158	A1	6/2017	WO	WO 2017/182468	A1	10/2017
WO	WO 2017/100431	A2	6/2017	WO	WO 2017/182585	A1	10/2017
WO	WO 2017/104404	A1	6/2017	WO	WO 2017/184334	A1	10/2017
WO	WO 2017/105251	A1	6/2017	WO	WO 2017/184768	A1	10/2017
WO	WO 2017/105350	A1	6/2017	WO	WO 2017/184786	A1	10/2017
WO	WO 2017/105991	A1	6/2017	WO	WO 2017/186550	A1	11/2017
WO	WO 2017/106414	A1	6/2017	WO	WO 2017/189308	A1	11/2017
WO	WO 2017/106528	A2	6/2017	WO	WO 2017/189336	A1	11/2017
WO	WO 2017/106537	A2	6/2017	WO	WO 2017/190041	A1	11/2017
WO	WO 2017/106569	A1	6/2017	WO	WO 2017/190257	A1	11/2017
WO	WO 2017/106616	A1	6/2017	WO	WO 2017/190664	A1	11/2017
WO	WO 2017/106657	A1	6/2017	WO	WO 2017/191210	A1	11/2017
WO	WO 2017/106767	A1	6/2017	WO	WO 2017/191274	A2	11/2017
WO	WO 2017/109134	A1	6/2017	WO	WO 2017/192172	A1	11/2017
WO	WO 2017/109757	A1	6/2017	WO	WO 2017/192512	A2	11/2017
WO	WO 2017/112620	A1	6/2017	WO	WO 2017/192544	A1	11/2017
WO	WO 2017/115268	A1	7/2017	WO	WO 2017/192573	A1	11/2017
WO	WO 2017/117395	A1	7/2017	WO	WO 2017/193029	A2	11/2017
WO	WO 2017/118598	A1	7/2017	WO	WO 2017/193053	A1	11/2017
WO	WO 2017/118720	A1	7/2017	WO	WO 2017/196768	A1	11/2017
WO	WO 2017/123609	A1	7/2017	WO	WO 2017/197038	A1	11/2017
WO	WO 2017/123910	A1	7/2017	WO	WO 2017/197238	A1	11/2017
WO	WO 2017/124086	A1	7/2017	WO	WO 2017/197301	A1	11/2017
WO	WO 2017/124100	A1	7/2017	WO	WO 2017/201476	A1	11/2017
WO	WO 2017/124652	A1	7/2017	WO	WO 2017/205290	A1	11/2017
WO	WO 2017/126987	A1	7/2017	WO	WO 2017/205423	A1	11/2017
WO	WO 2017/127807	A1	7/2017	WO	WO 2017/207589	A1	12/2017
WO	WO 2017/131237	A1	8/2017	WO	WO 2017/208247	A1	12/2017
WO	WO 2017/132112	A1	8/2017	WO	WO 2017/209809	A1	12/2017
WO	WO 2017/132580	A2	8/2017	WO	WO 2017/213896	A1	12/2017
WO	WO 2017/136520	A1	8/2017	WO	WO 2017/213898	A2	12/2017
WO	WO 2017/136629	A1	8/2017	WO	WO 2017/214460	A1	12/2017
WO	WO 2017/136794	A1	8/2017	WO	WO 2017/216392	A1	12/2017
WO	WO 2017/139264	A1	8/2017	WO	WO 2017/216771	A2	12/2017
WO	WO 2017/139505	A2	8/2017	WO	WO 2017/218185	A1	12/2017
WO	WO 2017/141173	A2	8/2017	WO	WO 2017/219027	A1	12/2017
WO	WO 2017/142835	A1	8/2017	WO	WO 2017/219033	A1	12/2017
WO	WO 2017/142923	A1	8/2017	WO	WO 2017/220751	A1	12/2017
WO	WO 2017/142999	A2	8/2017	WO	WO 2017/222370	A1	12/2017
WO	WO 2017/143042	A2	8/2017	WO	WO 2017/222773	A1	12/2017
WO	WO 2017/147056	A1	8/2017	WO	WO 2017/222834	A1	12/2017
WO	WO 2017/147278	A1	8/2017	WO	WO 2017/223107	A1	12/2017
WO	WO 2017/147432	A1	8/2017	WO	WO 2017/223330	A1	12/2017
WO	WO 2017/147446	A1	8/2017	WO	WO 2018/000657	A1	1/2018
WO	WO 2017/147555	A1	8/2017	WO	WO 2018/002719	A1	1/2018
WO	WO 2017/151444	A1	9/2017	WO	WO 2018/005117	A1	1/2018
WO	WO 2017/151719	A1	9/2017	WO	WO 2018/005289	A2	1/2018
WO	WO 2017/152015	A1	9/2017	WO	WO 2018/005691	A1	1/2018
WO	WO 2017/155717	A1	9/2017	WO	WO 2018/005782	A1	1/2018
WO	WO 2017/157422	A1	9/2017	WO	WO 2018/005873	A1	1/2018
WO	WO 2017/158153	A1	9/2017	WO	WO 2018/06693	A1	1/2018
WO	WO 2017/160689	A1	9/2017	WO	WO 2018/009520	A1	1/2018
WO	WO 2017/160752	A1	9/2017	WO	WO 2018/009562	A1	1/2018
WO	WO 2017/160890	A1	9/2017	WO	WO 2018/009822	A1	1/2018
WO	WO 2017/161068	A1	9/2017	WO	WO 2018/013821	A1	1/2018
WO	WO 2017/165741	A1	9/2017	WO	WO 2018/013932	A1	1/2018
WO	WO 2017/165826	A1	9/2017	WO	WO 2018/013990	A1	1/2018
WO	WO 2017/165862	A1	9/2017	WO	WO 2018/014384	A1	1/2018
WO	WO 2017/167712	A1	10/2017	WO	WO 2018/015444	A1	1/2018
WO	WO 2017/172644	A2	10/2017	WO	WO 2018/015936	A2	1/2018
WO	WO 2017/172645	A2	10/2017	WO	WO 2018/017754	A1	1/2018
WO	WO 2017/172860	A1	10/2017	WO	WO 2018/018979	A1	2/2018
WO	WO 2017/173004	A1	10/2017	WO	WO 2018/020248	A1	2/2018
WO	WO 2017/173054	A1	10/2017	WO	WO 2018/021878	A1	2/2018
WO	WO 2017/173092	A1	10/2017	WO	WO 2018/022480	A1	2/2018
WO	WO 2017/174329	A1	10/2017	WO	WO 2018/022634	A1	2/2018
WO	WO 2017/176529	A1	10/2017	WO	WO 2018/025206	A1	2/2018
WO	WO 2017/176806	A1	10/2017	WO	WO 2018/026723	A1	2/2018
				WO	WO 2018/026976	A1	2/2018
				WO	WO 2018/027078	A1	2/2018
				WO	WO 2018/030608	A1	2/2018
				WO	WO 2018/031683	A1	2/2018

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	WO 2018/035250	A1	2/2018	WO	WO 2018/154439	A1	8/2018
WO	WO 2018/035300	A1	2/2018	WO	WO 2018/154459	A1	8/2018
WO	WO 2018/035423	A1	2/2018	WO	WO 2018/154462	A2	8/2018
WO	WO 2018/035503	A1	2/2018	WO	WO 2018/156372	A1	8/2018
WO	WO 2018/039145	A1	3/2018	WO	WO 2018/156824	A1	8/2018
WO	WO 2018/039438	A1	3/2018	WO	WO 2018/161009	A1	9/2018
WO	WO 2018/039440	A1	3/2018	WO	WO 2018/161032	A1	9/2018
WO	WO 2018/039448	A1	3/2018	WO	WO 2018/165504	A1	9/2018
WO	WO 2018/045630	A1	3/2018	WO	WO 2018/165629	A1	9/2018
WO	WO 2018/048827	A1	3/2018	WO	WO 2018/170015	A1	9/2018
WO	WO 2018/049073	A1	3/2018	WO	WO 2018/170340	A1	9/2018
WO	WO 2018/049168	A1	3/2018	WO	WO 2018/175502	A2	9/2018
WO	WO 2018/051347	A1	3/2018	WO	WO 2018/176009	A1	9/2018
WO	WO 2018/058064	A1	3/2018	WO	WO 2018/177351	A1	10/2018
WO	WO 2018/062866	A2	4/2018	WO	WO 2018/179578	A1	10/2018
WO	WO 2018/064352	A1	4/2018	WO	WO 2018/183403	A1	10/2018
WO	WO 2018/064371	A1	4/2018	WO	WO 2018/189184	A1	10/2018
WO	WO 2018/064516	A1	4/2018	WO	WO 2018/191388	A1	10/2018
WO	WO 2018/067546	A1	4/2018	WO	WO 2018/195402	A1	10/2018
WO	WO 2018/067846	A1	4/2018	WO	WO 2018/195545	A2	10/2018
WO	WO 2018/068053	A2	4/2018	WO	WO 2018/195555	A1	10/2018
WO	WO 2018/069474	A1	4/2018	WO	WO 2018/197020	A1	11/2018
WO	WO 2018/071623	A2	4/2018	WO	WO 2018/197495	A1	11/2018
WO	WO 2018/071663	A1	4/2018	WO	WO 2018/202800	A1	11/2018
WO	WO 2018/071868	A1	4/2018	WO	WO 2018/204493	A1	11/2018
WO	WO 2018/071892	A1	4/2018	WO	WO 2018/208755	A1	11/2018
WO	WO 2018/074979	A1	4/2018	WO	WO 2018/208998	A1	11/2018
WO	WO 2018/079134	A1	5/2018	WO	WO 2018/209158	A2	11/2018
WO	WO 2018/080573	A1	5/2018	WO	WO 2018/209320	A1	11/2018
WO	WO 2018/081504	A1	5/2018	WO	WO 2018/213351	A1	11/2018
WO	WO 2018/081535	A2	5/2018	WO	WO 2018/213708	A1	11/2018
WO	WO 2018/081728	A1	5/2018	WO	WO 2018/213726	A1	11/2018
WO	WO 2018/083128	A2	5/2018	WO	WO 2018/213771	A1	11/2018
WO	WO 2018/083606	A1	5/2018	WO	WO 2018/213791	A1	11/2018
WO	WO 2018/085288	A1	5/2018	WO	WO 2018/217852	A1	11/2018
WO	WO 2018/085414	A1	5/2018	WO	WO 2018/217981	A1	11/2018
WO	WO 2018/086623	A1	5/2018	WO	WO 2018/218166	A1	11/2018
WO	WO 2018/089664	A1	5/2018	WO	WO 2018/218188	A2	11/2018
WO	WO 2018/093990	A1	5/2018	WO	WO 2018/218206	A1	11/2018
WO	WO 2018/098383	A1	5/2018	WO	WO 2018/226855	A1	12/2018
WO	WO 2018/098480	A1	5/2018	WO	WO 2019/005884	A1	1/2019
WO	WO 2018/098587	A1	6/2018	WO	WO 2019/005886	A1	1/2019
WO	WO 2018/099256	A1	6/2018	WO	WO 2019/010384	A1	1/2019
WO	WO 2018/103686	A1	6/2018	WO	WO 2019/023680	A1	1/2019
WO	WO 2018/106268	A1	6/2018	WO	WO 2019/042284	A1	3/2019
WO	WO 2018/107028	A1	6/2018	WO	WO 2019/051097	A1	3/2019
WO	WO 2018/107103	A1	6/2018	WO	WO 2019/075357	A1	4/2019
WO	WO 2018/107129	A1	6/2018	WO	WO 2019/079347	A1	4/2019
WO	WO 2018/108272	A1	6/2018	WO	WO 2019/084062	A1	5/2019
WO	WO 2018/109101	A1	6/2018	WO	WO 2019/090169	A1	5/2019
WO	WO 2018/111946	A1	6/2018	WO	WO 2019/090367	A1	5/2019
WO	WO 2018/111947	A1	6/2018	WO	WO 2019/092042	A1	5/2019
WO	WO 2018/112336	A1	6/2018	WO	WO 2019/118935	A1	6/2019
WO	WO 2018/112446	A2	6/2018	WO	WO 2019/118949	A1	6/2019
WO	WO 2018/119354	A1	6/2018	WO	WO 2019/123430	A1	6/2019
WO	WO 2018/119359	A1	6/2018	WO	WO 2019/126709	A1	6/2019
WO	WO 2018/120283	A1	7/2018	WO	WO 2019/139645	A2	7/2019
WO	WO 2018/130830	A1	7/2018	WO	WO 2019/139951	A1	7/2019
WO	WO 2018/135838	A2	7/2018	WO	WO 2019/147014	A1	8/2019
WO	WO 2018/136396	A2	7/2018	WO	WO 2019/161251	A1	8/2019
WO	WO 2018/138385	A1	8/2018	WO	WO 2019/168953	A1	9/2019
WO	WO 2018/142364	A1	8/2018	WO	WO 2019/183641	A1	9/2019
WO	WO 2018/148246	A1	8/2018	WO	WO 2019/204369	A1	10/2019
WO	WO 2018/148256	A1	8/2018	WO	WO 2019/217942	A1	11/2019
WO	WO 2018/148647	A2	8/2018	WO	WO 2019/226593	A1	11/2019
WO	WO 2018/149418	A1	8/2018	WO	WO 2019/226953	A1	11/2019
WO	WO 2018/149888	A1	8/2018	WO	WO 2019/236566	A1	12/2019
WO	WO 2018/149915	A1	8/2018	WO	WO 2019/241649	A1	12/2019
WO	WO 2018/152197	A1	8/2018	WO	WO 2020/014261	A1	1/2020
WO	WO 2018/152418	A1	8/2018	WO	WO 2020/028555	A2	2/2020
WO	WO 2018/154380	A1	8/2018	WO	WO 2020/028823	A1	2/2020
WO	WO 2018/154387	A1	8/2018	WO	WO 2020/041751	A1	2/2020
WO	WO 2018/154412	A1	8/2018	WO	WO 2020/047124	A1	3/2020
WO	WO 2018/154413	A1	8/2018	WO	WO 2020/051360	A1	3/2020
WO	WO 2018/154418	A1	8/2018	WO	WO 2020/081568	A1	4/2020
				WO	WO 2020/086908	A1	4/2020
				WO	WO 2020/092453	A1	5/2020
				WO	WO 2020/102659	A1	5/2020
				WO	WO 2020/102709	A1	5/2020

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	WO 2020/154500	A1	7/2020
WO	WO 2020/157008	A1	8/2020
WO	WO 2020/160071	A1	8/2020
WO	WO 2020/160481	A1	8/2020
WO	WO 2020/160517	A1	8/2020
WO	WO 2020/180975	A1	9/2020
WO	WO 2020/181178	A1	9/2020
WO	WO 2020/181180	A1	9/2020
WO	WO 2020/181193	A1	9/2020
WO	WO 2020/181195	A1	9/2020
WO	WO 2020/181202	A1	9/2020
WO	WO 2020/191153	A2	9/2020
WO	WO 2020/191171	A1	9/2020
WO	WO 2020/191233	A1	9/2020
WO	WO 2020/191234	A1	9/2020
WO	WO 2020/191239	A1	9/2020
WO	WO 2020/191241	A1	9/2020
WO	WO 2020/191242	A1	9/2020
WO	WO 2020/191243	A1	9/2020
WO	WO 2020/191245	A1	9/2020
WO	WO 2020/191246	A1	9/2020
WO	WO 2020/191248	A1	9/2020
WO	WO 2020/191249	A1	9/2020
WO	WO 2020/205681	A1	10/2020
WO	WO 2020/210751	A1	10/2020
WO	WO 2020/214842	A1	10/2020
WO	WO 2020/236982	A1	11/2020
WO	WO 2020/247587	A1	12/2020
WO	WO 2021/025750	A1	2/2021
WO	WO 2021/030666	A1	2/2021
WO	WO 2021/072328	A1	4/2021
WO	WO 2021/108717	A2	6/2021
WO	WO 2021/138469	A1	7/2021
WO	WO 2021/155065	A1	8/2021
WO	WO 2021/158921	A2	8/2021
WO	WO 2021/158995	A1	8/2021
WO	WO 2021/158999	A1	8/2021
WO	WO 2021/222318	A1	11/2021
WO	WO 2021/226558	A1	11/2021
WO	WO 2022/067130	A2	3/2022
WO	WO 2022/150790	A2	7/2022
WO	WO 2023/015309	A2	2/2023

OTHER PUBLICATIONS

Ran FA, Cong L, Yan WX, Scott DA, Gootenberg JS, Kriz AJ, Zetsche B, Shalem O, Wu X, Makarova KS, Koonin EV, Sharp PA, Zhang F. In vivo genome editing using *Staphylococcus aureus* Cas9. *Nature*. Apr. 9, 2015;520(7546):186-91. (Year: 2015).*

Fu et al. Chapter Two—Targeted Genome Editing in Human Cells Using CRISPR/Cas Nucleases and Truncated Guide RNAs Methods in Enzymology, Academic Press, vol. 546, 2014, pp. 21-45. (Year: 2014).*

Svitashev et al, Targeted Mutagenesis, Precise Gene Editing, and Site-Specific Gene Insertion in Maize Using Cas9 and Guide RNA, *Plant Physiology*, vol. 169, Issue 2, Oct. 2015, pp. 931-945 (Year: 2015).*

Hendel, A., Bak, R., Clark, J. et al. Chemically modified guide RNAs enhance CRISPR-Cas genome editing in human primary cells. *Nat Biotechnol* 33, 985-989 (2015) (Year: 2015).*

Fu et al. Improving CRISPR-Cas nuclease specificity using truncated guide RNAs. *Nat Biotechnol* 32, 279-284 (2014) (Year: 2014).*

Hwang WY, Fu Y, Reyon D, Maeder ML, Kaini P, et al. (2013) Heritable and Precise Zebrafish Genome Editing Using a CRISPR-Cas System. *PLOS One* 8(7): e68708. (Year: 2013).*

Courtney, D., Moore, J., Atkinson, S. et al. CRISPR/Cas9 DNA cleavage at SNP-derived PAM enables both in vitro and in vivo KRT12 mutation-specific targeting *Gene Ther* 23, 108-112 (2016) (Year: 2016).*

Shin, H., Wang, C., Lee, H. et al. CRISPR/Cas9 targeting events cause complex deletions and insertions at 17 sites in the mouse genome. *Nat Commun* 8, 15464 (2017) (Year: 2017).*

Ousterout DG, Kabadi AM, Thakore PI, Majoros WH, Reddy TE, Gersbach CA. Multiplex CRISPR/Cas9-based genome editing for correction of dystrophin mutations that cause Duchenne muscular dystrophy. *Nat Commun*. Feb. 18, 2015;6:6244. (Year: 2015).*

Zhang Y, Long C, Li H, McAnally JR, Baskin KK, Shelton JM, Bassel-Duby R, Olson EN. CRISPR-Cpf1 correction of muscular dystrophy mutations in human cardiomyocytes and mice. *Sci Adv*. Apr. 12, 2017;3(4):e1602814. (Year: 2017).*

U.S. Appl. No. 61/716,256, filed Oct. 19, 2012, Jinek et al.

U.S. Appl. No. 61/717,324, filed Oct. 23, 2012, Cho et al.

U.S. Appl. No. 61/734,256, filed Dec. 6, 2012, Chen et al.

U.S. Appl. No. 61/758,624, filed Jan. 30, 2013, Chen et al.

U.S. Appl. No. 61/761,046, filed Feb. 5, 2013, Knight et al.

U.S. Appl. No. 61/794,422, filed Mar. 15, 2013, Knight et al.

U.S. Appl. No. 61/803,599, filed Mar. 20, 2013, Kim et al.

U.S. Appl. No. 61/837,481, filed Jun. 20, 2013, Cho et al.

U.S. Appl. No. 61/838,178, filed Jun. 21, 2013, Joung et al.

U.S. Appl. No. 62/288,661, filed Jan. 29, 2016, Muir et al.

U.S. Appl. No. 62/357,332, filed Jun. 30, 2016, Liu et al.

[No Author Listed] NCBI Accession No. XP_015843220.1. C →U editing enzyme APOBEC-1 [*Peromyscus maniculatus bairdii*], XP002793540. Mar. 21, 2016.

[No Author Listed] NCBI Accession No. XP_021505673.1. C →U editing enzyme APOBEC-1 [*Meriones unguiculatus*], XP002793541. Jun. 27, 2017.

[No Author Listed], "Human genome." *Encyclopedia Britannica*. *Encyclopedia Britannica, Inc.* Published Feb. 15, 2019. Last accessed online via <https://www.britannica.com/science/human-genome> on Mar. 19, 2021. 2 pages.

[No Author Listed], EMBL Accession No. Q99ZW2. Nov. 2012. 2 pages.

[No Author Listed], Invitrogen Lipofectamine™ 2000 product sheets, 2002. 2 pages.

[No Author Listed], Invitrogen Lipofectamine™ 2000 product sheets, 2005. 3 pages.

[No Author Listed], Invitrogen Lipofectamine™ LTX product sheets, 2011. 4 pages.

[No Author Listed], Thermo Fisher Scientific—How Cationic Lipid Mediated Transfection Works, retrieved from the internet Aug. 27, 2015. 2 pages.

Abremski et al., Bacteriophage P1 site-specific recombination. Purification and properties of the Cre recombinase protein. *J Biol Chem*. Feb. 10, 1984;259(3):1509-14.

Abudayyeh et al., C2c2 is a single-component programmable RNA-guided RNA-targeting CRISPR effector. *Science* Aug. 2016;353(6299):aaf5573. DOI: 10.1126/science.aaf5573.

Abudayyeh et al., A cytosine deaminase for programmable single-base RNA editing. *Science*. Jul. 26, 2019;365(6451):382-386. doi: 10.1126/science.aax7063. Epub Jul. 11, 2019.

Abudayyeh et al., RNA targeting with CRISPR-Cas13. *Nature*. Oct. 12, 2017;550(7675):280-284. doi: 10.1038/nature24049. Epub Oct. 4, 2017.

Ada et al., Carbohydrate-protein conjugate vaccines. *Clin Microbiol Infect*. Feb. 2003;9(2):79-85. doi: 10.1046/j.1469-0691.2003.00530.x.

Adamala et al., Programmable RNA-binding protein composed of repeats of a single modular unit. *Proc Natl Acad Sci U S A*. May 10, 2016;113(19):E2579-88. doi: 10.1073/pnas.1519368113. Epub Apr. 26, 2016.

Adams et al., New biarsenical ligands and tetracysteine motifs for protein labeling in vitro and in vivo: synthesis and biological applications. *J Am Chem Soc*. May 29, 2002;124(21):6063-76. doi: 10.1021/ja017687n.

Adli, The CRISPR tool kit for genome editing and beyond. *Nat Commun*. May 15, 2018;9(1):1911. doi: 10.1038/s41467-018-04252-2.

Aguilo et al., Coordination of m(6)A mRNA Methylation and Gene Transcription by ZFP217 Regulates Pluripotency and Reprogramming. *Cell Stem Cell*. Dec. 3, 2015;17(6):689-704. doi: 10.1016/j.stem.2015.09.005. Epub Oct. 29, 2015.

Ahmad et al., Antibody-mediated specific binding and cytotoxicity of liposome-entrapped doxorubicin to lung cancer cells in vitro. *Cancer Res*. Sep. 1, 1992;52(17):4817-20.

(56)

References Cited

OTHER PUBLICATIONS

- Aida et al., Prime editing primarily incudes undesired outcomes in mice. *bioRxiv preprint and Supplemental Information*. Aug. 6, 2020. Retrieved from www.biorxiv.org. doi: 10.1101/2020.08.06.239723. 40 pages.
- Aihara et al., A conformational switch controls the DNA cleavage activity of lambda integrase. *Mol Cell*. Jul. 2003;12(1):187-98.
- Aik et al., Structure of human RNA ?-methyladenine demethylase ALKBH5 provides insights into its mechanisms of nucleic acid recognition and demethylation. *Nucleic Acids Res*. Apr. 2014;42(7):4741-54. doi: 10.1093/nar/gku085. Epub Jan. 30, 2014.
- Aird et al., Increasing Cas9-mediated homology-directed repair efficiency through covalent tethering of DNA repair template. *Commun Biol*. May 31, 2018;1:54. doi: 10.1038/s42003-018-0054-2.
- Akcakaya et al., In vivo CRISPR editing with no detectable genome-wide off-target mutations. *Nature*. Sep. 2018;561(7723):416-419. doi: 10.1038/s41586-018-0500-9. Epub Sep. 12, 2018. PMID: 30209390; PMCID: PMC6194229.
- Akins et al., Mitochondrial plasmids of *Neurospora*: integration into mitochondrial DNA and evidence for reverse transcription in mitochondria. *Cell*. Nov. 21, 1986;47(4):505-16. doi: 10.1016/0092-8674(86)90615-x.
- Akinsheye et al., Fetal hemoglobin in sickle cell anemia. *Blood*. Jul. 7, 2011;118(1):19-27. doi: 10.1182/blood-2011-03-325258. Epub Apr. 13, 2011.
- Akopian et al., Chimeric recombinases with designed DNA sequence recognition. *Proc Natl Acad Sci U S A*. Jul. 22, 2003;100(15):8688-91. Epub Jul. 1, 2003.
- Alarcón et al., HNRNPA2B1 Is a Mediator of m(6)A-Dependent Nuclear RNA Processing Events. *Cell*. Sep. 10, 2015;162(6):1299-308. doi: 10.1016/j.cell.2015.08.011. Epub Aug. 27, 2015.
- Alarcón et al., N6-methyladenosine marks primary microRNAs for processing. *Nature*. Mar. 26, 2015;519(7544):482-5. doi: 10.1038/nature14281. Epub Mar. 18, 2015.
- Alexander, HFE-associated hereditary hemochromatosis. *Genet Med*. May 2009;11(5):307-13. doi: 10.1097/GIM.0b013e31819d30f2.
- Alexandrov et al., Signatures of mutational processes in human cancer. *Nature*. Aug. 22, 2013;500(7463):415-21. doi: 10.1038/nature12477. Epub Aug. 14, 2013.
- Ali et al., Novel genetic abnormalities in Bernard-Soulier syndrome in India. *Ann Hematol*. Mar. 2014;93(3):381-4. doi: 10.1007/s00277-013-1895-x. Epub Sep. 1, 2013.
- Altschul et al., Basic local alignment search tool. *J Mol Biol*. Oct. 5, 1990;215(3):403-10. doi: 10.1016/S0022-2836(05)80360-2.
- Amato et al., Interpreting elevated fetal hemoglobin in pathology and health at the basic laboratory level: new and known Y-gene mutations associated with hereditary persistence of fetal hemoglobin. *Int J Lab Hematol*. Feb. 2014;36(1):13-9. doi: 10.1111/ijlh.12094. Epub Apr. 29, 2013.
- Ames et al., A eubacterial riboswitch class that senses the coenzyme tetrahydrofolate. *Chem Biol*. Jul. 30, 2010;17(7):681-5. doi: 10.1016/j.chembiol.2010.05.020.
- Anders et al., Chapter One: In Vitro Enzymology of Cas9. in *Methods in Enzymology*, eds Doudna et al. 2014: 546:1-20.
- Anders et al., Structural basis of PAM-dependent target DNA recognition by the Cas9 endonuclease. *Nature*. Sep. 25, 2014;513(7519):569-73. doi: 10.1038/nature13579. Epub Jul. 27, 2014.
- Anderson, Human gene therapy. *Science*. May 8, 1992;256(5058):808-13. doi: 10.1126/science.1589762.
- André et al., Axotomy-induced expression of calcium-activated chloride current in subpopulations of mouse dorsal root ganglion neurons. *J Neurophysiol*. Dec. 2003;90(6):3764-73. doi: 10.1152/jn.00449.2003. Epub Aug. 27, 2003.
- Anzalone et al., Genome editing with CRISPR-Cas nucleases, base editors, transposases and prime editors. *Nat Biotechnol*. Jul. 2020;38(7):824-844. doi: 10.1038/s41587-020-0561-9. Epub Jun. 22, 2020.
- Anzalone et al., Reprogramming eukaryotic translation with ligand-responsive synthetic RNA switches. *Nat Methods*. May 2016;13(5):453-8. doi: 10.1038/nmeth.3807. Epub Mar. 21, 2016.
- Anzalone et al., Search-and-replace genome editing without double-strand breaks or donor DNA. *Nature*. Dec. 2019;576(7785):149-157. doi: 10.1038/s41586-019-1711-4. Epub Oct. 21, 2019.
- Aplan, Causes of oncogenic chromosomal translocation. *Trends Genet*. Jan. 2006;22(1):46-55. doi: 10.1016/j.tig.2005.10.002. Epub Oct. 28, 2005.
- Araki et al., Comparative analysis of right element mutant lox sites on recombination efficiency in embryonic stem cells. *BMC Biotechnol*. Mar. 31, 2010;10:29. doi: 10.1186/1472-6750-10-29.
- Araki et al., Site-specific recombinase, R, encoded by yeast plasmid pSR1. *J Mol Biol*. May 5, 1992;225(1):25-37. doi: 10.1016/0022-2836(92)91023-i.
- Araki et al., Targeted integration of DNA using mutant lox sites in embryonic stem cells. *Nucleic Acids Res*. Feb. 15, 1997;25(4):868-72. doi: 10.1093/nar/25.4.868.
- Arambula et al., Surface display of a massively variable lipoprotein by a *Legionella* diversity-generating retroelement. *Proc Natl Acad Sci U S A*. May 14, 2013;110(20):8212-7. doi: 10.1073/pnas.1301366110. Epub Apr. 30, 2013.
- Arazoe et al., Targeted Nucleotide Editing Technologies for Microbial Metabolic Engineering. *Biotechnol J*. Sep. 2018;13(9):e1700596. doi: 10.1002/biot.201700596. Epub Jun. 19, 2018.
- Arbab et al., Determinants of Base Editing Outcomes from Target Library Analysis and Machine Learning. *Cell*. Jul. 23, 2020;182(2):463-480.e30. doi: 10.1016/j.cell.2020.05.037. Epub Jun. 12, 2020.
- Arezi et al., Novel mutations in Moloney Murine Leukemia Virus reverse transcriptase increase thermostability through tighter binding to template-primer. *Nucleic Acids Res*. Feb. 2009;37(2):473-81. doi: 10.1093/nar/gkn952. Epub Dec. 4, 2008.
- Arnold et al., Mutants of Tn3 resolvase which do not require accessory binding sites for recombination activity. *EMBO J*. Mar. 1, 1999;18(5):1407-14.
- Asante et al., A naturally occurring variant of the human prion protein completely prevents prion disease. *Nature*. Jun. 25, 2015;522(7557):478-81. doi: 10.1038/nature14510. Epub Jun. 10, 2015.
- Atkins et al., Ribosomal frameshifting and transcriptional slippage: From genetic steganography and cryptography to adventitious use. *Nucleic Acids Res*. Sep. 6, 2016;44(15):7007-78. doi: 10.1093/nar/gkw530. Epub Jul. 19, 2016.
- Auer et al., Highly efficient CRISPR/Cas9-mediated knock-in in zebrafish by homology-independent DNA repair. *Genome Res*. Jan. 2014;24(1):142-53. doi: 10.1101/gr.161638.113. Epub Oct. 31, 2013.
- Autieri et al., IRT-1, a novel interferon-gamma-responsive transcript encoding a growth-suppressing basic leucine zipper protein. *J Biol Chem*. Jun. 12, 1998;273(24):14731-7. doi: 10.1074/jbc.273.24.14731.
- Avidan et al., The processivity and fidelity of DNA synthesis exhibited by the reverse transcriptase of bovine leukemia virus. *Eur J Biochem*. Feb. 2002;269(3):859-67. doi: 10.1046/j.0014-2956.2001.02719.x.
- Babacic et al., CRISPR-cas gene-editing as plausible treatment of neuromuscular and nucleotide-repeat-expansion diseases: A systematic review. *PLoS One*. Feb. 22, 2019;14(2):e0212198. doi: 10.1371/journal.pone.0212198.
- Bacman et al., Specific elimination of mutant mitochondrial genomes in patient-derived cells by mitoTALENs. *Nat Med*. Sep. 2013;19(9):1111-3. doi: 10.1038/nm.3261. Epub Aug. 4, 2013.
- Badran et al., Continuous evolution of *Bacillus thuringiensis* toxins overcomes insect resistance. *Nature*. May 5, 2016;533(7601):58-63. doi: 10.1038/nature17938. Epub Apr. 27, 2016.
- Badran et al., Development of potent in vivo mutagenesis plasmids with broad mutational spectra. *Nat Commun*. Oct. 7, 2015;6:8425. doi: 10.1038/ncomms9425.
- Bagyinszky et al., Characterization of mutations in PRNP (prion) gene and their possible roles in neurodegenerative diseases. *Neuropsychiatr Dis Treat*. Aug. 14, 2018;14:2067-2085. doi: 10.2147/NDT.S165445.

(56)

References Cited

OTHER PUBLICATIONS

- Balakrishnan et al., Flap endonuclease I. *Annu Rev Biochem.* 2013;82:119-38. doi: 10.1146/annurev-biochem-072511-122603. Epub Feb. 28, 2013.
- Baldari et al., A novel leader peptide which allows efficient secretion of a fragment of human interleukin 1 beta in *Saccharomyces cerevisiae*. *EMBO J.* Jan. 1987;6(1):229-34.
- Banerjee et al., Cadmium inhibits mismatch repair by blocking the ATPase activity of the MSH2-MSH6 complex [published correction appears in *Nucleic Acids Res.* 2005;33(5):1738]. *Nucleic Acids Res.* 2005;33(4):1410-1419. Published Mar. 3, 2005. doi: 10.1093/nar/gki291.
- Banerji et al., A lymphocyte-specific cellular enhancer is located downstream of the joining region in immunoglobulin heavy chain genes. *Cell.* Jul. 1983;33(3):729-40. doi: 10.1016/0092-8674(83)90015-6.
- Bannert et al., Retroelements and the human genome: new perspectives on an old relation. *Proc Natl Acad Sci U S A.* Oct. 5, 2004;101 Suppl 2(Suppl 2):14572-9. doi: 10.1073/pnas.0404838101. Epub Aug. 13, 2004.
- Baranauskas et al., Generation and characterization of new highly thermostable and processive M-MuLV reverse transcriptase variants. *Protein Eng Des Sel.* Oct. 2012;25(10):657-68. doi: 10.1093/protein/gzs034. Epub Jun. 12, 2012.
- Barmania et al., C-C chemokine receptor type five (CCR5): An emerging target for the control of HIV infection. *Appl Transl Genom.* May 26, 2013;2:3-16. doi: 10.1016/j.atg.2013.05.004.
- Barnes et al., Repair and genetic consequences of endogenous DNA base damage in mammalian cells. *Annu Rev Genet.* 2004;38:445-76.
- Barnes et al., The fidelity of Taq polymerase catalyzing PCR is improved by an-terminal deletion. *Gene.* Mar. 1, 1992;112(1):29-35. doi: 10.1016/0378-1119(92)90299-5.
- Barrangou et al., CRISPR provides acquired resistance against viruses in prokaryotes. *Science.* Mar. 23, 2007;315(5819):1709-12.
- Barrangou, RNA-mediated programmable DNA cleavage. *Nat Biotechnol.* Sep. 2012;30(9):836-8. doi: 10.1038/nbt.2357.
- Bartlett et al., Efficient expression of protein coding genes from the murine U1 small nuclear RNA promoters. *Proc Natl Acad Sci U S A.* Aug. 20, 1996;93(17):8852-7. doi: 10.1073/pnas.93.17.8852.
- Bartosovic et al., N6-methyladenosine demethylase FTO targets pre-mRNAs and regulates alternative splicing and 3'-end processing. *Nucleic Acids Res.* Nov. 2, 2017;45(19):11356-11370. doi: 10.1093/nar/gkx778.
- Basha et al., Influence of cationic lipid composition on gene silencing properties of lipid nanoparticle formulations of siRNA in antigen-presenting cells. *Mol Ther.* Dec. 2011; 19(12):2186-200. doi: 10.1038/mt.2011.190. Epub Oct. 4, 2011.
- Basturea et al., Substrate specificity and properties of the *Escherichia coli* 16S rRNA methyltransferase, RsmE. *RNA.* Nov. 2007;13(11):1969-76. doi: 10.1261/rna.700507. Epub Sep. 13, 2007.
- Batey et al., Structure of a natural guanine-responsive riboswitch complexed with the metabolite hypoxanthine. *Nature.* Nov. 18, 2004;432(7015):411-5.
- Beale et al., Comparison of the differential context-dependence of DNA deamination by APOBEC enzymes: correlation with mutation spectra in vivo. *J Mol Biol.* Mar. 26, 2004;337(3):585-96.
- Bebenek et al., Error-prone polymerization by HIV-1 reverse transcriptase. Contribution of template-primer misalignment, miscoding, and termination probability to mutational hot spots. *J Biol Chem.* May 15, 1993;268(14):10324-34.
- Bedell et al., In vivo genome editing using a high-efficiency TALEN system. *Nature.* Nov. 1, 2012;491(7422):114-8. Doi: 10.1038/nature11537. Epub Sep. 23, 2012.
- Begley, Scientists unveil the 'most clever CRISPR gadget' so far. *STAT.* Apr. 20, 2016. <https://www.statnews.com/2016/04/20/clever-crispr-advance-unveiled/>.
- Behr, Gene transfer with synthetic cationic amphiphiles: prospects for gene therapy. *Bioconjug Chem.* Sep.-Oct. 1994;5(5):382-9. doi: 10.1021/bc00029a002.
- Bell et al., Ribozyme-catalyzed excision of targeted sequences from within RNAs. *Biochemistry.* Dec. 24, 2002;41(51):15327-33. doi: 10.1021/bi0267386.
- Belshaw et al., Controlling programmed cell death with a cyclophilin-cyclosporin-based chemical inducer of dimerization. *Chem Biol.* Sep. 1996;3(9):731-8. doi: 10.1016/s1074-5521(96)90249-5.
- Belshaw et al., Controlling protein association and subcellular localization with a synthetic ligand that induces heterodimerization of proteins. *Proc Natl Acad Sci U S A.* May 14, 1996;93(10):4604-7. doi: 10.1073/pnas.93.10.4604.
- Benarroch, HCN channels: function and clinical implications. *Neurology.* Jan. 15, 2013;80(3):304-10. doi: 10.1212/WNL.0b013e31827dec42.
- Bennett et al., Painful and painless channelopathies. *Lancet Neurol.* Jun. 2014;13(6):587-99. doi: 10.1016/S1474-4422(14)70024-9. Epub May 6, 2014.
- Bentin, T., A ribozyme transcribed by a ribozyme. *Artif DNA PNA XNA.* Apr. 2011;2(2):40-42. doi: 10.4161/adna.2.2.16852.
- Berger et al., Reverse transcriptase and its associated ribonuclease H: interplay of two enzyme activities controls the yield of single-stranded complementary deoxyribonucleic acid. *Biochemistry.* May 10, 1983;22(10):2365-72. doi: 10.1021/bi00279a010.
- Berges et al., Transduction of brain by herpes simplex virus vectors. *Mol Ther.* Jan. 2007;15(1):20-9. doi: 10.1038/sj.mt.6300018.
- Berkhout et al., Identification of an active reverse transcriptase enzyme encoded by a human endogenous HERV-K retrovirus. *J Virol.* Mar. 1999;73(3):2365-75. doi: 10.1128/JVI.73.3.2365-2375. 1999.
- Bernhart et al., Local RNA base pairing probabilities in large sequences. *Bioinformatics.* Mar. 1, 2006;22(5):614-5. doi: 10.1093/bioinformatics/btk014. Epub Dec. 20, 2005.
- Bernstein et al., Role for a bidentate ribonuclease in the initiation step of RNA interference. *Nature.* Jan. 18, 2001;409(6818):363-6. doi: 10.1038/35053110.
- Bershtein et al., Advances in laboratory evolution of enzymes. *Curr Opin; Chem Biol.* Apr. 2008;12(2):151-8. doi: 10.1016/j.cbpa.2008.01.027. Epub Mar. 7, 2008. Review
- Bertolotti et al., Toward genosafe endonuclease-boosted gene targeting using breakthrough CRISP/Cas9 for next generation stem cell gene therapy culminating in efficient ex vivo in vivo gene repair/genomic editing. *Molecular Therapy.* May 2015;23(Suppl1):S139. Abstract 350. 18th Ann Meeting of the American Society of Gene and Cell Therapy. ASGCT 2015. New Orleans, LA. May 13, 2015-May 16, 2015.
- Bertrand et al., Localization of ASH1 mRNA particles in living yeast. *Mol Cell.* Oct. 1998;2(4):437-45. doi: 10.1016/s1097-2765(00)80143-4.
- Bessen et al., High-resolution specificity profiling and off-target prediction for site-specific DNA recombinases. *Nat Commun.* Apr. 2, 2019;10(1):1937. doi: 10.1038/s41467-019-09987-0.
- Beumer et al., Efficient gene targeting in *Drosophila* with zinc-finger nucleases. *Genetics.* Apr. 2006; 172(4):2391-403. Epub Feb. 1, 2006.
- Bhagwat, DNA-cytosine deaminases: from antibody maturation to antiviral defense. *DNA Repair (Amst).* Jan. 5, 2004;3(1):85-9.
- Bi et al., Pseudo attP sites in favor of transgene integration and expression in cultured porcine cells identified by *Streptomyces* phage phiC31 integrase. *BMC Mol Biol.* Sep. 8, 2013;14:20. doi: 10.1186/1471-2199-14-20.
- Bibb et al., Integration and excision by the large serine recombinase phiRv1 integrase. *Mol Microbiol.* Mar. 2005;55(6):1896-910. doi: 10.1111/j.1365-2958.2005.04517.x.
- Billon et al., CRISPR-Mediated Base Editing Enables Efficient Disruption of Eukaryotic Genes through Induction of Stop Codons. *Mol Cell.* Sep. 21, 2017;67(6):1068-1079.e4. doi: 10.1016/j.molcel.2017.08.008. Epub Sep. 7, 2017.
- Birling et al., Site-specific recombinases for manipulation of the mouse genome. *Methods Mol Biol.* 2009;561:245-63. doi: 10.1007/978-1-60327-019-9_16.
- Biswas et al., A structural basis for allosteric control of DNA recombination by lambda integrase. *Nature.* Jun. 2, 2005;435(7045):1059-66. doi: 10.1038/nature03657.

(56)

References Cited

OTHER PUBLICATIONS

- Bitinaite et al., FokI dimerization is required for DNA cleavage. *Proc Natl Acad Sci U S A*. Sep. 1, 1998;95(18):10570-5.
- Blaese et al., Vectors in cancer therapy: how will they deliver? *Cancer Gene Ther*. Dec. 1995;2(4):291-7.
- Blain et al., Nuclease activities of Moloney murine leukemia virus reverse transcriptase. Mutants with altered substrate specificities. *J Biol Chem*. Nov. 5, 1993;268(31):23585-92.
- Blaisonneau et al., A circular plasmid from the yeast *Torulaspora delbrueckii*. *Plasmid*. 1997;38(3):202-9. doi: 10.1006/plas.1997.1315.
- Blau et al., A proliferation switch for genetically modified cells. *PNAS* Apr. 1, 1997 94 (7)3076-3081; <https://doi.org/10.1073/pnas.94.7.3076>.
- Bloom et al., Evolving strategies for enzyme engineering. *Curr Opin Struct Biol*. Aug. 2005;15(4):447-52.
- Boch, TALEs of genome targeting. *Nat Biotechnol*. Feb. 2011;29(2):135-6. Doi: 10.1038/nbt.1767.
- Bodi et al., Yeast m6A Methylated mRNAs Are Enriched on Translating Ribosomes during Meiosis, and under Rapamycin Treatment. *PLoS One*. Jul. 17, 2015;10(7):e0132090. doi: 10.1371/journal.pone.0132090.
- Boeckle et al., Melittin analogs with high lytic activity at endosomal pH enhance transfection with purified targeted PEI polyplexes. *J Control Release*. May 15, 2006;112(2):240-8. Epub Mar. 20, 2006.
- Boersma et al., Selection strategies for improved biocatalysts. *FEBS J*. May 2007;274(9):2181-95.
- Bogdanove et al., Engineering altered protein-DNA recognition specificity. *Nucleic Acids Res*. Jun. 1, 2018;46(10):4845-4871. doi: 10.1093/nar/gky289.
- Bohlke et al., Sense codon emancipation for proteome-wide incorporation of noncanonical amino acids: rare isoleucine codon AUA as a target for genetic code expansion. *FEMS Microbiol Lett*. Feb. 2014;351(2):133-44. doi: 10.1111/1574-6968.12371. Epub Jan. 27, 2014.
- Bolotin et al., Clustered regularly interspaced short palindrome repeats (CRISPRs) have spacers of extrachromosomal origin. *Microbiology*. Aug. 2005;151(Pt 8):2551-61.
- Bolusani et al., Evolution of variants of yeast site-specific recombinase Flp that utilize native genomic sequences as recombination target sites. *Nucleic Acids Res*. 2006;34(18):5259-69. Epub Sep. 26, 2006.
- Bondeson et al., Inversion of the IDS gene resulting from recombination with IDS-related sequences is a common cause of the Hunter syndrome. *Hum Mol Genet*. Apr. 1995;4(4):615-21. doi: 10.1093/hmg/4.4.615.
- Borchardt et al., Controlling mRNA stability and translation with the CRISPR endoribonuclease Csy4. *RNA*. Nov. 2015;21(11):1921-30. doi: 10.1261/rna.051227.115. Epub Sep. 9, 2015.
- Borman, Improved route to single-base genome editing. *Chemical & Engineering News*, Apr. 25, 2016;94(17)p. 5. <http://cen.acs.org/articles/94/i17/Improved-route-single-base-genome.html>.
- Bourinet et al., Silencing of the Cav3.2 T-type calcium channel gene in sensory neurons demonstrates its major role in nociception. *EMBO J*. Jan. 26, 2005;24(2):315-24. doi: 10.1038/sj.emboj.7600515. Epub Dec. 16, 2004.
- Boutabout et al., DNA synthesis fidelity by the reverse transcriptase of the yeast retrotransposon Ty1. *Nucleic Acids Res*. Jun. 1, 2001;29(11):2217-22. doi: 10.1093/nar/29.11.2217.
- Box et al., A multi-domain protein system based on the HC fragment of tetanus toxin for targeting DNA to neuronal cells. *J Drug Target*. Jul. 2003;11(6):333-43. doi: 10.1080/1061186310001634667.
- Branden and Tooze, *Introduction to Protein Structure*. 1999; 2nd edition. Garland Science Publisher: 3-12.
- Braun et al., Immunogenic duplex nucleic acids are nuclease resistant. *J Immunol*. Sep. 15, 1988;141(6):2084-9.
- Briner et al., Guide RNA functional modules direct Cas9 activity and orthogonality. *Mol Cell*. Oct. 23, 2014;56(2):333-339. doi: 10.1016/j.molcel.2014.09.019.
- Britt et al., Re-engineering plant gene targeting. *Trends Plant Sci*. Feb. 2003;8(2):90-5.
- Brouns et al., Small CRISPR RNAs guide antiviral defense in prokaryotes. *Science*. Aug. 15, 2008;321(5891):960-4. doi: 10.1126/science.1159689.
- Brown et al., A mammalian protein targeted by G1-arresting rapamycin-receptor complex. *Nature*. Jun. 30, 1994;369(6483):756-8. doi: 10.1038/369756a0.
- Brown et al., Characterization of the genetic elements required for site-specific integration of plasmid pSE211 in *Saccharopolyspora erythraea*. *J Bacteriol*. Apr. 1990;172(4):1877-88. doi: 10.1128/jb.172.4.1877-1888.1990.
- Brown et al., Serine recombinases as tools for genome engineering. *Methods*. Apr. 2011;53(4):372-9. doi: 10.1016/j.ymeth.2010.12.031. Epub Dec. 30, 2010.
- Brown et al., Structural insights into the stabilization of MALATI noncoding RNA by a bipartite triple helix. *Nat Struct Mol Biol*. Jul. 2014;21(7):633-40. doi: 10.1038/nsmb.2844. Epub Jun. 22, 2014.
- Brusse et al., Spinocerebellar ataxia associated with a mutation in the fibroblast growth factor 14 gene (SCA27): A new phenotype. *Mov Disord*. Mar. 2006;21(3):396-401.
- Brzezicha et al., Identification of human tRNA:m5C methyltransferase catalysing intron-dependent m5C formation in the first position of the anticodon of the pre-tRNA Leu (CAA). *Nucleic Acids Res*. 2006;34(20):6034-43. doi: 10.1093/nar/gkl1765. Epub Oct. 27, 2006.
- Buchholz et al., Alteration of Cre recombinase site specificity by substrate-linked protein evolution. *Nat Biotechnol*. Nov. 2001;19(11):1047-52.
- Buchschaecher et al., Human immunodeficiency virus vectors for inducible expression of foreign genes. *J Virol*. May 1992;66(5):2731-9. doi: 10.1128/JVI.66.5.2731-2739.1992.
- Buchwald et al., Long-term, continuous intravenous heparin administration by an implantable infusion pump in ambulatory patients with recurrent venous thrombosis. *Surgery*. Oct. 1980;88(4):507-16.
- Budisa et al., Residue-specific bioincorporation of non-natural, biologically active amino acids into proteins as possible drug carriers: structure and stability of the per-thiaproline mutant of annexin V. *Proc Natl Acad Sci U S A*. Jan. 20, 1998;95(2):455-9.
- Budker et al., Protein/amphipathic polyamine complexes enable highly efficient transfection with minimal toxicity. *Biotechniques*. Jul. 1997;23(1):139, 142-7. doi: 10.2144/97231r02.
- Budworth et al., A brief history of triplet repeat diseases. *Methods Mol Biol*. 2013; 1010:3-17. doi: 10.1007/978-1-62703-411-1_1.
- Bulow et al., Multienzyme systems obtained by gene fusion. *Trends Biotechnol*. Jul. 1991;9(7):226-31.
- Burke et al., Activating mutations of Tn3 resolvase marking interfaces important in recombination catalysis and its regulation. *Mol Microbiol*. Feb. 2004;51(4):937-48.
- Burke et al., RNA Aptamers to the Adenosine Moiety of S-adenosyl Methionine: Structural Inferences From Variations on a Theme and the Reproducibility of SELEX. *Nucleic Acids Res*. May 15, 1997;25(10):2020-4. doi: 10.1093/nar/25.10.2020.
- Burstein et al., New CRISPR-Cas systems from uncultivated microbes. *Nature*. Feb. 2017;542(7640):237-240.
- Burton et al., Gene delivery using herpes simplex virus vectors. *DNA Cell Biol*. Dec. 2002;21(12):915-36. doi: 10.1089/104454902762053864.
- Buskirk et al., Directed evolution of ligand dependence: small-molecule-activated protein splicing. *Proc Natl Acad Sci U S A*. Jul. 20, 2004;101(29):10505-10. Epub Jul. 9, 2004.
- Buskirk et al., In vivo evolution of an RNA-based transcriptional activator. *Chem Biol*. Jun. 2003; 10(6):533-40. doi: 10.1016/s1074-5521(03)00109-1.
- Butt et al., Efficient CRISPR/Cas9-Mediated Genome Editing Using a Chimeric Single-Guide RNA Molecule. *Front Plant Sci*. Aug. 24, 2017;8:1441(1-8). doi: 10.3389/fpls.2017.01441.
- Byrne et al., Multiplex gene regulation: a two-tiered approach to transgene regulation in transgenic mice. *Proc Natl Acad Sci U S A*. Jul. 1989;86(14):5473-7. doi: 10.1073/pnas.86.14.5473.
- Cade et al., Highly efficient generation of heritable zebrafish gene mutations using homo- and heterodimeric TALENs. *Nucleic Acids Res*. Sep. 2012;40(16):8001-10. Doi: 10.1093/nar/gks518. Epub Jun. 7, 2012.
- Cadwell et al., Randomization of genes by PCR mutagenesis. *PCR Methods Appl*. Aug. 1992;2(1):28-33. doi: 10.1101/gr.2.1.28.

(56)

References Cited

OTHER PUBLICATIONS

- Cai et al., Reconstruction of ancestral protein sequences and its applications. *BMC Evol Biol.* Sep. 17, 2004;4:33. doi: 10.1186/1471-2148-4-33.
- Calame et al., Transcriptional controlling elements in the immunoglobulin and T cell receptor loci. *Adv Immunol.* 1988;43:235-75. doi: 10.1016/s0065-2776(08)60367-3.
- Caldecott et al., Single-strand break repair and genetic disease. *Nat Rev Genet.* Aug. 2008;9(8):619-31. doi: 10.1038/nrg2380.
- Camarero et al., Biosynthesis of a Head-to-Tail Cyclized Protein with Improved Biological Activity. *J. Am. Chem. Soc.* May 29, 1999; 121(23):5597-5598. <https://doi.org/10.1021/ja990929n>.
- Cameron, Recent advances in transgenic technology. *Mol Biotechnol.* Jun. 1997;7(3):253-65.
- Camper et al., Postnatal repression of the alpha-fetoprotein gene is enhancer independent. *Genes Dev.* Apr. 1989;3(4):537-46. doi: 10.1101/gad.3.4.537.
- Camps et al., Targeted gene evolution in *Escherichia coli* using a highly error-prone DNA polymerase I. *Proc Natl Acad Sci U S A.* Aug. 19, 2003;100(17):9727-32. Epub Aug. 8, 2003.
- Canver et al., Customizing the genome as therapy for the ?-hemoglobinopathies. *Blood.* May 26, 2016;127(21):2536-45. doi: 10.1182/blood-2016-01-678128. Epub Apr. 6, 2016.
- Cargill et al., Characterization of single-nucleotide polymorphisms in coding regions of human genes. *Nat Genet.* Jul. 1999;22(3):231-8.
- Carlson et al., Negative selection and stringency modulation in phage-assisted continuous evolution. *Nat Chem Biol.* Mar. 2014;10(3):216-22. doi: 10.1038/nchembio.1453. Epub Feb. 2, 2014. With Supplementary Results.
- Caron et al., Intracellular delivery of a Tat-eGFP fusion protein into muscle cells. *Mol Ther.* Mar. 2001;3(3):310-8.
- Carr et al., Genome engineering. *Nat Biotechnol.* Dec. 2009;27(12):1151-62. doi: 10.1038/nbt.1590.
- Carroll et al., Gene targeting in *Drosophila* and *Caenorhabditis elegans* with zinc-finger nucleases. *Methods Mol Biol.* 2008;435:63-77. doi: 10.1007/978-1-59745-232-8_5.
- Carroll et al., Progress and prospects: zinc-finger nucleases as gene therapy agents. *Gene Ther.* Nov. 2008;15(22):1463-8. doi: 10.1038/gt.2008.145. Epub Sep. 11, 2008.
- Carroll, A CRISPR approach to gene targeting. *Mol Ther.* Sep. 2012;20(9):1658-60. doi: 10.1038/mt.2012.171.
- Carroll, Genome engineering with zinc-finger nucleases. *Genetics.* Aug. 2011;188(4):773-82. doi: 10.1534/genetics.111.131433. Review.
- Carvalho et al., Evolution in health and medicine Sackler colloquium: Genomic disorders: a window into human gene and genome evolution. *Proc Natl Acad Sci U S A.* Jan. 26, 2010;107 Suppl 1(Suppl 1):1765-71. doi: 10.1073/pnas.0906222107. Epub Jan. 13, 2010.
- Caspi et al., Distribution of split DnaE inteins in cyanobacteria. *Mol Microbiol.* Dec. 2003;50(5):1569-77. doi: 10.1046/j.1365-2958.2003.03825.x.
- Cattaneo et al., SEL1L affects human pancreatic cancer cell cycle and invasiveness through modulation of PTEN and genes related to cell-matrix interactions. *Neoplasia.* 2005;7(11):1030-1038.
- Cermak et al., Efficient design and assembly of custom TALEN and other TAL effector-based constructs for DNA targeting. *Nucleic Acids Res.* Jul. 2011;39(12):e82. Doi: 10.1093/nar/gkr218. Epub Apr. 14, 2011.
- Chadalavada et al., Wild-type is the optimal sequence of the HDV ribozyme under cotranscriptional conditions. *RNA.* Dec. 2007;13(12):2189-201. doi: 10.1261/rna.778107. Epub Oct. 23, 2007.
- Chadwick et al., In Vivo Base Editing of PCSK9 (Proprotein Convertase Subtilisin/Kexin Type 9) as a Therapeutic Alternative to Genome Editing. *Arterioscler Thromb Vasc Biol.* Sep. 2017;37(9):1741-1747. doi: 10.1161/ATVBAHA.117.309881. Epub Jul. 27, 2017.
- Chaikind et al., A programmable Cas9-serine recombinase fusion protein that operates on DNA sequences in mammalian cells. *Nucleic Acids Res.* Nov. 16, 2016;44(20):9758-9770. Epub Aug. 11, 2016.
- Chalberg et al., Integration specificity of phage phiC31 integrase in the human genome. *J Mol Biol.* Mar. 17, 2006;357(1):28-48. doi: 10.1016/j.jmb.2005.11.098. Epub Dec. 22, 2005.
- Chalberg et al., phiC31 integrase confers genomic integration and long-term transgene expression in rat retina. *Invest Ophthalmol Vis Sci.* Jun. 2005;46(6):2140-6. doi: 10.1167/iovs.04-1252.
- Chan et al., Molecular recording of mammalian embryogenesis. *Nature.* Jun. 2019;570(7759):77-82. doi: 10.1038/s41586-019-1184-5. Epub May 13, 2019.
- Chan et al., Novel selection methods for DNA-encoded chemical libraries. *Curr Opin Chem Biol.* 2015;26:55-61. doi: 10.1016/j.cbpa.2015.02.010.
- Chan et al., The choice of nucleotide inserted opposite abasic sites formed within chromosomal DNA reveals the polymerase activities participating in translesion DNA synthesis. *DNA Repair (Amst).* Nov. 2013;12(11):878-89. doi: 10.1016/j.dnarep.2013.07.008. Epub Aug. 26, 2013.
- Chapman et al., Playing the end game: DNA double-strand break repair pathway choice. *Mol Cell.* Aug. 24, 2012;47(4):497-510. doi: 10.1016/j.molcel.2012.07.029.
- Chari et al., Unraveling CRISPR-Cas9 genome engineering parameters via a library-on-library approach. *Nat Methods.* Sep. 2015;12(9):823-6. doi: 10.1038/nmeth.3473. Epub Jul. 13, 2015.
- Chapientier et al., Biotechnology: Rewriting a genome. *Nature.* Mar. 7, 2013;495(7439):50-1. doi: 10.1038/495050a.
- Chaturvedi et al., Stabilization of triple-stranded oligonucleotide complexes: use of probes containing alternating phosphodiester and stereo-uniform cationic phosphoramidate linkages. *Nucleic Acids Res.* Jun. 15, 1996;24(12):2318-23.
- Chavez et al., Highly efficient Cas9-mediated transcriptional programming. *Nat Methods.* Apr. 2015;12(4):326-8. doi: 10.1038/nmeth.3312. Epub Mar. 2, 2015.
- Chavez et al., Precise Cas9 targeting enables genomic mutation prevention. *bioRxiv.* Jun. 14, 2016; <http://dx.doi.org/10.1101/058974>. 6 pages. bioRxiv preprint first posted online Jun. 14, 2016.
- Chavez et al., Therapeutic applications of the ?C31 integrase system. *Curr Gene Ther.* Oct. 2011;11(5):375-81. Review.
- Chawla et al., An atlas of RNA base pairs involving modified nucleobases with optimal geometries and accurate energies. *Nucleic Acids Res.* Aug. 18, 2015;43(14):6714-29. doi: 10.1093/nar/gkv606. Epub Jun. 27, 2015.
- Chelico et al., Biochemical basis of immunological and retroviral responses to DNA-targeted cytosine deamination by activation-induced cytidine deaminase and APOBEC3G. *J Biol Chem.* Oct. 9, 2009;284(41):27761-5. doi: 10.1074/jbc.R109.052449. Epub Aug. 13, 2009.
- Chelico et al., Stochastic properties of processive cytidine DNA deaminases AID and APOBEC3G. *Philos Trans R Soc Lond B Biol Sci.* Mar. 12, 2009;364(1517):583-93. doi: 10.1098/rstb.2008.0195.
- Chen et al., Enhanced proofreading governs CRISPR-Cas9 targeting accuracy. *Nature.* Oct. 19, 2017;550(7676):407-410. doi: 10.1038/nature24268. Epub Sep. 20, 2017.
- Chen et al., A general strategy for the evolution of bond-forming enzymes using yeast display. *Proc Natl Acad Sci U S A.* Jul. 12, 2011;108(28):11399-404. doi: 10.1073/pnas.1101046108. Epub Jun. 22, 2011.
- Chen et al., Fusion protein linkers: property, design and functionality. *Adv Drug Deliv Rev.* Oct. 2013;65(10):1357-69. doi: 10.1016/j.addr.2012.09.039. Epub Sep. 29, 2012.
- Chen et al., Genome-wide CRISPR screen in a mouse model of tumor growth and metastasis. *Cell.* Mar. 12, 2015;160(6):1246-60. doi: 10.1016/j.cell.2015.02.038. Epub Mar. 5, 2015.
- Chen et al., Highly Efficient Mouse Genome Editing by CRISPR Ribonucleoprotein Electroporation of Zygotes. *J Biol Chem.* Jul. 8, 2016;291(28):14457-67. doi: 10.1074/jbc.M116.733154. Epub May 5, 2016.

(56)

References Cited

OTHER PUBLICATIONS

- Chen et al., m(6)A RNA methylation is regulated by microRNAs and promotes reprogramming to pluripotency. *Cell Stem Cell*. Mar. 5, 2015;16(3):289-301. doi: 10.1016/j.stem.2015.01.016. Epub Feb. 12, 2015.
- Chen et al., Structure of the DNA deaminase domain of the HIV-1 restriction factor APOBEC3G. *Nature*. Mar. 6, 2008;452(7183):116-9. doi: 10.1038/nature06638. Epub Feb. 20, 2008.
- Chen et al., Targeting genomic rearrangements in tumor cells through Cas9-mediated insertion of a suicide gene. *Nat Biotechnol*. Jun. 2017;35(6):543-550. doi: 10.1038/nbt.3843. Epub May 1, 2017.
- Cheng et al., Multiplexed activation of endogenous genes by CRISPR-on, an RNA-guided transcriptional activator system. *Cell Res*. Oct. 2013;23(10):1163-71. doi: 10.1038/cr.2013.122. Epub Aug. 27, 2013.
- Chesnoy et al., Structure and function of lipid-DNA complexes for gene delivery. *Annu Rev Biophys Biomol Struct*. 2000;29:27-47.
- Chester et al., The apolipoprotein B mRNA editing complex performs a multifunctional cycle and suppresses nonsense-mediated decay. *EMBO J*. Aug. 1, 2003;22(15):3971-82. doi: 10.1093/emboj/cdg369.
- Chew et al., A multifunctional AAV-CRISPR-Cas9 and its host response. *Nat Methods*. Oct. 2016;13(10):868-74. doi: 10.1038/nmeth.3993. Epub Sep. 5, 2016.
- Chew et al., A multifunctional AAV-CRISPR-Cas9 and its host response. *Nat Methods*. Oct. 2016; 13(10):868-74. doi: 10.1038/nmeth.3993. Epub Sep. 5, 2016. Supplementary Information.
- Chichili et al., Linkers in the structural biology of protein-protein interactions. *Protein Science*. 2013;22:153-67.
- Chin, Expanding and reprogramming the genetic code of cells and animals. *Annu Rev Biochem*. 2014;83:379-408. doi: 10.1146/annurev-biochem-060713-035737. Epub Feb. 10, 2014.
- Chipev et al., A leucine—proline mutation in the H1 subdomain of keratin 1 causes epidermolytic hyperkeratosis. *Cell*. Sep. 4, 1992;70(5):821-8.
- Cho et al., Analysis of off-target effects of CRISPR/Cas-derived RNA-guided endonucleases and nickases. *Genome Res*. Jan. 2014;24(1):132-41. doi: 10.1101/gr.162339.113. Epub Nov. 19, 2013.
- Cho et al., Site-specific recombination of bacteriophage P22 does not require integration host factor. *J Bacteriol*. Jul. 1999;181(14):4245-9. doi: 10.1128/JB.181.14.4245-4249.1999.
- Cho et al., Targeted genome engineering in human cells with the Cas9 RNA-guided endonuclease. *Nat Biotechnol*. Mar. 2013;31(3):230-2. doi: 10.1038/nbt.2507. Epub Jan. 29, 2013.
- Cho et al., The calcium-activated chloride channel anoctamin 1 acts as a heat sensor in nociceptive neurons. *Nat Neurosci*. May 27, 2012;15(7):1015-21. doi: 10.1038/nn.3111.
- Choe et al., Forging Ahead through Darkness: PCNA, Still the Principal Conductor at the Replication Fork. *Mol Cell*. Feb. 2, 2017;65(3):380-392. doi: 10.1016/j.molcel.2016.12.020.
- Choi et al., (6)-methyladenosine in mRNA disrupts tRNA selection and translation-elongation dynamics. *Nat Struct Mol Biol*. Feb. 2016;23(2):110-5. doi: 10.1038/nsmb.3148. Epub Jan. 11, 2016.
- Choi et al., Protein trans-splicing and characterization of a split family B-type DNA polymerase from the hyperthermophilic archaeal parasite *Nanoarchaeum equitans*. *J Mol Biol*. Mar. 10, 2006;356(5):1093-106. doi: 10.1016/j.jmb.2005.12.036. Epub Dec. 27, 2005.
- Choi et al., Translesion synthesis across abasic lesions by human B-family and Y-family DNA polymerases η , θ , ψ , δ , and REV1. *J Mol Biol*. Nov. 19, 2010;404(1):34-44. doi: 10.1016/j.jmb.2010.09.015. Epub Oct. 1, 2010.
- Chong et al., Modulation of protein splicing of the *Saccharomyces cerevisiae* vacuolar membrane ATPase intein. *J Biol Chem*. Apr. 24, 1998;273(17): 10567-77. doi: 10.1074/jbc.273.17.10567.
- Chong et al., Utilizing the C-terminal cleavage activity of a protein splicing element to purify recombinant proteins in a single chromatographic step. *Nucleic Acids Res*. Nov. 15, 1998;26(22):5109-15. doi: 10.1093/nar/26.22.5109.
- Chong et al., Protein splicing involving the *Saccharomyces cerevisiae* VMA intein. The steps in the splicing pathway, side reactions leading to protein cleavage, and establishment of an in vitro splicing system. *J Biol Chem*. Sep. 6, 1996;271(36):22159-68. doi: 10.1074/jbc.271.36.22159.
- Chong et al., Protein splicing of the *Saccharomyces cerevisiae* VMA intein without the endonuclease motifs. *J Biol Chem*. Jun. 20, 1997;272(25):15587-90. doi: 10.1074/jbc.272.25.15587.
- Chong et al., Single-column purification of free recombinant proteins using a self-cleavable affinity tag derived from a protein splicing element. *Gene*. Jun. 19, 1997;192(2):271-81. doi: 10.1016/s0378-1119(97)00105-4.
- Choudhury et al., CRISPR-dCas9 mediated TET1 targeting for selective DNA demethylation at BRCA1 promoter. *Oncotarget*. Jul. 19, 2016;7(29):46545-46556. doi: 10.18632/oncotarget.10234.
- Choudhury et al., Engineering RNA endonucleases with customized sequence specificities. *Nat Commun*. 2012;3:1147. doi: 10.1038/ncomms2154.
- Choulilika et al., Induction of homologous recombination in mammalian chromosomes by using the I-SceI system of *Saccharomyces cerevisiae*. *Mol Cell Biol*. Apr. 1995;15(4):1968-73. doi: 10.1128/MCB.15.4.1968.
- Christian et al., Targeting G with TAL effectors: a comparison of activities of TALENs constructed with NN and NK repeat variable di-residues. *PLoS One*. 2012;7(9):e45383. doi: 10.1371/journal.pone.0045383. Epub Sep. 24, 2012.
- Christiansen et al., Characterization of the lactococcal temperate phage TP901-1 and its site-specific integration. *J Bacteriol*. Feb. 1994;176(4):1069-76. doi: 10.1128/jb.176.4.1069-1076.1994.
- Chuai et al., DeepCRISPR: optimized CRISPR guide RNA design by deep learning. *Genome Biol*. Jun. 26, 2018;19(1):80. doi: 10.1186/s13059-018-1459-4.
- Chuai et al., In Silico Meets In Vivo: Towards Computational CRISPR-Based sgRNA Design. *Trends Biotechnol*. Jan. 2017;35(1):12-21. doi: 10.1016/j.tibtech.2016.06.008. Epub Jul. 11, 2016.
- Chuang et al., Novel Heterotypic Rox Sites for Combinatorial Dre Recombination Strategies. *G3 (Bethesda)*. Dec. 29, 2015;6(3):559-71. doi: 10.1534/g3.115.025841.
- Chujo et al., Trmt61B is a methyltransferase responsible for 1-methyladenosine at position 58 of human mitochondrial tRNAs. *RNA*. Dec. 2012;18(12):2269-76. doi: 10.1261/ma.035600.112. Epub Oct. 24, 2012.
- Clackson et al., Redesigning an FKBP-ligand interface to generate chemical dimerizers with novel specificity. *Proc Natl Acad Sci U S A*. Sep. 1, 1998;95(18): 10437-42. doi: 10.1073/pnas.95.18.10437.
- Clement et al., CRISPResso2 provides accurate and rapid genome editing sequence analysis. *Nat Biotechnol*. Mar. 2019;37(3):224-226. doi: 10.1038/s41587-019-0032-3.
- Cobb et al., Directed evolution as a powerful synthetic biology tool. *Methods*. Mar. 15, 2013;60(1):81-90. doi: 10.1016/j.jymeth.2012.03.009. Epub Mar. 23, 2012.
- Coffey et al., The Economic Impact of BSE on the U.S. Beef Industry: Product Value Losses, Regulatory Costs, and Consumer Reactions. Kansas State University Agricultural Experiment Station and Cooperative Extension Service. MF-2678. May 2005. 68 pages. Accessed via <https://bookstore.ksre.ksu.edu/pubs/MF2678.pdf>.
- Cokol et al., Finding nuclear localization signals. *EMBO Rep*. Nov. 2000;1(5):411-5. doi: 10.1093/embo-reports/kvd092.
- Cole et al., Reconstructing evolutionary adaptive paths for protein engineering. *Methods Mol Biol*. 2013;978:115-25. doi: 10.1007/978-1-62703-293-3_8.
- Cole-Strauss et al., Correction of the mutation responsible for sickle cell anemia by an RNA-DNA oligonucleotide. *Science*. Sep. 6, 1996;273(5280):1386-9.
- Collinge, Prion diseases of humans and animals: their causes and molecular basis. *Annu Rev Neurosci*. 2001;24:519-50. doi: 10.1146/annurev.neuro.24.1.519.

(56)

References Cited

OTHER PUBLICATIONS

- Conrad et al., A Kaposi's sarcoma virus RNA element that increases the nuclear abundance of intronless transcripts. *EMBO J.* May 18, 2005;24(10):1831-41. doi: 10.1038/sj.emboj.7600662. Epub Apr. 28, 2005.
- Conticello, The AID/APOBEC family of nucleic acid mutators. *Genome Biol.* 2008;9(6):229. doi: 10.1186/GB-2008-9-6-229. Epub Jun. 17, 2008.
- Costa et al., Frequent use of the same tertiary motif by self-folding RNAs. *EMBO J.* Mar. 15, 1995;14(6):1276-85.
- Cotton et al., Insertion of a Synthetic Peptide into a Recombinant Protein Framework: A Protein Biosensor. *J. Am. Chem. Soc.* Jan. 22, 1999; 121(5):1100-1. <https://doi.org/10.1021/ja983804b>.
- Covino et al., The CCL2/CCR2 Axis in the Pathogenesis of HIV-1 Infection: A New Cellular Target for Therapy? *Current Drug Targets* Dec. 2016;17(1):76-110. DOI: 10.2174/138945011701151217110917.
- Cox et al., An SCN9A channelopathy causes congenital inability to experience pain. *Nature.* Dec. 14, 2006;444(7121):894-8. doi: 10.1038/nature05413.
- Cox et al., Conditional gene expression in the mouse inner ear using Cre-loxP. *J Assoc Res Otolaryngol.* Jun. 2012;13(3):295-322. doi: 10.1007/s10162-012-0324-5. Epub Apr. 24, 2012.
- Cox et al., Congenital insensitivity to pain: novel SCN9A missense and in-frame deletion mutations. *Hum Mutat.* Sep. 2010;31(9):E1670-86. doi: 10.1002/humu.21325.
- Cox et al., RNA editing with CRISPR-Cas13. *Science.* Nov. 24, 2017;358(6366):1019-1027. doi: 10.1126/science.aag0180. Epub Oct. 25, 2017.
- Cox et al., Therapeutic genome editing: prospects and challenges. *Nat Med.* Feb. 2015;21(2):121-31. doi: 10.1038/nm.3793.
- Cox, Proteins pinpoint double strand breaks. *Elife.* Oct. 29, 2013;2:e01561. doi: 10.7554/eLife.01561.
- Crabtree et al., Three-part inventions: intracellular signaling and induced proximity. *Trends Biochem Sci.* Nov. 1996;21(11):418-22. doi: 10.1016/s0968-0004(96)20027-1.
- Cradick et al., CRISPR/Cas9 systems targeting β -globin and CCR5 genes have substantial off-target activity. *Nucleic Acids Res.* Nov. 1, 2013;41(20):9584-92. doi: 10.1093/nar/gkt714. Epub Aug. 11, 2013.
- Cradick et al., ZFN-site searches genomes for zinc finger nuclease target sites and off-target sites. *BMC Bioinformatics.* May 13, 2011;12:152. doi: 10.1186/1471-2105-12-152.
- Cradick et al., Zinc-finger nucleases as a novel therapeutic strategy for targeting hepatitis B virus DNAs. *Mol Ther.* May 2010;18(5):947-54. Doi: 10.1038/mt.2010.20. Epub Feb. 16, 2010.
- Crick, On protein synthesis. *Symp Soc Exp Biol.* 1958;12:138-63.
- Cronican et al., A class of human proteins that deliver functional proteins into mammalian cells in vitro and in vivo. *Chem Biol.* Jul. 29, 2011;18(7):833-8. doi: 10.1016/j.chembiol.2011.07.003.
- Cronican et al., Potent delivery of functional proteins into Mammalian cells in vitro and in vivo using a supercharged protein. *ACS Chem Biol.* Aug. 20, 2010;5(8):747-52. doi:10.1021/cb1001153.
- Crystal, Transfer of genes to humans: early lessons and obstacles to success. *Science.* Oct. 20, 1995;270(5235):404-10. doi: 10.1126/science.270.5235.404.
- Cui et al., m6A RNA Methylation Regulates the Self-Renewal and Tumorigenesis of Glioblastoma Stem Cells. *Cell Rep.* Mar. 14, 2017;18(11):2622-2634. doi: 10.1016/j.celrep.2017.02.059.
- Cui et al., Review of CRISPR/Cas9 sgRNA Design Tools. *Interdiscip Sci. Jun.* 2018;10(2):455-465. doi: 10.1007/s12539-018-0298-z. Epub Apr. 11, 2018.
- Cunningham et al., Ensembl 2015. *Nucleic Acids Res.* Jan. 2015;43(Database issue):D662-9. doi: 10.1093/nar/gku1010. Epub Oct. 28, 2014.
- Cupples et al., A set of lacZ mutations in *Escherichia coli* that allow rapid detection of each of the six base substitutions. *Proc Natl Acad Sci U S A.* Jul. 1989;86(14):5345-9.
- Dahlem et al., Simple methods for generating and detecting locus-specific mutations induced with TALENs in the zebrafish genome. *PLoS Genet.* 2012;8(8):e1002861. doi: 10.1371/journal.pgen.1002861. Epub Aug. 16, 2012.
- Dang et al., Optimizing sgRNA structure to improve CRISPR-Cas9 knockout efficiency. *Genome Biol.* Dec. 15, 2015;16:280. doi: 10.1186/s13059-015-0846-3.
- Das et al., The crystal structure of the monomeric reverse transcriptase from Moloney murine leukemia virus. *Structure.* May 2004;12(5):819-29. doi: 10.1016/j.str.2004.02.032.
- Dassa et al., Fractured genes: a novel genomic arrangement involving new split inteins and a new homing endonuclease family. *Nucleic Acids Res.* May 2009;37(8):2560-73. doi: 10.1093/nar/gkp095. Epub Mar. 5, 2009.
- Dassa et al., Trans protein splicing of cyanobacterial split inteins in endogenous and exogenous combinations. *Biochemistry.* Jan. 9, 2007;46(1):322-30. doi: 10.1021/bi0611762.
- Database EBI Accession No. ADE34233 Jan. 29, 2004.
- Database EBI Accession No. BFF09785. May 3, 20181. 2 pages.
- Database EBI Accession No. BGE38086. Jul. 2, 20195. 2 pages.
- Database UniProt Accession No. G8I3E0. Jan. 14, 2012.
- Datsenko et al., One-step inactivation of chromosomal genes in *Escherichia coli* K-12 using PCR products. *Proc Natl Acad Sci U S A.* Jun. 6, 2000;97(12):6640-5.
- Davidson et al., Viral vectors for gene delivery to the nervous system. *Nat Rev Neurosci.* May 2003;4(5):353-64. doi: 10.1038/nrn1104.
- Davis et al., Small molecule-triggered Cas9 protein with improved genome-editing specificity. *Nat Chem Biol.* May 2015;11(5):316-8. doi: 10.1038/nchembio.1793. Epub Apr. 6, 2015.
- Dean et al., Genetic restriction of HIV-1 infection and progression to AIDS by a deletion allele of the CKR5 structural gene. Hemophilia Growth and Development Study, Multicenter AIDS Cohort Study, Multicenter Hemophilia Cohort Study, San Francisco City Cohort, Alive Study. *Science.* Sep. 27, 1996;273(5283):1856-62. doi: 10.1126/science.273.5283.1856.
- DeKosky et al., Large-scale sequence and structural comparisons of human naive and antigen-experienced antibody repertoires. *Proc Natl Acad Sci U S A.* May 10, 2016;113(19):E2636-45. doi: 10.1073/pnas.1525510113. Epub Apr. 25, 2016.
- Delebecque et al., Organization of intracellular reactions with rationally designed RNA assemblies. *Science.* Jul. 22, 2011;333(6041):470-4. doi: 10.1126/science.1206938. Epub Jun. 23, 2011.
- Deng et al., Widespread occurrence of N6-methyladenosine in bacterial mRNA. *Nucleic Acids Res.* Jul. 27, 2015;43(13):6557-67. doi: 10.1093/nar/gkv596. Epub Jun. 11, 2015.
- Denizio et al., Harnessing natural DNA modifying activities for editing of the genome and epigenome. *Curr Opin Chem Biol.* Aug. 2018;45:10-17. doi: 10.1016/j.cbpa.2018.01.016. Epub Feb. 13, 2018.
- Deussing, Targeted mutagenesis tools for modelling psychiatric disorders. *Cell Tissue Res.* Oct. 2013;354(1):9-25. doi: 10.1007/s00441-013-1708-5. Epub Sep. 10, 2013.
- Dever et al., CRISPR/Cas9 γ -globin gene targeting in human haematopoietic stem cells. *Nature.* Nov. 17, 2016;539(7629):384-389. doi: 10.1038/nature20134. Epub Nov. 7, 2016.
- Deverman et al., Cre-dependent selection yields AAV variants for widespread gene transfer to the adult brain. *Nat Biotechnol.* Feb. 2016;34(2):204-9. doi: 10.1038/nbt.3440. Epub Feb. 1, 2016.
- Devigili et al., Paroxysmal itch caused by gain-of-function Nav1.7 mutation. *Pain.* Sep. 2014;155(9):1702-1707. doi: 10.1016/j.pain.2014.05.006. Epub May 10, 2014.
- Dianov et al., Mammalian base excision repair: the forgotten archangel. *Nucleic Acids Res.* Apr. 1, 2013;41(6):3483-90. doi: 10.1093/nar/gkt076. Epub Feb. 13, 2013.
- Dickey et al., Single-stranded DNA-binding proteins: multiple domains for multiple functions. *Structure.* Jul. 2, 2013;21(7):1074-84. doi: 10.1016/j.str.2013.05.013.
- Dickinson et al., Experimental interrogation of the path dependence and stochasticity of protein evolution using phage-assisted continuous evolution. *Proc Natl Acad Sci USA.* May 2013;110(22):9007-12.

(56)

References Cited

OTHER PUBLICATIONS

- Dillon, Regulating gene expression in gene therapy. *Trends Biotechnol.* May 1993;11(5):167-73. doi: 10.1016/0167-7799(93)90109-M.
- Ding et al., A Talen genome-editing system for generating human stem cell-based disease models. *Cell Stem Cell.* Feb. 7, 2013;12(2):238-51. Doi: 10.1016/j.stem.2012.11.011. Epub Dec. 13, 2012.
- Ding et al., Permanent alteration of PCSK9 with in vivo CRISPR-Cas9 genome editing. *Circ Res.* Aug. 15, 2014;115(5):488-92. doi: 10.1161/CIRCRESAHA.115.304351. Epub Jun. 10, 2014.
- Dingwall et al., Nuclear targeting sequences—a consensus? *Trends Biochem Sci.* Dec. 1991;16(12):478-81. doi: 10.1016/0968-0004(91)90184-w.
- Diver et al., Single-Step Synthesis of Cell-Permeable Protein Dimerizers That Activate Signal Transduction and Gene Expression. *J. Am. Chem. Soc.* Jun. 4, 1997;119(22):5106-5109. <https://doi.org/10.1021/ja963891c>.
- Dixon et al., Reengineering orthogonally selective riboswitches. *Proc Natl Acad Sci U S A.* Feb. 16, 2010;107(7):2830-5. doi: 10.1073/pnas.0911209107. Epub Jan. 26, 2010.
- Doench et al., Rational design of highly active sgRNAs for CRISPR-Cas9-mediated gene inactivation. *Nat Biotechnol.* Dec. 2014;32(12):1262-7. doi: 10.1038/nbt.3026. Epub Sep. 3, 2014.
- Dolan et al., Trans-splicing with the group I intron ribozyme from *Azoarcus*. *RNA.* Feb. 2014;20(2):202-13. doi: 10.1261/rna.041012.113. Epub Dec. 16, 2013.
- Doman et al., Evaluation and minimization of Cas9-independent off-target DNA editing by cytosine base editors. *Nat Biotechnol.* May 2020;38(5):620-628. doi: 10.1038/s41587-020-0414-6. Epub Feb. 10, 2020.
- Dominissini et al., Topology of the human and mouse m6A RNA methylomes revealed by m6A-seq. *Nature.* Apr. 29, 2012;485(7397):201-6. doi: 10.1038/nature11112.
- Dorgan et al., An enzyme-coupled continuous spectrophotometric assay for S-adenosylmethionine-dependent methyltransferases. *Anal Biochem.* Mar. 15, 2006;350(2):249-55. doi: 10.1016/j.ab.2006.01.004. Epub Feb. 7, 2006.
- Dormiani et al., Long-term and efficient expression of human β -globin gene in a hematopoietic cell line using a new site-specific integrating non-viral system. *Gene Ther.* Aug. 2015;22(8):663-74. doi: 10.1038/gt.2015.30. Epub Apr. 1, 2015.
- Dorr et al., Reprogramming the specificity of sortase enzymes. *Proc Natl Acad Sci U S A.* Sep. 16, 2014;111(37):13343-8. doi: 10.1073/pnas.1411179111. Epub Sep. 3, 2014.
- Doudna et al., Genome editing. The new frontier of genome engineering with CRISPR-Cas9. *Science.* Nov. 28, 2014;346(6213):1258096. doi: 10.1126/science.1258096.
- Doudna, The promise and challenge of therapeutic genome editing. *Nature.* Feb. 2020;578(7794):229-236. doi: 10.1038/s41586-020-1978-5. Epub Feb. 12, 2020.
- Dove et al., Conversion of the omega subunit of *Escherichia coli* RNA polymerase into a transcriptional activator or an activation target. *Genes Dev.* Mar. 1, 1998;12(5):745-54.
- Doyon et al., Directed evolution and substrate specificity profile of homing endonuclease I-SceI. *J Am Chem Soc.* Feb. 22, 2006;128(7):2477-84.
- Doyon et al., Heritable targeted gene disruption in zebrafish using designed zinc-finger nucleases. *Nat Biotechnol.* Jun. 2008;26(6):702-8. Doi: 10.1038/nbt1409. Epub May 25, 2008.
- Drake, A constant rate of spontaneous mutation in DNA-based microbes. *Proc Natl Acad Sci USA.* Aug. 15, 1991;88(16):7160-4.
- Dubois et al., Retroviral RNA Dimerization: From Structure to Functions. *Front Microbiol.* Mar. 22, 2018;9:527. doi: 10.3389/fmicb.2018.00527.
- Dunaime, Breakthrough method means CRISPR just got a lot more relevant to human health. *The Verge.* Apr. 20, 2016. <http://www.theverge.com/2016/4/20/11450262/crispr-base-editing-single-nucleotides-dna-gene-liu-harvard>.
- Dunbar et al., Gene therapy comes of age. *Science.* Jan. 12, 2018;359(6372):eaan4672. doi: 10.1126/science.aan4672.
- Dupuy et al., Le syndrome de De La Chapelle [De La Chapelle syndrome]. *Presse Med.* Mar. 3, 2001;30(8):369-72. French.
- Durai et al., A bacterial one-hybrid selection system for interrogating zinc finger-DNA interactions. *Comb Chem High Throughput Screen.* May 2006;9(4):301-11.
- Durai et al., Zinc finger nucleases: custom-designed molecular scissors for genome engineering of plant and mammalian cells. *Nucleic Acids Res.* Oct. 26, 2005;33(18):5978-90. doi: 10.1093/nar/gki912.
- During et al., Controlled release of dopamine from a polymeric brain implant: in vivo characterization. *Ann Neurol.* Apr. 1989;25(4):351-6.
- Edlund et al., Cell-specific expression of the rat insulin gene: evidence for role of two distinct 5' flanking elements. *Science.* Nov. 22, 1985;230(4728):912-6. doi: 10.1126/science.3904002.
- Edwards et al., An *Escherichia coli* tyrosine transfer RNA is a leucine-specific transfer RNA in the yeast *Saccharomyces cerevisiae*. *Proc Natl Acad Sci U S A.* Feb. 15, 1991;88(4):1153-6.
- Edwards et al., Crystal structures of the thi-box riboswitch bound to thiamine pyrophosphate analogs reveal adaptive RNA-small molecule recognition. *Structure.* Sep. 2006;14(9):1459-68.
- Eick et al., Robustness of Reconstructed Ancestral Protein Functions to Statistical Uncertainty. *Mol Biol Evol.* Feb. 1, 2017;34(2):247-261. doi: 10.1093/molbev/msw223.
- Eiler et al., Structural Basis for the Fast Self-Cleavage Reaction Catalyzed by the Twister Ribozyme. *Proc Natl Acad Sci U S A.* Sep. 9, 2014;111(36):13028-33. doi: 10.1073/pnas.1414571111. Epub Aug. 25, 2014.
- Eltoukhy et al., Nucleic acid-mediated intracellular protein delivery by lipid-like nanoparticles. *Biomaterials.* Aug. 2014;35(24):6454-61. doi: 10.1016/j.biomaterials.2014.04.014. Epub May 13, 2014.
- Emery et al., HCN2 ion channels play a central role in inflammatory and neuropathic pain. *Science.* Sep. 9, 2011;333(6048):1462-6. doi: 10.1126/science.1206243.
- Endo et al., Toward establishing an efficient and versatile gene targeting system in higher plants. *Biocatalysis and Agricultural Biotechnology* 2014;3,(1):2-6.
- Engel et al., The emerging role of mRNA methylation in normal and pathological behavior. *Genes Brain Behav.* Mar. 2018;17(3):e12428. doi: 10.1111/gbb.12428. Epub Nov. 17, 2017.
- Engelward et al., Base excision repair deficient mice lacking the Aag alkyladenine DNA glycosylase. *Proc Natl Acad Sci U S A.* Nov. 25, 1997;94(24):13087-92.
- England, Unnatural amino acid mutagenesis: a precise tool for probing protein structure and function. *Biochemistry.* Sep. 21, 2004;43(37):11623-9.
- Enyeart et al., Biotechnological applications of mobile group II introns and their reverse transcriptases: gene targeting, RNA-seq, and non-coding RNA analysis. *Mobile DNA* 5, 2 (2014). <https://doi.org/10.1186/1759-8753-5-2>. <https://doi.org/10.1186/1759-8753-5-2>.
- Epstein, HSV-1-based amplicon vectors: design and applications. *Gene Ther.* Oct. 2005;12 Suppl 1:S154-8. doi: 10.1038/sj.gt.3302617.
- Eriksson et al., Recurrent de novo point mutations in lamin A cause Hutchinson-Gilford progeria syndrome. *Nature.* May 15, 2003;423(6937):293-8. doi: 10.1038/nature01629. Epub Apr. 25, 2003. PMID: 12714972.
- Estacion et al., A sodium channel gene SCN9A polymorphism that increases nociceptor excitability. *Ann Neurol.* Dec. 2009;66(6):862-6. doi: 10.1002/ana.21895.
- Esvelt et al., A system for the continuous directed evolution of biomolecules. *Nature.* Apr. 28, 2011;472(7344):499-503. doi: 10.1038/nature09929. Epub Apr. 10, 2011.
- Esvelt et al., Genome-scale engineering for systems and synthetic biology. *Mol Syst Biol.* 2013;9:641. doi: 10.1038/msb.2012.66.
- Esvelt et al., Orthogonal Cas9 proteins for RNA-guided gene regulation and editing. *Nat Methods.* Nov. 2013;10(11):1116-21. doi: 10.1038/nmeth.2681. Epub Sep. 29, 2013.
- Evans et al., Protein trans-splicing and cyclization by a naturally split intein from the *dnaE* gene of *Synechocystis* species PCC6803. *J Biol Chem.* Mar. 31, 2000;275(13):9091-4. doi: 10.1074/jbc.275.13.9091.

(56)

References Cited

OTHER PUBLICATIONS

- Evans et al., Semisynthesis of cytotoxic proteins using a modified protein splicing element. *Protein Sci.* Nov. 1998;7(11):2256-64. doi: 10.1002/pro.5560071103.
- Evans et al., The cyclization and polymerization of bacterially expressed proteins using modified self-splicing inteins. *J Biol Chem.* Jun. 25, 1999;274(26):18359-63. doi: 10.1074/jbc.274.26.18359.
- Evans et al., The in vitro ligation of bacterially expressed proteins using an intein from *Methanobacterium thermoautotrophicum*. *J Biol Chem.* Feb. 12, 1999;274(7):3923-6. doi: 10.1074/jbc.274.7.3923.
- Fagerlund et al., The Cpf1 CRISPR-Cas protein expands genome-editing tools. *Genome Biology* Nov. 17, 2015;16:251. <https://doi.org/10.1186/s13059-015-0824-9>.
- Falnes et al., DNA repair by bacterial AlkB proteins. *Res Microbiol.* Oct. 2003;154(8):531-8. doi: 10.1016/S0923-2508(03)00150-5.
- Falnes et al., Repair of methyl lesions in DNA and RNA by oxidative demethylation. *Neuroscience.* Apr. 14, 2007;145(4):1222-32. doi: 10.1016/j.neuroscience.2006.11.018. Epub Dec. 18, 2006.
- Fang et al., Synthetic Studies Towards Halichondrins: Synthesis of the Left Halves of Norhalichondrins and Homohalichondrins. *Tetrahedron Letters* 1992;33(12):1557-1560.
- Farboud et al., Dramatic enhancement of genome editing by CRISPR/Cas9 through improved guide RNA design. *Genetics.* Apr. 2015;199(4):959-71. doi: 10.1534/genetics.115.175166. Epub Feb. 18, 2015.
- Farhood et al., Codelivery to mammalian cells of a transcriptional factor with cis-acting element using cationic liposomes. *Anal Biochem.* Feb. 10, 1995;225(1):89-93.
- Fawcett et al., Transposable elements controlling I-R hybrid dysgenesis in *D. melanogaster* are similar to mammalian LINEs. *Cell.* Dec. 26, 1986;47(6):1007-15. doi: 10.1016/0092-8674(86)90815-9.
- Feldstein et al., Two sequences participating in the autolytic processing of satellite tobacco ringspot virus complementary RNA. *Gene.* Oct. 15, 1989;82(1):53-61. doi: 10.1016/0378-1119(89)90029-2.
- Felletti et al., Twister Ribozymes as Highly Versatile Expression Platforms for Artificial Riboswitches. *Nat Commun.* Sep. 27, 2016;7:12834. doi: 10.1038/ncomms12834.
- Feng et al., Crystal structures of the human RNA demethylase Alkbh5 reveal basis for substrate recognition. *J Biol Chem.* Apr. 25, 2014;289(17):11571-11583. doi: 10.1074/jbc.M113.546168. Epub Mar. 10, 2014.
- Feng et al., Human L1 retrotransposon encodes a conserved endonuclease required for retrotransposition. *Cell.* Nov. 29, 1996;87(5):905-16. doi: 10.1016/S0092-8674(00)81997-2.
- Ferretti et al., Complete genome sequence of an M1 strain of *Streptococcus pyogenes*. *Proc Natl Acad Sci U S A.* Apr. 10, 2001;98(8):4658-63.
- Ferry et al., Rational design of inducible CRISPR guide RNAs for de novo assembly of transcriptional programs. *Nat Commun.* Mar. 3, 2017;8:14633. doi: 10.1038/ncomms14633.
- Feuk, Inversion variants in the human genome: role in disease and genome architecture. *Genome Med.* Feb. 12, 2010;2(2):11. doi: 10.1186/gm132.
- Filippov et al., A novel type of RNase III family proteins in eukaryotes. *Gene.* Mar. 7, 2000;245(1):213-21. doi: 10.1016/S0378-1119(99)00571-5.
- Fine et al., Trans-spliced Cas9 allows cleavage of HBB and CCR5 genes in human cells using compact expression cassettes. *Scientific Reports* 2015;5(1):Article No. 10777. doi:10.1038/srep10777. With Supplementary Information.
- Fire et al., Potent and specific genetic interference by double-stranded RNA in *Caenorhabditis elegans*. *Nature.* Feb. 19, 1998;391(6669):806-11. doi: 10.1038/35888.
- Fischbach et al., Directed evolution can rapidly improve the activity of chimeric assembly-line enzymes. *Proc Natl Acad Sci U S A.* Jul. 17, 2007;104(29):11951-6. doi: 10.1073/pnas.0705348104. Epub Jul. 9, 2007.
- Fischer et al., Cryptic epitopes induce high-titer humoral immune response in patients with cancer. *J Immunol.* Sep. 1, 2010;185(5):3095-102. doi: 10.4049/jimmunol.0902166. Epub Jul. 26, 2010.
- Fitzjohn, Diversitree: comparative phylogenetic analyses of diversification in R. *Methods in Ecology and Evolution.* Dec. 2012;3(6):1084-92. doi: 10.1111/j.2041-210X.2012.00234.x.
- Flajolet et al., Woodchuck hepatitis virus enhancer I and enhancer II are both involved in—myc2 activation in woodchuck liver tumors. *J Virol.* Jul. 1998;72(7):6175-80. doi: 10.1128/JVI.72.7.6175-6180.1998.
- Flaman et al., A rapid PCR fidelity assay. *Nucleic Acids Res.* Aug. 11, 1994;22(15):3259-60. doi: 10.1093/nar/22.15.3259.
- Flynn et al., CRISPR-mediated genotypic and phenotypic correction of a chronic granulomatous disease mutation in human iPS cells. *Exp Hematol.* Oct. 2015;43(10):838-848.e3. doi: 10.1016/j.exphem.2015.06.002. Epub Jun. 1, 2015. Including supplementary figures and data.
- Fogg et al., New applications for phage integrases. *J Mol Biol.* Jul. 29, 2014;426(15):2703-16. doi: 10.1016/j.jmb.2014.05.014. Epub May 22, 2014.
- Fogg et al., Genome Integration and Excision by a New *Streptomyces* Bacteriophage. *Joe. Appl Environ Microbiol.* Feb. 15, 2017;83(5):e02767-16. doi: 10.1128/AEM.02767-16.
- Fonfara et al., Phylogeny of Cas9 determines functional exchangeability of dual-RNA and Cas9 among orthologous type II CRISPR-Cas systems. *Nucleic Acids Res.* Feb. 2014;42(4):2577-90. doi: 10.1093/nar/gkt1074. Epub Nov. 22, 2013. Including Supplementary Information.
- Forster et al., Self-cleavage of virusoid RNA is performed by the proposed 55-nucleotide active site. *Cell.* Jul. 3, 1987;50(1):9-16. doi: 10.1016/0092-8674(87)90657-x.
- Fortini et al., Different DNA polymerases are involved in the short- and long-patch base excision repair in mammalian cells. *Biochemistry.* Mar. 17, 1998;37(11):3575-80. doi: 10.1021/bi972999h.
- Fouts et al., Sequencing *Bacillus anthracis* typing phages gamma and cherry reveals a common ancestry. *J Bacteriol.* May 2006;188(9):3402-8. doi: 10.1128/JB.188.9.3402-3408.2006.
- Freitas et al., Mechanisms and signals for the nuclear import of proteins. *Curr Genomics.* Dec. 2009;10(8):550-7. doi: 10.2174/138920209789503941.
- Freshney, *Culture of Animal Cells. A Manual of Basic Technique.* Alan R. Liss, Inc. New York. 1983;4.
- Fu et al., High-frequency off-target mutagenesis induced by CRISPR-Cas nucleases in human cells. *Nat Biotechnol.* Sep. 2013;31(9):822-6. doi: 10.1038/nbt.2623. Epub Jun. 23, 2013.
- Fu et al., Promises and Pitfalls of Intracellular Delivery of Proteins. *Bioconjugate Chemistry.* Aug. 2014;25:1602-8.
- Fuchs et al., Polyarginine as a multifunctional fusion tag. *Protein Sci.* Jun. 2005;14(6):1538-44.
- Fujisawa et al., Disease-associated mutations in CIAS1 induce cathepsin B-dependent rapid cell death of human THP-1 monocytic cells. *Blood.* Apr. 1, 2007;109(7):2903-11.
- Fukui et al., DNA Mismatch Repair in Eukaryotes and Bacteria. *J Nucleic Acids.* Jul. 27, 2010;2010. pii: 260512. doi: 10.4061/2010/260512.
- Fung et al., Repair at single targeted DNA double-strand breaks in pluripotent and differentiated human cells. *PLoS One.* 2011;6(5):e20514. doi: 10.1371/journal.pone.0020514. Epub May 25, 2011.
- Furukawa et al., In vitro selection of allosteric ribozymes that sense the bacterial second messenger c-di-GMP. *Methods Mol Biol.* 2014;1111:209-20. doi: 10.1007/978-1-62703-755-6_15.
- Fusi et al., In Silico Predictive Modeling of CRISPR/Cas9 guide efficiency. *Jun. 26, 2015; bioRxiv.* <http://dx.doi.org/10.1101/021568>.
- Gaj et al., 3rd. Genome engineering with custom recombinases. *Methods Enzymol.* 2014;546:79-91. doi: 10.1016/B978-0-12-801185-0.00004-0.
- Gaj et al., A comprehensive approach to zinc-finger recombinase customization enables genomic targeting in human cells. *Nucleic Acids Res.* Feb. 6, 2013;41(6):3937-46.

(56)

References Cited

OTHER PUBLICATIONS

- Gaj et al., Enhancing the specificity of recombinase-mediated genome engineering through dimer interface redesign. *J Am Chem Soc.* Apr. 2, 2014;136(13):5047-56. doi: 10.1021/ja4130059. Epub Mar. 20, 2014.
- Gaj et al., Expanding the scope of site-specific recombinases for genetic and metabolic engineering. *Biotechnol Bioeng.* Jan. 2014;111(1):1-15. doi: 10.1002/bit.25096. Epub Sep. 13, 2013.
- Gaj et al., Structure-guided reprogramming of serine recombinase DNA sequence specificity. *Proc Natl Acad Sci U S A.* Jan. 11, 2011;108(2):498-503. doi: 10.1073/pnas.1014214108. Epub Dec. 27, 2010.
- Gaj et al., ZFN, TALEN, and CRISPR/Cas-based methods for genome engineering. *Trends Biotechnol.* Jul. 2013;31(7):397-405. doi: 10.1016/j.tibtech.2013.04.004. Epub May 9, 2013.
- Gajula, Designing an Elusive CoG?GoC CRISPR Base Editor. *Trends Biochem Sci.* Feb. 2019;44(2):91-94. doi: 10.1016/j.tibs.2018.10.004. Epub Nov. 13, 2018.
- Gallo et al., A novel pathogenic PSEN1 mutation in a family with Alzheimer's disease: phenotypical and neuropathological features. *J Alzheimers Dis.* 2011;25(3):425-31. doi: 10.3233/JAD-2011-110185.
- Gangopadhyay et al., Precision Control of CRISPR-Cas9 Using Small Molecules and Light. *Biochemistry.* Jan. 29, 2019;58(4):234-244. doi: 10.1021/acs.biochem.8b01202. Epub Jan. 22, 2019.
- Gao et al., Cationic liposome-mediated gene transfer. *Gene Ther.* Dec. 1995;2(10):710-22.
- Gao et al., Self-processing of ribozyme-flanked RNAs into guide RNAs in vitro and in vivo for CRISPR-mediated genome editing. *J Integr Plant Biol.* Apr. 2014;56(4):343-9. doi: 10.1111/jipb.12152. Epub Mar. 6, 2014.
- Gapinske et al., CRISPR-SKIP: programmable gene splicing with single base editors. *Genome Biol.* Aug. 15, 2018;19(1):107. doi: 10.1186/s13059-018-1482-5.
- Garcia et al., Transglycosylation: a mechanism for RNA modification (and editing?). *Bioorg Chem.* Jun. 2005;33(3):229-51. doi: 10.1016/j.bioorg.2005.01.001. Epub Feb. 23, 2005.
- Gardlik et al., Vectors and delivery systems in gene therapy. *Med Sci Monit.* Apr. 2005;11(4):RA110-21. Epub Mar. 24, 2005.
- Garibyan et al., Use of the rpoB gene to determine the specificity of base substitution mutations on the *Escherichia coli* chromosome. *DNA Repair (Amst).* May 13, 2003;2(5):593-608.
- Garneau et al., The CRISPR/Cas bacterial immune system cleaves bacteriophage and plasmid DNA. *Nature.* Nov. 4, 2010;468(7320):67-71. doi: 10.1038/nature09523.
- Gasiunas et al., Cas9-crRNA ribonucleoprotein complex mediates specific DNA cleavage for adaptive immunity in bacteria. *Proc Natl Acad Sci U S A.* Sep. 25, 2012;109(39):E2579-86. Epub Sep. 4, 2012. Supplementary materials included.
- Gasiunas et al., RNA-dependent DNA endonuclease Cas9 of the CRISPR system: Holy Grail of genome editing? *Trends Microbiol.* Nov. 2013;21(11):562-7. doi: 10.1016/j.tim.2013.09.001. Epub Oct. 1, 2013.
- Gearing, Addgene blog. CRISPR 101: Cas9 nickase design and homology directed repair. 2018. pp. 1-12. <https://blog.addgene.org/crispr-101-cas9-nickase-design-and-homology-directed-repair>. Last retrieved online Jun. 25, 2021.
- Gehrke et al., An APOBEC3A-Cas9 base editor with minimized bystander and off-target activities. *Nat Biotechnol.* Nov. 2018;36(10):977-982. doi: 10.1038/nbt.4199. Epub Jul. 30, 2018. GenBank Accession No. J01600.1. Brooks et al., *E. coli* dam gene coding for DNA adenine methylase. Apr. 26, 1993. GenBank Accession No. U07651.1. Lu, *Escherichia coli* K12 negative regulator of replication initiation (seqA) gene, complete cds. Jul. 19, 1994. GenBank Submission; NIH/NCBI Accession No. 4UN5_B. Anders et al., Jul. 23, 2014. 5 pages. GenBank Submission; NIH/NCBI Accession No. AAA66622.1. Martinelli et al., May 18, 1995. 2 pages. GenBank Submission; NIH/NCBI Accession No. AGT42196. Farzadfar et al., Nov. 2, 2013. 2 pages. GenBank Submission; NIH/NCBI Accession No. AIT42264.1. Hyun et al., Oct. 15, 2014. 2 pages. GenBank Submission; NIH/NCBI Accession No. AKA60242.1. Tong et al., Apr. 5, 2015. 1 page. GenBank Submission; NIH/NCBI Accession No. AKQ21048.1. Gilles et al., Jul. 19, 2015. 1 page. GenBank Submission; NIH/NCBI Accession No. AKS40380.1. Nodvig et al., Aug. 2, 2015. 1 page. GenBank Submission; NIH/NCBI Accession No. APG80656.1. Burstein et al., Dec. 10, 2016. 1 pages. GenBank Submission; NIH/NCBI Accession No. AYD60528.1. Ram et al., Oct. 2, 2018. 1 page. GenBank Submission; NIH/NCBI Accession No. BDB43378. Zhang et al., Aug. 11, 2016. 2 pages. GenBank Submission; NIH/NCBI Accession No. J04623. Kita et al., Apr. 26, 1993. 2 pages. GenBank Submission; NIH/NCBI Accession No. KR710351.1. Sahni et al., Jun. 1, 2015. 2 pages. GenBank Submission; NIH/NCBI Accession No. NC_002737.1. Ferretti et al., Jun. 27, 2013. 1 page. GenBank Submission; NIH/NCBI Accession No. NC_015683.1. Trost et al., Jul. 6, 2013. 1 page. GenBank Submission; NIH/NCBI Accession No. NC_016782.1. Trost et al., Jun. 11, 2013. 1 page. GenBank Submission; NIH/NCBI Accession No. NC_016786.1. Trost et al., Aug. 28, 2013. 1 page. GenBank Submission; NIH/NCBI Accession No. NC_017053.1. Fittipaldi et al., Jul. 6, 2013. 1 page. GenBank Submission; NIH/NCBI Accession No. NC_017317.1. Trost et al., Jun. 11, 2013. 1 page. GenBank Submission; NIH/NCBI Accession No. NC_017861.1. Heidelberg et al., Jun. 11, 2013. 1 page. GenBank Submission; NIH/NCBI Accession No. NC_018010.1. Lucas et al., Jun. 11, 2013. 2 pages. GenBank Submission; NIH/NCBI Accession No. NC_018721.1. Feng et al., Jun. 11, 2013. 1 pages. GenBank Submission; NIH/NCBI Accession No. NC_021284.1. Ku et al., Jul. 12, 2013. 1 page. GenBank Submission; NIH/NCBI Accession No. NC_021314.1. Zhang et al., Jul. 15, 2013. 1 page. GenBank Submission; NIH/NCBI Accession No. NC_021846.1. Lo et al., Jul. 22, 2013. 1 page. GenBank Submission; NIH/NCBI Accession No. NM_002945.3. Weiser et al., Sep. 3, 2017. 5 pages. GenBank Submission; NIH/NCBI Accession No. NM_002947.4. Xiao et al., May 1, 2019. 4 pages. GenBank Submission; NIH/NCBI Accession No. NM_174936. Guo et al., Oct. 28, 2015. 6 pages. GenBank Submission; NIH/NCBI Accession No. NP_358988.1. Hoskins et al., Jan. 11, 2017. 2 pages. GenBank Submission; NIH/NCBI Accession No. NP_472073.1. Glaser et al., Jun. 27, 2013. 2 pages. GenBank Submission; NIH/NCBI Accession No. NP_628093.1. Hsiao et al., Aug. 3, 2016. 2 pages. GenBank Submission; NIH/NCBI Accession No. NP_955579.1. Chen et al., Aug. 13, 2018. 5 pages. GenBank Submission; NIH/NCBI Accession No. P42212. Prasher et al., Mar. 19, 2014. 7 pages. GenBank Submission; NIH/NCBI Accession No. QBJ66766. Duan et al., Aug. 12, 2020. 2 pages. GenBank Submission; NIH/NCBI Accession No. RFF81513.1. Zhou et al., Aug. 21, 2018. 2 pages. GenBank Submission; NIH/NCBI Accession No. SNX31424.1. Weckx, S., Feb. 16, 2018. 2 pages. GenBank Submission; NIH/NCBI Accession No. TGH57013. Xu et al., Apr. 9, 2019. 2 pages. GenBank Submission; NIH/NCBI Accession No. WP_002989955.1. No Author Listed, May 6, 2013. 1 page. GenBank Submission; NIH/NCBI Accession No. WP_010922251.1. No Author Listed, May 15, 2013. 1 page.

(56)

References Cited**OTHER PUBLICATIONS**

- GenBank Submission; NIH/NCBI, Accession No. WP_011054416.
1. No Author Listed, May 15, 2013. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_011284745.
1. No Author Listed, May 16, 2013. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_011285506.
1. No Author Listed, May 16, 2013. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_011527619.
1. No Author Listed, May 16, 2013. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_012560673.
1. No Author Listed, May 17, 2013. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_014407541.
1. No Author Listed, May 18, 2013. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_016631044.
1. Haft et al., Sep. 22, 2020. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_020905136.
1. No Author Listed, Jul. 25, 2013. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_023080005.
1. No Author Listed, Oct. 27, 2013. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_023610282.
1. No Author Listed, Nov. 27, 2013. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_030125963.
1. No Author Listed, Jul. 9, 2014. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_030126706.
1. No Author Listed, Jul. 9, 2014. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_031386437.
No Author Listed., Sep. 23, 2019. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_031488318.
1. No Author Listed., Aug. 5, 2014. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_031589969.
1. Haft et al., Oct. 9, 2019. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. WP_032460140.
1. No Author Listed, Oct. 4, 2014. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_032461047.
1. No Author Listed, Oct. 4, 2014. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_032462016.
1. Haft et al., Oct. 4, 2014. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_032462936.
1. No Author Listed, Oct. 4, 2014. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_032464890.
1. No Author Listed, Oct. 4, 2014. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_038431314.
1. No Author Listed, Dec. 26, 2014. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_038432938.
1. No Author Listed, Dec. 26, 2014. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_038434062.
1. No Author Listed, Dec. 26, 2014. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_044924278.
1. Haft et al., Oct. 9, 2019. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. WP_047338501.
1. Haft et al., Oct. 9, 2019. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. WP_048327215.
1. No Author Listed, Jun. 26, 2015. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_049519324.
1. No Author Listed, Jul. 20, 2015. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_060798984.
1. Haft et al., Oct. 9, 2019. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. WP_062913273.
1. Haft et al., Oct. 9, 2019. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. WP_072754838.
No Author Listed., Sep. 23, 2019. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_095142515.
1. No Author Listed., Sep. 23, 2019. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_118538418.
1. No Author Listed., Oct. 13, 2019. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_119223642.
1. No Author Listed., Oct. 13, 2019. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_119227726.
1. No Author Listed., Oct. 13, 2019. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_119623382.
1. No Author Listed., Oct. 13, 2019. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_132221894.
1. No Author Listed., Sep. 23, 2019. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. WP_133478044.
1. Haft et al., Oct. 9, 2019. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. YP_002342100.
1. Bernardini et al., Jun. 10, 2013. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. YP_002344900.
1. Gundogdu et al., Mar. 19, 2014. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. YP_009137104.
1. Davison, Aug. 13, 2018. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. YP_009283008.
1. Bernardini et al., Sep. 23, 2016. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. YP_820832.1.
Makarova et al., Aug. 27, 2013. 2 pages.
- George et al., Adenosine deaminases acting on RNA, RNA editing, and interferon action. *J Interferon Cytokine Res.* Jan. 2011;31(1):99-117. doi: 10.1089/jir.2010.0097. Epub Dec. 23, 2010. PMID: 21182352; PMCID: PMC3034097.
- Gerard et al., Influence on stability in *Escherichia coli* of the carboxy-terminal structure of cloned Moloney murine sarcoma virus reverse transcriptase. *DNA.* Aug. 1986;5(4):271-9. doi: 10.1089/dna.1986.5.271.
- Gerard et al., Purification and characterization of the DNA polymerase and RNase H activities in Moloney murine sarcoma-leukemia virus. *J Virol.* Apr. 1975;15(4):785-97. doi: 10.1128/JVI.15.4.785-797.1975.
- Gerard et al., The role of template-primer in protection of reverse transcriptase from thermal inactivation. *Nucleic Acids Res.* Jul. 15, 2002;30(14):3118-29. doi: 10.1093/nar/gkf417.
- Gerber et al., An adenosine deaminase that generates inosine at the wobble position of tRNAs. *Science.* Nov. 5, 1999;286(5442):1146-9. doi: 10.1126/science.286.5442.1146.
- Gersbach et al., Directed evolution of recombinase specificity by split gene reassembly. *Nucleic Acids Res.* Jul. 2010;38(12):4198-206. doi: 10.1093/nar/gkq125. Epub Mar. 1, 2010.
- Gersbach et al., Targeted plasmid integration into the human genome by an engineered zinc-finger recombinase. *Nucleic Acids Res.* Sep. 1, 2011;39(17):7868-78. doi: 10.1093/nar/gkr421. Epub Jun. 7, 2011.
- Ghahfarokhi et al., Blastocyst Formation Rate and Transgene Expression are Associated with Gene Insertion into Safe and Non-Safe Harbors in the Cattle Genome. *Sci Rep.* Nov. 13, 2017;7(1):15432. doi: 10.1038/s41598-017-15648-3.
- Gibson et al., Enzymatic assembly of DNA molecules up to several hundred kilobases. *Nat Methods.* May 2009;6(5):343-5. doi: 10.1038/nmeth.1318. Epub Apr. 12, 2009.
- Gil, Position-dependent sequence elements downstream of AAUAAA are required for efficient rabbit beta-globin mRNA 3' end formation. *Cell.* May 8, 1987;49(3):399-406. doi: 10.1016/0092-8674(87)90292-3.
- Gilbert et al., CRISPR-mediated modular RNA-guided regulation of transcription in eukaryotes. *Cell.* 2013 154(2):442-51.
- Gillerson et al., Image-based analysis of lipid nanoparticle-mediated siRNA delivery, intracellular trafficking and endosomal escape. *Nat Biotechnol.* Jul. 2013;31(7):638-46. doi: 10.1038/nbt.2612. Epub Jun. 23, 2013.
- Glasgow et al., DNA-binding properties of the Hin recombinase. *J Biol Chem.* Jun. 15, 1989;264(17):10072-82.
- Glassner et al., Generation of a strong mutator phenotype in yeast by imbalanced base excision repair. *Proc Natl Acad Sci U S A.* Aug. 18, 1998;95(17):9997-10002.
- Goldberg et al., Epigenetics: a landscape takes shape. *Cell.* Feb. 23, 2007;128(4):635-8. doi: 10.1016/j.cell.2007.02.006.
- Goldberg et al., Loss-of-function mutations in the Nav1.7 gene underlie congenital indifference to pain in multiple human populations. *Clin Genet.* Apr. 2007;71(4):311-9. doi: 10.1111/j.1399-0004.2007.00790.x.
- Gong et al., Active DNA demethylation by oxidation and repair. *Cell Res.* Dec. 2011;21(12):1649-51. doi: 10.1038/cr.2011.140. Epub Aug. 23, 2011.

(56)

References Cited

OTHER PUBLICATIONS

- Gonzalez et al., An iCRISPR platform for rapid, multiplexable, and inducible genome editing in human pluripotent stem cells. *Cell Stem Cell*. Aug. 7, 2014;15(2):215-26. doi: 10.1016/j.stem.2014.05.018. Epub Jun. 12, 2014.
- Goodnough et al., Development of a delivery vehicle for intracellular transport of botulinum neurotoxin antagonists. *FEBS Lett*. Feb. 27, 2002;513(2-3):163-8.
- Gordley et al., Evolution of programmable zinc finger-recombinases with activity in human cells. *J Mol Biol*. Mar. 30, 2007;367(3):802-13. Epub Jan. 12, 2007.
- Gordley et al., Synthesis of programmable integrases. *Proc Natl Acad Sci U S A*. Mar. 31, 2009;106(13):5053-8. doi: 10.1073/pnas.0812502106. Epub Mar. 12, 2009.
- Gou et al., Designing single guide RNA for CRISPR-Cas9 base editor by deep learning. Peer reviewed Thesis/Dissertation. UCLA Electronic Theses and Dissertations. Jan. 1, 2019. Retrieved from the Internet via <https://escholarship.org/uc/item/7vf9z54t>. Last accessed on Apr. 29, 2021.
- Grainge et al., The integrase family of recombinase: organization and function of the active site. *Mol Microbiol*. Aug. 1999;33(3):449-56.
- Gregory et al., Integration site for Streptomyces phage phiBT1 and development of site-specific integrating vectors. *J Bacteriol*. Sep. 2003;185(17):5320-3. doi: 10.1128/jb.185.17.5320-5323.2003.
- Griffiths, Endogenous retroviruses in the human genome sequence. *Genome Biol*. 2001;2(6):Reviews1017. doi: 10.1186/GB-2001-2-6-reviews1017. Epub Jun. 5, 2001.
- Grindley et al., Mechanisms of site-specific recombination. *Annu Rev Biochem*. 2006;75:567-605. doi: 10.1146/annurev.biochem.73.011303.073908.
- Grishok et al., Genes and Mechanisms Related to RNA Interference Regulate Expression of the Small Temporal RNAs that Control *C. elegans* Developmental Timing. *Jul. 13, 2001;106(1):p. 23-4.*
- Groher et al., Synthetic riboswitches—A tool comes of age. *Biochim Biophys Acta*. Oct. 2014;1839(10):964-973. doi: 10.1016/j.bbaggm.2014.05.005. Epub May 17, 2014.
- Groth et al., Construction of transgenic *Drosophila* by using the site-specific integrase from phage phiC31. *Genetics*. Apr. 2004;166(4):1775-82. doi: 10.1534/genetics.166.4.1775.
- Groth et al., Phage integrases: biology and applications. *J Mol Biol*. Jan. 16, 2004;335(3):667-78.
- Gruber et al., Strategies for measuring evolutionary conservation of RNA secondary structures. *BMC Bioinformatics*. Feb. 26, 2008;9:122. doi: 10.1186/1471-2105-9-122.
- Gruber et al., The Vienna RNA websuite. *Nucleic Acids Res*. Jul. 1, 2008;36(Web Server issue):W70-4. doi: 10.1093/nar/gkn188. Epub Apr. 19, 2008.
- Grunebaum et al., Recent advances in understanding and managing adenosine deaminase and purine nucleoside phosphorylase deficiencies. *Curr Opin Allergy Clin Immunol*. Dec. 2013;13(6):630-8. doi: 10.1097/ACI.0000000000000006.
- Grünwald et al., Transcriptome-wide off-target RNA editing induced by CRISPR-guided DNA base editors. *Nature*. May 2019;569(7756):433-437. doi: 10.1038/s41586-019-1161-z. Epub Apr. 17, 2019.
- Guedon et al., Current gene therapy using viral vectors for chronic pain. *Mol Pain*. May 13, 2015;11:27. doi: 10.1186/s12990-015-0018-1.
- Guilinger et al., Broad specificity profiling of TALENs results in engineered nucleases with improved DNA-cleavage specificity. *Nat Methods*. Apr. 2014;11(4):429-35. doi: 10.1038/nmeth.2845. Epub Feb. 16, 2014.
- Guilinger et al., Fusion of catalytically inactive Cas9 to FokI nuclease improves the specificity of genome modification. *Nat Biotechnol*. Jun. 2014;32(6):577-82. doi: 10.1038/nbt.2909. Epub Apr. 25, 2014.
- Gumulya et al., Exploring the past and the future of protein evolution with ancestral sequence reconstruction: the 'retro' approach to protein engineering. *Biochem J*. Jan. 1, 2017;474(1):1-19. doi: 10.1042/BCJ20160507.
- Guo et al., Evolution of Tetrahymena ribozyme mutants with increased structural stability. *Nat Struct Biol*. Nov. 2002;9(11):855-61. doi: 10.1038/nsb850.
- Guo et al., Facile functionalization of FK506 for biological studies by the thiol-ene 'click' reaction. *RSC Advances*. 2014;22:11400-3.
- Guo et al., Protein tolerance to random amino acid change. *Proc Natl Acad Sci U S A*. Jun. 22, 2004;101(25):9205-10. Epub Jun. 14, 2004.
- Guo et al., Structure of Cre recombinase complexed with DNA in a site-specific recombination synapse. *Nature*. Sep. 4, 1997;389(6646):40-6.
- Gupta et al., Cross-talk between cognate and noncognate RpoE sigma factors and Zn(2+)-binding anti-sigma factors regulates photooxidative stress response in *Azospirillum brasilense*. *Antioxid Redox Signal*. Jan. 1, 2014;20(1):42-59. doi: 10.1089/ars.2013.5314. Epub Jul. 19, 2013.
- Gupta et al., Sequences in attB that affect the ability of phiC31 integrase to synapse and to activate DNA cleavage. *Nucleic Acids Res*. 2007;35(10):3407-19. doi: 10.1093/nar/gkm206. Epub May 3, 2007.
- Guzman et al., Tight regulation, modulation, and high-level expression by vectors containing the arabinose PBAD promoter. *J Bacteriol*. 1995;177(14):4121-4130.
- Haapaniemi et al., CRISPR-Cas9 genome editing induces a p53-mediated DNA damage response. *Nat Med*. Jul. 2018;24(7):927-930. doi: 10.1038/s41591-018-0049-z. Epub Jun. 11, 2018.
- Haeussler et al., Evaluation of off-target and on-target scoring algorithms and integration into the guide RNA selection tool CRISPOR. *Genome Biol*. Jul. 5, 2016;17(1):148. doi: 10.1186/s13059-016-1012-2.
- Hale et al., RNA-guided RNA cleavage by a CRISPR RNA-Cas protein complex. *Cell*. Nov. 25, 2009;139(5):945-56. doi: 10.1016/j.cell.2009.07.040.
- Halmi et al., Targeted CRISPR/dCas9-mediated reactivation of epigenetically silenced genes suggests limited escape from the inactive X chromosome. 2nd Intl Conf on Epigenetics and Bioengineering. Oct. 4, 2018; Retrieved from the Internet: <https://aiche.confex.com/aiche/epibiol8/webprogram/paper544785.html>. Retrieved Jun. 29, 2020.
- Halperin et al., CRISPR-guided DNA polymerases enable diversification of all nucleotides in a tunable window. *Nature*. Aug. 2018;560(7717):248-252. doi: 10.1038/s41586-018-0384-8. Epub Aug. 1, 2018.
- Halvas et al., Role of murine leukemia virus reverse transcriptase deoxyribonucleoside triphosphate-binding site in retroviral replication and in vivo fidelity. *J Virol*. Nov. 2000;74(22):10349-58. doi: 10.1128/jvi.74.22.10349-10358.2000.
- Hamano-Takaku et al., A mutant *Escherichia coli* tyrosyl-tRNA synthetase utilizes the unnatural amino acid azatyrine more efficiently than tyrosine. *J Biol Chem*. Dec. 22, 2000;275(51):40324-8.
- Han, New CRISPR/Cas9-based Tech Edits Single Nucleotides Without Breaking DNA. *Genome Web*. Apr. 20, 2016. <https://www.genomeweb.com/gene-silencing/gene-editing/new-crisprcas9-based-tech-edits-single-nucleotides-without-breaking-dna>.
- Handa et al., Template-assisted synthesis of adenine-mutagenized cDNA by a retroelement protein complex. *Nucleic Acids Res*. Oct. 12, 2018;46(18):9711-9725. doi: 10.1093/nar/gky620.
- Hanson et al., Codon optimality, bias and usage in translation and mRNA decay. *Nat Rev Mol Cell Biol*. Jan. 2018;19(1):20-30. doi: 10.1038/nrm.2017.91. Epub Oct. 11, 2017.
- Harms et al., Evolutionary biochemistry: revealing the historical and physical causes of protein properties. *Nat Rev Genet*. Aug. 2013;14(8):559-71. doi: 10.1038/nrg3540.
- Harrington et al., A thermostable Cas9 with increased lifetime in human plasma. *Nat Commun*. Nov. 10, 2017;8(1):1424. doi: 10.1038/s41467-017-01408-4. Posted May 16, 2017 as bioRxiv preprint. Doi.org/10.1101/138867.

(56)

References Cited

OTHER PUBLICATIONS

- Harris et al., RNA Editing Enzyme APOBEC1 and Some of Its Homologs Can Act as DNA Mutators. *Mol Cell*. Nov. 2002;10(5):1247-53.
- Hartung et al., Correction of metabolic, craniofacial, and neurologic abnormalities in MPS I mice treated at birth with adeno-associated virus vector transducing the human alpha-L-iduronidase gene. *Mol Ther*. Jun. 2004;9(6):866-75.
- Hartung et al., Cre mutants with altered DNA binding properties. *J Biol Chem*. Sep. 4, 1998;273(36):22884-91.
- Hasadsri et al., Functional protein delivery into neurons using polymeric nanoparticles. *J Biol Chem*. Mar. 13, 2009;284(11):6972-81. doi: 10.1074/jbc.M805956200. Epub Jan. 7, 2009.
- Hasegawa et al., Spontaneous mutagenesis associated with nucleotide excision repair in *Escherichia coli*. *Genes Cells*. May 2008;13(5):459-69. doi: 10.1111/j.1365-2443.2008.01185.x.
- Hayes et al., Stop codons preceded by rare arginine codons are efficient determinants of SsrA tagging in *Escherichia coli*. *Proc Natl Acad Sci U S A*. Mar. 19, 2002;99(6):3440-5. Epub Mar. 12, 2002.
- Hector et al., CDKL5 variants: Improving our understanding of a rare neurologic disorder. *Neurol Genet*. Dec. 15, 2017;3(6):e200. doi: 10.1212/NXG.0000000000000200.
- Held et al., In vivo correction of murine hereditary tyrosinemia type I by phiC31 integrase-mediated gene delivery. *Mol Ther*. Mar. 2005;11(3):399-408. doi: 10.1016/j.ymthe.2004.11.001.
- Heller et al., Replisome assembly and the direct restart of stalled replication forks. *Nat Rev Mol Cell Biol*. Dec. 2006;7(12):932-43. Epub Nov. 8, 2006.
- Hendricks et al., The *S. cerevisiae* Mag1 3-methyladenine DNA glycosylase modulates susceptibility to homologous recombination. *DNA Repair (Amst)*. 2002;1(8):645-659.
- Hemonat et al., Use of adeno-associated virus as a mammalian DNA cloning vector: transduction of neomycin resistance into mammalian tissue culture cells. *Proc Natl Acad Sci U S A*. Oct. 1984;81(20):6466-70. doi: 10.1073/pnas.81.20.6466.
- Herschhorn et al., Retroviral reverse transcriptases. *Cell Mol Life Sci*. Aug. 2010;67(16):2717-47. doi: 10.1007/s00018-010-0346-2. Epub Apr. 1, 2010.
- Herzig et al., A Novel Leu92 Mutant of HIV-1 Reverse Transcriptase with a Selective Deficiency in Strand Transfer Causes a Loss of Viral Replication. *J Virol*. Aug. 2015;89(16):8119-29. doi: 10.1128/JVI.00809-15. Epub May 20, 2015.
- Hess et al., Directed evolution using dCas9-targeted somatic hypermutation in mammalian cells. *Nat Methods*. Dec. 2016;13(12):1036-1042. doi: 10.1038/nmeth.4038. Epub Oct. 31, 2016.
- Hickford et al., Antitumour polyether macrolides: four new halichondrins from the New Zealand deep-water marine sponge *Lissodendoryx* sp. *Bioorg Med Chem*. Mar. 15, 2009;17(6):2199-203. doi: 10.1016/j.bmc.2008.10.093. Epub Nov. 19, 2008.
- Higgs et al., Genetic complexity in sickle cell disease. *Proc Natl Acad Sci U S A*. Aug. 19, 2008;105(33):11595-6. doi: 10.1073/pnas.0806633105. Epub Aug. 11, 2008.
- Hill et al., Functional analysis of conserved histidines in ADP-glucose pyrophosphorylase from *Escherichia coli*. *Biochem Biophys Res Commun*. Mar. 17, 1998;244(2):573-7.
- Hille et al., The Biology of CRISPR-Cas: Backward and Forward. *Cell*. Mar. 8, 2018;172(6):1239-1259. doi: 10.1016/j.cell.2017.11.032.
- Hilton et al., Enabling functional genomics with genome engineering. *Genome Res*. Oct. 2015;25(10):1442-55. doi: 10.1101/gr.190124.115.
- Hirano et al., Site-specific recombinases as tools for heterologous gene integration. *Appl Microbiol Biotechnol*. Oct. 2011;92(2):227-39. doi: 10.1007/s00253-011-3519-5. Epub Aug. 7, 2011. Review.
- Hirano et al., Structural Basis for the Altered PAM Specificities of Engineered CRISPR-Cas9. *Mol Cell*. Mar. 17, 2016;61(6):886-94. doi: 10.1016/j.molcel.2016.02.018.
- Hoang et al., UFBoot2: Improving the Ultrafast Bootstrap Approximation. *Mol Biol Evol*. Feb. 1, 2018;35(2):518-522. doi: 10.1093/molbev/msx281.
- Hockemeyer et al., Efficient targeting of expressed and silent genes in human ESCs and iPSCs using zinc-finger nucleases. *Nat Biotechnol*. Sep. 2009;27(9):851-7. doi: 10.1038/nbt.1562. Epub Aug. 13, 2009.
- Hoernes et al., Translating the epitranscriptome. *Wiley Interdiscip Rev RNA*. Jan. 2017;8(1):e1375. doi: 10.1002/wrna.1375. Epub Jun. 27, 2016.
- Hoess et al., DNA specificity of the Cre recombinase resides in the 25 kDa carboxyl domain of the protein. *J Mol Biol*. Dec. 20, 1990;216(4):873-82. doi: 10.1016/S0022-2836(99)80007-2.
- Holden et al., Crystal structure of the anti-viral APOBEC3G catalytic domain and functional implications. *Nature*. Nov. 6, 2008;456(7218):121-4. doi: 10.1038/nature07357. Epub Oct. 12, 2008.
- Hollis et al., Phage integrases for the construction and manipulation of transgenic mammals. *Reprod Biol Endocrinol*. Nov. 7, 2003;1:79. doi: 10.1186/1477-7827-1-79.
- Holsinger et al., Signal transduction in T lymphocytes using a conditional allele of Sos. *Proc Natl Acad Sci U S A*. Oct. 10, 1995;92(21):9810-4. doi: 10.1073/pnas.92.21.9810.
- Holt et al., Human hematopoietic stem/progenitor cells modified by zinc-finger nucleases targeted to CCR5 control HIV-1 in vivo. *Nat Biotechnol*. Aug. 2010;28(8):839-47. doi: 10.1038/nbt.1663. Epub Jul. 2, 2010.
- Hondares et al., Peroxisome Proliferator-activated Receptor (PPAR) Induces PPAR Coactivator 1 (PGC-1) Gene Expression and Contributes to Thermogenic Activation of Brown Fat. *J Biol. Chem*. Oct. 2011;286(50):43112-22. doi: 10.1074/jbc.M111.252775.
- Hoogenboom et al., Natural and designer binding sites made by phage display technology. *Immunol Today*. Aug. 2000;21(8):371-8.
- Horvath et al., CRISPR/Cas, the immune system of bacteria and archaea. *Science*. Jan. 8, 2010;327(5962):167-70. doi: 10.1126/science.1179555.
- Horvath et al., Diversity, Activity, and Evolution of CRISPR Loci in *Streptococcus thermophilus*. *J Bacteriol*. Feb. 2008;190(4):1401-12. doi: 10.1128/JB.01415-07. Epub Dec. 7, 2007.
- Hotta et al., [Neurotropic viruses—classification, structure and characteristics]. *Nihon Rinsho*. Apr. 1997;55(4):777-82. Japanese.
- Houdebine, The methods to generate transgenic animals and to control transgene expression. *J Biotechnol*. Sep. 25, 2002;98(2-3):145-60.
- Housden et al., Identification of potential drug targets for tuberous sclerosis complex by synthetic screens combining CRISPR-based knockouts with RNAi. *Sci Signal*. Sep. 8, 2015;8(393):rs9. doi: 10.1126/scisignal.aab3729.
- Howard et al., Intracerebral drug delivery in rats with lesion-induced memory deficits. *J Neurosurg*. Jul. 1989;71(1):105-12.
- Hower et al., Shape-based peak identification for ChIP-Seq. *BMC Bioinformatics*. Jan. 12, 2011;12:15. doi: 10.1186/1471-2105-12-15.
- Hsu et al., DNA targeting specificity of RNA-guided Cas9 nucleases. *Nat Biotechnol*. Sep. 2013;31(9):827-32. doi: 10.1038/nbt.2647. Epub Jul. 21, 2013.
- Hsu et al., DNA targeting specificity of RNA-guided Cas9 nucleases. *Nat Biotechnol*. Sep. 2013;31(9):827-32. doi: 10.1038/nbt.2647. Epub Jul. 21, 2013. Supplementary Information. 27 pages.
- Hu et al., Chemical Biology Approaches to Genome Editing: Understanding, Controlling, and Delivering Programmable Nucleases. *Cell Chem Biol*. Jan. 21, 2016;23(1):57-73. doi: 10.1016/j.chembiol.2015.12.009.
- Hu et al., Evolved Cas9 variants with broad PAM compatibility and high DNA specificity. *Nature*. Apr. 5, 2018;556(7699):57-63 and Extended/Supplementary Data. doi: 10.1038/nature26155. Epub Feb. 28, 2018. 21 pages.
- Hua et al., Expanding the base editing scope in rice by using Cas9 variants. *Plant Biotechnol J*. Feb. 2019;17(2):499-504. doi: 10.1111/pbi.12993. Epub Oct. 5, 2018.
- Hua et al., Precise A•T to G•C Base Editing in the Rice Genome. *Mol Plant*. Apr. 2, 2018;11(4):627-630. doi: 10.1016/j.molp.2018.02.007. Epub Feb. 21, 2018.
- Huang et al., Circularly permuted and PAM-modified Cas9 variants broaden the targeting scope of base editors. *Nat Biotechnol*. Jun. 2019;37(6):626-631. doi: 10.1038/s41587-019-0134-y. Epub May 20, 2019. Including Supplementary Information.

(56)

References Cited

OTHER PUBLICATIONS

- Huang et al., Heritable gene targeting in zebrafish using customized TALENs. *Nat Biotechnol.* Aug. 5, 2011;29(8):699-700. doi: 10.1038/nbt.1939.
- Huggins et al., Flap endonuclease 1 efficiently cleaves base excision repair and DNA replication intermediates assembled into nucleosomes. *Mol Cell.* Nov. 2002;10(5):1201-11. doi: 10.1016/s1097-2765(02)00736-0.
- Humbert et al., Targeted gene therapies: tools, applications, optimization. *Crit Rev Biochem Mol Biol.* May-Jun. 2012;47(3):264-81. doi: 10.3109/10409238.2012.658112.
- Hung et al., Protein localization in disease and therapy. *J Cell Sci.* Oct. 15, 2011;124(Pt 20):3381-92. doi: 10.1242/jcs.089110.
- Hurt et al., Highly specific zinc finger proteins obtained by directed domain shuffling and cell-based selection. *Proc Natl Acad Sci U S A.* Oct. 14, 2003;100(21):12271-6. Epub Oct. 3, 2003.
- Husimi, Selection and evolution of bacteriophages in cellstat. *Adv Biophys.* ; 1989;25:1-43. Review.
- Hwang et al., Efficient In Vivo Genome Editing Using RNA-Guided Nucleases. *Nat Biotechnol.* Mar. 2013; 31(3): 227-229. doi: 10.1038/nbt.2501. Epub Jan. 29, 2013.
- Hwang et al., Web-based design and analysis tools for CRISPR base editing. *BMC Bioinformatics.* Dec. 27, 2018;19(1):542. doi: 10.1186/s12859-018-2585-4.
- Ibba et al., Relaxing the substrate specificity of an aminoacyl-tRNA synthetase allows in vitro and in vivo synthesis of proteins containing unnatural amino acids. *FEBS Lett.* May 15, 1995;364(3):272-5.
- Ibba et al., Substrate specificity is determined by amino acid binding pocket size in *Escherichia coli* phenylalanyl-tRNA synthetase. *Biochemistry.* Jun. 14, 1994;33(23):7107-12.
- Ihry et al., p53 inhibits CRISPR-Cas9 engineering in human pluripotent stem cells. *Nat Med.* Jul. 2018;24(7):939-946. doi: 10.1038/s41591-018-0050-6. Epub Jun. 11, 2018.
- Iida et al., A site-specific, conservative recombination system carried by bacteriophage P1. Mapping the recombinase gene *cin* and the cross-over sites *cix* for the inversion of the C segment. *EMBO J.* 1982;1(11):1445-53.
- Iida et al., The Min DNA inversion enzyme of plasmid p15B of *Escherichia coli* 15T-: a new member of the Din family of site-specific recombinases. *Mol Microbiol.* Jun. 1990;4(6):991-7. doi: 10.1111/j.1365-2958.1990.tb00671.x.
- Ikedobi et al., Mutation analysis of 24 known cancer genes in the NCI-60 cell line set. *Mol Cancer Ther.* Nov. 2006;5(11):2606-12. Epub Nov. 6, 2006.
- Imanishi et al., Detection of N6-methyladenosine based on the methyl-sensitivity of MazF RNA endonuclease. *Chem Commun (Camb).* Nov. 30, 201;53(96):12930-12933. doi: 10.1039/c7cc07699a.
- Imburgio et al., Studies of promoter recognition and start site selection by T7 RNA polymerase using a comprehensive collection of promoter variants. *Biochemistry.* Aug. 29, 2000;39(34):10419-30.
- Ingram, A specific chemical difference between the globins of normal human and sickle-cell anaemia haemoglobin. *Nature.* Oct. 13, 1956;178(4537):792-4. doi: 10.1038/178792a0.
- Irion et al., Identification and targeting of the ROSA26 locus in human embryonic stem cells. *Nat Biotechnol.* Dec. 2007;25(12):1477-82. doi: 10.1038/nbt1362. Epub Nov. 25, 2007.
- Irrthum et al., Congenital hereditary lymphedema caused by a mutation that inactivates VEGFR3 tyrosine kinase. *Am J Hum Genet.* Aug. 2000;67(2):295-301. Epub Jun. 9, 2000.
- Ishino et al., Nucleotide sequence of the *iap* gene, responsible for alkaline phosphatase isozyme conversion in *Escherichia coli*, and identification of the gene product. *J Bacteriol.* Dec. 1987;169(12):5429-33.
- Iwai et al., Circular beta-lactamase: stability enhancement by cyclizing the backbone. *FEBS Lett.* Oct. 8, 1999;459(2):166-72. doi: 10.1016/s0014-5793(99)01220-x.
- Iwai et al., Highly efficient protein trans-splicing by a naturally split DnaE intein from *Nostoc punctiforme*. *FEBS Lett.* Mar. 20, 2006;580(7):1853-8. doi: 10.1016/j.febslet.2006.02.045. Epub Feb. 24, 2006.
- Jaffrey et al., Emerging links between m6A and misregulated mRNA methylation in cancer. *Genome Med.* Jan. 12, 2017;9(1):2. doi: 10.1186/s13073-016-0395-8.
- Jamieson et al., Drug discovery with engineered zinc-finger proteins. *Nat Rev Drug Discov.* May 2003;2(5):361-8.
- Jansen et al., Backbone and nucleobase contacts to glucosamine-6-phosphate in the glmS ribozyme. *Nat Struct Mol Biol.* Jun. 2006;13(6):517-23. Epub May 14, 2006.
- Jansen et al., Identification of genes that are associated with DNA repeats in prokaryotes. *Mol Microbiol.* Mar. 2002;43(6):1565-75.
- Jardine et al., HIV-1 Vaccines. Priming a broadly neutralizing antibody response to HIV-1 using a germline-targeting immunogen. *Science.* Jul. 10, 2015;349(6244):156-61. doi: 10.1126/science.aac5894. Epub Jun. 18, 2015.
- Jasin et al., Repair of strand breaks by homologous recombination. *Cold Spring Harb Perspect Biol.* Nov. 1, 2013;5(11):a012740. doi: 10.1101/cshperspect.a012740.
- Jemielity et al., Novel "anti-reverse" cap analogs with superior translational properties. *RNA.* Sep. 2003;9(9):1108-22. doi: 10.1261/rna.5430403.
- Jenkins et al., Comparison of a preQ1 riboswitch aptamer in metabolite-bound and free states with implications for gene regulation. *J Biol Chem.* Jul. 15, 2011;286(28):24626-37. doi: 10.1074/jbc.M111.230375. Epub May 18, 2011.
- Jeong et al., Measurement of deoxyinosine adduct: Can it be a reliable tool to assess oxidative or nitrosative DNA damage? *Toxicol Lett.* Oct. 17, 2012;214(2):226-33. doi: 10.1016/j.toxlet.2012.08.013. Epub Aug. 23, 2012.
- Jiang et al., Chemical modifications of adenine base editor mRNA and guide RNA expand its application scope. *Nat Commun.* Apr. 24, 2020;11(1):1979. doi: 10.1038/s41467-020-15892-8.
- Jiang et al., CRISPR-Cas9 Structures and Mechanisms. *Annu Rev Biophys.* May 22, 2017;46:505-529. doi: 10.1146/annurev-biophys-062215-010822. Epub Mar. 30, 2017.
- Jiang et al., Prime editing efficiently generates W542L and S621I double mutations in two ALS genes of maize. *bioRxiv preprint.* Jul. 6, 2020. Retrieved from www.biorxiv.org. doi: 10.1101/2020.07.06.188896. 15 pages.
- Jiang et al., Structural Biology. A Cas9-guide RNA Complex Preorganized for Target DNA Recognition. *Science.* Jun. 26, 2015;348(6242):1477-81. doi: 10.1126/science.aab1452.
- Jiang et al., Structures of a CRISPR-Cas9 R-loop complex primed for DNA cleavage. *Science.* Feb. 19, 2016;351(6275):867-71. doi: 10.1126/science.aad8282. Epub Jan. 14, 2016.
- Jin et al., Cytosine, but not adenine, base editors induce genome-wide off-target mutations in rice. *Science.* Apr. 19, 2019;364(6437):292-295. doi: 10.1126/science.aaw7166. Epub Feb. 28, 2019.
- Jinek et al., Structures of Cas9 endonucleases reveal RNA-mediated conformational activation. *Science.* Mar. 14, 2014;343(6176):1247997. doi: 10.1126/science.1247997. Epub Feb. 6, 2014.
- Jiricny, The multifaceted mismatch-repair system. *Nat Rev Mol Cell Biol.* May 2006;7(5):335-46. doi: 10.1038/nrm1907.
- Johann et al., GLVR1, a receptor for gibbon ape leukemia virus, is homologous to a phosphate permease of *Neurospora crassa* and is expressed at high levels in the brain and thymus. *J Virol.* Mar. 1992;66(3):1635-40. doi: 10.1128/JVI.66.3.1635-1640.1992.
- Johansson et al., RNA Recognition by the MS2 Phage Coat Protein. *Seminars in Virology.* 1997;8(3):176-85. https://doi.org/10.1006/smv.1997.0120.
- Johansson et al., Selenocysteine in proteins-properties and biotechnological use. *Biochim Biophys Acta.* Oct. 30, 2005;1726(1):1-13. Epub Jun. 1, 2005.
- Johns et al., The promise and peril of continuous in vitro evolution. *J Mol Evol.* Aug. 2005;61(2):253-63. Epub Jun. 27, 2005.
- Johnson et al., Trans insertion-splicing: ribozyme-catalyzed insertion of targeted sequences into RNAs. *Biochemistry.* Aug. 9, 2005;44(31):10702-10. doi: 10.1021/bi0504815.

(56)

References Cited

OTHER PUBLICATIONS

- Joho et al., Identification of a region of the bacteriophage T3 and T7 RNA polymerases that determines promoter specificity. *J Mol Biol.* Sep. 5, 1990;215(1):31-9.
- Jore et al., Structural basis for CRISPR RNA-guided DNA recognition by Cascade. *Nat Struct Mol Biol.* May 2011;18(5):529-36. doi: 10.1038/nsmb.2019. Epub Apr. 3, 2011.
- Joung et al., TALENs: a widely applicable technology for targeted genome editing. *Nat Rev Mol Cell Biol.* Jan. 2013;14(1):49-55. doi: 10.1038/nrm3486. Epub Nov. 21, 2012.
- Joyce et al., Amplification, mutation and selection of catalytic RNA. *Gene.* Oct. 15, 1989;82(1):83-7. doi: 10.1016/0378-1119(89)90033-4.
- Jusiak et al., Comparison of Integrases Identifies Bxb1-GA Mutant as the Most Efficient Site-Specific Integrase System in Mammalian Cells. *ACS Synth Biol.* Jan. 18, 2019;8(1):16-24. doi: 10.1021/acssynbio.8b00089. Epub Jan. 9, 2019.
- Jyothy et al., Translocation Down syndrome. *Indian J Med Sci.* Mar. 2002;56(3):122-6.
- Kacian et al., Purification of the DNA polymerase of avian myeloblastosis virus. *Biochim Biophys Acta.* Sep. 24, 1971;246(3):365-83. doi: 10.1016/0005-2787(71)90773-8.
- Kaczmarczyk et al., Manipulating the Prion Protein Gene Sequence and Expression Levels with CRISPR/Cas9. *PLoS One.* Apr. 29, 2016;11(4):e0154604. doi: 10.1371/journal.pone.0154604.
- Kadoch et al., Reversible disruption of mSWI/SNF (BAF) complexes by the SS18-SSX oncogenic fusion in synovial sarcoma. *Cell.* Mar. 28, 2013;153(1):71-85. doi: 10.1016/j.cell.2013.02.036.
- Kahmann et al., G inversion in bacteriophage Mu DNA is stimulated by a site within the invertase gene and a host factor. *Cell.* Jul. 1985;41(3):771-80. doi: 10.1016/s0092-8674(85)80058-1.
- Kaiser et al., Gene therapy. Putting the fingers on gene repair. *Science.* Dec. 23, 2005;310(5756):1894-6.
- Kakiyama et al., A peptide release system using a photo-cleavable linker in a cell array format for cell-toxicity analysis. *Polymer J.* Feb. 27, 2013;45:535-9.
- Kalyaanamoorthy et al., ModelFinder: fast model selection for accurate phylogenetic estimates. *Nat Methods.* Jun. 2017; 14(6):587-589. doi: 10.1038/nmeth.4285. Epub May 8, 2017.
- Kandavelou et al., Targeted manipulation of mammalian genomes using designed zinc finger nucleases. *Biochem Biophys Res Commun.* Oct. 9, 2009;388(1):56-61. doi: 10.1016/j.bbrc.2009.07.112. Epub Jul. 25, 2009.
- Kang et al., Structural Insights into riboswitch control of the biosynthesis of queuosine, a modified nucleotide found in the anticodon of tRNA. *Mol Cell.* Mar. 27, 2009;33(6):784-90. doi: 10.1016/j.molcel.2009.02.019. Epub Mar. 12, 2009.
- Kang et al., Precision genome engineering through adenine base editing in plants. *Nat Plants.* Jul. 2018;4(7):427-431. doi: 10.1038/s41477-018-0178-x. Epub Jun. 4, 2018. Erratum in: *Nat Plants.* Sep. 2018;4(9):730.
- Kao et al., Cleavage specificity of *Saccharomyces cerevisiae* flap endonuclease 1 suggests a double-flap structure as the cellular substrate. *J Biol Chem.* Apr. 26, 2002;277(17):14379-89. doi: 10.1074/jbc.M110662200. Epub Feb. 1, 2002.
- Kappel et al., Regulating gene expression in transgenic animals. *Curr Opin Biotechnol.* Oct. 1992;3(5):548-53.
- Karimova et al., Discovery of Nigri/nox and Panto/pox site-specific recombinase systems facilitates advanced genome engineering. *Sci Rep.* Jul. 22, 2016;6:30130. doi: 10.1038/srep30130.
- Karimova et al., Vika/vox, a novel efficient and specific Cre/loxP-like site-specific recombination system. *Nucleic Acids Res.* Jan. 2013;41(2):e37. doi: 10.1093/nar/gks1037. Epub Nov. 9, 2012.
- Karpenshif et al., From yeast to mammals: recent advances in genetic control of homologous recombination. *DNA Repair (Amst).* Oct. 1, 2012;11(10):781-8. doi: 10.1016/j.dnarep.2012.07.001. Epub Aug. 11, 2012. Review.
- Karpinsky et al., Directed evolution of a recombinase that excises the provirus of most HIV-1 primary isolates with high specificity. *Nat Biotechnol.* Apr. 2016;34(4):401-9. doi: 10.1038/nbt.3467. Epub Feb. 22, 2016.
- Katafuchi et al., DNA polymerases involved in the incorporation of oxidized nucleotides into DNA: their efficiency and template base preference. *Mutat Res.* Nov. 28, 2010;703(1):24-31. doi: 10.1016/j.mrgentox.2010.06.004. Epub Jun. 11, 2010.
- Kato et al., Improved purification and enzymatic properties of three forms of reverse transcriptase from avian myeloblastosis virus. *J Virol Methods.* Dec. 1984;9(4):325-39. doi: 10.1016/0166-0934(84)90058-2.
- Katoh et al., MAFFT multiple sequence alignment software version 7: improvements in performance and usability. *Mol Biol Evol.* Apr. 2013;30(4):772-80. doi: 10.1093/molbev/mst010. Epub Jan. 16, 2013.
- Kaufman et al., Translational efficiency of polycistronic mRNAs and their utilization to express heterologous genes in mammalian cells. *EMBO J.* Jan. 1987;6(1):187-93.
- Kavli et al., Excision of cytosine and thymine from DNA by mutants of human uracil-DNA glycosylase. *EMBO J.* Jul. 1, 1996;15(13):3442-7.
- Kawarasaki et al., Enhanced crossover SCRATCHY: construction and high-throughput screening of a combinatorial library containing multiple non-homologous crossovers. *Nucleic Acids Res.* Nov. 1, 2003;31(21):e126.
- Kay et al., Viral vectors for gene therapy: the art of turning infectious agents into vehicles of therapeutics. *Nat Med.* Jan. 2001;7(1):33-40.
- Kaya et al., A bacterial Argonaute with noncanonical guide RNA specificity. *Proc. Natl. Acad. Sci. USA.* Apr. 2016;113(15):4057-62.
- Keijzers et al., Human exonuclease 1 (EXO1) activity characterization and its function on flap structures. *Biosci Rep.* Apr. 25, 2015;35(3):e00206. doi: 10.1042/BSR20150058.
- Kellendonk et al., Regulation of Cre recombinase activity by the synthetic steroid RU 486. *Nucleic Acids Res.* Apr. 15, 1996;24(8):1404-11.
- Kelman, PCNA: structure, functions and interactions. *Oncogene.* Feb. 13, 1997;14(6):629-40. doi: 10.1038/sj.onc.1200886.
- Keravala et al., A diversity of serine phage integrases mediate site-specific recombination in mammalian cells. *Mol Genet Genomics.* Aug. 2006;276(2):135-46. doi: 10.1007/s00438-006-0129-5. Epub May 13, 2006.
- Kessel et al., Murine developmental control genes. *Science.* Jul. 27, 1990;249(4967):374-9. doi: 10.1126/science.1974085.
- Kessler et al., Gene delivery to skeletal muscle results in sustained expression and systemic delivery of a therapeutic protein. *Proc Natl Acad Sci U S A.* Nov. 26, 1996;93(24):14082-7. doi: 10.1073/pnas.93.24.14082.
- Ketha et al., Application of bioinformatics-coupled experimental analysis reveals a new transport-competent nuclear localization signal in the nucleoprotein of Influenza A virus strain. *BMC Cell Biol.* Apr. 28, 2008; 9:22. <https://doi.org/10.1186/1471-2121-9-22>.
- Kiga et al., An engineered *Escherichia coli* tyrosyl-tRNA synthetase for site-specific incorporation of an unnatural amino acid into proteins in eukaryotic translation and its application in a wheat germ cell-free system. *Proc Natl Acad Sci U S A.* Jul. 23, 2002;99(15):9715-20. Epub Jul. 3, 2002.
- Kilbride et al., Determinants of product topology in a hybrid Cre-Tn3 resolvase site-specific recombination system. *J Mol Biol.* Jan. 13, 2006;355(2):185-95. Epub Nov. 9, 2005.
- Kilcher et al., Brochothrix thermosphacta bacteriophages feature heterogeneous and highly mosaic genomes and utilize unique prophage insertion sites. *J Bacteriol.* Oct. 2010; 192(20):5441-53. doi: 10.1128/JB.00709-10. Epub Aug. 13, 2010.
- Kim et al., DJ-1, a novel regulator of the tumor suppressor PTEN. *Cancer Cell.* 2005;7(3):263-273.
- Kim et al., Genome-wide target specificity of CRISPR RNA-guided adenine base editors. *Nat Biotechnol.* Apr. 2019;37(4):430-435. doi: 10.1038/s41587-019-0050-1. Epub Mar. 4, 2019.
- Kim et al., A library of TAL effector nucleases spanning the human genome. *Nat Biotechnol.* Mar. 2013;31(3):251-8. Doi: 10.1038/nbt.2517. Epub Feb. 17, 2013.

(56)

References Cited

OTHER PUBLICATIONS

- Kim et al., An anionic human protein mediates cationic liposome delivery of genome editing proteins into mammalian cells. *Nat Commun.* Jul. 2, 2019;10(1):2905. doi: 10.1038/s41467-019-10828-3.
- Kim et al., Evaluating and Enhancing Target Specificity of Gene-Editing Nucleases and Deaminases. *Annu Rev Biochem.* Jun. 20, 2019;88:191-220. doi: 10.1146/annurev-biochem-013118-111730. Epub Mar. 18, 2019.
- Kim et al., Genome-wide target specificities of CRISPR RNA-guided programmable deaminases. *Nat Biotechnol.* May 2017;35(5):475-480. doi: 10.1038/nbt.3852. Epub Apr. 10, 2017.
- Kim et al., High cleavage efficiency of a 2A peptide derived from porcine teschovirus-1 in human cell lines, zebrafish and mice. *PLoS One.* 2011;6(4):e18556. doi: 10.1371/journal.pone.0018556. Epub Apr. 29, 2011.
- Kim et al., High-throughput analysis of the activities of xCas9, SpCas9-NG and SpCas9 at matched and mismatched target sequences in human cells. *Nat Biomed Eng.* Jan. 2020;4(1):111-124. doi: 10.1038/s41551-019-0505-1. Epub Jan. 14, 2020.
- Kim et al., Highly efficient RNA-guided base editing in mouse embryos. *Nat Biotechnol.* May 2017;35(5):435-437. doi: 10.1038/nbt.3816. Epub Feb. 27, 2017.
- Kim et al., Highly efficient RNA-guided genome editing in human cells via delivery of purified Cas9 ribonucleoproteins. *Genome Res.* Jun. 2014;24(6):1012-9. doi: 10.1101/gr.171322.113. Epub Apr. 2, 2014.
- Kim et al., In vivo genome editing with a small Cas9 orthologue derived from *Campylobacter jejuni*. *Nat Commun.* Feb. 21, 2017;8:14500. doi: 10.1038/ncomms14500. PMID: 28220790; PMCID: PMC5473640.
- Kim et al., Increasing the genome-targeting scope and precision of base editing with engineered Cas9-cytidine deaminase fusions. *Nat Biotechnol.* Apr. 2017;35(4):371-376. doi: 10.1038/nbt.3803. Epub Feb. 13, 2017.
- Kim et al., Mycobacteriophage Bxb1 integrates into the *Mycobacterium smegmatis* groEL1 gene. *Mol Microbiol.* Oct. 2003;50(2):463-73. doi: 10.1046/j.1365-2958.2003.03723.x.
- Kim et al., Rescue of high-specificity Cas9 variants using sgRNAs with matched 5' nucleotides. *Genome Biol.* Nov. 15, 2017;18(1):218. doi: 10.1186/s13059-017-1355-3.
- Kim et al., Structural and kinetic characterization of *Escherichia coli* TadA, the wobble-specific tRNA deaminase. *Biochemistry.* May 23, 2006;45(20):6407-16. doi: 10.1021/bi0522394. PMID: 16700551.
- Kim et al., TALENs and ZFNs are associated with different mutations/signatures. *Nat Methods.* Mar. 2013;10(3):185. doi: 10.1038/nmeth.2364. Epub Feb. 10, 2013.
- Kim et al., Targeted genome editing in human cells with zinc finger nucleases constructed via modular assembly. *Genome Res.* Jul. 2009;19(7):1279-88. doi: 10.1101/gr.089417.108. Epub May 21, 2009.
- Kim et al., The role of apolipoprotein E in Alzheimer's disease. *Neuron.* Aug. 13, 2009;63(3):287-303. doi: 10.1016/j.neuron.2009.06.026.
- Kim et al., Transcriptional repression by zinc finger peptides. Exploring the potential for applications in gene therapy. *J Biol Chem.* Nov. 21, 1997;272(47):29795-800.
- Kitamura et al., Uracil DNA glycosylase counteracts APOBEC3G-induced hypermutation of hepatitis B viral genomes: excision repair of covalently closed circular Dna. *PLoS Pathog.* 2013;9(5):e1003361. doi: 10.1371/journal.ppat.1003361. Epub May 16, 2013.
- Klapacz et al., Frameshift mutagenesis and microsatellite instability induced by human alkyladenine DNA glycosylase. *Mol Cell.* Mar. 26, 2010;37(6):843-53. doi: 10.1016/j.molcel.2010.01.038.
- Klauser et al., An engineered small RNA-mediated genetic switch based on a ribozyme expression platform. *Nucleic Acids Res.* May 1, 2013;41(10):5542-52. doi: 10.1093/nar/gkt253. Epub Apr. 12, 2013.
- Kleinstiver et al., Engineered CRISPR-Cas9 nucleases with altered PAM specificities. *Nature.* Jul. 23, 2015;523(7561):481-5 and Supplementary Materials. doi: 10.1038/nature14592. Epub Jun. 22, 2015. 27 pages.
- Kleinstiver et al., Monomeric site-specific nucleases for genome editing. *Proc Natl Acad Sci U S A.* May 22, 2012;109(21):8061-6. doi: 10.1073/pnas.1117984109. Epub May 7, 2012.
- Klippel et al., Isolation and characterization of unusual gin mutants. *EMBO J.* Dec. 1, 1988;7(12):3983-9.
- Klippel et al., The DNA invertase Gin of phage Mu: formation of a covalent complex with DNA via a phosphoserine at amino acid position 9. *EMBO J.* Apr. 1988;7(4):1229-37.
- Klompe et al., Transposon-encoded CRISPR-Cas systems direct RNA-guided DNA integration. *Nature.* Jul. 2019;571(7764):219-225. doi: 10.1038/s41586-019-1323-z. Epub Jun. 12, 2019.
- Knott et al., Guide-bound structures of an RNA-targeting A-cleaving CRISPR-Cas13a enzyme. *Nat Struct Mol Biol.* Oct. 2017;24(10):825-833. doi: 10.1038/nsmb.3466. Epub Sep. 11, 2017.
- Koblan et al., Improving cytidine and adenine base editors by expression optimization and ancestral reconstruction. *Nat Biotechnol.* Oct. 2018;36(9):843-846. doi: 10.1038/nbt.4172. Epub May 29, 2018.
- Kobori et al., Deep Sequencing Analysis of Aptazyme Variants Based on a Pistol Ribozyme. *ACS Synth Biol.* Jul. 21, 2017;6(7):1283-1288. doi: 10.1021/acssynbio.7b00057. Epub Apr. 14, 2017.
- Kohli et al., A portable hot spot recognition loop transfers sequence preferences from APOBEC family members to activation-induced cytidine deaminase. *J Biol Chem.* Aug. 21, 2009;284(34):22898-904. doi: 10.1074/jbc.M109.025536. Epub Jun. 26, 2009.
- Kohli et al., Local sequence targeting in the AID/APOBEC family differentially impacts retroviral restriction and antibody diversification. *J Biol Chem.* Dec. 24, 2010;285(52):40956-64. doi: 10.1074/jbc.M110.177402. Epub Oct. 6, 2010.
- Kolot et al., Site promiscuity of coliphage HK022 integrase as a tool for gene therapy. *Gene Ther.* Jul. 2015;22(7):521-7. doi: 10.1038/gt.2015.9. Epub Mar. 12, 2015.
- Kolot et al., Site-specific recombination in mammalian cells expressing the Int recombinase of bacteriophage HK022. *Mol Biol Rep.* Aug. 1999;26(3):207-13. doi: 10.1023/a:1007096701720.
- Komor et al., CRISPR-Based Technologies for the Manipulation of Eukaryotic Genomes. *Cell.* Jan. 12, 2017;168(1-2):20-36. doi: 10.1016/j.cell.2016.10.044.
- Komor et al., Improved base excision repair inhibition and bacteriophage Mu Gam protein yields C:G-to-T:A base editors with higher efficiency and product purity. *Sci Adv.* Aug. 30, 2017;3(8):eaao4774. doi: 10.1126/sciadv.aao4774. eCollection Aug. 2017.
- Komor, Editing the Genome Without Double-Stranded DNA Breaks. *ACS Chem Biol.* Feb. 16, 2018;13(2):383-388. doi: 10.1021/acscchembio.7b00710. Epub Oct. 9, 2017.
- Konermann et al., Genome-scale transcriptional activation by an engineered CRISPR-Cas9 complex. *Nature.* Jan. 29, 2015;517(7536):583-8. doi: 10.1038/nature14136. Epub Dec. 10, 2014.
- Koonin et al., Diversity, classification and evolution of CRISPR-Cas systems. *Curr Opin Microbiol.* 2017;37:677-78. doi:10.1016/j.mib.2017.05.008.
- Kosicki et al., Repair of double-strand breaks induced by CRISPR-Cas9 leads to large deletions and complex rearrangements. *Nat Biotechnol.* Sep. 2018;36(8):765-771. doi: 10.1038/nbt.4192. Epub Jul. 16, 2018.
- Kotewicz et al., Cloning and overexpression of Moloney murine leukemia virus reverse transcriptase in *Escherichia coli*. *Gene.* 1985;35(3):249-58. doi: 10.1016/0378-1119(85)90003-4.
- Kotewicz et al., Isolation of cloned Moloney murine leukemia virus reverse transcriptase lacking ribonuclease H activity. *Nucleic Acids Res.* Jan. 11, 1988;16(1):265-77. doi: 10.1093/nar/16.1.265.
- Kotin, Prospects for the use of adeno-associated virus as a vector for human gene therapy. *Hum Gene Ther.* Jul. 1994;5(7):793-801. doi: 10.1089/hum.1994.5.7-793.
- Kouzminova et al., Patterns of chromosomal fragmentation due to uracil-DNA incorporation reveal a novel mechanism of replication-

(56)

References Cited

OTHER PUBLICATIONS

- dependent double-stranded breaks. *Mol Microbiol.* Apr. 2008;68(1):202-15. doi: 10.1111/j.1365-2958.2008.06149.x.
- Kowal et al., Exploiting unassigned codons in *Micrococcus luteus* for tRNA-based amino acid mutagenesis. *Nucleic Acids Res.* Nov. 15, 1997;25(22):4685-9.
- Kowalski et al., Delivering the Messenger: Advances in Technologies for Therapeutic mRNA Delivery. *Mol Ther.* Apr. 10, 2019;27(4):710-728. doi: 10.1016/j.ymthe.2019.02.012. Epub Feb. 19, 2019.
- Kozak, An analysis of 5'-noncoding sequences from 699 vertebrate messenger RNAs. *Nucleic Acids Res.* Oct. 26, 1987;15(20):8125-48. doi: 10.1093/nar/15.20.8125.
- Kremer et al., Adenovirus and adeno-associated virus mediated gene transfer. *Br Med Bull.* Jan. 1995;51(1):31-44. doi: 10.1093/oxfordjournals.bmb.a072951.
- Krokan et al., Uracil in DNA—occurrence, consequences and repair. *Oncogene.* Dec. 16, 2002;21(58):8935-48. doi: 10.1038/sj.onc.1205996.
- Krokan et al., Base excision repair. *Cold Spring Harb Perspect Biol.* Apr. 1, 2013;5(4):a012583. doi: 10.1101/cshperspect.a012583.
- Krzywkowski et al., Limited reverse transcriptase activity of phi29 DNA polymerase. *Nucleic Acids Res.* Apr. 20, 2018;46(7):3625-3632. doi: 10.1093/nar/gky190.
- Kumar et al., Gene therapy for chronic neuropathic pain: how does it work and where do we stand today? *Pain Med.* May 2011;12(5):808-22. doi: 10.1111/j.1526-4637.2011.01120.x.
- Kumar et al., Structural and functional consequences of the mutation of a conserved arginine residue in alphaA and alphaB crystal-lins. *J Biol Chem.* Aug. 20, 1999;274(34):24137-41.
- Kundu et al., Leucine to proline substitution by SNP at position 197 in Caspase-9 gene expression leads to neuroblastoma: a bioinformatics analysis. *3 Biotech.* 2013; 3:225-34.
- Kunkel et al., Eukaryotic Mismatch Repair in Relation to DNA Replication. *Annu Rev Genet.* 2015;49:291-313. doi: 10.1146/annurev-genet-112414-054722.
- Kunz et al., DNA Repair in mammalian cells: Mismatched repair: variations on a theme. *Cell Mol Life Sci.* Mar. 2009;66(6):1021-38. doi: 10.1007/s00018-009-8739-9.
- Kurjan et al., Structure of a yeast pheromone gene (MF alpha): a putative alpha-factor precursor contains four tandem copies of mature alpha-factor. *Cell.* Oct. 1982;30(3):933-43. doi: 10.1016/0092-8674(82)90298-7.
- Kury et al., De Novo Disruption of the Proteasome Regulatory Subunit PSMD12 Causes a Syndromic Neurodevelopmental Disorder. *Am J Hum Genet.* Feb. 2, 2017;100(2):352-363. doi: 10.1016/j.ajhg.2017.01.003. Epub Jan. 26, 2017.
- Kuscu et al., CRISPR-Cas9-AID base editor is a powerful gain-of-function screening tool. *Nat Methods.* Nov. 29, 2016;13(12):983-984. doi: 10.1038/nmeth.4076.
- Kuscu et al., CRISPR-STOP: gene silencing through base-editing-induced nonsense mutations. *Nat Methods.* Jul. 2017;14(7):710-712. doi: 10.1038/nmeth.4327. Epub Jun. 5, 2017.
- Kuscu et al., Genome-wide analysis reveals characteristics of off-target sites bound by the Cas9 endonuclease. *Nat Biotechnol.* Jul. 2014;32(7):677-83. doi: 10.1038/nbt.2916. Epub May 18, 2014.
- Kwart et al., Precise and efficient scarless genome editing in stem cells using CORRECT. *Nat Protoc.* Feb. 2017;12(2):329-354. doi: 10.1038/nprot.2016.171. Epub Jan. 19, 2017.
- Kweon et al., Fusion guide RNAs for orthogonal gene manipulation with Cas9 and Cpf1. *Nat Commun.* Nov. 23, 2017;8(1):1723. doi: 10.1038/s41467-017-01650-w. Erratum in: *Nat Commun.* Jan. 16, 2018;9(1):303.
- Kwon et al., Chemical basis of glycine riboswitch cooperativity. *RNA.* Jan. 2008;14(1):25-34. Epub Nov. 27, 2007.
- Köhler et al., A possible approach to site-specific insertion of two different unnatural amino acids into proteins in mammalian cells via nonsense suppression. *Chem Biol.* Nov. 2003; 10(11):1095-102.
- Köhler et al., Complete set of orthogonal 21st aminoacyl-tRNA synthetase-amber, ochre and opal suppressor tRNA pairs: concomi-
- tant suppression of three different termination codons in an mRNA in mammalian cells. *Nucleic Acids Res.* Dec. 1, 2004;32(21):6200-11. Print 2004.
- Kügler et al., Human synapsin 1 gene promoter confers highly neuron-specific long-term transgene expression from an adenoviral vector in the adult rat brain depending on the transduced area. *Gene Ther.* Feb. 2003; 10(4):337-47. doi: 10.1038/sj.gt.3301905.
- Lada et al., Mutator effects and mutation signatures of editing deaminases produced in bacteria and yeast. *Biochemistry (Mosc).* Jan. 2011;76(1):131-46.
- Lakich et al., Inversions disrupting the factor VIII gene are a common cause of severe haemophilia A. *Nat Genet.* Nov. 1993;5(3):236-41. doi: 10.1038/ng1193-236.
- Lancaster et al., Limited trafficking of a neurotropic virus through inefficient retrograde axonal transport and the type I interferon response. *PLoS Pathog.* Mar. 5, 2010;6(3):e1000791. doi: 10.1371/journal.ppat.1000791.
- Landrum et al., ClinVar: public archive of relationships among sequence variation and human phenotype. *Nucleic Acids Res.* Jan. 2014;42(Database issue):D980-5. doi: 10.1093/nar/gkt1113. Epub Nov. 14, 2013.
- Langer et al., Chemical and Physical Structure of Polymers as Carriers for Controlled Release of Bioactive Agents: A Review. *Journal of Macromolecular Science*, 2006;23(1):61-126. DOI: 10.1080/07366578308079439.
- Langer et al., New methods of drug delivery. *Science.* Sep. 28, 1990;249(4976):1527-33.
- Lapinaite et al., DNA capture by a CRISPR-Cas9-guided adenine base editor. *Science.* Jul. 31, 2020;369(6503):566-571. doi: 10.1126/science.abb1390.
- Larson et al., CRISPR interference (CRISPRi) for sequence-specific control of gene expression. *Nat Protoc.* Nov. 2013;8(11):2180-96. doi: 10.1038/nprot.2013.132. Epub Oct. 17, 2013.
- Lau et al., Molecular basis for discriminating between normal and damaged bases by the human alkyladenine glycosylase, AAG. *Proc Natl Acad Sci U S A.* Dec. 5, 2000;97(25):13573-8.
- Lavergne et al., Defects in type IIA von Willebrand disease: a cysteine 509 to arginine substitution in the mature von Willebrand factor disrupts a disulphide loop involved in the interaction with platelet glycoprotein Ib-IX. *Br J Haematol.* Sep. 1992;82(1):66-72.
- Lawrence et al., Supercharging proteins can impart unusual resilience. *J Am Chem Soc.* Aug. 22, 2007;129(33):10110-2. Epub Aug. 1, 2007.
- Lawyer et al., High-level expression, purification, and enzymatic characterization of full-length *Thermus aquaticus* DNA polymerase and a truncated form deficient in 5' to 3' exonuclease activity. *PCR Methods Appl.* May 1993;2(4):275-87. doi: 10.1101/gr.2.4.275.
- Lazar et al., Transforming growth factor alpha: mutation of aspartic acid 47 and leucine 48 results in different biological activities. *Mol Cell Biol.* Mar. 1988;8(3):1247-52.
- Lazarevic et al., Nucleotide sequence of the *Bacillus subtilis* temperate bacteriophage SPbetac2. *Microbiology (Reading).* May 1999;145 (Pt 5):1055-1067. doi: 10.1099/13500872-145-5-1055.
- Le Grice et al., Purification and characterization of recombinant equine infectious anemia virus reverse transcriptase. *J Virol.* Dec. 1991;65(12):7004-7. doi: 10.1128/JVI.65.12.7004-7007.1991.
- Leaver-Fay et al., ROSETTA3: an object-oriented software suite for the simulation and design of macromolecules. *Methods Enzymol.* 2011;487:545-74. doi: 10.1016/B978-0-12-381270-4.00019-6.
- Lecote et al., A population-based experimental model for protein evolution: effects of mutation rate and selection stringency on evolutionary outcomes. *Biochemistry.* Feb. 26, 2013;52(8):1490-9. doi: 10.1021/bi3016185. Epub Feb. 14, 2013.
- Ledford, Gene-editing hack yields pinpoint precision. *Nature.* Apr. 20, 2016. <http://www.nature.com/news/gene-editing-hack-yields-pinpoint-precision-1.19773>.
- Lee et al., A chimeric thyroid hormone receptor constitutively bound to DNA requires retinoid X receptor for hormone-dependent transcriptional activation in yeast. *Mol Endocrinol.* Sep. 1994;8(9):1245-52.
- Lee et al., A monoclonal antibody that targets a NaV1.7 channel voltage sensor for pain and itch relief. *Cell.* Jun. 5, 2014;157(6):1393-

(56)

References Cited

OTHER PUBLICATIONS

1404. doi: 10.1016/j.cell.2014.03.064. Epub May 22, 2014. Retraction in: Cell. Jun. 25, 2020;181(7):1695.
- Lee et al., An allosteric self-splicing ribozyme triggered by a bacterial second messenger. Science. Aug. 13, 2010;329(5993):845-8. doi: 10.1126/science.1190713.
- Lee et al., Failure to detect DNA-guided genome editing using *Neisseria meningitidis* Argonaute. Nat Biotechnol. Nov. 28, 2016;35(11):17-18. doi: 10.1038/nbt.3753.
- Lee et al., Group I Intron-Based Therapeutics Through Trans-Splicing Reaction. Prog Mol Biol Transl Sci. 2018;159:79-100. doi: 10.1016/bs.pmbts.2018.07.001. Epub Aug. 9, 2018.
- Lee et al., PIK3CA gene is frequently mutated in breast carcinomas and hepatocellular carcinomas. Oncogene. Feb. 17, 2005;24(8):1477-80.
- Lee et al., Recognition of liposomes by cells: in vitro binding and endocytosis mediated by specific lipid headgroups and surface charge density. Biochim Biophys Acta. Jan. 31, 1992;1103(2):185-97.
- Lee et al., Ribozyme Mediated gRNA Generation for In Vitro and In Vivo CRISPR/Cas9 Mutagenesis. PLoS One. Nov. 10, 2016;11(11):e0166020. doi: 10.1371/journal.pone.0166020. eCollection 2016.
- Lee et al., Simultaneous targeting of linked loci in mouse embryos using base editing. Sci Rep. Feb. 7, 2019;9(1):1662. doi: 10.1038/s41598-018-33533-5.
- Lee et al., Site-specific integration of mycobacteriophage L5: integration-proficient vectors for *Mycobacterium smegmatis*, *Mycobacterium tuberculosis*, and bacille Calmette-Guérin. Proc Natl Acad Sci U S A. Apr. 15, 1991;88(8):3111-5. doi: 10.1073/pnas.88.8.3111.
- Lee et al., Synthetically modified guide RNA and donor DNA are a versatile platform for CRISPR-Cas9 engineering. Elife. May 2, 2017;6:e25312. doi: 10.7554/eLife.25312.
- Lee et al., Targeted chromosomal deletions in human cells using zinc finger nucleases. Genome Res. Jan. 2010 20: 81-89; Published in Advance Dec. 1, 2009, doi:10.1101/gr.099747.109.
- Lee et al., Targeting fidelity of adenine and cytosine base editors in mouse embryos. Nat Commun. Nov. 15, 2018;9(1):4804. doi: 10.1038/s41467-018-07322-7.
- Lee et al., Transcriptional regulation and its misregulation in disease. Cell. Mar. 14, 2013;152(6):1237-51. doi: 10.1016/j.cell.2013.02.014.
- Lei et al., Efficient targeted gene disruption in *Xenopus* embryos using engineered transcription activator-like effector nucleases (TALENs). Proc Natl Acad Sci U S A. Oct. 23, 2012;109(43):17484-9. Doi: 10.1073/pnas.1215421109. Epub Oct. 8, 2012.
- Lei et al., Site-specificity of serine integrase demonstrated by the attB sequence preference of ϕ BT1 integrase. FEBS Lett. Apr. 2018;592(8):1389-1399. doi: 10.1002/1873-3468.13023. Epub Mar. 25, 2018.
- Leipold et al., A de novo gain-of-function mutation in SCN11A causes loss of pain perception. Nat Genet. Nov. 2013;45(11):1399-404. doi: 10.1038/ng.2767. Epub Sep. 15, 2013.
- Lemos et al., CRISPR/Cas9 cleavages in budding yeast reveal templated insertions and strand-specific insertion/deletion profiles. Proc Natl Acad Sci U S A. Feb. 27, 2018;115(9):E2040-E2047. doi: 10.1073/pnas.1716855115. Epub Feb. 13, 2018.
- Lenk et al., Pathogenic mechanism of the FIG4 mutation responsible for Charcot-Marie-Tooth disease CMT4J. PLoS Genet. Jun. 2011;7(6):e1002104. doi: 10.1371/journal.pgen.1002104. Epub Jun. 2, 2011.
- Levy et al., Cytosine and adenine base editing of the brain, liver, retina, heart and skeletal muscle of mice via adeno-associated viruses. Nat Biomed Eng. 2020;4(1):97-110. doi:10.1038/s41551-019-0501-5.
- Levy et al., Inhibition of calcification of bioprosthetic heart valves by local controlled-release diphosphonate. Science. Apr. 12, 1985;228(4696):190-2.
- Levy et al., Membrane-associated guanylate kinase dynamics reveal regional and developmental specificity of synapse stability. J Physiol. Mar. 1, 2017;595(5):1699-1709. doi: 10.1113/JP273147. Epub Jan. 18, 2017.
- Lew et al., Protein splicing in vitro with a semisynthetic two-component minimal intein. J Biol Chem. Jun. 26, 1998;273(26):15887-90. doi: 10.1074/jbc.273.26.15887.
- Lewis et al., A serum-resistant cytofectin for cellular delivery of antisense oligodeoxynucleotides and plasmid DNA. Proc Natl Acad Sci U S A. Apr. 16, 1996;93(8):3176-81.
- Lewis et al., Building the Class 2 CRISPR-Cas Arsenal. Mol Cell. 2017;65(3):377-379.
- Lewis et al., Codon 129 polymorphism of the human prion protein influences the kinetics of amyloid formation. J Gen Virol. Aug. 2006;87(Pt 8):2443-9.
- Lewis et al., Cytosine deamination and the precipitous decline of spontaneous mutation during Earth's history. Proc Natl Acad Sci U S A. Jul. 19, 2016;113(29):8194-9. doi: 10.1073/pnas.1607580113. Epub Jul. 5, 2016.
- Lewis et al., RNA modifications and structures cooperate to guide RNA-protein interactions. Nat Rev Mol Cell Biol. Mar. 2017;18(3):202-210. doi: 10.1038/nrm.2016.163. Epub Feb. 1, 2017.
- Li et al., A Radioactivity-Based Assay for Screening Human m6A-RNA Methyltransferase, METTL3-METTL14 Complex, and Demethylase ALKBH5. J Biomol Screen. Mar. 2016;21(3):290-7. doi: 10.1177/1087057115623264. Epub Dec. 23, 2015.
- Li et al., Base editing with a Cpf1-cytidine deaminase fusion. Nat Biotechnol. Apr. 2018;36(4):324-327. doi: 10.1038/nbt.4102. Epub Mar. 19, 2018.
- Li et al., Current approaches for engineering proteins with diverse biological properties. Adv Exp Med Biol. 2007;620:18-33.
- Li et al., Disruption of splicing-regulatory elements using CRISPR/Cas9 to rescue spinal muscular atrophy in human iPSCs and mice. National Science Review. Jan. 1, 2020:92-101. DOI: 10.1093/nsr/nwz131. Retrieved from the Internet via <https://academic.oup.com/nsr/article-pdf/7/1/92/33321439/nwz131.pdf>. Last accessed Apr. 28, 2021.
- Li et al., Fast and accurate short read alignment with Burrows-Wheeler transform. Bioinformatics. Jul. 15, 2009;25(14):1754-60. doi: 10.1093/bioinformatics/btp324. Epub May 18, 2009.
- Li et al., Generation of Targeted Point Mutations in Rice by a Modified CRISPR/Cas9 System. Mol Plant. Mar. 6, 2017;10(3):526-529. doi: 10.1016/j.molp.2016.12.001. Epub Dec. 8, 2016.
- Li et al., Highly efficient and precise base editing in discarded human triploid embryos. Protein Cell. Aug. 19, 2017. doi: 10.1007/s13238-017-0458-7. [Epub ahead of print].
- Li et al., Lagging strand DNA synthesis at the eukaryotic replication fork involves binding and stimulation of FEN-1 by proliferating cell nuclear antigen. J Biol Chem. Sep. 22, 1995;270(38):22109-12. doi: 10.1074/jbc.270.38.22109.
- Li et al., Loss of post-translational modification sites in disease. Pac Symp Biocomput. 2010:337-47. doi: 10.1142/9789814295291_0036.
- Li et al., Modularly assembled designer TAL effector nucleases for targeted gene knockout and gene replacement in eukaryotes. Nucleic Acids Res. Aug. 2011;39(14):6315-25. doi: 10.1093/nar/gkr188. Epub Mar. 31, 2011.
- Li et al., Multiplex and homologous recombination-mediated genome editing in *Arabidopsis* and *Nicotiana benthamiana* using guide RNA and Cas9. Nat Biotechnol. Aug. 2013;31(8):688-91. doi: 10.1038/nbt.2654.
- Li et al., Precise Modifications of Both Exogenous and Endogenous Genes in Rice by Prime Editing. Mol Plant. May 4, 2020;13(5):671-674. doi: 10.1016/j.molp.2020.03.011. Epub Mar. 25, 2020.
- Li et al., Protein trans-splicing as a means for viral vector-mediated in vivo gene therapy. Hum Gene Ther. Sep. 2008;19(9):958-64. doi: 10.1089/hum.2008.009.
- Li et al., RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome. BMC Bioinformatics. Aug. 4, 2011;12:323. doi: 10.1186/1471-2105-12-323.
- Li et al., TAL nucleases (TALENs): hybrid proteins composed of TAL effectors and FokI DNA-cleavage domain. Nucleic Acids Res. Jan. 2011;39(1):359-72. doi: 10.1093/nar/gkq704. Epub Aug. 10, 2010.

(56)

References Cited

OTHER PUBLICATIONS

- Li, Mechanisms and functions of DNA mismatch repair. *Cell Res.* Jan. 2008;18(1):85-98. doi: 10.1038/cr.2007.115.
- Liang et al., Correction of β -thalassemia mutant by base editor in human embryos. *Protein Cell.* Nov. 2017;8(11):811-822. doi: 10.1007/s13238-017-0475-6. Epub Sep. 23, 2017.
- Liang et al., Homology-directed repair is a major double-strand break repair pathway in mammalian cells. *Proc Natl Acad Sci U S A.* Apr. 28, 1998;95(9):5172-7. doi: 10.1073/pnas.95.9.5172.
- Liang et al., Rapid and highly efficient mammalian cell engineering via Cas9 protein transfection. *Send to; J Biotechnol.* Aug. 20, 2015;208:44-53. doi: 10.1016/j.jbiotec.2015.04.024.
- Lieber et al., Mechanism and regulation of human non-homologous DNA end-joining. *Nat Rev Mol Cell Biol.* Sep. 2003;4(9):712-20.
- Liefke et al., The oxidative demethylase ALKBH3 marks hyperactive gene promoters in human cancer cells. *Genome Med.* Jun. 30, 2015;7(1):66. doi: 10.1186/s13073-015-0180-0.
- Lienert et al., Two- and three-input TALE-based and logic computation in embryonic stem cells. *Nucleic Acids Res.* Nov. 2013;41(21):9967-75. doi: 10.1093/nar/gkt758. Epub Aug. 27, 2013.
- Lilley, D.M. The Varkud Satellite Ribozyme. *RNA.* Feb. 2004;10(2):151-8. doi: 10.1261/rna.5217104.
- Lim et al., Crystal structure of the moloney murine leukemia virus RNase H domain. *J Virol.* Sep. 2006;80(17):8379-89. doi: 10.1128/JVI.00750-06.
- Lim et al., Viral vectors for neurotrophic factor delivery: a gene therapy approach for neurodegenerative diseases of the CNS. *Pharmacol Res.* Jan. 2010;61(1):14-26. doi: 10.1016/j.phrs.2009.10.002. Epub Oct. 17, 2009.
- Lin et al., Enhanced homology-directed human genome engineering by controlled timing of CRISPR/Cas9 delivery. *Elife.* Dec. 15, 2014;3:e04766. doi: 10.7554/eLife.04766.
- Lin et al., Prime genome editing in rice and wheat. *Nat Biotechnol.* May 2020;38(5):582-585 and Supplemental Info. doi: 10.1038/s41587-020-0455-x. Epub Mar. 16, 2020. 8 pages.
- Lin et al., Prime genome editing in rice and wheat. *Nat Biotechnol.* May 2020;38(5):582-585. doi: 10.1038/s41587-020-0455-x. Epub Mar. 16, 2020.
- Lin et al., The human REV1 gene codes for a DNA template-dependent dCMP transferase. *Nucleic Acids Res.* Nov. 15, 1999;27(22):4468-75. doi: 10.1093/nar/27.22.4468.
- Link et al., Engineering ligand-responsive gene-control elements: lessons learned from natural riboswitches. *Gene Ther.* Oct. 2009;16(10):1189-201. doi: 10.1038/gt.2009.81. Epub Jul. 9, 2009. Review.
- Liu et al., C2c1-sgRNA Complex Structure Reveals RNA-Guided DNA Cleavage Mechanism. *Molecular Cell* Jan. 2017;65(2):310-22.
- Liu et al., Split dnaE genes encoding multiple novel inteins in *Trichodesmium erythraeum*. *J Biol Chem.* Jul. 18, 2003;278(29):26315-8. doi: 10.1074/jbc.C300202200. Epub May 24, 2003.
- Liu et al., A METTL3-METTL14 complex mediates mammalian nuclear RNA N6-adenosine methylation. *Nat Chem Biol.* Feb. 2014;10(2):93-5. doi: 10.1038/nchembio.1432. Epub Dec. 6, 2013.
- Liu et al., Adding new chemistries to the genetic code. *Annu Rev Biochem.* 2010;79:413-44. doi: 10.1146/annurev.biochem.052308.105824.
- Liu et al., Apolipoprotein E and Alzheimer disease: risk, mechanisms and therapy. *Nat Rev Neurol.* Feb. 2013;9(2):106-18. doi: 10.1038/nrneurol.2012.263. Epub Jan. 8, 2013.
- Liu et al., Balancing AID and DNA repair during somatic hypermutation. *Trends Immunol.* Apr. 2009;30(4):173-81. doi: 10.1016/j.it.2009.01.007.
- Liu et al., Calcineurin is a common target of cyclophilin-cyclosporin A and FKBP-FK506 complexes. *Cell.* Aug. 23, 1991;66(4):807-15. doi: 10.1016/0092-8674(91)90124-h.
- Liu et al., CasX enzymes comprise a distinct family of RNA-guided genome editors. *Nature.* Feb. 2019;566(7743):218-223. doi: 10.1038/s41586-019-0908-x. Epub Feb. 4, 2019. Author manuscript entitled CRISPR-CasX is an RNA-dominated enzyme active for human genome editing.
- Liu et al., Cell-penetrating peptide-mediated delivery of TALEN proteins via bioconjugation for genome engineering. *PLoS One.* Jan. 20, 2014;9(1):e85755. doi: 10.1371/journal.pone.0085755. eCollection 2014.
- Liu et al., Computational approaches for effective CRISPR guide RNA design and evaluation. *Comput Struct Biotechnol J.* Nov. 29, 2019;18:35-44. doi: 10.1016/j.csbj.2019.11.006.
- Liu et al., Direct Promoter Repression by BCL11A Controls the Fetal to Adult Hemoglobin Switch. *Cell.* Apr. 5, 2018;173(2):430-442.e17. doi: 10.1016/j.cell.2018.03.016. Epub Mar. 29, 2018.
- Liu et al., Distance determination by GIY-YIG intron endonucleases: discrimination between repression and cleavage functions. *Nucleic Acids Res.* Mar. 31, 2006;34(6):1755-64. Print 2006.
- Liu et al., Editing DNA Methylation in the Mammalian Genome. *Cell.* Sep. 22, 2016;167(1):233-247.e17. doi: 10.1016/j.cell.2016.08.056.
- Liu et al., Engineering a tRNA and aminoacyl-tRNA synthetase for the site-specific incorporation of unnatural amino acids into proteins in vivo. *Proc Natl Acad Sci U S A.* Sep. 16, 1997;94(19):10092-7.
- Liu et al., Fast Colorimetric Sensing of Adenosine and Cocaine Based on a General Sensor Design Involving Aptamers and Nanoparticles. *Angew Chem.* Dec. 16, 2006;45(1):90-4. DOI: 10.1002/anie.200502589.
- Liu et al., Fast Colorimetric Sensing of Adenosine and Cocaine Based on a General Sensor Design Involving Aptamers and Nanoparticles. *Angew Chem.* 2006;118(1):96-100.
- Liu et al., Flap endonuclease 1: a central component of DNA metabolism. *Annu Rev Biochem.* 2004;73:589-615. doi:10.1146/annurev.biochem.73.012803.092453.
- Liu et al., Functional Nucleic Acid Sensors. *Chem Rev.* May 2009;109(5):1948-98. doi: 10.1021/cr030183i.
- Liu et al., Genetic incorporation of unnatural amino acids into proteins in mammalian cells. *Nat Methods.* Mar. 2007;4(3):239-44. Epub Feb. 25, 2007.
- Liu et al., Highly efficient RNA-guided base editing in rabbit. *Nat Commun.* Jul. 13, 2018;9(1):2717. doi: 10.1038/s41467-018-05232-2.
- Liu et al., (6)-methyladenosine-dependent RNA structural switches regulate RNA-protein interactions. *Nature.* Feb. 26, 2015;518(7540):560-4. doi: 10.1038/nature14234.
- Liu et al., Probing N6-methyladenosine RNA modification status at single nucleotide resolution in mRNA and long noncoding RNA. *RNA.* Dec. 2013;19(12):1848-56. doi: 10.1261/ma.041178.113. Epub Oct. 18, 2013.
- Liu et al., Reverse transcriptase of foamy virus. Purification of the enzymes and immunological identification. *Arch Virol.* 1977;55(3):187-200. doi: 10.1007/BF01319905.
- Liu et al., Reverse transcriptase-mediated tropism switching in *Bordetella* bacteriophage. *Science.* Mar. 15, 2002;295(5562):2091-4. doi: 10.1126/science.1067467.
- Liu et al., *Saccharomyces cerevisiae* flap endonuclease I uses flap equilibration to maintain triplet repeat stability. *Mol Cell Biol.* May 2004;24(9):4049-64. doi: 10.1128/MCB.24.9.4049-4064.2004.
- Liu et al., The Molecular Architecture for RNA-Guided RNA Cleavage by Cas13a. *Cell.* Aug. 10, 2017;170(4):714-726.e10. doi: 10.1016/j.cell.2017.06.050. Epub Jul. 27, 2017.
- Loessner et al., Complete nucleotide sequence, molecular analysis and genome structure of bacteriophage A118 of *Listeria monocytogenes*: implications for phage evolution. *Mol Microbiol.* Jan. 2000;35(2):324-40. doi: 10.1046/j.1365-2958.2000.01720.x.
- Lombardo et al., Gene editing in human stem cells using zinc finger nucleases and integrase-defective lentiviral vector delivery. *Nat Biotechnol.* Nov. 2007;25(11):1298-306. Epub Oct. 28, 2007.
- Lopez-Girona et al., Cereblon is a direct protein target for immunomodulatory and antiproliferative activities of lenalidomide and pomalidomide. *Leukemia.* Nov. 2012;26(11):2326-35. doi: 10.1038/leu.2012.119. Epub May 3, 2012.
- Losey et al., Crystal structure of *Staphylococcus aureus* tRNA adenosine deaminase tadA in complex with RNA. *Nature Struct. Mol. Biol.* Feb. 2006;13(2):153-9.

(56)

References Cited

OTHER PUBLICATIONS

- Lu et al., Precise Editing of a Target Base in the Rice Genome Using a Modified CRISPR/Cas9 System. *Mol Plant*. Mar. 6, 2017;10(3):523-525. doi: 10.1016/j.molp.2016.11.013. Epub Dec. 6, 2016.
- Luan et al., Reverse transcription of R2Bm RNA is primed by a nick at the chromosomal target site: a mechanism for non-LTR retrotransposition. *Cell*. Feb. 26, 1993;72(4):595-605. doi: 10.1016/0092-8674(93)90078-5.
- Luckow et al., High level expression of nonfused foreign genes with Autographa californica nuclear polyhedrosis virus expression vectors. *Virology*. May 1989;170(1):31-9. doi: 10.1016/0042-6822(89)90348-6.
- Lukacsovich et al., Repair of a specific double-strand break generated within a mammalian chromosome by yeast endonuclease I-SceI. *Nucleic Acids Res*. Dec. 25, 1994;22(25):5649-57. doi: 10.1093/nar/22.25.5649.
- Lundberg et al., Delivery of short interfering RNA using endosomolytic cell-penetrating peptides. *FASEB J*. Sep. 2007;21(11):2664-71. Epub Apr. 26, 2007.
- Lundquist et al., Site-directed mutagenesis and characterization of uracil-DNA glycosylase inhibitor protein. Role of specific carboxylic amino acids in complex formation with *Escherichia coli* uracil-DNA glycosylase. *J Biol Chem*. Aug. 22, 1997;272(34):21408-19.
- Lynch, Evolution of the mutation rate. *Trends Genet*. Aug. 2010;26(8):345-52. doi: 10.1016/j.tig.2010.05.003. Epub Jun. 30, 2010.
- Lyons et al., Efficient Recognition of an Unpaired Lesion by a DNA Repair Glycosylase. *J. Am. Chem. Soc.*, 2009;131(49):17742-3. DOI: 10.1021/ja908378y.
- Lüke et al., Partial purification and characterization of the reverse transcriptase of the simian immunodeficiency virus TYO-7 isolated from an African green monkey. *Biochemistry*. Feb. 20, 1990;29(7):1764-9. doi: 10.1021/bi00459a015.
- Ma et al., Identification of pseudo attP sites for phage phiC31 integrase in bovine genome. *Biochem Biophys Res Commun*. Jul. 7, 2006;345(3):984-8. doi: 10.1016/j.bbrc.2006.04.145. Epub May 3, 2006.
- Ma et al., In vitro protein engineering using synthetic tRNA(Ala) with different anticodons. *Biochemistry*. Aug. 10, 1993;32(31):7939-45.
- Ma et al., PhiC31 integrase induces efficient site-specific recombination in the Capra hircus genome. *DNA Cell Biol*. Aug. 2014;33(8):484-91. doi: 10.1089/dna.2013.2124. Epub Apr. 22, 2014.
- Ma et al., Single-Stranded DNA Cleavage by Divergent CRISPR-Cas9 Enzymes. *Mol Cell*. Nov. 5, 2015;60(3):398-407. doi: 10.1016/j.molcel.2015.10.030.
- Ma et al., Targeted AID-mediated mutagenesis (TAM) enables efficient genomic diversification in mammalian cells. *Nature Methods*. Oct. 2016; 13:1029-35. doi:10.1038/nmeth.4027.
- Maas et al., Identification and characterization of a human tRNA-specific adenosine deaminase related to the ADAR family of pre-mRNA editing enzymes. *Proc Natl Acad Sci U S A*. Aug. 3, 1999;96(16):8895-900. doi: 10.1073/pnas.96.16.8895.
- Macbeth et al., Inositol hexakisphosphate is bound in the ADAR2 core and required for RNA editing. *Science*. Sep. 2, 2005;309(5740):1534-9. doi: 10.1126/science.1113150.
- Macrae et al., Ribonuclease revisited: structural insights into ribonuclease III family enzymes. *Curr Opin Struct Biol*. Feb. 2007;17(1):138-45. doi: 10.1016/j.sbi.2006.12.002. Epub Dec. 27, 2006.
- Madura et al., Structural basis for ineffective T-cell responses to MHC anchor residue-improved "heteroclitic" peptides. *Eur J Immunol*. Feb. 2015;45(2):584-91. doi: 10.1002/eji.201445114. Epub Dec. 28, 2014.
- Maeder et al., CRISPR RNA-guided activation of endogenous human genes. *Nat Methods*. Oct. 2013;10(10):977-9. doi: 10.1038/nmeth.2598. Epub Jul. 25, 2013.
- Maeder et al., Rapid "open-source" engineering of customized zinc-finger nucleases for highly efficient gene modification. *Mol Cell*. Jul. 25, 2008;31(2):294-301. doi:10.1016/j.molcel.2008.06.016.
- Maeder et al., Robust, synergistic regulation of human gene expression using TALE activators. *Nat Methods*. Mar. 2013;10(3):243-5. doi: 10.1038/nmeth.2366. Epub Feb. 10, 2013.
- Magin et al., Corf, the Rev/Rex homologue of HTDV/HERV-K, encodes an arginine-rich nuclear localization signal that exerts a trans-dominant phenotype when mutated. *Virology*. Aug. 15, 2000;274(1):11-6. doi: 10.1006/viro.2000.0438.
- Maizels et al., Initiation of homologous recombination at DNA nicks. *Nucleic Acids Res*. Aug. 21, 2018;46(14):6962-6973. doi: 10.1093/nar/gky588.
- Maji et al., A High-Throughput Platform to Identify Small-Molecule Inhibitors of CRISPR-Cas9. *Cell*. May 2, 2019;177(4):1067-1079. e19. doi: 10.1016/j.cell.2019.04.009.
- Makarova et al., An updated evolutionary classification of CRISPR-Cas systems. *Nat Rev Microbiol*. Nov. 2015;13(11):722-36. doi: 10.1038/nrmicro3569. Epub Sep. 28, 2015.
- Makarova et al., Classification and Nomenclature of CRISPR-Cas Systems: Where from Here? *Crispr J*. Oct. 2018;1(5):325-336. doi: 10.1089/crispr.2018.0033.
- Makarova et al., Evolution and classification of the CRISPR-Cas systems. *Nat Rev Microbiol*. Jun. 2011;9(6):467-77. doi: 10.1038/nrmicro2577. Epub May 9, 2011.
- Makarova et al., Prokaryotic homologs of Argonaute proteins are predicted to function as key components of a novel system of defense against mobile genetic elements. *Biology Direct* 2009;4:29. doi: 10.1186/1745-6150-4-29.
- Makeyev et al., Evolutionary potential of an RNA virus. *J Virol*. Feb. 2004;78(4):2114-20.
- Malashkevich et al., Crystal structure of tRNA adenosine deaminase TadA from *Escherichia coli*. Deposited: 2005-03-10 Released: Mar. 10, 2006 doi:10.2210/pdb1z3a/pdb (2006).
- Mali et al., CAS9 transcriptional activators for target specificity screening and paired nickases for cooperative genome engineering. *Nat Biotechnol*. Sep. 2013;31(9):833-8. doi: 10.1038/nbt.2675. Epub Aug. 1, 2013.
- Mandal et al., Riboswitches Control Fundamental Biochemical Pathways in *Bacillus Subtilis* and Other Bacteria. *Cell*. May 30, 2003;113(5):577-86. doi: 10.1016/s0092-8674(03)00391-x.
- Mani et al., Design, engineering, and characterization of zinc finger nucleases. *Biochem Biophys Res Commun*. Sep. 23, 2005;335(2):447-57.
- Marceau, Functions of single-strand DNA-binding proteins in DNA replication, recombination, and repair. *Methods Mol Biol*. 2012;922:1-21. doi: 10.1007/978-1-62703-032-8_1.
- Maresca et al., Obligate ligation-gated recombination (ObLiGaRe): custom-designed nuclease-mediated targeted integration through nonhomologous end joining. *Genome Res*. Mar. 2013;23(3):539-46. Doi: 10.1101/gr.145441.112. Epub Nov. 14, 2012.
- Marioni et al., DNA methylation age of blood predicts all-cause mortality in later life. *Genome Biol*. Jan. 30, 2015;16:25. doi: 10.1186/s13059-015-0584-6.
- Marquart et al., Predicting base editing outcomes with an attention-based deep learning algorithm trained on high-throughput target library screen. *bioRxiv*. Jul. 5, 2020. DOI:10.1101/2020.07.05.186544. Retrieved from the Internet via <https://www.biorxiv.org/content/10.1101/2020.07.05.186544v1.full.pdf> lased accessed on Apr. 28, 2021.
- Marraffini et al., CRISPR interference limits horizontal gene transfer in staphylococci by targeting DNA. *Science*. Dec. 19, 2008;322(5909):1843-5. doi: 10.1126/science.1165771.
- Martinez et al., Hypermutagenesis of RNA using human immunodeficiency virus type 1 reverse transcriptase and biased dNTP concentrations. *Proc Natl Acad Sci U S A*. Dec. 6, 1994;91(25):11787-91. doi: 10.1073/pnas.91.25.11787.
- Martsof et al., Complete trisomy 17p a relatively new syndrome. *Ann Genet*. 1988;31(3):172-4.
- Martz, L., Nav-i-gating antibodies for pain. *Science-Business eXchange*. Jun. 12, 2014;7(662):1-2. doi: 10.1038/scibx.2014.662.

(56)

References Cited

OTHER PUBLICATIONS

- Maruyama et al., Increasing the efficiency of precise genome editing with CRISPR-Cas9 by inhibition of nonhomologous end joining. *Nat Biotechnol.* May 2015;33(5):538-42. doi: 10.1038/nbt.3190. Epub Mar. 23, 2015.
- Marzec et al., Prime Editing: A New Way for Genome Editing. *Trends Cell Biol.* Apr. 2020;30(4):257-259. doi: 10.1016/j.tcb.2020.01.004. Epub Jan. 27, 2020.
- Mascola et al., HIV-1 neutralizing antibodies: understanding nature's pathways. *Immunol Rev.* Jul. 2013;254(1):225-44. doi: 10.1111/imr.12075.
- Mathys et al., Characterization of a self-splicing mini-intein and its conversion into autocatalytic- and C-terminal cleavage elements: facile production of protein building blocks for protein ligation. *Gene.* Apr. 29, 1999;231(1-2):1-13. doi: 10.1016/S0378-1119(99)00103-1.
- Matsuura et al., A gene essential for the site-specific excision of actinophage r4 prophage genome from the chromosome of a lysogen. *J Gen Appl Microbiol.* 1995;41(1):53-61.
- Matthews, Structures of human ADAR2 bound to dsRNA reveal base-flipping mechanism and basis for site selectivity. *Nat Struct Mol Biol.* May 2016;23(5):426-33. doi: 10.1038/nsmb.3203. Epub Apr. 11, 2016.
- May et al., Emergent lineages of mumps virus suggest the need for a polyvalent vaccine. *Int J Infect Dis.* Jan. 2018;66:1-4. doi: 10.1016/j.ijid.2017.09.024. Epub Oct. 4, 2017.
- Mccarroll et al., Copy-number variation and association studies of human disease. *Nat Genet.* Jul. 2007;39(7 Suppl):S37-42. doi: 10.1038/ng2080.
- McDonald et al., Characterization of mutations at the mouse phenylalanine hydroxylase locus. *Genomics.* Feb. 1, 1997;39(3):402-5. doi: 10.1006/geno.1996.4508.
- McInerney et al., Error Rate Comparison during Polymerase Chain Reaction by DNA Polymerase. *Mol Biol Int.* 2014;2014:287430. doi: 10.1155/2014/287430. Epub Aug. 17, 2014.
- McKenna et al., Recording development with single cell dynamic lineage tracing. *Development.* Jun. 27, 2019;146(12):dev169730. doi: 10.1242/dev.169730.
- McKenna et al., Whole-organism lineage tracing by combinatorial and cumulative genome editing. *Science.* Jul. 29, 2016;353(6298):aaf7907. doi: 10.1126/science.aaf7907. Epub May 26, 2016.
- McNaughton et al., Mammalian cell penetration, siRNA transfection, and DNA transfection by supercharged proteins. *Proc Natl Acad Sci U S A.* Apr. 14, 2009;106(15):6111-6. doi: 10.1073/pnas.0807883106. Epub Mar. 23, 2009.
- Mead et al., A novel protective prion protein variant that colocalizes with kuru exposure. *Engl J Med.* Nov. 19, 2009;361(21):2056-65. doi: 10.1056/NEJMoa0809716.
- Mei et al., Recent Progress in CRISPR/Cas9 Technology. *J Genet Genomics.* Feb. 20, 2016;43(2):63-75. doi: 10.1016/j.jgg.2016.01.001. Epub Jan. 18, 2016.
- Meinke et al., Cre Recombinase and Other Tyrosine Recombinases. *Chem Rev.* Oct. 26, 2016;116(20):12785-12820. doi: 10.1021/acs.chemrev.6b00077. Epub May 10, 2016.
- Meng et al., Targeted gene inactivation in zebrafish using engineered zinc-finger nucleases. *Nat Biotechnol.* Jun. 2008;26(6):695-701. doi: 10.1038/nbt1398. Epub May 25, 2008.
- Menéndez-Arias, Mutation rates and intrinsic fidelity of retroviral reverse transcriptases. *Viruses.* Dec. 2009;1(3):1137-65. doi: 10.3390/v1031137. Epub Dec. 4, 2009.
- Mercer et al., Chimeric TALE recombinases with programmable DNA sequence specificity. *Nucleic Acids Res.* Nov. 2012;40(21):11163-72. doi: 10.1093/nar/gks875. Epub Sep. 26, 2012.
- Mertens et al., Site-specific recombination in bacteriophage Mu: characterization of binding sites for the DNA invertase Gin. *EMBO J.* Apr. 1988;7(4):1219-27.
- Meyer et al., Breathing life into polycations: functionalization with pH-responsive endosomolytic peptides and polyethylene glycol enables siRNA delivery. *J Am Chem Soc.* Mar. 19, 2008;130(11):3272-3. doi: 10.1021/ja710344v. Epub Feb. 21, 2008.
- Meyer et al., Comprehensive analysis of mRNA methylation reveals enrichment in 3' UTRs and near stop codons. *Cell.* Jun. 22, 2012;149(7):1635-46. doi: 10.1016/j.cell.2012.05.003. Epub May 17, 2012.
- Meyer et al., Confirmation of a second natural preQ1 aptamer class in Streptococcaceae bacteria. *RNA.* Apr. 2008;14(4):685-95. doi: 10.1261/rna.937308. Epub Feb. 27, 2008.
- Meyer et al., Library generation by gene shuffling. *Curr Protoc Mol Biol.* Jan. 6, 2014;105:Unit 15.12.. doi: 10.1002/0471142727.mb1512s105.
- Meyer et al., Ribosome biogenesis factor Tsr3 is the aminocarboxypropyl transferase responsible for 18S rRNA hypermodification in yeast and humans. *Nucleic Acids Res.* May 19, 2016;44(9):4304-16. doi: 10.1093/nar/gkw244. Epub Apr. 15, 2016.
- Meyer et al., The dynamic epitranscriptome: N6-methyladenosine and gene expression control. *Nat Rev Mol Cell Biol.* May 2014;15(5):313-26. doi: 10.1038/nrm3785. Epub Apr. 9, 2014.
- Michel et al., Mitochondrial class II introns encode proteins related to the reverse transcriptases of retroviruses. *Nature.* Aug. 15-21, 1985;316(6029):641-3. doi: 10.1038/316641a0.
- Midoux et al., Chemical vectors for gene delivery: a current review on polymers, peptides and lipids containing histidine or imidazole as nucleic acids carriers. *Br J Pharmacol.* May 2009;157(2):166-78. doi: 10.1111/j.1476-5381.2009.00288.x.
- Mihai et al., PTEN inhibition improves wound healing in lung epithelia through changes in cellular mechanics that enhance migration. *Am J Physiol Lung Cell Mol Physiol.* 2012;302(3):L287-L299.
- Mijakovic et al., Bacterial single-stranded DNA-binding proteins are phosphorylated on tyrosine. *Nucleic Acids Res.* Mar. 20, 2006;34(5):1588-96. doi: 10.1093/nar/gkj514.
- Miller et al., A TALE nuclease architecture for efficient genome editing. *Nat Biotechnol.* Feb. 2011;29(2):143-8. doi:10.1038/nbt.1755. Epub Dec. 22, 2010.
- Miller et al., An improved zinc-finger nuclease architecture for highly specific genome editing. *Nat Biotechnol.* Jul. 2007;25(7):778-85. Epub Jul. 1, 2007.
- Miller et al., Construction and properties of retrovirus packaging cells based on gibbon ape leukemia virus. *J Virol.* May 1991;65(5):2220-4. doi: 10.1128/JVI.65.5.2220-2224.1991.
- Miller et al., Continuous evolution of SpCas9 variants compatible with non-G PAMs. *Nat Biotechnol.* Apr. 2020;38(4):471-481. doi: 10.1038/s41587-020-0412-8. Epub Feb. 10, 2020.
- Miller et al., Phage-assisted continuous and non-continuous evolution. *Nat Protoc.* Dec. 2020;15(12):4101-4127. doi: 10.1038/s41596-020-00410-3. Epub Nov. 16, 2020.
- Miller, Human gene therapy comes of age. *Nature.* Jun. 11, 1992;357(6378):455-60. doi: 10.1038/357455a0.
- Mills et al., Protein splicing in trans by purified- and C-terminal fragments of the *Mycobacterium tuberculosis* RecA intein. *Proc Natl Acad Sci U S A.* Mar. 31, 1998;95(7):3543-8. doi: 10.1073/pnas.95.7.3543.
- Minoche et al., Evaluation of genomic high-throughput sequencing data generated on Illumina HiSeq and genome analyzer systems. *Genome Biol.* Nov. 8, 2011;12(11):R112. doi: 10.1186/GB-2011-12-11-r112.
- Minoretto et al., A W148R mutation in the human FOXD4 gene segregating with dilated cardiomyopathy, obsessive-compulsive disorder, and suicidality. *Int J Mol Med.* Mar. 2007;19(3):369-72.
- Mir et al., Two Active Site Divalent Ions in the Crystal Structure of the Hammerhead Ribozyme Bound to a Transition State Analogue. *Biochemistry.* Feb. 2, 2016;55(4):633-6. doi: 10.1021/acs.biochem.5b01139. Epub Jan. 9, 2016.
- Mir et al., Type II-C CRISPR-Cas9 Biology, Mechanism, and Application. *ACS Chem Biol.* Feb. 16, 2018;13(2):357-365. doi: 10.1021/acschembio.7b00855. Epub Dec. 20, 2017.
- Mishina et al., Conditional gene targeting on the pure C57BL/6 genetic background. *Neurosci Res.* Jun. 2007;58(2):105-12. doi: 10.1016/j.neures.2007.01.004. Epub Jan. 18, 2007.
- Mitani et al., Delivering therapeutic genes—matching approach and application. *Trends Biotechnol.* May 1993;11(5):162-6. doi: 10.1016/0167-7799(93)90108-L.

(56)

References Cited

OTHER PUBLICATIONS

- Mitton-Fry et al., Poly(A) tail recognition by a viral RNA element through assembly of a triple helix. *Science*. Nov. 26, 2010;330(6008):1244-7. doi: 10.1126/science.1195858.
- Miyaoka et al., Systematic quantification of HDR and NHEJ reveals effects of locus, nuclease, and cell type on genome-editing. *Sci Rep*. Mar. 31, 2016;6:23549. doi: 10.1038/srep23549.
- Moede et al., Identification of a nuclear localization signal, RRMKWK, in the homeodomain transcription factor PDX-1. *FEBS Lett*. Nov. 19, 1999;461(3):229-34. doi: 10.1016/S0014-5793(99)01446-5.
- Mohr et al., A Reverse Transcriptase-Cas1 Fusion Protein Contains a Cas6 Domain Required for Both CRISPR RNA Biogenesis and RNA Spacer Acquisition. *Mol Cell*. Nov. 5, 2018;72(4):700-714.e8. doi: 10.1016/j.molcel.2018.09.013. Epub Oct. 18, 2018. Including Supplemental Information.
- Mohr et al., Thermostable group II intron reverse transcriptase fusion proteins and their use in cDNA synthesis and next-generation RNA sequencing. *RNA*. Jul. 2013;19(7):958-70. doi: 10.1261/rna.039743.113. Epub May 22, 2013.
- Mojica et al., Intervening sequences of regularly spaced prokaryotic repeats derive from foreign genetic elements. *J Mol Evol*. Feb. 2005;60(2):174-82.
- Mok et al., A bacterial cytidine deaminase toxin enables CRISPR-free mitochondrial base editing. *Nature*. Jul. 2020;583(7817):631-637. doi: 10.1038/s41586-020-2477-4. Epub Jul. 8, 2020.
- Mol et al., Crystal structure and mutational analysis of human uracil-DNA glycosylase: structural basis for specificity and catalysis. *Cell*. Mar. 24, 1995;80(6):869-78. doi: 10.1016/0092-8674(95)90290-2.
- Mol et al., Crystal structure of human uracil-DNA glycosylase in complex with a protein inhibitor: protein mimicry of DNA. *Cell*. Sep. 8, 1995;82(5):701-8.
- Molla et al., CRISPR/Cas-Mediated Base Editing: Technical Considerations and Practical Applications. *Trends Biotechnol*. Oct. 2019;37(10):1121-1142. doi: 10.1016/j.tibtech.2019.03.008. Epub Apr. 14, 2019.
- Monahan et al., Site-specific incorporation of unnatural amino acids into receptors expressed in Mammalian cells. *Chem Biol*. Jun. 2003;10(6):573-80.
- Monot et al., The specificity and flexibility of 11 reverse transcription priming at imperfect T-tracts. *PLoS Genet*. May 2013;9(5):e1003499. doi: 10.1371/journal.pgen.1003499. Epub May 9, 2013.
- Montange et al., Structure of the S-adenosylmethionine riboswitch regulatory mRNA element. *Nature*. Jun. 29, 2006;441(7097):1172-5.
- Moore et al., Improved somatic mutagenesis in zebrafish using transcription activator-like effector nucleases (TALENs). *PLoS One*. 2012;7(5):e37877. Doi: 10.1371/journal.pone.0037877. Epub May 24, 2012.
- Mootz et al., Conditional protein splicing: a new tool to control protein structure and function in vitro and in vivo. *J Am Chem Soc*. Sep. 3, 2003;125(35):10561-9.
- Mootz et al., Protein splicing triggered by a small molecule. *J Am Chem Soc*. Aug. 7, 2002;124(31):9044-5.
- Morbitz et al., Assembly of custom TALE-type DNA binding domains by modular cloning. *Nucleic Acids Res*. Jul. 2011;39(13):5790-9. doi: 10.1093/nar/gkr151. Epub Mar. 18, 2011.
- Moreno-Mateos et al., CRISPRscan: designing highly efficient sgRNAs for CRISPR-Cas9 targeting in vivo. *Nat Methods*. Oct. 2015;12(10):982-8. doi: 10.1038/nmeth.3543. Epub Aug. 31, 2015.
- Morita et al., The site-specific recombination system of actinophage TG1. *FEMS Microbiol Lett*. Aug. 2009;297(2):234-40. doi: 10.1111/j.1574-6968.2009.01683.x.
- Morris et al., A peptide carrier for the delivery of biologically active proteins into mammalian cells. *Nat Biotechnol*. Dec. 2001;19(12):1173-6.
- Moscou et al., A simple cipher governs DNA recognition by TAL effectors. *Science*. Dec. 11, 2009;326(5959):1501. doi: 10.1126/science.1178817.
- Mougiakos et al., Characterizing a thermostable Cas9 for bacterial genome editing and silencing. *Nat Commun*. Nov. 21, 2017;8(1):1647. doi: 10.1038/s41467-017-01591-4.
- Muir et al., Expressed protein ligation: a general method for protein engineering. *Proc Natl Acad Sci U S A*. Jun. 9, 1998;95(12):6705-10. doi: 10.1073/pnas.95.12.6705.
- Muller et al., Nucleotide exchange and excision technology (NExT) DNA shuffling: a robust method for DNA fragmentation and directed evolution. *Nucleic Acids Res*. Aug. 1, 2005;33(13):e117. doi: 10.1093/nar/gni116. PMID: 16061932; PMCID: PMC1182171.
- Muller, U.F., Design and Experimental Evolution of trans-Splicing Group I Intron Ribozymes. *Molecules*. Jan. 2, 2017;22(1):75. doi: 10.3390/molecules22010075.
- Mullins et al., Transgenesis in nonmurine species. *Hypertension*. Oct. 1993;22(4):630-3.
- Mumtsidu et al., Structural features of the single-stranded DNA-binding protein of Epstein-Barr virus. *J Struct Biol*. Feb. 2008;161(2):172-87. doi: 10.1016/j.jsb.2007.10.014. Epub Nov. 1, 2007.
- Murphy, Phage recombinases and their applications. *Adv Virus Res*. 2012;83:367-414. doi: 10.1016/B978-0-12-394438-2.00008-6. Review.
- Mussolino et al., A novel TALE nuclease scaffold enables high genome editing activity in combination with low toxicity. *Nucleic Acids Res*. Nov. 2011;39(21):9283-93. Doi: 10.1093/nar/gkr597. Epub Aug. 3, 2011.
- Mussolino et al., TALE nucleases: tailored genome engineering made easy. *Curr Opin Biotechnol*. Oct. 2012;23(5):644-50. doi: 10.1016/j.copbio.2012.01.013. Epub Feb. 17, 2012.
- Muzyczka et al., Adeno-associated virus (AAV) vectors: will they work? *J Clin Invest*. Oct. 1994;94(4):1351. doi: 10.1172/JCI117468.
- Myerowitz et al., The major defect in Ashkenazi Jews with Tay-Sachs disease is an insertion in the gene for the alpha-chain of beta-hexosaminidase. *J Biol Chem*. Dec. 15, 1988;263(35):18587-9.
- Myers et al., Insulin signal transduction and the IRS proteins. *Annu Rev Pharmacol Toxicol*. 1996;36:615-58. doi: 10.1146/annurev.pa.36.040196.003151.
- Nabel et al., Direct gene transfer for immunotherapy and immunization. *Trends Biotechnol*. May 1993;11(5):211-5. doi: 10.1016/0167-7799(93)90117-R.
- Nahar et al., A G-quadruplex motif at the 3' end of sgRNAs improves CRISPR-Cas9 based genome editing efficiency. *Chem Commun (Camb)*. Mar. 7, 2018;54(19):2377-2380. doi: 10.1039/c7cc08893k. Epub Feb. 16, 2018.
- Nahvi et al., Coenzyme B12 riboswitches are widespread genetic control elements in prokaryotes. *Nucleic Acids Res*. Jan. 2, 2004;32(1):143-50.
- Nakamura et al., Codon usage tabulated from international DNA sequence databases: status for the year 2000. *Nucleic Acids Res*. Jan. 1, 2000;28(1):292. doi: 10.1093/nar/28.1.292.
- Naorem et al., DGR mutagenic transposition occurs via hypermutagenic reverse transcription primed by nicked template RNA. *Proc Natl Acad Sci U S A*. Nov. 21, 2017;114(47):E10187-E10195. doi: 10.1073/pnas.1715952114. Epub Nov. 6, 2017.
- Narayanan et al., Clamping down on weak terminal base pairs: oligonucleotides with molecular caps as fidelity-enhancing elements at the 5'- and 3'-terminal residues. *Nucleic Acids Res*. May 20, 2004;32(9):2901-11. Print 2004.
- Navaratnam et al., An overview of cytidine deaminases. *Int J Hematol*. Apr. 2006;83(3):195-200.
- NCBI Reference Sequence: NM_002427.3. Wu et al., May 3, 2014. 5 pages.
- Neel et al., Riboswitches: Classification, function and in silico approach, *International Journal of Pharma Sciences and Research*. 2010;1(9):409-420.
- Nelson et al., Filamentous phage DNA cloning vectors: a noninfective mutant with a nonpolar deletion in gene III. *Virology*. 1981;108(2): 338-50.
- Nern et al., Multiple new site-specific recombinases for use in manipulating animal genomes. *Proc Natl Acad Sci U S A*. Aug. 23, 2011;108(34):14198-203. doi: 10.1073/pnas.1111704108. Epub Aug. 9, 2011.

(56)

References Cited

OTHER PUBLICATIONS

- Nguyen et al., Evolutionary drivers of thermoadaptation in enzyme catalysis. *Science*. Jan. 20, 2017;355(6322):289-294. doi: 10.1126/science.aah3717. Epub Dec. 22, 2016.
- Nguyen et al., IQ-TREE: a fast and effective stochastic algorithm for estimating maximum-likelihood phylogenies. *Mol Biol Evol*. Jan. 2015;32(1):268-74. doi: 10.1093/molbev/msu300. Epub Nov. 3, 2014.
- Ni et al., Nucleic acid aptamers: clinical applications and promising new horizons. *Curr Med Chem*. 2011;18(27):4206-14. Review.
- Nishida et al., Targeted nucleotide editing using hybrid prokaryotic and vertebrate adaptive immune systems. *Science*. Sep. 16, 2016;353(6305):1248. pii: aaf8729. doi: 10.1126/science.aaf8729. Epub Aug. 4, 2016.
- Nishikura, Functions and regulation of RNA editing by ADAR deaminases. *Annu Rev Biochem*. 2010;79:321-349. doi:10.1146/annurev-biochem-060208-105251.
- Nishimasu et al., Crystal structure of Cas9 in complex with guide RNA and target DNA. *Cell*. Feb. 27, 2014;156(5):935-49. doi: 10.1016/j.cell.2014.02.001. Epub Feb. 13, 2014.
- Nishimasu et al., Crystal Structure of *Staphylococcus aureus* Cas9. *Cell*. Aug. 27, 2015;162(5):1113-26. doi: 10.1016/j.cell.2015.08.007.
- Nishimasu et al., Engineered CRISPR-Cas9 nuclease with expanded targeting space. *Science*. Sep. 21, 2018;361(6408):1259-1262. doi: 10.1126/science.aas9129. Epub Aug. 30, 2018.
- Nomura et al., Controlling Mammalian Gene Expression by Allosteric Hepatitis Delta Virus Ribozymes. *ACS Synth Biol*. Dec. 20, 2013;2(12):684-9. doi: 10.1021/sb400037a. Epub May 22, 2013.
- Noris et al., A phenylalanine-55 to serine amino-acid substitution in the human glycoprotein IX leucine-rich repeat is associated with Bernard-Soulier syndrome. *Br J Haematol*. May 1997;97(2):312-20.
- Nottingham et al., RNA-seq of human reference RNA samples using a thermostable group II intron reverse transcriptase. *RNA*. Apr. 2016;22(4):597-613. doi: 10.1261/rna.055558.115. Epub Jan. 29, 2016.
- Nowak et al., Characterization of single-stranded DNA-binding proteins from the psychrophilic bacteria *Desulfotalea psychrophila*, *Flavobacterium psychrophilum*, *Psychrobacter arcticus*, *Psychrobacter cryohalolentis*, *Psychromonas ingrahamii*, *Psychroflexus torquus*, and *Photobacterium profundum*. *BMC Microbiol*. Apr. 14, 2014;14:91. doi: 10.1186/1471-2180-14-91.
- Nowak et al., Guide RNA Engineering for Versatile Cas9 Functionality. *Nucleic Acids Res*. Nov. 16, 2016;44(20):9555-9564. doi: 10.1093/nar/gkw908. Epub Oct. 12, 2016.
- Nowak et al., Structural analysis of monomeric retroviral reverse transcriptase in complex with an RNA/DNA hybrid. *Nucleic Acids Res*. Apr. 1, 2013;41(6):3874-87. doi: 10.1093/nar/gkt053. Epub Feb. 4, 2013.
- Numrych et al., A comparison of the effects of single-base and triple-base changes in the integrase arm-type binding sites on the site-specific recombination of bacteriophage lambda. *Nucleic Acids Res*. Jul. 11, 1990;18(13):3953-9. doi: 10.1093/nar/18.13.3953.
- Nyerges et al., A highly precise and portable genome engineering method allows comparison of mutational effects across bacterial species. *Proc Natl Acad Sci U S A*. Mar. 1, 2016;113(9):2502-7. doi: 10.1073/pnas.1520040113. Epub Feb. 16, 2016.
- O'Connell et al., Programmable RNA recognition and cleavage by CRISPR/Cas9. *Nature*. Dec. 11, 2014;516(7530):263-6. doi: 10.1038/nature13769. Epub Sep. 28, 2014.
- O'Maille et al., Structure-based combinatorial protein engineering (SCOPE). *J Mol Biol*. Aug. 23, 2002;321(4):677-91.
- Oakes et al., CRISPR-Cas9 Circular Permutants as Programmable Scaffolds for Genome Modification. *Cell*. Jan. 10, 2019;176(1-2):254-267.e16. doi: 10.1016/j.cell.2018.11.052.
- Oakes et al., Profiling of engineering hotspots identifies an allosteric CRISPR-Cas9 switch. *Nat Biotechnol*. Jun. 2016;34(6):646-51. doi: 10.1038/nbt.3528. Epub May 2, 2016.
- Oakes et al., Protein engineering of Cas9 for enhanced function. *Methods Enzymol*. 2014;546:491-511.
- Odsbu et al., Specific -terminal interactions of the *Escherichia coli* SeqA protein are required to form multimers that restrain negative supercoils and form foci. *Genes Cells*. Nov. 2005; 10(11):1039-49.
- Oemig et al., Solution structure of DnaE intein from *Nostoc punctiforme*: structural basis for the design of a new split intein suitable for site-specific chemical modification. *FEBS Lett*. May 6, 2009;583(9):1451-6.
- Offord, *Advances in Genome Editing*. The Scientist, Apr. 20, 2016. <http://www.the-scientist.com/?articles.view/articleNo/45903/title/Advances-in-Genome-Editing/>.
- Ohe et al., Purification and properties of xanthine dehydrogenase from *Streptomyces cyanogenus*. *J Biochem*. Jul. 1979;86(1):45-53.
- Olivares et al., Site-specific genomic integration produces therapeutic Factor IX levels in mice. *Nat Biotechnol*. Nov. 2002;20(11):1124-8. doi: 10.1038/nbt753. Epub Oct. 15, 2002.
- Olorunniji et al., Purification and In Vitro Characterization of Zinc Finger Recombinases. *Methods Mol Biol*. 2017;1642:229-245. doi: 10.1007/978-1-4939-7169-5_15.
- Olorunniji et al., Site-specific recombinases: molecular machines for the Genetic Revolution. *Biochem J*. Mar. 15, 2016;473(6):673-84. doi: 10.1042/BJ20151112.
- Olorunniji et al., Synapsis and catalysis by activated Tn3 resolvase mutants. *Nucleic Acids Res*. Dec. 2008;36(22):7181-91. doi: 10.1093/nar/gkn885. Epub Nov. 10, 2008.
- Orlando et al., Zinc-finger nuclease-driven targeted integration into mammalian genomes using donors with limited chromosomal homology. *Nucleic Acids Res*. Aug. 2010;38(15):e152. doi: 10.1093/nar/gkq512. Epub Jun. 8, 2010.
- Ortiz-Urda et al., Stable nonviral genetic correction of inherited human skin disease. *Nat Med*. Oct. 2002;8(10):1166-70. doi: 10.1038/nm766. Epub Sep. 16, 2002. Erratum in: *Nat Med*. Feb. 2003;9(2):237.
- Osborn et al., Base Editor Correction of COL7A1 in Recessive Dystrophic Epidermolysis Bullosa Patient-Derived Fibroblasts and iPSCs. *J Invest Dermatol*. Feb. 2020;140(2):338-347.e5. doi: 10.1016/j.jid.2019.07.701. Epub Aug. 19, 2019.
- Osborn et al., TALEN-based gene correction for epidermolysis bullosa. *Mol Ther*. Jun. 2013;21(6):1151-9. doi: 10.1038/mt.2013.56. Epub Apr. 2, 2013.
- Ostermeier et al., A combinatorial approach to hybrid enzymes independent of DNA homology. *Nat Biotechnol*. Dec. 1999;17(12):1205-9.
- Ostertag et al., Biology of mammalian L1 retrotransposons. *Annu Rev Genet*. 2001;35:501-38. doi: 10.1146/annurev.genet.35.102401.091032.
- Otomo et al., Improved segmental isotope labeling of proteins and application to a larger protein. *J Biomol NMR*. Jun. 1999;14(2):105-14. doi: 10.1023/a:1008308128050.
- Otomo et al., NMR observation of selected segments in a larger protein: central-segment isotope labeling through intein-mediated ligation. *Biochemistry*. Dec. 7, 1999;38(49):16040-4. doi: 10.1021/bi991902j.
- Otto et al., The probability of fixation in populations of changing size. *Genetics*. Jun. 1997; 146(2):723-33.
- Packer et al., Methods for the directed evolution of proteins. *Nat Rev Genet*. Jul. 2015;16(7):379-94. doi: 10.1038/nrg3927. Epub Jun. 9, 2015.
- Packer et al., Phage-assisted continuous evolution of proteases with altered substrate specificity. *Nat Commun*. Oct. 16, 2017;8(1):956. doi: 10.1038/s41467-017-01055-9.
- Paige et al., RNA mimics of green fluorescent protein. *Science*. Jul. 29, 2011;333(6042):642-6. doi:10.1126/science.1207339.
- Paiva et al., Targeted protein degradation: elements of PROTAC design. *Curr Opin Chem Biol*. Jun. 2019;50:111-119. doi: 10.1016/j.cbpa.2019.02.022. Epub Apr. 17, 2019.
- Pan et al., Biological and biomedical applications of engineered nucleases. *Mol Biotechnol*. Sep. 2013;55(1):54-62. doi: 10.1007/s12033-012-9613-9.
- Park et al., Digenome-seq web tool for profiling CRISPR specificity. *Nat Methods*. May 30, 2017;14(6):548-549. doi: 10.1038/nmeth.4262.

(56)

References Cited

OTHER PUBLICATIONS

- Park et al., Highly efficient editing of the β -globin gene in patient-derived hematopoietic stem and progenitor cells to treat sickle cell disease. *Nucleic Acids Res.* Sep. 5, 2019;47(15):7955-7972. doi: 10.1093/nar/gkz475.
- Park et al., Sendai virus, an RNA virus with no risk of genomic integration, delivers CRISPR/Cas9 for efficient gene editing. *Mol Ther Methods Clin Dev.* Aug. 24, 2016;3:16057. doi: 10.1038/mtm.2016.57.
- Parker et al., Admixture mapping identifies a quantitative trait locus associated with FEV1/FVC in the COPD Gene Study. *Genet Epidemiol.* Nov. 2014;38(7):652-9. doi: 10.1002/gepi.21847. Epub Aug. 11, 2014.
- Patel et al., Flap endonucleases pass 5'-flaps through a flexible arch using a disorder-thread-order mechanism to confer specificity for free 5'-ends. *Nucleic Acids Res.* May 2012;40(10):4507-19. doi: 10.1093/nar/gks051. Epub Feb. 8, 2012.
- Pattanayak et al., Determining the specificities of TALENs, Cas9, and other genome-editing enzymes. *Methods Enzymol.* 2014;546:47-78. doi: 10.1016/B978-0-12-801185-0.00003-9.
- Pattanayak et al., High-throughput profiling of off-target DNA cleavage reveals RNA-programmed Cas9 nuclease specificity. *Nat Biotechnol.* Sep. 2013;31(9):839-43. doi: 10.1038/nbt.2673. Epub Aug. 11, 2013.
- Pattanayak et al., Revealing off-target cleavage specificities of zinc-finger nucleases by in vitro selection. *Nat Methods.* Aug. 7, 2011;8(9):765-70. doi: 10.1038/nmeth.1670.
- Pavletich et al., Zinc finger-DNA recognition: crystal structure of a Zif268-DNA complex at 2.1 Å. *Science.* May 10, 1991;252(5007):809-17.
- Pawson et al., Protein phosphorylation in signaling—50 years and counting. *Trends Biochem Sci.* Jun. 2005;30(6):286-90. doi: 10.1016/j.tibs.2005.04.013.
- Pearl, Structure and function in the uracil-DNA glycosylase superfamily. *Mutat Res.* Aug. 30, 2000;460(3-4):165-81.
- Peck et al., Directed evolution of a small-molecule-triggered intein with improved splicing properties in mammalian cells. *Chem Biol.* May 27, 2011;18(5):619-30. doi: 10.1016/j.chembiol.2011.02.014.
- Pellenz et al., New human chromosomal safe harbor sites for genome engineering with CRISPR/Cas9, TAL effector and homing endonucleases. Aug. 20, 2018. bioRxiv doi: <https://doi.org/10.1101/396390>.
- Pelletier, CRISPR-Cas systems for the study of the immune function. Nov. 15, 2016. <https://doi.org/10.1002/9780470015902.a0026896>.
- Pennisi et al., The CRISPR craze. *Science.* Aug. 23, 2013;341(6148):833-6. doi: 10.1126/science.341.6148.833.
- Pennisi et al., The tale of the TALEs. *Science.* Dec. 14, 2012;338(6113):1408-11. doi: 10.1126/science.338.6113.1408.
- Perach et al., Catalytic features of the recombinant reverse transcriptase of bovine leukemia virus expressed in bacteria. *Virology.* Jun. 20, 1999;259(1):176-89. doi: 10.1006/viro.1999.9761.
- Perez et al., Establishment of HIV-1 resistance in CD4+ T cells by genome editing using zinc-finger nucleases. *Nat Biotechnol.* Jul. 2008;26(7):808-16. doi: 10.1038/nbt.1410. Epub Jun. 29, 2008.
- Perez-Pinera et al., Advances in targeted genome editing. *Curr Opin Chem Biol.* Aug. 2012;16(3-4):268-77. doi: 10.1016/j.cbpa.2012.06.007. Epub Jul. 20, 2012.
- Perez-Pinera et al., RNA-guided gene activation by CRISPR-Cas9-based transcription factors. *Nat Methods.* Oct. 2013;10(10):973-6. doi: 10.1038/nmeth.2600. Epub Jul. 25, 2013.
- Perler et al., Protein splicing and autoprolysis mechanisms. *Curr Opin Chem Biol.* Oct. 1997;1(3):292-9. doi: 10.1016/s1367-5931(97)80065-8.
- Perler et al., Protein splicing elements: inteins and exteins—a definition of terms and recommended nomenclature. *Nucleic Acids Res.* Apr. 11, 1994;22(7):1125-7. doi: 10.1093/nar/22.7.1125.
- Perler, InBase, the New England Biolabs Intein Database. *Nucleic Acids Res.* Jan. 1, 1999;27(1):346-7. doi: 10.1093/nar/27.1.346.
- Perler, Protein splicing of inteins and hedgehog autoprolysis: structure, function, and evolution. *Cell.* Jan. 9, 1998;92(1):1-4. doi: 10.1016/s0092-8674(00)80892-2.
- Perreault et al., Mixed deoxyribo- and ribo-oligonucleotides with catalytic activity. *Nature.* Apr. 5, 1990;344(6266):565-7. doi: 10.1038/344565a0.
- Petek et al., Frequent endonuclease cleavage at off-target locations in vivo. *Mol Ther.* May 2010;18(5):983-6. doi: 10.1038/mt.2010.35. Epub Mar. 9, 2010.
- Petersen-Mahrt et al., AID mutates *E. coli* suggesting a DNA deamination mechanism for antibody diversification. *Nature.* Jul. 4, 2002;418(6893):99-103.
- Petolino et al., Editing Plant Genomes: a new era of crop improvement. *Plant Biotechnol J.* Feb. 2016;14(2):435-6. doi: 10.1111/pbi.12542.
- Peyrottes et al., Oligodeoxynucleoside phosphoramidates (P-NH2): synthesis and thermal stability of duplexes with DNA and RNA targets. *Nucleic Acids Res.* May 15, 1996;24(10):1841-8.
- Phillips, The challenge of gene therapy and DNA delivery. *J Pharm Pharmacol.* Sep. 2001;53(9):1169-74.
- Pickart et al., Ubiquitin: structures, functions, mechanisms. *Biochim Biophys Acta.* Nov. 29, 2004;1695(1-3):55-72. doi: 10.1016/j.bbamer.2004.09.019.
- Pinkert et al., An albumin enhancer located 10 kb upstream functions along with its promoter to direct efficient, liver-specific expression in transgenic mice. *Genes Dev.* May 1987;1(3):268-76. doi: 10.1101/gad.1.3.268.
- Pirakitikulr et al., PCRless library mutagenesis via oligonucleotide recombination in yeast. *Protein Sci.* Dec. 2010;19(12):2336-46. doi: 10.1002/pro.513.
- Plasterk et al., DNA inversions in the chromosome of *Escherichia coli* and in bacteriophage Mu: relationship to other site-specific recombination systems. *Proc Natl Acad Sci U S A.* Sep. 1983;80(17):5355-8.
- Plosky et al., CRISPR-Mediated Base Editing without DNA Double-Strand Breaks. *Mol Cell.* May 19, 2016;62(4):477-8. doi: 10.1016/j.molcel.2016.05.006.
- Pluciennik et al., PCNA function in the activation and strand direction of MutL? endonuclease in mismatch repair. *Proc Natl Acad Sci U S A.* Sep. 14, 2010;107(37):16066-71. doi: 10.1073/pnas.1010662107. Epub Aug. 16, 2010.
- Poller et al., A leucine-to-proline substitution causes a defective alpha 1-antichymotrypsin allele associated with familial obstructive lung disease. *Genomics.* Sep. 1993;17(3):740-3.
- Popp et al., Sortagging: a versatile method for protein labeling. *Nat Chem Biol.* Nov. 2007;3(11):707-8. doi: 10.1038/nchembio.2007.31. Epub Sep. 23, 2007.
- Porteus, Design and testing of zinc finger nucleases for use in mammalian cells. *Methods Mol Biol.* 2008;435:47-61. doi: 10.1007/978-1-59745-232-8_4.
- Posnick et al., Imbalanced base excision repair increases spontaneous mutation and alkylation sensitivity in *Escherichia coli*. *J Bacteriol.* Nov. 1999;181(21):6763-71.
- Pospisilová et al., Hydrolytic cleavage of N6-substituted adenine derivatives by eukaryotic adenine and adenosine deaminases. *Biosci Rep.* 2008;28(6):335-347. doi: 10.1042/BSR20080081.
- Pourcel et al., CRISPR elements in *Yersinia pestis* acquire new repeats by preferential uptake of bacteriophage DNA, and provide additional tools for evolutionary studies. *Microbiology.* Mar. 2005;151(Pt 3):653-63.
- Prasad et al., Rev1 is a base excision repair enzyme with 5'-deoxyribose phosphate lyase activity. *Nucleic Acids Res.* Dec. 15, 2016;44(22):10824-10833. doi: 10.1093/nar/gkw869. Epub Sep. 28, 2016.
- Prorocic et al., Zinc-finger recombinase activities in vitro. *Nucleic Acids Res.* Nov. 2011;39(21):9316-28. doi: 10.1093/nar/gkr652. Epub Aug. 17, 2011.
- Proudfoot et al., Zinc finger recombinases with adaptable DNA sequence specificity. *PLoS One.* Apr. 29, 2011;6(4):e19537. doi: 10.1371/journal.pone.0019537.
- Pruschy et al., Mechanistic studies of a signaling pathway activated by the organic dimerizer FK1012. *Chem Biol.* Nov. 1994;1(3):163-72. doi: 10.1016/1074-5521(94)90006-x.

(56)

References Cited

OTHER PUBLICATIONS

- Prykhozhij et al., CRISPR multitargeter: a web tool to find common and unique CRISPR single guide RNA targets in a set of similar sequences. *PLoS One*. Mar. 5, 2015;10(3):e0119372. doi: 10.1371/journal.pone.0119372. eCollection 2015.
- Pu et al., Evolution of a split RNA polymerase as a versatile biosensor platform. *Nat Chem Biol*. Apr. 2017;13(4):432-438. doi: 10.1038/nchembio.2299. Epub Feb. 13, 2017.
- Putnam et al., Protein mimicry of DNA from crystal structures of the uracil-DNA glycosylase inhibitor protein and its complex with *Escherichia coli* uracil-DNA glycosylase. *J Mol Biol*. Mar. 26, 1999;287(2):331-46.
- Qi et al., Engineering naturally occurring trans-acting non-coding RNAs to sense molecular signals. *Nucleic Acids Res*. Jul. 2012;40(12):5775-86. doi: 10.1093/nar/gks168. Epub Mar. 1, 2012.
- Qi et al., Repurposing CRISPR as an RNA-guided platform for sequence-specific control of gene expression. *Cell*. Feb. 28, 2013;152(5):1173-83. doi: 10.1016/j.cell.2013.02.022.
- Qu et al., Global mapping of binding sites for phic31 integrase in transgenic maden-darby bovine kidney cells using ChIP-seq. *Hereditas*. Jan. 14, 2019;156:3. doi: 10.1186/s41065-018-0079-z.
- Queen et al., Immunoglobulin gene transcription is activated by downstream sequence elements. *Cell*. Jul. 1983;33(3):741-8. doi: 10.1016/0092-8674(83)90016-8.
- Radany et al., Increased spontaneous mutation frequency in human cells expressing the phage PBS2-encoded inhibitor of uracil-DNA glycosylase. *Mutat Res*. Sep. 15, 2000;461(1):41-58. doi: 10.1016/S0921-8777(00)00040-9.
- Raghavan et al., Abstract 27: Therapeutic Targeting of Human Lipid Genes with in vivo CRISPR-Cas9 Genome Editing. Oral Abstract Presentations: Lipoprotein Metabolism and Therapeutic Targets. *Arterioscler Thromb Vasc Biol*. 2015;35(Suppl. 1):Abstract 27. 5 pages.
- Raillard et al., Targeting sites within HIV-1 cDNA with a DNA-cleaving ribozyme. *Biochemistry*. Sep. 10, 1996;35(36):11693-701. doi: 10.1021/bi960845g.
- Raina et al., PROTAC-induced BET protein degradation as a therapy for castration-resistant prostate cancer. *Proc Natl Acad Sci U S A*. Jun. 28, 2016;113(26):7124-9. doi: 10.1073/pnas.1521738113. Epub Jun. 6, 2016.
- Rakonjac et al., Roles of PIII in filamentous phage assembly. *J Mol Biol*. 1998; 282(1):25-41.
- Ramakrishna et al., Gene disruption by cell-penetrating peptide-mediated delivery of Cas9 protein and guide RNA. *Genome Res*. Jun. 2014;24(6):1020-7. doi: 10.1101/gr.171264.113. Epub Apr. 2, 2014.
- Ramamurthy et al., Identification of immunogenic B-cell epitope peptides of rubella virus E1 glycoprotein towards development of highly specific immunoassays and/or vaccine. Conference Abstract. 2019.
- Ramirez et al., Engineered zinc finger nickases induce homology-directed repair with reduced mutagenic effects. *Nucleic Acids Res*. Jul. 2012;40(12):5560-8. doi: 10.1093/nar/gks179. Epub Feb. 28, 2012.
- Ramirez et al., Unexpected failure rates for modular assembly of engineered zinc fingers. *Nat Methods*. May 2008;5(5):374-5. doi: 10.1038/nmeth0508-374.
- Ran et al., Double Nicking by RNA-guided CRISPR Cas9 for Enhanced Genome Editing Specificity. *Cell*. Sep. 12, 2013;154(6):1380-9. doi: 10.1016/j.cell.2013.08.021. Epub Aug. 29, 2013.
- Ran et al., Genome engineering using the CRISPR-Cas9 system. *Nat Protoc*. Nov. 2013;8(11):2281-308. doi: 10.1038/nprot.2013.143. Epub Oct. 24, 2013.
- Ran et al., In vivo genome editing using *Staphylococcus aureus* Cas9. *Nature*. Apr. 9, 2015;520(7546):186-91. doi: 10.1038/nature14299. Epub Apr. 1, 2015.
- Ranzau et al., Genome, Epigenome, and Transcriptome Editing via Chemical Modification of Nucleobases in Living Cells. *Biochemistry*. Feb. 5, 2019;58(5):330-335. doi: 10.1021/acs.biochem.8b00958. Epub Dec. 12, 2018.
- Rashel et al., A novel site-specific recombination system derived from bacteriophage phiMR11. *Biochem Biophys Res Commun*. Apr. 4, 2008;368(2):192-8. doi: 10.1016/j.bbrc.2008.01.045. Epub Jan. 22, 2008.
- Rasila et al., Critical evaluation of random mutagenesis by error-prone polymerase chain reaction protocols, *Escherichia coli* mutator strain, and hydroxylamine treatment. *Anal Biochem*. May 1, 2009;388(1):71-80. doi: 10.1016/j.ab.2009.02.008. Epub Feb. 10, 2009.
- Raskin et al., Substitution of a single bacteriophage T3 residue in bacteriophage T7 RNA polymerase at position 748 results in a switch in promoter specificity. *J Mol Biol*. Nov. 20, 1992;228(2):506-15.
- Raskin et al., T7 Rna polymerase mutants with altered promoter specificities. *Proc Natl Acad Sci U S A*. Apr. 15, 1993;90(8):3147-51.
- Rath et al., Fidelity of end joining in mammalian episomes and the impact of Metnase on joint processing. *BMC Mol Biol*. Mar. 22, 2014;15:6. doi: 10.1186/1471-2199-15-6.
- Ravishankar et al., X-ray analysis of a complex of *Escherichia coli* uracil DNA glycosylase (EcUDG) with a proteinaceous inhibitor. The structure elucidation of a prokaryotic UDG. *Nucleic Acids Res*. 26 (21): 4880-4887 (1998).
- Ray et al., A compendium of RNA-binding motifs for decoding gene regulation. *Nature*. Jul. 11, 2013;499(7457):172-7. doi: 10.1038/nature12311.
- Ray et al., Homologous recombination: ends as the means. *Trends Plant Sci*. Oct. 2002;7(10):435-40.
- Rebar et al., Phage display methods for selecting zinc finger proteins with novel DNA-binding specificities. *Methods Enzymol*. 1996;267:129-49.
- Rees et al., Analysis and minimization of cellular RNA editing by DNA adenine base editors. *Sci Adv*. May 8, 2019;5(5):eaax5717. doi: 10.1126/sciadv.aax5717.
- Rees et al., Base editing: precision chemistry on the genome and transcriptome of living cells. *Nat Rev Genet*. Dec. 2018;19(12):770-788. doi: 10.1038/s41576-018-0059-1.
- Rees et al., Development of hRad51-Cas9 nickase fusions that mediate HDR without double-stranded breaks. *Nat Commun*. May 17, 2019;10(1):2212. doi: 10.1038/s41467-019-09983-4.
- Rees et al., Improving the DNA specificity and applicability of base editing through protein engineering and protein delivery. *Nat Commun*. Jun. 6, 2017;8:15790. doi: 10.1038/ncomms15790.
- Relph et al., Recent developments and current status of gene therapy using viral vectors in the United Kingdom. *BMJ*. 2004;329(7470):839-842. doi: 10.1136/bmj.329.7470.839.
- Remy et al., Gene transfer with a series of lipophilic DNA-binding molecules. *Bioconjug Chem*. Nov.-Dec. 1994;5(6):647-54. doi: 10.1021/bc00030a021.
- Ren et al., In-line Alignment and Mg²⁺ Coordination at the Cleavage Site of the env22 Twister Ribozyme. *Nat Commun*. Nov. 20, 2014;5:5534. doi: 10.1038/ncomms6534.
- Ren et al., Pistol Ribozyme Adopts a Pseudoknot Fold Facilitating Site-Specific In-Line Cleavage. *Nat Chem Biol*. Sep. 2016;12(9):702-8. doi: 10.1038/nchembio.2125. Epub Jul. 11, 2016.
- Reynaud et al., What role for AID: mutator, or assembler of the immunoglobulin mutasome? *Nat Immunol*. Jul. 2003;4(7):631-8.
- Reyon et al., Flash assembly of TALENs for high-throughput genome editing. *Nat Biotechnol*. May 2012;30(5):460-5. doi: 10.1038/nbt.2170.
- Ribeiro et al., Protein Engineering Strategies to Expand CRISPR-Cas9 Applications. *Int J Genomics*. Aug. 2, 2018;2018:1652567. doi: 10.1155/2018/1652567.
- Richter et al., Phage-assisted evolution of an adenine base editor with improved Cas domain compatibility and activity. *Nat Biotechnol*. Jul. 2020;38(7):883-891. doi: 10.1038/s41587-020-0453-z. Epub Mar. 16, 2020.
- Riechmann et al., The C-terminal domain of TolA is the coreceptor for filamentous phage infection of *E. coli*. *Cell*. 1997; 90(2):351-60. PMID:9244308.
- Ringrose et al., The Kw recombinase, an integrase from *Kluyveromyces waltii*. *Eur J Biochem*. Sep. 15, 1997;248(3):903-12. doi: 10.1111/j.1432-1033.1997.00903.x.

(56)

References Cited

OTHER PUBLICATIONS

- Risso et al., Hyperstability and substrate promiscuity in laboratory resurrections of Precambrian β -lactamases. *J Am Chem Soc.* Feb. 27, 2013;135(8):2899-902. doi: 10.1021/ja311630a. Epub Feb. 14, 2013.
- Ritchie et al., limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res.* Apr. 20, 2015;43(7):e47. doi: 10.1093/nar/gkv007. Epub Jan. 20, 2015.
- Robertson et al., DNA repair in mammalian cells: Base excision repair: the long and short of it. *Cell Mol Life Sci.* Mar. 2009;66(6):981-93. doi: 10.1007/s00018-009-8736-z.
- Robertson et al., Selection in vitro of an RNA enzyme that specifically cleaves single-stranded DNA. *Nature.* Mar. 29, 1990;344(6265):467-8. doi: 10.1038/344467a0.
- Robinson et al., The protein tyrosine kinase family of the human genome. *Oncogene.* Nov. 20, 2000;19(49):5548-57. doi: 10.1038/sj.onc.1203957.
- Rogozin et al., Evolution and diversification of lamprey antigen receptors: evidence for involvement of an AID-APOBEC family cytosine deaminase. *Nat Immunol.* Jun. 2007;8(6):647-56. doi: 10.1038/ni1463. Epub Apr. 29, 2007.
- Rong et al., Homologous recombination in human embryonic stem cells using CRISPR/Cas9 nickase and a long DNA donor template. *Protein Cell.* Apr. 2014;5(4):258-60. doi: 10.1007/s13238-014-0032-5.
- Rongrong et al., Effect of deletion mutation on the recombination activity of Cre recombinase. *Acta Biochim Pol.* 2005;52(2):541-4. Epub May 15, 2005.
- Roth et al., A widespread self-cleaving ribozyme class is revealed by bioinformatics. *Nat Chem Biol.* Jan. 2014;10(1):56-60. doi: 10.1038/nchembio.1386. Epub Nov. 17, 2013.
- Roth et al., Purification and characterization of murine retroviral reverse transcriptase expressed in *Escherichia coli*. *J Biol Chem.* Aug. 5, 1985;260(16):9326-35.
- Rouet et al., Expression of a site-specific endonuclease stimulates homologous recombination in mammalian cells. *Proc Natl Acad Sci U S A.* Jun. 21, 1994;91(13):6064-8. doi: 10.1073/pnas.91.13.6064.
- Rouet et al., Introduction of double-strand breaks into the genome of mouse cells by expression of a rare-cutting endonuclease. *Mol Cell Biol.* Dec. 1994;14(12):8096-106. doi: 10.1128/mcb.14.12.8096.
- Rouet et al., Receptor-Mediated Delivery of CRISPR-Cas9 Endonuclease for Cell-Type-Specific Gene Editing. *J Am Chem Soc.* May 30, 2018;140(21):6596-6603. doi: 10.1021/jacs.8b01551. Epub May 18, 2018.
- Roundtree et al., YTHDC1 mediates nuclear export of N6-methyladenosine methylated mRNAs. *Elife.* Oct. 6, 2017;6:e31311. doi: 10.7554/eLife.31311.
- Rowland et al., Regulatory mutations in Sin recombinase support a structure-based model of the synaptoneme. *Mol Microbiol.* Oct. 2009;74(2):282-98. doi: 10.1111/j.1365-2958.2009.06756.x. Epub Jun. 8, 2009.
- Rowland et al., Sin recombinase from *Staphylococcus aureus*: synaptic complex architecture and transposon targeting. *Mol Microbiol.* May 2002;44(3):607-19. doi: 10.1046/j.1365-2958.2002.02897.x.
- Rowley, Chromosome translocations: dangerous liaisons revisited. *Nat Rev Cancer.* Dec. 2001;1(3):245-50. doi: 10.1038/35106108.
- Rubio et al., Transfer RNA travels from the cytoplasm to organelles. *Wiley Interdiscip Rev RNA.* Nov.-Dec. 2011;2(6):802-17. doi: 10.1002/wrna.93. Epub Jul. 11, 2011.
- Rudolph et al., Synthetic riboswitches for the conditional control of gene expression in *Streptomyces coelicolor*. *Microbiology.* Jul. 2013; 159(Pt 7):1416-22. doi: 10.1099/mic.0.067322-0. Epub May 15, 2013.
- Rutherford et al., Attachment site recognition and regulation of directionality by the serine integrases. *Nucleic Acids Res.* Sep. 2013;41(17):8341-56. doi: 10.1093/nar/gkt580. Epub Jul. 2, 2013.
- Ryu et al., Adenine base editing in mouse embryos and an adult mouse model of Duchenne muscular dystrophy. *Nat Biotechnol.* Jul. 2018;36(6):536-539. doi: 10.1038/nbt.4148. Epub Apr. 27, 2018.
- Rüfer et al., Non-contact positions impose site selectivity on Cre recombinase. *Nucleic Acids Res.* Jul. 1, 2002;30(13):2764-71. doi: 10.1093/nar/gkf399.
- Sadelain et al., Safe harbours for the integration of new DNA in the human genome. *Nat Rev Cancer.* Dec. 1, 2011;12(1):51-8. doi: 10.1038/nrc3179.
- Sage et al., Proliferation of functional hair cells in vivo in the absence of the retinoblastoma protein. *Science.* Feb. 18, 2005;307(5712):1114-8. Epub Jan. 13, 2005.
- Sale et al., Y-family DNA polymerases and their role in tolerance of cellular DNA damage. *Nat Rev Mol Cell Biol.* Feb. 23, 2012;13(3):141-52. doi: 10.1038/nrm3289.
- Saleh-Gohari et al., Conservative homologous recombination preferentially repairs DNA double-strand breaks in the S phase of the cell cycle in human cells. *Nucleic Acids Res.* Jul. 13, 2004;32(12):3683-8. Print 2004.
- Samal et al., Cationic polymers and their therapeutic potential. *Chem Soc Rev.* Nov. 7, 2012;41(21):7147-94. doi: 10.1039/c2cs35094g. Epub Aug. 10, 2012.
- Samanta et al., A reverse transcriptase ribozyme. *Elife.* Sep. 26, 2017;6:e31153. doi: 10.7554/eLife.31153.
- Samulski et al., Helper-free stocks of recombinant adeno-associated viruses: normal integration does not require viral gene expression. *J Virol.* Sep. 1989;63(9):3822-8. doi: 10.1128/JVI.63.9.3822-3828.1989.
- Sander et al., CRISPR-Cas systems for editing, regulating and targeting genomes. *Nat Biotechnol.* Apr. 2014;32(4):347-55. doi: 10.1038/nbt.2842. Epub Mar. 2, 2014.
- Sander et al., In silico abstraction of zinc finger nuclease cleavage profiles reveals an expanded landscape of off-target sites. *Nucleic Acids Res.* Oct. 2013;41(19):e181. doi: 10.1093/nar/gkt716. Epub Aug. 14, 2013.
- Sander et al., Targeted gene disruption in somatic zebrafish cells using engineered TALENs. *Nat Biotechnol.* Aug. 5, 2011;29(8):697-8. doi: 10.1038/nbt.1934.
- Sang et al., A unique uracil-DNA binding protein of the uracil DNA glycosylase superfamily. *Nucleic Acids Res.* Sep. 30, 2015;43(17):8452-63. doi: 10.1093/nar/gkv854. Epub Aug. 24, 2015.
- Sang, Prospects for transgenesis in the chick. *Mech Dev.* Sep. 2004;121(9):1179-86.
- Sanjana et al., A transcription activator-like effector toolbox for genome engineering. *Nat Protoc.* Jan. 5, 2012;7(1):171-92. doi: 10.1038/nprot.2011.431.
- Santiago et al., Targeted gene knockout in mammalian cells by using engineered zinc-finger nucleases. *Proc Natl Acad Sci U S A.* Apr. 15, 2008;105(15):5809-14. doi: 10.1073/pnas.0800940105. Epub Mar. 21, 2008.
- Santoro et al., Directed evolution of the site specificity of Cre recombinase. *Proc Natl Acad Sci U S A.* Apr. 2, 2002;99(7):4185-90. Epub Mar. 19, 2002.
- Saparbav et al., Excision of hypoxanthine from DNA containing dIMP residues by the *Escherichia coli*, yeast, rat, and human alkylpurine DNA glycosylases. *Proc Natl Acad Sci U S A.* Jun. 21, 1994;91(13):5873-7. doi: 10.1073/pnas.91.13.5873.
- Sapranaukas et al., The *Streptococcus thermophilus* CRISPR/Cas system provides immunity in *Escherichia coli*. *Nucleic Acids Res.* Nov. 2011;39(21):9275-82. doi: 10.1093/nar/gkr606. Epub Aug. 3, 2011.
- Sapunar et al., Dorsal root ganglion—a potential new therapeutic target for neuropathic pain. *J Pain Res.* 2012;5:31-8. doi: 10.2147/JPR.S26603. Epub Feb. 16, 2012.
- Saraconi et al., The RNA editing enzyme APOBEC1 induces somatic mutations and a compatible mutational signature is present in esophageal adenocarcinomas. *Genome Biol.* Jul. 31, 2014;15(7):417. doi: 10.1186/s13059-014-0417-z.
- Sarkar et al., HIV-1 proviral DNA excision using an evolved recombinase. *Science.* Jun. 29, 2007;316(5833):1912-5. doi: 10.1126/science.1141453.
- Sashital et al., Mechanism of foreign DNA selection in a bacterial adaptive immune system. *Mol Cell.* Jun. 8, 2012;46(5):606-15. doi: 10.1016/j.molcel.2012.03.020. Epub Apr. 19, 2012.

(56)

References Cited

OTHER PUBLICATIONS

- Sasidharan et al., The selection of acceptable protein mutations. PNAS; Jun. 12, 2007;104(24):10080-5. www.pnas.org/cgi/doi/10.1073.pnas.0703737104.
- Satomura et al., Precise genome-wide base editing by the CRISPR Nickase system in yeast. Sci Rep. May 18, 2017;7(1):2095. doi: 10.1038/s41598-017-02013-7.
- Sauadek et al., A preliminary trial of the programmable implantable medication system for insulin delivery. Engl J Med. Aug. 31, 1989;321(9):574-9.
- Sauer et al., DNA recombination with a heterospecific Cre homolog identified from comparison of the pac-cl regions of P1-related phages. Nucleic Acids Res. Nov. 18, 2004;32(20):6086-95. doi: 10.1093/nar/gkh941.
- Savic et al., Covalent linkage of the DNA repair template to the CRISPR-Cas9 nuclease enhances homology-directed repair. Elife. May 29, 2018;7:e33761. doi: 10.7554/eLife.33761.
- Saville et al., A site-specific self-cleavage reaction performed by a novel RNA in *Neurospora* mitochondria. Cell. May 18, 1990;61(4):685-96. doi: 10.1016/0092-8674(90)90480-3.
- Savva et al., The structural basis of specific base-excision repair by uracil-DNA glycosylase. Nature. Feb. 9, 1995;373(6514):487-93. doi: 10.1038/373487a0.
- Schaaper et al., Base selection, proofreading, and mismatch repair during DNA replication in *Escherichia coli*. J Biol Chem. Nov. 15, 1993;268(32):23762-5.
- Schaaper et al., Spectra of spontaneous mutations in *Escherichia coli* strains defective in mismatch correction: the nature of in vivo DNA replication errors. Proc Natl Acad Sci U S A. Sep. 1987;84(17):6220-4.
- Schaefer et al., Understanding RNA modifications: the promises and technological bottlenecks of the 'epitranscriptome'. Open Biol. May 2017;7(5):170077. doi: 10.1098/rsob.170077.
- Schek et al., Definition of the upstream efficiency element of the simian virus 40 late polyadenylation signal by using in vitro analyses. Mol Cell Biol. Dec. 1992;12(12):5386-93. doi: 10.1128/mcb.12.12.5386.
- Schenk et al., MPDU1 mutations underlie a novel human congenital disorder of glycosylation, designated type If. J Clin Invest. Dec. 2001; 108(11):1687-95. doi: 10.1172/JCI13419.
- Schmitz et al., Behavioral abnormalities in prion protein knockout mice and the potential relevance of PrP(C) for the cytoskeleton. Prion. 2014;8(6):381-6. doi: 10.4161/19336896.2014.983746.
- Schriefer et al., Low pressure DNA shearing: a method for random DNA sequence analysis. Nucleic Acids Res. Dec. 25, 1990;18(24):7455-6.
- Schultz et al., Expression and secretion in yeast of a 400-kDa envelope glycoprotein derived from Epstein-Barr virus. Gene. 1987;54(1):113-23. doi: 10.1016/0378-1119(87)90353-2.
- Schultz et al., Oligo-2'-fluoro-2'-deoxynucleotide N3' → P5' phosphoramidates: synthesis and properties. Nucleic Acids Res. Aug. 1, 1996;24(15):2966-73.
- Schwank et al., Functional repair of CFTR by CRISPR/Cas9 in intestinal stem cell organoids of cystic fibrosis patients. Cell Stem Cell. Dec. 5, 2013;13(6):653-8. doi:10.1016/j.stem.2013.11.002.
- Schwartz et al., Post-translational enzyme activation in an animal via optimized conditional protein splicing. Nat Chem Biol. Jan. 2007;3(1):50-4. Epub Nov. 26, 2006.
- Schwarze et al., In vivo protein transduction: delivery of a biologically active protein into the mouse. Science. Sep. 3, 1999;285(5433):1569-72.
- Scimmenti et al., Directed evolution of a recombinase for improved genomic integration at a native human sequence. Nucleic Acids Res. Dec. 15, 2001;29(24):5044-51.
- SCORE Results for Lueticken et al., Complete genome sequence of a *Streptococcus dysgalactiae* subsp. RT equisimilis strain possessing Lancefield's group A antigen. RL Submitted to the EMBL/GenBank/DBJ databases. May 2012. 3 pages.
- SCORE Results for Okumura et al., Evolutionary paths of streptococcal and staphylococcal superantigens. RL BMC Genomics. 2012;13:404-404. 3 pages.
- Scott et al., Production of cyclic peptides and proteins in vivo. Proc Natl Acad Sci U S A. Nov. 23, 1999;96(24):13638-43. doi: 10.1073/pnas.96.24.13638.
- Sebastián-Martín et al., Transcriptional inaccuracy threshold attenuates differences in RNA-dependent DNA synthesis fidelity between retroviral reverse transcriptases. Sci Rep. Jan. 12, 2018;8(1):627. doi: 10.1038/s41598-017-18974-8.
- Seed, An LFA-3 cDNA encodes a phospholipid-linked membrane protein homologous to its receptor CD2. Nature. Oct. 29-Nov. 4, 1987;329(6142):840-2. doi: 10.1038/329840a0.
- Sefton et al., Implantable pumps. Crit Rev Biomed Eng. 1987;14(3):201-40.
- Segal et al., Toward controlling gene expression at will: selection and design of zinc finger domains recognizing each of the 5'-GNN-3' DNA target sequences. Proc Natl Acad Sci U S A. Mar. 16, 1999;96(6):2758-63.
- Sells et al., Delivery of protein into cells using polycationic liposomes. Biotechniques. Jul. 1995;19(1):72-6, 78.
- Semenova et al., Interference by clustered regularly interspaced short palindromic repeat (CRISPR) RNA is governed by a seed sequence. Proc Natl Acad Sci U S A. Jun. 21, 2011;108(25):10098-103. doi: 10.1073/pnas.1104144108. Epub Jun. 6, 2011.
- Semple et al., Rational design of cationic lipids for siRNA delivery. Nat Biotechnol. Feb. 2010;28(2):172-6. doi: 10.1038/nbt.1602. Epub Jan. 17, 2010.
- Serganov et al., Coenzyme recognition and gene regulation by a flavin mononucleotide riboswitch. Nature. Mar. 12, 2009;458(7235):233-7. doi: 10.1038/nature07642. Epub Jan. 25, 2009.
- Serganov et al., Structural basis for discriminative regulation of gene expression by adenine- and guanine-sensing mRNAs. Chem Biol. Dec. 2004;11(12):1729-41.
- Serganov et al., Structural basis for gene regulation by a thiamine pyrophosphate-sensing riboswitch. Nature. Jun. 2, 2006;441(7097):1167-71. Epub May 21, 2006.
- Seripa et al., The missing ApoE allele. Ann Hum Genet. Jul. 2007;71(Pt 4):496-500. Epub Jan. 22, 2007.
- Serrano-Heras et al., Protein p56 from the *Bacillus subtilis* phage phi29 inhibits DNA-binding ability of uracil-DNA glycosylase. Nucleic Acids Res. 2007;35(16):5393-401. Epub Aug. 13, 2007.
- Setten et al., The current state and future directions of RNAi-based therapeutics. Nat Rev Drug Discov. Jun. 2019;18(6):421-446. doi: 10.1038/s41573-019-0017-4.
- Severinov et al., Expressed protein ligation, a novel method for studying protein-protein interactions in transcription. J Biol Chem. Jun. 26, 1998;273(26):16205-9. doi: 10.1074/jbc.273.26.16205.
- Sha et al., Monobodies and other synthetic binding proteins for expanding protein science. Protein Sci. May 2017;26(5):910-924. doi: 10.1002/pro.3148. Epub Mar. 24, 2017.
- Shah et al., Inteins: nature's gift to protein chemists. Chem Sci. 2014;5(1):446-461.
- Shah et al., Kinetic control of one-pot trans-splicing reactions by using a wild-type and designed split intein. Angew Chem Int Ed Engl. Jul. 11, 2011;50(29):6511-5. doi: 10.1002/anie.201102909. Epub Jun. 8, 2011.
- Shah et al., Protospacer recognition motifs: mixed identities and functional diversity. RNA Biol. May 2013;10(5):891-9. doi: 10.4161/rna.23764. Epub Feb. 12, 2013.
- Shah et al., Target-specific variants of FLP recombinase mediate genome engineering reactions in mammalian cells. FEBS J. Sep. 2015;282(17):3323-33. doi: 10.1111/febs.13345. Epub Jul. 1, 2015.
- Shaikh et al., Chimeras of the FLP and Cre recombinases: tests of the mode of cleavage by FLP and Cre. J Mol Biol. Sep. 8, 2000;302(1):27-48.
- Shalem et al., Genome-scale CRISPR-Cas9 knockout screening in human cells. Science. Jan. 3, 2014;343(6166):84-7. doi: 10.1126/science.1247005. Epub Dec. 12, 2013.
- Sharbeen et al., Ectopic restriction of DNA repair reveals that UNG2 excises AID-induced uracils predominantly or exclusively during G1 phase. J Exp Med. May 7, 2012;209(5):965-74. doi: 10.1084/jem.20112379. Epub Apr. 23, 2012.

(56)

References Cited

OTHER PUBLICATIONS

- Sharer et al., The ARF-like 2 (ARL2)-binding protein, BART. Purification, cloning, and initial characterization. *J Biol Chem.* Sep. 24, 1999;274(39):27553-61. doi: 10.1074/jbc.274.39.27553.
- Sharma et al., Efficient introduction of aryl bromide functionality into proteins in vivo. *FEBS Lett.* Feb. 4, 2000;467(1):37-40.
- Sharma et al., Identification of novel methyltransferases, Bmt5 and Bmt6, responsible for the m3U methylations of 25S rRNA in *Saccharomyces cerevisiae*. *Nucleic Acids Res.* Mar. 2014;42(5):3246-60. doi: 10.1093/nar/gkt1281. Epub Dec. 11, 2013.
- Sharon et al., Functional Genetic Variants Revealed by Massively Parallel Precise Genome Editing. *Cell.* Oct. 4, 2018;175(2):544-557.e16. doi: 10.1016/j.cell.2018.08.057. Epub Sep. 2, 2018.
- Shaw et al., Implications of human genome architecture for rearrangement-based disorders: the genomic basis of disease. *Hum Mol Genet.* Apr. 1, 2004;13 Spec No. 1:R57-64. doi: 10.1093/hmg/ddh073. Epub Feb. 5, 2004.
- Shcherbakova et al., Near-infrared fluorescent proteins for multicolor in vivo imaging. *Nat Methods.* Aug. 2013;10(8):751-4. doi: 10.1038/nmeth.2521. Epub Jun. 16, 2013.
- Shechner et al., Multiplexable, locus-specific targeting of long RNAs with CRISPR-Display. *Nat Methods.* Jul. 2015;12(7):664-70. doi: 10.1038/nmeth.3433. Epub Jun. 1, 2015.
- Shee et al., Engineered proteins detect spontaneous DNA breakage in human and bacterial cells. *Elife.* Oct. 29, 2013;2:e01222. doi: 10.7554/eLife.01222.
- Shen et al., Herpes simplex virus 1 (HSV-1) for cancer treatment. *Cancer Gene Ther.* Nov. 2006;13(11):975-92. doi: 10.1038/sj.cgt.7700946. Epub Apr. 7, 2006.
- Shen, Data processing, Modeling and Analysis scripts for CRISPR-inDelphi. GitHub—maxwshen/indelfi-dataprocessinganalysis at 6b68e3cec73c9358fef6e5f178a935f3c2a4118f. Apr. 10, 2018. Retrieved online via <https://github.com/maxwshen/indelfi-dataprocessinganalysis/tree/6b68e3cec73c9358fef6e5f178a935f3c2a4118f> Last retrieved on Jul. 26, 2021. 2 pages.
- Sheridan, First CRISPR-Cas patent opens race to stake out intellectual property. *Nat Biotechnol.* 2014;32(7):599-601.
- Sheridan, Gene therapy finds its niche. *Nat Biotechnol.* Feb. 2011;29(2):121-8. doi: 10.1038/nbt.1769.
- Shi et al., Structural basis for targeted DNA cytosine deamination and mutagenesis by APOBEC3A and APOBEC3B. *Nat Struct Mol Biol.* Feb. 2017;24(2):131-139. doi: 10.1038/nsmb.3344. Epub Dec. 19, 2016.
- Shi et al., YTHDF3 facilitates translation and decay of N6-methyladenosine-modified RNA. *Cell Res.* Mar. 2017;27(3):315-328. doi: 10.1038/cr.2017.15. Epub Jan. 20, 2017.
- Shimantani et al., Targeted base editing in rice and tomato using a CRISPR-Cas9 cytidine deaminase fusion. *Nat Biotechnol.* May 2017;35(5):441-443. doi: 10.1038/nbt.3833. Epub Mar. 27, 2017.
- Shimajima et al., Spinocerebellar ataxias type 27 derived from a disruption of the fibroblast growth factor 14 gene with mimicking phenotype of paroxysmal non-kinesigenic dyskinesia. *Brain Dev.* Mar. 2012;34(3):230-3. doi: 10.1016/j.braindev.2011.04.014. Epub May 19, 2011.
- Shindo et al., A Comparison of Two Single-Stranded DNA Binding Models by Mutational Analysis of APOBEC3G. *Biology (Basel).* Aug. 2, 2012;1(2):260-76. doi: 10.3390/biology1020260.
- Shmakov et al., Discovery and Functional Characterization of Diverse Class 2 CRISPR Cas Systems. *Molecular Cell.* Nov. 2015;60(3):385-97.
- Shmakov et al., Diversity and evolution of class 2 CRISPR-Cas systems. *Nat Rev Microbiol.* Mar. 2017;15(3):169-182. doi: 10.1038/nrmicro.2016.184. Epub Jan. 23, 2017.
- Shultz et al., A genome-wide analysis of FRT-like sequences in the human genome. *PLoS One.* Mar. 23, 2011;6(3):e18077. doi: 10.1371/journal.pone.0018077.
- Siebert et al., An improved PCR method for walking in uncloned genomic DNA. *Nucleic Acids Res.* Mar. 25, 1995;23(6):1087-8.
- Silas et al., Direct CRISPR spacer acquisition from RNA by a natural reverse transcriptase-Cas1 fusion protein. *Science.* Feb. 26, 2016;351(6276):aad4234. doi: 10.1126/science.aad4234.
- Silva et al., Selective disruption of the DNA polymerase III α - β complex by the umuD gene products. *Nucleic Acids Res.* Jul. 2012;40(12):5511-22. doi: 10.1093/nar/gks229. Epub Mar. 9, 2012.
- Simonelli et al., Base excision repair intermediates are mutagenic in mammalian cells. *Nucleic Acids Res.* Aug. 2, 2005;33(14):4404-11. Print 2005.
- Singh et al., Cross-talk between diverse serine integrases. *J Mol Biol.* Jan. 23, 2014;426(2):318-31. doi: 10.1016/j.jmb.2013.10.013. Epub Oct. 22, 2013.
- Singh et al., Real-time observation of DNA recognition and rejection by the RNA-guided endonuclease Cas9. *Nat Commun.* Sep. 14, 2016;7:12778. doi: 10.1038/ncomms12778.
- Singh et al., Real-time observation of DNA target interrogation and product release by the RNA-guided endonuclease CRISPR Cpf1 (Cas12a). *Proc Natl Acad Sci U S A.* May 22, 2018;115(21):5444-5449. doi: 10.1073/pnas.1718686115. Epub May 7, 2018.
- Sirk et al., Expanding the zinc-finger recombinase repertoire: directed evolution and mutational analysis of serine recombinase specificity determinants. *Nucleic Acids Res.* Apr. 2014;42(7):4755-66. doi: 10.1093/nar/gkt1389. Epub Jan. 21, 2014.
- Siu et al., Riboregulated toehold-gated gRNA for programmable CRISPR-Cas9 function. *Nat Chem Biol.* Mar. 2019;15(3):217-220. doi: 10.1038/s41589-018-0186-1. Epub Dec. 10, 2018.
- Sivalingam et al., Biosafety assessment of site-directed transgene integration in human umbilical cord-lining cells. *Mol Ther.* Jul. 2010;18(7):1346-56. doi: 10.1038/mt.2010.61. Epub Apr. 27, 2010.
- Sjoberg et al., The consensus coding sequences of human breast and colorectal cancers. *Science.* Oct. 13, 2006;314(5797):268-74. Epub Sep. 7, 2006.
- Skretas et al., Regulation of protein activity with small-molecule-controlled inteins. *Protein Sci.* Feb. 2005;14(2):523-32. Epub Jan. 4, 2005.
- Sledz et al., Structural insights into the molecular mechanism of the m(6)A writer complex. *Elife.* Sep. 14, 2016;5:e18434. doi: 10.7554/eLife.18434.
- Slupphaug et al., A nucleotide-flipping mechanism from the structure of human uracil-DNA glycosylase bound to DNA. *Nature.* Nov. 7, 1996;384(6604):87-92. doi: 10.1038/384087a0.
- Smargon et al., Cas13b Is a Type VI-B CRISPR-Associated RNA-Guided RNase Differentially Regulated by Accessory Proteins Csx27 and Csx28. *Mol Cell.* Feb. 16, 2017;65(4):618-630.e7. doi: 10.1016/j.molcel.2016.12.023. Epub Jan. 5, 2017.
- Smith et al., Diversity in the serine recombinases. *Mol Microbiol.* Apr. 2002;44(2):299-307. Review.
- Smith et al., Expression of a dominant negative retinoic acid receptor γ in *Xenopus* embryos leads to partial resistance to retinoic acid. *Roux Arch Dev Biol.* Mar. 1994;203(5):254-265. doi: 10.1007/BF00360521.
- Smith et al., Herpesvirus transport to the nervous system and back again. *Annu Rev Microbiol.* 2012;66:153-76. doi: 10.1146/annurev-micro-092611-150051. Epub Jun. 15, 2012.
- Smith et al., Production of human beta interferon in insect cells infected with a baculovirus expression vector. *Mol Cell Biol.* Dec. 1983;3(12):2156-65. doi: 10.1128/mcb.3.12.2156.
- Smith et al., Single-step purification of polypeptides expressed in *Escherichia coli* as fusions with glutathione S-transferase. *Gene.* Jul. 15, 1988;67(1):31-40. doi: 10.1016/0378-1119(88)90005-4.
- Smith, Filamentous fusion phage: novel expression vectors that display cloned antigens on the virion surface. *Science.* Jun. 14, 1985;228(4705):1315-7.
- Smith, Phage-encoded Serine Integrases and Other Large Serine Recombinases. *Microbiol Spectr.* Aug. 2015;3(4). doi: 10.1128/microbiolspec.MDNA3-0059-2014.
- Somanathan et al., AAV vectors expressing LDLR gain-of-function variants demonstrate increased efficacy in mouse models of familial hypercholesterolemia. *Circ Res.* Aug. 29, 2014;115(6):591-9. doi: 10.1161/CIRCRESAHA.115.304008. Epub Jul. 14, 2014.
- Sommerfelt et al., Receptor interference groups of 20 retroviruses plating on human cells. *Virology.* May 1990;176(1):58-69. doi: 10.1016/0042-6822(90)90230-o.

(56)

References Cited

OTHER PUBLICATIONS

- Song et al., Delivery of CRISPR/Cas systems for cancer gene therapy and immunotherapy. *Adv Drug Deliv Rev.* Jan. 2021; 168:158-180. doi: 10.1016/j.addr.2020.04.010. Epub May 1, 2020.
- Southworth et al., Control of protein splicing by intein fragment reassembly. *EMBO J.* Feb. 16, 1998;17(4):918-26. doi: 10.1093/emboj/17.4.918.
- Southworth et al., Purification of proteins fused to either the amino or carboxy terminus of the *Mycobacterium xenopi* gyrase A intein. *Biotechniques.* Jul. 1999;27(1):110-4, 116, 118-20. doi: 10.2144/99271st04.
- Spencer et al., A general strategy for producing conditional alleles of Src-like tyrosine kinases. *Proc Natl Acad Sci U S A.* Oct. 10, 1995;92(21):9805-9. doi: 10.1073/pnas.92.21.9805.
- Spencer et al., Controlling signal transduction with synthetic ligands. *Science.* Nov. 12, 1993;262(5136):1019-24. doi: 10.1126/science.7694365.
- Spencer et al., Functional analysis of Fas signaling in vivo using synthetic inducers of dimerization. *Curr Biol.* Jul. 1, 1996;6(7):839-47. doi: 10.1016/s0960-9822(02)00607-3.
- Srivastava et al., An inhibitor of nonhomologous end-joining abrogates double-strand break repair and impedes cancer progression. *Cell.* Dec. 21, 2012;151(7):1474-87. doi: 10.1016/j.cell.2012.11.054.
- Stadtman, Selenocysteine. *Annu Rev Biochem.* 1996;65:83-100.
- Stamos et al., Structure of a Thermostable Group II Intron Reverse Transcriptase with Template-Primer and Its Functional and Evolutionary Implications. *Mol Cell.* Dec. 7, 2017;68(5):926-939.e4. doi: 10.1016/j.molcel.2017.10.024. Epub Nov. 16, 2017.
- Steele et al., The prion protein knockout mouse: a phenotype under challenge. *Prion.* Apr.-Jun. 2007;1(2):83-93. doi: 10.4161/pr.1.2.4346. Epub Apr. 25, 2007.
- Steiner et al., The neurotropic herpes viruses: herpes simplex and varicella-zoster. *Lancet Neurol.* Nov. 2007;6(11):1015-28. doi: 10.1016/S1474-4422(07)70267-3.
- Stella et al., Structure of the Cpf1 endonuclease R-loop complex after target DNA cleavage. *Nature.* Jun. 22, 2017;546(7659):559-563. doi: 10.1038/nature22398. Epub May 31, 2017.
- Stenglein et al., APOBEC3 proteins mediate the clearance of foreign DNA from human cells. *Nat Struct Mol Biol.* Feb. 2010;17(2):222-9. doi: 10.1038/nsmb.1744. Epub Jan. 10, 2010.
- Stenson et al., Human Gene Mutation Database: towards a comprehensive central mutation database. *J Med Genet.* Feb. 2008;45(2):124-6. doi:10.1136/jmg.2007.055210.
- Stenson et al., The Human Gene Mutation Database: towards a comprehensive repository of inherited mutation data for medical research, genetic diagnosis and next-generation sequencing studies. *Hum Genet.* Jun. 2017;136(6):665-677. doi: 10.1007/s00439-017-1779-6. Epub Mar. 27, 2017.
- Stephens et al., The landscape of cancer genes and mutational processes in breast cancer. *Nature.* Jun. 2012;486:400-404. doi: 10.1038/nature11017.
- Sternberg et al., Conformational control of DNA target cleavage by CRISPR-Cas9. *Nature.* Nov. 5, 2015;527(7576):110-3. doi: 10.1038/nature15544. Epub Oct. 28, 2015.
- Sternberg et al., DNA interrogation by the CRISPR RNA-guided endonuclease Cas9. *Nature.* Mar. 6, 2014;507(7490):62-7. doi: 10.1038/nature13011. Epub Jan. 29, 2014.
- Stevens et al., A promiscuous split intein with expanded protein engineering applications. *Proc Natl Acad Sci U S A.* Aug. 8, 2017;114(32):8538-8543. doi: 10.1073/pnas.1701083114. Epub Jul. 24, 2017.
- Stevens et al., Design of a Split Intein with Exceptional Protein-Splicing Activity. *J Am Chem Soc.* Feb. 24, 2016;138(7):2162-5. doi: 10.1021/jacs.5b13528. Epub Feb. 8, 2016.
- Stockwell et al., Probing the role of homomeric and heteromeric receptor interactions in TGF-beta signaling using small molecule dimerizers. *Curr Biol.* Jun. 18, 1998;8(13):761-70. doi: 10.1016/s0960-9822(98)70299-4.
- Strecker et al., Engineering of CRISPR-Cas12b for human genome editing. *Nat Commun.* Jan. 22, 2019;10(1):212. doi: 10.1038/s41467-018-08224-4.
- Strecker et al., RNA-guided DNA insertion with CRISPR-associated transposases. *Science.* Jul. 5, 2019;365(6448):48-53. doi: 10.1126/science.aax9181. Epub Jun. 6, 2019.
- Strutt et al., RNA-dependent RNA targeting by CRISPR-Cas9. *Elife.* Jan. 5, 2018;7:e32724. doi: 10.7554/eLife.32724.
- Sudarsan et al., An mRNA structure in bacteria that controls gene expression by binding lysine. *Genes Dev.* Nov. 1, 2003;17(21):2688-97.
- Sudarsan et al., Riboswitches in eubacteria sense the second messenger cyclic di-GMP. *Science.* Jul. 18, 2008;321(5887):411-3. doi: 10.1126/science.1159519.
- Suess et al., A theophylline responsive riboswitch based on helix slipping controls gene expression in vivo. *Nucleic Acids Res.* Mar. 5, 2004;32(4):1610-4.
- Suh et al., Restoration of visual function in adult mice with an inherited retinal disease via adenine base editing. *Nat Biomed Eng.* Feb. 2021;5(2):169-178. doi: 10.1038/s41551-020-00632-6. Epub Oct. 19, 2020.
- Sullenger et al., Ribozyme-mediated repair of defective mRNA by targeted, trans-splicing. *Nature.* Oct. 13, 1994;371(6498):619-22. doi: 10.1038/371619a0.
- Sun et al., Optimized TAL effector nucleases (TALENs) for use in treatment of sickle cell disease. *Mol Biosyst.* Apr. 2012;8(4):1255-63. doi: 10.1039/c2mb05461b. Epub Feb. 3, 2012.
- Sun et al., The CRISPR/Cas9 system for gene editing and its potential application in pain research. *Transl Periop & Pain Med.* Aug. 3, 2016;1(3):22-33.
- Suzuki et al., Crystal structures reveal an elusive functional domain of pyrrolysyl-tRNA synthetase. *Nat Chem Biol.* Dec. 2017;13(12):1261-1266. doi: 10.1038/nchembio.2497. Epub Oct. 16, 2017.
- Suzuki et al., VCre/VloxP and SCre/SloxP: new site-specific recombination systems for genome engineering. *Nucleic Acids Res.* Apr. 2011;39(8):e49. doi: 10.1093/nar/gkq1280. Epub Feb. 1, 2011.
- Swarts et al., Argonaute of the archaeon *Pyrococcus furiosus* is a DNA-guided nuclease that targets cognate DNA. *Nucleic Acids Res.* May 26, 2015;43(10):5120-9. doi: 10.1093/nar/gkv415. Epub Apr. 29, 2015.
- Swarts et al., DNA-guided DNA interference by a prokaryotic Argonaute. *Nature.* Mar. 13, 2014;507(7491):258-61. doi: 10.1038/nature12971. Epub Feb. 16, 2014.
- Swarts et al., The evolutionary journey of Argonaute proteins. *Nat Struct Mol Biol.* Sep. 2014;21(9):743-53. doi: 10.1038/nsmb.2879.
- Szczepiek et al., Structure-based redesign of the dimerization interface reduces the toxicity of zinc-finger nucleases. *Nat Biotechnol.* Jul. 2007;25(7):786-93. Epub Jul. 1, 2007.
- Tagalakis et al., Lack of RNA-DNA oligonucleotide (chimeraplast) mutagenic activity in mouse embryos. *Mol Reprod Dev.* Jun. 2005;71(2):140-4.
- Tahara et al., Potent and Selective Inhibitors of 8-Oxoguanine DNA Glycosylase. *J Am Chem Soc.* Feb. 14, 2018;140(6):2105-2114. doi: 10.1021/jacs.7b09316. Epub Feb. 5, 2018.
- Tajiri et al., Functional cooperation of MutT, MutM and MutY proteins in preventing mutations caused by spontaneous oxidation of guanine nucleotide in *Escherichia coli*. *Mutat Res.* May 1995;336(3):257-67. doi: 10.1016/0921-8777(94)00062-b.
- Takimoto et al., Stereochemical basis for engineered pyrrolysyl-tRNA synthetase and the efficient in vivo incorporation of structurally divergent non-native amino acids. *ACS Chem Biol.* Jul. 15, 2011;6(7):733-43. doi: 10.1021/cb200057a. Epub May 5, 2011.
- Tambunan et al., Vaccine Design for H5N1 Based on B- and T-cell Epitope Predictions. *Bioinform Biol Insights.* Apr. 28, 2016;10:27-35. doi: 10.4137/BBI.S38378.
- Tanenbaum et al., A protein-tagging system for signal amplification in gene expression and fluorescence imaging. *Cell.* Oct. 23, 2014;159(3):635-46. doi: 10.1016/j.cell.2014.09.039. Epub Oct. 9, 2014.
- Tanese et al., Expression of enzymatically active reverse transcriptase in *Escherichia coli*. *Proc Natl Acad Sci U S A.* Aug. 1985;82(15):4944-8. doi: 10.1073/pnas.82.15.4944.

(56)

References Cited

OTHER PUBLICATIONS

- Tang et al., Aptazyme-embedded guide RNAs enable ligand-responsive genome editing and transcriptional activation. *Nat Commun.* Jun. 28, 2017;8:15939. doi: 10.1038/ncomms15939.
- Tang et al., Rewritable multi-event analog recording in bacterial and mammalian cells. *Science.* Apr. 13, 2018;360(6385):eaap8992. doi: 10.1126/science.aap8992. Epub Feb. 15, 2018.
- Tassabehji, Williams-Beuren syndrome: a challenge for genotype-phenotype correlations. *Hum Mol Genet.* Oct. 15, 2003;12 Spec No. 2:R229-37. doi: 10.1093/hmg/ddg299. Epub Sep. 2, 2003.
- Taube et al., Reverse transcriptase of mouse mammary tumour virus: expression in bacteria, purification and biochemical characterization. *Biochem J.* Feb. 1, 1998;329 (Pt 3):579-87. doi: 10.1042/bj3290579. Erratum in: *Biochem J.* Jun. 15, 1998;332(Pt 3):808.
- Tebas et al., Gene editing of CCR5 in autologous CD4 T cells of persons infected with HIV. *Engl J Med.* Mar. 6, 2014;370(10):901-10. doi: 10.1056/NEJMoa1300662.
- Tee et al., Polishing the craft of genetic diversity creation in directed evolution. *Biotechnol Adv.* Dec. 2013;31(8):1707-21. doi: 10.1016/j.biotechadv.2013.08.021. Epub Sep. 6, 2013.
- Telenti et al., The *Mycobacterium xenopi* GyrA protein splicing element: characterization of a minimal intein. *J Bacteriol.* Oct. 1997;179(20):6378-82. doi: 10.1128/jb.179.20.6378-6382.1997.
- Telesnitsky et al., RNase H domain mutations affect the interaction between Moloney murine leukemia virus reverse transcriptase and its primer-template. *Proc Natl Acad Sci U S A.* Feb. 15, 1993;90(4):1276-80. doi: 10.1073/pnas.90.4.1276.
- Teng et al., Mutational analysis of apolipoprotein B mRNA editing enzyme (APOBEC1): structure-function relationships of RNA editing and dimerization. *J Lipid Res.* Apr. 1999;40(4):623-35.
- Tessarollo et al., Targeted mutation in the neurotrophin-3 gene results in loss of muscle sensory neurons. *Proc Natl Acad Sci U S A.* Dec. 6, 1994;91(25):11844-8.
- Tesson et al., Knockout rats generated by embryo microinjection of TALENs. *Nat Biotechnol.* Aug. 5, 2011;29(8):695-6. doi: 10.1038/nbt.1940.
- Thompson et al., Cellular uptake mechanisms and endosomal trafficking of supercharged proteins. *Chem Biol.* Jul. 27, 2012;19(7):831-43. doi: 10.1016/j.chembiol.2012.06.014.
- Thompson et al., Engineering and identifying supercharged proteins for macromolecule delivery into mammalian cells. *Methods Enzymol.* 2012;503:293-319. doi: 10.1016/B978-0-12396962-0.00012-4.
- Thomson et al., Mutational analysis of loxP sites for efficient Cre-mediated insertion into genomic DNA. *Genesis.* Jul. 2003;36(3):162-7. doi: 10.1002/gene.10211.
- Thorpe et al., Functional correction of episomal mutations with short DNA fragments and RNA-DNA oligonucleotides. *J Gene Med.* Mar. 2002-Apr;4(2):195-204.
- Thuronyi et al., Continuous evolution of base editors with expanded target compatibility and improved activity. *Nat Biotechnol.* Sep. 2019;37(9):1070-1079. doi: 10.1038/s41587-019-0193-0. Epub Jul. 22, 2019.
- Thyagarajan et al., Creation of engineered human embryonic stem cell lines using phiC31 integrase. *Stem Cells.* Jan. 2008;26(1):119-26. doi: 10.1634/stemcells.2007-0283. Epub Oct. 25, 2007.
- Thyagarajan et al., Mammalian genomes contain active recombinase recognition sites. *Gene.* Feb. 22, 2000;244(1-2):47-54.
- Thyagarajan et al., Site-specific genomic integration in mammalian cells mediated by phage phiC31 integrase. *Mol Cell Biol.* Jun. 2001;21(12):3926-34.
- Tinland et al., The T-DNA-linked VirD2 protein contains two distinct functional nuclear localization signals. *Proc Natl Acad Sci U S A.* Aug. 15, 1992;89(16):7442-6. doi: 10.1073/pnas.89.16.7442.
- Tirumalai et al., Recognition of core-type DNA sites by lambda integrase. *J Mol Biol.* Jun. 12, 1998;279(3):513-27.
- Tom et al., Mechanism whereby proliferating cell nuclear antigen stimulates flap endonuclease 1. *J Biol Chem.* Apr. 7, 2000;275(14):10498-505. doi: 10.1074/jbc.275.14.10498.
- Tone et al., Single-stranded DNA binding protein Gp5 of *Bacillus subtilis* phage ϕ 29 is required for viral DNA replication in growth-temperature dependent fashion. *Biosci Biotechnol Biochem.* 2012;76(12):2351-3. doi: 10.1271/bbb.120587. Epub Dec. 7, 2012.
- Toor et al., Crystal structure of a self-spliced group II intron. *Science.* Apr. 4, 2008;320(5872):77-82. doi: 10.1126/science.1153803.
- Toro et al., On the Origin and Evolutionary Relationships of the Reverse Transcriptases Associated With Type III CRISPR-Cas Systems. *Front Microbiol.* Jun. 15, 2018;9:1317. doi: 10.3389/fmicb.2018.01317.
- Toro et al., The Reverse Transcriptases Associated with CRISPR-Cas Systems. *Sci Rep.* Aug. 2, 2017;7(1):7089. doi: 10.1038/s41598-017-07828-y.
- Torres et al., Non-integrative lentivirus drives high-frequency Cre-mediated cassette exchange in human cells. *PLoS One.* 2011;6(5):e19794. doi: 10.1371/journal.pone.0019794. Epub May 23, 2011.
- Tourdot et al., A general strategy to enhance immunogenicity of low-affinity HLA-A2.1-associated peptides: implication in the identification of cryptic tumor epitopes. *Eur J Immunol.* Dec. 2000;30(12):3411-21.
- Townsend et al., Role of HFE in iron metabolism, hereditary haemochromatosis, anaemia of chronic disease, and secondary iron overload. *Lancet.* Mar. 2, 2002;359(9308):786-90. doi: 10.1016/S0140-6736(02)07885-6.
- Tracewell et al., Directed enzyme evolution: climbing fitness peaks one amino acid at a time. *Curr Opin Chem Biol.* Feb. 2009;13(1):3-9. doi: 10.1016/j.cbpa.2009.01.017. Epub Feb. 25, 2009.
- Tratschin et al., A human parvovirus, adeno-associated virus, as a eucaryotic vector: transient expression and encapsidation of the procaryotic gene for chloramphenicol acetyltransferase. *Mol Cell Biol.* Oct. 1984;4(10):2072-81. doi: 10.1128/mcb.4.10.2072.
- Tratschin et al., Adeno-associated virus vector for high-frequency integration, expression, and rescue of genes in mammalian cells. *Mol Cell Biol.* Nov. 1985;5(11):3251-60. doi: 10.1128/mcb.5.11.3251.
- Trausch et al., The structure of a tetrahydrofolate-sensing riboswitch reveals two ligand binding sites in a single aptamer. *Structure.* Oct. 12, 2011;19(10):1413-23. doi: 10.1016/j.str.2011.06.019. Epub Sep. 8, 2011.
- Traxler et al., A genome-editing strategy to treat β -hemoglobinopathies that recapitulates a mutation associated with a benign genetic condition. *Nat Med.* Sep. 2016;22(9):987-90. doi: 10.1038/nm.4170. Epub Aug. 15, 2016.
- Trudeau et al., On the Potential Origins of the High Stability of Reconstructed Ancestral Proteins. *Mol Biol Evol.* Oct. 2016;33(10):2633-41. doi: 10.1093/molbev/msw138. Epub Jul. 12, 2016.
- Truong et al., Development of an intein-mediated split-Cas9 system for gene therapy. *Nucleic Acids Res.* Jul. 27, 2015;43(13):6450-8. doi: 10.1093/nar/gkv601. Epub Jun. 16, 2015. With Supplementary Data.
- Tsai et al., CIRCLE-seq: a highly sensitive in vitro screen for genome-wide CRISPR-Cas9 nuclease off-targets. *Nat Methods.* Jun. 2017;14(6):607-614. doi: 10.1038/nmeth.4278. Epub May 1, 2017.
- Tsai et al., Dimeric CRISPR RNA-guided FokI nucleases for highly specific genome editing. *Nat Biotechnol.* Jun. 2014;32(6):569-76. doi: 10.1038/nbt.2908. Epub Apr. 25, 2014.
- Tsai et al., GUIDE-seq enables genome-wide profiling of off-target cleavage by CRISPR-Cas nucleases. *Nat Biotechnol.* Feb. 2015;33(2):187-97. doi: 10.1038/nbt.3117. Epub Dec. 16, 2014.
- Tsang et al., Specialization of the DNA-cleaving activity of a group I ribozyme through in vitro evolution. *J Mol Biol.* Sep. 13, 1996;262(1):31-42. doi: 10.1006/jmbi.1996.0496.
- Tsutakawa et al., Human flap endonuclease structures, DNA double-base flipping, and a unified understanding of the FEN1 superfamily. *Cell.* Apr. 15, 2011;145(2):198-211. doi: 10.1016/j.cell.2011.03.004.

(56)

References Cited

OTHER PUBLICATIONS

- Turan et al., Recombinase-mediated cassette exchange (RMCE)—a rapidly-expanding toolbox for targeted genomic modifications. *Gene*. Feb. 15, 2013;515(1):1-27. doi: 10.1016/j.gene.2012.11.016. Epub Nov. 29, 2012.
- Turan et al., Recombinase-mediated cassette exchange (RMCE): traditional concepts and current challenges. *J Mol Biol*. Mar. 25, 2011;407(2): 193-221. doi: 10.1016/j.jmb.2011.01.004. Epub Jan. 15, 2011.
- Turan et al., Site-specific recombinases: from tag-and-target- to tag-and-exchange-based genomic modifications. *FASEB J*. Dec. 2011;25(12):4088-107. doi: 10.1096/fj.11-186940. Epub Sep. 2, 2011. Review.
- Tycko et al., Pairwise library screen systematically interrogates *Staphylococcus aureus* Cas9 specificity in human cells. bioRxiv. doi: <https://doi.org/10.1101/269399> Posted Feb. 22, 2018.
- UNIPROT Submission; UniProt, Accession No. P01011. Last modified Jun. 11, 2014, version 2. 15 pages.
- UNIPROT Submission; UniProt, Accession No. P01011. Last modified Sep. 18, 2013, version 2. 15 pages.
- UNIPROT Submission; UniProt, Accession No. P04264. Last modified Jun. 11, 2014, version 6. 15 pages.
- UNIPROT Submission; UniProt, Accession No. P04275. Last modified Jul. 9, 2014, version 107. 29 pages.
- UniProt A0A1V6. Dec. 11, 2019.
- UNIPROT Submission; Accession No. F0NH53. May 3, 2011. 4 pages.
- UNIPROT Submission; Accession No. F0NN87. May 3, 2011. 4 pages.
- UNIPROT Submission; Accession No. G3ECR1.2. No Author Listed., Aug. 12, 2020, 8 pages.
- UNIPROT Submission; Accession No. P0DOC6. No Author Listed., Oct. 5, 2016. 5 pages.
- UNIPROT Submission; Accession No. T0D7A2. Oct. 16, 2013. 10 pages.
- Urnov et al., Genome editing with engineered zinc finger nucleases. *Nat Rev Genet*. Sep. 2010;11(9):636-46. doi: 10.1038/nrg2842.
- Urnov et al., Highly efficient endogenous human gene correction using designed zinc-finger nucleases. *Nature*. Jun. 2, 2005;435(7042):646-51. Epub Apr. 3, 2005.
- Usman et al., Exploiting the chemical synthesis of RNA. *Trends Biochem Sci*. Sep. 1992;17(9):334-9. doi: 10.1016/0968-0004(92)90306-t.
- Vagner et al., Efficiency of homologous DNA recombination varies along the *Bacillus subtilis* chromosome. *J Bacteriol*. Sep. 1988;170(9):3978-82.
- Van Brunt et al., Genetically Encoded Azide Containing Amino Acid in Mammalian Cells Enables Site-Specific Antibody-Drug Conjugates Using Click Cycloaddition Chemistry. *Bioconjug Chem*. Nov. 18, 2015;26(11):2249-60. doi: 10.1021/acs.bioconjchem.5b00359. Epub Sep. 11, 2015.
- Van Overbeek et al., DNA Repair Profiling Reveals Nonrandom Outcomes at Cas9-Mediated Breaks. *Mol Cell*. Aug. 18, 2016;63(4):633-646. doi: 10.1016/j.molcel.2016.06.037. Epub Aug. 4, 2016.
- Vanamee et al., FokI requires two specific DNA sites for cleavage. *J Mol Biol*. May 25, 2001;309(1):69-78.
- Varga et al., Progressive vascular smooth muscle cell defects in a mouse model of Hutchinson-Gilford progeria syndrome. *Proc Natl Acad Sci U S A*. Feb. 28, 2006;103(9):3250-5. doi: 10.1073/pnas.0600012103. Epub Feb. 21, 2006.
- Venken et al., Genome-wide manipulations of *Drosophila melanogaster* with transposons, FLP recombinase, and ΦC31 integrase. *Methods Mol Biol*. 2012;859:203-28. doi: 10.1007/978-1-61779-603-6_12.
- Verma, The reverse transcriptase. *Biochim Biophys Acta*. Mar. 21, 1977;473(1):1-38. doi: 10.1016/0304-419x(77)90005-1.
- Vigne et al., Third-generation adenovectors for gene therapy. *Restor Neurol Neurosci*. Jan. 1, 1995;8(1):35-6. doi: 10.3233/RNN-1995-81208.
- Vik et al., Endonuclease V cleaves at inosines in RNA. *Nat Commun*. 2013;4:2271. doi: 10.1038/ncomms3271.
- Vilenchik et al., Endogenous DNA double-strand breaks: production, fidelity of repair, and induction of cancer. *Proc Natl Acad Sci U S A*. Oct. 28, 2003;100(22):12871-6. doi: 10.1073/pnas.2135498100. Epub Oct. 17, 2003.
- Vitreschak et al., Regulation of the vitamin B12 metabolism and transport in bacteria by a conserved RNA structural element. *RNA*. Sep. 2003;9(9):1084-97.
- Voigt et al., Rational evolutionary design: the theory of in vitro protein evolution. *Adv Protein Chem*. 2000;55:79-160.
- Vriend et al., Nick-initiated homologous recombination: Protecting the genome, one strand at a time. *DNA Repair (Amst)*. Feb. 2017;50:1-13. doi: 10.1016/j.dnarep.2016.12.005. Epub Dec. 29, 2016.
- Wacey et al., Disentangling the perturbational effects of amino acid substitutions in the DNA-binding domain of p53. *Hum Genet*. Jan. 1999;104(1):15-22.
- Wadia et al., Modulation of cellular function by TAT mediated transduction of full length proteins. *Curr Protein Pept Sci*. Apr. 2003;4(2):97-104.
- Wadia et al., Transducible TAT-HA fusogenic peptide enhances escape of TAT-fusion proteins after lipid raft macropinocytosis. *Nat Med*. Mar. 2004;10(3):310-5. Epub Feb. 8, 2004.
- Wah et al., Structure of FokI has implications for DNA cleavage. *Proc Natl Acad Sci U S A*. Sep. 1, 1998;95(18):10564-9.
- Wals et al., Unnatural amino acid incorporation in *E. coli*: current and future applications in the design of therapeutic proteins. *Front Chem*. Apr. 1, 2014;2:15. doi: 10.3389/fchem.2014.00015. eCollection 2014.
- Wan et al., Material solutions for delivery of CRISPR/Cas-based genome editing tools: Current status and future outlook. *Materials Today*. Jun. 2019;26:40-66. doi: 10.1016/j.mattod.2018.12.003.
- Wang et al., CRISPR-Cas9 and CRISPR-Assisted Cytidine Deaminase Enable Precise and Efficient Genome Editing in *Klebsiella pneumoniae*. *Appl Environ Microbiol*. 2018;84(23):e01834-18. Published Nov. 15, 2018. doi:10.1128/AEM.01834-18.
- Wang et al., AID upmutants isolated using a high-throughput screen highlight the immunity/cancer balance limiting DNA deaminase activity. *Nat Struct Mol Biol*. Jul. 2009;16(7):769-76. doi: 10.1038/nsmb.1623. Epub Jun. 21, 2009.
- Wang et al., Continuous directed evolutions of proteins with improved soluble expression. *Nature Chemical Biology*. Nat Publishing Group. Aug. 20, 2018; 14(10):972-980.
- Wang et al., CRISPR-Cas9 Targeting of PCSK9 in Human Hepatocytes In Vivo—Brief Report. *Arterioscler Thromb Vasc Biol*. May 2016;36(5):783-6. doi: 10.1161/ATVBAHA.116.307227. Epub Mar. 3, 2016.
- Wang et al., Efficient delivery of genome-editing proteins using bioreducible lipid nanoparticles. *Proc Natl Acad Sci U S A*. Feb. 29, 2016. pii: 201520244. [Epub ahead of print].
- Wang et al., Enhanced base editing by co-expression of free uracil DNA glycosylase inhibitor. *Cell Res*. Oct. 2017;27(1):1289-92. doi: 10.1038/cr.2017.111. Epub Aug. 29, 2017.
- Wang et al., Evolution of new nonantibody proteins via iterative somatic hypermutation. *Proc Natl Acad Sci U S A*. Nov. 30, 2004;101(48):16745-9. Epub Nov. 19, 2004.
- Wang et al., Expanding the genetic code. *Annu Rev Biophys Biomol Struct*. 2006;35:225-49. Review.
- Wang et al., Genetic screens in human cells using the CRISPR-Cas9 system. *Science*. Jan. 3, 2014;343(6166):80-4. doi: 10.1126/science.1246981. Epub Dec. 12, 2013.
- Wang et al., Highly efficient CRISPR/HDR-mediated knock-in for mouse embryonic stem cells and zygotes. *Biotechniques*. 2015;59:201-2;204;206-8.
- Wang et al., (6-methyladenosine) Modulates Messenger RNA Translation Efficiency. *Cell*. Jun. 4, 2015;161(6):1388-99. doi: 10.1016/j.cell.2015.05.014.
- Wang et al., N6-methyladenosine-dependent regulation of messenger RNA stability. *Nature*. Jan. 2, 2014;505(7481):117-20. doi: 10.1038/nature12730. Epub Nov. 27, 2013.
- Wang et al., Nucleation, propagation and cleavage of target RNAs in Ago silencing complexes. *Nature*. Oct. 8, 2009;461(7265):754-61. doi: 10.1038/nature08434.

(56)

References Cited

OTHER PUBLICATIONS

- Wang et al., One-step generation of mice carrying mutations in multiple genes by CRISPR/Cas-mediated genome engineering. *Cell*. May 9, 2013;153(4):910-8. doi: 10.1016/j.cell.2013.04.025. Epub May 2, 2013.
- Wang et al., Optimized paired-sgRNA/Cas9 cloning and expression cassette triggers high-efficiency multiplex genome editing in kiwi-fruit. *Plant Biotechnol J*. Aug. 2018;16(8):1424-1433. doi: 10.1111/pbi.12884. Epub Feb. 6, 2018.
- Wang et al., Programming cells by multiplex genome engineering and accelerated evolution. *Nature*. Aug. 13, 2009;460(7257):894-8. Epub Jul. 26, 2009.
- Wang et al., Reading RNA methylation codes through methyl-specific binding proteins. *RNA Biol*. 2014;11(6):669-72. doi: 10.4161/rna.28829. Epub Apr. 24, 2014.
- Wang et al., Recombinase technology: applications and possibilities. *Plant Cell Rep*. Mar. 2011;30(3):267-85. doi: 10.1007/s00299-010-0938-1. Epub Oct. 24, 2010.
- Wang et al., Riboswitches that sense S-adenosylhomocysteine and activate genes involved in coenzyme recycling. *Mol Cell*. Mar. 28, 2008;29(6):691-702. doi: 10.1016/j.molcel.2008.01.012.
- Wang et al., *Staphylococcus aureus* protein SAUGI acts as a uracil-DNA glycosylase inhibitor. *Nucleic Acids Res*. Jan. 2014;42(2):1354-64. doi: 10.1093/nar/gkt964. Epub Oct. 22, 2013.
- Wang et al., Structural basis of (6)-adenosine methylation by the METTL3-METTL14 complex. *Nature*. Jun. 23, 2016;534(7608):575-8. doi: 10.1038/nature18298. Epub May 25, 2016.
- Wang et al., Targeted gene addition to a predetermined site in the human genome using a ZFN-based nicking enzyme. *Genome Res*. Jul. 2012;22(7):1316-26. doi: 10.1101/gr.122879.111. Epub Mar. 20, 2012.
- Wang et al., Uracil-DNA glycosylase inhibitor gene of bacteriophage PBS2 encodes a binding protein specific for uracil-DNA glycosylase. *J Biol Chem*. Jan. 15, 1989;264(2):1163-71.
- Warren et al., A chimeric Cre recombinase with regulated directionality. *Proc Natl Acad Sci USA*. Nov. 25, 2008;105(47):18278-83. doi: 10.1073/pnas.0809949105. Epub Nov. 14, 2008.
- Warren et al., Mutations in the amino-terminal domain of lambda-integrase have differential effects on integrative and excisive recombination. *Mol Microbiol*. Feb. 2005;55(4):1104-12.
- Watowich, The erythropoietin receptor: molecular structure and hematopoietic signaling pathways. *J Investig Med*. Oct. 2011;59(7):1067-72. doi: 10.2310/JIM.0b013e31820fb28c.
- Waxman et al., Regulating excitability of peripheral afferents: emerging ion channel targets. *Nat Neurosci*. Feb. 2014;17(2):153-63. doi: 10.1038/nn.3602. Epub Jan. 28, 2014.
- Weber et al., Assembly of designer TAL effectors by Golden Gate cloning. *PLoS One*. 2011;6(5):e19722. doi: 10.1371/journal.pone.0019722. Epub May 19, 2011.
- Weill et al., DNA polymerases in adaptive immunity. *Nat Rev Immunol*. Apr. 2008;8(4):302-12. doi: 10.1038/nri2281. Epub Mar. 14, 2008.
- Weinberg et al., New Classes of Self-Cleaving Ribozymes Revealed by Comparative Genomics Analysis. *Nat Chem Biol*. Aug. 2015;11(8):606-10. doi: 10.1038/nchembio.1846. Epub Jul. 13, 2015.
- Weinberg et al., The aptamer core of SAM-IV riboswitches mimics the ligand-binding site of SAM-I riboswitches. *RNA*. May 2008;14(5):822-8. doi: 10.1261/rna.988608. Epub Mar. 27, 2008.
- Weinberger et al., Disease-causing mutations C277R and C277Y modify gating of human CIC-1 chloride channels in myotonia congenita. *J Physiol*. Aug. 1, 2012;590(Pt 15):3449-64. doi: 10.1113/jphysiol.2012.232785. Epub May 28, 2012.
- Weinert et al., Unbiased detection of CRISPR off-targets in vivo using DISCOVER-Seq. *Science*. Apr. 19, 2019;364(6437):286-289. doi: 10.1126/science.aav9023. Epub Apr. 18, 2019.
- Weiss et al., Loss-of-function mutations in sodium channel Nav1.7 cause anosmia. *Nature*. Apr. 14, 2011;472(7342):186-90. doi: 10.1038/nature09975. Epub Mar. 23, 2011.
- Wen et al., Inclusion of a universal tetanus toxoid CD4(+) T cell epitope P2 significantly enhanced the immunogenicity of recombinant rotavirus ?VP8* subunit parenteral vaccines. *Vaccine*. Jul. 31, 2014;32(35):4420-4427. doi: 10.1016/j.vaccine.2014.06.060. Epub Jun. 21, 2014.
- West et al., Gene expression in adeno-associated virus vectors: the effects of chimeric mRNA structure, helper virus, and adenovirus VA1 RNA. *Virology*. Sep. 1987;160(1):38-47. doi: 10.1016/0042-6822(87)90041-9.
- Wharton et al., A new-specificity mutant of 434 repressor that defines an amino acid-base pair contact. *Nature*. Apr. 30-May 6, 1987;326(6116):888-91.
- Wharton et al., Changing the binding specificity of a repressor by redesigning an alpha-helix. *Nature*. Aug. 15-21, 1985;316(6029):601-5.
- Wheeler et al., The thermostability and specificity of ancient proteins. *Curr Opin Struct Biol*. Jun. 2016;38:37-43. doi: 10.1016/j.sbi.2016.05.015. Epub Jun. 9, 2016.
- Wiedenheft et al., RNA-guided genetic silencing systems in bacteria and archaea. *Nature*. Feb. 15, 2012;482(7385):331-8. doi: 10.1038/nature10886. Review.
- Wienert et al., KLF1 drives the expression of fetal hemoglobin in British HPFH. *Blood*. Aug. 10, 2017;130(6):803-807. doi: 10.1182/blood-2017-02-767400. Epub Jun. 28, 2017.
- Wijesinghe et al., Efficient deamination of 5-methylcytosines in DNA by human APOBEC3A, but not by AID or APOBEC3G. *Nucleic Acids Res*. Oct. 2012;40(18):9206-17. doi: 10.1093/nar/gks685. Epub Jul. 13, 2012.
- Wijnker et al., Managing meiotic recombination in plant breeding. *Trends Plant Sci*. Dec. 2008;13(12):640-6. doi: 10.1016/j.tplants.2008.09.004. Epub Oct. 22, 2008.
- Williams et al., Assessing the accuracy of ancestral protein reconstruction methods. *PLoS Comput Biol*. Jun. 23, 2006;2(6):e69. doi: 10.1371/journal.pcbi.0020069. Epub Jun. 23, 2006.
- Wilson et al., Assessing annotation transfer for genomics: quantifying the relations between protein sequence, structure and function through traditional and probabilistic scores. *J Mol Biol* 2000;297:233-49.
- Wilson et al., Formation of infectious hybrid virions with gibbon ape leukemia virus and human T-cell leukemia virus retroviral envelope glycoproteins and the gag and pol proteins of Moloney murine leukemia virus. *J Virol*. May 1989;63(5):2374-8. doi: 10.1128/JVI.63.5.2374-2378.1989.
- Wilson et al., In Vitro Selection of Functional Nucleic Acids. *Annu Rev Biochem*. 1999;68:611-47. doi: 10.1146/annurev.biochem.68.1.611.
- Wilson et al., Kinase dynamics. Using ancient protein kinases to unravel a modern cancer drug's mechanism. *Science*. Feb. 20, 2015;347(6224):882-6. doi: 10.1126/science.aaa1823.
- Wilson et al., Programmable m6A modification of cellular RNAs with a Cas13-directed methyltransferase. *Nat Biotechnol*. Dec. 2020;38(12):1431-1440. doi: 10.1038/s41587-020-0572-6. Epub Jun. 29, 2020.
- Winkler et al., An mRNA structure that controls gene expression by binding FMN. *Proc Natl Acad Sci U S A*. Dec. 10, 2002;99(25):15908-13. Epub Nov. 27, 2002.
- Winkler et al., Control of gene expression by a natural metabolite-responsive ribozyme. *Nature*. Mar. 18, 2004;428(6980):281-6.
- Winkler et al., Thiamine derivatives bind messenger RNAs directly to regulate bacterial gene expression. *Nature*. Oct. 31, 2002;419(6910):952-6. Epub Oct. 16, 2002.
- Winoto et al., A novel, inducible and T cell-specific enhancer located at the 3' end of the T cell receptor alpha locus. *EMBO J*. Mar. 1989;8(3):729-33.
- Winter et al., Drug Development. Phthalimide conjugation as a strategy for in vivo target protein degradation. *Science*. Jun. 19, 2015;348(6241):1376-81. doi: 10.1126/science.aab1433. Epub May 21, 2015.
- Winter et al., Targeted exon skipping with AAV-mediated split adenine base editors. *Cell Discov*. Aug. 20, 2019;5:41. doi: 10.1038/s41421-019-0109-7.

(56)

References Cited

OTHER PUBLICATIONS

- Wold, Replication protein A: a heterotrimeric, single-stranded DNA-binding protein required for eukaryotic DNA metabolism. *Annu Rev Biochem.* 1997;66:61-92. doi: 10.1146/annurev.biochem.66.1.61.
- Wolf et al., tadA, an essential tRNA-specific adenosine deaminase from *Escherichia coli*. *EMBO J.* Jul. 15, 2002;21(14):3841-51.
- Wolfe et al., Analysis of zinc fingers optimized via phage display: evaluating the utility of a recognition code. *J Mol Biol.* Feb. 5, 1999;285(5):1917-34.
- Wong et al., A statistical analysis of random mutagenesis methods used for directed protein evolution. *J Mol Biol.* Jan. 27, 2006;355(4):858-71. Epub Nov. 17, 2005.
- Wong et al., The Diversity Challenge in Directed Protein Evolution. *Comb Chem High Throughput Screen.* May 2006;9(4):271-88.
- Wood et al., A genetic system yields self-cleaving inteins for bioseparations. *Nat Biotechnol.* Sep. 1999;17(9):889-92. doi: 10.1038/12879.
- Wood et al., Targeted genome editing across species using ZFNs and TALENs. *Science.* Jul. 15, 2011;333(6040):307. doi: 10.1126/science.1207773. Epub Jun. 23, 2011.
- Woods et al., The phenotype of congenital insensitivity to pain due to the NaV1.9 variant p.L811P. *Eur J Hum Genet.* May 2015;23(5):561-3. doi: 10.1038/ejhg.2014.166. Epub Aug. 13, 2014.
- Wright et al., Continuous in vitro evolution of catalytic function. *Science.* Apr. 25, 1997;276(5312):614-7.
- Wright et al., Rational design of a split-Cas9 enzyme complex. *Proc Natl Acad Sci U S A.* Mar. 10, 2015;112(10):2984-9. doi: 10.1073/pnas.1501698112. Epub Feb. 23, 2015.
- Wu et al., Correction of a genetic disease in mouse via use of CRISPR-Cas9. *Cell Stem Cell.* Dec. 5, 2013;13(6):659-62. doi: 10.1016/j.stem.2013.10.016.
- Wu et al., Genome-wide binding of the CRISPR endonuclease Cas9 in mammalian cells. *Nat Biotechnol.* Jul. 2014;32(7):670-6. doi: 10.1038/nbt.2889. Epub Apr. 20, 2014.
- Wu et al., Protein trans-splicing and functional mini-inteins of a cyanobacterial dnaB intein. *Biochim Biophys Acta.* Sep. 8, 1998;1387(1-2):422-32. doi: 10.1016/S0167-4838(98)00157-5.
- Wu et al., Protein trans-splicing by a split intein encoded in a split DnaE gene of *Synechocystis* sp. PCC6803. *Proc Natl Acad Sci U S A.* Aug. 4, 1998;95(16):9226-31. doi: 10.1073/pnas.95.16.9226.
- Wu et al., Readers, writers and erasers of N6-methylated adenosine modification. *Curr Opin Struct Biol.* Dec. 2017;47:67-76. doi: 10.1016/j.sbi.2017.05.011. Epub Jun. 16, 2017.
- Xiang et al., RNA m6A methylation regulates the ultraviolet-induced DNA damage response. *Nature.* Mar. 23, 2017;543(7646):573-576. doi: 10.1038/nature21671. Epub Mar. 15, 2017.
- Xiao et al., Genetic incorporation of multiple unnatural amino acids into proteins in mammalian cells. *Angew Chem Int Ed Engl.* Dec. 23, 2013;52(52):14080-3. doi: 10.1002/anie.201308137. Epub Nov. 8, 2013.
- Xiao et al., Nuclear m(6)A Reader YTHDC1 Regulates mRNA Splicing. *Mol Cell.* Feb. 18, 2016;61(4):507-519. doi: 10.1016/j.molcel.2016.01.012. Epub Feb. 11, 2016.
- Xie et al., Adjusting the attB site in donor plasmid improves the efficiency of Φ C31 integrase system. *DNA Cell Biol.* Jul. 2012;31(7):1335-40. doi: 10.1089/dna.2011.1590. Epub Apr. 10, 2012.
- Xiong et al., Origin and evolution of retroelements based upon their reverse transcriptase sequences. *EMBO J.* Oct. 1990;9(10):3353-62.
- Xu et al., Accuracy and efficiency define Bxb1 integrase as the best of fifteen candidate serine recombinases for the integration of DNA into the human genome. *BMC Biotechnol.* Oct. 20, 2013;13:87. doi: 10.1186/1472-6750-13-87.
- Xu et al., Chemical ligation of folded recombinant proteins: segmental isotopic labeling of domains for NMR studies. *Proc Natl Acad Sci U S A.* Jan. 19, 1999;96(2):388-93. doi: 10.1073/pnas.96.2.388.
- Xu et al., Multiplex nucleotide editing by high-fidelity Cas9 variants with improved efficiency in rice. *BMC Plant Biol.* 2019;19(1):511. Published Nov. 21, 2019. doi: 10.1186/s12870-019-2131-1. Includes supplementary data and materials.
- Xu et al., Protein splicing: an analysis of the branched intermediate and its resolution by succinimide formation. *EMBO J.* Dec. 1, 1994;13(23):5517-22.
- Xu et al., PTMD: A Database of Human Disease-associated Post-translational Modifications. *Genomics Proteomics Bioinformatics.* Aug. 2018;16(4):244-251. doi: 10.1016/j.gpb.2018.06.004. Epub Sep. 21, 2018.
- Xu et al., Sequence determinants of improved CRISPR sgRNA design. *Genome Res.* Aug. 2015;25(8):1147-57. doi: 10.1101/gr.191452.115. Epub Jun. 10, 2015.
- Xu et al., Structures of human ALKBH5 demethylase reveal a unique binding mode for specific single-stranded N6-methyladenosine RNA demethylation. *J Biol Chem.* Jun. 20, 2014;289(25):17299-311. doi: 10.1074/jbc.M114.550350. Epub Apr. 28, 2014.
- Xu et al., The mechanism of protein splicing and its modulation by mutation. *EMBO J.* Oct. 1, 1996;15(19):5146-53.
- Yahata et al., Unified, Efficient, and Scalable Synthesis of Halichondrins: Zirconium/Nickel-Mediated One-Pot Ketone Synthesis as the Final Coupling Reaction. *Angew Chem Int Ed Engl.* Aug. 28, 2017;56(36):10796-10800. doi: 10.1002/anie.201705523. Epub Jul. 28, 2017.
- Yamamoto et al., The ons and offs of inducible transgenic technology: a review. *Neurobiol Dis.* Dec. 2001;8(6):923-32.
- Yamamoto et al., Virological and immunological bases for HIV-1 vaccine design. *Uirusu* 2007;57(2):133-139. <https://doi.org/10.2222/jsv.57.133>.
- Yamano et al., Crystal Structure of Cpf1 in Complex with Guide RNA and Target DNA. *Cell.* May 5, 2016;165(4):949-62 and Supplemental Info. doi: 10.1016/j.cell.2016.04.003. Epub Apr. 21, 2016.
- Yamano et al., Crystal Structure of Cpf1 in Complex with Guide RNA and Target DNA. *Cell.* May 5, 2016;165(4):949-62. doi: 10.1016/j.cell.2016.04.003. Epub Apr. 21, 2016.
- Yamazaki et al., Segmental Isotope Labeling for Protein NMR Using Peptide Splicing. *J. Am. Chem. Soc.* May 22, 1998;120(22):5591-2. <https://doi.org/10.1021/ja9807760>.
- Yan et al., Cas13d Is a Compact RNA-Targeting Type VI Crispr Effector Positively Modulated by a WYL-Domain-Containing Accessory Protein. *Mol Cell.* Apr. 19, 2018;70(2):327-339.e5. doi: 10.1016/j.molcel.2018.02.028. Epub Mar. 15, 2018.
- Yan et al., Functionally diverse type V CRISPR-Cas systems. *Science.* Jan. 4, 2019;363(6422):88-91. doi: 10.1126/science.aav7271. Epub Dec. 6, 2018.
- Yang et al., Highly Efficient A•T to G•C Base Editing by Cas9n-Guided tRNA Adenosine Deaminase in Rice. *Mol Plant.* Apr. 2, 2018;11(4):631-634. doi: 10.1016/j.molp.2018.02.008. Epub Feb. 22, 2018.
- Yang et al., APOBEC: From mutator to editor. *J Genet Genomics.* Sep. 20, 2017;44(9):423-437. doi: 10.1016/j.jgg.2017.04.009. Epub Aug. 7, 2017.
- Yang et al., Construction of an integration-proficient vector based on the site-specific recombination mechanism of enterococcal temperate phage phiFC1. *J Bacteriol.* Apr. 2002;184(7):1859-64. doi: 10.1128/jb.184.7.1859-1864.2002.
- Yang et al., Engineering and optimising deaminase fusions for genome editing. *Nat Commun.* Nov. 2, 2016;7:13330. doi: 10.1038/ncomms13330.
- Yang et al., Genome editing with targeted deaminases. *BioRxiv.* Preprint. First posted online Jul. 28, 2016.
- Yang et al., Increasing targeting scope of adenosine base editors in mouse and rat embryos through fusion of TadA deaminase with Cas9 variants. *Protein Cell.* Sep. 2018;9(9):814-819. doi: 10.1007/s13238-018-0568-x.
- Yang et al., Mutations in SCN9A, encoding a sodium channel alpha subunit, in patients with primary erythralgia. *J Med Genet.* Mar. 2004;41(3):171-4. doi: 10.1136/jmg.2003.012153.
- Yang et al., New CRISPR-Cas systems discovered. *Cell Res.* Mar. 2017;27(3):313-314. doi: 10.1038/cr.2017.21. Epub Feb. 21, 2017.
- Yang et al., One Prime for All Editing. *Cell.* Dec. 12, 2019;179(7):1448-1450. doi: 10.1016/j.cell.2019.11.030.

(56)

References Cited

OTHER PUBLICATIONS

- Yang et al., PAM-dependent Target DNA Recognition and Cleavage by C2C1 CRISPR-Cas endonuclease. *Cell* Dec. 2016;167(7):1814-28.
- Yang et al., Permanent genetic memory with >1-byte capacity. *Nat Methods*. Dec. 2014;11(12):1261-6. doi: 10.1038/nmeth.3147. Epub Oct. 26, 2014.
- Yang et al., Preparation of RNA-directed DNA polymerase from spleens of Balb-c mice infected with Rauscher leukemia virus. *Biochem Biophys Res Commun*. Apr. 28, 1972;47(2):505-11. doi: 10.1016/0006-291x(72)90743-7.
- Yang et al., Small-molecule control of insulin and PDGF receptor signaling and the role of membrane attachment. *Curr Biol*. Jan. 1, 1998;8(1):11-8. doi: 10.1016/s0960-9822(98)70015-6.
- Yang, Development of Human Genome Editing Tools for the Study of Genetic Variations and Gene Therapies. Doctoral Dissertation. Harvard University. 2013. Accessible via nrs.harvard.edu/urn-3:HUL.InstRepos:11181072. 277 pages.
- Yang, Nucleases: diversity of structure, function and mechanism. *Q Rev Biophys*. Feb. 2011;44(1):1-93. doi: 10.1017/S0033583510000181. Epub Sep. 21, 2010.
- Yasui et al., Miscoding Properties of 2'-Deoxyinosine, a Nitric Oxide-Derived DNA Adduct, during Translesion Synthesis Catalyzed by Human DNA Polymerases. *J Molec Biol*. Apr. 4, 2008;377(4):1015-23.
- Yasui, Alternative excision repair pathways. *Cold Spring Harb Perspect Biol*. Jun. 1, 2013;5(6):a012617. doi: 10.1101/cshperspect.a012617.
- Yasukawa et al., Characterization of Moloney murine leukaemia virus/avian myeloblastosis virus chimeric reverse transcriptases. *J Biochem*. Mar. 2009;145(3):315-24. doi: 10.1093/jb/mvn166. Epub Dec. 6, 2008.
- Yazaki et al., Hereditary systemic amyloidosis associated with a new apolipoprotein AII stop codon mutation Stop78Arg. *Kidney Int*. Jul. 2003;64(1):11-6.
- Yeh et al., In vivo base editing of post-mitotic sensory cells. *Nat Commun*. Jun. 5, 2018;9(1):2184. doi: 10.1038/s41467-018-04580-3.
- Yeh et al., In vivo base editing restores sensory transduction and transiently improves auditory function in a mouse model of recessive deafness. *Sci Transl Med*. Jun. 3, 2020;12(546):eaay9101. doi: 10.1126/scitranslmed.aay9101.
- Yin et al., Genome editing with Cas9 in adult mice corrects a disease mutation and phenotype. *Nat Biotechnol*. Jun. 2014;32(6):551-3. doi: 10.1038/nbt.2884. Epub Mar. 30, 2014.
- Yokoe et al., Spatial dynamics of GFP-tagged proteins investigated by local fluorescence enhancement. *Nat Biotechnol*. Oct. 1996;14(10):1252-6. doi: 10.1038/nbt1096-1252.
- Young et al., Beyond the canonical 20 amino acids: expanding the genetic lexicon. *J Biol Chem*. Apr. 9, 2010;285(15):11039-44. doi: 10.1074/jbc.R109.091306. Epub Feb. 10, 2010.
- Yu et al., Circular permutation: a different way to engineer enzyme structure and function. *Trends Biotechnol*. Jan. 2011;29(1):18-25. doi: 10.1016/j.tibtech.2010.10.004. Epub Nov. 17, 2010.
- Yu et al., Progress towards gene therapy for HIV infection. *Gene Ther*. Jan. 1994;1(1):13-26.
- Yu et al., Small molecules enhance CRISPR genome editing in pluripotent stem cells. *Cell Stem Cell*. Feb. 5, 2015;16(2):142-7. doi: 10.1016/j.stem.2015.01.003.
- Yuan et al., Laboratory-directed protein evolution. *Microbiol Mol Biol Rev*. 2005; 69(3):373-92. PMID: 16148303.
- Yuan et al., Tetrameric structure of a serine integrase catalytic domain. *Structure*. Aug. 6, 2008;16(8):1275-86. doi: 10.1016/j.str.2008.04.018.
- Yuen et al., Control of transcription factor activity and osteoblast differentiation in mammalian cells using an evolved small-molecule-dependent intein. *J Am Chem Soc*. Jul. 12, 2006;128(27):8939-46.
- Zakas et al., Enhancing the pharmaceutical properties of protein drugs by ancestral sequence reconstruction. *Nat Biotechnol*. Jan. 2017;35(1):35-37. doi: 10.1038/nbt.3677. Epub Sep. 26, 2016.
- Zalatan et al., Engineering complex synthetic transcriptional programs with CRISPR RNA scaffolds. *Cell*. Jan. 15, 2015;160(1-2):339-50. doi: 10.1016/j.cell.2014.11.052. Epub Dec. 18, 2014.
- Zelphati et al., Intracellular delivery of proteins with a new lipid-mediated delivery system. *J Biol Chem*. Sep. 14, 2001;276(37):35103-10. Epub Jul. 10, 2001.
- Zeng et al., Correction of the Marfan Syndrome Pathogenic FBN1 Mutation by Base Editing in Human Cells and Heterozygous Embryos. *Mol Ther*. Nov. 7, 2018;26(11):2631-2637. doi: 10.1016/j.ymthe.2018.08.007. Epub Aug. 14, 2018.
- Zetsche et al., A split-Cas9 architecture for inducible genome editing and transcription modulation. *Nat Biotechnol*. Feb. 2015;33(2):139-42. doi: 10.1038/nbt.3149.
- Zettler et al., The naturally split Npu DnaE intein exhibits an extraordinarily high rate in the protein trans-splicing reaction. *FEBS Lett*. Mar. 4, 2009;583(5):909-14. doi: 10.1016/j.febslet.2009.02.003. Epub Feb. 10, 2009.
- Zhang et al., Π -Clamp-mediated cysteine conjugation. *Nat Chem*. Feb. 2016;8(2):120-8. doi: 10.1038/nchem.2413. Epub Dec. 21, 2015.
- Zhang et al., A new strategy for the site-specific modification of proteins in vivo. *Biochemistry*. Jun. 10, 2003;42(22):6735-46.
- Zhang et al., Circular intronic long noncoding RNAs. *Mol Cell*. Sep. 26, 2013;51(6):792-806. doi: 10.1016/j.molcel.2013.08.017. Epub Sep. 12, 2013.
- Zhang et al., Comparison of non-canonical PAMs for CRISPR/Cas9-mediated DNA cleavage in human cells. *Sci Rep*. Jun. 2014;4:5405.
- Zhang et al., Conditional gene manipulation: Cre-ating a new biological era. *J Zhejiang Univ Sci B*. Jul. 2012;13(7):511-24. doi: 10.1631/jzus.B1200042. Review.
- Zhang et al., Copy number variation in human health, disease, and evolution. *Annu Rev Genomics Hum Genet*. 2009;10:451-81. doi: 10.1146/annurev.genom.9.081307.164217.
- Zhang et al., CRISPR/Cas9 for genome editing: progress, implications and challenges. *Hum Mol Genet*. Sep. 15, 2014;23(R1):R40-6. doi: 10.1093/hmg/ddu125. Epub Mar. 20, 2014.
- Zhang et al., Efficient construction of sequence-specific TAL effectors for modulating mammalian transcription. *Nat Biotechnol*. Feb. 2011;29(2):149-53. doi: 10.1038/nbt.1775. Epub Jan. 19, 2011.
- Zhang et al., Global analysis of small RNA and mRNA targets of Hfq. *Mol Microbiol*. Nov. 2003;50(4):1111-24. doi: 10.1046/j.1365-2958.2003.03734.x.
- Zhang et al., Myoeediting: Toward Prevention of Muscular Dystrophy by Therapeutic Genome Editing. *Physiol Rev*. Jul. 1, 2018;98(3):1205-1240. doi: 10.1152/physrev.00046.2017.
- Zhang et al., Programmable base editing of zebrafish genome using a modified CRISPR-Cas9 system. *Nat Commun*. Jul. 25, 2017;8(1):118. doi: 10.1038/s41467-017-00175-6.
- Zhang et al., Reversible RNA Modification N1-methyladenosine (m1A) in mRNA and tRNA. *Genomics Proteomics Bioinformatics*. Jun. 2018;16(3):155-161. doi: 10.1016/j.gpb.2018.03.003. Epub Jun. 14, 2018.
- Zhang et al., Ribozymes and Riboswitches: Modulation of RNA Function by Small Molecules. *Biochemistry*. Nov. 2, 2010;49(43):9123-31. doi: 10.1021/bi1012645.
- Zhang et al., Stabilized plasmid-lipid particles for regional gene therapy: formulation and transfection properties. *Gene Ther*. Aug. 1999;6(8):1438-47.
- Zhao et al., An ultraprocessive, accurate reverse transcriptase encoded by a metazoan group II intron. *RNA*. Feb. 2018;24(2):183-195. doi: 10.1261/rna.063479.117. Epub Nov. 6, 2017.
- Zhao et al., Crystal structures of a group II intron maturase reveal a missing link in spliceosome evolution. *Nat Struct Mol Biol*. Jun. 2016;23(6):558-65. doi: 10.1038/nsmb.3224. Epub May 2, 2016.
- Zhao et al., Post-transcriptional gene regulation by mRNA modifications. *Nat Rev Mol Cell Biol*. Jan. 2017;18(1):31-42. doi: 10.1038/nrm.2016.132. Epub Nov. 3, 2016.
- Zheng et al., ALKBH5 is a mammalian RNA demethylase that impacts RNA metabolism and mouse fertility. *Mol Cell*. Jan. 10, 2013;49(1):18-29. doi: 10.1016/j.molcel.2012.10.015. Epub Nov. 21, 2012.

(56)

References Cited

OTHER PUBLICATIONS

- Zheng et al., DNA editing in DNA/RNA hybrids by adenosine deaminases that act on RNA. *Nucleic Acids Res.* Apr. 7, 2017;45(6):3369-3377. doi: 10.1093/nar/gkx050.
- Zheng et al., Highly efficient base editing in bacteria using a Cas9-cytidine deaminase fusion. *Commun Biol.* Apr. 19, 2018;1:32. doi: 10.1038/s42003-018-0035-5.
- Zheng et al., Structural basis for the complete resistance of the human prion protein mutant G127V to prion disease. *Sci Rep.* Sep. 4, 2018;8(1):13211. doi: 10.1038/s41598-018-31394-6.
- Zhong et al., Rational Design of Aptazyme Riboswitches for Efficient Control of Gene Expression in Mammalian Cells. *Elife.* Nov. 2, 2016;5:e18858. doi: 10.7554/eLife.18858.
- Zhou et al., Dynamic m(6)A mRNA methylation directs translational control of heat shock response. *Nature.* Oct. 22, 2015;526(7574):591-4. doi: 10.1038/nature15377. Epub Oct. 12, 2015.
- Zhou et al., GISSD: Group I Intron Sequence and Structure Database. *Nucleic Acids Res.* Jan. 2008;36(Database issue):D31-7. doi: 10.1093/nar/gkm766. Epub Oct. 16, 2007.
- Zhou et al., Off-target RNA mutation induced by DNA base editing and its elimination by mutagenesis. *Nature.* Jul. 2019;571(7764):275-278. doi: 10.1038/s41586-019-1314-0. Epub Jun. 10, 2019.
- Zhou et al., Protective V127 prion variant prevents prion disease by interrupting the formation of dimer and fibril from molecular dynamics simulations. *Sci Rep.* Feb. 24, 2016;6:21804. doi: 10.1038/srep21804.
- Zhou et al., Seamless Genetic Conversion of SMN2 to SMN1 via CRISPR/Cpf1 and Single-Stranded Oligodeoxynucleotides in Spinal Muscular Atrophy Patient-Specific Induced Pluripotent Stem Cells. *Hum Gene Ther.* Nov. 2018;29(11):1252-1263. doi: 10.1089/hum.2017.255. Epub May 9, 2018.
- Zielenski, Genotype and phenotype in cystic fibrosis. *Respiration.* 2000;67(2):117-33. doi: 10.1159/000029497.
- Zimmerly et al., An Unexplored Diversity of Reverse Transcriptases in Bacteria. *Microbiol Spectr.* Apr. 2015;3(2):MDNA3-0058-2014. doi: 10.1128/microbiolspec.MDNA3-0058-2014.
- Zimmerly et al., Group II intron mobility occurs by target DNA-primed reverse transcription. *Cell.* Aug. 25, 1995;82(4):545-54. doi: 10.1016/0092-8674(95)90027-6.
- Zimmermann et al., Molecular interactions and metal binding in the theophylline-binding core of an RNA aptamer. *RNA.* May 2000;6(5):659-67.
- Zong et al., Precise base editing in rice, wheat and maize with a Cas9-cytidine deaminase fusion. *Nat Biotechnol.* May 2017;35(5):438-440. doi: 10.1038/nbt.3811. Epub Feb. 27, 2017.
- Zorko et al., Cell-penetrating peptides: mechanism and kinetics of cargo delivery. *Adv Drug Deliv Rev.* Feb. 28, 2005;57(4):529-45. Epub Jan. 22, 2005.
- Zou et al., Gene targeting of a disease-related gene in human induced pluripotent stem and embryonic stem cells. *Cell Stem Cell.* Jul. 2, 2009;5(1):97-110. doi: 10.1016/j.stem.2009.05.023. Epub Jun. 18, 2009.
- Zufferey et al., Woodchuck hepatitis virus posttranscriptional regulatory element enhances expression of transgenes delivered by retroviral vectors. *J Virol.* Apr. 1999;73(4):2886-92. doi: 10.1128/JVI.73.4.2886-2892.1999.
- Zuker et al., Optimal computer folding of large RNA sequences using thermodynamics and auxiliary information. *Nucleic Acids Res.* Jan. 10, 1981;9(1):133-48. doi: 10.1093/nar/9.1.133.
- Zuo et al., Cytosine base editor generates substantial off-target single-nucleotide variants in mouse embryos. *Science.* Apr. 19, 2019;364(6437):289-292. doi: 10.1126/science.aav9973. Epub Feb. 28, 2019.
- Zuris et al., Cationic lipid-mediated delivery of proteins enables efficient protein-based genome editing in vitro and in vivo. *Nat Biotechnol.* 2015;33:73-80.
- U.S. Appl. No. 61/874,682, filed Sep. 6, 2013, Liu et al.
- U.S. Appl. No. 61/874,746, filed Sep. 6, 2013, Liu et al.
- Arakawa et al., A method to convert mRNA into a gRNA library for CRISPR/Cas9 editing of any organism. *Sci Adv.* Aug. 24, 2016;2(8):e1600699. doi: 10.1126/sciadv.1600699.
- Arbab et al., Cloning-free CRISPR. *Stem Cell Reports.* Nov. 10, 2015;5(5):908-917. doi: 10.1016/j.stemcr.2015.09.022. Epub Oct. 29, 2015.
- Bae et al., Microhomology-based choice of Cas9 nuclease target sites. *Nat Methods.* Jul. 2014;11(7):705-6. doi: 10.1038/nmeth.3015.
- Biehls et al., DNA Double-Strand Break Resection Occurs during Non-homologous End Joining in G1 but Is Distinct from Resection during Homologous Recombination. *Mol Cell.* Feb. 16, 2017;65(4):671-684.e5. doi: 10.1016/j.molcel.2016.12.016. Epub Jan. 26, 2017.
- Ceccaldi et al., Repair Pathway Choices and Consequences at the Double-Strand Break. *Trends Cell Biol.* Jan. 2016;26(1):52-64. doi: 10.1016/j.tcb.2015.07.009. Epub Oct. 1, 2015.
- Christian et al., Targeting DNA double-strand breaks with TAL effector nucleases. *Genetics.* Oct. 2010;186(2):757-61. doi: 10.1534/genetics.110.120717. Epub Jul. 26, 2010.
- Chu et al., Increasing the efficiency of homology-directed repair for CRISPR-Cas9-induced precise gene editing in mammalian cells. *Nat Biotech.* Feb. 13, 2015;33:543-8.
- Chylinski et al., The tracrRNA and Cas9 families of type II CRISPR-Cas immunity systems. *RNA Biol.* May 2013;10(5):726-37. doi: 10.4161/rna.24321. Epub Apr. 5, 2013.
- Cong et al., Multiplex genome engineering using CRISPR/Cas systems. *Science.* Feb. 15, 2013;339(6121):819-23. doi: 10.1126/science.1231143. Epub Jan. 3, 2013.
- Cornu et al., Refining strategies to translate genome editing to the clinic. *Nat Med.* Apr. 3, 2017;23(4):415-423. doi: 10.1038/nm.4313.
- Davis et al., DNA double strand break repair via non-homologous end-joining. *Transl Cancer Res.* Jun. 2013;2(3):130-143. doi: 10.3978/j.issn.2218-676X.2013.04.02.
- Deltcheva et al., CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III. *Nature.* Mar. 31, 2011;471(7340):602-7. doi: 10.1038/nature09886.
- Deriano et al., Modernizing the nonhomologous end-joining repertoire: alternative and classical NHEJ share the stage. *Annu Rev Genet.* 2013;47:433-55. doi: 10.1146/annurev-genet-110711-155540. Epub Sep. 11, 2013.
- Dicarlo et al., Genome engineering in *Saccharomyces cerevisiae* using CRISPR-Cas systems. *Nucleic Acids Research.* Apr. 2013;41(7):4336-43.
- Dicarlo et al., Safeguarding CRISPR-Cas9 gene drives in yeast. *Nat Biotechnol.* Dec. 2015;33(12):1250-1255. doi: 10.1038/nbt.3412. Epub Nov. 16, 2015.
- Doench et al., Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. *Nat Biotechnol.* Feb. 2016;34(2):184-191. doi: 10.1038/nbt.3437.
- Evers et al., CRISPR knockout screening outperforms shRNA and CRISPRi in identifying essential genes. *Nat Biotechnol.* Jun. 2016;34(6):631-3. doi: 10.1038/nbt.3536. Epub Apr. 25, 2016.
- Fu et al., Improving CRISPR-Cas nuclease specificity using truncated guide RNAs. *Nat Biotechnol.* Mar. 2014;32(3):279-84. doi: 10.1038/nbt.2808. Epub Jan. 26, 2014.
- Heidenreich et al., Non-homologous end joining as an important mutagenic process in cell cycle-arrested cells. *EMBO J.* May 1, 2003;22(9):2274-83. doi: 10.1093/emboj/cdg203.
- Hu et al., Evolved Cas9 variants with broad PAM compatibility and high DNA specificity. *Nature.* Apr. 5, 2018;556(7699):57-63. doi: 10.1038/nature26155. Epub Feb. 28, 2018.
- Hwang et al., Efficient genome editing in zebrafish using a CRISPR-Cas system. *Nat Biotechnol.* Mar. 2013;31(3):227-9. doi: 10.1038/nbt.2501. Epub Jan. 29, 2013.
- Invitation to Pay Additional Fees mailed Mar. 8, 2019 in connection with PCT/US2018/065886.
- International Search Report and Written Opinion mailed Apr. 25, 2019 in connection with PCT/US2018/065886.
- Jiang et al., RNA-guided editing of bacterial genomes using CRISPR-Cas systems. *Nat Biotechnol.* Mar. 2013;31(3):233-9. doi: 10.1038/nbt.2508. Epub Jan. 29, 2013.

(56)

References Cited

OTHER PUBLICATIONS

- Jinek et al., A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science*. Aug. 17, 2012;337(6096):816-21. doi: 10.1126/science.1225829. Epub Jun. 28, 2012.
- Jinek et al., RNA-programmed genome editing in human cells. *Elife*. Jan. 29, 2013;2:e00471. doi: 10.7554/eLife.00471.
- Kim et al., Hybrid restriction enzymes: zinc finger fusions to Fok I cleavage domain. *Proc Natl Acad Sci U S A*. Feb. 6, 1996;93(3):1156-60. doi: 10.1073/pnas.93.3.1156.
- Kim et al., In vivo high-throughput profiling of CRISPR-Cpf1 activity. *Nat Methods*. Feb. 2017;14(2):153-159. doi: 10.1038/nmeth.4104. Epub Dec. 19, 2016.
- Kleinstiver et al., Broadening the targeting range of *Staphylococcus aureus* CRISPR-Cas9 by modifying PAM recognition. *Nat Biotechnol*. Dec. 2015;33(12):1293-1298. doi: 10.1038/nbt.3404. Epub Nov. 2, 2015.
- Kleinstiver et al., Engineered CRISPR-Cas9 nucleases with altered PAM specificities. *Nature*. Jul. 23, 2015;523(7561):481-5. doi: 10.1038/nature14592. Epub Jun. 22, 2015.
- Kleinstiver et al., High-fidelity CRISPR-Cas9 nucleases with no detectable genome-wide off-target effects. *Nature*. Jan. 28, 2016;529(7587):490-5. doi: 10.1038/nature16526. Epub Jan. 6, 2016.
- Koike-Yusa et al., Genome-wide recessive genetic screening in mammalian cells with a lentiviral CRISPR-guide RNA library. *Nat Biotechnol*. Mar. 2014;32(3):267-73. doi: 10.1038/nbt.2800. Epub Dec. 23, 2013.
- Komor et al., Programmable editing of a target base in genomic DNA without double-stranded DNA cleavage. *Nature*. Apr. 20, 2016;533(7603):420-4. doi: 10.1038/nature17946.
- Kraft et al., Deletions, Inversions, Duplications: Engineering of Structural Variants using CRISPR/Cas in Mice. *Cell Rep*. Feb. 10, 2015;10(5):833-839. doi: 10.1016/j.celrep.2015.01.016. Epub Feb. 7, 2015.
- Landrum et al., ClinVar: public archive of interpretations of clinically relevant variants. *Nucleic Acids Res*. Jan. 4, 2016;44(D1):D862-8. doi: 10.1093/nar/gkv1222. Epub Nov. 17, 2015.
- Long et al., Postnatal genome editing partially restores dystrophin expression in a mouse model of muscular dystrophy. *Science*. Jan. 22, 2016;351(6271):400-3. doi: 10.1126/science.aad5725. Epub Dec. 31, 2015.
- Mali et al., Cas9 as a versatile tool for engineering biology. *Nat Methods*. Oct. 2013; 10(10):957-63. doi: 10.1038/nmeth.2649.
- Mali et al., RNA-guided human genome engineering via Cas9. *Science*. Feb. 15, 2013;339(6121):823-6. doi: 10.1126/science.1232033. Epub Jan. 3, 2013.
- Mandal et al., Efficient ablation of genes in human hematopoietic stem and effector cells using CRISPR/Cas9. *Cell Stem Cell*. Nov. 6, 2014;15(5):643-52. doi: 10.1016/j.stem.2014.10.004. Epub Nov. 6, 2014.
- McVey et al., MMEJ repair of double-strand breaks (director's cut): deleted sequences and alternative endings. *Trends Genet*. Nov. 2008;24(11):529-38. doi: 10.1016/j.tig.2008.08.007. Epub Sep. 21, 2008.
- Nakade et al., Microhomology-mediated end-joining-dependent integration of donor DNA in cells and animals using TALENs and CRISPR/Cas9. *Nat Commun*. Nov. 20, 2014;5:5560. doi: 10.1038/ncomms6560.
- Oh et al., Positional cloning of a gene for Hermansky-Pudlak syndrome, a disorder of cytoplasmic organelles. *Nat Genet*. Nov. 1996;14(3):300-6. doi: 10.1038/ng1196-300.
- Orthwein et al., A mechanism for the suppression of homologous recombination in G1 cells. *Nature*. Dec. 17, 2015;528(7582):422-6. doi: 10.1038/nature16142. Epub Dec. 9, 2015.
- Paquet et al., Efficient introduction of specific homozygous and heterozygous mutations using CRISPR/Cas9. *Nature*. May 5, 2016;533(7601): 125-9. doi: 10.1038/nature17664. Epub Apr. 27, 2016.
- Richardson et al., Enhancing homology-directed genome editing by catalytically active and inactive CRISPR-Cas9 using asymmetric donor DNA. *Nat Biotechnol*. Mar. 2016;34(3):339-44. doi: 10.1038/nbt.3481. Epub Jan. 20, 2016.
- Sakuma et al., MMEJ-assisted gene knock-in using TALENs and CRISPR-Cas9 with the PITCh systems. *Nat Protoc*. Jan. 2016;11(1):118-33. doi: 10.1038/nprot.2015.140. Epub Dec. 17, 2015.
- Shalem et al., High-throughput functional genomics using CRISPR-Cas9. *Nat Rev Genet*. May 2015;16(5):299-311. doi: 10.1038/nrg3899. Epub Apr. 9, 2015.
- Shen et al., Predictable and precise template-free CRISPR editing of pathogenic variants. *Nature*. Nov. 2018;563(7733):646-651. doi: 10.1038/s41586-018-0686-x. Epub Nov. 7, 2018. Erratum in: *Nature*. Mar. 2019;567(7746):E1-E2.
- Sherwood et al., Discovery of directional and nondirectional pioneer transcription factors by modeling DNase profile magnitude and shape. *Nat Biotechnol*. Feb. 2014;32(2):171-178. doi: 10.1038/nbt.2798. Epub Jan. 19, 2014.
- Shin et al., CRISPR/Cas9 targeting events cause complex deletions and insertions at 17 sites in the mouse genome. *Nat Commun*. May 31, 2017;8:15464. doi: 10.1038/ncomms15464.
- Slaymaker et al., Rationally engineered Cas9 nucleases with improved specificity. *Science*. Jan. 1, 2016;351(6268):84-8. doi: 10.1126/science.aad5227. Epub Dec. 1, 2015.
- Sterne-Weiler et al., Exon identity crisis: disease-causing mutations that disrupt the splicing code. *Genome Biol*. Jan. 23, 2014;15(1):201. doi: 10.1186/gb4150.
- Surun et al., High Efficiency Gene Correction in Hematopoietic Cells by Donor-Template-Free CRISPR/Cas9 Genome Editing. *Mol Ther Nucleic Acids*. Mar. 2, 2018;10:1-8. doi: 10.1016/j.omtn.2017.11.001. Epub Nov. 10, 2017. PMID: 29499925; PMCID: PMC5723376.
- Suzuki et al., In vivo genome editing via CRISPR/Cas9 mediated homology-independent targeted integration. *Nature*. Dec. 1, 2016;540(7631): 144-149. doi: 10.1038/nature20565. Epub Nov. 16, 2016.
- Tang et al., Evaluation of Bioinformatic Programmes for the Analysis of Variants within Splice Site Consensus Regions. *Adv Bioinformatics*. 2016;2016:5614058. doi: 10.1155/2016/5614058. Epub May 24, 2016.
- Urasaki et al., Functional dissection of the Tol2 transposable element identified the minimal cis-sequence and a highly repetitive sequence in the subterminal region essential for transposition. *Genetics*. Oct. 2006;174(2):639-49. doi: 10.1534/genetics.106.060244. Epub Sep. 7, 2006.
- Yu et al., Synthesis-dependent microhomology-mediated end joining accounts for multiple types of repair junctions. *Nucleic Acids Res*. Sep. 2010;38(17):5706-17. doi: 10.1093/nar/gkq379. Epub May 11, 2010.
- Zetsche et al., Cpf1 is a single RNA-guided endonuclease of a class 2 CRISPR-Cas system. *Cell*. Oct. 22, 2015;163(3):759-71. doi: 10.1016/j.cell.2015.09.038. Epub Sep. 25, 2015.
- [No Author Listed], "Lambda DNA" from Catalog & Technical Reference. New England Biolabs Inc. 2002/2003. pp. 133 and 270-273.
- [No Author Listed], *Mus musculus* (Mouse). UniProtKB Accession No. P51908 (ABEC1_MOUSE). Oct. 1, 1996. 10 pages.
- Asokan et al., The AAV vector toolkit: poised at the clinical crossroads. *Mol Ther*. Apr. 2012;20(4):699-708. doi: 10.1038/mt.2011.287. Epub Jan. 24, 2012.
- Auricchio et al., Exchange of surface proteins impacts on viral vector cellular specificity and transduction characteristics: the retina as a model. *Hum Mol Genet*. Dec. 15, 2001;10(26):3075-81. doi: 10.1093/hmg/10.26.3075.
- Baba et al., Construction of *Escherichia coli* K-12 in-frame, single-gene knockout mutants: the Keio collection. *Mol Syst Biol*. 2006;2:2006.0008. doi: 10.1038/msb4100050. Epub Feb. 21, 2006.
- Badran et al., In vivo continuous directed evolution. *Curr Opin Chem Biol*. Feb. 2015;24:1-10. doi: 10.1016/j.cbpa.2014.09.040. Epub Nov. 7, 2014.
- Banno et al., Deaminase-mediated multiplex genome editing in *Escherichia coli*. *Nat Microbiol*. Apr. 2018;3(4):423-429. doi: 10.1038/s41564-017-0102-6. Epub Feb. 5, 2018.

(56)

References Cited

OTHER PUBLICATIONS

- Bass, B.L., RNA editing by adenosine deaminases that act on RNA. *Annu Rev Biochem.* 2002;71:817-46. doi: 10.1146/annurev.biochem.71.110601.135501. Epub Nov. 9, 2001.
- Bothmer et al., Characterization of the interplay between DNA repair and CRISPR/Cas9-induced DNA lesions at an endogenous locus. *Nat Commun.* Jan. 9, 2017;8:13905. doi: 10.1038/ncomms13905.
- Brierley et al., Viral RNA pseudoknots: versatile motifs in gene expression and replication. *Nat Rev Microbiol.* Aug. 2007;5(8):598-610. doi: 10.1038/nrmicro1704.
- Canny et al., Inhibition of 53BP1 Favors Homology-Dependent DNA Repair and Increases CRISPR-Cas9 Genome-Editing Efficiency. *Nat Biotechnol.* Jan. 2018;36(1):95-102. doi: 10.1038/nbt.4021. Epub Nov. 27, 2017.
- Cao et al., Rapamycin reverses cellular phenotypes and enhances mutant protein clearance in Hutchinson-Gilford progeria syndrome cells. *Sci Transl Med.* Jun. 29, 2011;3(89):89ra58. doi: 10.1126/scitranslmed.3002346.
- Chatterjee et al., Robust Genome Editing of Single-Base PAM Targets; with Engineered ScCas9 Variants. *bioRxiv.* doi: 10.1101/620351. Posted Apr. 26, 2019.
- Chen et al., Alterations in PMS2, MSH2 and MLH1 expression in human prostate cancer. *Int J Oncol.* May 2003;22(5):1033-43.
- Choudhury et al., CRISPR/Cas9 recombineering-mediated deep mutational scanning of essential genes in *Escherichia coli*. *Mol Syst Biol.* Mar. 2020; 16(3):e9265. doi: 10.15252/msb.20199265.
- Davis et al., Assaying Repair at DNA Nicks. *Methods Enzymol.* 2018;601:71-89. doi: 10.1016/bs.mie.2017.12.001. Epub Feb. 1, 2018.
- Davis et al., Homology-directed repair of DNA nicks via pathways distinct from canonical double-strand break repair. *Proc Natl Acad Sci U S A.* Mar. 12, 2014;111(10):E924-32. doi: 10.1073/pnas.1400236111. Epub Feb. 20, 2014.
- Davis et al., Two Distinct Pathways Support Gene Correction by Single-Stranded Donors at DNA Nicks. *Cell Rep.* Nov. 8, 2016;17(7):1872-1881. doi: 10.1016/j.celrep.2016.10.049.
- De Sandre-Giovannoli et al., Lamin A truncation in Hutchinson-Gilford progeria. *Science.* Jun. 27, 2003;300(5628):2055. doi: 10.1126/science.1084125. Epub Apr. 17, 2003.
- Drenth et al., Mutations in sodium-channel gene SCN9A cause a spectrum of human genetic pain disorders. *J Clin Invest.* Dec. 2007;117(12):3603-9. doi: 10.1172/JCI33297.
- Drost et al., Inactivation of DNA mismatch repair by variants of uncertain significance in the PMS2 gene. *Hum Mutat.* Nov. 2013;34(11):1477-80. doi: 10.1002/humu.22426. Epub Sep. 11, 2013.
- Duan et al., Enhancement of muscle gene delivery with pseudotyped adeno-associated virus type 5 correlates with myoblast differentiation. *J Virol.* Aug. 2001;75(16):7662-71. doi: 10.1128/JVI.75.16.7662-7671.2001.
- Dugar et al., CRISPR RNA-Dependent Binding and Cleavage of Endogenous RNAs by the *Campylobacter jejuni* Cas9. *Mol Cell.* Mar. 1, 2018;69(5):893-905.e7. doi: 10.1016/j.molcel.2018.01.032.
- Eisenberg et al., A-to-I RNA editing—immune protector and transcriptome diversifier. *Nat Rev Genet.* Aug. 2018; 19(8):473-490. doi: 10.1038/s41576-018-0006-1.
- Filippova et al., Guide RNA modification as a way to improve CRISPR/Cas9-based genome-editing systems. *Biochimie.* Dec. 2019;167:49-60. doi: 10.1016/j.biochi.2019.09.003. Epub Sep. 4, 2019.
- Friedman, J. H., Greedy function approximation: A gradient boosting machine. *Ann. Statist.* Oct. 2001;29(5):1189-232. doi: 10.1214/aos/1013203451.
- GENBANK Submission; NIH/NCBI, Accession No. NG_008692.2. McClintock et al., Aug. 27, 2018. 33 pages.
- GENBANK Submission; NIH/NCBI, Accession No. NP_001075493.1. Schiaffella et al., Jun. 24, 2018. 2 pages.
- GENBANK Submission; NIH/NCBI, Accession No. NP_001157741.1. Zeng et al., Sep. 17, 2018. 3 pages.
- GENBANK Submission; NIH/NCBI, Accession No. NP_001157742.1. Zeng et al., Oct. 21, 2018. 3 pages.
- GENBANK Submission; NIH/NCBI, Accession No. NP_033040.2. Liu et al., Jun. 23, 2018. 2 pages.
- GENBANK Submission; NIH/NCBI, Accession No. XP_003314669.1. No Author Listed, Mar. 20, 2018. 2 pages.
- GENBANK Submission; NIH/NCBI, Accession No. XP_026671085.1. No Author Listed, Oct. 17, 2018. 1 page.
- Gutschner et al., Post-translational Regulation of Cas9 during G1 Enhances Homology-Directed Repair. *Cell Rep.* Feb. 16, 2016;14(6):1555-1566. doi: 10.1016/j.celrep.2016.01.019. Epub Feb. 4, 2016.
- Halbert et al., Repeat transduction in the mouse lung by using adeno-associated virus vectors with different serotypes. *J Virol.* Feb. 2000;74(3):1524-32. doi: 10.1128/jvi.74.3.1524-1532.2000.
- Hardt et al., Missense variants in hMLH1 identified in patients from the German HNPCC consortium and functional studies. *Fam Cancer.* Jun. 2011;10(2):273-84. doi: 10.1007/s10689-011-9431-4.
- Harmen et al., DNA mismatch repair and oligonucleotide end-protection promote base-pair substitution distal from a CRISPR/Cas9-induced DNA break. *Nucleic Acids Res.* Apr. 6, 2018;46(6):2945-2955. doi: 10.1093/nar/gky076.
- Harrington et al., Programmed DNA destruction by miniature CRISPR-Cas14 enzymes. *Science.* Nov. 16, 2018;362(6416):839-842. doi: 10.1126/science.aav4294. Epub Oct. 18, 2018.
- Hart et al., High-Resolution CRISPR Screens Reveal Fitness Genes and Genotype-Specific Cancer Liabilities. *Cell.* Dec. 3, 2015;163(6):1515-26. doi: 10.1016/j.cell.2015.11.015. Epub Nov. 25, 2015.
- Hilbers et al., New developments in structure determination of pseudoknots. *Biopolymers.* 1998;48(2-3):137-53. doi: 10.1002/(SICI)1097-0282(1998)48:2<137::AID-BIP4>3.0.CO;2-H.
- Huang et al., Gain-of-function mutations in sodium channel Na(v)1.9 in painful neuropathy. *Brain.* Jun. 2014;137(Pt 6):1627-42. doi: 10.1093/brain/awu079. Epub Apr. 27, 2014.
- Isaacs et al., Engineered riboregulators enable post-transcriptional control of gene expression. *Nat Biotechnol.* Jul. 2004;22(7):841-7. doi: 10.1038/nbt986. Epub Jun. 20, 2004.
- Ishizuka et al., Loss of ADAR1 in tumours overcomes resistance to immune checkpoint blockade. *Nature.* Jan. 2019;565(7737):43-48. doi: 10.1038/s41586-018-0768-9. Epub Dec. 17, 2018.
- Jia et al., The MLH1 ATPase domain is needed for suppressing aberrant formation of interstitial telomeric sequences. *DNA Repair (Amst).* May 2018;65:20-25. doi: 10.1016/j.dnarep.2018.03.002. Epub Mar. 7, 2018.
- Kan et al., Mechanisms of precise genome editing using oligonucleotide donors. *Genome Res.* Jul. 2017;27(7):1099-1111. doi: 10.1101/gr.214775.116. Epub Mar. 29, 2017.
- Kim et al., Adenine base editors catalyze cytosine conversions in human cells. *Nat Biotechnol.* Oct. 2019;37(10):1145-1148. doi: 10.1038/s41587-019-0254-4. Epub Sep. 23, 2019.
- Kim et al., Predicting the efficiency of prime editing guide RNAs in human cells. *Nat Biotechnol.* Feb. 2021;39(2):198-206. doi: 10.1038/s41587-020-0677-y. Epub Sep. 21, 2020.
- Kim et al., RAD51 mutants cause replication defects and chromosomal instability. *Mol Cell Biol.* Sep. 2012;32(18):3663-80. doi: 10.1128/MCB.00406-12. Epub Jul. 9, 2012.
- Knott et al., CRISPR-Cas guides the future of genetic engineering. *Science.* Aug. 31, 2018;361(6405):866-869. doi: 10.1126/science.aat5011.
- Kweon et al., A CRISPR-based base-editing screen for the functional assessment of BRCA1 variants. *Oncogene.* Jan. 2020;39(1):30-35. doi: 10.1038/s41388-019-0968-2. Epub Aug. 29, 2019.
- Li et al., Programmable Single and Multiplex Base-Editing in *Bombyx mori* Using RNA-Guided Cytidine Deaminases. *G3 (Bethesda).* May 4, 2018;8(5):1701-1709. doi: 10.1534/g3.118.200134.
- Liao et al., One-step assembly of large CRISPR arrays enables multi-functional targeting and reveals constraints on array design. *bioRxiv.* May 2, 2018. doi: 10.1101/312421. 45 pages.
- Lin et al., [Construction and evaluation of DnaB split intein high expression vector and a six amino acids cyclic peptide library]. *Sheng Wu Gong Cheng Xue Bao.* Nov. 2008;24(11):1924-30. Chinese.

(56)

References Cited

OTHER PUBLICATIONS

- Lindahl, T., Instability and decay of the primary structure of DNA. *Nature*. Apr. 22, 1993;362(6422):709-15. doi: 10.1038/362709a0.
- Liu et al., Human BRCA2 protein promotes RAD51 filament formation on RPA-covered single-stranded DNA. *Nat Struct Mol Biol*. Oct. 2010; 17(10):1260-2. doi: 10.1038/nsmb.1904. Epub Aug. 22, 2010.
- Liu et al., Improving Editing Efficiency for the Sequences with NGH PAM Using xCas9- Derived Base Editors. *Mol Ther Nucleic Acids*. Sep. 6, 2019;17:626-635. doi: 10.1016/j.omtn.2019.06.024. Epub Jul. 12, 2019.
- Liu et al., Intrinsic Nucleotide Preference of Diversifying Base Editors Guides Antibody Ex Vivo Affinity Maturation. *Cell Rep*. Oct. 23, 2018;25(4):884-892.e3. doi: 10.1016/j.celrep.2018.09.090.
- Ma et al., Human RAD52 interactions with replication protein A and the RAD51 presynaptic complex. *J Biol Chem*. Jul. 14, 2017;292(28):11702-11713. doi: 10.1074/jbc.M117.794545. Epub May 27, 2017.
- Marsden et al., The Tumor-Associated Variant RAD51 G151D Induces a Hyper-Recombination Phenotype. *PLoS Genet*. Aug. 11, 2016;12(8):e1006208. doi: 10.1371/journal.pgen.1006208.
- Mason et al., Non-enzymatic roles of human RAD51 at stalled replication forks. *bioRxiv*. Jul. 31, 2019; doi.org/10.1101/359380. 36 pages. bioRxiv preprint first posted online Jul. 31, 2019.
- Mootz et al., Protein splicing triggered by a small molecule. *J Am Chem Soc*. Aug. 7, 2002;124(31):9044-5 and Supporting Information. doi: 10.1021/ja0267690. 4 pages.
- Murugan et al., The Revolution Continues: Newly Discovered Systems Expand the CRISPR-Cas Toolkit. *Mol Cell*. Oct. 5, 2017;68(1):15-25. doi: 10.1016/j.molcel.2017.09.007.
- Nelson et al., The unstable repeats—three evolving faces of neurological disease. *Neuron*. Mar. 6, 2013;77(5):825-43. doi: 10.1016/j.neuron.2013.02.022.
- Noack et al., Epitranscriptomics: A New Regulatory Mechanism of Brain Development and Function. *Front Neurosci*. Feb. 20, 2018;12:85. doi: 10.3389/fnins.2018.00085. 9 pages.
- Pellegrini et al., Insights into DNA recombination from the structure of a RAD51-BRCA2 complex. *Nature*. Nov. 21, 2002;420(6913):287-93. doi: 10.1038/nature01230. Epub Nov. 10, 2002.
- Perez-Palma et al., Simple ClinVar: an interactive web server to explore and retrieve gene and disease variants aggregated in ClinVar database. *Nucleic Acids Res*. Jul. 2, 2019;47(W1):W99-W105. doi: 10.1093/nar/gkz411.
- Prasad et al., Visualizing the assembly of human Rad51 filaments on double-stranded DNA. *J Mol Biol*. Oct. 27, 2006;363(3):713-28. doi: 10.1016/j.jmb.2006.08.046. Epub Aug. 22, 2006.
- Rajagopal et al., High-throughput mapping of regulatory DNA. *Nat Biotechnol*. Feb. 2016;34(2):167-74. doi: 10.1038/nbt.3468. Epub Jan. 25, 2016.
- Richardson et al., CRISPR-Cas9 genome editing in human cells occurs via the Fanconi anemia pathway. *Nat Genet*. Aug. 2018;50(8):1132-1139. doi: 10.1038/s41588-018-0174-0. Epub Jul. 27, 2018.
- Richardson et al., Frequent chromosomal translocations induced by DNA double-strand breaks. *Nature*. Jun. 8, 2000;405(6787):697-700. doi: 10.1038/35015097.
- Safari et al., CRISPR Cpf1 proteins: structure, function and implications for genome editing. *Cell Biosci*. May 9, 2019;9:36. doi: 10.1186/s13578-019-0298-7.
- San Filippo et al., Mechanism of eukaryotic homologous recombination. *Annu Rev Biochem*. 2008;77:229-57. doi: 10.1146/annurev.biochem.77.061306.125255.
- Schlacher et al., Double-strand break repair-independent role for BRCA2 in blocking stalled replication fork degradation by MRE11. *Cell*. May 13, 2011;145(4):529-42. doi: 10.1016/j.cell.2011.03.041. Erratum in: *Cell*. Jun. 10, 2011;145(6):993.
- Shen et al., Efficient genome modification by CRISPR-Cas9 nickase with minimal off-target effects. *Nat Methods*. Apr. 2014; 11(4):399-402. doi: 10.1038/nmeth.2857. Epub Mar. 2, 2014.
- Song et al., RS-1 enhances CRISPR/Cas9- and TALEN-mediated knock-in efficiency. *Nat Commun*. Jan. 28, 2016;7:10548. doi: 10.1038/ncomms10548.
- Stark et al., ATP hydrolysis by mammalian RAD51 has a key role during homology-directed DNA repair. *J Biol Chem*. Jun. 7, 2002;277(23):20185-94. doi: 10.1074/jbc.M112132200. Epub Mar. 28, 2002.
- Tan et al., Engineering of high-precision base editors for site-specific single nucleotide replacement. *Nat Commun*. Jan. 25, 2019;10(1):439. doi: 10.1038/s41467-018-08034-8. Erratum in: *Nat Commun*. May 1, 2019;10(1):2019.
- Thompson et al., The Future of Multiplexed Eukaryotic Genome Engineering. *ACS Chem Biol*. Feb. 16, 2018;13(2):313-325. doi: 10.1021/acscchembio.7b00842. Epub Dec. 28, 2017.
- Trojan et al., Functional analysis of hMLH1 variants and HNPCC-related mutations using a human expression system. *Gastroenterology*. Jan. 2002;122(1):211-9. doi: 10.1053/gast.2002.30296.
- Tsai et al., GUIDE-seq enables genome-wide profiling of off-target cleavage by CRISPR-Cas nucleases. *Nat Biotechnol*. Feb. 2015;33(2):187-197. doi: 10.1038/nbt.3117. Epub Dec. 16, 2014.
- Vakulskas et al., A high-fidelity Cas9 mutant delivered as a ribonucleoprotein complex enables efficient gene editing in human hematopoietic stem and progenitor cells. *Nat Med*. Aug. 2018;24(8):1216-1224. doi: 10.1038/s41591-018-0137-0. Epub Aug. 6, 2018.
- Van Den Oord et al., Pixel Recurrent Neural Networks. *Proceedings of the 33rd International Conference on Machine Learning. Journal of Machine Learning Research*. Aug. 19, 2016. vol. 48. 11 pages.
- Varshney et al., The regulation and functions of DNA and RNA G-quadruplexes. *Nat Rev Mol Cell Biol*. Aug. 2020;21(8):459-474. doi: 10.1038/s41580-020-0236-x. Epub Apr. 20, 2020.
- Wills et al., Pseudoknot-dependent read-through of retroviral gag termination codons: importance of sequences in the spacer and loop 2. *EMBO J*. Sep. 1, 1994;13(17):4137-44. doi: 10.1002/j.1460-2075.1994.tb06731.x.
- Wu et al., A novel SCN9A mutation responsible for primary erythromelalgia and is resistant to the treatment of sodium channel blockers. *PLoS One*. 2013;8(1):e55212. doi: 10.1371/journal.pone.0055212. Epub Jan. 31, 2013. 15 pages.
- Yamane et al., Deep-sequencing identification of the genomic targets of the cytidine deaminase AID and its cofactor RPA in B lymphocytes. *Nat Immunol*. Jan. 2011;12(1):62-9. doi: 10.1038/ni.1964. Epub Nov. 28, 2010.
- Yang et al., A Tale of Two Moieties: Rapidly Evolving CRISPR/Cas-Based Genome Editing. *Trends Biochem Sci*. Oct. 2020;45(10):874-888. doi: 10.1016/j.tibs.2020.06.003. Epub Jun. 30, 2020.
- Yang et al., BRCA2 function in DNA binding and recombination from a BRCA2-DSS1-ssDNA structure. *Science*. Sep. 13, 2002;297(5588):1837-48. doi: 10.1126/science.297.5588.1837.
- Yang et al., Genome-wide inactivation of porcine endogenous retroviruses (PERVs). *Science*. Nov. 27, 2015;350(6264):1101-4. doi: 10.1126/science.aad1191. Epub Oct. 11, 2015.
- Yang et al., The BRCA2 homologue Brh2 nucleates RAD51 filament formation at a dsDNA-ssDNA junction. *Nature*. Feb. 10, 2005;433(7026):653-7. doi: 10.1038/nature03234.
- Yu et al., Dynamic control of Rad51 recombinase by self-association and interaction with BRCA2. *Mol Cell*. Oct. 2003;12(4):1029-41. doi: 10.1016/s1097-2765(03)00394-0.
- Zhang et al., Efficient precise knockin with a double cut HDR donor after CRISPR/Cas9-mediated double-stranded DNA cleavage. *Genome Biol*. Feb. 20, 2017;18(1):35. doi: 10.1186/s13059-017-1164-8.
- Zolotukhin et al., Production and purification of serotype 1, 2, and 5 recombinant adeno-associated viral vectors. *Methods*. Oct. 2002;28(2):158-67. doi: 10.1016/s1046-2023(02)00220-7.
- International Preliminary Report on Patentability mailed Jun. 25, 2020 in connection with PCT/US2018/065886.
- [No Author Listed] HyPhy—Hypothesis testing using Phylogenies. Last modified Apr. 21, 2017. Accessed online via http://hyphy.org/w/index.php/Main_Page on Apr. 28, 2021.
- [No Author Listed] NCBI Reference Sequence: WP_087959824.1. Oct. 9, 2019. 2 pages.
- [No Author Listed] Score result for SEQ 355 to W02017032580. Muir et al. 2016.

(56)

References Cited

OTHER PUBLICATIONS

- [No Author Listed], Gag-Pol polyprotein. UniProtKB/Swiss-Prot No. P03355.5. Sep. 18, 2019. 18 pages.
- [No Author Listed], *Homo sapiens* signal transducer and activator of transcription 3 (STAT3), transcript variant 1, mRNA. NCBI Ref Seq No. NM_139276.2. Retrieved from https://www.ncbi.nlm.nih.gov/nucleotide/nm_139276.2. Feb. 26, 2020. 8 pages.
- Addgene Plasmid # 44246. pCas9-humanized, 2017, Stanley Qi.
- Addgene Plasmid # 73021. PCMV-BE3, 2017, David Liu.
- Addgene Plasmid # 79620. pcDNA3.1_pCMV-nCas-PmCDA1-ugi pH1-gRNA(HPRT), 2017, Akihiko Kondo.
- Ai et al., C-terminal Loop Mutations Determine Folding and Secretion Properties of PCSK9. *iMedPub J: Biochem Mol Biol J*. Nov. 5, 2016;2(3):17. doi: 10.21767/2471-8084.100026. 12 pages.
- Amrann et al., Tightly regulated tac promoter vectors useful for the expression of unfused and fused proteins in *Escherichia coli*. *Gene*. Sep. 30, 1988;69(2):301-15.
- Anzalone et al., Search-and-replace genome editing without double-strand breaks or donor DNA. *Nature*. Dec. 2019;576(7785):149-157 and Suppl. Info. doi: 10.1038/s41586-019-1711-4. Epub Oct. 21, 2019. 72 pages.
- Basila et al., Minimal 2'-O-methyl phosphorothioate linkage modification pattern of synthetic guide RNAs for increased stability and efficient CRISPR-Cas9 gene editing avoiding cellular toxicity. *PLoS One*. Nov. 27, 2017;12(11):e0188593. doi: 10.1371/journal.pone.0188593.
- Beaudry et al., Directed evolution of an RNA enzyme. *Science*. Jul. 31, 1992;257(5070):635-41. doi: 10.1126/science.1496376.
- Bertsimas et al., Simulated annealing. *Statistical Science*. Feb. 1993;8(1):10-15. doi: 10.1214/ss/1177011077.
- Bibikova et al., Targeted chromosomal cleavage and mutagenesis in *Drosophila* using zinc-finger nucleases. *Genetics*. Jul. 2002;161(3):1169-75. doi: 10.1093/genetics/161.3.1169.
- Blauw et al., SMN1 gene duplications are associated with sporadic ALS. *Neurology*. Mar. 13, 2012;78(11):776-80. doi: 10.1212/WNL.0b013e318249f697. Epub Feb. 8, 2012.
- Bogdanove et al., TAL effectors: customizable proteins for DNA targeting. *Science*. Sep. 30, 2011;333(6051):1843-6. doi: 10.1126/science.1204094.
- Brutlag et al., Improved sensitivity of biological sequence database searches. *Comput Appl Biosci*. Jul. 1990;6(3):237-45. doi: 10.1093/bioinformatics/6.3.237.
- Buckley et al., Targeting the von Hippel-Lindau E3 ubiquitin ligase using small molecules to disrupt the VHL/HIF-1 α interaction. *J Am Chem Soc*. Mar. 14, 2012;134(10):4465-8. doi: 10.1021/ja209924v. Epub Feb. 27, 2012.
- Böck et al., Selenocysteine: the 21st amino acid. *Mol Microbiol*. Mar. 1991;5(3):515-20.
- Canchaya et al., Genome analysis of an inducible prophage and prophage remnants integrated in the *Streptococcus pyogenes* strain SF370. *Virology*. Oct. 25, 2002;302(2):245-58. doi: 10.1006/viro.2002.1570.
- Carlier et al., Burkholderia cenocepacia H111 Rhy-family protein. *Apr*. 16, 2015. Retrieved from the Internet via <https://www.ebi.ac.uk/ena/browser/api/embl/CDN65395.1?lineLimit=1000>. Last retrieved Apr. 26, 2021.
- Carlier et al., Genome Sequence of Burkholderia cenocepacia H111, a Cystic Fibrosis Airway Isolate. *Genome Announc*. Apr. 10, 2014;2(2):e00298-14. doi: 10.1128/genomeA.00298-14.
- Cartegni et al., Determinants of exon 7 splicing in the spinal muscular atrophy genes, SMN1 and SMN2. *Am J Hum Genet*. Jan. 2006;78(1):63-77. doi: 10.1086/498853. Epub Nov. 16, 2005.
- Chang et al., Degradation of survival motor neuron (SMN) protein is mediated via the ubiquitin/proteasome pathway. *Neurochem Int*. Dec. 2004;45(7):1107-12. doi: 10.1016/j.neuint.2004.04.005.
- Chatterjee et al., A Cas9 with PAM recognition for adenine dinucleotides. *Nat Commun*. May 18, 2020;11(1):2474. doi: 10.1038/s41467-020-16117-8.
- Chen et al., Dynamic imaging of genomic loci in living human cells by an optimized CRISPR/Cas system. *Cell*. Dec. 19, 2013;155(7):1479-91. doi: 10.1016/j.cell.2013.12.001. Erratum in: *Cell*. Jan. 16, 2014;156(1-2):373.
- Cheng et al., [Cloning, expression and activity identification of human innate immune protein apolipoprotein B mRNA editing enzyme catalytic subunit 3A(APOBEC3A)]. Xi Bao Yu Fen Zi Mian Yi Xue Za Zhi. *Chinese Journal of Cellular and Molecular Immunology*. Feb. 2017;33(2):179-84. Chinese.
- Cho et al., A degron created by SMN2 exon 7 skipping is a principal contributor to spinal muscular atrophy severity. *Genes Dev*. Mar. 1, 2010;24(5):438-42. doi: 10.1101/gad.1884910.
- Chung-II et al., Artificial control of gene expression in mammalian cells by modulating RNA interference through aptamer-small molecule interaction. *RNA*. May 2006;12(5):710-6. Epub Apr. 10, 2006.
- Corcia et al., The importance of the SMN genes in the genetics of sporadic ALS. *Amyotroph Lateral Scler*. Oct. -Dec. 2009-Dec.;10(5-6):436-40. doi: 10.3109/17482960902759162.
- Corti et al., Genetic correction of human induced pluripotent stem cells from patients with spinal muscular atrophy. *Sci Transl Med*. Dec. 19, 2012;4(165):165ra162. doi: 10.1126/scitranslmed.3004108.
- Cucchiari et al., Enhanced expression of the central survival of motor neuron (SMN) protein during the pathogenesis of osteoarthritis. *J Cell Mol Med*. Jan. 2014;18(1):115-24. doi: 10.1111/jcmm.12170. Epub Nov. 17, 2013.
- Cui et al., Targeted integration in rat and mouse embryos with zinc-finger nucleases. *Nat Biotechnol*. Jan. 2011;29(1):64-7. doi: 10.1038/nbt.1731. Epub Dec. 12, 2010.
- D'Adda di Fagagna et al., The Gam protein of bacteriophage Mu is an orthologue of eukaryotic Ku. *EMBO Rep*. Jan. 2003;4(1):47-52.
- D'Ydewalle et al., The Antisense Transcript SMN-AS1 Regulates SMN Expression and Is a Novel Therapeutic Target for Spinal Muscular Atrophy. *Neuron*. Jan. 4, 2017;93(1):66-79 and Supplemental Information. doi: 10.1016/j.neuron.2016.11.033. Epub Dec. 22, 2016.
- Dahlman et al., Orthogonal gene knockout and activation with a catalytically active Cas9 nuclease. *Nat Biotechnol*. Nov. 2015;33(11):1159-61. doi: 10.1038/nbt.3390.
- De Felipe et al., Co-translational, intraribosomal cleavage of polypeptides by the foot-and-mouth disease virus 2A peptide. *J Biol Chem*. Mar. 28, 2003;278(13):11441-8. doi: 10.1074/jbc.M211644200. Epub Jan. 8, 2003.
- De La Peña et al., The Hammerhead Ribozyme: A Long History for a Short RNA. *Molecules*. Jan. 4, 2017;22(1):78. doi: 10.3390/molecules22010078.
- De Souza, Primer: genome editing with engineered nucleases. *Nat Methods*. Jan. 2012;9(1):27.
- De Wit et al., The Human CD4+ T Cell Response against Mumps Virus Targets a Broadly Recognized Nucleoprotein Epitope. *J Virol*. Mar. 5, 2019;93(6):e01883-18. doi: 10.1128/JVI.01883-18.
- Dickinson et al., A system for the continuous directed evolution of proteases rapidly reveals drug-resistance mutations. *Nat Commun*. Oct. 30, 2014;5:5352. doi: 10.1038/ncomms6352.
- Dumas et al., Designing logical codon reassignment—Expanding the chemistry in biology. *Chem Sci*. Jan. 1, 2015;6(1):50-69. doi: 10.1039/c4sc01534g. Epub Jul. 14, 2014. Review.
- East-Seletsky et al., Two distinct RNase activities of CRISPR-C2c2 enable guide-RNA processing and RNA detection. *Nature*. Oct. 2016;538(7624):270-3.
- Ekstrand et al., Frequent alterations of the PI3K/AKT/mTOR pathways in hereditary nonpolyposis colorectal cancer. *Fam Cancer*. Jun. 2010;9(2):125-9. doi: 10.1007/s10689-009-9293-1.
- Entin-Meer et al., The role of phenylalanine-119 of the reverse transcriptase of mouse mammary tumour virus in DNA synthesis, ribose selection and drug resistance. *Biochem J*. Oct. 15, 2002;367(Pt 2):381-91. doi: 10.1042/BJ20020712.
- Fang et al., The Menu of Features that Define Primary MicroRNAs and Enable De Novo Design of MicroRNA Genes. *Mol Cell*. Oct. 1, 2015;60(1):131-45. doi: 10.1016/j.molcel.2015.08.015. Epub Sep. 24, 2015.

(56)

References Cited

OTHER PUBLICATIONS

- Gao et al., DNA-guided genome editing using the *Natronobacterium gregoryi* Argonaute. *Nat Biotechnol.* Jul. 2016;34(7):768-73. doi: 10.1038/nbt.3547. Epub May 2, 2016.
- Gao et al., Treatment of autosomal dominant hearing loss by in vivo delivery of genome editing agents. *Nature.* Jan. 11, 2018;553(7687):217-221. doi: 10.1038/nature25164. Epub Dec. 20, 2017.
- Gaudelli et al., Directed evolution of adenine base editors with increased activity and therapeutic application. *Nat Biotechnol.* Jul. 2020;38(7):892-900. doi: 10.1038/s41587-020-0491-6. Epub Apr. 13, 2020.
- Gaudelli et al., Programmable base editing of A•T to G•C in genomic DNA without DNA cleavage. *Nature.* Nov. 23, 2017;551(7681):464-471. doi: 10.1038/nature24644. Epub Oct. 25, 2017. Erratum in: *Nature.* May 2, 2018.
- Geisberg et al., Global analysis of mRNA isoform half-lives reveals stabilizing and destabilizing elements in yeast. *Cell.* Feb. 13, 2014;156(4):812-24. doi: 10.1016/j.cell.2013.12.026.
- GENBANK Submission; NIH/NCBI, Accession No. NC_000001.11. Gregory et al., Jun. 6, 2016. 3 pages.
- GENBANK Submission; NIH/NCBI, Accession No. NM_206933.2. Khalaileh et al., Sep. 16, 2018. 12 pages.
- GENBANK Submission; NIH/NCBI, Accession No. NP_996816.2. Fu et al., Sep. 22, 2019. 9 pages.
- Grati et al., Localization of PDZD7 to the stereocilia ankle-link associates this scaffolding protein with the Usher syndrome protein network. *J Neurosci.* Oct. 10, 2012;32(41):14288-93. doi: 10.1523/JNEUROSCI.3071-12.2012.
- Green et al., Characterization of the mechanical unfolding of RNA pseudoknots. *J Mol Biol.* Jan. 11, 2008;375(2):511-28. doi: 10.1016/j.jmb.2007.05.058. Epub May 26, 2007.
- Hagen et al., A high rate of polymerization during synthesis of mouse mammary tumor virus DNA alleviates hypermutation by APOBEC3 proteins. *PLoS Pathog.* Feb. 15, 2019;15(2):e1007533. doi: 10.1371/journal.ppat.1007533.
- Hawley-Nelson et al., Transfection of Cultured Eukaryotic Cells Using Cationic Lipid Reagents. *Curr Prot Mol Biol.* Jan. 2008;9.4. 1-9.4.17. doi: 10.102/0471142727.mb0904s81. 17 pages.
- Hendel et al., Chemically modified guide RNAs enhance CRISPR-Cas genome editing in human primary cells. *Nat Biotechnol.* Sep. 2015;33(9):985-989. doi: 10.1038/nbt.3290. Epub Jun. 29, 2015. Author Manuscript. 14 pages.
- Heyer et al., Regulation of homologous recombination in eukaryotes. *Annu Rev Genet.* 2010;44:113-39. doi: 10.1146/annurev-genet-051710-150955. Author Manuscript. 33 pages.
- Hockemeyer et al., Genetic engineering of human pluripotent cells using TALE nucleases. *Nat Biotechnol.* Jul. 7, 2011;29(8):731-4. doi: 10.1038/nbt.1927.
- Hou et al., Efficient genome engineering in human pluripotent stem cells using Cas9 from *Neisseria meningitidis*. *Proc Natl Acad Sci U S A.* Sep. 24, 2013;110(39):15644-9. doi: 10.1073/pnas.1313587110. Epub Aug. 12, 2013.
- Houck-Loomis et al., An equilibrium-dependent retroviral mRNA switch regulates translational recoding. *Nature.* Nov. 27, 2011;480(7378):561-4. doi: 10.1038/nature10657.
- Houseley et al., The many pathways of RNA degradation. *Cell.* Feb. 20, 2009;136(4):763-76. doi: 10.1016/j.cell.2009.01.019.
- Hänsel-Hertsch et al., DNA G-quadruplexes in the human genome: detection, functions and therapeutic potential. *Nat Rev Mol Cell Biol.* May 2017;18(5):279-284. doi: 10.1038/nrm.2017.3. Epub Feb. 22, 2017.
- Ibrahim et al., RNA recognition by 3'-to-5' exonucleases: the substrate perspective. *Biochim Biophys Acta.* Apr. 2008;1779(4):256-65. doi: 10.1016/j.bbagg.2007.11.004. Epub Dec. 3, 2007.
- Iyama et al., DNA repair mechanisms in dividing and non-dividing cells. *DNA Repair (Amst).* Aug. 2013;12(8):620-36. doi: 10.1016/j.dnarep.2013.04.015. Epub May 16, 2013.
- Jakimo et al., A Cas9 with Complete PAM Recognition for Adenine Dinucleotides. *bioRxiv preprint.* Sep. 27, 2018. doi.org/10.1101/429654. 29 pages.
- Jeggo, DNA breakage and repair. *Adv Genet.* 1998;38:185-218. doi: 10.1016/s0065-2660(08)60144-3.
- Jost et al., Titrating gene expression using libraries of systematically attenuated CRISPR guide RNAs. *Nat Biotechnol.* Mar. 2020;38(3):355-364. doi: 10.1038/s41587-019-0387-5. Epub Jan. 13, 2020.
- King et al., No gain, no pain: Nav1.7 as an analgesic target. *ACS Chem Neurosci.* Sep. 17, 2014;5(9):749-51. doi: 10.1021/cn500171p. Epub Aug. 11, 2014.
- Klein et al., Cocystal structure of a class I preQ1 riboswitch reveals a pseudoknot recognizing an essential hypermodified nucleobase. *Nat Struct Mol Biol.* Mar. 2009;16(3):343-4. doi: 10.1038/nsmb.1563. Epub Feb. 22, 2009.
- Kleiner et al., In vitro selection of a DNA-templated small-molecule library reveals a class of macrocyclic kinase inhibitors. *J Am Chem Soc.* Aug. 25, 2010;132(33):11779-91. doi: 10.1021/ja104903x.
- Klement et al., Discrimination between bacteriophage T3 and T7 promoters by the T3 and T7 RNA polymerases depends primarily upon a three base-pair region located 10 to 12 base-pairs upstream from the start site. *J Mol Biol.* Sep. 5, 1990;215(1):21-9.
- Konishi et al., Amino acid substitutions away from the RNase H catalytic site increase the thermal stability of Moloney murine leukemia virus reverse transcriptase through RNase H inactivation. *Biochem Biophys Res Commun.* Nov. 14, 2014;454(2):269-74. doi: 10.1016/j.bbrc.2014.10.044. Epub Oct. 17, 2014.
- Ku et al., Nucleic Acid Aptamers: An Emerging Tool for Biotechnology and Biomedical Sensing. *Sensors (Basel).* Jul. 6, 2015;15(7):16281-313. doi: 10.3390/s150716281.
- Kwok et al., G-Quadruplexes: Prediction, Characterization, and Biological Application. *Trends Biotechnol.* Oct. 2017;35(10):997-1013. doi: 10.1016/j.tibtech.2017.06.012. Epub Jul. 26, 2017.
- Langer et al., Chemical and Physical Structure of Polymers as Carriers for Controlled Release of Bioactive Agents: A Review. *J Macromol Sci, Part C, 1983;23(1):61-126.* doi: 10.1080/07366578308079439.
- Le et al., SMNDelta7, the major product of the centromeric survival motor neuron (SMN2) gene, extends survival in mice with spinal muscular atrophy and associates with full-length SMN. *Hum Mol Genet.* Mar. 15, 2005;14(6):845-57. doi: 10.1093/hmg/ddi078. Epub Feb. 9, 2005.
- Lee et al., Single C-to-T substitution using engineered APOBEC3G-nCas9 base editors with minimum genome- and transcriptome-wide off-target effects. *Sci Adv.* Jul. 15, 2020;6(29):eaba1773. doi: 10.1126/sciadv.aba1773. 13 pages.
- Lee et al., Single C-to-T substitution using engineered APOBEC3G-nCas9 base editors with minimum genome- and transcriptome-wide off-target effects. *Sci Adv.* Jul. 15, 2020;6(29):eaba1773. doi: 10.1126/sciadv.aba1773. 13 pages.
- Lefebvre et al., Identification and characterization of a spinal muscular atrophy-determining gene. *Cell.* Jan. 13, 1995;80(1):155-65. doi: 10.1016/0092-8674(95)90460-3.
- Lesinski et al., The potential for targeting the STAT3 pathway as a novel therapy for melanoma. *Future Oncol.* Jul. 2013;9(7):925-7. doi: 10.2217/fon.13.83. Author Manuscript. 4 pages.
- Lin et al., Base editing-mediated splicing correction therapy for spinal muscular atrophy. *Cell Res.* Jun. 2020;30(6):548-550. doi: 10.1038/s41422-020-0304-y. Epub Mar. 24, 2020.
- Liu et al., Usherin is required for maintenance of retinal photoreceptors and normal development of cochlear hair cells. *Proc Natl Acad Sci U S A.* Mar. 13, 2007;104(11):4413-8. doi: 10.1073/pnas.0610950104. Epub Mar. 5, 2007.
- Lorson et al., A single nucleotide in the SMN gene regulates splicing and is responsible for spinal muscular atrophy. *Proc Natl Acad Sci U S A.* May 25, 1999;96(11):6307-11. doi: 10.1073/pnas.96.11.6307.
- Lutz et al., Postsymptomatic restoration of SMN rescues the disease phenotype in a mouse model of severe spinal muscular atrophy. *J Clin Invest.* Aug. 2011;121(8):3029-41. doi: 10.1172/JCI57291. Epub Jul. 25, 2011.
- MacFadden et al., Mechanism and structural diversity of exoribonuclease-resistant RNA structures in flaviviral RNAs. *Nat Commun.* Jan. 9, 2018;9(1):119. doi: 10.1038/s41467-017-02604-y.
- Maerker et al., A novel Usher protein network at the periciliary reloading point between molecular transport machineries in verte-

(56)

References Cited

OTHER PUBLICATIONS

- brate photoreceptor cells. *Hum Mol Genet.* Jan. 1, 2008;17(1):71-86. doi: 10.1093/hmg/ddm285. Epub Sep. 28, 2007.
- Mangeot et al., Genome editing in primary cells and in vivo using viral-derived Nanoblades loaded with Cas9-sgRNA ribonucleoproteins. *Nat Commun.* Jan. 3, 2019;10(1):45. doi: 10.1038/s41467-018-07845-z.
- Marcovitz et al., Frustration in protein-DNA binding influences conformational switching and target search kinetics. *Proc Natl Acad Sci U S A.* Nov. 1, 2011;108(44):17957-62. doi: 10.1073/pnas.1109594108. Epub Oct. 14, 2011.
- Mendell et al., Single-Dose Gene-Replacement Therapy for Spinal Muscular Atrophy. *N Engl J Med.* Nov. 2, 2017;377(18):1713-1722. doi: 10.1056/NEJMoal706198.
- Millevoi et al., G-quadruplexes in RNA biology. *Wiley Interdiscip Rev RNA.* Jul.-Aug. 2012;3(4):495-507. doi: 10.1002/wrna.1113. Epub Apr. 4, 2012.
- Monani et al., A single nucleotide difference that alters splicing patterns distinguishes the SMA gene SMN1 from the copy gene SMN2. *Hum Mol Genet.* Jul. 1999;8(7):1177-83. doi: 10.1093/hmg/8.7.1177.
- Murray et al., Selective vulnerability of motor neurons and disassociation of pre- and post-synaptic pathology at the neuromuscular junction in mouse models of spinal muscular atrophy. *Hum Mol Genet.* Apr. 1, 2008;17(7):949-62. doi: 10.1093/hmg/ddm367. Epub Dec. 8, 2007.
- Nelson et al., In vivo genome editing improves muscle function in a mouse model of Duchenne muscular dystrophy. *Science.* Jan. 22, 2016;351(6271):403-7. doi: 10.1126/science.aad5143. Epub Dec. 31, 2015.
- Nomura et al., Synthetic mammalian riboswitches based on guanine aptazyme. *Chem Commun (Camb).* Jul. 21, 2012;48(57):7215-7. doi: 10.1039/c2cc33140c. Epub Jun. 13, 2012.
- Ottesen, ISS-N1 makes the First FDA-approved Drug for Spinal Muscular Atrophy. *Transl Neurosci.* Jan. 26, 2017;8:1-6. doi: 10.1515/tnci-2017-0001.
- Pandey et al., Effect of loops and G-quartets on the stability of RNA G-quadruplexes. *J Phys Chem B.* Jun. 13, 2013;117(23):6896-905. doi: 10.1021/jp401739m. Epub May 29, 2013. Supplementary Information, 21 pages.
- Parente et al., Advances in spinal muscular atrophy therapeutics. *Ther Adv Neurol Disord.* Feb. 5, 2018;11:1756285618754501. doi: 10.1177/1756285618754501. 13 pages.
- Passini et al., Antisense oligonucleotides delivered to the mouse CNS ameliorate symptoms of severe spinal muscular atrophy. *Sci Transl Med.* Mar. 2, 2011;3(72):72ra18. doi: 10.1126/scitranslmed.3001777.
- Pendse et al., Exon 13-skipped USH2A protein retains functional integrity in mice, suggesting an exo-skipping therapeutic approach to treat USH2A-associated disease. *bioRxiv preprint.* Feb. 4, 2020. Retrieved from www.biorxiv.org. doi: 10.1101/2020.02.04.934240. 34 pages.
- Pendse et al., In Vivo Assessment of Potential Therapeutic Approaches for USH2A-Associated Diseases. *Adv Exp Med Biol.* 2019;1185:91-96. doi: 10.1007/978-3-030-27378-1_15.
- Pfeiffer et al., Mechanisms of DNA double-strand break repair and their potential to induce chromosomal aberrations. *Mutagenesis.* Jul. 2000;15(4):289-302. doi: 10.1093/mutage/15.4.289.
- Pieken et al., Kinetic characterization of ribonuclease-resistant 2'-modified hammerhead ribozymes. *Science.* Jul. 19, 1991;253(5017):314-7. doi: 10.1126/science.1857967.
- Pijlman et al., A highly structured, nuclease-resistant, noncoding RNA produced by flaviviruses is required for pathogenicity. *Cell Host Microbe.* Dec. 11, 2008;4(6):579-91. doi: 10.1016/j.chom.2008.10.007.
- Piotukh et al., Directed evolution of sortase A mutants with altered substrate selectivity profiles. *J Am Chem Soc.* Nov. 9, 2011;133(44):17536-9. doi: 10.1021/ja205630g. Epub Oct. 13, 2011.
- Porensky et al., A single administration of morpholino antisense oligomer rescues spinal muscular atrophy in mouse. *Hum Mol Genet.* Apr. 1, 2012;21(7):1625-38. doi: 10.1093/hmg/ddr600. Epub Dec. 20, 2011.
- Prashant et al., CAS9 transcriptional activators for target specificity screening and paired nickases for cooperative genome engineering. *Nature Biotechnology* 2013;31(9):833-8.
- Ramos et al., Age-dependent SMN expression in disease-relevant tissue and implications for SMA treatment. *J Clin Invest.* Nov. 1, 2019;129(11):4817-4831. doi: 10.1172/JCI124120.
- Reiners et al., Scaffold protein harmonin (USH1C) provides molecular links between Usher syndrome type 1 and type 2. *Hum Mol Genet.* Dec. 15, 2005;14(24):3933-43. doi: 10.1093/hmg/ddi417. Epub Nov. 21, 2005.
- Robert et al., Virus-Like Particles Derived from HIV-1 for Delivery of Nuclear Proteins: Improvement of Production and Activity by Protein Engineering. *Mol Biotechnol.* Jan. 2017;59(1):9-23. doi: 10.1007/s12033-016-9987-1.
- Rodriguez-Muela et al., Single-Cell Analysis of SMN Reveals Its Broader Role in Neuromuscular Disease. *Cell Rep.* Feb. 7, 2017;18(6):1484-1498 and Supplemental Information. doi: 10.1016/j.celrep.2017.01.035.
- Saayman et al., The therapeutic application of CRISPR/Cas9 technologies for HIV. *Expert Opin Biol Ther.* Jun. 2015;15(6):819-30. doi: 10.1517/14712598.2015.1036736. Epub Apr. 12, 2015.
- Sadowski, The F1p recombinase of the 2-microns plasmid of *Saccharomyces cerevisiae*. *Prog Nucleic Acid Res Mol Biol.* 1995;51:53-91.
- Sanjurjo-Soriano et al., Genome Editing in Patient iPSCs Corrects the Most Prevalent USH2A Mutations and Reveals Intriguing Mutant mRNA Expression Profiles. *Mol Ther Methods Clin Dev.* Nov. 27, 2019;17:156-173. doi: 10.1016/j.omtm.2019.11.016.
- Schechner et al., Multiplexable, locus-specific targeting of long RNAs with CRISPR-Display. *Nat Methods.* Jul. 2015;12(7):664-70. doi: 10.1038/nmeth.3433. Epub Jun. 1, 2015. Author manuscript entitled CRISPR Display: A modular method for locus-specific targeting of long noncoding RNAs and synthetic RNA devices in vivo.
- Schrank et al., Inactivation of the survival motor neuron gene, a candidate gene for human spinal muscular atrophy, leads to massive cell death in early mouse embryos. *Proc Natl Acad Sci USA.* Sep. 2, 1997;94(18):9920-5. doi: 10.1073/pnas.94.18.9920.
- Schöller et al., Interactions, localization, and phosphorylation of the m6A generating METTL3-METTL14-WTAP complex. *RNA.* Apr. 2018;24(4):499-512. doi: 10.1261/rna.064063.117. Epub Jan. 18, 2018.
- Score Results for Shimomura et al., Complete Genome Sequencing and Analysis of a Lancefield Group G Rt *Streptococcus dysgalactiae* Subsp. *equisimilis* Strain Causing Streptococcal RT Toxic Shock Syndrome (Stss). *RL BMC Genomics.* 2011;12:17-17. 3 pages.
- Shingledecker et al., Molecular dissection of the *Mycobacterium tuberculosis* RecA intein: design of a minimal intein and of a trans-splicing system involving two intein fragments. *Gene.* Jan. 30, 1998;207(2):187-95. doi: 10.1016/s0378-1119(97)00624-0.
- Singh et al., Splicing of a critical exon of human Survival Motor Neuron is regulated by a unique silencer element located in the last intron. *Mol Cell Biol.* Feb. 2006;26(4):1333-46. doi: 10.1128/MCB.26.4.1333-1346.2006.
- Sorusch et al., Characterization of the ternary Usher syndrome SANS/ush2a/whirlin protein complex. *Hum Mol Genet.* Mar. 15, 2017;26(6):1157-1172. doi: 10.1093/hmg/ddx027.
- Steckelberg et al., A folded viral noncoding RNA blocks host cell exoribonucleases through a conformationally dynamic RNA structure. *Proc Natl Acad Sci U S A.* Jun. 19, 2018;115(25):6404-6409. doi: 10.1073/pnas.1802429115. Epub Jun. 4, 2018.
- Sumner et al., Two breakthrough gene-targeted treatments for spinal muscular atrophy: challenges remain. *J Clin Invest.* Aug. 1, 2018;128(8):3219-3227. doi: 10.1172/JCI121658. Epub Jul. 9, 2018.
- Tabebordbar et al., In vivo gene editing in dystrophic mouse muscle and muscle stem cells. *Science.* Jan. 22, 2016;351(6271):407-411. doi: 10.1126/science.aad5177. Epub Dec. 31, 2015.
- Talbot et al., Spinal muscular atrophy. *Semin Neurol.* Jun. 2001;21(2):189-97. doi: 10.1055/s-2001-15264.

(56)

References Cited

OTHER PUBLICATIONS

- Uniprot Consortium, UniProt: the universal protein knowledgebase. *Nucleic Acids Res.* Mar. 16, 2018;46(5):2699. doi: 10.1093/nar/gky092. Erratum for: *Nucleic Acids Res.* Jan. 4, 2017;45(D1):D158-D169.
- Van Brunt et al., *Molecular Farming: Transgenic Animals as Bioreactors*. Biotechnology (NY). 1988;6(10):1149-1154. doi: 10.1038/nbt1088-1149.
- Van Duyne et al., Teaching Cre to follow directions. *Proc Natl Acad Sci U S A.* Jan. 6, 2009;106(1):4-5. doi: 10.1073/pnas.0811624106. Epub Dec. 31, 2008.
- Van Swieten et al., A mutation in the fibroblast growth factor 14 gene is associated with autosomal dominant cerebellar ataxia [corrected]. *Am J Hum Genet.* Jan. 2003;72(1):191-9. Epub Dec. 13, 2002.
- Van Wijk et al., Identification of 51 novel exons of the Usher syndrome type 2A (USH2A) gene that encode multiple conserved functional domains and that are mutated in patients with Usher syndrome type II. *Am J Hum Genet.* Apr. 2004;74(4):738-44. doi: 10.1086/383096. Epub Mar. 10, 2004.
- Vellore et al., A group II intron-type open reading frame from the thermophile *Bacillus* (*Geobacillus*) *stearothermophilus* encodes a heat-stable reverse transcriptase. *Appl Environ Microbiol.* Dec. 2004;70(12):7140-7. doi: 10.1128/AEM.70.12.7140-7147.2004.
- Vidal et al., Yeast forward and reverse 'n'-hybrid systems. *Nucleic Acids Res.* Feb. 15, 1999;27(4):919-29. doi: 10.1093/nar/27.4.919.
- Villiger et al., Treatment of a metabolic liver disease by in vivo genome base editing in adult mice. *Nat Med.* Oct. 2018;24(10):1519-1525. doi: 10.1038/s41591-018-0209-1. Epub Oct. 8, 2018.
- Walton et al., Unconstrained genome targeting with near-PAMless engineered CRISPR-Cas9 variants. *Science.* Apr. 17, 2020;368(6488):290-296. doi: 10.1126/science.aba8853. Epub Mar. 26, 2020.
- Wirth et al., Mildly affected patients with spinal muscular atrophy are partially protected by an increased SMN2 copy number. *Hum Genet.* May 2006;119(4):422-8. doi: 10.1007/s00439-006-0156-7. Epub Mar. 1, 2006.
- Woo et al., Gene activation of SMN by selective disruption of lncRNA-mediated recruitment of PRC2 for the treatment of spinal muscular atrophy. *Proc Natl Acad Sci U S A.* Feb. 21, 2017;114(8):E1509-E1518. doi: 10.1073/pnas.1616521114. Epub Feb. 13, 2017.
- Wu et al., Human single-stranded DNA binding proteins: guardians of genome stability. *Acta Biochim Biophys Sin (Shanghai).* Jul. 2016;48(7):671-7. doi: 10.1093/abbs/gmw044. Epub May 23, 2016.
- Wu et al., MLV based viral-like-particles for delivery of toxic proteins and nuclear transcription factors. *Biomaterials.* Sep. 2014;35(29):8416-26. doi: 10.1016/j.biomaterials.2014.06.006. Epub Jul. 3, 2014.
- Wu et al., Widespread Influence of 3'-End Structures on Mammalian mRNA Processing and Stability. *Cell.* May 18, 2017;169(5):905-917.e11. doi: 10.1016/j.cell.2017.04.036.
- Yi et al., Engineering of TEV protease variants by yeast ER sequestration screening (YESS) of combinatorial libraries. *Proc Natl Acad Sci U S A.* Apr. 30, 2013;110(18):7229-34. doi: 10.1073/pnas.1215994110. Epub Apr. 15, 2013.
- Yu et al., Cytosine base editors with minimized unguided DNA and RNA off-target events and high on-target activity. *Nat Commun.* Apr. 28, 2020;11(1):2052. doi: 10.1038/s41467-020-15887-5.
- Zhang et al., Large genomic fragment deletions and insertions in mouse using CRISPR/Cas9. *PLoS One.* Mar. 24, 2015;10(3):e0120396. doi: 10.1371/journal.pone.0120396. 14 pages.
- [No Author Listed], MutL homolog 1. UniProtKB Acc. No. F1MPG0. May 3, 2011. Accessible at <https://rest.uniprot.org/unisave/F1MPG0?format=txt&versions=1>. 1 page.
- Acharya et al., hMSH2 forms specific mismatch-binding complexes with hMSH3 and hMSH6. *Proc Natl Acad Sci U S A.* Nov. 26, 1996;93(24):13629-34. doi: 10.1073/pnas.93.24.13629.
- Edraki et al., A Compact, High-Accuracy Cas9 with a Dinucleotide PAM for In Vivo Genome Editing. *Mol Cell.* Feb. 21, 2019;73(4):714-726.e4 and Supplemental Info. doi: 10.1016/j.molcel.2018.12.003. Epub Dec. 20, 2018.
- Fang et al., Human strand-specific mismatch repair occurs by a bidirectional mechanism similar to that of the bacterial reaction. *J Biol Chem.* Jun. 5, 1993;268(16):11838-44.
- Feng et al., Efficient genome editing in plants using a CRISPR/Cas system. *Cell Res.* Oct. 2013;23(10):1229-32. doi: 10.1038/cr.2013.114. Epub Aug. 20, 2013.
- Fishel et al., The human mutator gene homolog MSH2 and its association with hereditary nonpolyposis colon cancer. *Cell.* Dec. 3, 1993;75(5):1027-38. doi: 10.1016/0092-8674(93)90546-3. Erratum in: *Cell.* Apr. 8, 1994;77(1):1 p following 166.
- Geng et al., In vitro studies of DNA mismatch repair proteins. *Anal Biochem.* Jun. 15, 2011;413(2):179-84. doi: 10.1016/j.ab.2011.02.017. Epub Feb. 15, 2011.
- Genschel et al., Human exonuclease I is required for 5' and 3' mismatch repair. *J Biol Chem.* Apr. 12, 2002;277(15):13302-11. doi: 10.1074/jbc.M111854200. Epub Jan. 24, 2002.
- Genschel et al., Isolation of MutSbeta from human cells and comparison of the mismatch repair specificities of MutSbeta and MutSalphalpha. *J Biol Chem.* Jul. 31, 1998;273(31):19895-901. doi: 10.1074/jbc.273.31.19895. Erratum in: *J Biol Chem.* Oct. 9, 1998;273(41):27034.
- Gueneau et al., Structure of the MutLalpha C-terminal domain reveals how Mlh1 contributes to Pms1 endonuclease site. *Nat Struct Mol Biol.* Apr. 2013;20(4):461-8. doi: 10.1038/nsmb.2511. Epub Feb. 24, 2013.
- Guerrette et al., The interaction of the human MutL homologues in hereditary nonpolyposis colon cancer. *J Biol Chem.* Mar. 5, 1999;274(10):6336-41. doi: 10.1074/jbc.274.10.6336.
- Gupta et al., Mechanism of mismatch recognition revealed by human MutSbeta bound to unpaired DNA loops. *Nat Struct Mol Biol.* Dec. 18, 2011;19(1):72-8. doi: 10.1038/nsmb.2175.
- Iaccarino et al., hMSH2 and hMSH6 play distinct roles in mismatch binding and contribute differently to the ATPase activity of hMutSalphalpha. *EMBO J.* May 1, 1998;17(9):2677-86. doi: 10.1093/emboj/17.9.2677.
- Iyer et al., DNA mismatch repair: functions and mechanisms. *Chem Rev.* Feb. 2006;106(2):302-23. doi: 10.1021/cr0404794.
- Kadyrov et al., Endonucleolytic function of MutLalpha in human mismatch repair. *Cell.* Jul. 28, 2006;126(2):297-308. doi: 10.1016/j.cell.2006.05.039.
- Kunkel et al., DNA mismatch repair. *Annu Rev Biochem.* 2005;74:681-710. doi: 10.1146/annurev.biochem.74.082803.133243.
- Lahue et al., DNA mismatch correction in a defined system. *Science.* Jul. 14, 1989;245(4914):160-4. doi: 10.1126/science.2665076.
- Leach et al., Mutations of a mutS homolog in hereditary nonpolyposis colorectal cancer. *Cell.* Dec. 17, 1993;75(6):1215-25. doi: 10.1016/0092-8674(93)90330-s.
- Longworth, Expanding the Enzymatic Activity of the Programmable Endonuclease Cas9 in Zebrafish. Thesis. Rice University. Houston, TX. May 17, 2019. 41 pages.
- Lujan et al., Heterogeneous polymerase fidelity and mismatch repair bias genome variation and composition. *Genome Res.* Nov. 2014;24(11):1751-64. doi: 10.1101/gr.178335.114. Epub Sep. 12, 2014.
- Lyu et al., Virus-Like Particle Mediated CRISPR/Cas9 Delivery for Efficient and Safe Genome Editing. *Life (Basel).* Dec. 21, 2020;10(12):366. doi: 10.3390/life10120366.
- Mahoney et al., The Next Immune-Checkpoint Inhibitors: PD-1/PD-L1 Blockade in Melanoma. *Clin Ther.* Apr. 1, 2015;37(4):764-82. doi: 10.1016/j.clinthera.2015.02.018. Epub Mar. 29, 2015.
- Micozzi et al., Human cytidine deaminase: a biochemical characterization of its naturally occurring variants. *Int J Biol Macromol.* Feb. 2014;63:64-74. doi: 10.1016/j.ijbiomac.2013.10.029. Epub Oct. 29, 2013. Erratum in: *Int J Biol Macromol.* Feb. 2014;63:262.
- Parsons et al., Hypermutability and mismatch repair deficiency in RER+ tumor cells. *Cell.* Dec. 17, 1993;75(6):1227-36. doi: 10.1016/0092-8674(93)90331-j.

(56)

References Cited

OTHER PUBLICATIONS

- Petit et al., Powerful mutators lurking in the genome. *Philos Trans R Soc Lond B Biol Sci.* Mar. 12, 2009;364(1517):705-15. doi: 10.1098/rstb.2008.0272.
- Plotz et al., N-terminus of hMLH1 confers interaction of hMutLalpha and hMutLbeta with hMutSalpha. *Nucleic Acids Res.* Jun. 15, 2003;31(12):3217-26. doi: 10.1093/nar/gkg420.
- Räschle et al., Mutations within the hMLH1 and hPMS2 subunits of the human MutLalpha mismatch repair factor affect its ATPase activity, but not its ability to interact with hMutSalpha. *J Biol Chem.* Jun. 14, 2002;277(24):21810-20. doi: 10.1074/jbc.M108787200. Epub Apr. 10, 2002.
- Shcherbakova et al., Mutator phenotypes conferred by MLH1 overexpression and by heterozygosity for mlh1 mutations. *Mol Cell Biol.* Apr. 1999;19(4):3177-83. doi: 10.1128/MCB.19.4.3177.
- Strand et al., Destabilization of tracts of simple repetitive DNA in yeast by mutations affecting DNA mismatch repair. *Nature.* Sep. 16, 1993;365(6443):274-6. doi: 10.1038/365274a0. Erratum in: *Nature* Apr. 7, 1994;368(6471):569.
- Su et al., Mismatch specificity of methyl-directed DNA mismatch correction in vitro. *J Biol Chem.* May 15, 1988;263(14):6829-35. Erratum in: *J Biol Chem* Aug. 5, 1988;263(22):11015.
- Sugawara et al., Heteroduplex rejection during single-strand annealing requires Sgs1 helicase and mismatch repair proteins Msh2 and Msh6 but not Pms1. *Proc Natl Acad Sci U S A.* Jun. 22, 2004;101(25):9315-20. doi: 10.1073/pnas.0305749101. Epub Jun. 15, 2004.
- Supek et al., Differential DNA mismatch repair underlies mutation rate variation across the human genome. *Nature.* May 7, 2015;521(7550):81-4. doi: 10.1038/nature14173. Epub Feb. 23, 2015.
- Thomas et al., Heteroduplex repair in extracts of human HeLa cells. *J Biol Chem.* Feb. 25, 1991;266(6):3744-51.
- Tomer et al., Contribution of human mlh1 and pms2 ATPase activities to DNA mismatch repair. *J Biol Chem.* Jun. 14, 2002;277(24):21801-9. doi: 10.1074/jbc.M111342200. Epub Mar. 15, 2002.
- Tran et al., Hypermutability of homonucleotide runs in mismatch repair and DNA polymerase proofreading yeast mutants. *Mol Cell Biol.* May 1997;17(5):2859-65. doi: 10.1128/MCB.17.5.2859.
- Umar et al., DNA loop repair by human cell extracts. *Science.* Nov. 4, 1994;266(5186):814-6. doi: 10.1126/science.7973637.
- Warren et al., Structure of the human MutSalpha DNA lesion recognition complex. *Mol Cell.* May 25, 2007;26(4):579-92. doi: 10.1016/j.molcel.2007.04.018.
- Xi et al., C-terminal Loop Mutations Determine Folding and Secretion Properties of PCSK9. *Biochem Mol Biol J.* 2016;2(3):17. doi: 10.21767/2471-8084.100026. 12 pages.
- Zhang et al., Reconstitution of 5'-directed human mismatch repair in a purified system. *Cell.* Sep. 9, 2005;122(5):693-705. doi: 10.1016/j.cell.2005.06.027.
- [No Author Listed] NCBI Reference Sequence: WP_032188360.1. Apr. 6, 2015. 1 page.
- [No Author Listed], tRNA-specific adenosine deaminase [*Escherichia coli*]. GenBank Acc. No. CTS26096.1. Accessible at <https://www.ncbi.nlm.nih.gov/protein/CTS26096.1>. Aug. 22, 2015. 1 page.
- Alizadeh et al., HR9: An Important Cell Penetrating Peptide for Delivery of HCV NS3 DNA into HEK-293T Cells. *Avicenna J Med Biotechnol.* Jan.-Mar. 2020(1):44-51.
- Avidan et al., Expression and characterization of a recombinant novel reverse transcriptase of a porcine endogenous retrovirus. *Virology.* Mar. 15, 2003;307(2):341-57. doi: 10.1016/S0042-6822(02)00131-9.
- Bae et al., Heteroclitic CD33 peptide with enhanced anti-acute myeloid leukemic immunogenicity. *Clin Cancer Res.* Oct. 15, 2004;10(20):7043-52. doi: 10.1158/1078-0432.CCR-04-0322.
- Baños-Sanz et al., Crystal structure and functional insights into uracil-DNA glycosylase inhibition by phage Φ 29 DNA mimic protein p56. *Nucleic Acids Res.* Jul. 2013;41(13):6761-73. doi: 10.1093/nar/gkt395. Epub May 13, 2013.
- Cheriyian et al., Faster protein splicing with the Nostoc punctiforme DnaE intein using non-native extein residues. *J Biol Chem.* Mar. 1, 2013;288(9):6202-11. doi: 10.1074/jbc.M112.433094. Epub Jan. 10, 2013.
- Choi et al., Optimization of AAV expression cassettes to improve packaging capacity and transgene expression in neurons. *Mol Brain.* Mar. 1, 2014;7:17. doi: 10.1186/1756-6606-7-17.
- Damdindorj et al., A comparative analysis of constitutive promoters located in adeno-associated viral vectors. *PLoS One.* Aug. 29, 2014;9(8):e106472. doi: 10.1371/journal.pone.0106472.
- Ding et al., Gene therapy for cardiovascular disease. *Journal of Shanghai University (Natural Science Edition)*. 2016;3:270-9. DOI: 10.3969/j.issn.1007-2861.2016.03.013.
- Ekman et al., CRISPR-Cas9-Mediated Genome Editing Increases Lifespan and Improves Motor Deficits in a Huntington's Disease Mouse Model. *Mol Ther Nucleic Acids.* Sep. 6, 2019;17:829-839. doi: 10.1016/j.omtn.2019.07.009. Epub Jul. 26, 2019.
- Eriksen et al., Occlusion of the Ribosome Binding Site Connects the Translational Initiation Frequency, mRNA Stability and Premature Transcription Termination. *Front Microbiol.* Mar. 14, 2017;8:362. doi: 10.3389/fmicb.2017.00362.
- Fikes et al., Design of multi-epitope, analogue-based cancer vaccines. *Expert Opin Biol Ther.* Sep. 2003;3(6):985-93. doi: 10.1517/14712598.3.6.985.
- GenBank Submission; NIH/NCBI, Accession No. NP_060228.2. Bi et al., Dec. 21, 2005. 1 page.
- GenBank Submission; NIH/NCBI, Accession No. NP_062826.2. Bokar et al., Sep. 18, 2004. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. NP_066012.1. Ota et al., Apr. 3, 2005. 2 pages.
- GenBank Submission; NIH/NCBI, Accession No. WP_042518169.1. No Author, Feb. 10, 2015. 1 page.
- Hizi et al., Retroviral reverse transcriptases (other than those of HIV-1 and murine leukemia virus): a comparison of their molecular and biochemical properties. *Virus Res.* Jun. 2008;134(1-2):203-20. doi: 10.1016/j.virusres.2007.12.008. Epub Mar. 3, 2008.
- Houghton et al., Immunological validation of the EpiOptimizer program for streamlined design of heteroclitic epitopes. *Vaccine.* Jul. 20, 2007;25(29):5330-42. doi: 10.1016/j.vaccine.2007.05.008. Epub Jun. 4, 2007.
- Hwang et al., Heritable and precise zebrafish genome editing using a CRISPR-Cas system. *PLoS One.* Jul. 9, 2013;8(7):e68708. doi: 10.1371/journal.pone.0068708.
- Karimian et al., CRISPR/Cas9 novel therapeutic road for the treatment of neurodegenerative diseases. *Life Sci.* Oct. 15, 2020;259:118165. doi: 10.1016/j.lfs.2020.118165. Epub Jul. 29, 2020.
- Kirshenboim et al., Expression and characterization of a novel reverse transcriptase of the LTR retrotransposon Tfl. *Virology.* Sep. 30, 2007;366(2):263-76. doi: 10.1016/j.virol.2007.04.002. Epub May 23, 2007.
- Mariani et al., Species-specific exclusion of APOBEC3G from HIV-1 virions by Vif. *Cell.* Jul. 11, 2003;114(1):21-31. doi: 10.1016/S0092-8674(03)00515-4.
- Misra et al., An enzymatically active chimeric HIV-1 reverse transcriptase (RT) with the RNase-H domain of murine leukemia virus RT exists as a monomer. *J Biol Chem.* Apr. 17, 1998;273(16):9785-9. doi: 10.1074/jbc.273.16.9785.
- Nguyen Tran et al., Engineering domain-inlaid SaCas9 adenine base editors with reduced RNA off-targets and increased on-target DNA editing. *Nat Commun.* Sep. 25, 2020;11(1):4871. doi: 10.1038/s41467-020-18715-y.
- Niemeyer, C.M., Semisynthetic DNA-protein conjugates for biosensing and nanofabrication. *Angew Chem Int Ed Engl.* Feb. 8, 2010;49(7):1200-16. doi: 10.1002/anie.200904930.
- Nowak et al., Ty3 reverse transcriptase complexed with an RNA-DNA hybrid shows structural and functional asymmetry. *Nat Struct Mol Biol.* Apr. 2014;21(4):389-96. doi: 10.1038/nsmb.2785. Epub Mar. 9, 2014. Author Manuscript, 22 pages.
- Raaijmakers et al., CRISPR/Cas Applications in Myotonic Dystrophy: Expanding Opportunities. *Int J Mol Sci.* Jul. 27, 2019;20(15):3689. doi: 10.3390/ijms20153689.

(56)

References Cited

OTHER PUBLICATIONS

- Riddle et al., Frameshift suppression: a nucleotide addition in the anticodon of a glycine transfer RNA. *Nat New Biol.* Apr. 25, 1973;242(121):230-4. doi: 10.1038/newbio242230a0.
- Riddle et al., Frameshift suppressors. II. Genetic mapping and dominance studies. *J Mol Biol.* May 28, 1972;66(3):483-93. doi: 10.1016/0022-2836(72)90428-7.
- Riddle et al., Suppressors of frameshift mutations in *Salmonella typhimurium*. *J Mol Biol.* Nov. 28, 1970;54(1):131-44. doi: 10.1016/0022-2836(70)90451-1.
- Sadowski et al., The sequence-structure relationship and protein function prediction. *Curr Opin Struct Biol.* Jun. 2009;19(3):357-62. doi: 10.1016/j.sbi.2009.03.008. Epub May 4, 2009.
- Score Results for US 2014-0186919 A1 to Zhang et al. Aug. 2, 20148. 3 pages.
- Seffernick et al., Melamine deaminase and atrazine chlorohydrolase: 98 percent identical but functionally different. *J Bacteriol.* Apr. 2001; 183(8):2405-10. doi: 10.1128/JB.183.8.2405-2410.2001.
- Shalaby et al., Tissue-Specific Delivery of CRISPR Therapeutics: Strategies and Mechanisms of Non-Viral Vectors. *Int J Mol Sci.* Oct. 5, 2020;21(19):7353. doi: 10.3390/ijms21197353.
- Simon et al., Retrons and their applications in genome engineering. *Nucleic Acids Res.* Dec. 2, 2019;47(21):11007-11019. doi: 10.1093/nar/gkz865.
- Singh et al., Protein Engineering Approaches in the Post-Genomic Era. *Curr Protein Pept Sci.* 2018;19(1):5-15. doi: 10.2174/1389203718666161117114243.
- Studebaker et al., Depletion of uracil-DNA glycosylase activity is associated with decreased cell proliferation. *Biochem Biophys Res Commun.* Aug. 26, 2005;334(2):509-15. doi: 10.1016/j.bbrc.2005.06.118.
- Suh et al., Publisher Correction: Restoration of visual function in adult mice with an inherited retinal disease via adenine base editing. *Nat Biomed Eng.* Nov. 2020;4(11):1119. doi: 10.1038/s41551-020-00652-2. Erratum for: *Nat Biomed Eng.* Oct. 19, 2020.
- Tang et al., Identification of Dehalobacter reductive dehalogenases that catalyze dechlorination of chloroform, 1,1,1-trichloroethane and 1,1-dichloroethane. *Philos Trans R Soc Lond B Biol Sci.* Mar. 11, 2013;368(1616):20120318. doi: 10.1098/rstb.2012.0318.
- Toro et al., Comprehensive phylogenetic analysis of bacterial reverse transcriptases. *PLoS One.* Nov. 25, 2014;9(11):e114083. doi: 10.1371/journal.pone.0114083.
- Witkowski et al., Conversion of a beta-ketoacyl synthase to a malonyl decarboxylase by replacement of the active-site cysteine with glutamine. *Biochemistry.* Sep. 7, 1999;38(36):11643-50. doi: 10.1021/bi990993h.
- Yin et al., Optimizing genome editing strategy by primer-extension-mediated sequencing. *Cell Discov.* Mar. 26, 2019;5:18. doi: 10.1038/s41421-019-0088-8.
- Zhang et al., Propagated Perturbations from a Peripheral Mutation Show Interactions Supporting WW Domain Thermostability. *Structure.* Nov. 6, 2018;26(11):1474-1485.e5. doi: 10.1016/j.str.2018.07.014. Epub Sep. 6, 2018.
- [No Author Listed], CMP/dCMP-type deaminase domain-containing protein. Uniprot Accession No. A0A2Z6RZE9. Oct. 10, 2018. Accessible at <https://www.uniprot.org/uniprotkb/A0A2Z6RZE9/entry>. 8 pages.
- [No Author Listed], dCas9-5xPlat2AflD-P2A-scFvGcN4sfGFPETET1CD [Cloning vector pPlatTET-gRNA2]. GenBank No. BAY54124. Apr. 18, 2017. 5 pages.
- [No Author Listed], tRNA-specific adenosine deaminase [Candidatus *Moranella endobia* PCVAL]. GenBank Acc. No. AGJ61179.1. Accessible at <https://www.ncbi.nlm.nih.gov/protein/AGJ61179>. Jan. 30, 2014. 3 pages.
- [No Author Listed], tRNA-specific adenosine deaminase 2 [Terapene triunguis]. GenBank Acc. No. XP_024075810.1. Accessible at https://www.ncbi.nlm.nih.gov/protein/XP_024075810. Jul. 15, 2019. 2 pages.
- [No Author Listed], tRNA-specific adenosine deaminase TAD2 isoform X1 [*Oryza sativa japonica* Group]. GenBank Acc. No. XP_15631651.1. Accessible at <https://www.ncbi.nlm.nih.gov/protein/1002254769?sat=58&satkey=133677684>. Aug. 7, 2018. 2 pages.
- [No Author Listed], tRNA-specific adenosine deaminase TAD2 isoform X2 [*Panicum hallii*]. GenBank Acc. No. XP_025793740.1. Accessible at <https://www.ncbi.nlm.nih.gov/protein/025793740>. Jul. 27, 2018. 1 page.
- [No Author Listed], tRNA-specific adenosine deaminase. Uniprot Accession No. A0A1U7M801. May 10, 2017. Accessible at <https://www.uniprot.org/uniprotkb/A0A1U7M801/history>. 3 pages.
- [No Author Listed], tRNA-specific adenosine deaminase. Uniprot Accession No. A0A1Z4VPW4. Sep. 27, 2017 Accessible at <https://www.uniprot.org/uniprotkb/A0A1Z4VPW4/history>. 3 pages.
- [No Author Listed], tRNA-specific adenosine deaminase. Uniprot Accession No. A0A1Z9LY19. Oct. 25, 2017. Accessible at <https://www.uniprot.org/uniprotkb/A0A1Z9LY19/entry>. 12 pages.
- [No Author Listed], tRNA-specific adenosine deaminase. Uniprot Accession No. A0A2P5TOZ9. May 23, 20183. Accessible at <https://www.uniprot.org/uniprotkb/A0A2P5TOZ9/entry>. 10 pages.
- [No Author Listed], tRNA-specific adenosine deaminase. Uniprot Accession No. A0A4P6PH16. Jul. 31, 2019. Accessible at <https://www.uniprot.org/uniprotkb/A0A4P6PH16/entry>. 12 pages.
- [No Author Listed], tRNA-specific adenosine deaminase. Uniprot Accession No. A0A520SVM3. Oct. 16, 2019. Accessible at <https://www.uniprot.org/uniprotkb/A0A520SVM3/entry>. 10 pages.
- [No Author Listed], tRNA-specific adenosine deaminase. Uniprot Accession No. U2JU00. Nov. 13, 2013. Accessible at <https://www.uniprot.org/uniprotkb/U2JU00/entry>. 11 pages.
- Alves et al., Immunogenicity of the carcinoembryonic antigen derived peptide 694 in HLA-A2 healthy donors and colorectal carcinoma patients. *Cancer Immunol Immunother.* Nov. 2007;56(11):1795-805. doi: 10.1007/s00262-007-0323-2. Epub Apr. 20, 2007.
- Asemisen et al., Identification of a highly immunogenic HLA-A*01-binding T cell epitope of WT1. *Clin Cancer Res.* Dec. 15, 2006;12(24):7476-82. doi: 10.1158/1078-0432.CCR-06-1337.
- Attia et al., Autoimmunity correlates with tumor regression in patients with metastatic melanoma treated with anti-cytotoxic T-lymphocyte antigen-4. *J Clin Oncol.* Sep. 1, 2005;23(25):6043-53. doi: 10.1200/JCO.2005.06.205. Epub Aug. 8, 2005.
- Aurisicchio et al., A novel minigene scaffold for therapeutic cancer vaccines. *Oncoimmunology.* Jan. 1, 2014;3(1):e27529. doi: 10.4161/onci.27529. Epub Jan. 16, 2014.
- Bae et al., Identification of novel CD33 antigen-specific peptides for the generation of cytotoxic T lymphocytes against acute myeloid leukemia. *Cell Immunol.* Jan. 2004;227(1):38-50. doi: 10.1016/j.cellimm.2004.01.002.
- Bakker et al., Analogues of CTL epitopes with improved MHC class-I binding capacity elicit anti-melanoma CTL recognizing the wild-type epitope. *Int J Cancer.* Jan. 27, 1997;70(3):302-9. doi: 10.1002/(sici)1097-0215(19970127)70:3<302::aid-ijc10>3.0.co;2-h.
- Banerjee et al., Viral glycoproteins: biological role and application in diagnosis. *Virusdisease.* Mar. 2016;27(1):1-11. doi: 10.1007/s13337-015-0293-5. Epub Jan. 18, 2016.
- Barve et al., Induction of immune responses and clinical efficacy in a phase II trial of IDM-2101, a 10-epitope cytotoxic T-lymphocyte vaccine, in metastatic non-small-cell lung cancer. *J Clin Oncol.* Sep. 20, 2008;26(27):4418-25. doi: 10.1200/JCO.2008.16.6462.
- Benlalam et al., Identification of five new HLA-B*3501-restricted epitopes derived from common melanoma-associated antigens, spontaneously recognized by tumor-infiltrating lymphocytes. *J Immunol.* Dec. 1, 2003;171(11):6283-9. doi: 10.4049/jimmunol.171.11.6283.
- Bernatchez et al., Altered decamer and nonamer from an HLA-A0201-restricted epitope of Survivin differentially stimulate T-cell responses in different individuals. *Vaccine.* Apr. 5, 2011;29(16):3021-30. doi: 10.1016/j.vaccine.2011.01.115. Epub Feb. 12, 2011.
- Bioley et al., Melan-A/MART-1-specific CD4 T cells in melanoma patients: identification of new epitopes and ex vivo visualization of specific T cells by MHC class II tetramers. *J Immunol.* Nov. 15, 2006;177(10):6769-79. doi: 10.4049/jimmunol.177.10.6769.
- Blanchet et al., A new generation of Melan-A/MART-1 peptides that fulfill both increased immunogenicity and high resistance to bio-

(56)

References Cited

OTHER PUBLICATIONS

- degradation: implication for molecular anti-melanoma immunotherapy. *J Immunol.* Nov. 15, 2001;167(10):5852-61. doi: 10.4049/jimmunol.167.10.5852.
- Bolukbasi et al., DNA-binding-domain fusions enhance the targeting range and precision of Cas9. *Nat Methods.* Dec. 2015;12(12):1150-6. doi: 10.1038/nmeth.3624. Epub Oct. 19, 2015.
- Borbulevych et al., Increased immunogenicity of an anchor-modified tumor-associated antigen is due to the enhanced stability of the peptide/MHC complex: implications for vaccine design. *J Immunol.* Apr. 15, 2005;174(8):4812-20. doi: 10.4049/jimmunol.174.8.4812.
- Brichard et al., A tyrosinase nonapeptide presented by HLA-B44 is recognized on a human melanoma by autologous cytolytic T lymphocytes. *Eur J Immunol.* Jan. 1996;26(1):224-30. doi: 10.1002/eji.1830260135.
- Cacabelos et al., Chapter 1—The Epigenetic Machinery in the Life Cycle and Pharmacoeugenetics. *Pharmacoeugenetics.* vol. 10 in *Translational Epigenetics.* 2019:1-100. doi: <https://doi.org/10.1016/B978-0-12-813939-4.00001-2>. 7 pages.
- Campbell et al., Gesicle-Mediated Delivery of CRISPR/Cas9 Ribonucleoprotein Complex for Inactivating the HIV Provirus. *Mol Ther.* Jan. 2, 2019;27(1):151-163. doi: 10.1016/j.ymthe.2018.10.002. Epub Oct. 11, 2018.
- Campi et al., CD4(+) T cells from healthy subjects and colon cancer patients recognize a carcinoembryonic antigen-specific immunodominant epitope. *Cancer Res.* Dec. 1, 2003;63(23):8481-6.
- Casnici et al., Immunologic evaluation of peptides derived from BCR/ABL-out-of-frame fusion protein in Hla A2.1 transgenic mice. *J Immunother.* May 2012;35(4):321-8. doi: 10.1097/CJI.0b013e3182562d37.
- Casnici et al., Out of frame peptides from BCR/ABL alternative splicing are immunogenic in HLA A2.1 transgenic mice. *Cancer Lett.* Apr. 8, 2009;276(1):61-7. doi: 10.1016/j.canlet.2008.10.032. Epub Dec. 4, 2008.
- Castelli et al., Mass spectrometric identification of a naturally processed melanoma peptide recognized by CD8+ cytotoxic T lymphocytes. *J Exp Med.* Jan. 1, 1995;181(1):363-8. doi: 10.1084/jem.181.1.363.
- Castelli et al., Novel HLA-Cw8-restricted T cell epitopes derived from tyrosinase-related protein-2 and gp100 melanoma antigens. *J Immunol.* Feb. 1, 1999;162(3):1739-48.
- Castle et al., Exploiting the mutanome for tumor vaccination. *Cancer Res.* Mar. 1, 2012;72(5):1081-91. doi: 10.1158/0008-5472.CAN-11-3722. Epub Jan. 11, 2012.
- Cervera et al., Generation of HIV-1 Gag VLPs by transient transfection of HEK 293 suspension cell cultures using an optimized animal-derived component free medium. *J Biotechnol.* Jul. 20, 2013;166(4):152-65. doi: 10.1016/j.jbiotec.2013.05.001. Epub May 17, 2013.
- Chen et al., Identification of NY-ESO-1 peptide analogues capable of improved stimulation of tumor-reactive Ctl. *J Immunol.* Jul. 15, 2000;165(2):948-55. doi: 10.4049/jimmunol.165.2.948.
- Cho et al., Optimized peptide vaccines eliciting extensive CD8 T-cell responses with therapeutic antitumor effects. *Cancer Res.* Dec. 1, 2009;69(23):9012-9. doi: 10.1158/0008-5472.CAN-9-2019. Epub Nov. 10, 2009.
- Choi et al., Lentivirus pre-packed with Cas9 protein for safer gene editing. *Gene Ther.* Jul. 2016;23(7):627-33. doi: 10.1038/gt.2016.27. Epub Apr. 7, 2016.
- Christensen et al., Melan-A/MART1 analog peptide triggers anti-melanoma T-cells through crossreactivity with HMI.24. *J Immunother.* Jul.-Aug. 2009;32(6):613-21. doi: 10.1097/CJI.0b013e3181a95198.
- Coey, Sumoylation of thymine DNA glycosylase occurs efficiently and weakens DNA binding but does not regulate enzymatic turnover. *Dissertation.* 2017. 178 pages.
- Correale et al., In vitro generation of human cytotoxic T lymphocytes specific for peptides derived from prostate-specific antigen. *J Natl Cancer Inst.* Feb. 19, 1997;89(4):293-300. doi: 10.1093/jnci/89.4.293.
- Cox et al., Identification of a peptide recognized by five melanoma-specific human cytotoxic T cell lines. *Science.* Apr. 29, 1994;264(5159):716-9. doi: 10.1126/science.7513441.
- Cronin et al., Altering the tropism of lentiviral vectors through pseudotyping. *Curr Gene Ther.* Aug. 2005;5(4):387-98. doi: 10.2174/1566523054546224. Erratum in: *Curr Gene Ther.* Oct. 2005;5(5):531. Author Manuscript, 19 pages.
- Crosti et al., Identification of novel subdominant epitopes on the carcinoembryonic antigen recognized by CD4+ T cells of lung cancer patients. *J Immunol.* Apr. 15, 2006;176(8):5093-9. doi: 10.4049/jimmunol.176.8.5093.
- Dalet et al., An antigenic peptide produced by reverse splicing and double asparagine deamidation. *Proc Natl Acad Sci U S A.* Jul. 11, 2011;108(29):E323-31. doi: 10.1073/pnas.1101892108. Epub Jun. 13, 2011.
- David et al., Viral Vectors: The Road to Reducing Genotoxicity. *Toxicol Sci.* Feb. 2017;155(2):315-325. doi: 10.1093/toxsci/kfw220. Epub Nov. 1, 2016.
- Depontieu et al., Identification of tumor-associated, MHC class II-restricted phosphopeptides as targets for immunotherapy. *Proc Natl Acad Sci U S A.* Jul. 21, 2009;106(29):12073-8. doi: 10.1073/pnas.0903852106. Epub Jul. 6, 2009.
- Di Stasi et al., Review of the Results of WT1 Peptide Vaccination Strategies for Myelodysplastic Syndromes and Acute Myeloid Leukemia from Nine Different Studies. *Front Immunol.* Feb. 4, 2015;6:36. doi: 10.3389/fimmu.2015.00036.
- Duan et al., Immune rejection of mouse tumors expressing mutated self. *Cancer Res.* Apr. 15, 2009;69(8):3545-53. doi: 10.1158/0008-5472.CAN-08-2779. Epub Apr. 7, 2009. Author Manuscript. 18 pages.
- Duportet et al., A platform for rapid prototyping of synthetic gene networks in mammalian cells. *Nucleic Acids Res.* Dec. 1, 2014;42(21):13440-51. doi: 10.1093/nar/gku1082. Epub Nov. 5, 2014.
- Fontana et al., Rabies virus-like particles expressed in HEK293 cells. *Vaccine.* May 19, 2014;32(24):2799-804. doi: 10.1016/j.vaccine.2014.02.031. Epub Mar. 12, 2014.
- Fonteneau et al., The Tumor Antigen NY-ESO-1 Mediates Direct Recognition of Melanoma Cells by CD4+ T Cells after Intercellular Antigen Transfer. *J Immunol.* Jan. 1, 2016;196(1):64-71. doi: 10.4049/jimmunol.1402664. Epub Nov. 25, 2015.
- Fourcade et al., PD-1 and Tim-3 regulate the expansion of tumor antigen-specific CD8+ T cells induced by melanoma vaccines. *Cancer Res.* Feb. 15, 2014;74(4):1045-55. doi: 10.1158/0008-5472.CAN-13-2908. Epub Dec. 16, 2013.
- Fridman et al., An efficient T-cell epitope discovery strategy using in silico prediction and the iTopia assay platform. *Oncimmunology.* Nov. 1, 2012;1(8):1258-1270. doi: 10.4161/onci.21355.
- Fujiki et al., Identification and characterization of a WT1 (Wilms Tumor Gene) protein-derived HLA-DRB1*0405-restricted 16-mer helper peptide that promotes the induction and activation of WT1-specific cytotoxic T lymphocytes. *J Immunother.* Apr. 2007;30(3):282-93. doi: 10.1097/01.cji.0000211337.91513.94.
- Gee et al., Extracellular nanovesicles for packaging of CRISPR-Cas9 protein and sgRNA to induce therapeutic exon skipping. *Nat Commun.* Mar. 13, 2020;11(1):1334. doi: 10.1038/s41467-020-14957-y.
- GenBank Access No. BAP64357. Aug. 1, 2013 1 page.
- Geynisman et al., A randomized pilot phase I study of modified carcinoembryonic antigen (CEA) peptide (CAP1-6D)/montanide/GM-CSF-vaccine in patients with pancreatic adenocarcinoma. *J Immunother Cancer.* Jun. 27, 2013;1:8. doi: 10.1186/2051-1426-1-8.
- Ghosh et al., Synapsis in phage Bxb1 integration: selection mechanism for the correct pair of recombination sites. *J Mol Biol.* Jun. 3, 2005;349(2):331-48. doi: 10.1016/j.jmb.2005.03.043. Epub Apr. 7, 2005.
- Girard-Gagnepain et al., Baboon envelope pseudotyped LVs outperform VSV-G-LVs for gene transfer into early-cytokine-stimulated and resting HSCs. *Blood.* Aug. 21, 2014;124(8):1221-31. doi: 10.1182/blood-2014-02-558163. Epub Jun. 20, 2014.

(56)

References Cited

OTHER PUBLICATIONS

- Godefroy et al., Identification of two Melan-A CD4+ T cell epitopes presented by frequently expressed MHC class II alleles. *Clin Immunol.* Oct. 2006; 121(1):54-62. doi: 10.1016/j.clim.2006.05.007. Epub Jun. 30, 2006.
- Graff-Dubois et al., Generation of CTL recognizing an HLA-A*0201-restricted epitope shared by MAGE-A1, -A2, -A3, -A4, -A6, -A10, and -A12 tumor antigens: implication in a broad-spectrum tumor immunotherapy. *J Immunol.* Jul. 1, 2002;169(1):575-80. doi: 10.4049/jimmunol.169.1.575.
- Gross et al., High vaccination efficiency of low-affinity epitopes in antitumor immunotherapy. *J Clin Invest.* Feb. 2004;113(3):425-33. doi: 10.1172/JCI19418.
- Guevara-Patiño et al., Optimization of a self antigen for presentation of multiple epitopes in cancer immunity. *J Clin Invest.* May 2006;116(5):1382-90. doi: 10.1172/JCI25591. Epub Apr. 13, 2006.
- Guibinga et al., Cell surface heparan sulfate is a receptor for attachment of envelope protein-free retrovirus-like particles and VSV-G pseudotyped MLV-derived retrovirus vectors to target cells. *Mol Ther.* May 2002;5(5 Pt 1):538-46. doi: 10.1006/mthe.2002.0578.
- Gulley et al., Combining a Recombinant Cancer Vaccine with Standard Definitive Radiotherapy in Patients with Localized Prostate Cancer. *Clin Cancer Res.* May 2, 2005;11(9):3353-62. doi: 10.1158/1078-0432.CCR-04-2062.
- Guo et al., Direct recognition and lysis of leukemia cells by WT1-specific CD4+ T lymphocytes in an HLA class II-restricted manner. *Blood.* Aug. 15, 2005;106(4): 1415-8. doi: 10.1182/blood-2005-01-0413. Epub Apr. 21, 2005.
- Haeussler et al., Genome Editing with CRISPR-Cas9: Can It Get Any Better? *J Genet Genomics.* May 20, 2016;43(5):239-50. doi: 10.1016/j.jgg.2016.04.008. Epub Apr. 24, 2016. Author Manuscript. 22 pages.
- Herbst-Kralovetz et al., Norwalk virus-like particles as vaccines. *Expert Rev Vaccines.* Mar. 2010;9(3):299-307. doi: 10.1586/erv.09.163. Author Manuscript, 16 pages.
- Hirohashi et al., An HLA-A24-restricted cytotoxic T lymphocyte epitope of a tumor-associated protein, survivin. *Clin Cancer Res.* Jun. 2002;8(6):1731-9.
- Hong et al., Novel recombinant hepatitis B virus vectors efficiently deliver protein and RNA encoding genes into primary hepatocytes. *J Virol.* Jun. 2013;87(12):6615-24. doi: 10.1128/JVI.03328-12. Epub Apr. 3, 2013.
- Indikova et al., Highly efficient 'hit-and-run' genome editing with unconcentrated lentivectors carrying Vpr.ProT.Cas9 protein produced from RRE-containing transcripts. *Nucleic Acids Res.* Aug. 20, 2020;48(14):8178-8187. doi: 10.1093/nar/gkaa561.
- Jacobs et al., DNA glycosylases: in DNA repair and beyond. *Chromosoma.* Feb. 2012;121(1):1-20. doi: 10.1007/s00412-011-0347-4. Epub Nov. 3, 2011. 20 pages.
- Jalaguier et al., Efficient production of HIV-1 virus-like particles from a mammalian expression vector requires the N-terminal capsid domain. *PLoS One.* 2011;6(11):e28314. doi: 10.1371/journal.pone.0028314. Epub Nov. 30, 2011.
- Jaramillo et al., Identification of HLA-A3-restricted CD8+ T cell epitopes derived from mamaglobin-A, a tumor-associated antigen of human breast cancer. *Int J Cancer.* Dec. 10, 2002;102(5):499-506. doi: 10.1002/ijc.10736.
- Jeong et al., Current Status and Challenges of DNA Base Editing Tools. *Mol Ther.* Sep. 2, 2020;28(9):1938-1952. doi: 10.1016/j.ymthe.2020.07.021. Epub Jul. 23, 2020.
- Joglekar et al., Pseudotyped Lentiviral Vectors: One Vector, Many Guises. *Hum Gene Ther Methods.* Dec. 2017;28(6):291-301. doi: 10.1089/hgtb.2017.084. Epub Sep. 4, 2017.
- Kaczmarczyk et al., Protein delivery using engineered virus-like particles. *Proc Natl Acad Sci U S A.* Oct. 11, 2011;108(41):16998-7003. doi: 10.1073/pnas.1101874108. Epub Sep. 26, 2011.
- Kang et al., Chimeric rabies virus-like particles containing membrane-anchored GM-CSF enhances the immune response against rabies virus. *Viruses.* Mar. 11, 2015;7(3):1134-52. doi: 10.3390/v7031134.
- Kang et al., Identification of a tyrosinase epitope recognized by HLA-A24-restricted, tumor-infiltrating lymphocytes. *J Immunol.* Aug. 1, 1995;155(3):1343-8.
- Karbach et al., Long-term complete remission following radiosurgery and immunotherapy in a melanoma patient with brain metastasis: immunologic correlates. *Cancer Immunol Res.* May 2014;2(5):404-9. doi: 10.1158/2326-6066.CIR-13-0200. Epub Feb. 5, 2014.
- Kato et al., A lentiviral strategy for highly efficient retrograde gene transfer by pseudotyping with fusion envelope glycoprotein. *Hum Gene Ther.* Feb. 2011;22(2):197-206. doi: 10.1089/hum.2009.179. Epub Jan. 27, 2011.
- Kato et al., Selective neural pathway targeting reveals key roles of thalamostriatal projection in the control of visual discrimination. *J Neurosci.* Nov. 23, 2011;31(47):17169-79. doi: 10.1523/JNEUROSCI.4005-11.2011.
- Kawakami et al., Identification of a human melanoma antigen recognized by tumor-infiltrating lymphocytes associated with in vivo tumor rejection. *Proc Natl Acad Sci U S A.* Jul. 5, 1994;91(14):6458-62. doi: 10.1073/pnas.91.14.6458.
- Kawakami et al., Identification of new melanoma epitopes on melanosomal proteins recognized by tumor infiltrating T lymphocytes restricted by HLA-A1, -A2, and -A3 alleles. *J Immunol.* Dec. 15, 1998;161(12):6985-92.
- Kawakami et al., Identification of the immunodominant peptides of the MART-1 human melanoma antigen recognized by the majority of HLA-A2-restricted tumor infiltrating lymphocytes. *J Exp Med.* Jul. 1, 1994;180(1):347-52. doi: 10.1084/jem.180.1.347.
- Kawakami et al., Recognition of multiple epitopes in the human melanoma antigen gp100 by tumor-infiltrating T lymphocytes associated with in vivo tumor regression. *J Immunol.* Apr. 15, 1995;154(8):3961-8.
- Kawashima et al., Identification of gp100-derived, melanoma-specific cytotoxic T-lymphocyte epitopes restricted by HLA-A3 supertype molecules by primary in vitro immunization with peptide-pulsed dendritic cells. *Int J Cancer.* Nov. 9, 1998;78(4):518-24. doi: 10.1002/(sici)1097-0215(19981109)78:4<518::aid-ijc20>3.0.co;2-0.
- Kawashima et al., Identification of HLA-A3-restricted cytotoxic T lymphocyte epitopes from carcinoembryonic antigen and HER-2/neu by primary in vitro immunization with peptide-pulsed dendritic cells. *Cancer Res.* Jan. 15, 1999;59(2):431-5.
- Kawashima et al., The multi-epitope approach for immunotherapy for cancer: identification of several CTL epitopes from various tumor-associated antigens expressed on solid epithelial tumors. *Hum Immunol.* Jan. 1998;59(1):1-14. doi: 10.1016/s0198-8859(97)00255-3.
- Kemmler et al., Elevated tumor-associated antigen expression suppresses variant peptide vaccine responses. *J Immunol.* Nov. 1, 2011;187(9):4431-9. doi: 10.4049/jimmunol.1101555. Epub Sep. 21, 2011.
- Kittlesen et al., Human melanoma patients recognize an HLA-A1-restricted CTL epitope from tyrosinase containing two cysteine residues: implications for tumor vaccine development. *J Immunol.* Mar. 1, 1998;160(5):2099-106. Erratum in: *J Immunol* Mar. 1, 1999;162(5):3106. Shabanowitz JA [corrected to Shabanowitz JJ].
- Kizer et al., Application of functional genomics to pathway optimization for increased isoprenoid production. *Appl Environ Microbiol.* May 2008;74(10):3229-41. doi: 10.1128/AEM.02750-07. Epub Mar. 14, 2008.
- Kobayashi et al., CD4+ T cells from peripheral blood of a melanoma patient recognize peptides derived from nonmutated tyrosinase. *Cancer Res.* Jan. 15, 1998;58(2):296-301.
- Kobayashi et al., Identification of an antigenic epitope for helper T lymphocytes from carcinoembryonic antigen. *Clin Cancer Res.* Oct. 2002;8(10):3219-25.
- Kobayashi et al., Identification of helper T-cell epitopes that encompass or lie proximal to cytotoxic T-cell epitopes in the gp100 melanoma tumor antigen. *Cancer Res.* Oct. 15, 2001;61(20):7577-84.
- Kotterman et al., Engineering adeno-associated viruses for clinical gene therapy. *Nat Rev Genet.* Jul. 2014;15(7):445-51. doi: 10.1038/nrg3742. Epub May 20, 2014.

(56)

References Cited

OTHER PUBLICATIONS

- Kueh et al., The new editor-targeted genome engineering in the absence of homology-directed repair. *Cell Death Discov.* Jun. 13, 2016;2:16042. doi: 10.1038/cddiscovery.2016.42.
- Kurt et al., CRISPR C-to-G base editors for inducing targeted DNA transversions in human cells. *Nat Biotechnol.* Jan. 2021;39(1):41-46. doi: 10.1038/s41587-020-0609-x. Epub Jul. 20, 2020.
- Kushnir et al., Virus-like particles as a highly efficient vaccine platform: diversity of targets and production systems and advances in clinical development. *Vaccine.* Dec. 17, 2012;31(1):58-83. doi: 10.1016/j.vaccine.2012.10.083. Epub Nov. 6, 2012.
- Lally et al., Unmasking cryptic epitopes after loss of immunodominant tumor antigen expression through epitope spreading. *Int J Cancer.* Sep. 2001;93(6):841-7. doi: 10.1002/ijc.1420.
- Lapointe et al., Retrovirally transduced human dendritic cells can generate T cells recognizing multiple MHC class I and class II epitopes from the melanoma antigen glycoprotein 100. *J Immunol.* Oct. 15, 2001;167(8):4758-64. doi: 10.4049/jimmunol.167.8.4758.
- Larrieu et al., A HLA-Cw*0701 restricted Melan-A/MART1 epitope presented by melanoma tumor cells to CD8+ tumor infiltrating lymphocytes. *Cancer Immunol Immunother.* May 2008;57(5):745-52. doi: 10.1007/s00262-007-0436-7. Epub Dec. 21, 2007.
- Latham et al., Formation of wild-type and chimeric influenza virus-like particles following simultaneous expression of only four structural proteins. *J Virol.* Jul. 2001;75(13):6154-65. doi: 10.1128/JVI.75.13.6154-6165.2001.
- Lennerz et al., The response of autologous T cells to a human melanoma is dominated by mutated neoantigens. *Proc Natl Acad Sci U S A.* Nov. 1, 2005;102(44):16013-8. doi: 10.1073/pnas.0500090102. Epub Oct. 24, 2005.
- Li et al., Base-Resolution Mapping Reveals Distinct m1A Methylation in Nuclear- and Mitochondrial-Encoded Transcripts. *Mol Cell.* Dec. 7, 2017;68(5):993-1005.e9. doi: 10.1016/j.molcel.2017.10.019. Epub Nov. 5, 2017.
- Li et al., Expression and self-assembly of empty virus-like particles of hepatitis E virus. *J Virol.* Oct. 1997;71(10):7207-13. doi: 10.1128/JVI.71.10.7207-7213.1997.
- Lin et al., HLA-DPB1*05: 01-restricted WT1332-specific TCR-transduced CD4+ T lymphocytes display a helper activity for WT1-specific CTL induction and a cytotoxicity against leukemia cells. *J Immunother.* Apr. 2013;36(3):159-70. doi: 10.1097/CJI.0b013e3182873581.
- Lu, Periodic Chart of Amino Acid PDF. Accessed on the internet at https://figshare.com/articles/figure/periodic_chart_of_amino_acid_pdf/3445001/1. Posted Jun. 21, 2016. www.bachem.com. 1 page.
- Ludwig et al., Virus-like particles-universal molecular toolboxes. *Curr Opin Biotechnol.* Dec. 2007;18(6):537-45. doi: 10.1016/j.copbio.2007.10.013.
- Lueck et al., Engineered transfer RNAs for suppression of premature termination codons. *Nat Commun.* Feb. 18, 2019;10(1):822. doi: 10.1038/s41467-019-08329-4.
- Lupetti et al., Translation of a retained intron in tyrosinase-related protein (TRP) 2 mRNA generates a new cytotoxic T lymphocyte (CTL)-defined and shared human melanoma antigen not expressed in normal cells of the melanocytic lineage. *J Exp Med.* Sep. 21, 1998;188(6):1005-16. doi: 10.1084/jem.188.6.1005.
- Lyu et al., Delivering Cas9/sgRNA ribonucleoprotein (RNP) by lentiviral capsid-based bionanoparticles for efficient 'hit-and-run' genome editing. *Nucleic Acids Res.* Sep. 26, 2019;47(17):e99. doi: 10.1093/nar/gkz605.
- Maetzig et al., Retroviral protein transfer: falling apart to make an impact. *Curr Gene Ther.* Oct. 2012;12(5):389-409. doi: 10.2174/156652312802762581.
- Mandic et al., The alternative open reading frame of LAGE-1 gives rise to multiple promiscuous HLA-DR-restricted epitopes recognized by T-helper 1-type tumor-reactive CD4+ T cells. *Cancer Res.* Oct. 1, 2003;63(19):6506-15.
- Mangeot et al., A universal transgene silencing method based on RNA interference. *Nucleic Acids Res.* Jul. 12, 2004;32(12):e102. doi: 10.1093/nar/gnh105.
- Mangeot et al., Development of minimal lentivirus vectors derived from simian immunodeficiency virus (SIVmac251) and their use for gene transfer into human dendritic cells. *J Virol.* Sep. 2000;74(18):8307-15. doi: 10.1128/jvi.74.18.8307-8315.2000.
- Mangeot et al., Protein transfer into human cells by VSV-G-induced nanovesicles. *Mol Ther.* Sep. 2011;19(9):1656-66. doi: 10.1038/mt.2011.138. Epub Jul. 12, 2011.
- Meng et al., Identification of an HLA-DPB1*0501 restricted Melan-A/MART-1 epitope recognized by CD4+ T lymphocytes: prevalence for immunotherapy in Asian populations. *J Immunother.* Sep. 2011;34(7):525-34. doi: 10.1097/CJI.0b013e318226bd45. Author Manuscript. 16 pages.
- Michaux et al., A spliced antigenic peptide comprising a single spliced amino acid is produced in the proteasome by reverse splicing of a longer peptide fragment followed by trimming. *J Immunol.* Feb. 15, 2014;192(4):1962-71. doi: 10.4049/jimmunol.1302032. Epub Jan. 22, 2014.
- Milone et al., Clinical use of lentiviral vectors. *Leukemia.* Jul. 2018;32(7):1529-1541. doi: 10.1038/s41375-018-0106-0. Epub Mar. 22, 2018.
- Momose et al., Diving into marine genomics with CRISPR/Cas9 systems. *Mar Genomics.* Dec. 2016;30:55-65. doi: 10.1016/j.margen.2016.10.003. Epub Oct. 12, 2016.
- Morel et al., A tyrosinase peptide presented by HLA-B35 is recognized on a human melanoma by autologous cytotoxic T lymphocytes. *Int J Cancer.* Dec. 10, 1999;83(6):755-9. doi: 10.1002/(sici)1097-0215(19991210)83:6<755::aid-ijc10>3.0.co;2-s.
- Mselli-Lakhal et al., Gene transfer system derived from the caprine arthritis-encephalitis lentivirus. *J Virol Methods.* Sep. 2006;136(1-2):177-84. doi: 10.1016/j.jviromet.2006.05.006. Epub Jun. 21, 2006.
- Murawski et al., Newcastle disease virus-like particles containing respiratory syncytial virus G protein induced protection in BALB/c mice, with no evidence of immunopathology. *J Virol.* Jan. 2010;84(2):1110-23. doi: 10.1128/JVI.01709-09. Epub Nov. 4, 2009.
- Naskalska et al., Virus Like Particles as Immunogens and Universal Nanocarriers. *Pol J Microbiol.* 2015;64(1):3-13.
- Negre et al., Characterization of novel safe lentiviral vectors derived from simian immunodeficiency virus (SIVmac251) that efficiently transduce mature human dendritic cells. *Gene Ther.* Oct. 2000;7(19):1613-23. doi: 10.1038/sj.gt.3301292.
- Noppen et al., Naturally processed and concealed HLA-A2.1-restricted epitopes from tumor-associated antigen tyrosinase-related protein-2. *Int J Cancer.* Jul. 15, 2000;87(2):241-6.
- Nukaya et al., Identification of HLA-A24 epitope peptides of carcinoembryonic antigen which induce tumor-reactive cytotoxic T lymphocyte. *Int J Cancer.* Jan. 5, 1999;80(1):92-7. doi: 10.1002/(sici)1097-0215(19990105)80:1<92::aid-ijc18>3.0.co;2-m.
- Ogasawara et al., Recombinant viral-like particles of parvovirus B19 as antigen carriers of anthrax protective antigen. *In Vivo.* May-Jun. 2006;20(3):319-24.
- Ohnishi et al., HLA class I-restricted lysis of leukemia cells by a CD8(+) cytotoxic T-lymphocyte clone specific for WT1 peptide. *Blood.* Jan. 1, 2000;95(1):286-93.
- Oka et al., WT1 peptide vaccine for the treatment of cancer. *Curr Opin Immunol.* Apr. 2008;20(2):211-20. doi: 10.1016/j.coi.2008.04.009. Epub May 24, 2008.
- Olsen, J.C., Gene transfer vectors derived from equine infectious anemia virus. *Gene Ther.* Nov. 1998;5(11):1481-7. doi: 10.1038/sj.gt.3300768.
- Olson et al., HLA-A2-restricted T-cell epitopes specific for prostatic acid phosphatase. *Cancer Immunol Immunother.* Jun. 2010;59(6):943-53. doi: 10.1007/s00262-010-0820-6. Epub Feb. 6, 2010.
- Osen et al., Screening of human tumor antigens for CD4 T cell epitopes by combination of HLA-transgenic mice, recombinant adenovirus and antigen peptide libraries. *PLoS One.* Nov. 30, 2010;5(11):e14137. doi: 10.1371/journal.pone.0014137.
- Parkhurst et al., Identification of a shared HLA-A*0201-restricted T-cell epitope from the melanoma antigen tyrosinase-related protein 2 (TRP2). *Cancer Res.* Nov. 1, 1998;58(21):4895-901.
- Parkhurst et al., Induction of CD4+ Th1 lymphocytes that recognize known and novel class II MHC restricted epitopes from the melanoma antigen gp100 by stimulation with recombinant protein. *J*

(56)

References Cited

OTHER PUBLICATIONS

- Immunother. Mar.-Apr. 2004;27(2):79-91. doi: 10.1097/00002371-200403000-00001. Author Manuscript. 22 pages.
- Paschen et al., Detection of spontaneous CD4+ T-cell responses in melanoma patients against a tyrosinase-related protein-2-derived epitope identified in HLA-DRB1*0301 transgenic mice. *Clin Cancer Res.* Jul. 15, 2005;11(14):5241-7. doi: 10.1158/1078-0432.CCR-05-0170.
- Pavlov et al., Roles of DNA polymerases in replication, repair, and recombination in eukaryotes. *Int Rev Cytol.* 2006;255:41-132. doi: 10.1016/S0074-7696(06)55002-8.
- Pinilla et al., Combinatorial peptide libraries as an alternative approach to the identification of ligands for tumor-reactive cytolytic T lymphocytes. *Cancer Res.* Jul. 1, 2001;61(13):5153-60.
- Pinilla-Ibarz et al., Improved human T-cell responses against synthetic HLA-0201 analog peptides derived from the WT1 oncoprotein. *Leukemia.* Nov. 2006;20(11):2025-33. doi: 10.1038/sj.leu.2404380. Epub Aug. 31, 2006.
- Prather et al., De novo biosynthetic pathways: rational design of microbial chemical factories. *Curr Opin Biotechnol.* Oct. 2008;19(5):468-74. doi: 10.1016/j.copbio.2008.07.009. Epub Sep. 5, 2008.
- Quan et al., Influenza M1 VLPs containing neuraminidase induce heterosubtypic cross-protection. *Virology.* Sep. 1, 2012;430(2):127-35. doi: 10.1016/j.virol.2012.05.006. Epub Jun. 2, 2012.
- Rasmussen et al., Characterization of virus-like particles produced by a recombinant baculovirus containing the gag gene of the bovine immunodeficiency-like virus. *Virology.* Oct. 1990; 178(2):435-51. doi: 10.1016/0042-6822(90)90341-n.
- Remington et al., Complete nucleotide sequence of a neuropathogenic variant of Friend murine leukemia virus PVC-211. *Nucleic Acids Res.* Jun. 25, 1992;20(12):3249. doi: 10.1093/nar/20.12.3249.
- Renner et al., Intact Viral Particle Counts Measured by Flow Virometry Provide Insight into the Infectivity and Genome Packaging Efficiency of Moloney Murine Leukemia Virus. *J Virol.* Jan. 6, 2020;94(2):e01600-19. doi: 10.1128/JVI.01600-19.
- Riley et al., Identification of a new shared HLA-A2.1 restricted epitope from the melanoma antigen tyrosinase. *J Immunother.* May 20-Jun. 2001;24(3):212-20.
- Rimoldi et al., Efficient simultaneous presentation of NY-ESO-1/LAGE-1 primary and nonprimary open reading frame-derived CTL epitopes in melanoma. *J Immunol.* Dec. 15, 2000;165(12):7253-61. doi: 10.4049/jimmunol.165.12.7253.
- Robbins et al., Multiple HLA class II-restricted melanocyte differentiation antigens are recognized by tumor-infiltrating lymphocytes from a patient with melanoma. *J Immunol.* Nov. 15, 2002;169(10):6036-47. doi: 10.4049/jimmunol.169.10.6036.
- Robbins et al., The intronic region of an incompletely spliced gp100 gene transcript encodes an epitope recognized by melanoma-reactive tumor-infiltrating lymphocytes. *J Immunol.* Jul. 1, 1997;159(1):303-8.
- Rohovie et al., Virus-like particles: Next-generation nanoparticles for targeted therapeutic delivery. *Bioeng Transl Med.* Jan. 19, 2017;2(1):43-57. doi: 10.1002/btm2.10049.
- Rosenberg et al., Immunologic and therapeutic evaluation of a synthetic peptide vaccine for the treatment of patients with metastatic melanoma. *Nat Med.* Mar. 1998;4(3):321-7. doi: 10.1038/nm0398-321.
- Rubio-Godoy et al., Toward synthetic combinatorial peptide libraries in positional scanning format (PS-SCL)-based identification of CD8+ Tumor-reactive T-Cell Ligands: a comparative analysis of PS-SCL recognition by a single tumor-reactive CD8+ cytolytic T-lymphocyte clone. *Cancer Res.* Apr. 1, 2002;62(7):2058-63.
- Ruiz et al., Identification and characterization of a T-helper peptide from carcinoembryonic antigen. *Clin Cancer Res.* Apr. 15, 2004;10(8):2860-7. doi: 10.1158/1078-0432.ccr-03-0476.
- Rusk, Cas9 and the importance of asymmetry. *Nat Methods.* Apr. 2016;13(4):286-7. doi: 10.1038/nmeth.3826.
- Saenger et al., Improved tumor immunity using anti-tyrosinase related protein-1 monoclonal antibody combined with DNA vaccines in murine melanoma. *Cancer Res.* Dec. 1, 2008;68(23):9884-91. doi: 10.1158/0008-5472.CAN-08-2233. Author Manuscript. 19 pages.
- Saenz et al., Feline immunodeficiency virus-based lentiviral vectors. *Cold Spring Harb Protoc.* Jan. 1, 2012;2012(1):71-6. doi: 10.1101/pdb.ip067579.
- Saenz et al., Production, harvest, and concentration of feline immunodeficiency virus-based lentiviral vector from cells grown in CF10 or CF2 devices. *Cold Spring Harb Protoc.* Jan. 1, 2012;2012(1):118-23. doi: 10.1101/pdb.prot067546.
- Sakuma et al., Multiplex genome engineering in human cells using all-in-one CRISPR/Cas9 vector system. *Sci Rep.* Jun. 23, 2014;4:5400. doi: 10.1038/srep05400.
- Schneider et al., Overlapping peptides of melanocyte differentiation antigen Melan-A/MART-1 recognized by autologous cytolytic T lymphocytes in association with HLA-B45.1 and HLA-A2.1. *Int J Cancer.* Jan. 30, 1998;75(3):451-8. doi: 10.1002/(sici)1097-0215(19980130)75:3<451::aid-ijc20>3.0.co;2-a.
- Sensi et al., Identification of a novel gp100/pMel17 peptide presented by HLA-A*6801 and recognized on human melanoma by cytolytic T cell clones. *Tissue Antigens.* Apr. 2002;59(4):273-9. doi: 10.1034/j.1399-0039.2002.590404.x.
- Shang et al., The spontaneous CD8+ T-cell response to HLA-A2-restricted NY-ESO-1b peptide in hepatocellular carcinoma patients. *Clin Cancer Res.* Oct. 15, 2004;10(20):6946-55. doi: 10.1158/1078-0432.CCR-04-0502.
- Sharma et al., Noninfectious virus-like particles produced by Moloney murine leukemia virus-based retrovirus packaging cells deficient in viral envelope become infectious in the presence of lipofection reagents. *Proc Natl Acad Sci U S A.* Sep. 30, 1997;94(20):10803-8. doi: 10.1073/pnas.94.20.10803.
- Shellenberger et al., A recombinant polypeptide extends the in vivo half-life of peptides and proteins in a tunable manner. *Nat Biotechnol.* Dec. 2009;27(12):1186-90. doi: 10.1038/nbt.1588.
- Shen et al., Identification of a MHC class-II restricted epitope in carcinoembryonic antigen. *Cancer Immunol Immunother.* May 2004;53(5):391-403. doi: 10.1007/s00262-003-0455-y. Epub Nov. 18, 2003.
- Shim et al., Nonviral Delivery Systems for Cancer Gene Therapy: Strategies and Challenges. *Curr Gene Ther.* 2018;18(1):3-20. doi: 10.2174/1566523218666180119121949.
- Skipper et al., An HLA-A2-restricted tyrosinase antigen on melanoma cells results from posttranslational modification and suggests a novel pathway for processing of membrane proteins. *J Exp Med.* Feb. 1, 1996;183(2):527-34. doi: 10.1084/jem.183.2.527.
- Skipper et al., Shared epitopes for HLA-A3-restricted melanoma-reactive human CTL include a naturally processed epitope from Pmel-17/gp100. *J Immunol.* Dec. 1, 1996;157(11):5027-33.
- Slansky et al., Enhanced antigen-specific antitumor immunity with altered peptide ligands that stabilize the MHC-peptide-TCR complex. *Immunity.* Oct. 2000; 13(4):529-38. doi: 10.1016/s1074-7613(00)00052-2.
- Slingluff et al., Clinical and immunologic results of a randomized phase II trial of vaccination using four melanoma peptides either administered in granulocyte-macrophage colony-stimulating factor in adjuvant or pulsed on dendritic cells. *J Clin Oncol.* Nov. 1, 2003;21(21):4016-26. doi: 10.1200/JCO.2003.10.005.
- Slingluff et al., Immunologic and clinical outcomes of vaccination with a multipeptide melanoma peptide vaccine plus low-dose interleukin-2 administered either concurrently or on a delayed schedule. *J Clin Oncol.* Nov. 15, 2004;22(22):4474-85. doi: 10.1200/JCO.2004.10.212.
- Stevens et al., Design of a Split Intein with Exceptional Protein Splicing Activity. *J Am Chem Soc.* Feb. 24, 2016;138(7):2162-5. doi: 10.1021/jacs.5b13528. Epub Feb. 8, 2016. Abstract Only. 1 page.
- Tang et al., The *Arabidopsis* TRM61/TRM6 complex is a bona fide tRNA N1-methyladenosine methyltransferase. *J Exp Bot.* May 30, 2020;71(10):3024-3036. doi: 10.1093/jxb/eraa100.
- Tangri et al., Structural features of peptide analogs of human histocompatibility leukocyte antigen class I epitopes that are more potent and immunogenic than wild-type peptide. *J Exp Med.* Sep. 17, 2001;194(6):833-46. doi: 10.1084/jem.194.6.833.

(56)

References Cited

OTHER PUBLICATIONS

- Tomé-Amat et al., Secreted production of assembled Norovirus virus-like particles from *Pichia pastoris*. *Microb Cell Fact.* Sep. 10, 2014;13:134. doi: 10.1186/s12934-014-0134-z.
- Topalian et al., Melanoma-specific CD4+ T cells recognize nonmutated HLA-DR-restricted tyrosinase epitopes. *J Exp Med.* May 1, 1996;183(5):1965-71. doi: 10.1084/jem.183.5.1965.
- Touloukian et al., Expression of a "self"-antigen by human tumor cells enhances tumor antigen-specific CD4(+) T-cell function. *Cancer Res.* Sep. 15, 2002;62(18):5144-7. Author Manuscript. 11 pages.
- Touloukian et al., Identification of a MHC class II-restricted human gp100 epitope using DR4-IE transgenic mice. *J Immunol.* Apr. 1, 2000;164(7):3535-42. doi: 10.4049/jimmunol.164.7.3535.
- Touloukian et al., Normal tissue depresses while tumor tissue enhances human T cell responses in vivo to a novel self/tumor melanoma antigen, OA1. *J Immunol.* Feb. 1, 2003;170(3):1579-85. doi: 10.4049/jimmunol.170.3.1579.
- Trojan et al., Generation of cytotoxic T lymphocytes against native and altered peptides of human leukocyte antigen-A*0201 restricted epitopes from the human epithelial cell adhesion molecule. *Cancer Res.* Jun. 15, 2001;61(12):4761-5.
- Tsai et al., Identification of subdominant CTL epitopes of the GP100 melanoma-associated tumor antigen by primary in vitro immunization with peptide-pulsed dendritic cells. *J Immunol.* Feb. 15, 1997;158(4):1796-802.
- Tsang et al., A human cytotoxic T-lymphocyte epitope and its agonist epitope from the nonvariable number of tandem repeat sequence of MUC-1. *Clin Cancer Res.* Mar. 15, 2004;10(6):2139-49. doi: 10.1158/1078-0432.ccr-1011-03.
- Tsang et al., Generation of human cytotoxic T cells specific for human carcinoembryonic antigen epitopes from patients immunized with recombinant vaccinia-CEA vaccine. *J Natl Cancer Inst.* Jul. 5, 1995;87(13):982-90. doi: 10.1093/jnci/87.13.982.
- Tsuboi et al., Enhanced induction of human WT1-specific cytotoxic T lymphocytes with a 9-mer WT1 peptide modified at HLA-A*2402-binding residues. *Cancer Immunol Immunother.* Dec. 2002;51(11-12):614-20. doi: 10.1007/s00262-002-0328-9. Epub Oct. 18, 2002.
- Tuorto et al., Genome recoding by tRNA modifications. *Open Biol.* Dec. 2016;6(12):160287. doi: 10.1098/rsob.160287.
- Tycko et al., Methods for Optimizing CRISPR-Cas9 Genome Editing Specificity. *Mol Cell.* Aug. 4, 2016;63(3):355-70. doi: 10.1016/j.molcel.2016.07.004.
- Valmori et al., Analysis of the cytolytic T lymphocyte response of melanoma patients to the naturally HLA-A*0201-associated tyrosinase peptide 368-376. *Cancer Res.* Aug. 15, 1999;59(16):4050-5.
- Valmori et al., Enhanced generation of specific tumor-reactive CTL in vitro by selected Melan-A/MART-1 immunodominant peptide analogues. *J Immunol.* Feb. 15, 1998;160(4):1750-8.
- Valmori et al., Naturally occurring human lymphocyte antigen-A2 restricted CD8+ T-cell response to the cancer testis antigen NY-ESO-1 in melanoma patients. *Cancer Res.* Aug. 15, 2000;60(16):4499-506.
- Vigneron et al., A peptide derived from melanocytic protein gp100 and presented by HLA-B35 is recognized by autologous cytolytic T lymphocytes on melanoma cells. *Tissue Antigens.* Feb. 2005;65(2):156-62. doi: 10.1111/j.1399-0039.2005.00365.x.
- Vigneron et al., An antigenic peptide produced by peptide splicing in the proteasome. *Science.* Apr. 23, 2004;304(5670):587-90. doi: 10.1126/science.1095522.
- Visseren et al., Affinity, specificity and T-cell-receptor diversity of melanoma-specific CTL generated in vitro against a single tyrosinase epitope. *Int J Cancer.* Sep. 17, 1997;72(6):1122-8. doi: 10.1002/(sici)1097-0215(19970917)72:6<1122::aid-ijc30>3.0.co;2-3.
- Voelkel et al., Protein transduction from retroviral Gag precursors. *Proc Natl Acad Sci U S A.* Apr. 27, 2010;107(17):7805-10. doi: 10.1073/pnas.0914517107. Epub Apr. 12, 2010.
- Volpe et al., Alternative BCR/ABL splice variants in Philadelphia chromosome-positive leukemias result in novel tumor-specific fusion proteins that may represent potential targets for immunotherapy approaches. *Cancer Res.* Jun. 1, 2007;67(11):5300-7. doi: 10.1158/0008-5472.CAN-06-3737.
- Voutev et al., Bxb1 phage recombinase assists genome engineering in *Drosophila melanogaster*. *Biotechniques.* Jan. 1, 2017;62(1):37-38. doi: 10.2144/000114494.
- Walpita et al., Mammalian Cell-Derived Respiratory Syncytial Virus-Like Particles Protect the Lower as well as the Upper Respiratory Tract. *PLoS One.* Jul. 14, 2015;10(7):e0130755. doi: 10.1371/journal.pone.0130755.
- Walton et al., Spontaneous CD8 T cell responses against the melanocyte differentiation antigen RAB38/NY-MEL-1 in melanoma patients. *J Immunol.* Dec. 1, 2006;177(11):8212-8. doi: 10.4049/jimmunol.177.11.8212.
- Wang et al., CRISPR/Cas9 in Genome Editing and Beyond. *Annu Rev Biochem.* Jun. 2, 2016;85:227-64. doi: 10.1146/annurev-biochem-060815-014607. Epub Apr. 25, 2016.
- Wang et al., CRISPR-Based Therapeutic Genome Editing: Strategies and In Vivo Delivery by AAV Vectors. *Cell.* Apr. 2, 2020;181(1):136-150. doi: 10.1016/j.cell.2020.03.023.
- Wang et al., Recognition of an antigenic peptide derived from tyrosinase-related protein-2 by CTL in the context of HLA-A31 and -A33. *J Immunol.* Jan. 15, 1998;160(2):890-7.
- Wang et al., Recognition of breast cancer cells by CD8+ cytotoxic T-cell clones specific for NY-BR-1. *Cancer Res.* Jul. 1, 2006;66(13):6826-33. doi: 10.1158/0008-5472.CAN-05-3529.
- Wang et al., Utilization of an alternative open reading frame of a normal gene in generating a novel human cancer antigen. *J Exp Med.* Mar. 1, 1996;183(3):1131-40. doi: 10.1084/jem.183.3.1131.
- Wang et al., Virus-like particles for the prevention of human papillomavirus-associated malignancies. *Expert Rev Vaccines.* Feb. 2013;12(2):129-41. doi: 10.1586/erv.12.151. Author Manuscript, 22 pages.
- Wei et al., Systemic nanoparticle delivery of CRISPR-Cas9 ribonucleoproteins for effective tissue specific genome editing. *Nat Commun.* Jun. 26, 2020;11(1):3232. doi: 10.1038/s41467-020-17029-3.
- Wölfel et al., Two tyrosinase nonapeptides recognized on HLA-A2 melanomas by autologous cytolytic T lymphocytes. *Eur J Immunol.* Mar. 1994;24(3):759-64. doi: 10.1002/eji.1830240340.
- Yang et al., HIV-1 virus-like particles produced by stably transfected *Drosophila* S2 cells: a desirable vaccine component. *J Virol.* Jul. 2012;86(14):7662-76. doi: 10.1128/JVI.07164-11. Epub May 2, 2012.
- Yee et al., A general method for the generation of high-titer, pantropic retroviral vectors: highly efficient infection of primary hepatocytes. *Proc Natl Acad Sci U S A.* Sep. 27, 1994;91(20):9564-8. doi: 10.1073/pnas.91.20.9564.
- Yu et al., Poor immunogenicity of a self/tumor antigen derives from peptide-MHC-I instability and is independent of tolerance. *J Clin Invest.* Aug. 2004; 114(4):551-9. doi: 10.1172/JCI21695.
- Zarour et al., Melan-A/MART-1(51-73) represents an immunogenic HLA-DR4-restricted epitope recognized by melanoma-reactive CD4(+) T cells. *Proc Natl Acad Sci U S A.* Jan. 4, 2000;97(1):400-5. doi: 10.1073/pnas.97.1.400.
- Zeltins, A., Construction and characterization of virus-like particles: a review. *Mol Biotechnol.* Jan. 2013;53(1):92-107. doi: 10.1007/s12033-012-9598-4.
- Zhang et al., Cell-specific targeting of lentiviral vectors mediated by fusion proteins derived from Sindbis virus, vesicular stomatitis virus, or avian sarcoma/leukosis virus. *Retrovirology.* Jan. 25, 2010;7:3. doi: 10.1186/1742-4690-7-3.
- Zhao et al., Glycosylase base editors enable C-to-A and C-to-G base changes. *Nat Biotechnol.* Jan. 2021;39(1):35-40. doi: 10.1038/s41587-020-0592-2. Epub Jul. 20, 2020. Erratum in: *Nat Biotechnol.* Jan. 2021;39(1):115. doi: 10.1038/s41587-020-0648-3.
- Zhao et al., Study on p21 gene knock out in G401 cell line by using CRISPR/Cas9 system. *Tianjin Med J.* Oct. 2016;44(10):1190-1194.

* cited by examiner

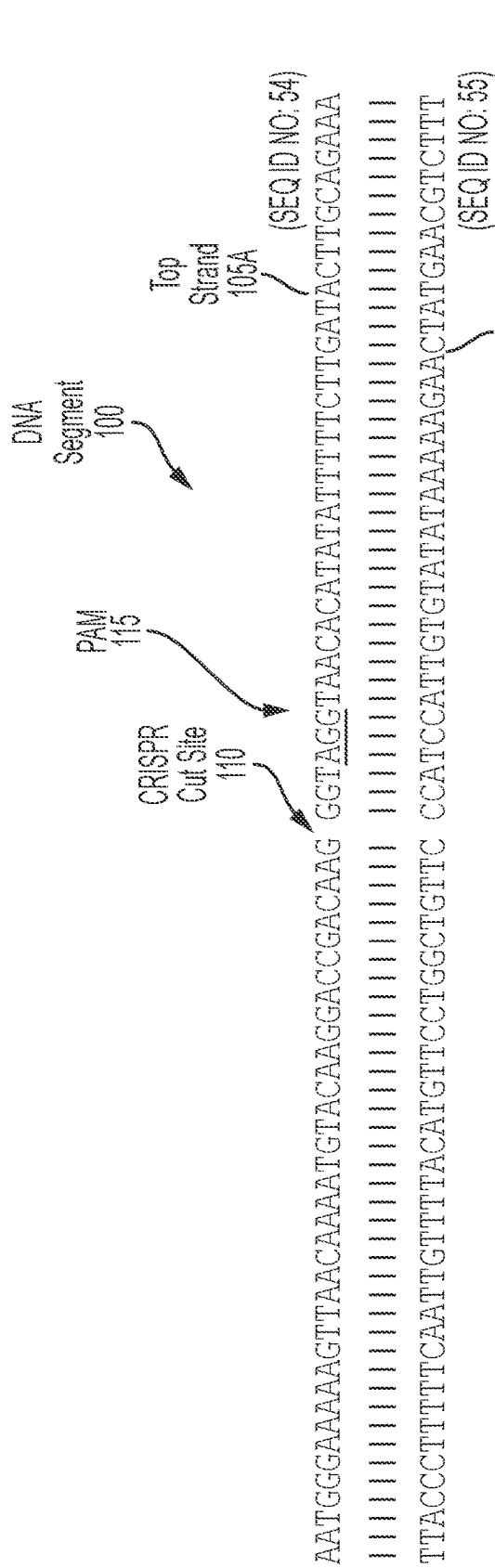


FIG. 1

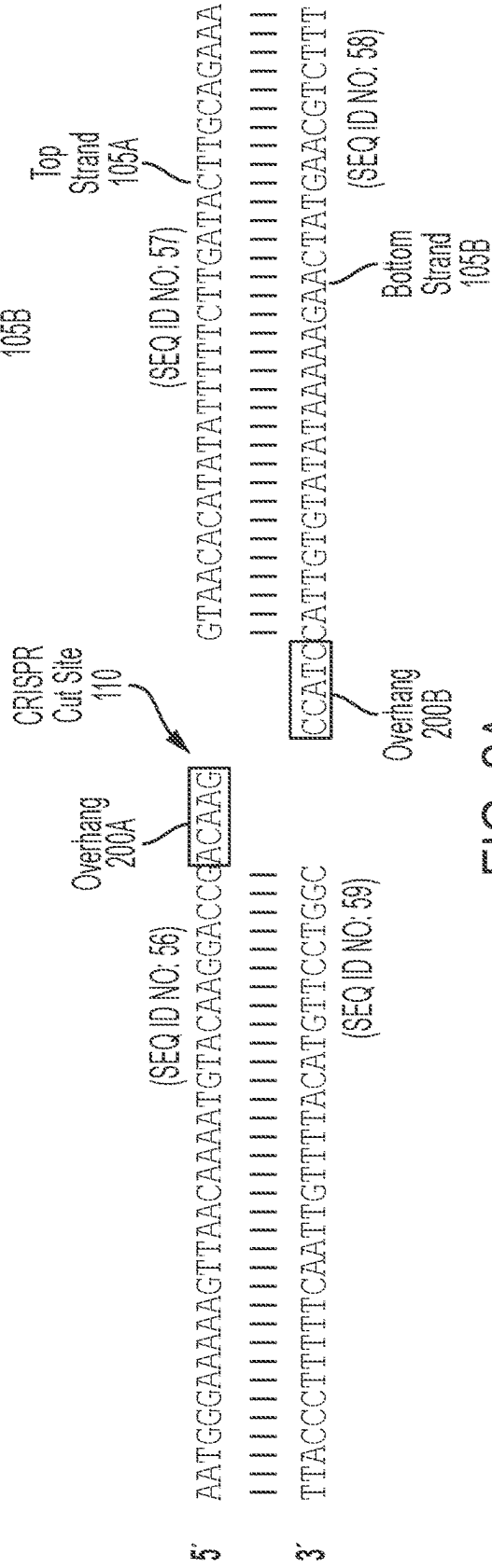


FIG. 2A

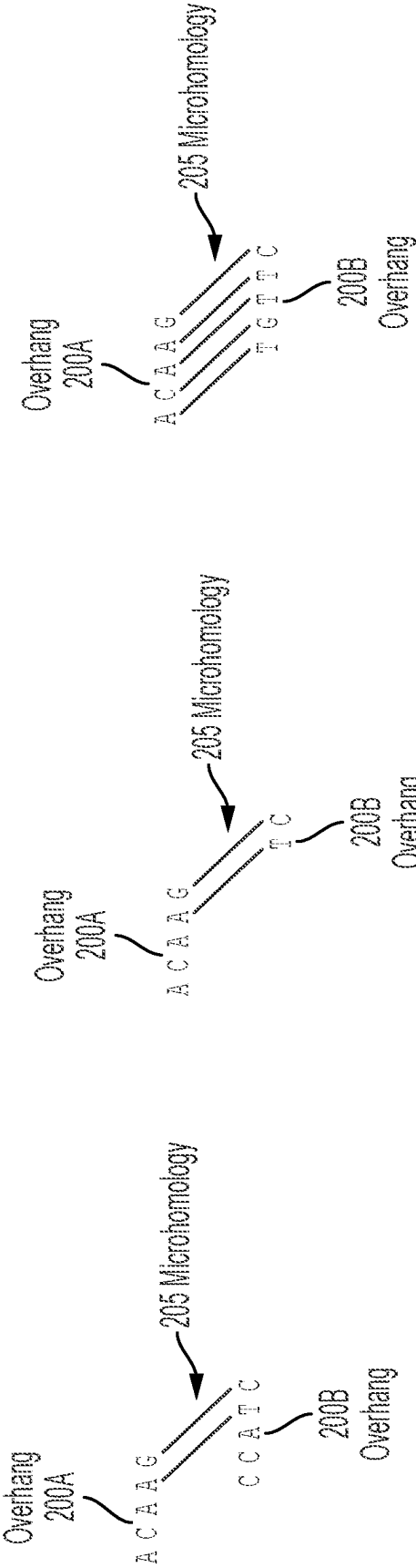


FIG. 2D

FIG. 2C

FIG. 2B

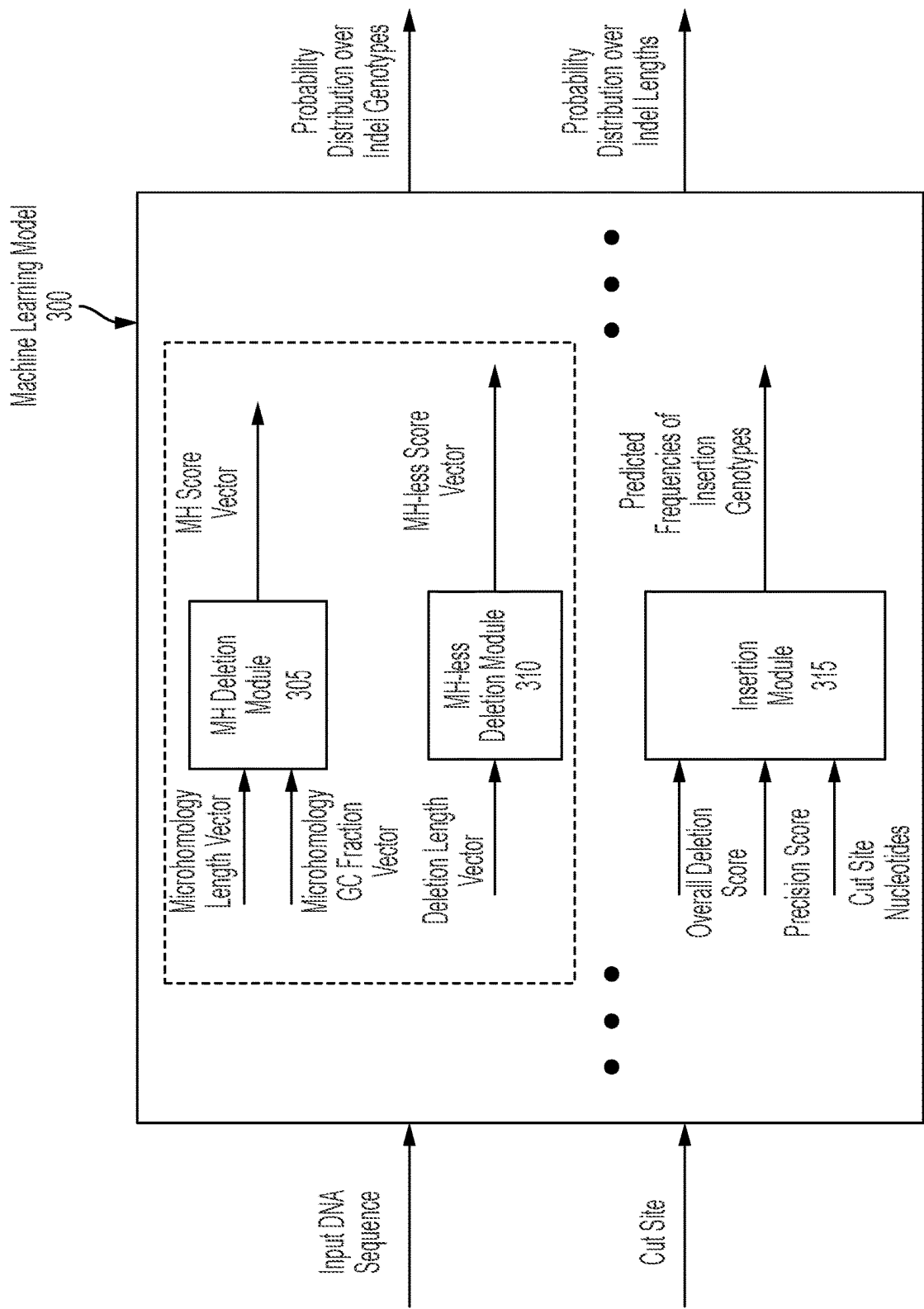


FIG. 3A

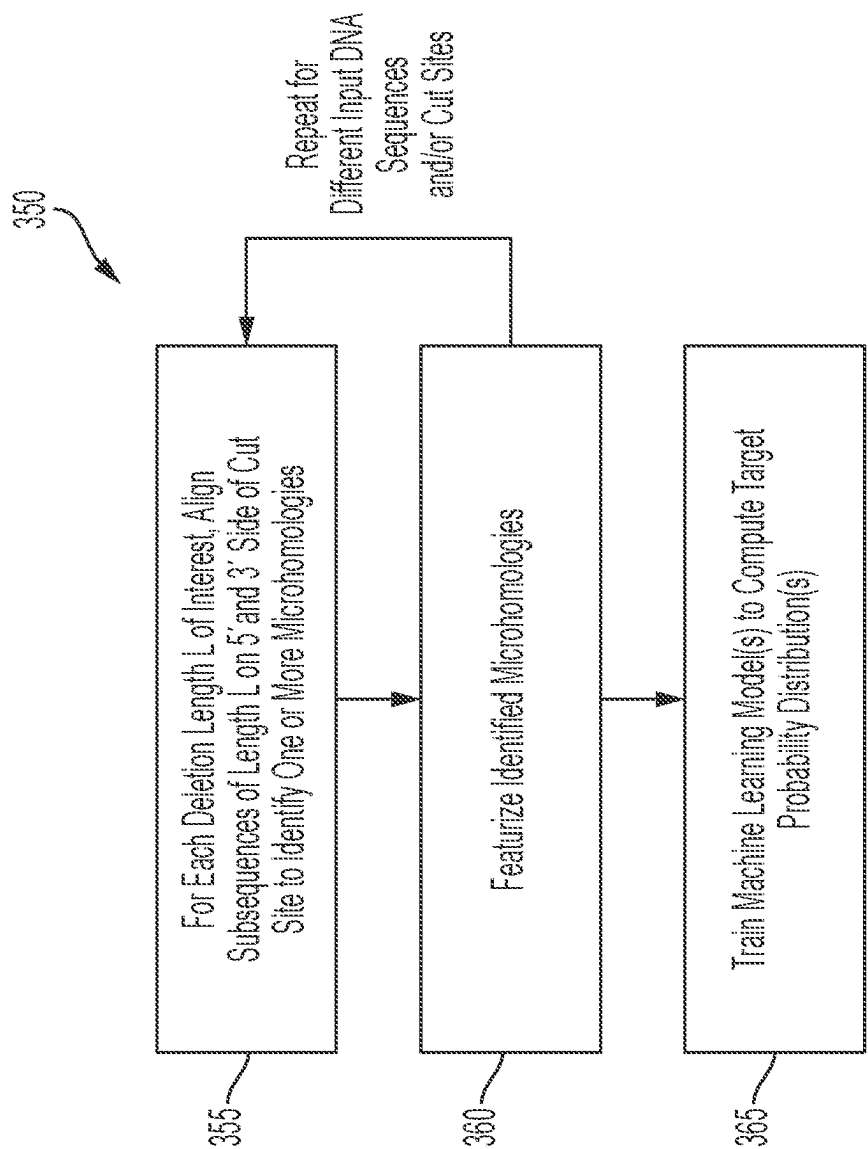


FIG. 3B

Neural
Network
400A

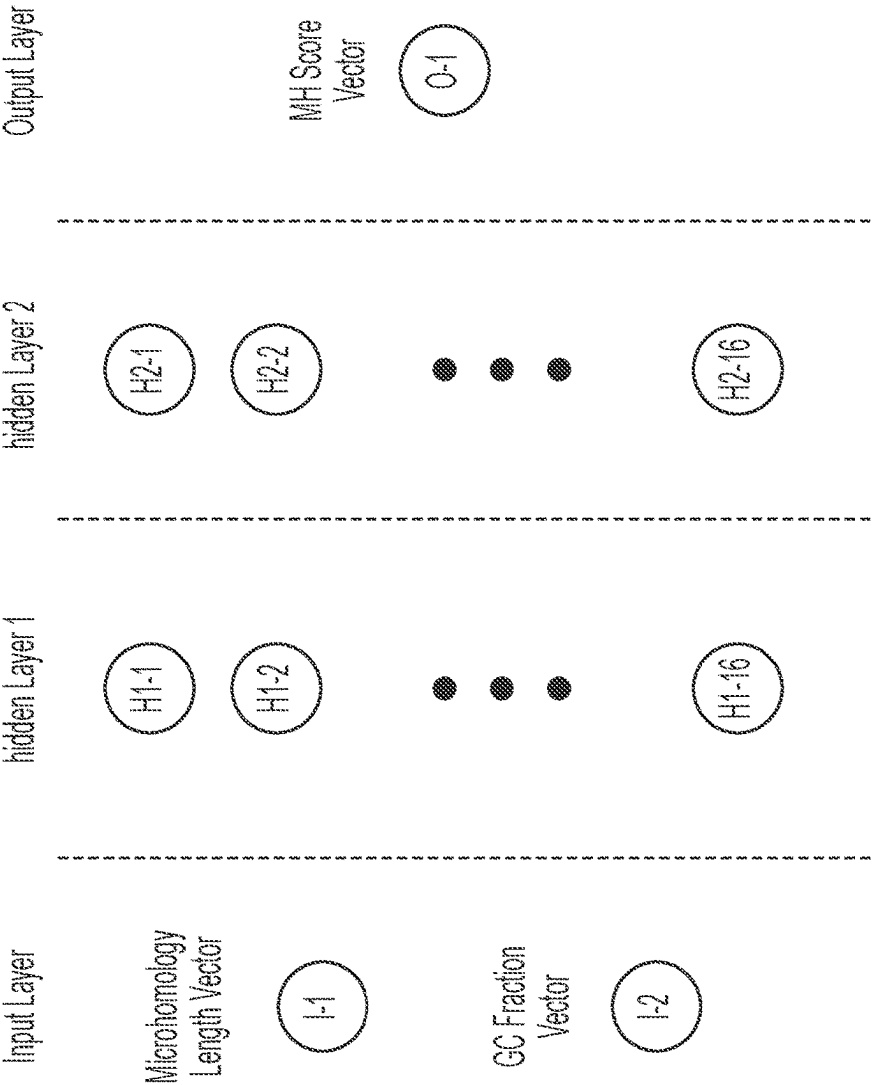


FIG. 4A

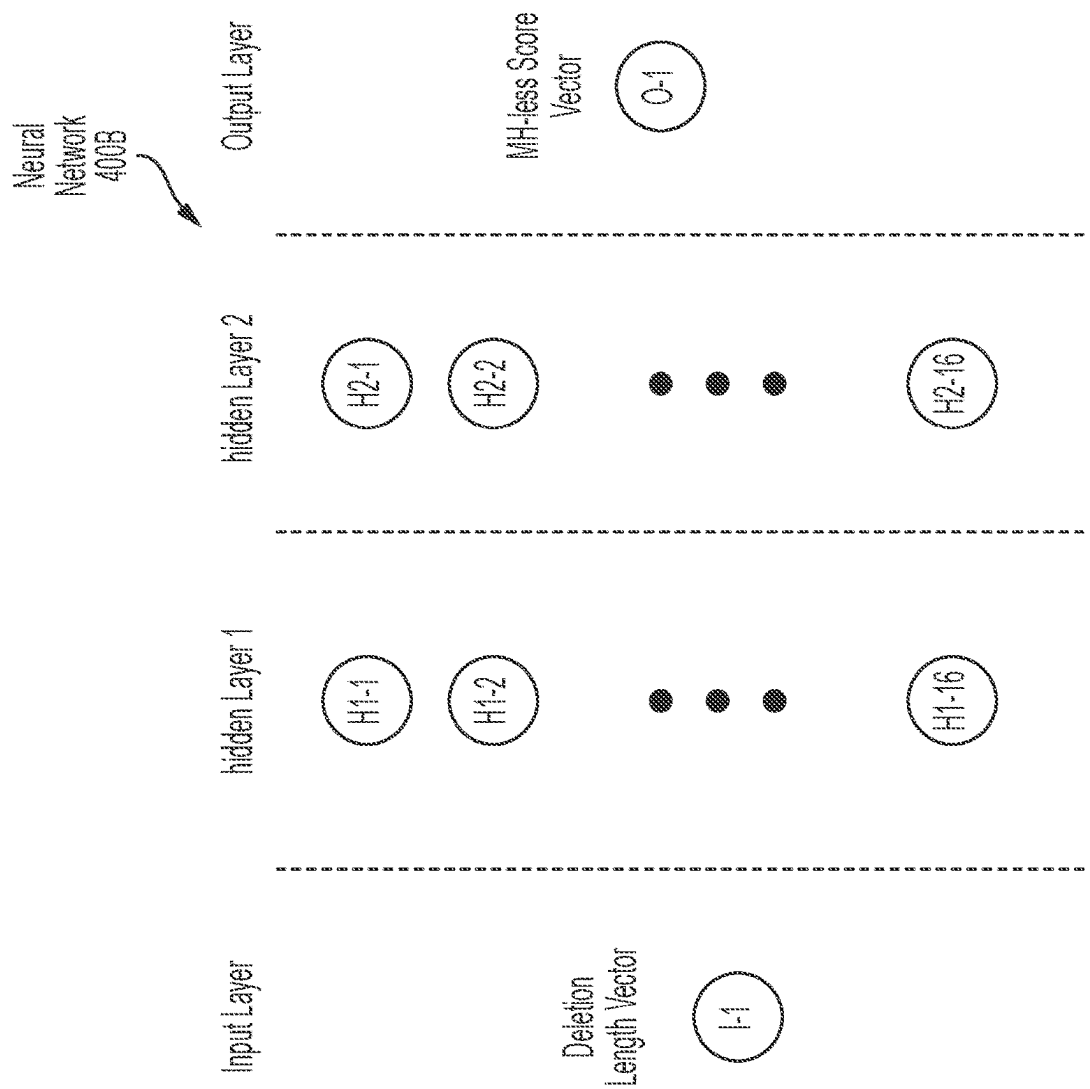


FIG. 4B

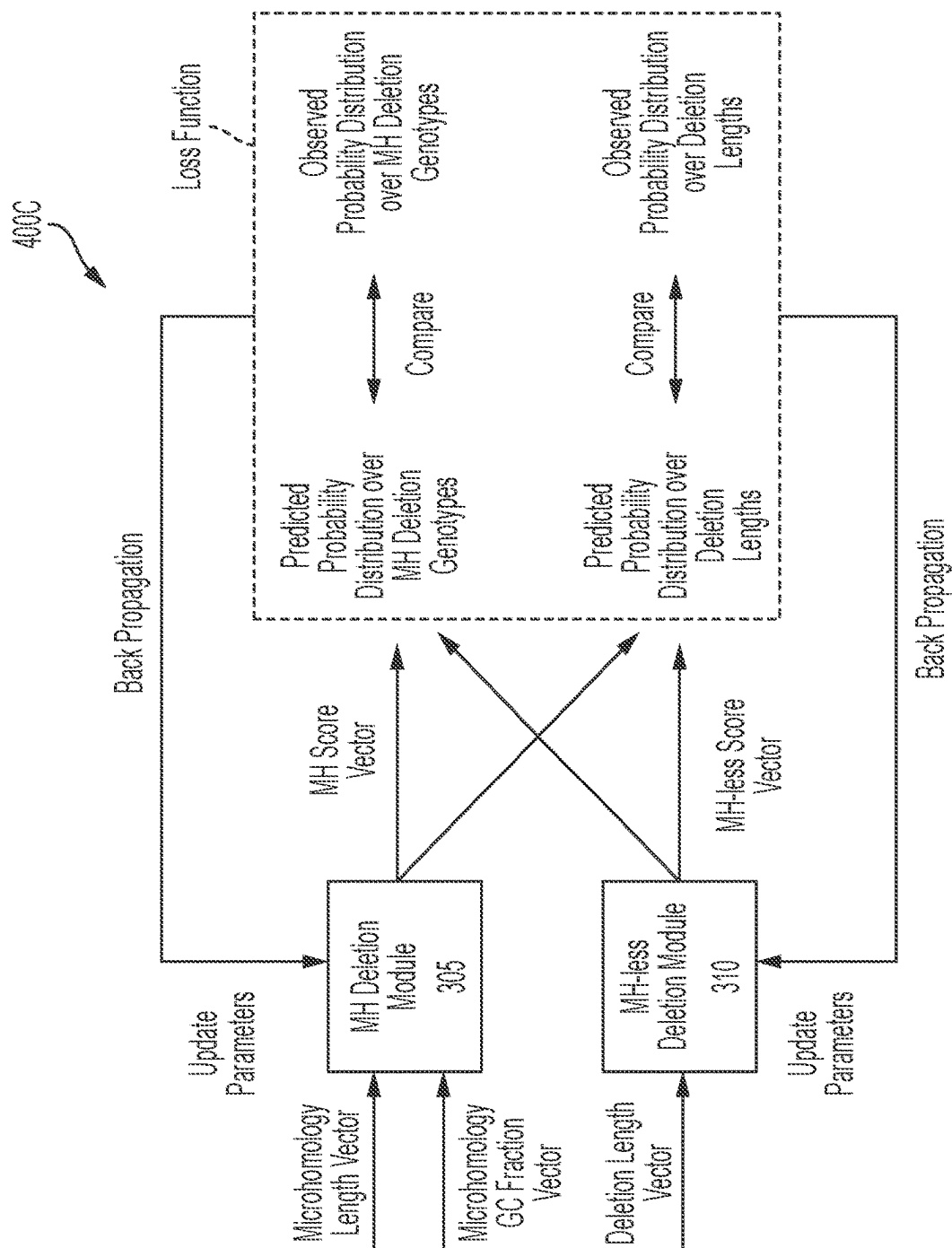


FIG. 4C

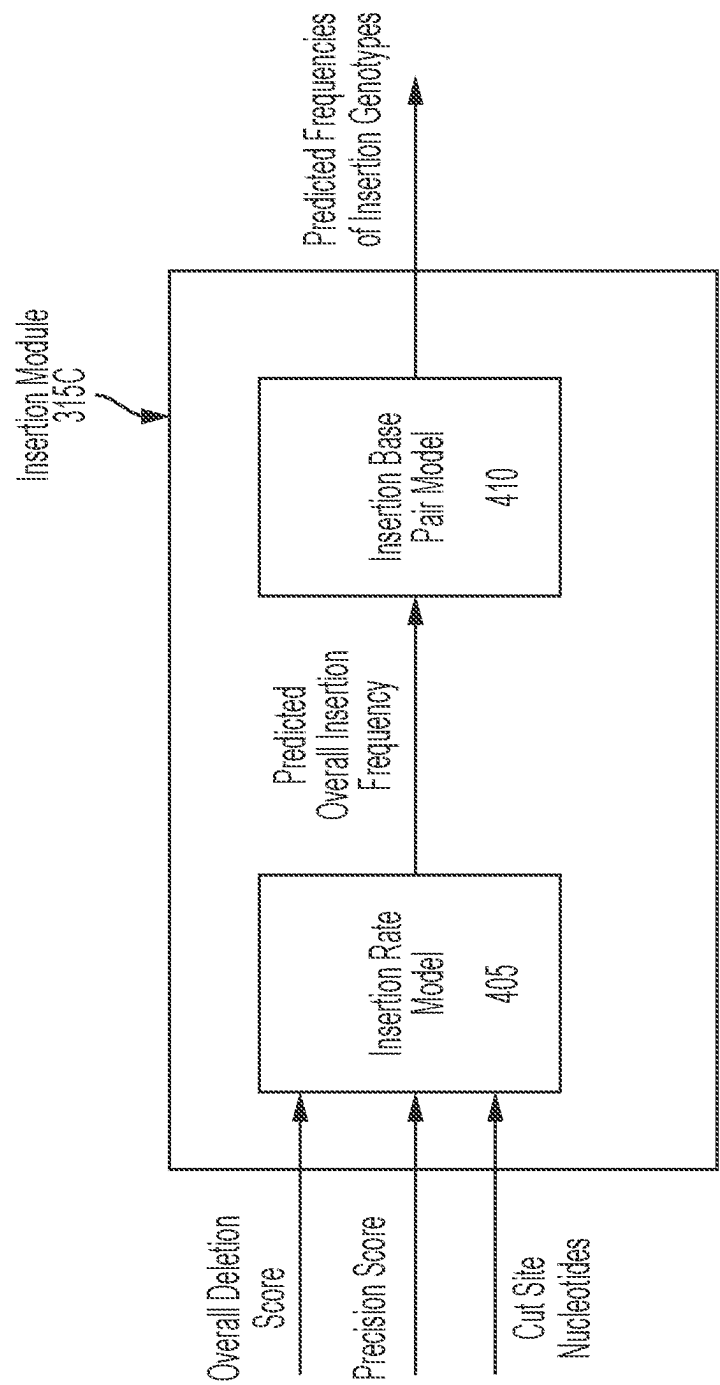


FIG. 4D

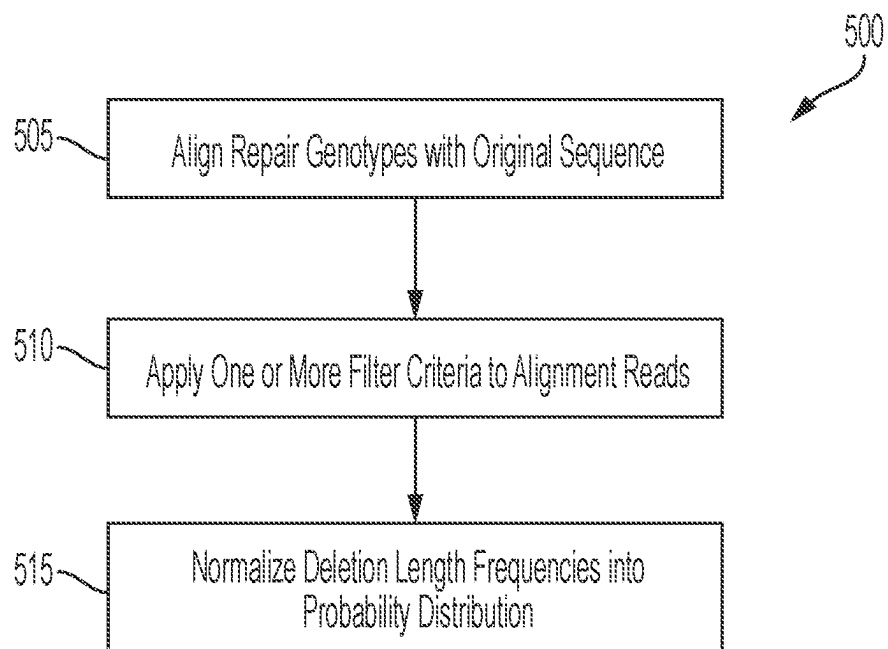


FIG. 5

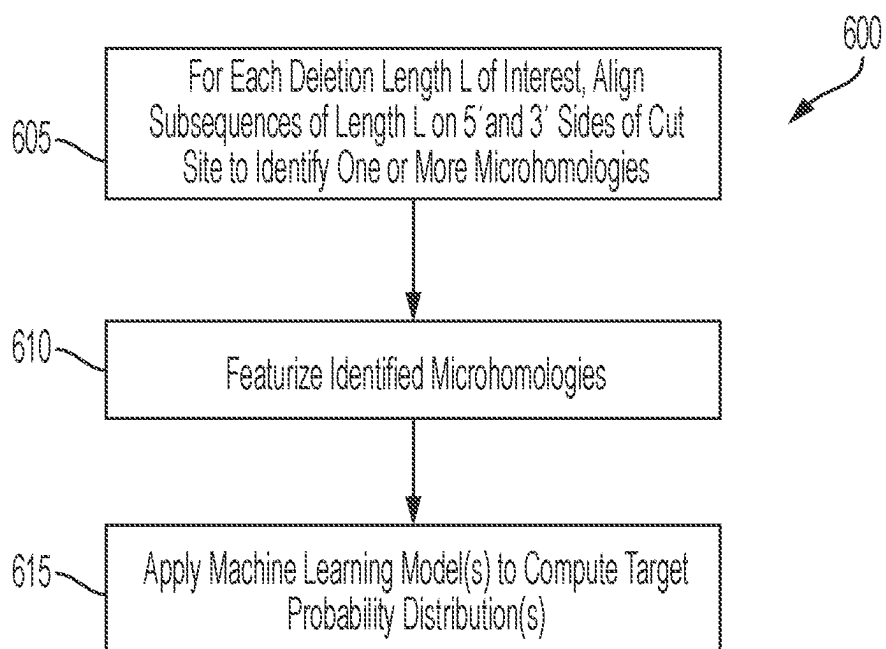


FIG. 6

CGT C GT CGGT
TGC TG C TGCCA

Blunt-end cut

ACGTA CGTACCGGT
TGCATG CATGCCA

1 base pair staggered cut

(SEQ ID NO: 60)

ACGTACCGTACCGGT
TGCATGGCATGCCA

Polymerase fill-in and
ligation

(SEQ ID NO: 61)

(SEQ ID NO: 62)

ACGTACCGTACCGGT
||||| |||||
ACGTAC-GTACCGGT

(SEQ ID NO: 63)

Alignment of repair
product (top) to original
sequence (bottom)
shows a 1 base pair
insertion of "C" which
was originally directly on
the 5' side (left side) of
the cut site

FIG. 7

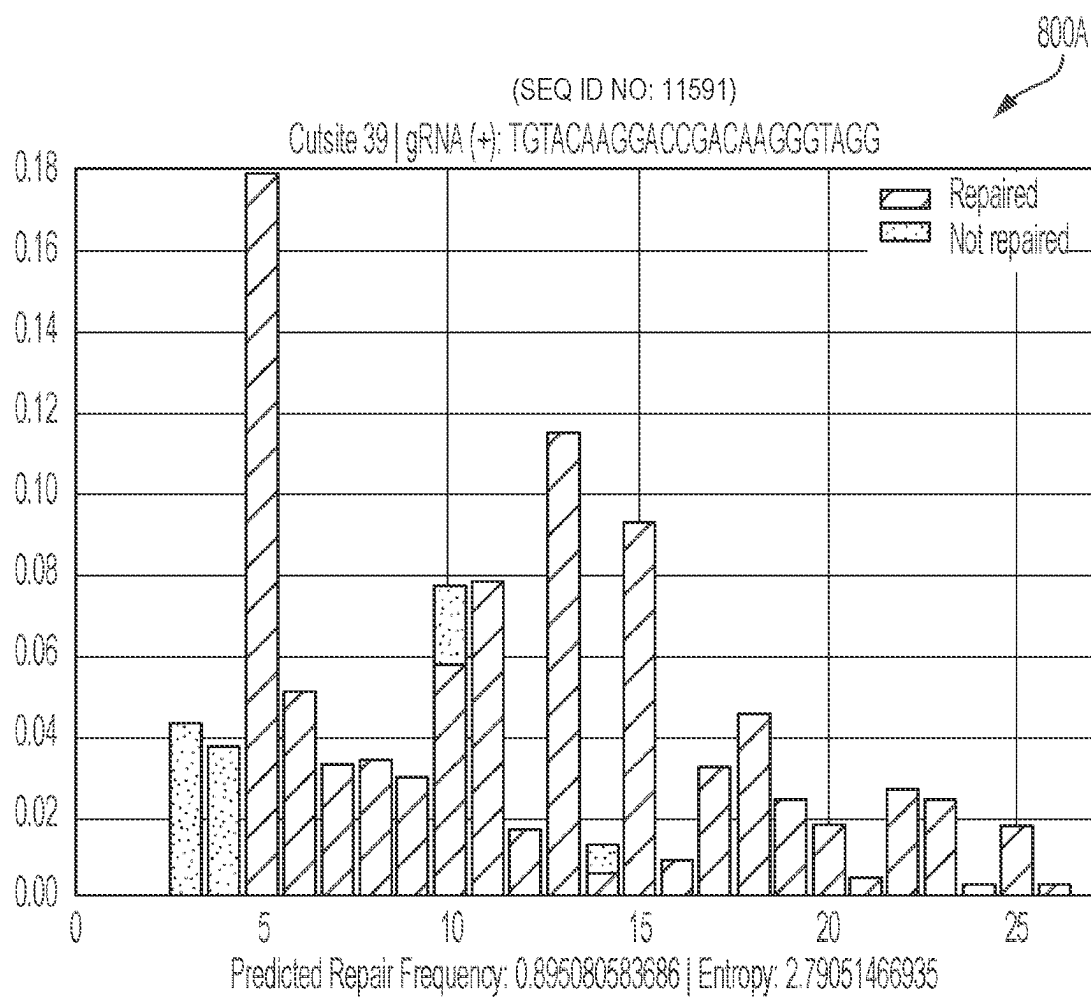


FIG. 8A

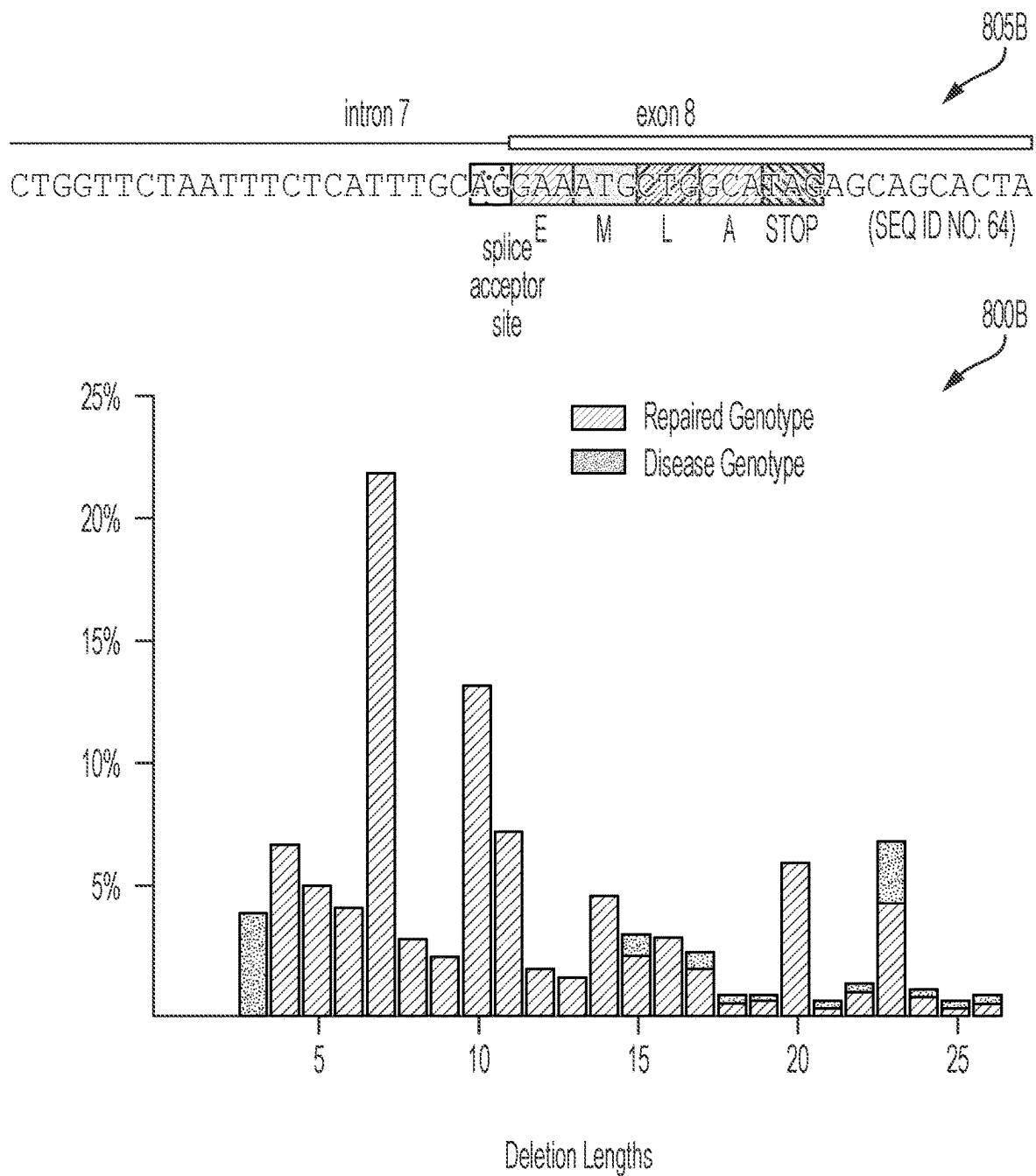


FIG. 8B

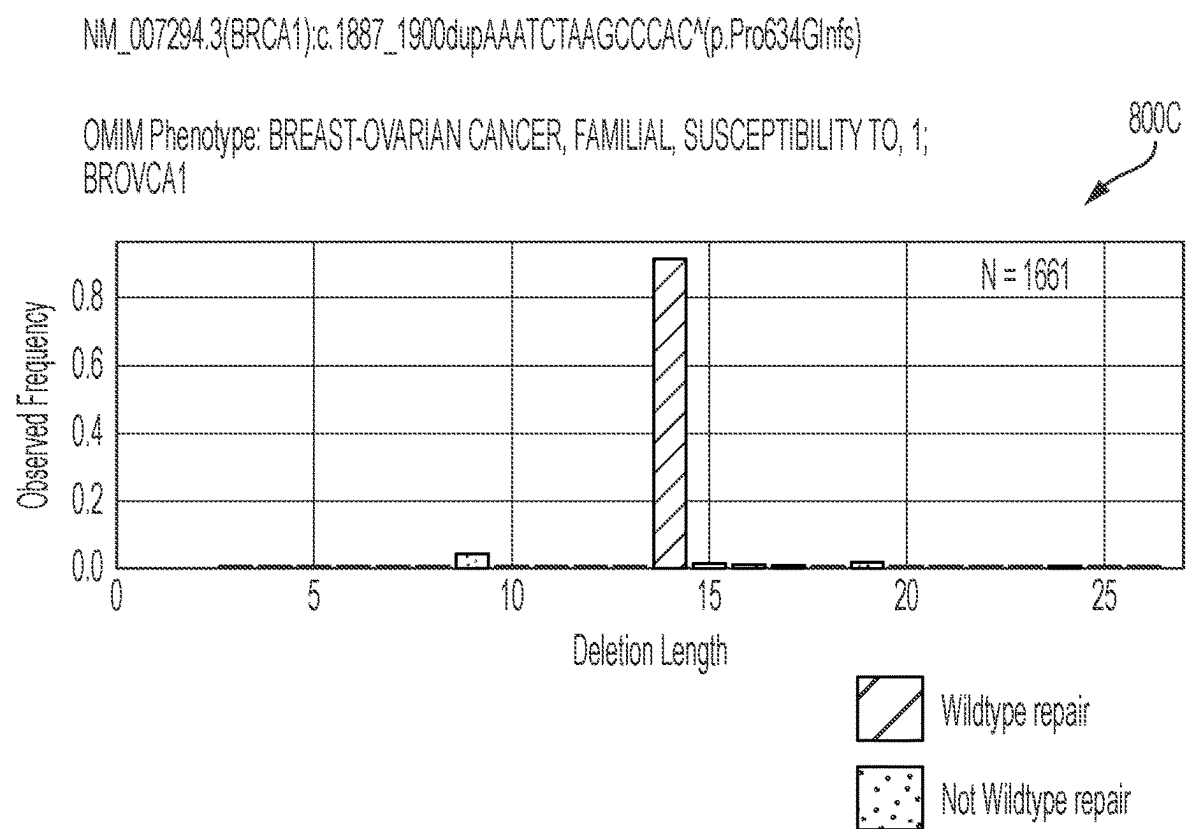


FIG. 8C

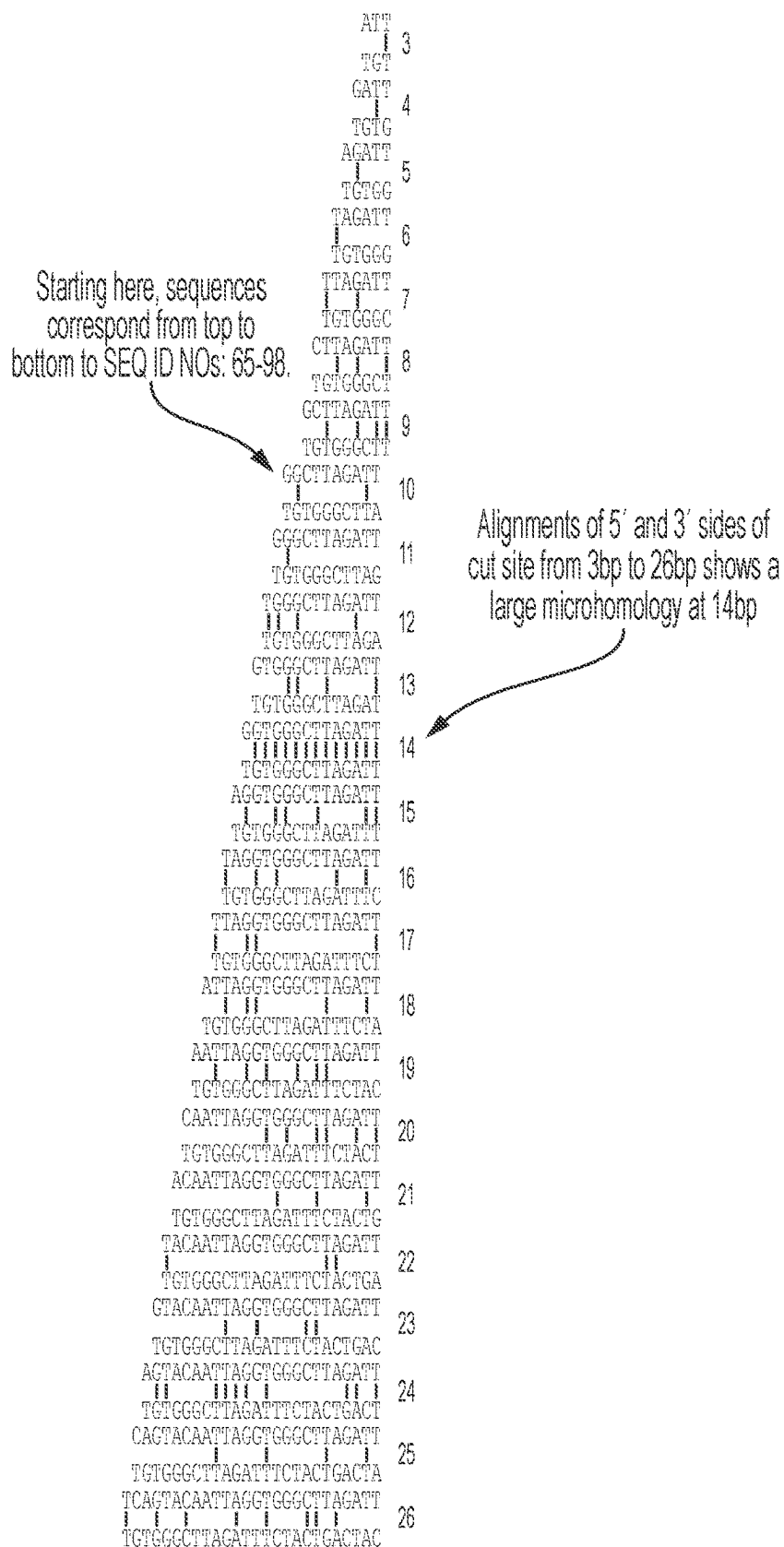


FIG. 8D

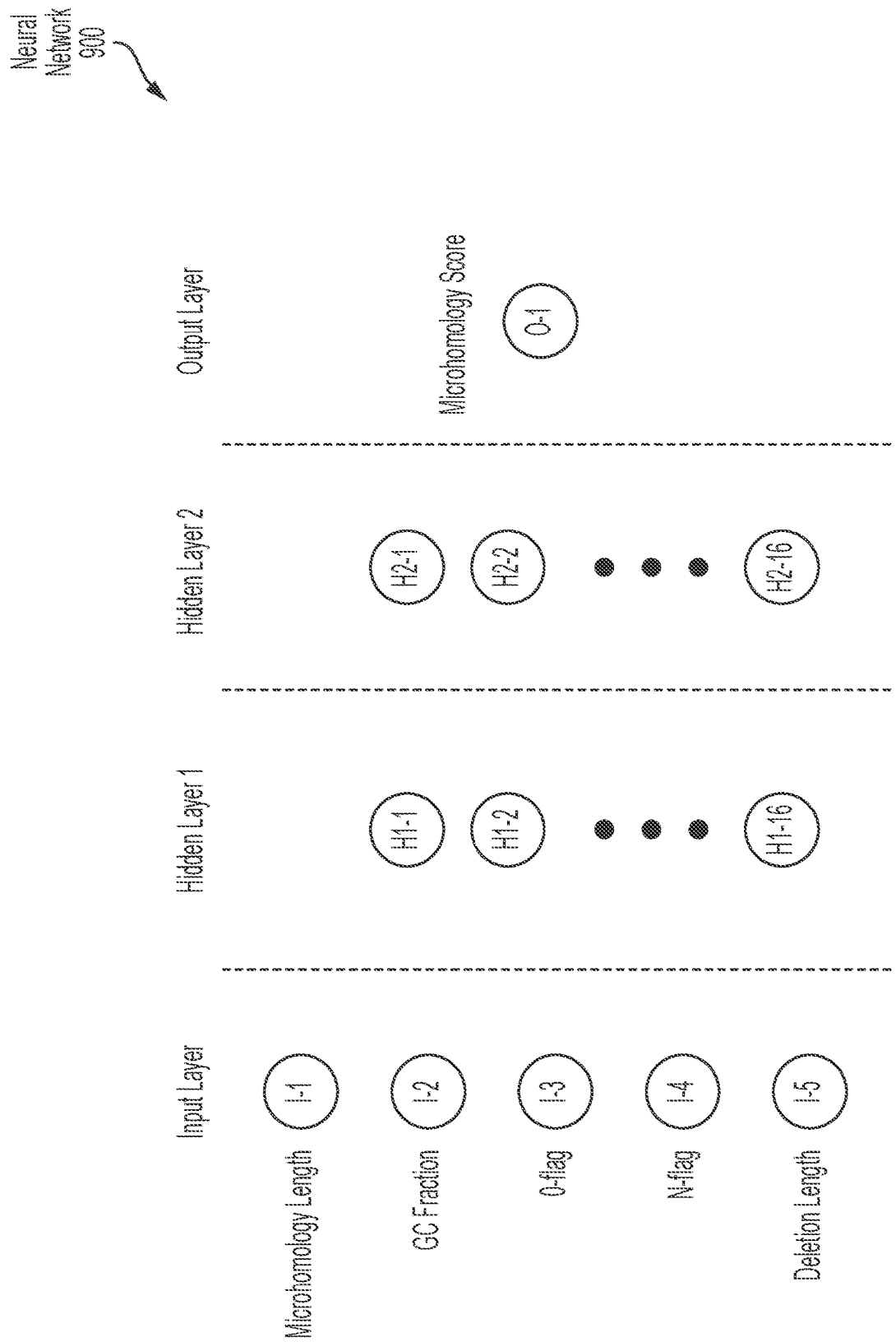


FIG. 9

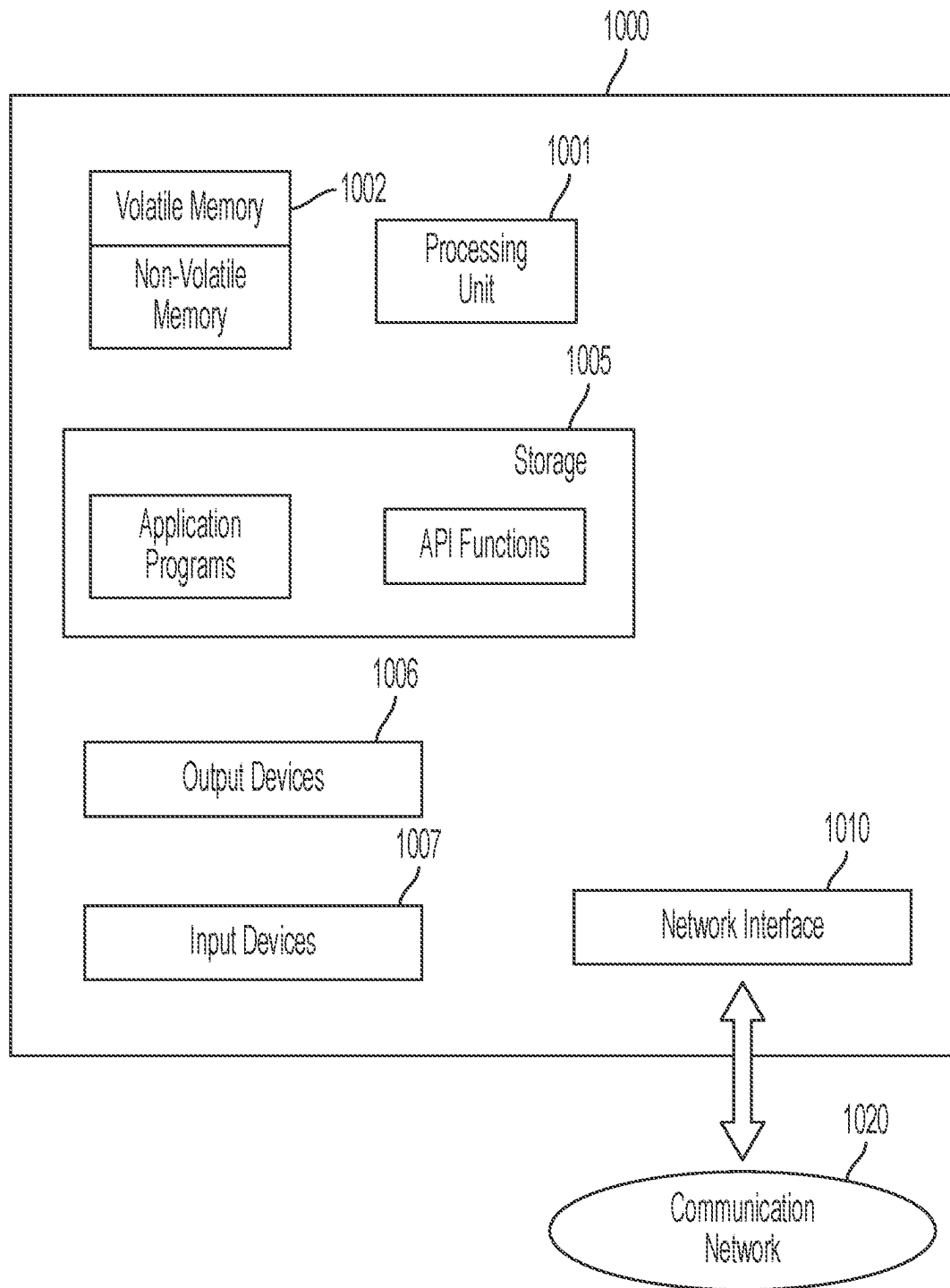


FIG. 10

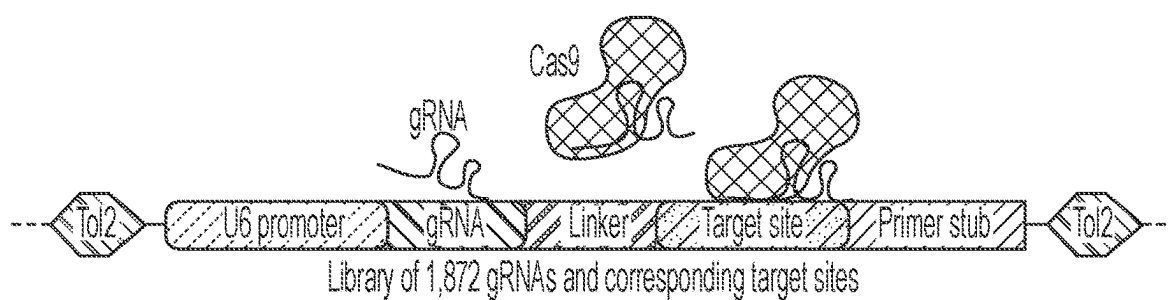


FIG. 11A

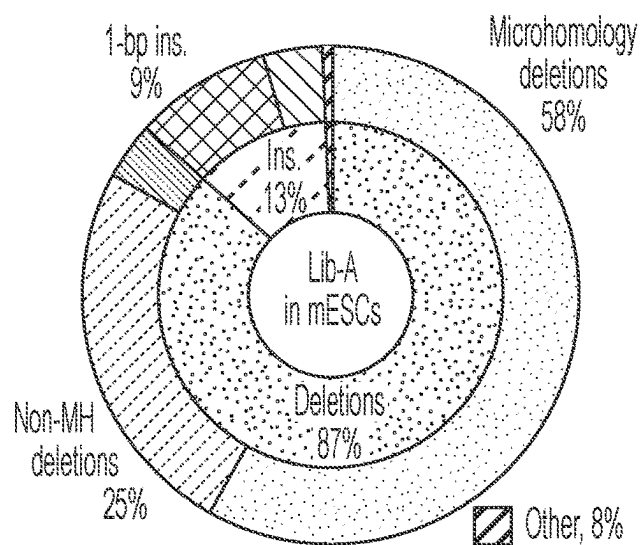


FIG. 11B

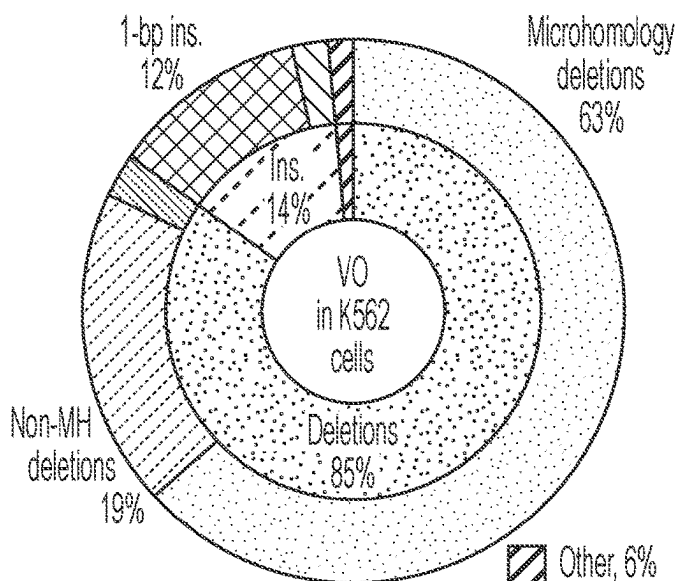


FIG. 11C

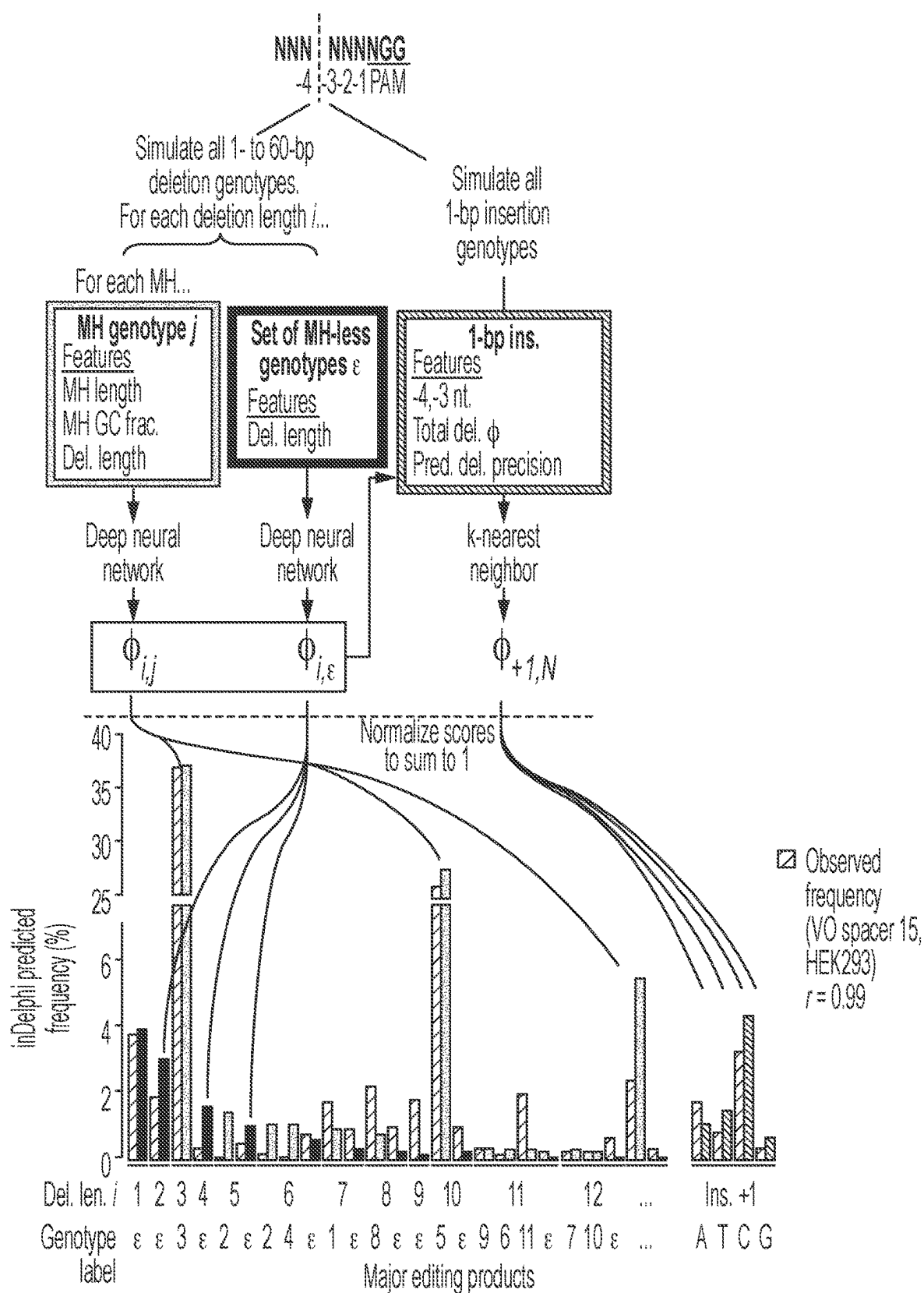


FIG. 12A

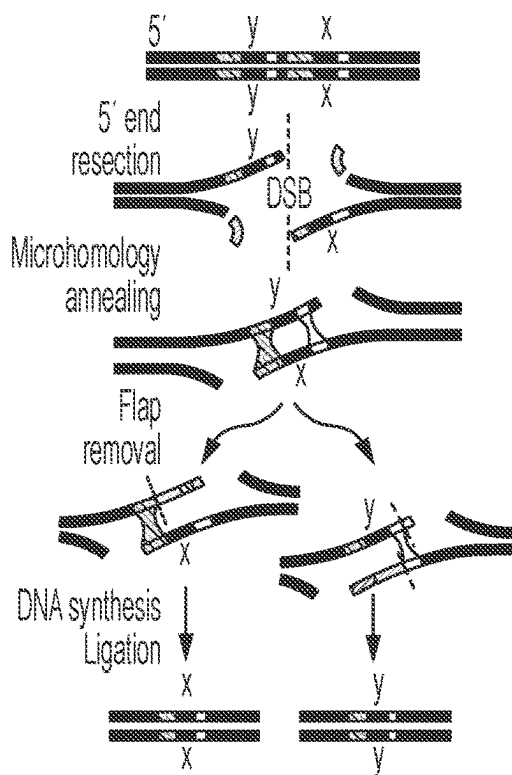


FIG. 12B

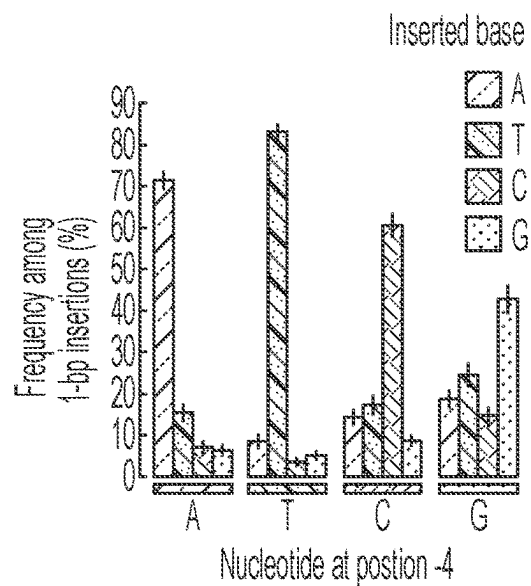


FIG. 12C

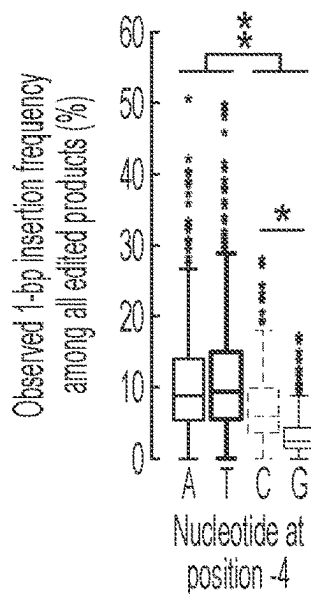


FIG. 12D

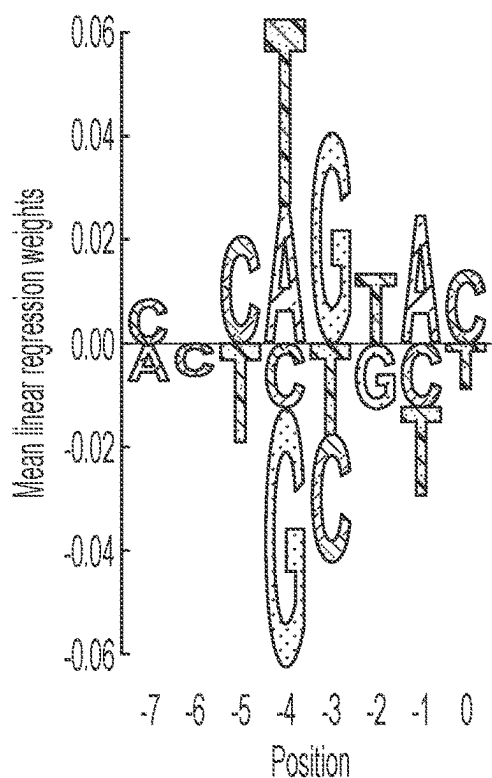


FIG. 12E

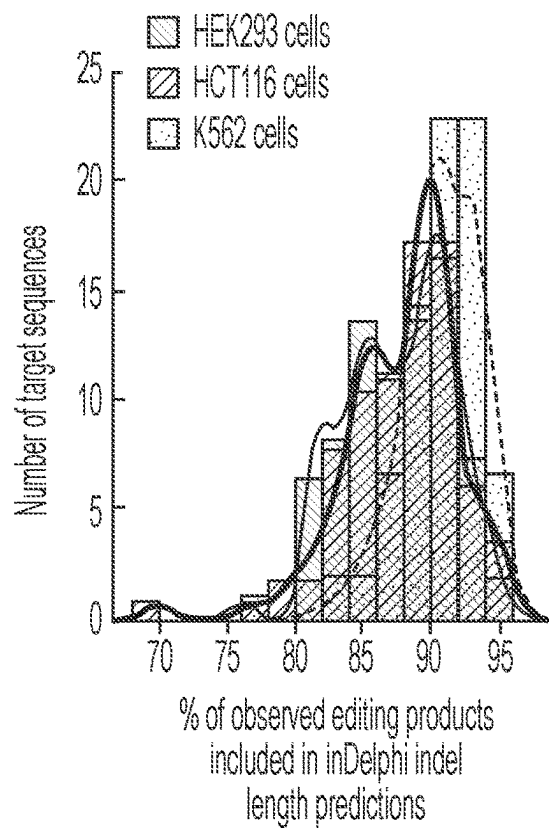


FIG. 13A

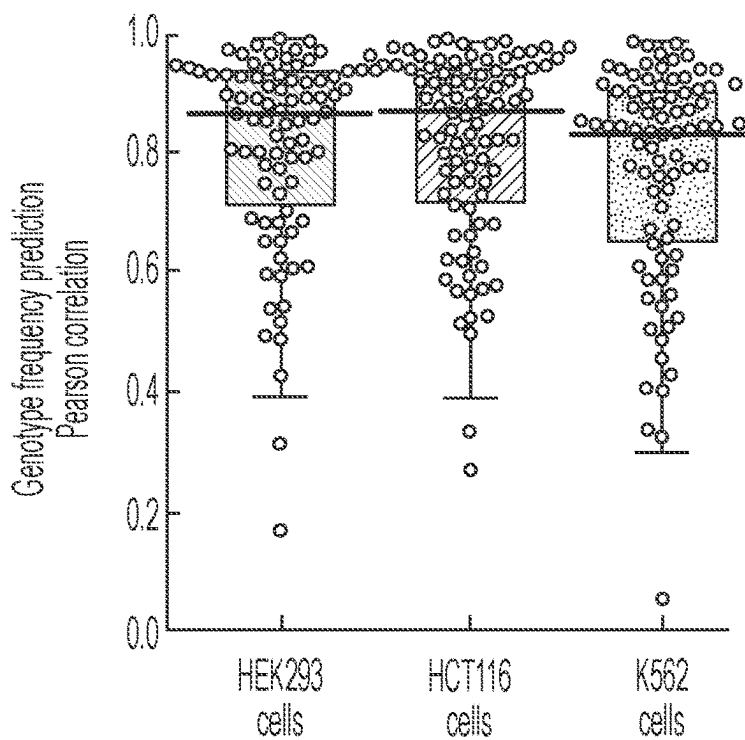


FIG. 13B

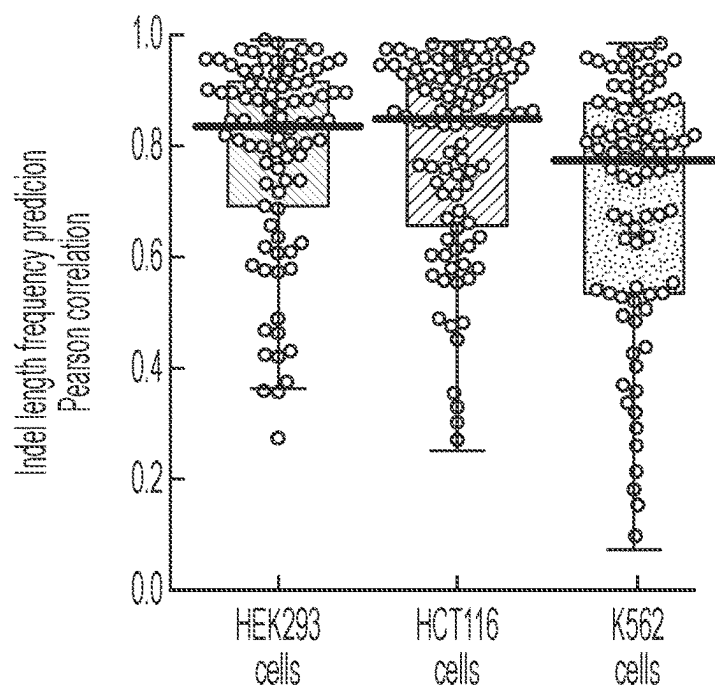


FIG. 13C

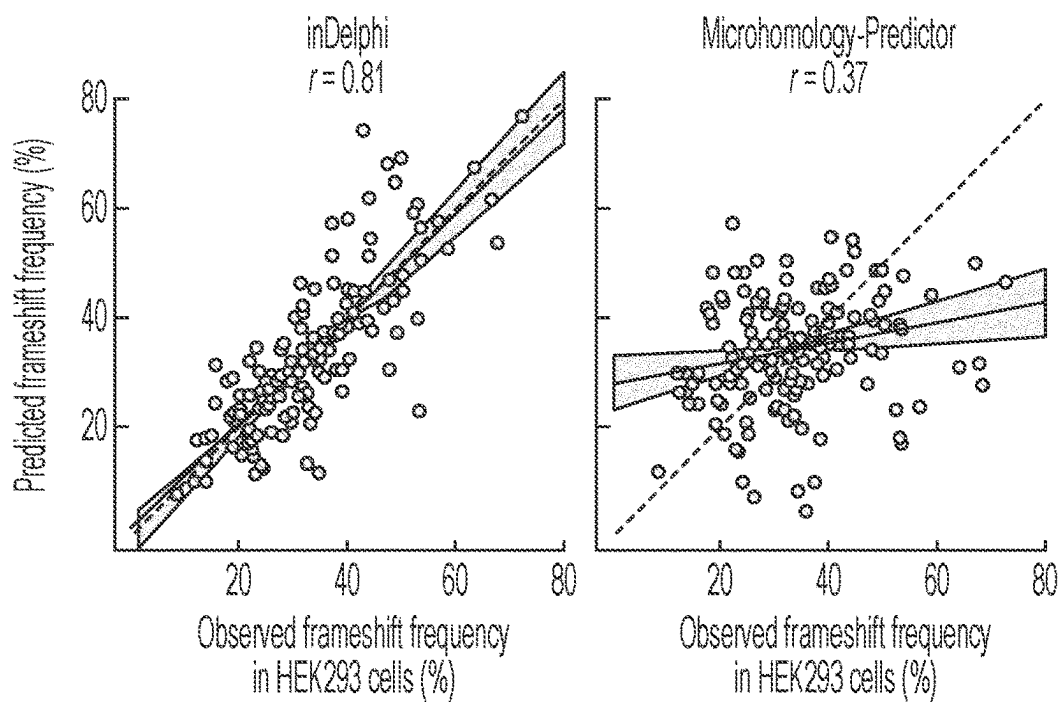


FIG. 13D

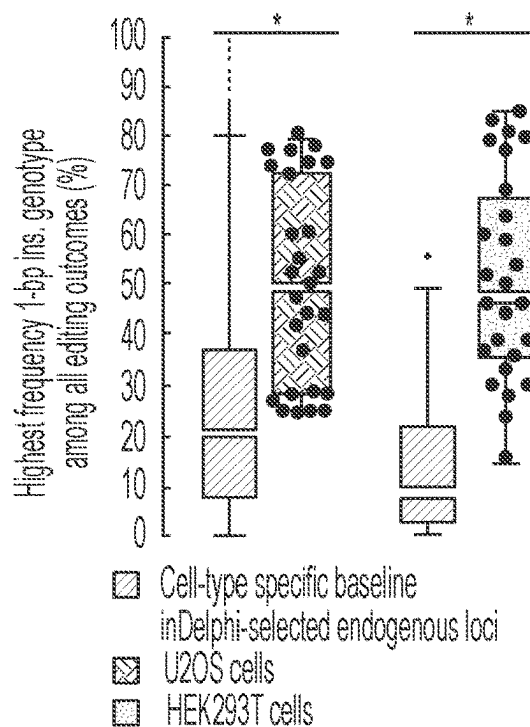


FIG. 13E

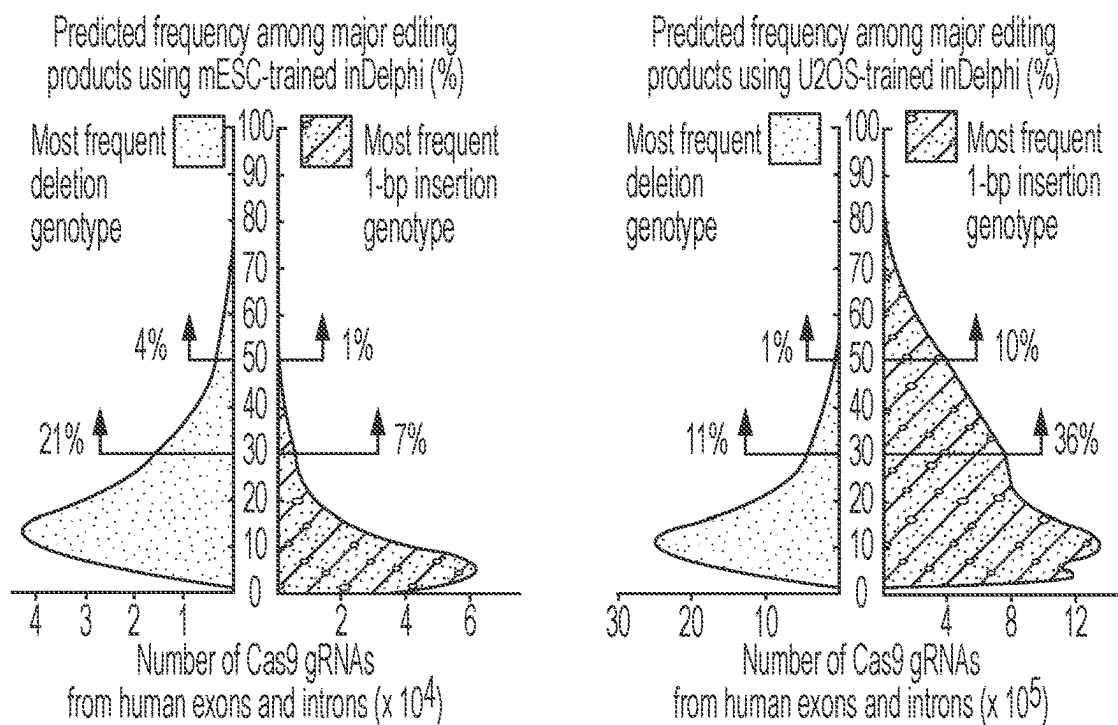


FIG. 13F

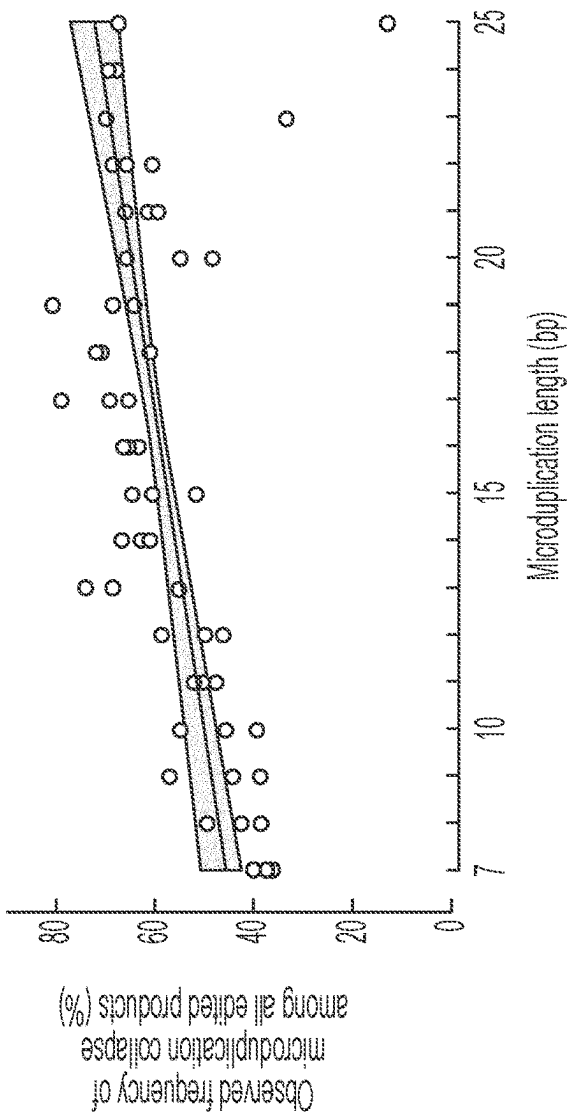


FIG. 14B

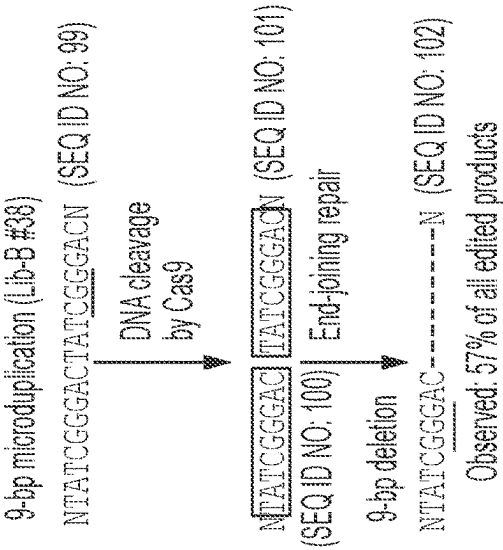


FIG. 14A

From top to bottom: SEQ ID NOs: 103 - 105

GCCCCACAAAGCGGACCCAGCCAGCNN NNTAGGGAATAAGGCCAACTGCACCCC
 AGTGCACGAGTGACGGGATAATGTGNN NNCAGGTAATTGATAGGCTTCCAACTTA
 CCGAGACCTACGCGAACCGAAATCGNN NNCCGGTGCGCAAGCAGGGCTGGCGTCC

PAM

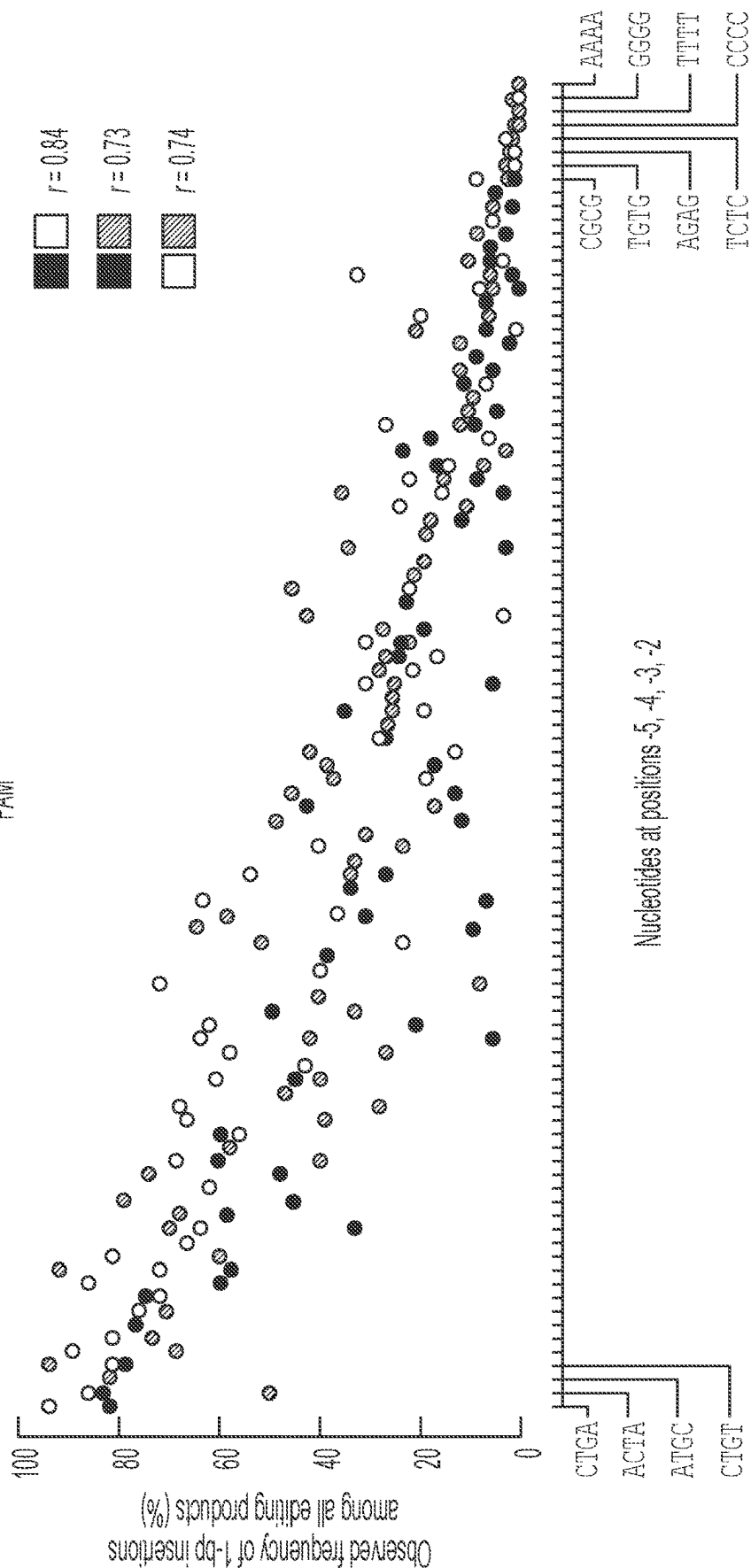


FIG. 14C

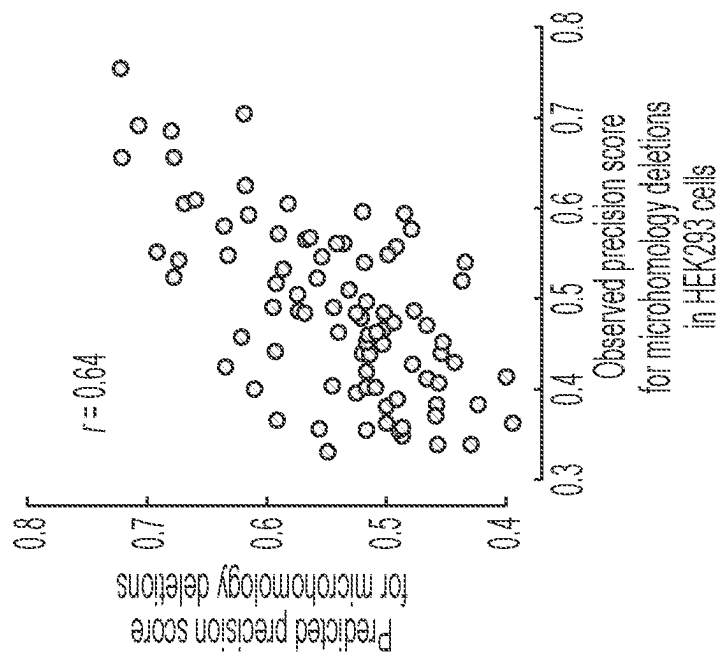


FIG. 14E

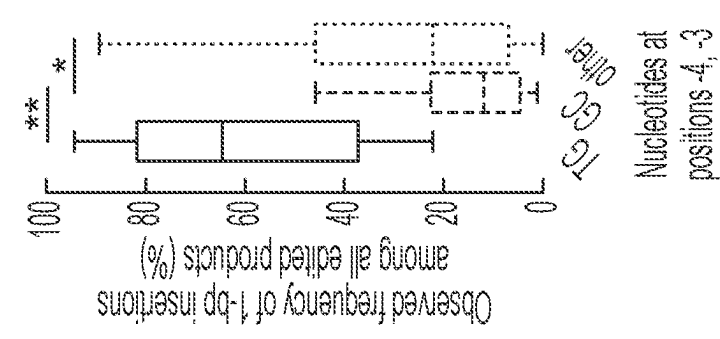


FIG. 14D

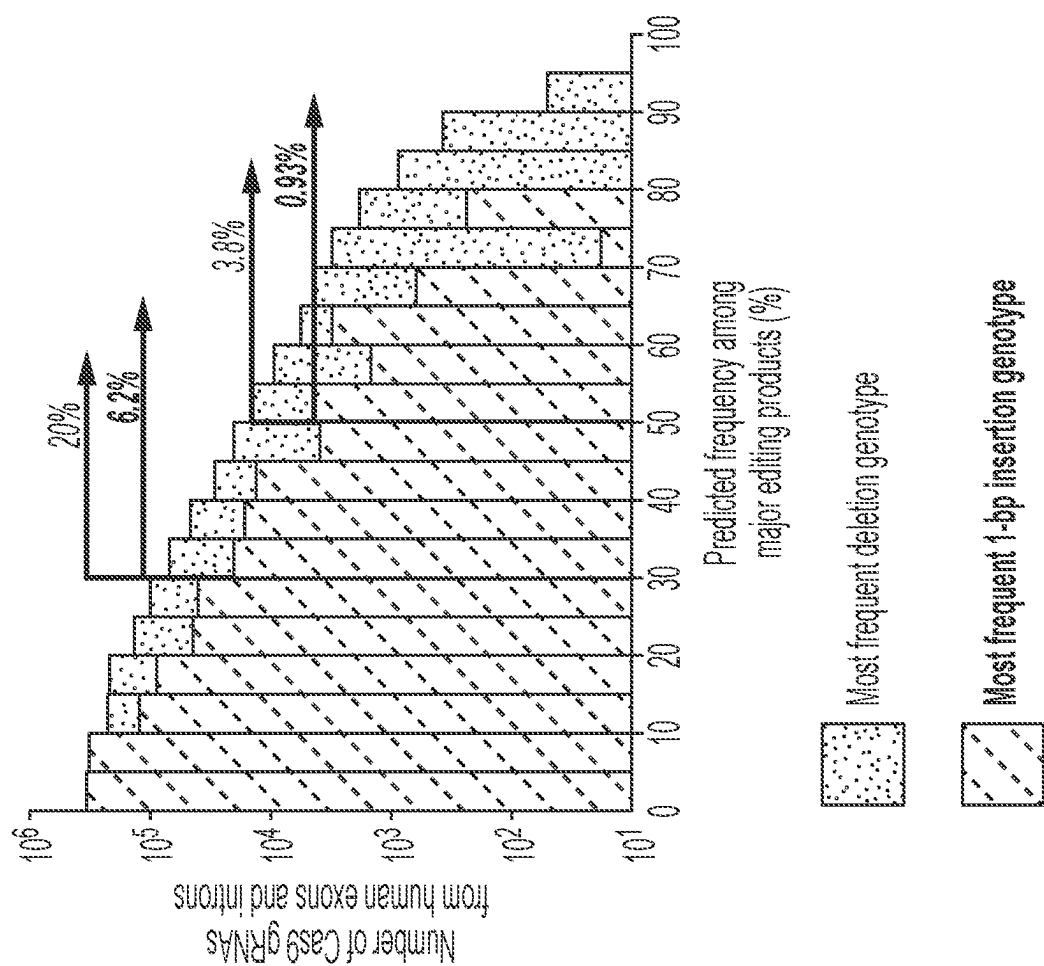


FIG. 14F

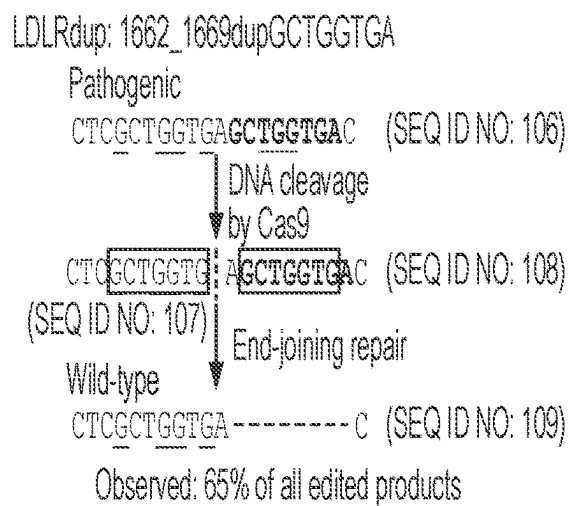


FIG. 15A

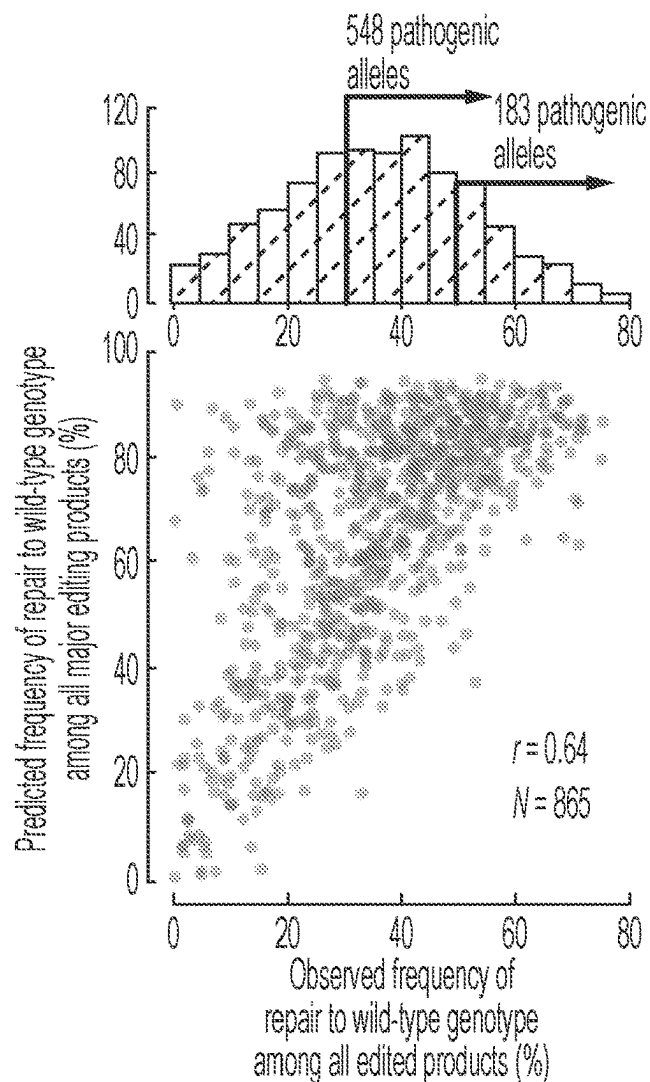


FIG. 15B

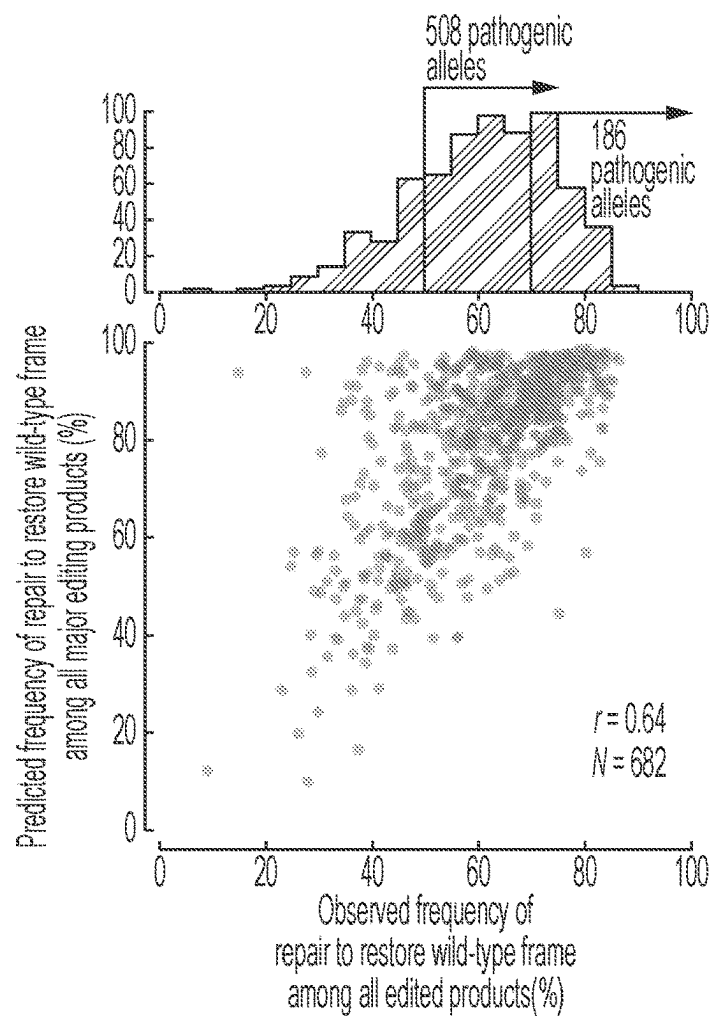


FIG. 15C

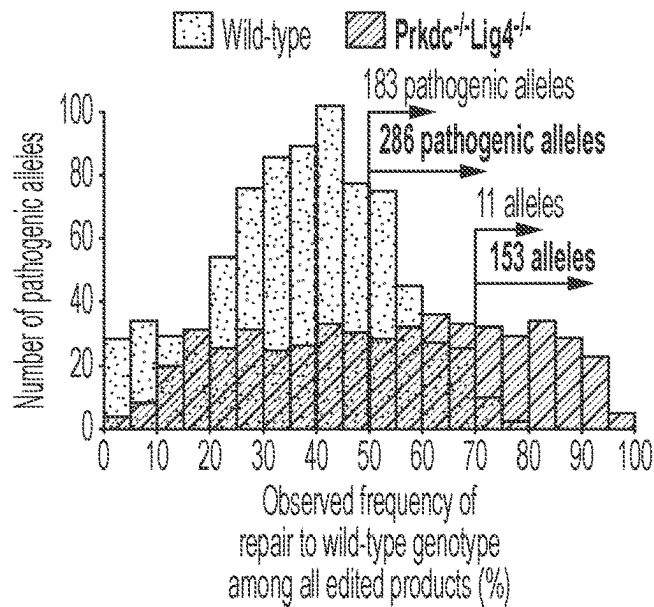


FIG. 15D

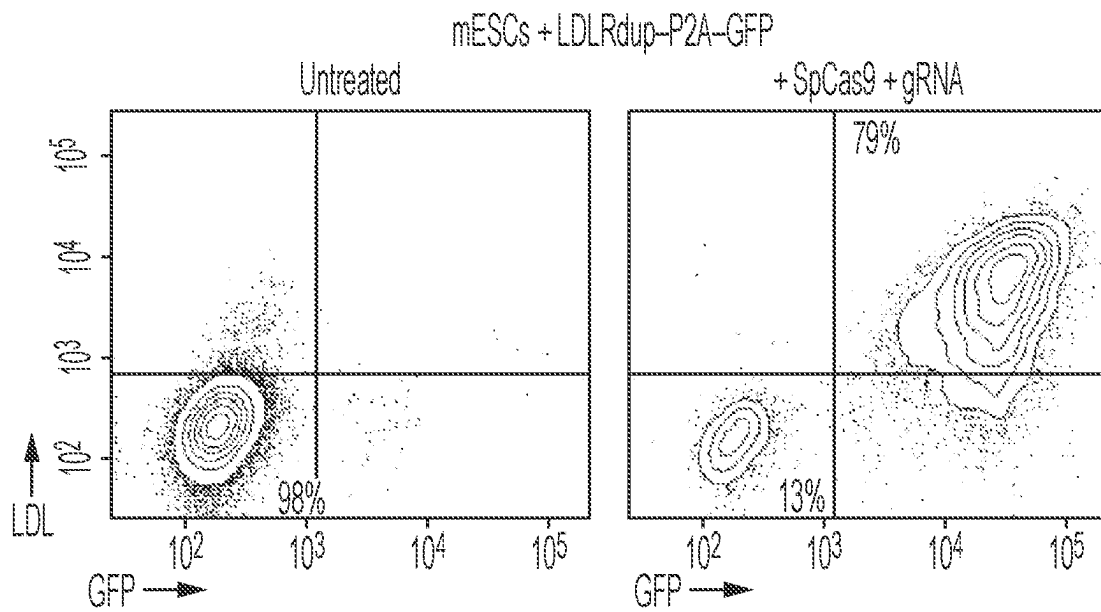


FIG. 15E

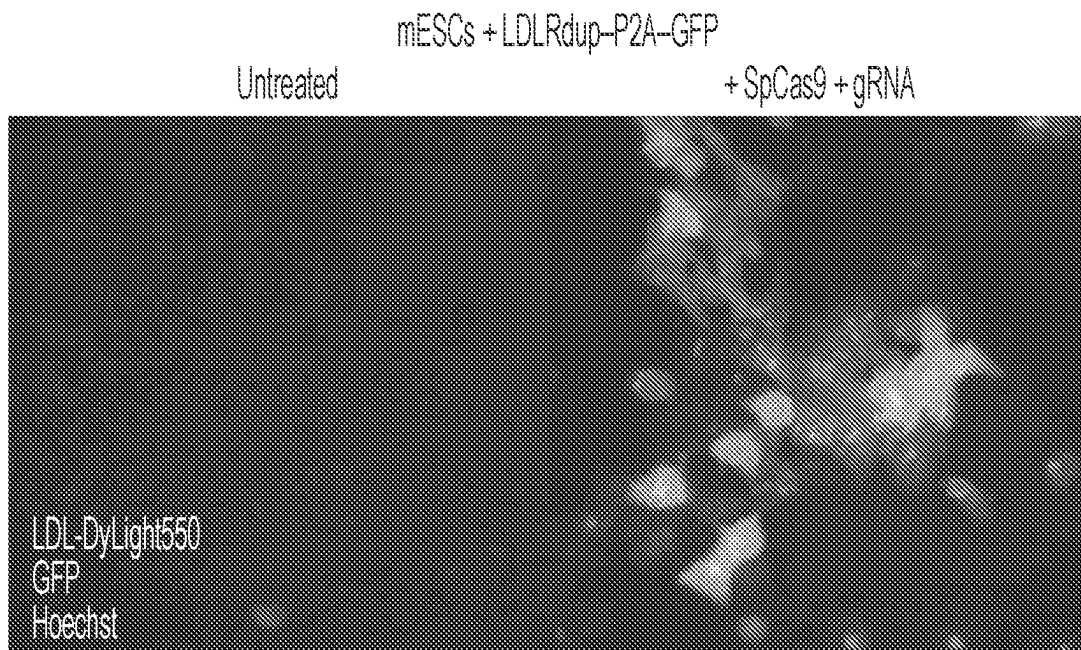


FIG. 15F

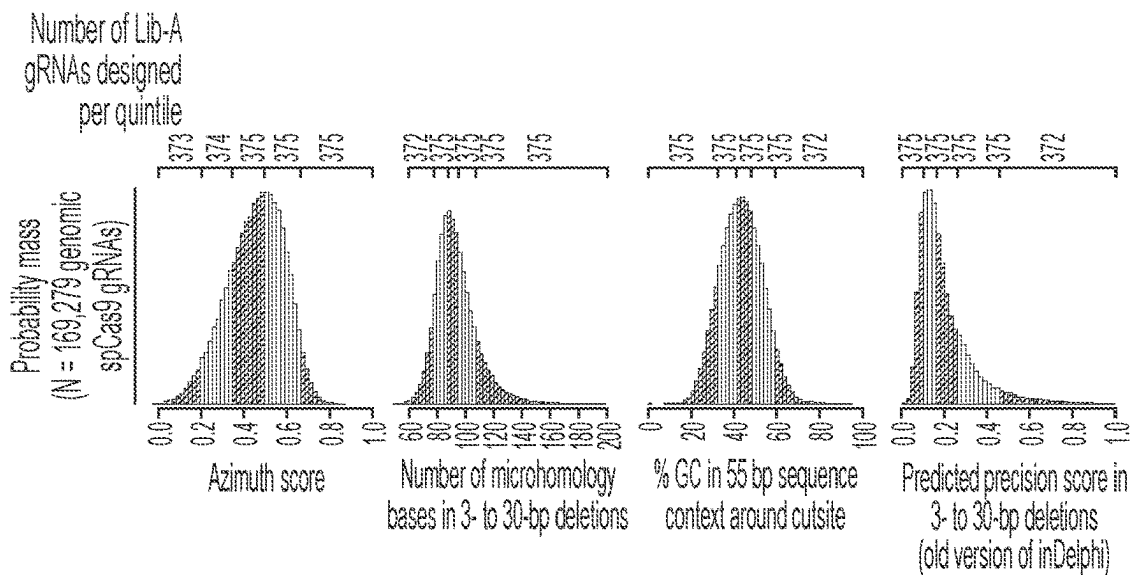


FIG. 16A

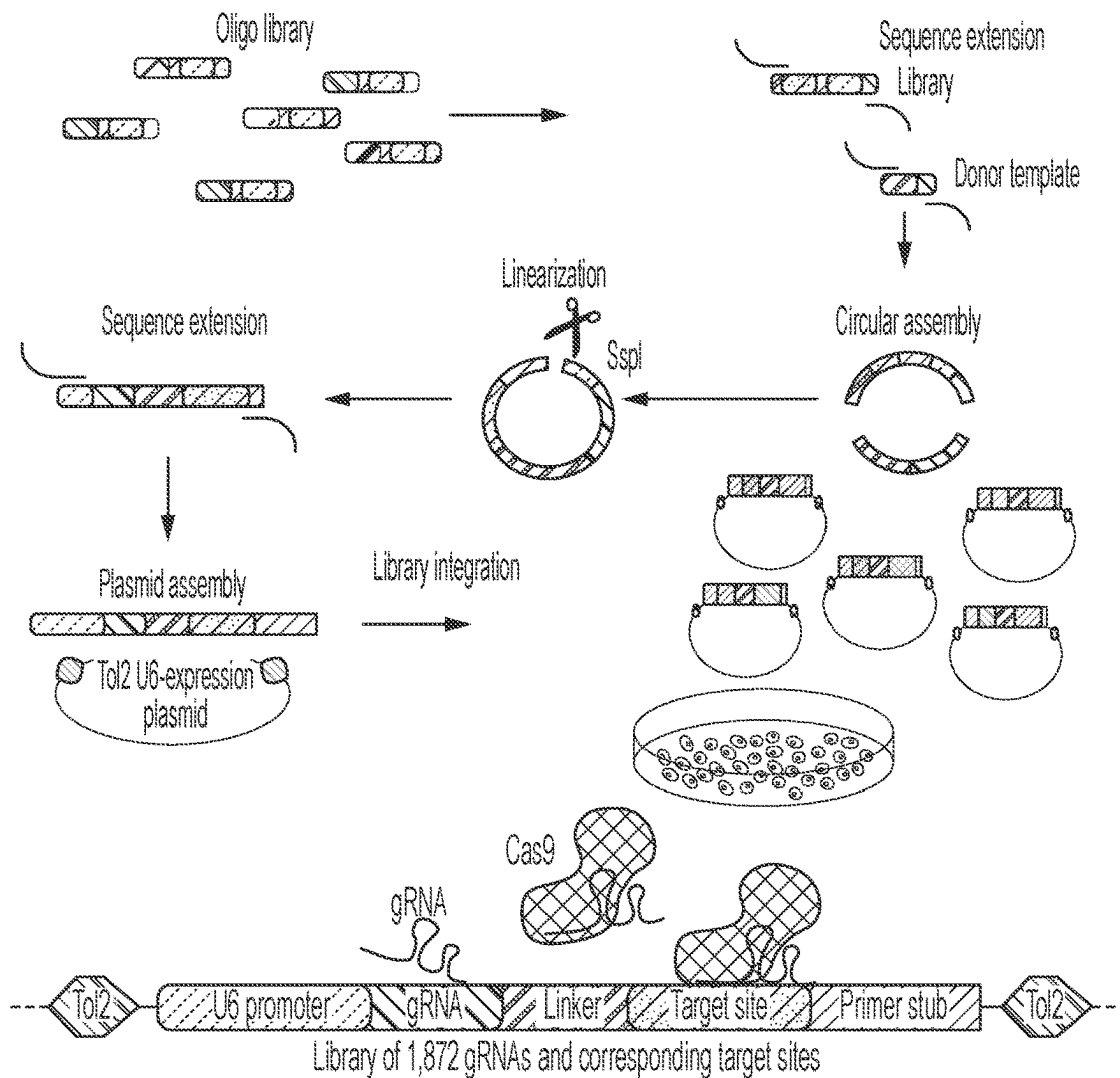


FIG. 16B

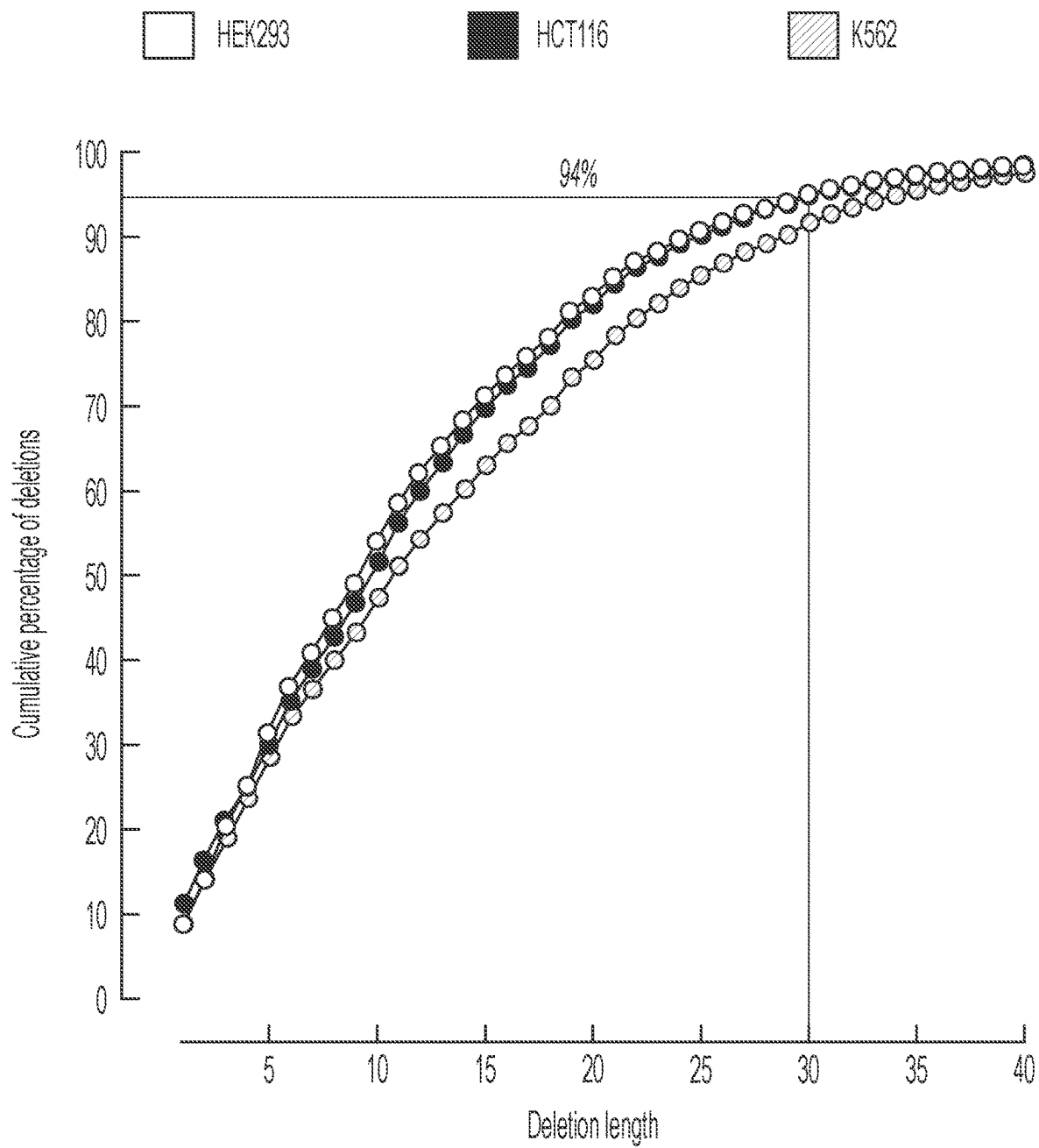


FIG. 16C

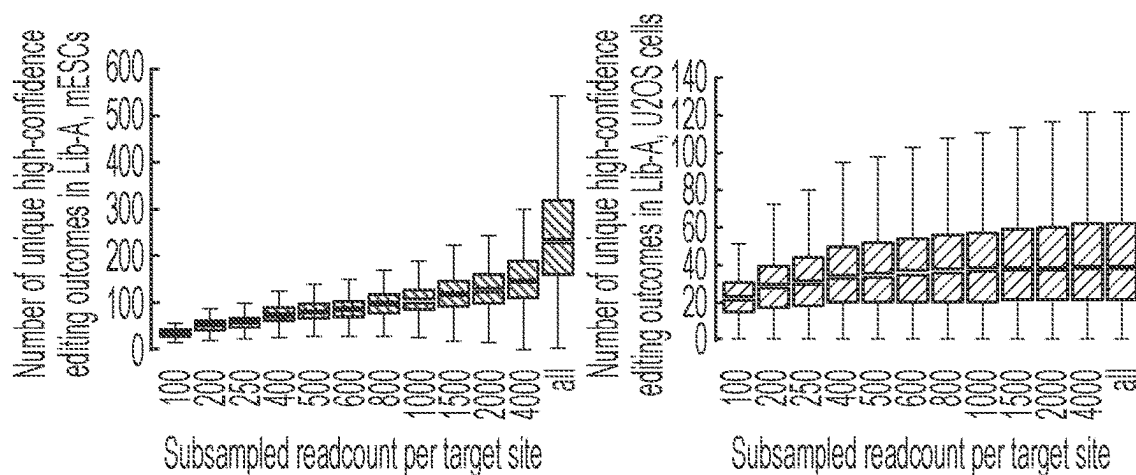


FIG. 16D

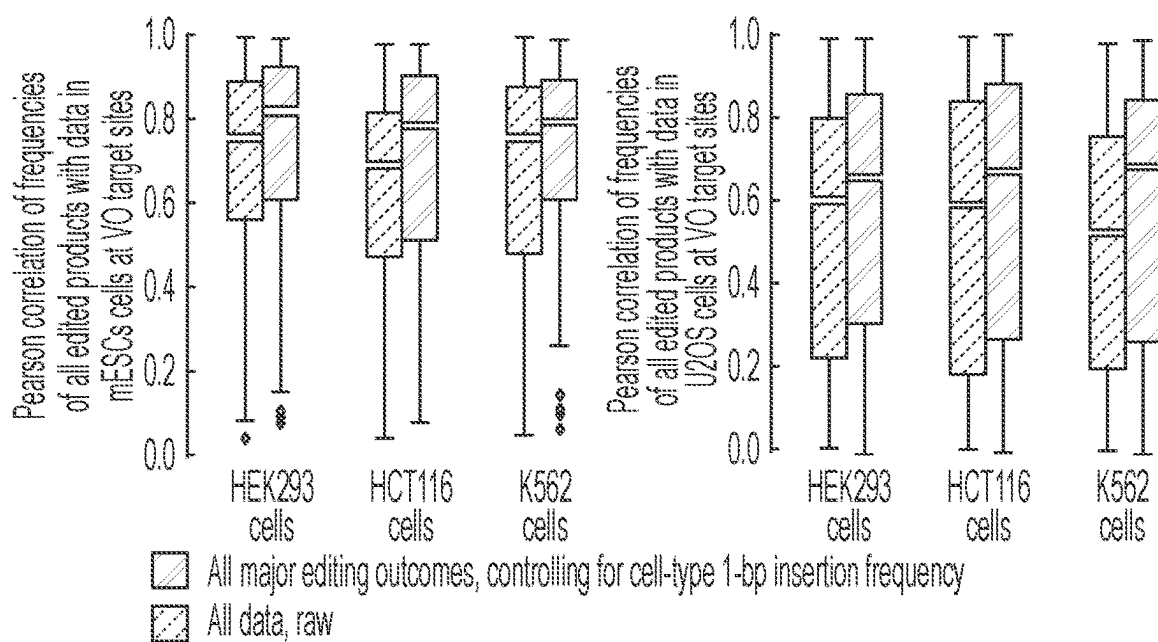


FIG. 16E

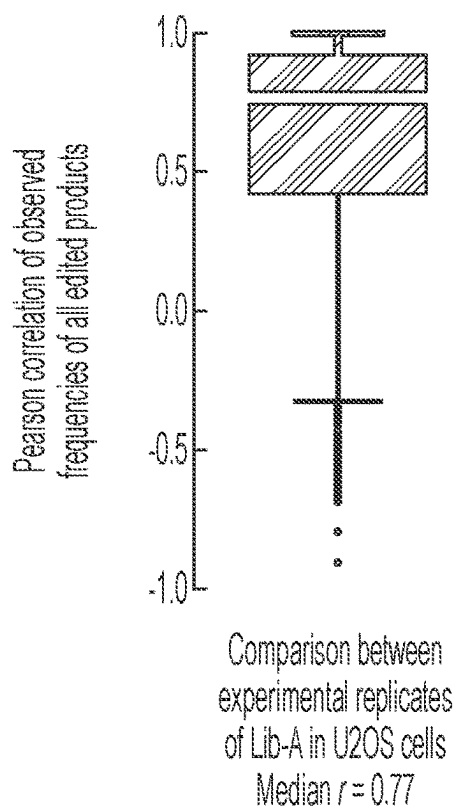
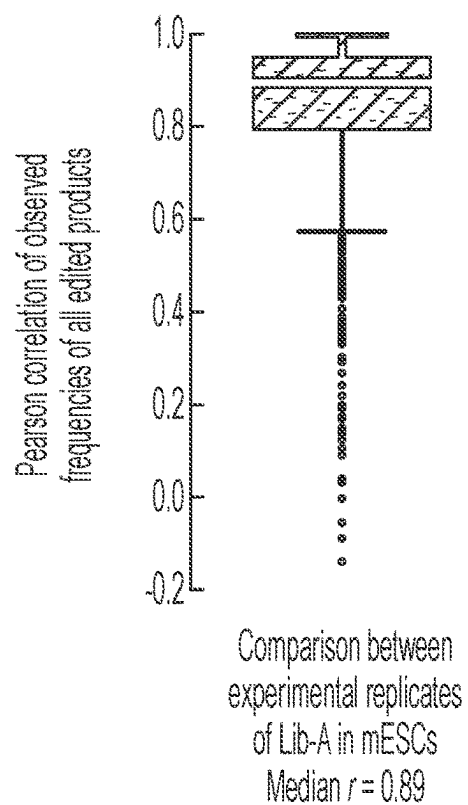


FIG. 16F

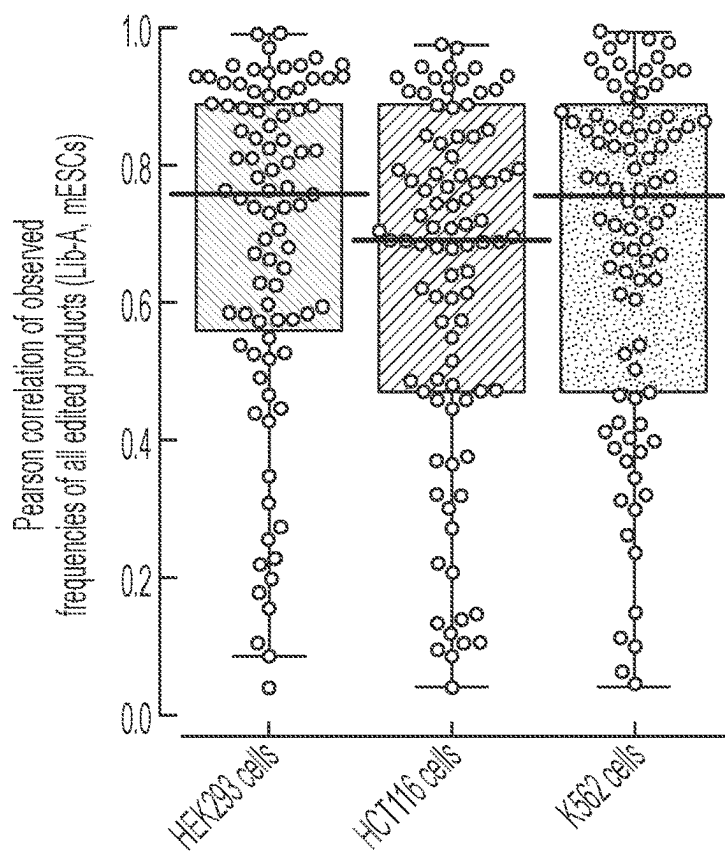


FIG. 17A

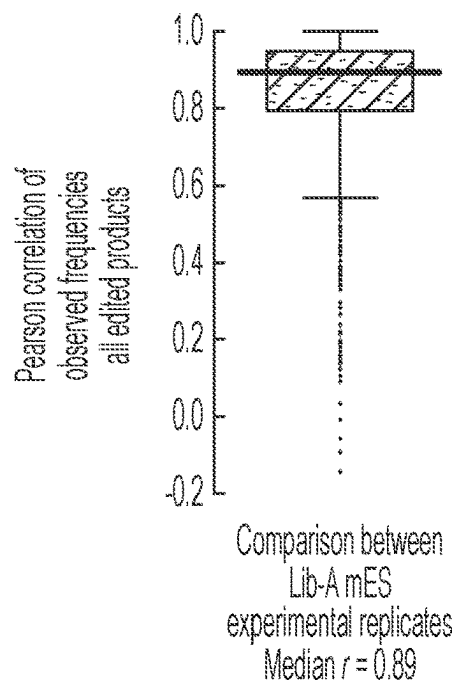


FIG. 17B

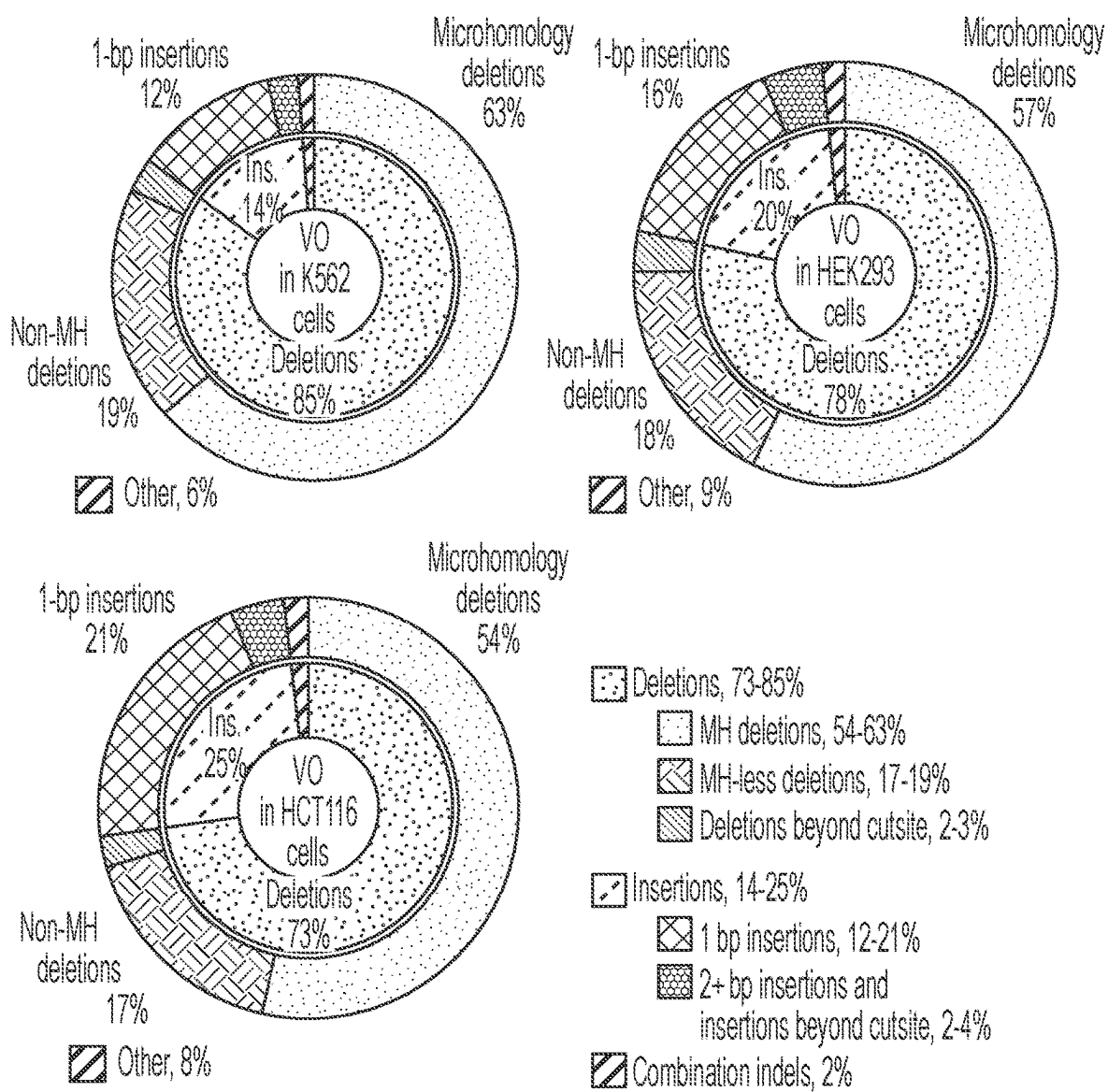


FIG. 17C

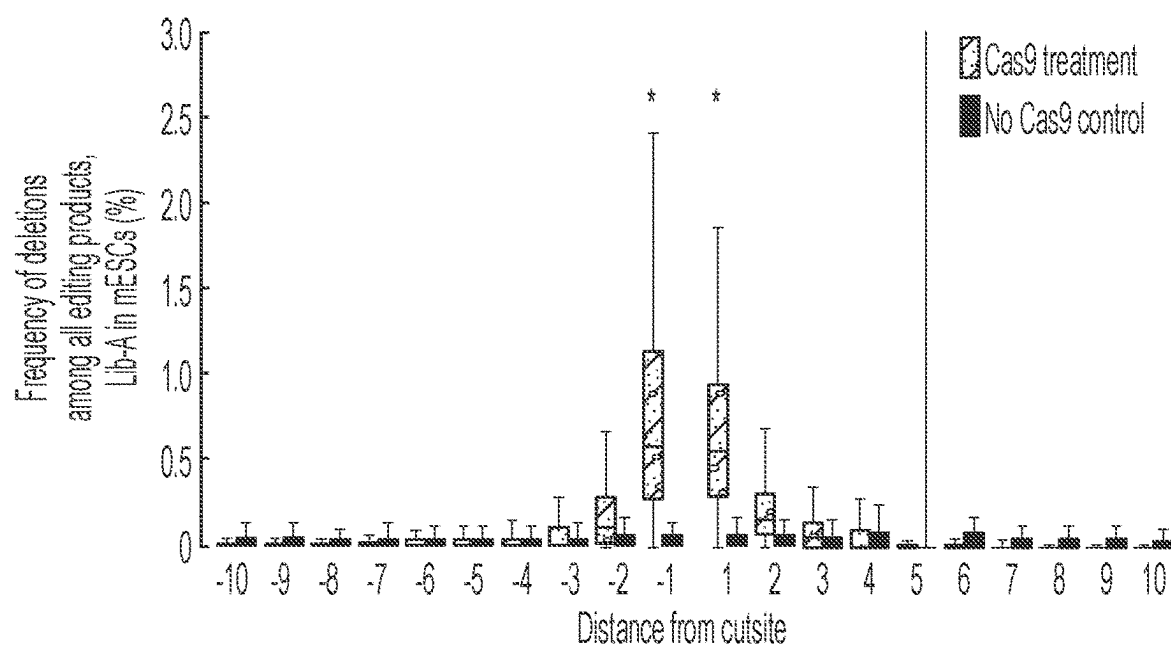
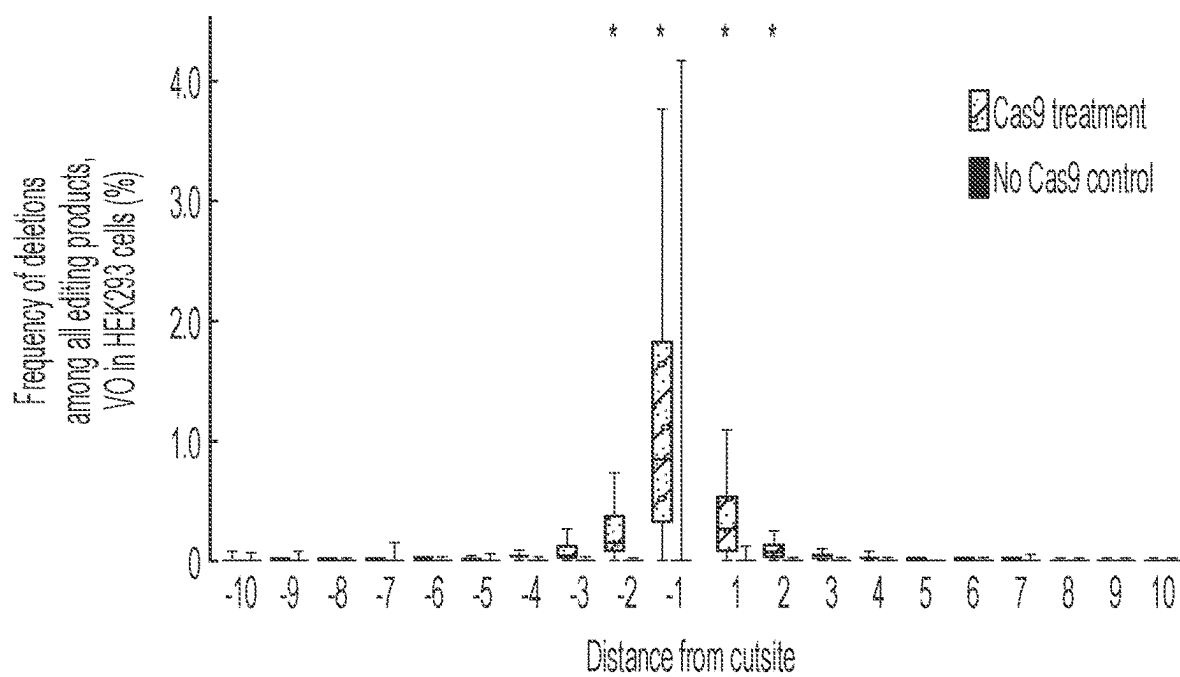


FIG. 17D

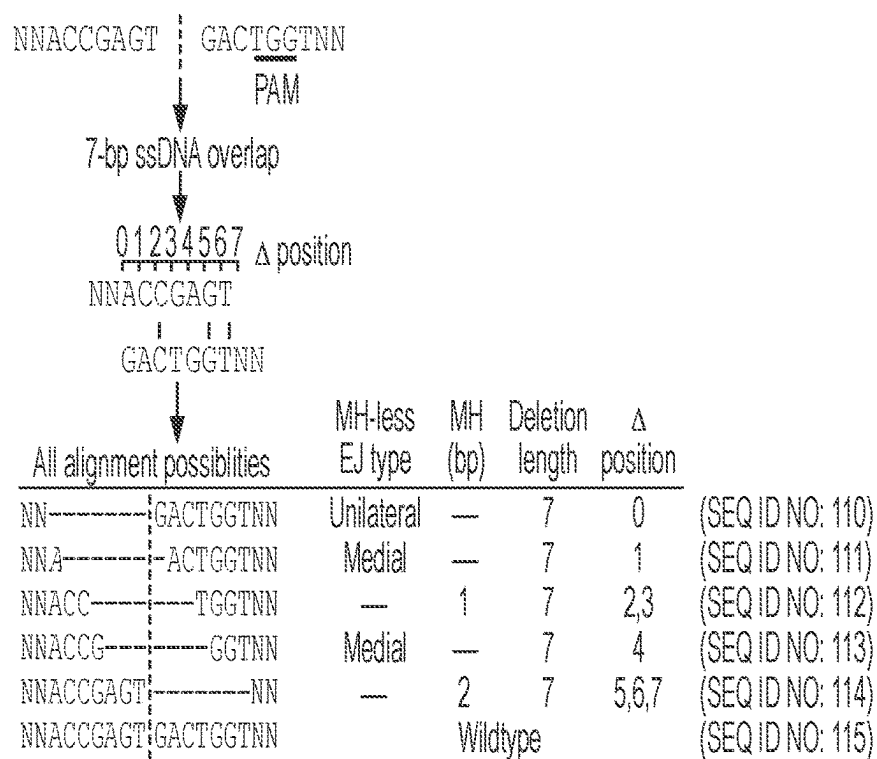


FIG. 18A

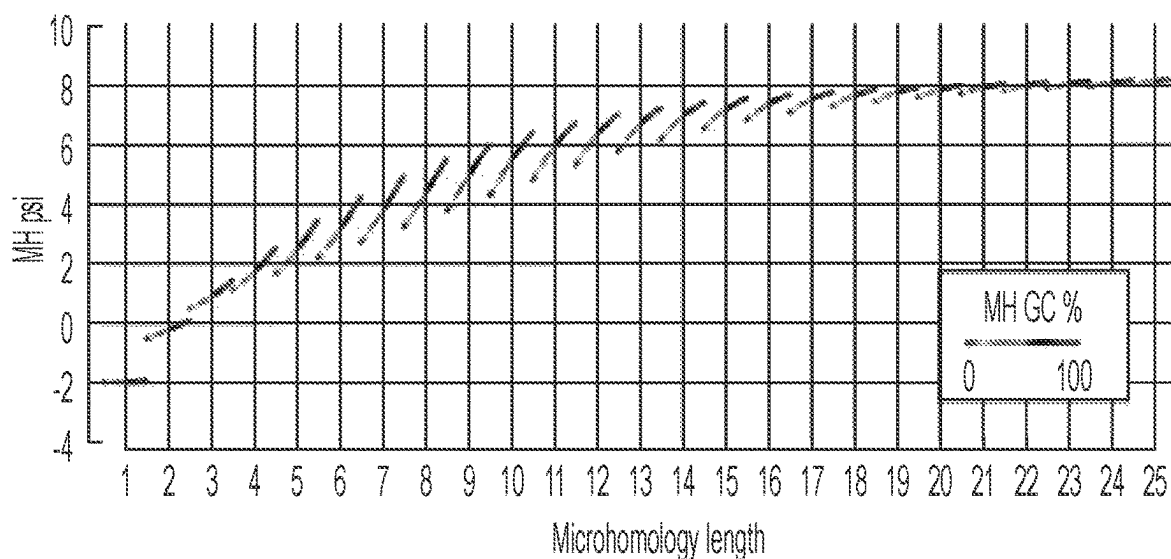


FIG. 18B

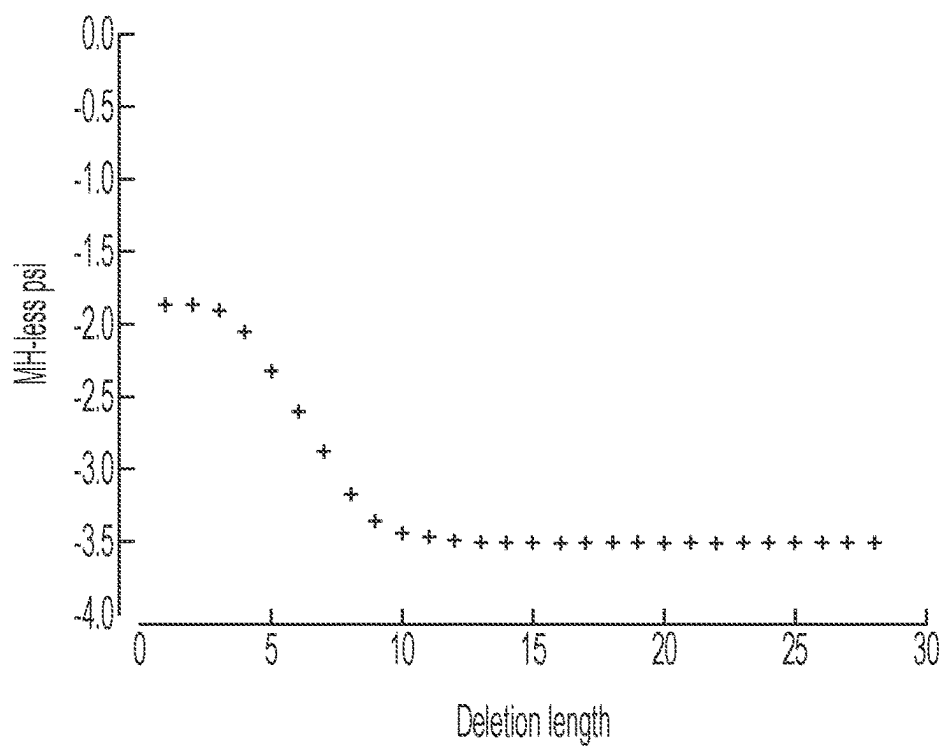


FIG. 18C

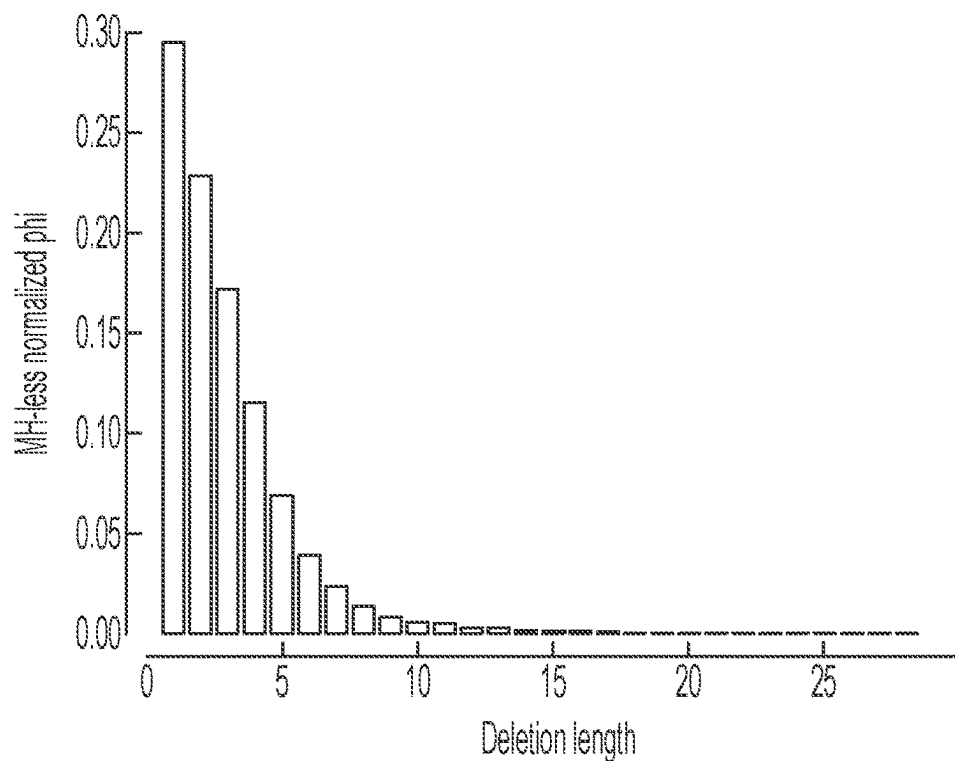


FIG. 18D

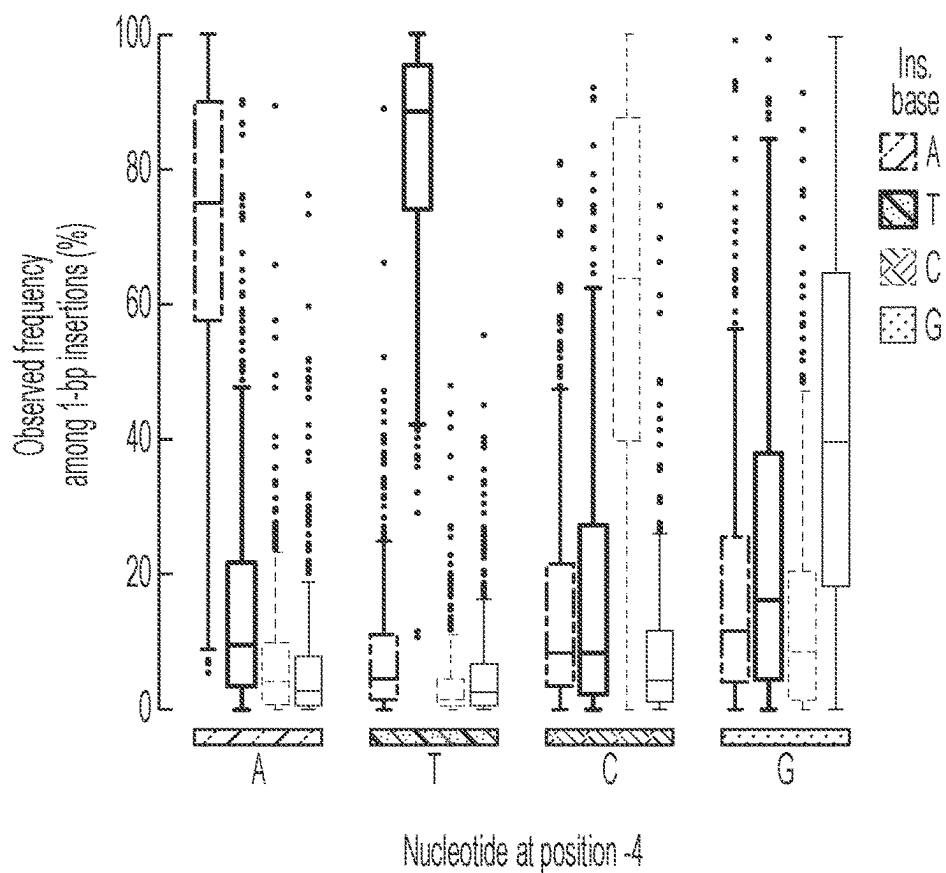


FIG. 18E

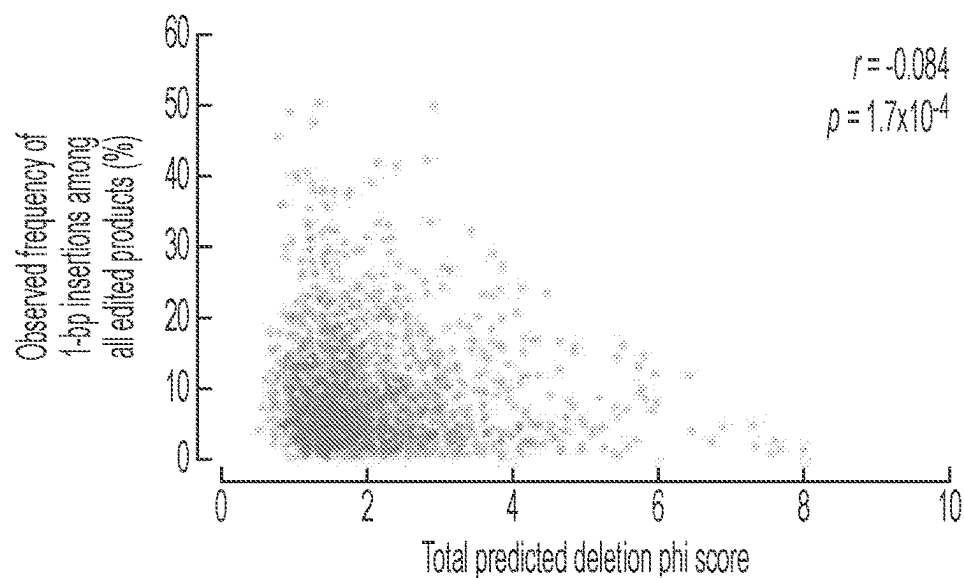


FIG. 18F

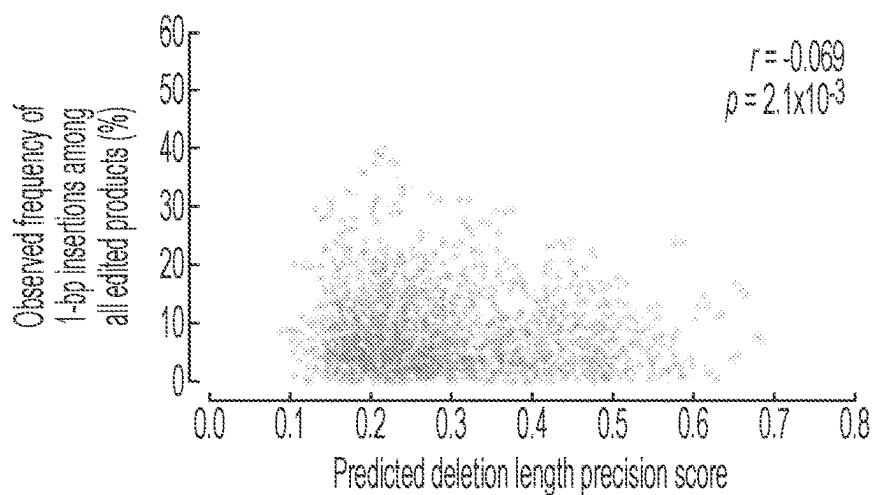


FIG. 18G

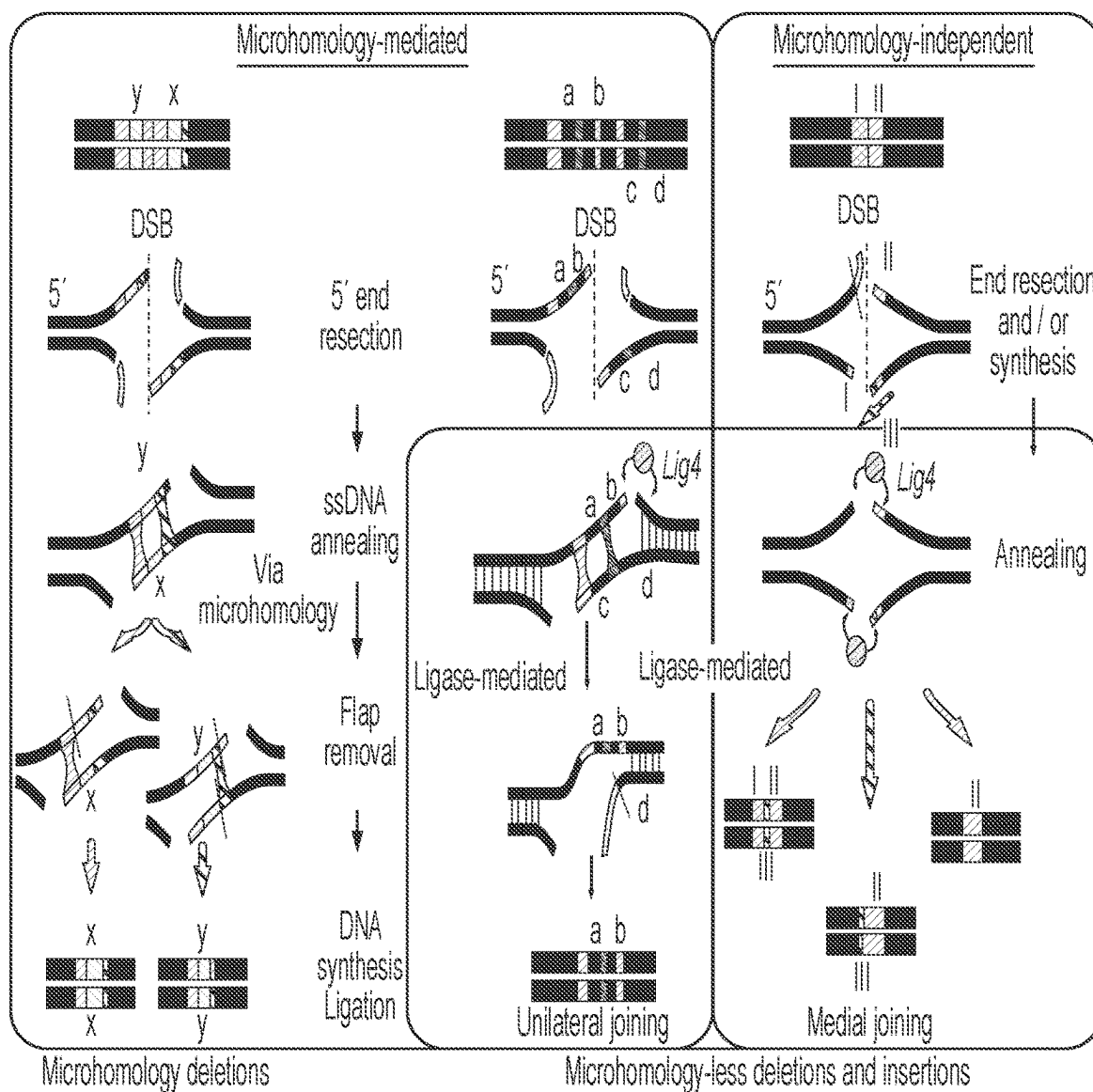


FIG. 18H

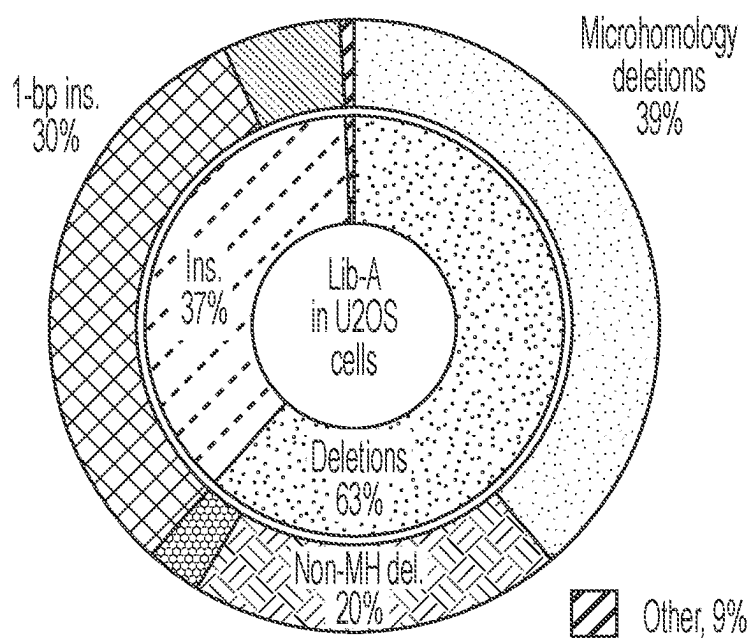


FIG. 18I

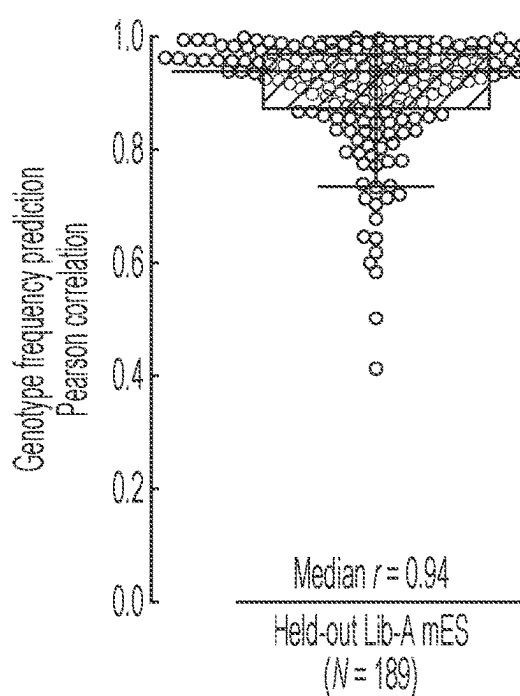


FIG. 19A

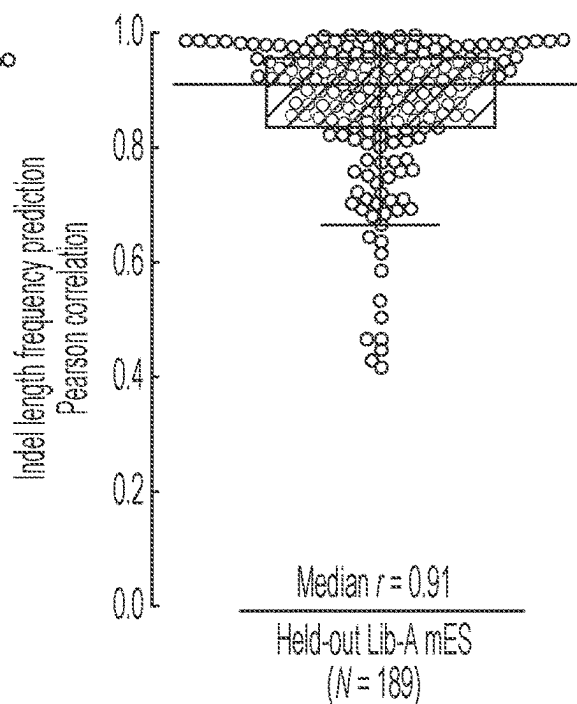


FIG. 19B

- 1,000,294 SpCas9 gRNA target sites in human exons
- 995,300 shuffled SpCas9 gRNA target sites in human exons
- 740,759 SpCas9 gRNA target sites in human introns

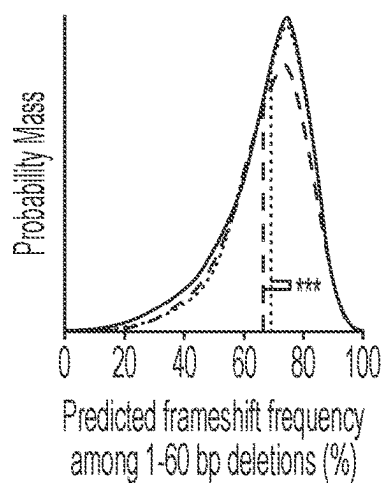


FIG. 19C

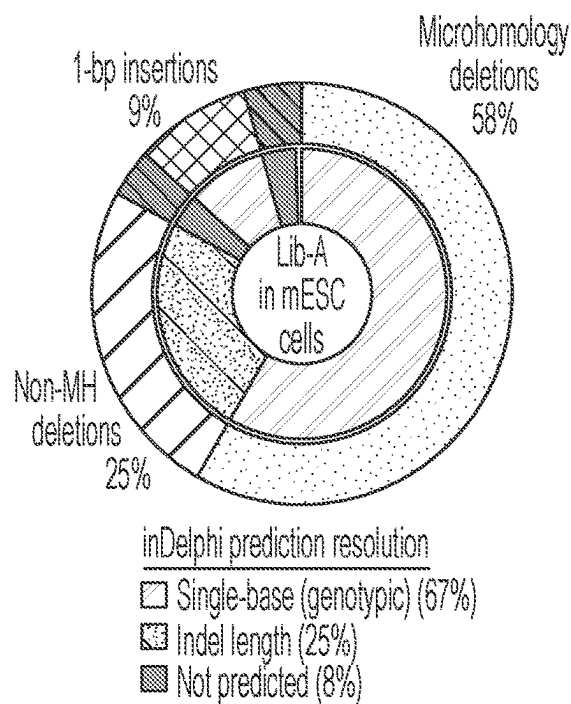


FIG. 19D

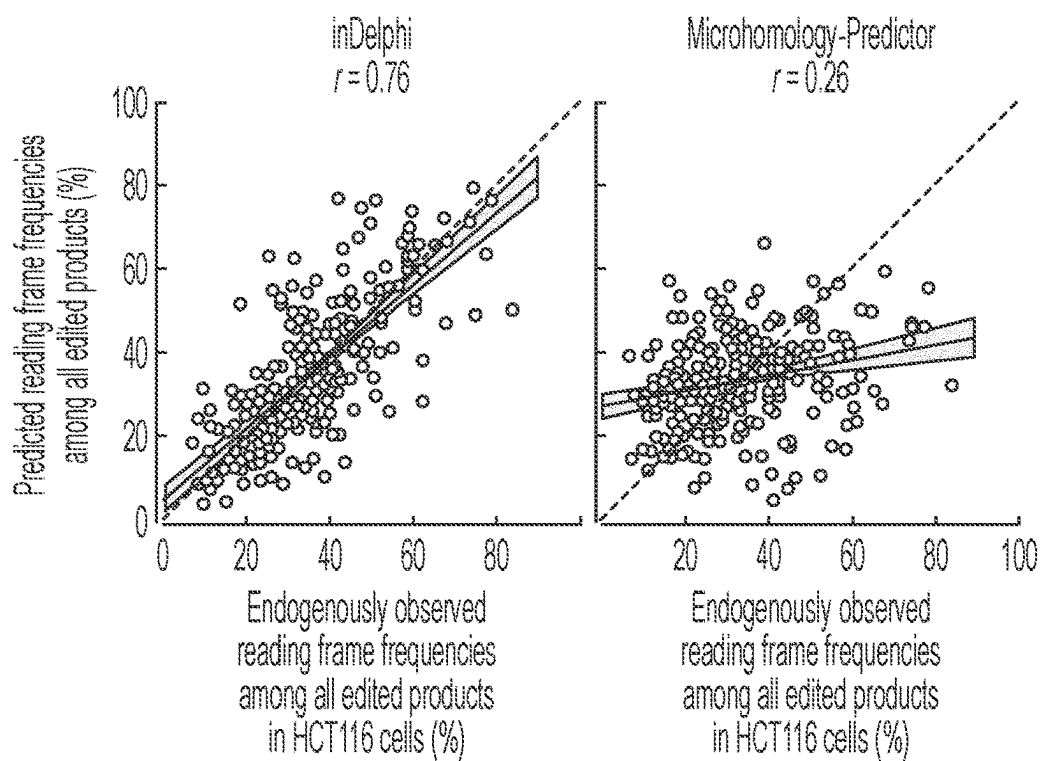


FIG. 19E

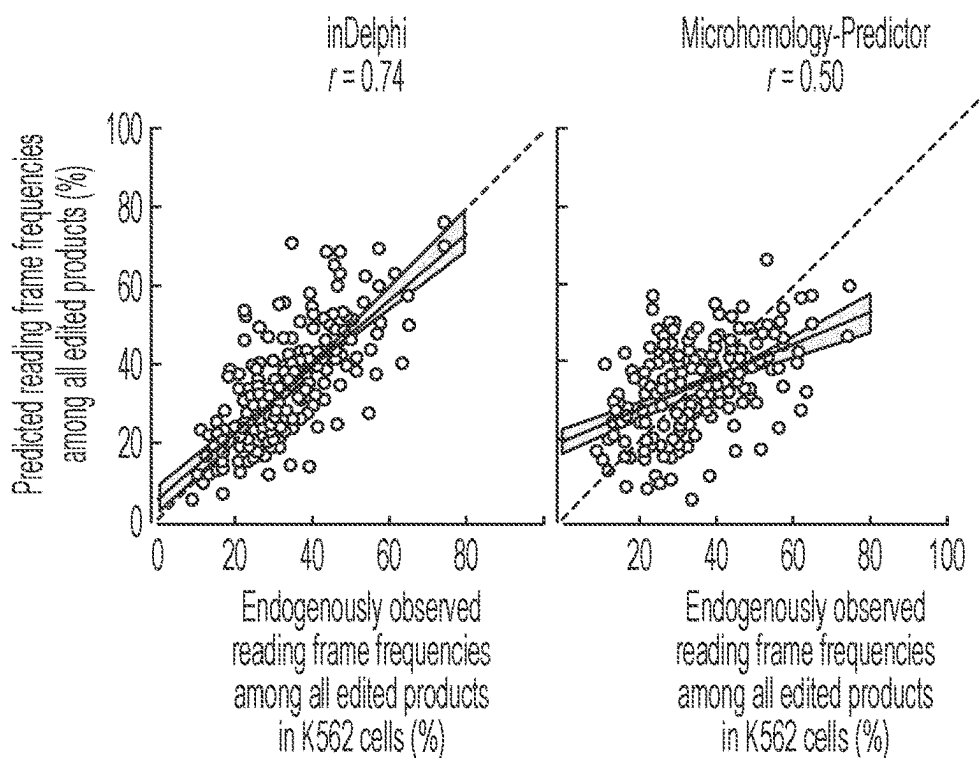


FIG. 19F

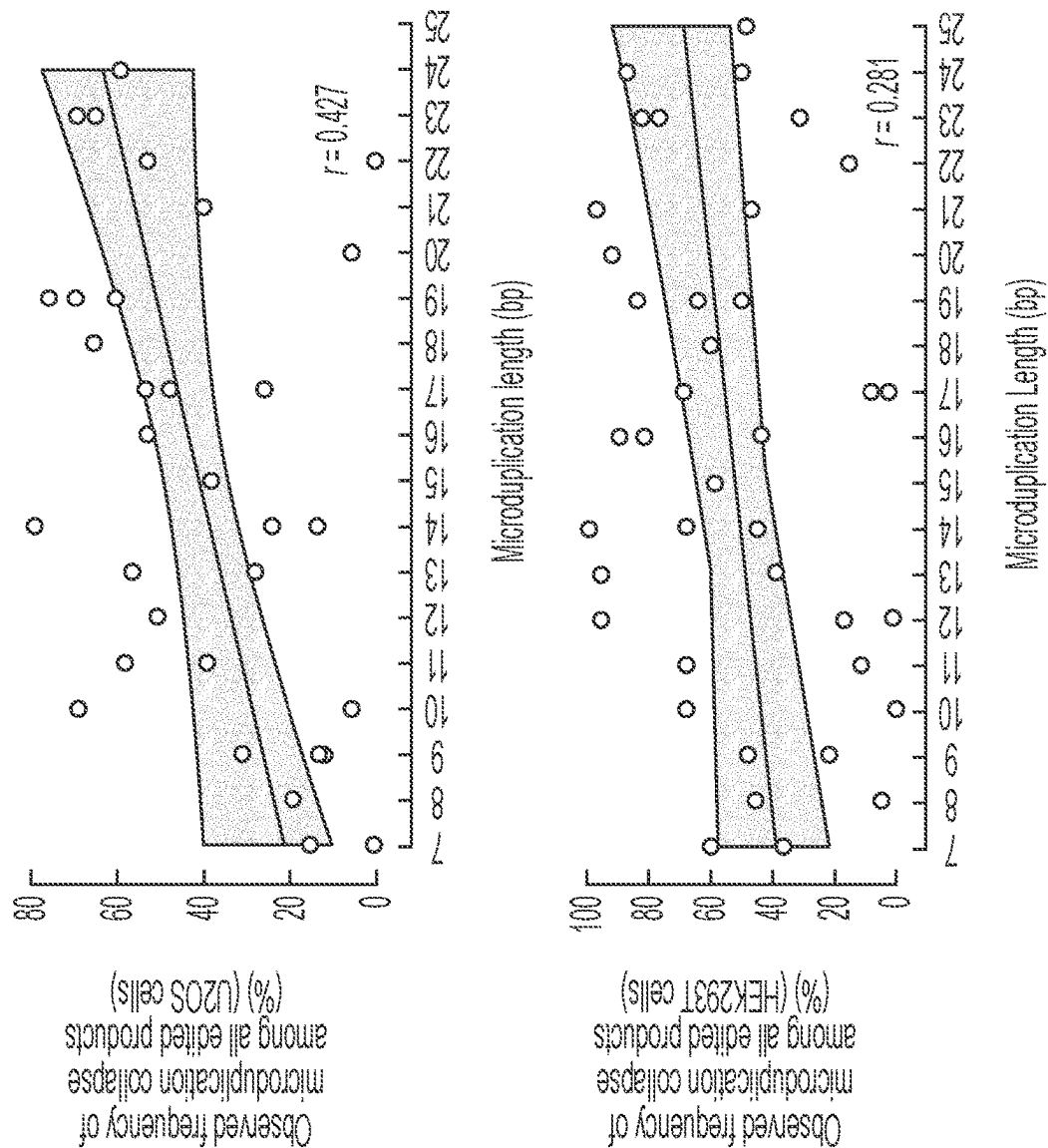


FIG. 20A

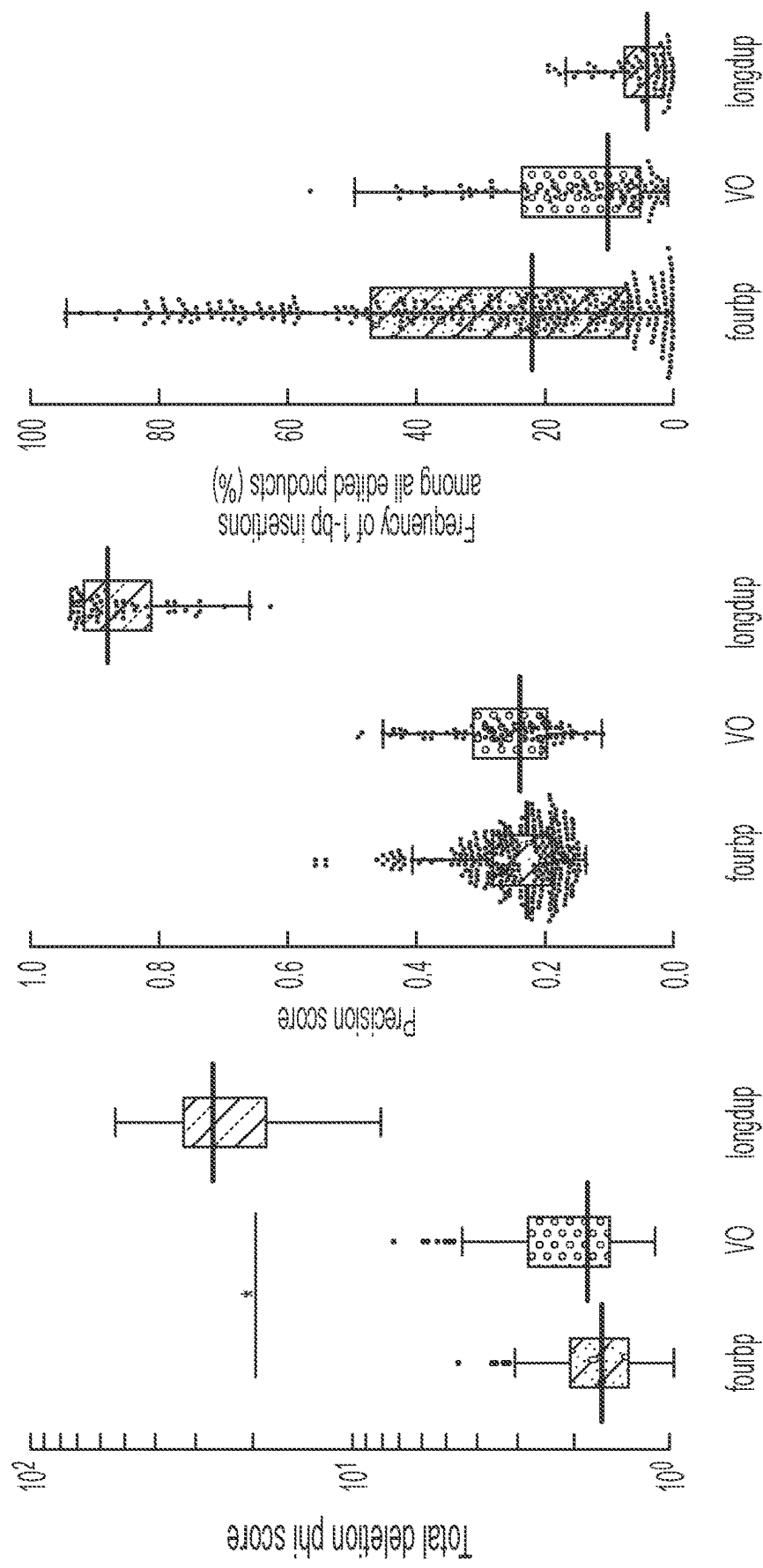


FIG. 20B

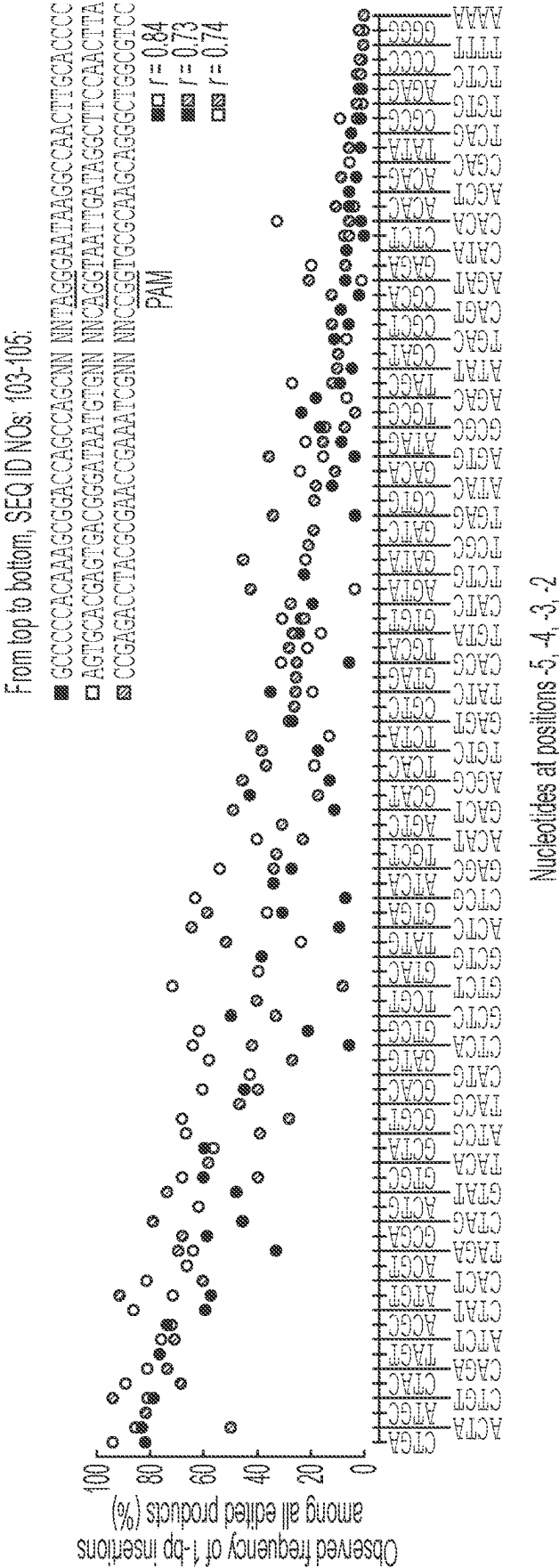


FIG. 20C

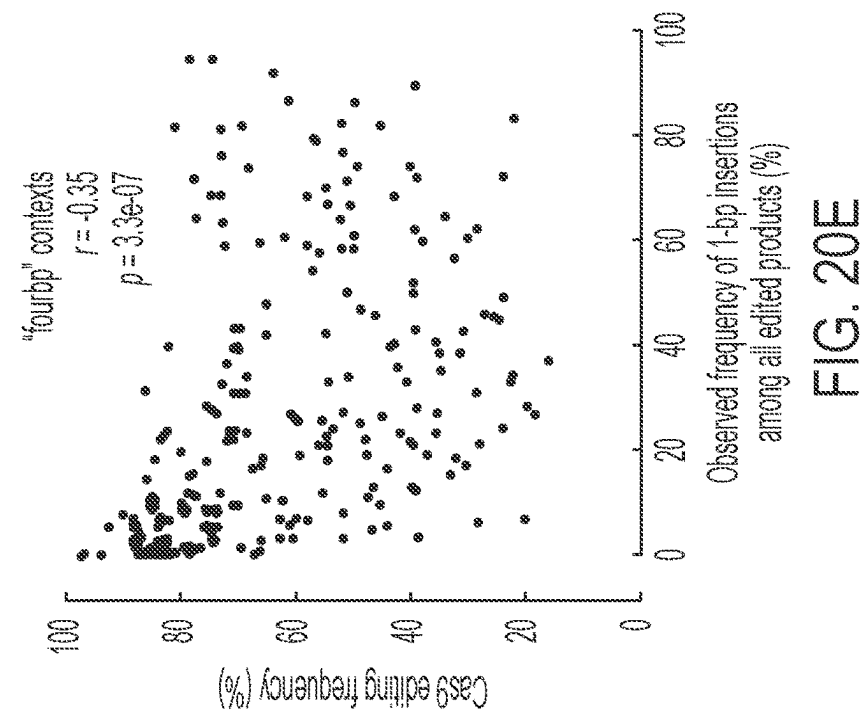


FIG. 20E

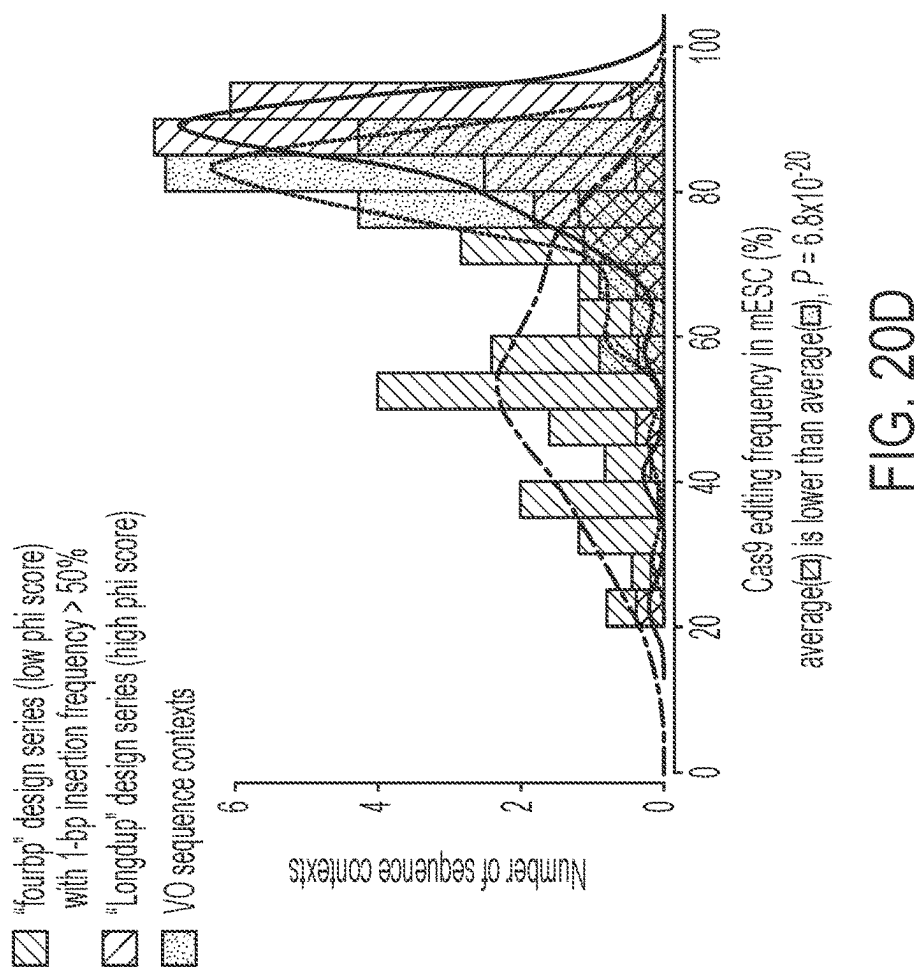


FIG. 20D

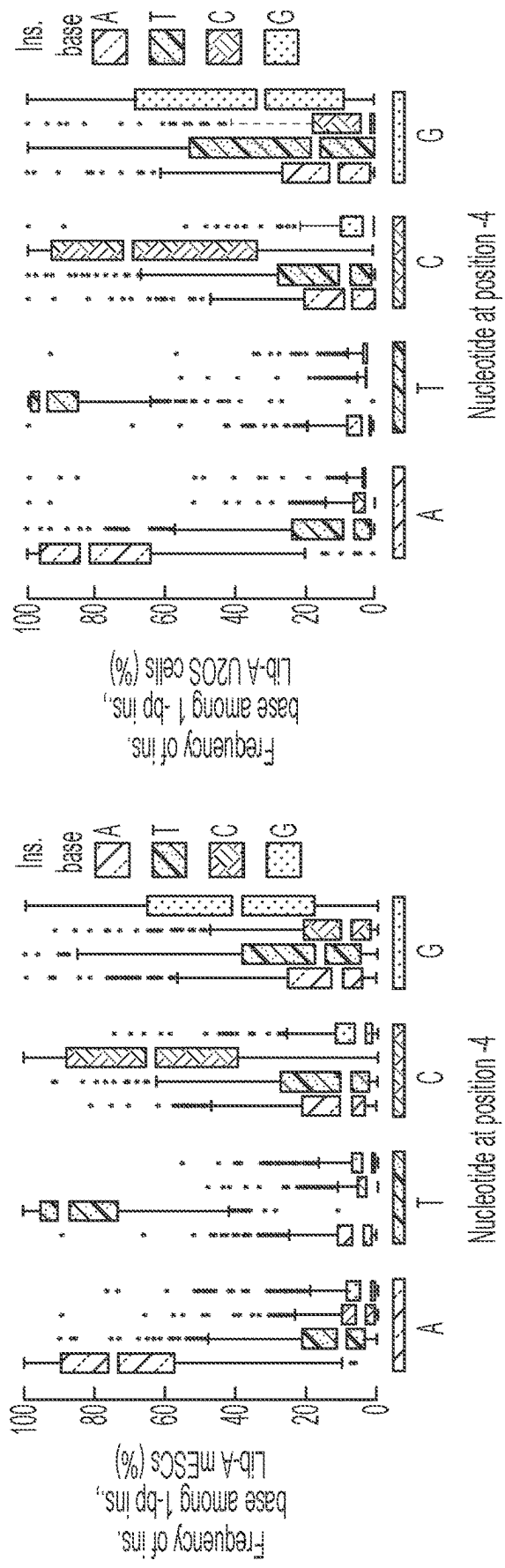


FIG. 20F

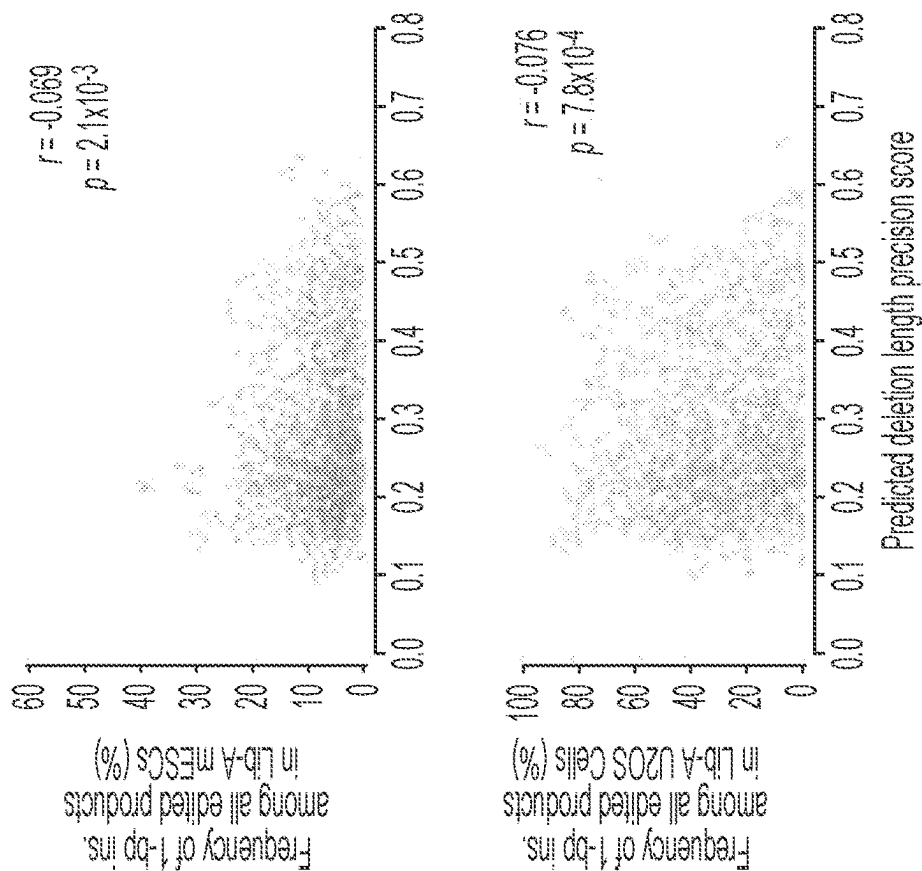


FIG. 20H

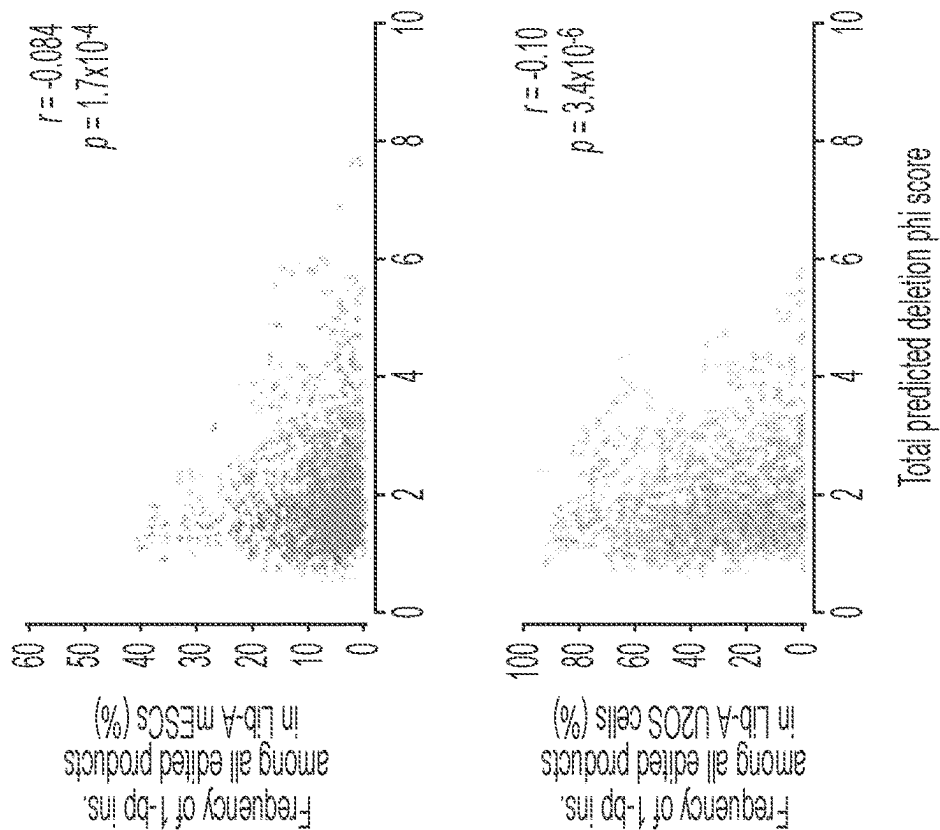


FIG. 20G

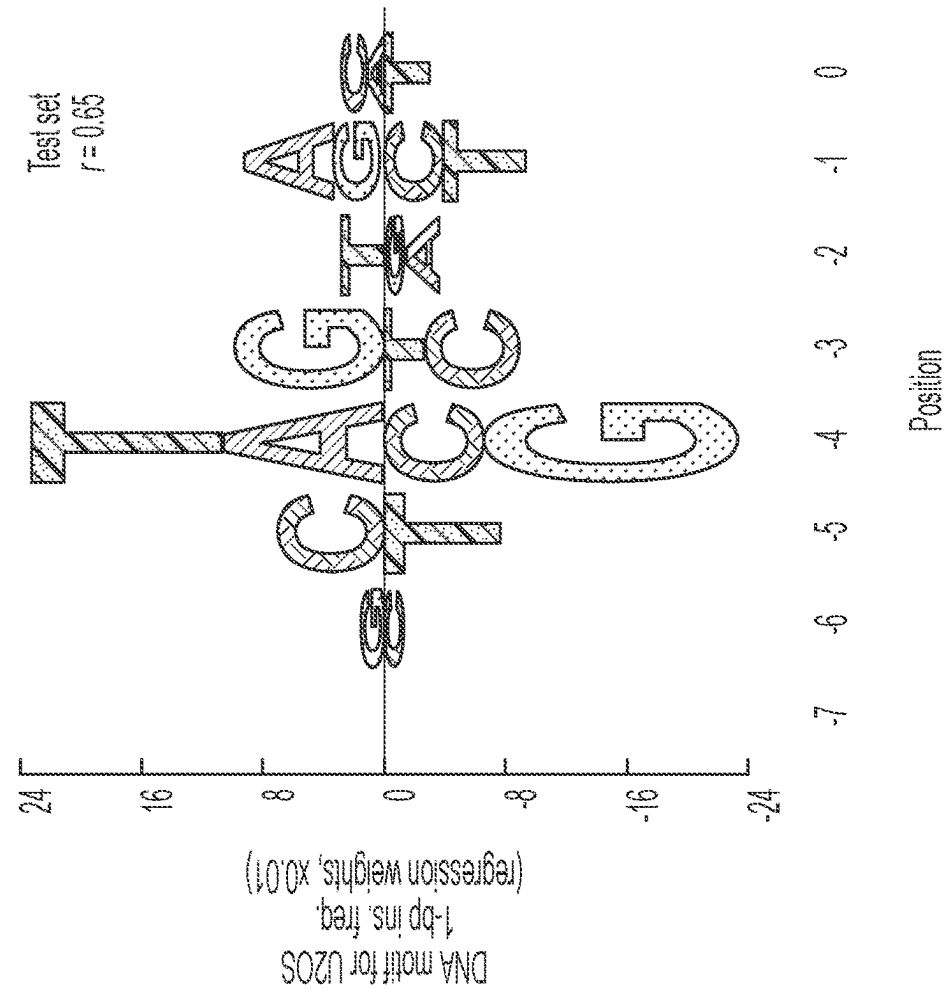


FIG. 20J

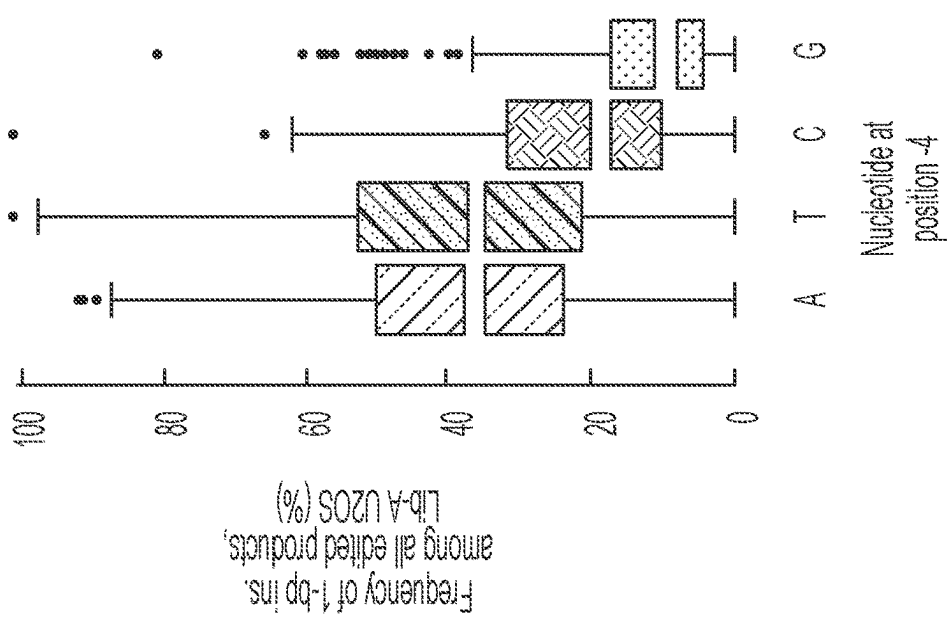
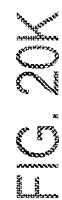


FIG. 20I



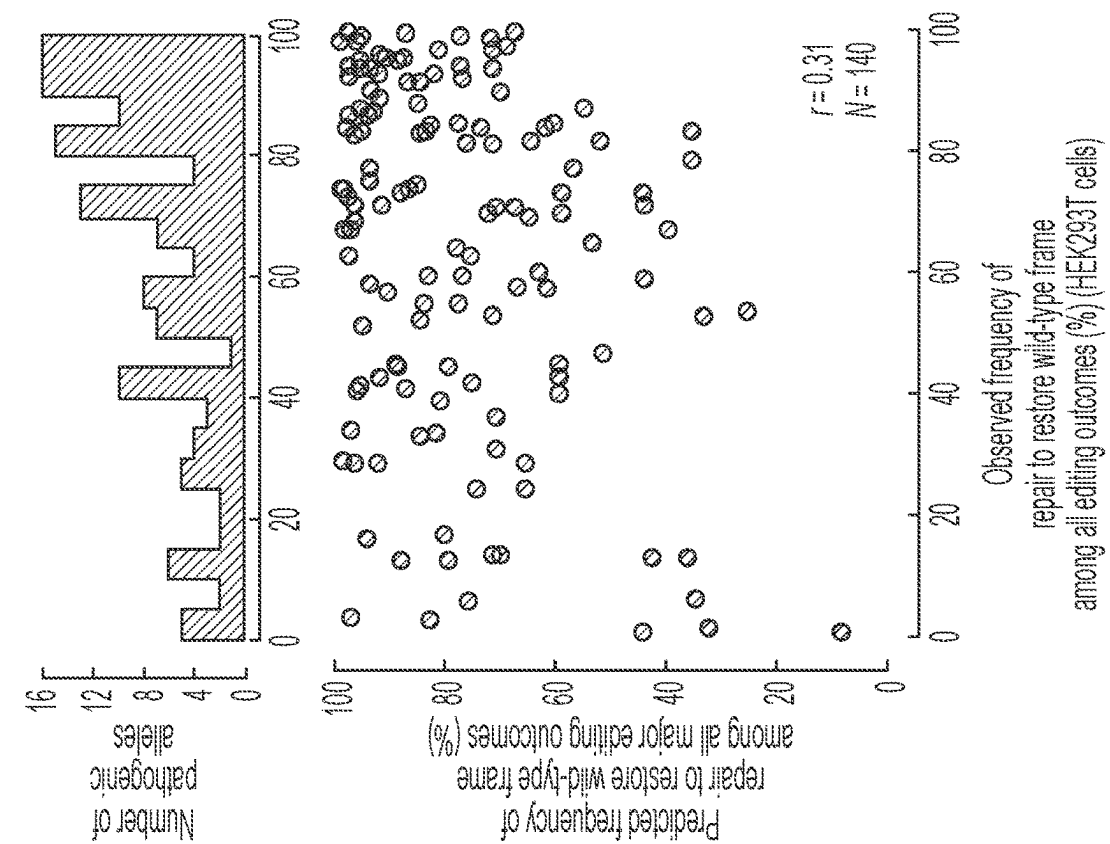


FIG. 21A

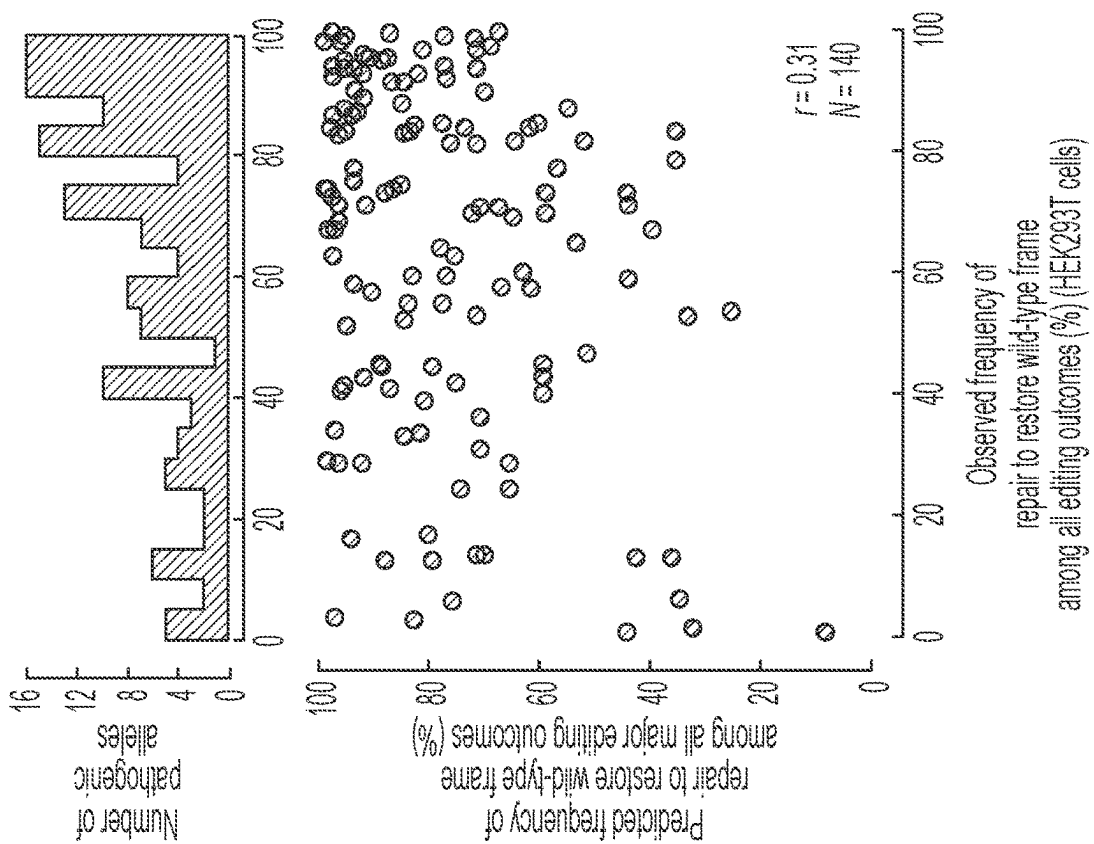
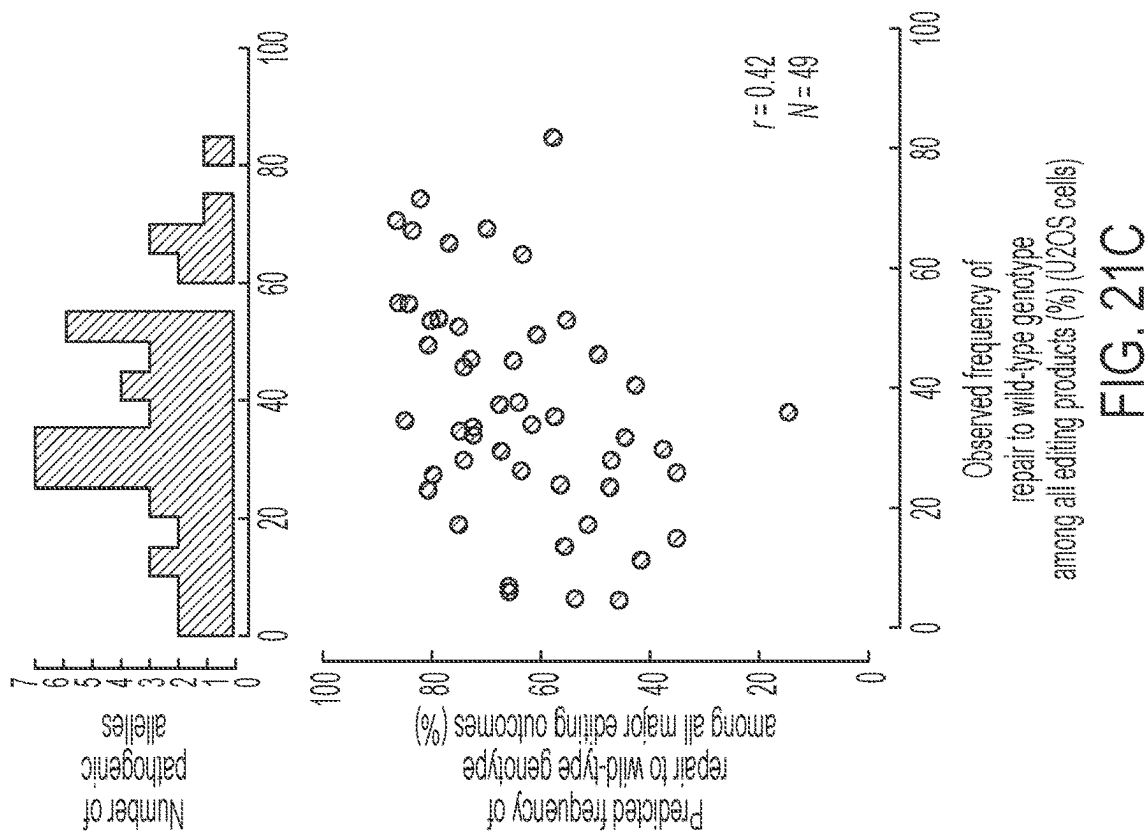
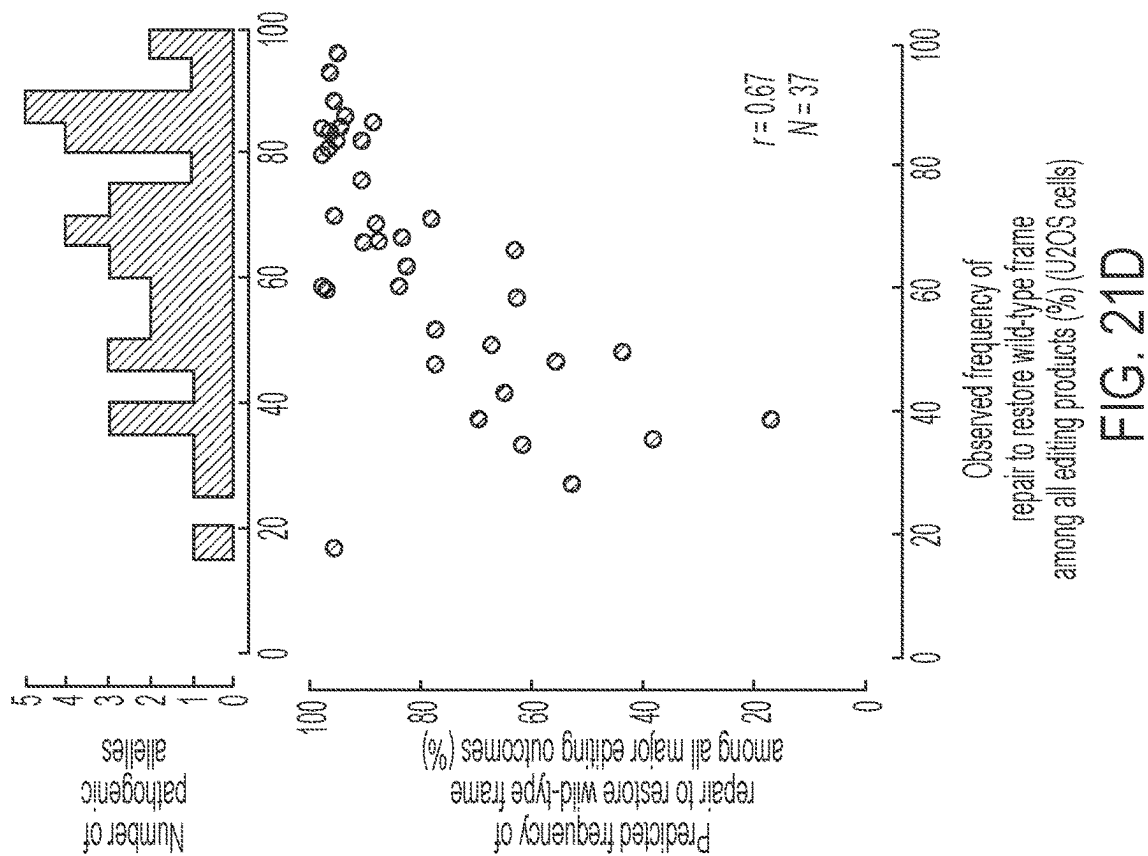


FIG. 21B



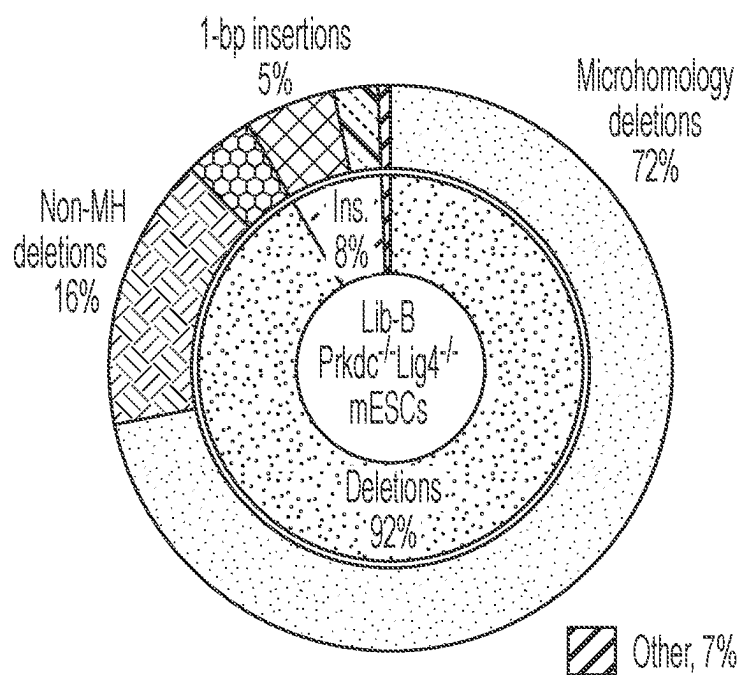


FIG. 22A

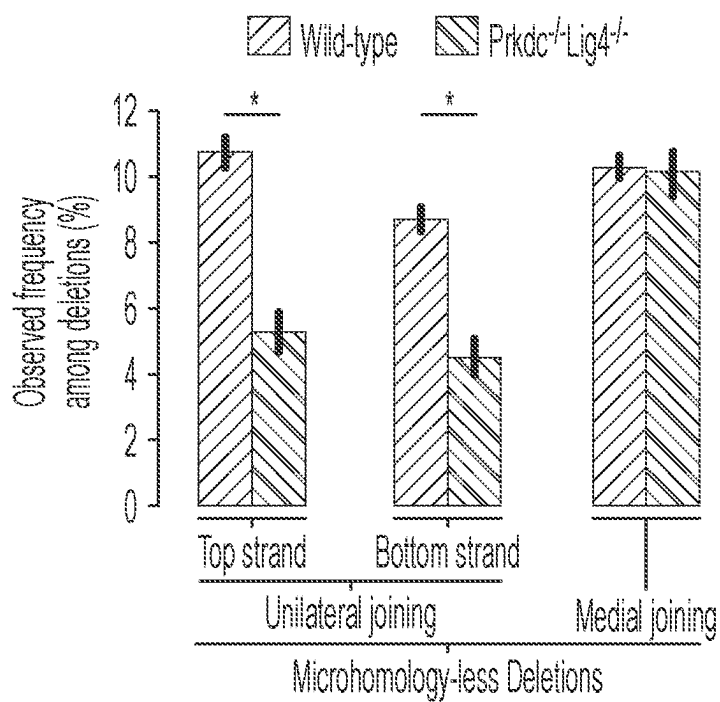


FIG. 22B

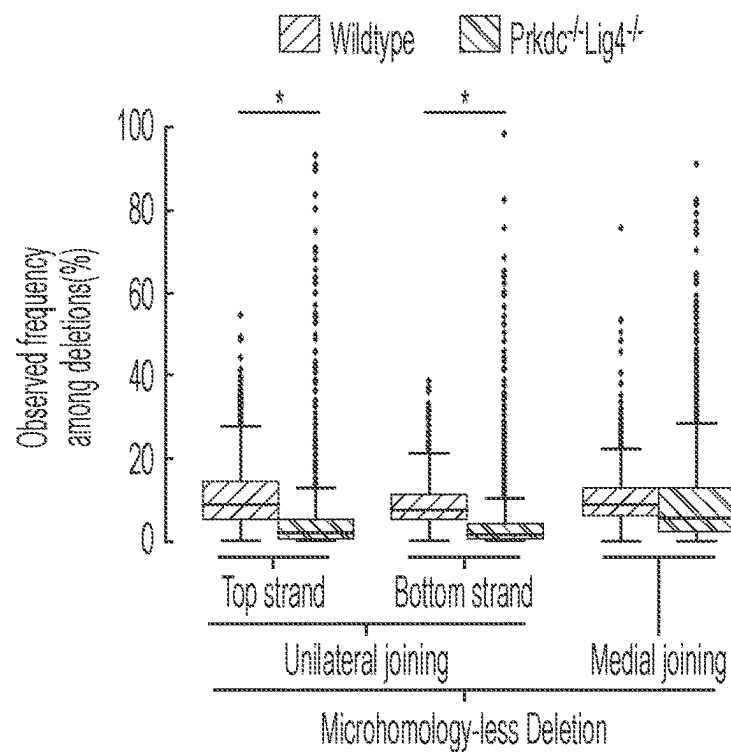


FIG. 22C

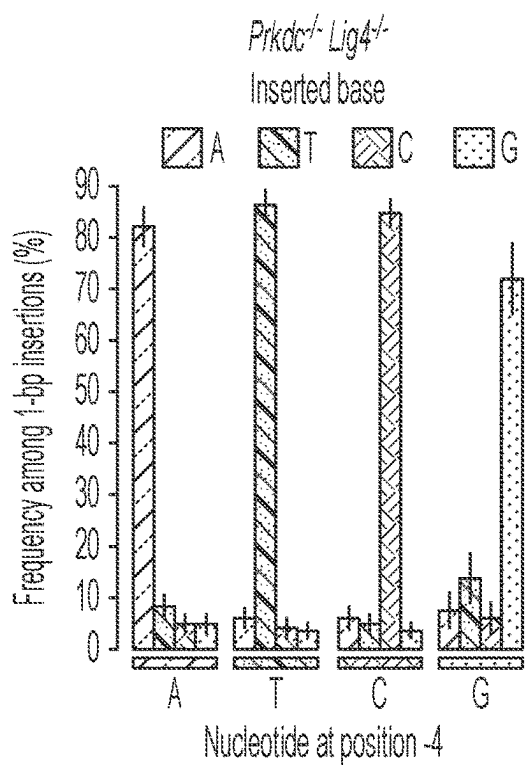


Fig 22D

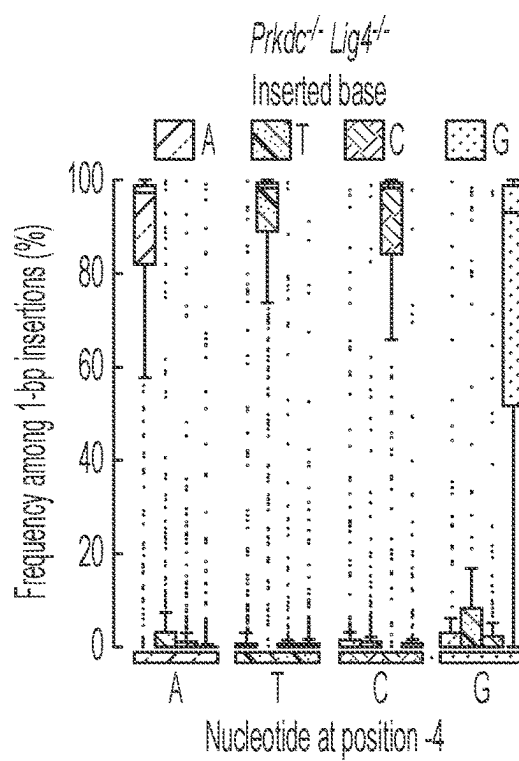


Fig 22E

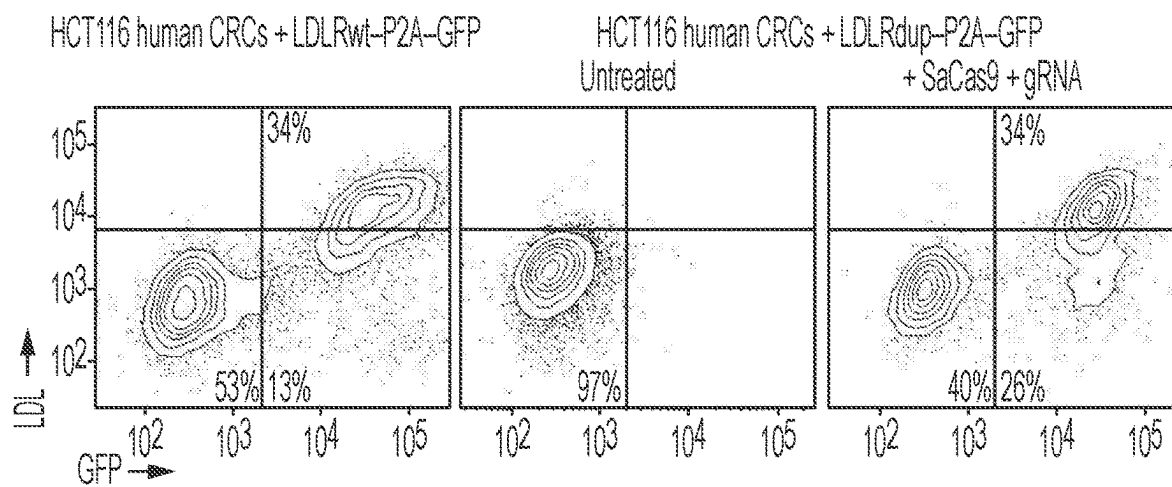


FIG. 23A

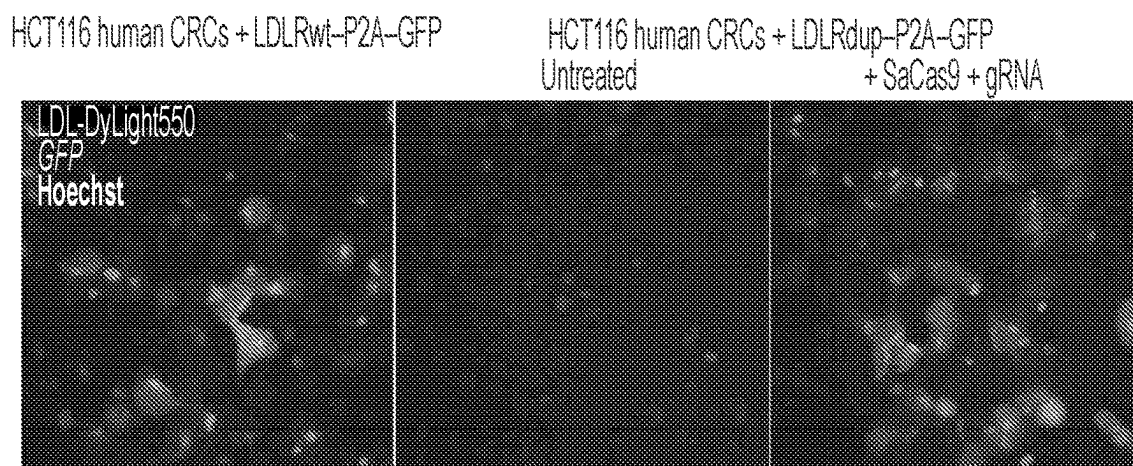


FIG. 23B

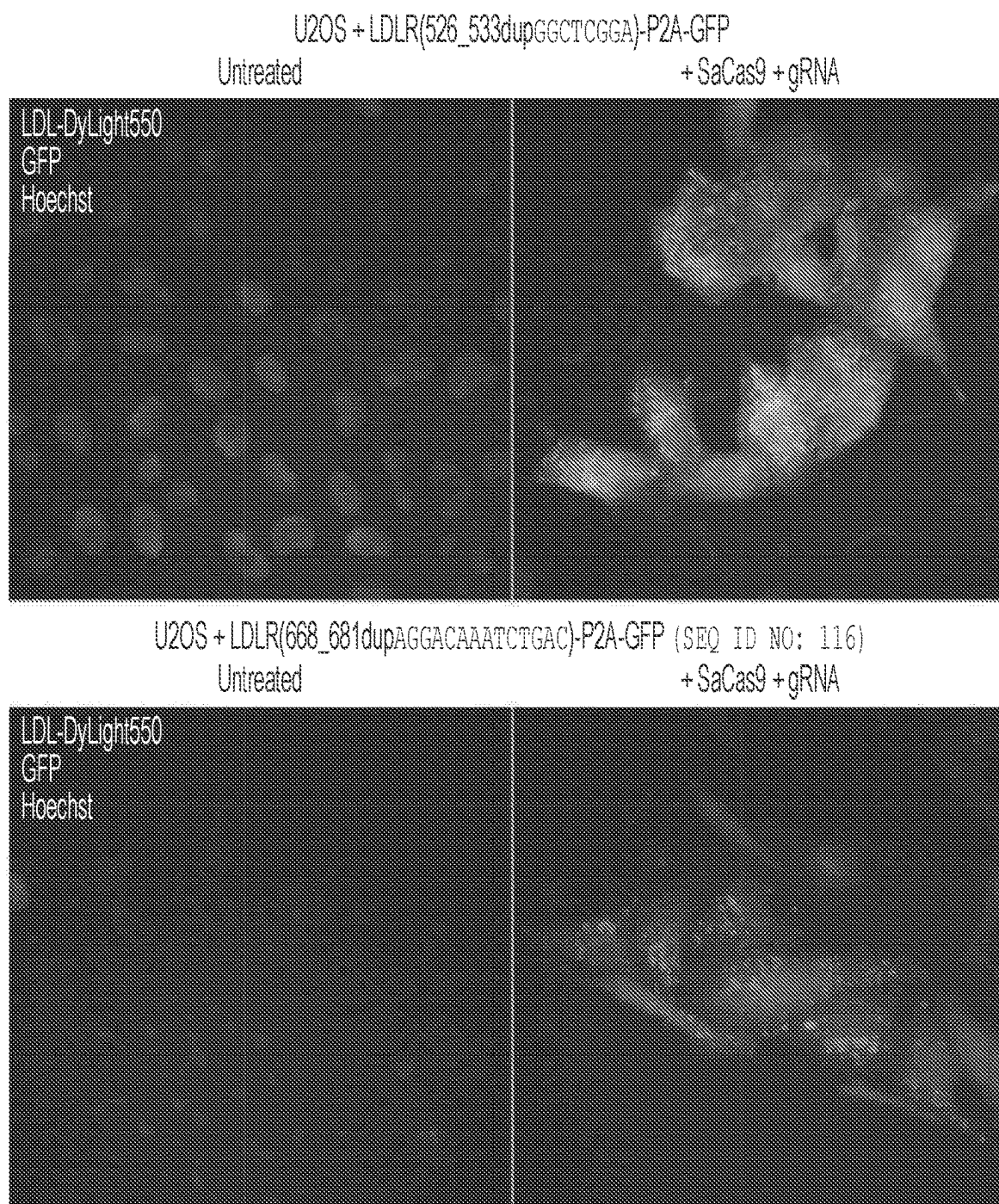


FIG. 23C

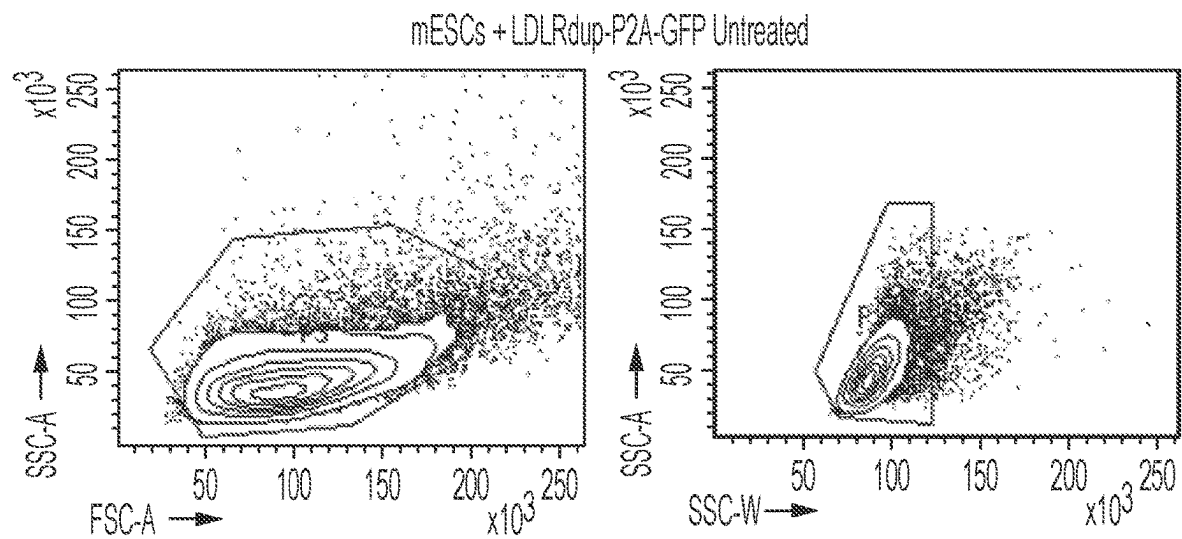


FIG. 23D

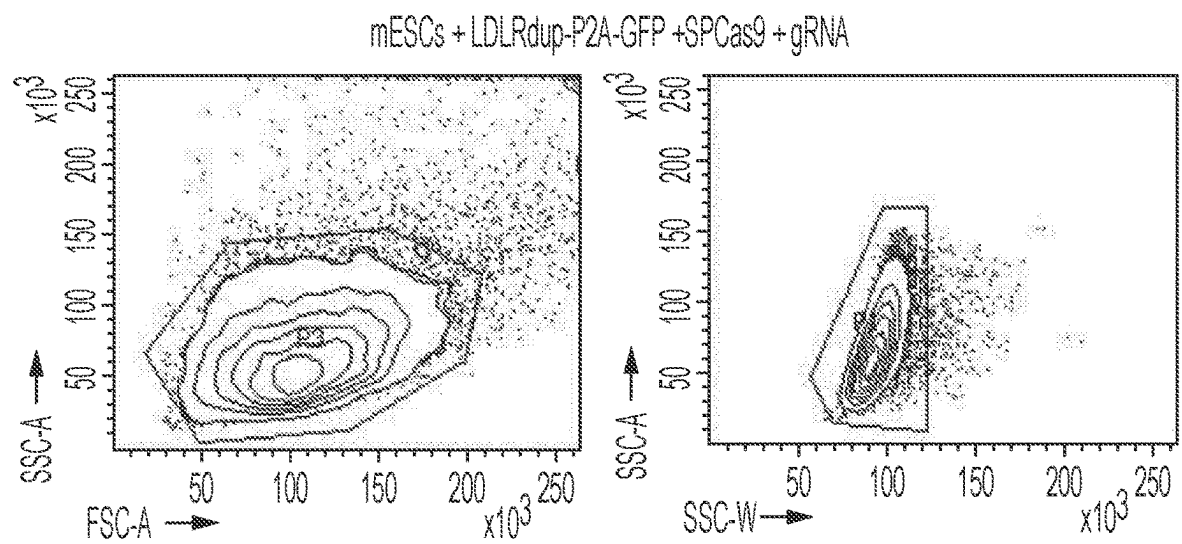


FIG. 23E

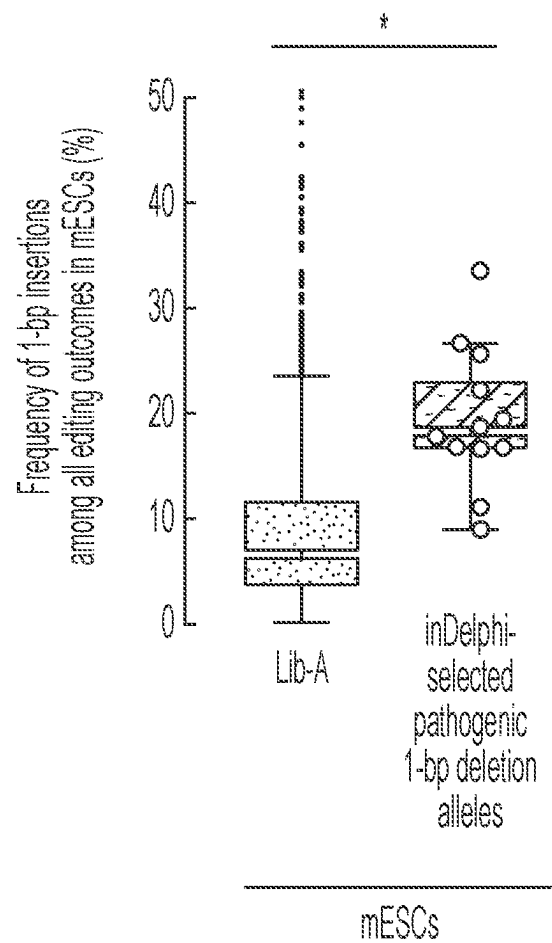


FIG. 23F

Pathogenic 1-bp deletion genotype (hg19)	Observed frequency of 1-bp insertions in all edited products (%)	Observed frequency of repair to wild- type frame in all edited products (%)	Observed frequency of repair to wild- type genotype in all edited products (%)	Observed frequency of repair to wild-type amino acid sequence all edited products (%)
KLHL7 chr7:23205401	33	55	33	33
NF2 chr22:30038270	17	60	1*	17
NBN chr8:90995063	19	30	17	18
PAX2 chr10:102541067	12	29	0*	11
LDLR chr19:11231158	19	49	13	13
MAN2B1 chr19:12767768	20	50	13	13
MLPH chr2:238436125	22	48	0*	22
SH3BP2 chr4:2824671	26	49	2*	26
COL4A4 chr2:227942769	16	49	16	16
NIPBL chr5:36961683	28	51	0*	28
UGT1A4 chr2:234627639	18	55	17	17
MFRP chr11:119216271	9	42	2*	2

FIG. 23G

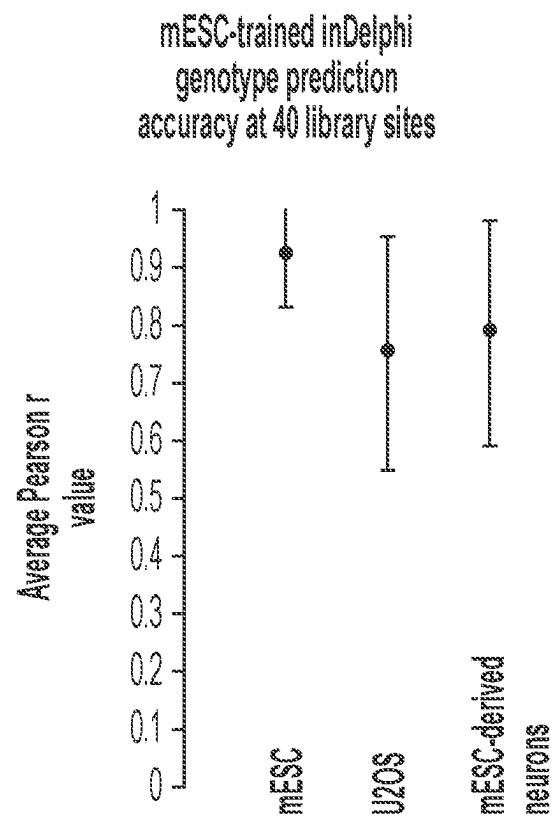
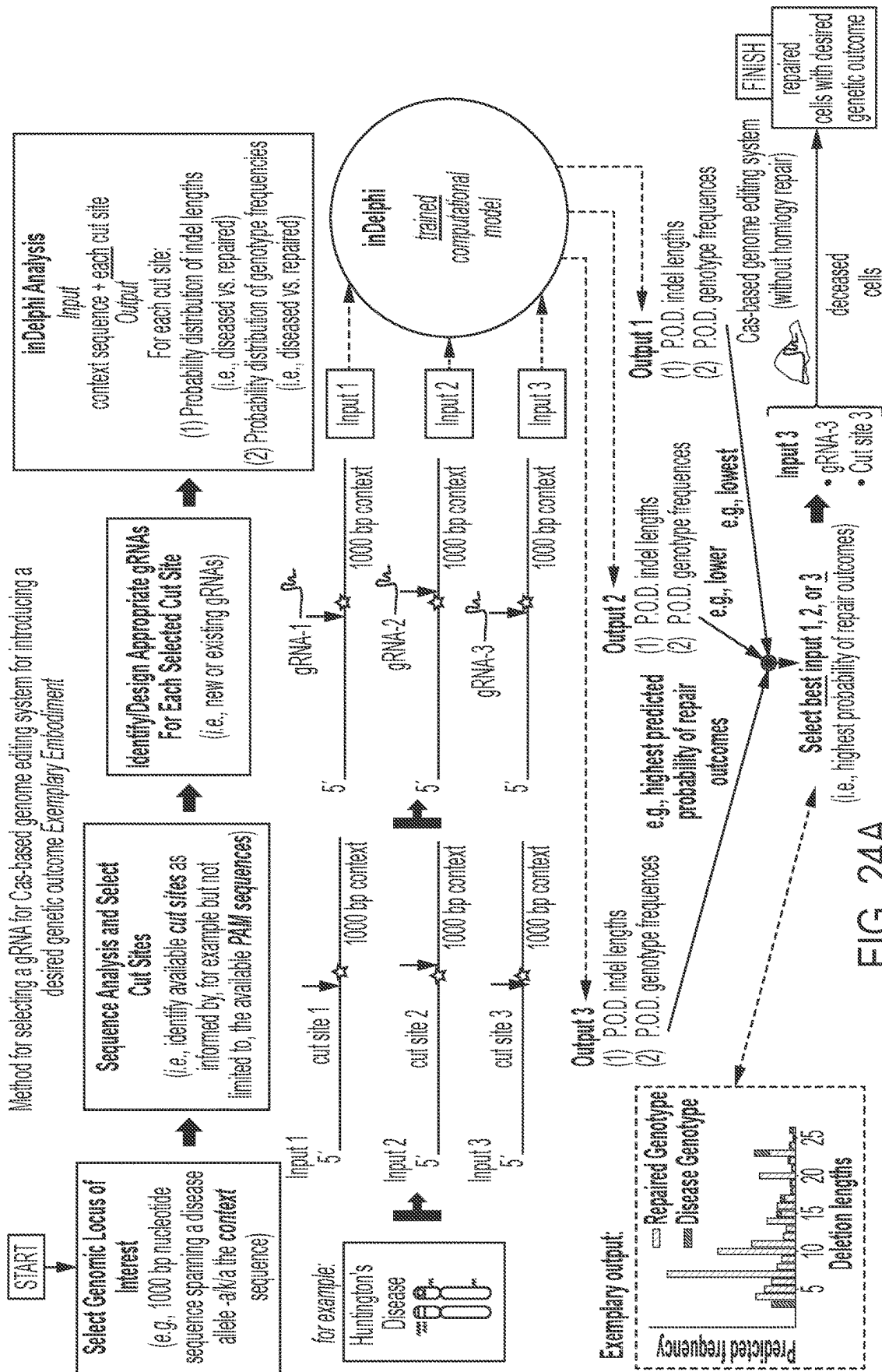
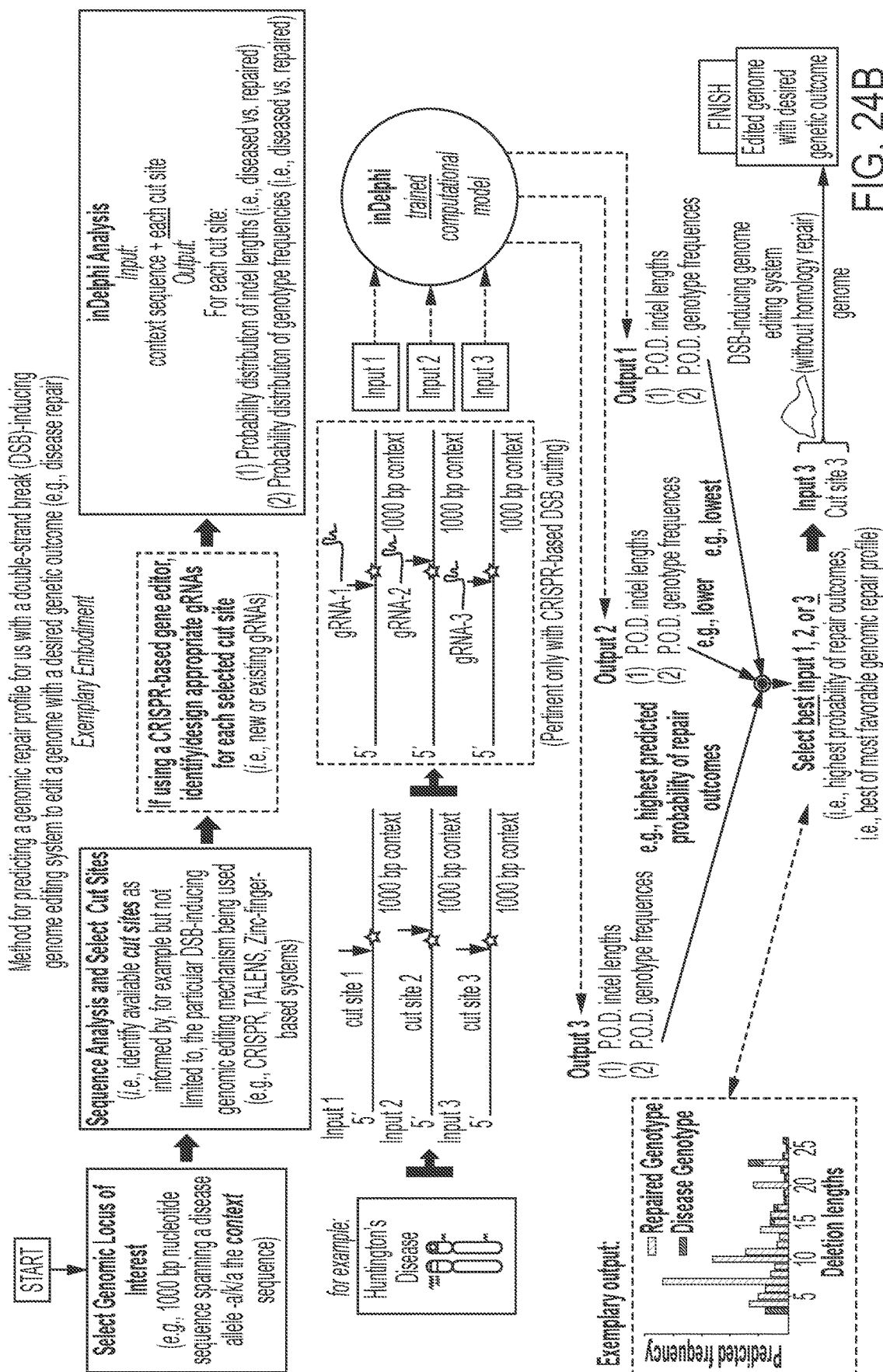


FIG. 23H



B
24
G
F

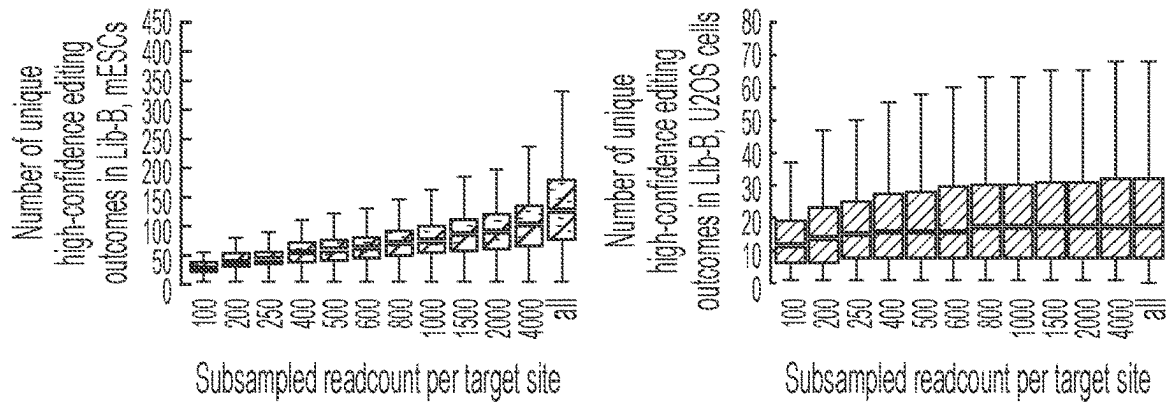


FIG. 25A

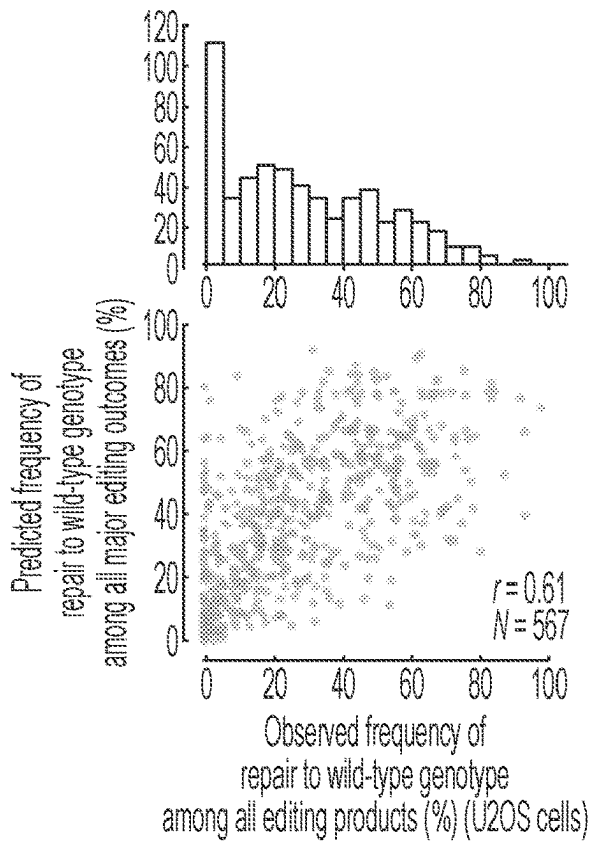


FIG. 25B

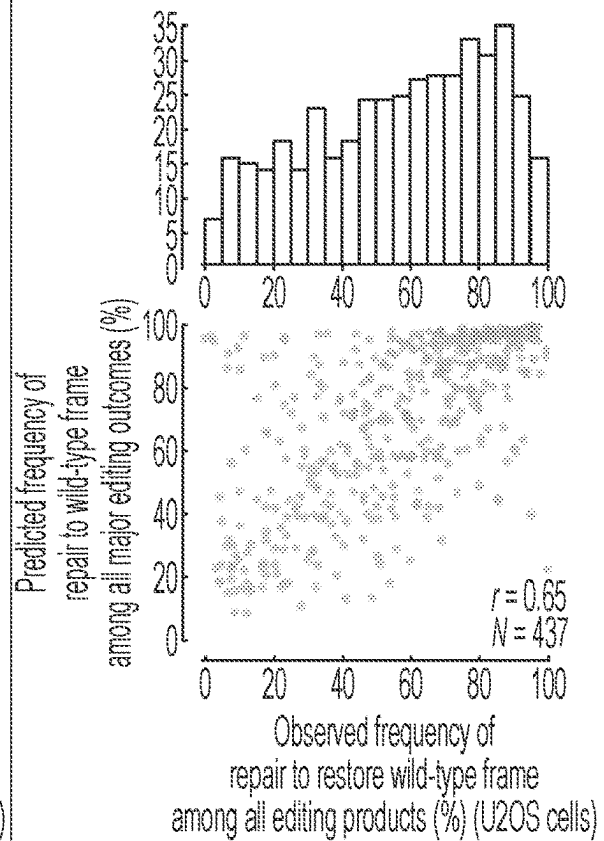


FIG. 25C

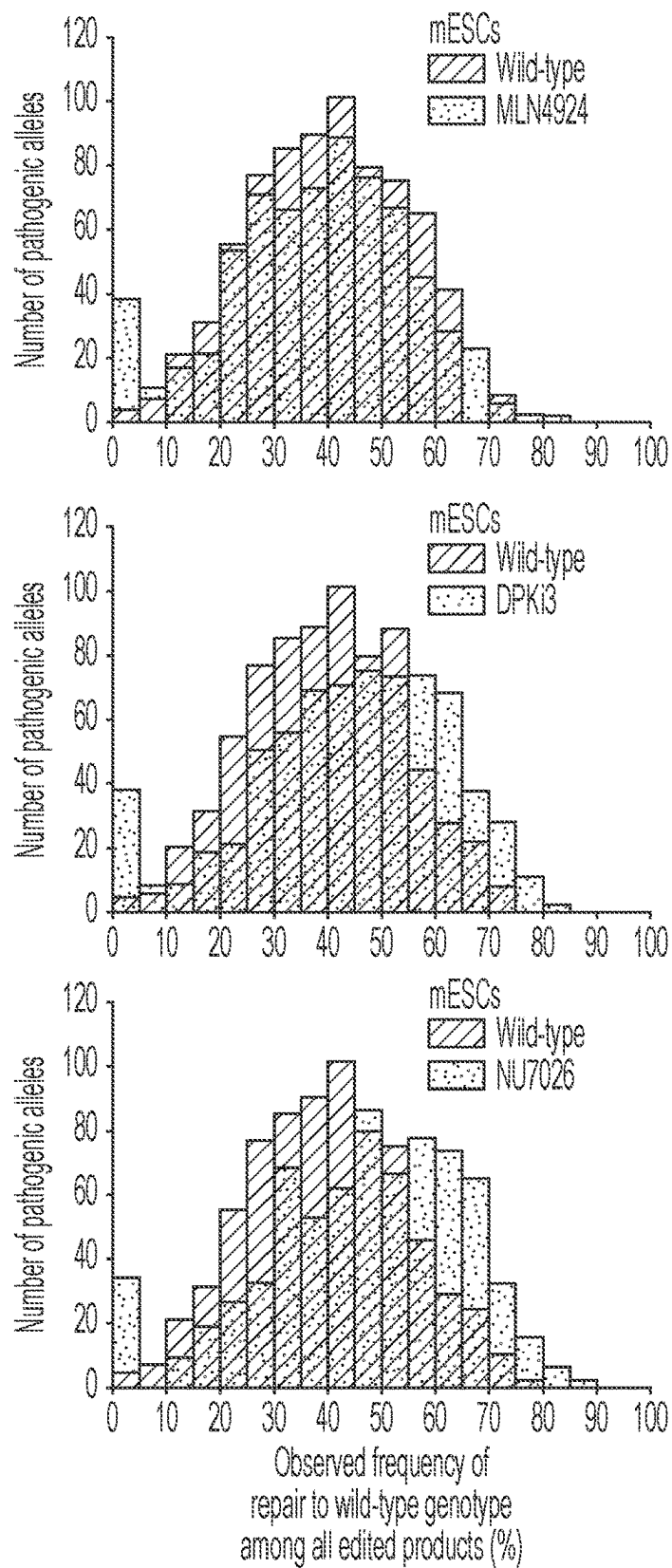


FIG 25D

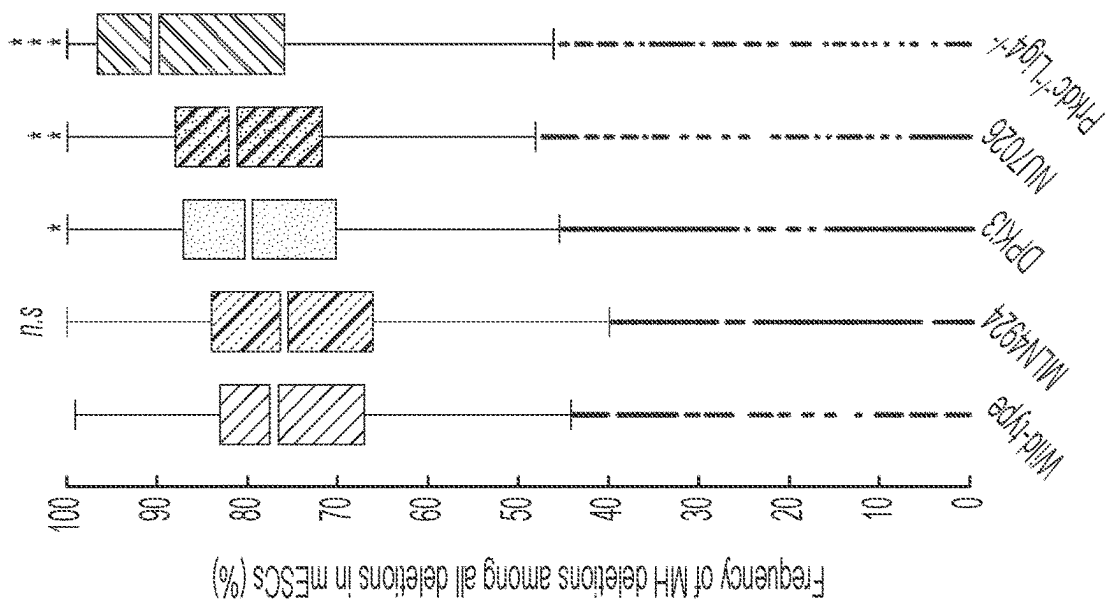


FIG. 26A

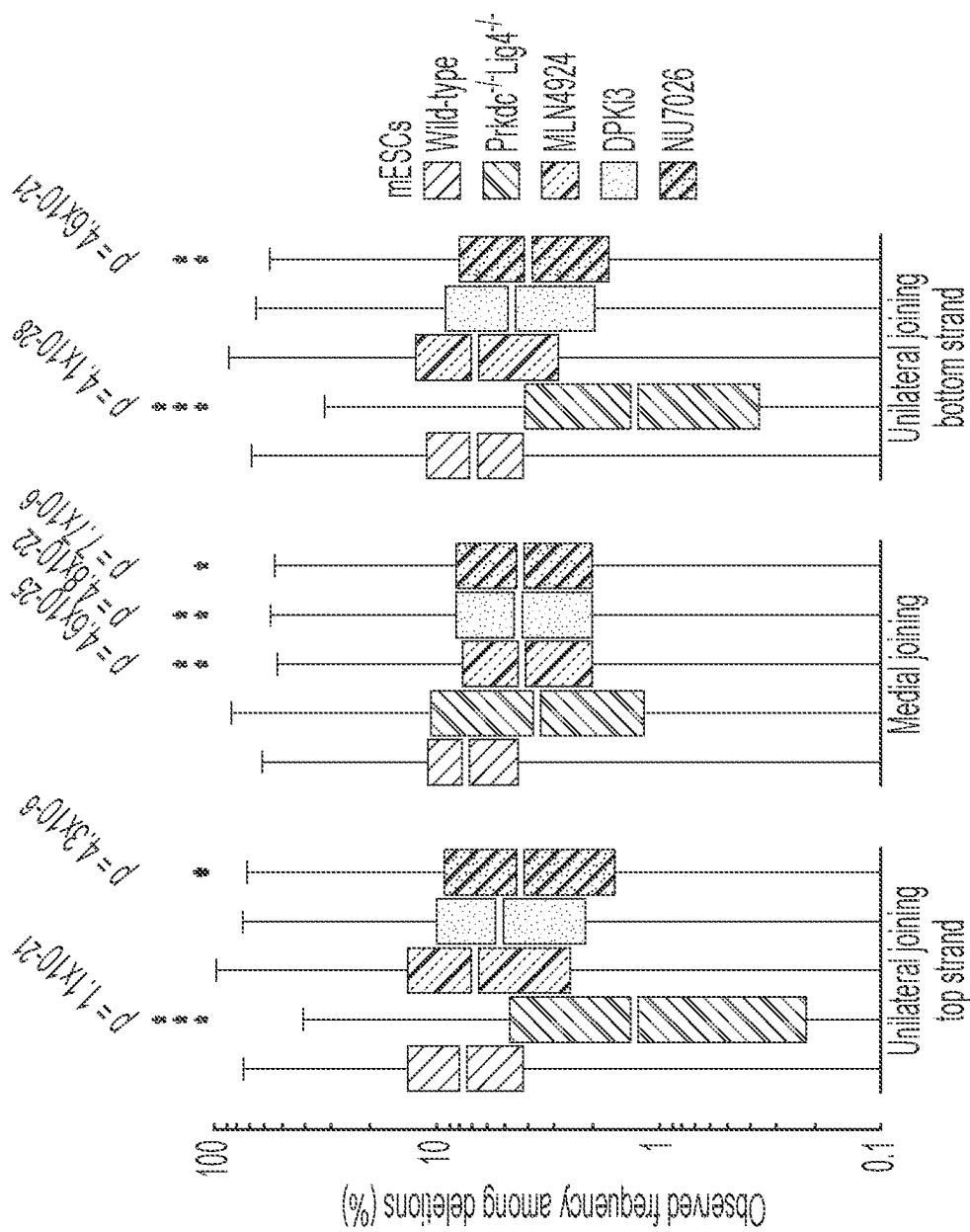


FIG. 26B

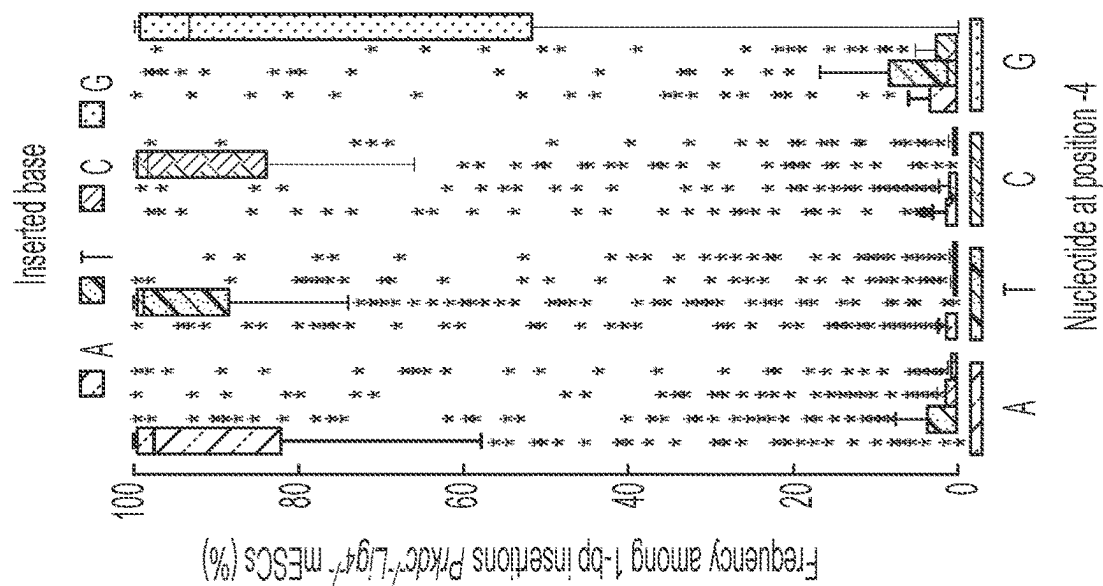


FIG. 26C

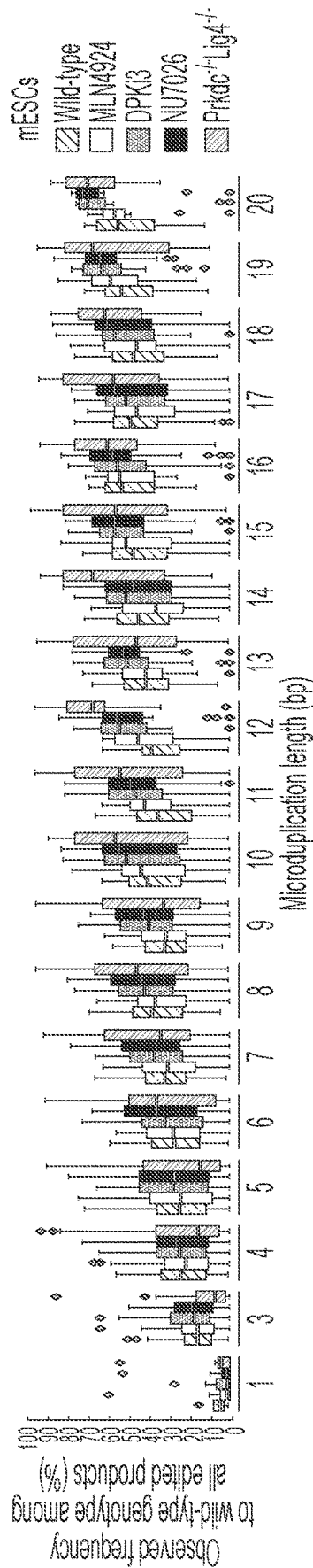


FIG. 26D

	Prkdc ^{-/-} Lig4 ^{-/-}	MLN4924	DPK3	NU7026
Prkdc ^{-/-} Lig4 ^{-/-}	-	0.09	0.16	0.18
MLN4924		-	0.77	0.73
DPK3			-	0.81
NU7026				-

FIG. 26E

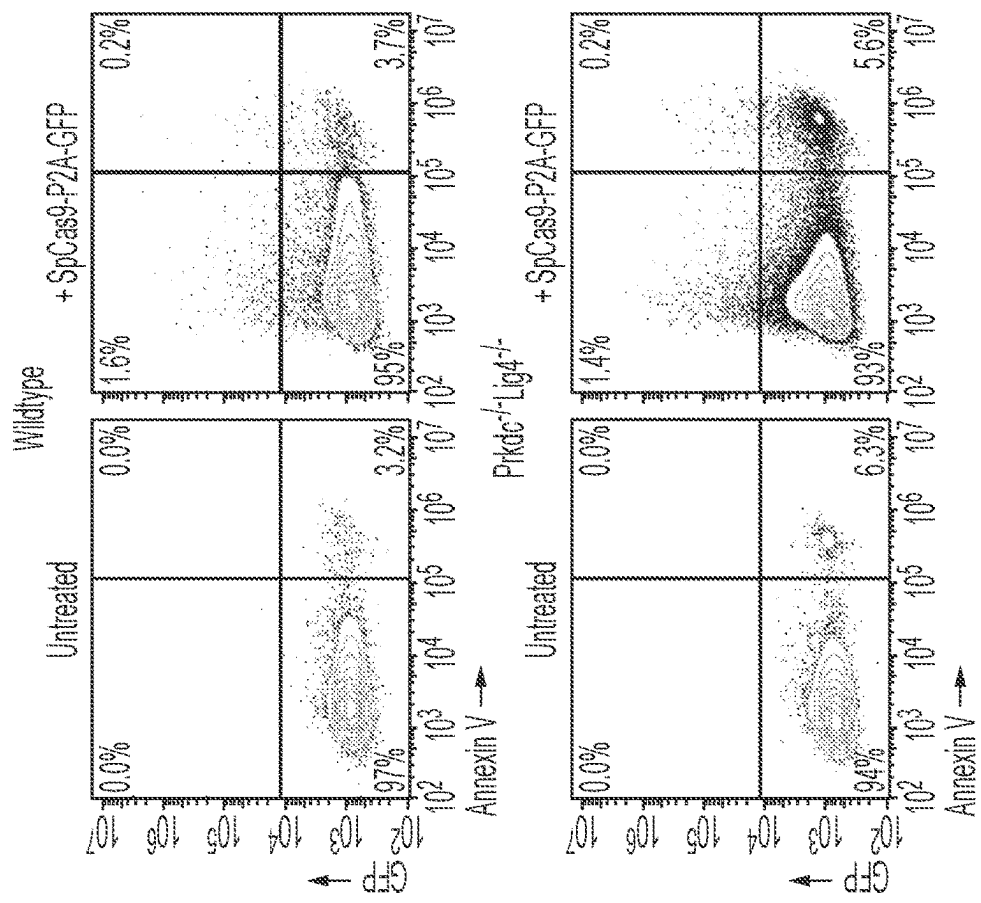


FIG. 26F

	Untreated	GFP-	GFP+
Annexin V , Wild-type	3.3 ± 0.3	4.8 ± 0.1	13.4 ± 0.8
Annexin V , Prkdc ^{-/-} Lig4 ^{-/-}	6.0 ± 0.7	6.0 ± 0.6	10.9 ± 0.9

FIG. 26G

1

SYSTEMS AND METHODS FOR PREDICTING REPAIR OUTCOMES IN GENETIC ENGINEERING

GOVERNMENT SUPPORT

This invention was made with government support under HG008754, EB022376, GM118062, HG009490, and DK101684 awarded by the National Institutes of Health, and HR0011-17-2-0049 awarded by the Defense Advanced Research Projects Agency. The government has certain rights in the invention.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

The contents of the electronic sequence listing (B119570051US01-SUBSEQ-ARM.txt; Size: 2,771,180 bytes; and Date of Creation: Nov. 7, 2023) is herein incorporated by reference in its entirety.

RELATED APPLICATIONS

This application is a national stage filing under 35 U.S.C. § 371 of international PCT application, PCT/US2018/065886, filed Dec. 15, 2018, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application, U.S. Ser. No. 62/599,623, filed on Dec. 15, 2017, and to U.S. Provisional Patent Application, U.S. Ser. No. 62/669,771, filed May 10, 2018, each of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

CRISPR (clustered regularly interspaced short palindromic repeats)-Cas9 has revolutionized genome editing, providing powerful research tools and promising agents for the potential treatment of genetic diseases¹⁻³. The DNA-targeting capabilities of Cas9 have been improved by the development of gRNA design principles⁴, modeling of factors leading to off-target DNA cleavage, enhancement of Cas9 sequence fidelity by modifications to the nuclease^{5,6} and gRNA⁷, and the evolution or engineering of Cas9 variants with alternative PAM sequences⁸⁻¹⁰. Similarly, control over the product distribution of genome editing has been advanced by the development of base editing to achieve precise and efficient single-nucleotide mutations^{7,11,12}, and the improvement of template-directed homology-directed repair (HDR) of double strand breaks¹³⁻¹⁵.

Non-template directed repair systems, including non-homologous end-joining (NHEJ) and microhomology-mediated end-joining (MMEJ), are major pathways involved in the repair of Cas9-mediated double-strand breaks that can result in highly heterogeneous repair outcomes that generate hundreds of genotypes following DNA cleavage at a single site. While end-joining repair of Cas9-mediated double-stranded DNA breaks has been harnessed to facilitate knock-in of DNA templates¹⁸⁻²¹ or deletion of intervening sequence between two cleavage sites²², NHEJ and MMEJ are not generally considered useful for precision genome editing applications. Recent work has found that the heterogeneous distribution of Cas9-mediated editing products at a given target site is reproducible and dependent on local sequence context^{20,21}, but no general methods have been described to predict genotypic products following Cas9-induced double-stranded DNA breaks.

2

The generally accepted view is that DNA double-strand break repair (i.e., template-free, non-homology-dependent repair) following cleavage by genome editing systems produces stochastic and heterogeneous repair products and is therefore impractical for applications beyond gene disruption. Further, template-free repair processes (e.g., MMEJ and NHEJ) following DNA double-strand break, despite being more efficient than homology-based repair, are generally not viewed as feasible solutions to precision repair applications, such as restoring the function of a defective gene with a gain-of-function genetic change. Accordingly, methods and solutions enabling the judicious application of template-free genome editing systems, including CRISPR/Cas, TALEN, or Zinc-Finger genome editing systems, would significantly advance the field of genome editing.

SUMMARY OF THE INVENTION

The present inventors have unexpectedly found through computational analyses that template-free DNA/genome editing systems, e.g., CRISPR/Cas9, Cas-based, Cpf1-based, or other DSB (double-strand break)-based genome editing systems, produce a predictable set of repair genotypes thereby enabling the use of such editing systems for applications involving or requiring precise manipulation of DNA, e.g., the correction of a disease-causing genetic mutation or modifying a wildtype sequence to confer a genetic advantage. This finding is contrary to the accepted view that DNA double-strand break repair (i.e., template-free, non-homology-dependent repair) following cleavage by genome editing systems produces stochastic and heterogeneous repair products and are therefore impractical for applications beyond gene disruption. Thus, the specification describes and discloses in various aspects and embodiments computational-based methods and systems for practically harnessing the innate efficiencies of template-free DNA repair systems for carrying out precise DNA and/or genomic editing without the reliance upon homology-based repair.

Accordingly, the specification provides in one aspect a method of introducing a desired genetic change in a nucleotide sequence using a double-strand break (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for introducing the desired genetic change into the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby introducing the desired genetic change in the nucleotide sequence at the cut site.

In another aspect, the specification provides a method of treating a genetic disease by correcting a disease-causing mutation using a double-strand break (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence comprising a disease-causing mutation; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for correcting the disease-causing mutation in the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby correcting the disease-causing mutation and treating the disease.

In yet another aspect, the specification provides a method of altering a genetic trait by introducing a genetic change in a nucleotide sequence using a double-strand break (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide

sequence; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for introducing the genetic change into the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby introducing the desired genetic change in the nucleotide sequence at the cut site and consequently altering the associated genetic trait.

In another aspect, the specification provides a method of selecting a guide RNA for use in a Cas-genome editing system capable of introducing a genetic change into a nucleotide sequence of a target genomic location, the method comprising: identifying in a nucleotide sequence of a target genomic location one or more available cut sites for a Cas-based genome editing system; and analyzing the nucleotide sequence and cut site with a computational model to identify a guide RNA capable of introducing the genetic change into the nucleotide sequence of the target genomic location.

In still another aspect, the specification provides a method of introducing a genetic change in the genome of a cell with a Cas-based genome editing system comprising: selecting a guide RNA for use in the Cas-based genome editing system in accordance with the method of the above aspect; and contacting the genome of the cell with the guide RNA and the Cas-based genome editing system, thereby introducing the genetic change.

In various embodiments, the cut sites available in the nucleotide sequence are a function of the particular DSB-inducing genome editing system in use, e.g., a Cas-based genome editing system.

In certain embodiments, the nucleotide sequence is a genome of a cell.

In certain other embodiments, the method for introducing the desired genetic change is done in vivo within a cell or an organism (e.g., a mammal), or ex vivo within a cell isolated or separated from an organism (e.g., an isolated mammalian cancer cell), or in vitro on an isolated nucleotide sequence outside the context of a cell.

In various embodiments, the DSB-inducing genome editing system can be a Cas-based genome editing system, e.g., a type II Cas-based genome editing system. In other embodiments, the DSB-inducing genome editing system can be a TALENS-based editing system or a Zinc-Finger-based genome editing system. In still other embodiments, the DSB-inducing genome editing system can be any such endonuclease-based system which catalyzes the formation of a double-strand break at a specific one or more cut sites.

In embodiments involving a Cas-based genome editing system, the method can further comprise selecting a cognate guide RNA capable of directing a double-strand break at the optimal cut site by the Cas-based genome editing system.

In certain embodiments, the guide RNA is selected from the group consisting the guide RNA sequences listed in any of Tables 1-6. In various embodiments, the guide RNA can be known or can be newly designed.

In various embodiments, the double-strand brake (DSB)-inducing genome editing system is capable of editing the genome without homology-directed repair.

In other embodiments, the double-strand brake (DSB)-inducing genome editing system comprises a type I Cas RNA-guided endonuclease, or a variant or orthologue thereof.

In still other embodiments, the double-strand brake (DSB)-inducing genome editing system comprises a type II Cas RNA-guided endonuclease, or a functional variant or orthologue thereof.

The double-strand brake (DSB)-inducing genome editing system may comprise a Cas9 RNA-guided endonuclease, or a variant or orthologue thereof in certain embodiments.

In still other embodiments, the double-strand brake (DSB)-inducing genome editing system can comprise a Cpf1 RNA-guided endonuclease, or a variant or orthologue thereof.

In yet further embodiments, the double-strand brake (DSB)-inducing genome editing system can comprise a *Streptococcus pyogenes* Cas9 (SpCas9), *Staphylococcus pyogenes* Cas9 (SpCas9), *Staphylococcus aureus* Cas (Sa-Cas9), *Francisella novicida* Cas9 (FnCas9), or a functional variant or orthologue thereof.

In various embodiments, the desired genetic change to be introduced into the nucleotide sequence, e.g., a genome, is to a correction to a genetic mutation. In embodiments, the genetic mutation is a single-nucleotide polymorphism, a deletion mutation, an insertion mutation, or a microduplication error.

In still other embodiments, the genetic change can comprise a 2-60-bp deletion or a 1-bp insertion.

The genetic change in other embodiments can comprise a deletion of between 2-20, or 4-40, or 8-80, or 16-160, or 32-320, 64-640, or up to 1000, 2000, 3000, 4000, 5000, 6000, 7000, 8000, 9000, or 10,000 or more nucleotides. Preferably, the deletion can restore the function of a defective gene, e.g., a gain-of-function frameshift genetic change.

In other embodiments, the desired genetic change is a desired modification to a wildtype gene that confers and/or alters one or more traits, e.g., conferring increased resistance to a pathogen or altering a monogenic trait (e.g., eye color) or polygenic trait (e.g., height or weight).

In embodiments involving correcting a disease-causing mutation, the disease can be a monogenic disease. Such monogenic diseases can include, for example, sickle cell disease, cystic fibrosis, polycystic kidney disease, Tay-Sachs disease, achondroplasia, beta-thalassemia, Hurler syndrome, severe combined immunodeficiency, hemophilia, glycogen storage disease Ia, and Duchenne muscular dystrophy.

In any of the above aspects and embodiments, the step of identifying the available cut sites can involve identifying one or more PAM sequences in the case of a Cas-based genome editing system.

In various embodiments of the above methods, the computational model used to analyze the nucleotide sequence is a deep learning computational model, or a neural network model having one or more hidden layers. In various embodiments, the computational model is trained with experimental data to predict the probability of distribution of indel lengths for any given nucleotide sequence and cut site. In still other embodiments, the computational model is trained with experimental data to predict the probability of distribution of genotype frequencies for any given nucleotide sequence and cut site.

In various embodiments, the computational model comprises one or more training modules for evaluating experimental data.

In various embodiments, the computational model can comprise: a first training module for computing a microhomology score matrix; a second training module for computing a microhomology independent score matrix; and a third training module for computing a probability distribution over 1-bp insertions, wherein once trained with experimental data the computational model computes a probability distribution over indel genotypes and a probability distribution over indel lengths for any given input nucleotide sequence and cut site.

In other embodiments, the computational model predicts genomic repair outcomes for any given input nucleotide sequence and cut site.

In various embodiments, the genomic repair outcomes can comprise microhomology deletions, microhomology-less deletions, and/or 1-bp insertions.

In still other embodiments, the computational model can comprise one or more modules each comprising one more input features selected from the group consisting of: a target site nucleotide sequence; a cut site; a PAM-sequence; microhomology lengths relative at a cut site, % GC content at a cut site; and microhomology deletion lengths at a cut site, and type of DSB-genome editing system.

In various embodiments, the nucleotide sequence analyzed by the computational model is between about 25-100 nucleotides, 50-200 nucleotides, 100-400 nucleotides, 200-800 nucleotides, 400-1600 nucleotides, 800-3200 nucleotides, and 1600-6400 nucleotide, or even up to 7K, 8K, 9K, 10K, 11K, 12K, 13K, 14K, 15K, 16K, 17K, 18K, 19K, 20K nucleotides, or more in length.

In another aspect, the specification relates to guide RNAs which are identified by various methods described herein. In certain embodiments, the guide RNAs can be any of those presented in Tables 1-6, the contents of which form part of this specification.

According to various embodiments, the RNA can be purely ribonucleic acid molecules. However, in other embodiments, the RNA guides can comprise one or more naturally-occurring or non-naturally occurring modifications. In various embodiments, the modifications can including, but are not limited to, nucleoside analogs, chemically modified bases, intercalated bases, modified sugars, and modified phosphate group linkers. In certain embodiments, the guide RNAs can comprise one or more phosphorothioate and/or 5'-N-phosphoramidite linkages.

In still other aspects, the specification discloses vectors comprising one or more nucleotide sequences disclosed herein, e.g., vectors encoding one or more guide RNAs, one or more target nucleotide sequences which are being edited, or a combination thereof. The vectors may comprise naturally occurring sequences, or non-naturally occurring sequences, or a combination thereof.

In still other aspects, the specification discloses host cells comprising the herein disclosed vectors encoding one more nucleotide sequences embodied herein, e.g., one or more guide RNAs, one or more target nucleotide sequences which are being edited, or a combination thereof.

In other aspects, the specification discloses a Cas-based genome editing system comprising a Cas protein (or homolog, variant, or orthologue thereof) complexed with at least one guide RNA. In certain embodiments, the guide RNA can be any of those disclosed in Tables 1-6, or a functional variant thereof.

In still other aspects, the specification provides a Cas-based genome editing system comprising an expression vector having at least one expressible nucleotide sequence encoding a Cas protein (or homolog, variant, or orthologue thereof) and at least one other expressible nucleotide sequence encoding a guide RNA, wherein the guide RNA can be identified by the methods disclosed herein for selecting a guide RNA.

In yet another aspect, the specification provides a Cas-based genome editing system comprising an expression vector having at least one expressible nucleotide sequence encoding a Cas protein (or homolog, variant, or orthologue thereof) and at least one other expressible nucleotide

sequence encoding a guide RNA, wherein the guide RNA can be identified by the methods disclosed herein for selecting a guide RNA.

In still a further aspect, the specification provides a library for training a computational model for selecting a guide RNA sequence for use with a Cas-based genome editing system capable of introducing a genetic change into a genome without homology-directed repair, wherein the library comprises a plurality of vectors each comprising a first nucleotide sequence of a target genomic location having a cut site and a second nucleotide sequence encoding a cognate guide RNA capable of directing a Cas-based genome editing system to carry out a double-strand break at the cut site of the first nucleotide sequence.

In another aspect, the specification provides a library and its use for training a computational model for selecting an optimized cut site for use with a DSB-based genome editing system (e.g., Cas-based system, TALAN-based system, or a Zinc-Finger-based system) that is capable of introducing a desired genetic change into a nucleotide sequence (e.g., a genome) at the selected cut site without homology-directed repair, wherein the library comprises a plurality of vectors each comprising a nucleotide sequence having a cut site, and optionally a second nucleotide sequence encoding a cognate guide RNA (in embodiments involving a Cas-based genome editing system).

In a still further aspect, the specification discloses a computational model.

In certain embodiments, the computational model can predict and/or compute an optimized or preferred cut site for a DSB-based genome editing system for introducing a genetic change into a nucleotide sequence. In preferred embodiments, the repair does not require homology-based repair mechanisms.

In certain other embodiments, the computational model can predict and/or compute an optimized or preferred cut site for a Cas-based genome editing system for introducing a genetic change into a nucleotide sequence. In preferred embodiments, the repair does not require homology-based repair mechanisms.

In still other embodiments, the computation model provides for the selection of a optimized or preferred guide RNA for use with a Cas-based genome editing system for introducing a genetic change in a genome. In preferred embodiments, the repair does not require homology-based repair mechanisms.

In various embodiments, the computational model is a neural network model having one or more hidden layers.

In other embodiments, the computational model is a deep learning computational model.

In various embodiments, that the DSB-based genome editing system (e.g., a Cas-based genome editing system) edits the genome without relying on homology-based repair.

In various embodiments, that computational model is trained with experimental data to predict the probability of distribution of indel lengths for any given nucleotide sequence and cut site. In other embodiments, computational model is trained with experimental data to predict the probability of distribution of genotype frequencies for any given nucleotide sequence and cut site.

In embodiments, the computational model comprises one or more training modules for evaluating experimental data.

In an embodiment, the computational model comprises: a first training module (305) for computing a microhomology score matrix (305); a second training module (310) for computing a microhomology independent score matrix; and a third training module (315) for computing a probability

distribution over 1-bp insertions, wherein once trained with experimental data the computational model computes a probability distribution over indel genotypes and a probability distribution over indel lengths for any given input nucleotide sequence and cut site.

In certain embodiments, the computational model predicts genomic repair outcomes for any given input nucleotide sequence (i.e., context sequence) and cut site.

In certain embodiments, the genomic repair outcomes comprise microhomology deletions, microhomology-less deletions, and 1-bp insertions.

In various embodiments, the one or more modules each comprising one more input features selected from the group consisting of: a target site nucleotide sequence; a cut site; a PAM-sequence; microhomology lengths relative at a cut site, % GC content at a cut site; and microhomology deletion lengths at a cut site.

In certain embodiments, the nucleotide sequence analyzed by the computational model is between about 25-100 nucleotides, 50-200 nucleotides, 100-400 nucleotides, 200-800 nucleotides, 400-1600 nucleotides, 800-3200 nucleotides, and 1600-6400 nucleotide, or more.

In yet another aspect, the specification discloses a method for training a computational model, comprising: (i) preparing a library comprising a plurality of nucleic acid molecules each encoding a nucleotide target sequence and a cognate guide RNA, wherein each nucleotide target sequence comprises a cut site; (ii) introducing the library into a plurality of host cells; (iii) contacting the library in the host cells with a Cas-based genome editing system to produce a plurality of genomic repair products; (iv) determining the sequences of the genomic repair products; and (iv) training the computational model with input data that comprises at least the sequences of the nucleotide target sequence and/or the genomic repair products and the cut sites.

In still another aspect, the specification discloses a method for training a computational model, comprising: (i) preparing a library comprising a plurality of nucleic acid molecules each encoding a nucleotide target sequence and a cut site; (ii) introducing the library into a plurality of host cells; (iii) contacting the library in the host cells with a DSB-based genome editing system to produce a plurality of genomic repair products; (iv) determining the sequences of the genomic repair products; and (iv) training the computational model with input data that comprises at least the sequences of the nucleotide target sequence and/or the genomic repair products and the cut sites.

In certain embodiments, the trained computational models disclosed herein are capable of computing a probability of distribution of indel lengths for any given input nucleotide sequence and input cut site, and/or a probability of distribution of genotype frequencies for any given input nucleotide sequence and input cut site.

In embodiments relating to Cas-based genomic editing systems, the trained computational model is capable of selecting a guide RNA for use with a Cas-based genome editing system for introducing a genetic change into a genome.

The computational model provides a means to produce precision genetic change with a DSB-based genomic editing system. The genetic changes can include microhomology deletion, microhomology-less deletion, and 1-bp insertion. In certain embodiments, the genetic change corrects a disease-causing mutation. In other embodiments, the genetic change modifies a wildtype sequence, which may confer a change in a genetic trait (e.g., a monogenic or polygenic trait). The disease-causing mutation that can be corrected

using the computational model with a DSB-based genomic editing system can include, but is not limited to, sickle cell disease, cystic fibrosis, polycystic kidney disease, Tay-Sachs disease, achondroplasia, beta-thalassemia, Hurler syndrome, severe combined immunodeficiency, hemophilia, glycogen storage disease Ia, or Duchenne muscular dystrophy.

In another aspect, the disclosure provides a method for selecting one or more guide RNAs (gRNAs) from a plurality of gRNAs for CRISPR, comprising acts of: for at least one gRNA of the plurality of gRNAs, using a local DNA sequence and a cut site targeted by the at least one gRNA to predict a frequency of one or more repair genotypes resulting from template-free repair following application of CRISPR with the at least one gRNA; and determining whether to select the at least one gRNA based at least in part on the predicted frequency of the one or more repaired genotypes.

In embodiments, the one or more repair genotypes correspond to one or more healthy alleles of a gene related to a disease. In other embodiments, the predicted frequency of the one or more repair genotypes is at least about 30%, or at least about 40%, or at least about 50%, or more.

In certain embodiments, the step of predicting the frequency of the one or more repair genotypes comprises: for each deletion length of a plurality of deletion lengths, aligning subsequences of that deletion length on 5' and 3' sides of the cut site to identify one or more longest microhomologies; featurizing the identified microhomologies; applying a machine learning model to compute a frequency distribution over the plurality of deletion lengths; and using frequency distribution over the plurality of deletion lengths to determine the frequency of the one or more repair genotypes.

In certain embodiments, the plurality of gRNAs comprise gRNAs for CRISPR/Cas9, and the application of CRISPR comprises application of CRISPR/Cas9.

In yet another aspect, the system comprises: at least one processor; and at least one computer-readable storage medium having encoded thereon instructions which, when executed, cause the at least one processor to perform a herein disclosed computational method.

A method for editing a nucleotide sequence using a DSB-based genomic editing system that introduces a genetic change at a cut site in a nucleotide sequence, wherein the cut site location is informed by a computational model that computes a frequency distribution over the plurality of deletion lengths and/or a frequency distribution of one or more repaired genotypes over the deletion lengths.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows an illustrative DNA segment **100**, in accordance with some embodiments.

FIGS. 2A-D show an illustrative matching of 3' ends of top and bottom strands of a DNA segment at a cut site and an illustrative repair product, in accordance with some embodiments.

FIG. 3A shows an illustrative machine learning model **300**, in accordance with some embodiments.

FIG. 3B shows an illustrative process **350** for building one or more machine learning models for predicting frequencies of deletion genotypes and/or deletion lengths, in accordance with some embodiments.

FIG. 4A shows an illustrative neural network **400A** for computing microhomology (MH) scores, in accordance with some embodiments.

FIG. 4B shows an illustrative neural network **400B** for computing MH-less scores, in accordance with some embodiments.

FIG. 4C shows an illustrative process **400C** for training two neural networks jointly, in accordance with some embodiments.

FIG. 4D shows an illustrative implementation of the insertion module **315** shown in FIG. 3A, in accordance with some embodiments.

FIG. 5 shows an illustrative process **500** for processing data collected from CRISPR/Cas9 experiments, in accordance with some embodiments.

FIG. 6 shows an illustrative process **600** for using a machine learning model to predict frequencies of indel genotypes and/or indel lengths, in accordance with some embodiments.

FIG. 7 shows illustrative examples of a blunt-end cut and a staggered cut, in accordance with some embodiments.

FIG. 8A shows an illustrative plot **800A** of predicted repair genotypes, in accordance with some embodiments.

FIG. 8B shows another illustrative plot **800B** of predicted repair genotypes, in accordance with some embodiments.

FIG. 8C shows another illustrative plot **800C** of predicted repair genotypes, in accordance with some embodiments.

FIG. 8D shows a microhomology identified in the example of FIG. 8C, in accordance with some embodiments.

FIG. 9 shows another illustrative neural network **900** for computing a frequency distribution over deletion lengths, in accordance with some embodiments.

FIG. 10 shows, schematically, an illustrative computer **1000** on which any aspect of the present disclosure may be implemented.

FIGS. 11A-11C show a high-throughput assessment of Cas9-mediated DNA repair products. FIG. 11A, A genome-integrated screening library approach for monitoring Cas9 editing products at thousands of target sequences. FIG. 11B, Frequency of Cas9-mediated repair products by class from 1,996 Lib-A target sequences in mouse embryonic stem cells (mESCs).

FIG. 11C, Distribution of Cas9-mediated repair products by class in 88 VO target sequences in K562 cells.

FIGS. 12A-12E show modeling of Cas9-mediated indels by inDelphi. FIG. 12A, Schematic of computational flow for inDelphi modeling. inDelphi separates Cas9-mediated editing products by indel type and uses machine learning tools trained on experimental Lib-A editing products to predict relative frequencies of editing products for any target site. Major editing products include 1-to 60-bp MH deletions, 1-to 60-bp MH-less deletions, and 1-bp insertions. FIG. 12B, Mechanism depicting microhomology-mediated end-joining repair, which yields distinct repair outcomes that reflect which microhomologous bases are used during repair.

FIG. 12C, Observed mean frequency of 1-bp insertion genotypes among 1,981 Lib-A target sequences with varying -4 nucleotides. Error bars show 95% C.I. on sample mean with 1000-fold bootstrapping. Data distributions are shown in FIGS. 18A-18H. FIG. 12D, Comparison of observed 1-bp insertion frequencies among all Cas9-edited products from 1,996 Lib-A target sequences. The box denotes the 25th, 50th, and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as fliers. * $P=5.4 \times 10^{-36}$; ** $P=8.6 \times 10^{-70}$, two-sided two-sample T-test, test statistic=-13.0 and -18.4, degrees of freedom=777 and 1,994; Hedges' g effect size=0.94 and 0.85, for*and**respectively. FIG. 12E, Motif representation of base identities that impact the frequency of 1-bp insertions in Lib-A data. Only bases with non-zero linear regression

weights in 10,000-fold iterative cross-validation are shown. Median held-out Pearson correlation 0.62, total N=1996.

FIGS. 13A-13F show that Cas9-mediated editing outcomes are accurately predicted by inDelphi. FIG. 13A, Histogram of the observed fraction of Cas9-mediated editing products whose indel length is included in inDelphi predictions in endogenous VO target sequences in HEK293 (N=86), HCT116 (N=91), and K562 (N=82) cells. FIG. 13B, Distribution of Pearson correlation values comparing inDelphi predictions to observed product frequencies in VO sequence contexts in HEK293T (N=86), HCT116 (N=91), and K562 (N=82) cells. The box denotes the 25th, 50th, and 75th percentiles, and whiskers show 1.5 times the interquartile range.

FIG. 13C, Distribution of Pearson correlation values comparing inDelphi predictions to observed indel length frequencies in VO sequence contexts in HEK293 (N=86), HCT116 (N=91), and K562 (N=82) cells. Box plot as in (FIG. 13B). FIG. 13D, Comparison of inDelphi and Microhomology-Predictor frameshift predictions to observed frameshift frequencies among 86 VO target sequences in HEK293 cells. The error band represents the 95% C.I. around the regression estimate with 1,000-fold bootstrapping. FIG. 13E shows 1-bp insertion frequencies among edited outcomes in U2OS and HEK293T cells (n=27 and 26 observations, baseline n=1,958 and 89 target sites, $P=4.2 \times 10^{-8}$ and 8.1×10^{-12} , respectively), two-sided Welch's t-test. FIG. 13F shows smoothed predicted distribution of the highest frequency indel among major editing outcomes (+1 to -60 indels) for SpCas9 gRNAs targeting the human genome.

FIGS. 14A-14F show high-precision, template-free Cas9 nuclease-mediated deletion and insertion. FIG. 14A, Schematic of deletion repair at a designed Lib-B target sequence with a 9-bp microduplication and strong sequence microhomology. FIG. 14B, Observed frequency of microduplication collapse among all edited products at 56 Lib-B target sequences designed with 7- to 25-bp microduplications. The error band represents the 95% C.I. around the regression estimate with 1,000-fold bootstrapping. FIG. 14C, Observed frequencies of 1-bp insertions among 205 sequence contexts designed to vary base identity at positions -5 to -2 (relative to the PAM at positions 0-2) in three surrounding low-microhomology sequence contexts. The X-axis is sorted by median 1-bp insertion frequency; see FIGS. 20A-20E for the complete axis. FIG. 14D, Comparison of the observed 1-bp insertion frequency at 205 Lib-B designed sequences as in (FIG. 14C) with varying positions -4 and -3. The box denotes the 25th, 50th, and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as fliers. * $P=0.03$; ** $P=2.98 \times 10^{-7}$, two-sided two-sample T-test, test statistic=-2.2 and -6.5, degrees of freedom=185 and 32, Hedges' g effect size=0.58 and 2.3, for*and**respectively. FIG. 14E, Comparison of predicted precision scores to observed precision scores for microhomology deletions in 86 VO target sites in HEK293 cells. FIG. 14F, Distribution of the predicted frequency of the most frequent deletion and insertion outcomes among major editing outcomes (1-bp insertions, 1- to 60-bp MH deletions, and 1- to 60-bp MH-less deletions) at 1,063,802 Cas9 gRNAs targeting human exons and introns.

FIGS. 15A-15F show precise template-free Cas9-mediated editing of pathogenic alleles to wild-type genotypes. FIG. 15A, Using Cas9-nuclease to correct a pathogenic LDLR allele to wild-type. FIG. 15B, Comparison among ClinVar/HGMD pathogenic alleles of observed and predicted frequencies of repair to wild-type alleles, accompa-

nied by a histogram of observed frequencies. Major editing products include 1-bp insertions, 1- to 60-bp MH deletions, and 1- to 60-bp MH-less deletions. FIG. 15C, Comparison of observed and predicted frequencies of frameshift repair to the wild-type frame among ClinVar/HGMD pathogenic alleles, accompanied by a histogram of observed frequencies. Major editing products are defined as in (FIG. 15B).

FIG. 15D, Histograms of observed frequencies of repair to the wild-type genotype for wild-type mESCs and Prdkc-/-Lig4-/- mESCs at Lib-B pathogenic microduplication alleles with predicted repair frequency $\geq 50\%$ among all major editing products, defined as in (FIG. 15B). Dashed lines indicate sample means which differ significantly. $P=7.8 \times 10^{-12}$; two-sided two-sample T-test, test statistic = -6.9, degrees of freedom = 1,297, Hedges' g effect size = 0.39. FIG. 15E, Flow cytometry contour plots showing GFP fluorescence and LDL-DyLight550 uptake in mESCs containing the LDLRdup1662_1669dupGCTGGTGA-P2A-GFP allele (LDLRdup-P2A-GFP) and treated with SpCas9 and gRNA when denoted. FIG. 15F, Fluorescence microscopy of mESCs containing the LDLRdup1662_1669dupGCTGGTGA-P2A-GFP allele treated with SpCas9 and gRNA, or untreated.

FIGS. 16A-16F show design and cloning of a high-throughput library to assess CRISPR-Cas9-mediated editing products. FIG. 16A, From left to right, distributions of predicted Cas9 on-target efficiency (Azimuth score), number of nucleotides participating in microhomology in 3-30-bp deletions, GC content, and estimated precision of deletion outcomes derived from 169,279 potential SpCas9 gRNA target sites in the human genome with quintiles marked as used to design Lib-A. FIG. 16B, Schematic of the cloning process used to clone Lib-A and Lib-B. The cloning process involves ordering a library of oligonucleotides pairing a gRNA protospacer with its 55-bp target sequence, centered on an NGG PAM. To insert the gRNA hairpin between the gRNA protospacer and the target site, the library undergoes an intermediate Gibson Assembly circularization step, restriction enzyme linearization, and Gibson Assembly into a plasmid backbone containing a U6-promoter to facilitate gRNA expression, a hygromycin resistance cassette, and flanking Tol2 transposon sites to facilitate integration into the genome. FIG. 16C, Analysis of cumulative percentage of all CRISPR-Cas9-mediated deletions from VO target sequences in HEK293 (N=89), HCT116 (N=92), and K562 (N=86) that delete up to the reported number of nucleotides (X-axis). 94% of deletions are 30-bp or shorter. FIG. 16D shows the number of unique high-confidence editing outcomes called by simulating data subsampling in data in lib-A (n=2,000 target sites) in mESCs (combined data from n=3 independent biological replicates) and U2OS cells (combined data from n=2 independent biological replicates). For 'all', the original non-subsampled data are presented. Each box depicts data for 2,000 target sites. Outliers are not depicted. FIG. 16E shows Pearson's r of genotype frequencies comparing lib-A in mESCs and U2OS cells with endogenous data in HEK293 (n=87 target sites), HCT116 (n=88), and K562 (n=86) cells. Outliers are depicted as diamonds. 1-bp insertion frequency adjustment was performed at each target site by proportionally scaling them to be equal between two cell types. FIG. 16F shows Pearson's r of genotype frequencies at lib-A target sites, comparing two independent biological replicate experiments in mESCs (n=1,861 target sites, median r=0.89) and U2OS cells (n=1,921, median r=0.77). Outliers are depicted as diamonds. Box plots denote the 25th, 50th and 75th percentiles and whiskers show 1.5 times the interquartile range.

FIGS. 17A-17D show that high-throughput CRISPR-Cas9 editing outcome screening yields replicate-consistent data that is concordant with the repair spectrum at endogenous human genomic loci. FIG. 17A, Box and swarm plot of the Pearson correlation of the genotypic product frequency spectra at VO target sequences comparing Lib-A in mESCs with endogenous data in HEK293 (N=87), HCT116 (N=88), and K562 (N=86). Each dot represents a target sequence, the box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range. FIG. 17B, Pearson correlation of the genotypic product frequency spectra at 1,861 Lib-A target sequences comparing two biological replicate experiments in mESCs. Median r=0.89. The box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as fliers. FIG. 17C, Distribution of Cas9-mediated genotypic products by repair category in endogenous data at VO target sequences in K562 (N=88), HCT116 (N=92), and HEK293 (N=89). FIG. 17D, Frequencies of deletions occurring beyond the Cas9 cutsite by distance as measured by the number of bases between the deleted base nearest to the cutsite and the two bases immediately surrounding the cutsite. Cutsite and distances are oriented with the NGG PAM on the positive side. * $P < 1 \times 10^{-5}$ for the Pearson correlation between a specific deletion frequency distribution and Cas9 editing rates across target sequences (VO-HEK293T N=96, Lib-A mESC N=2000). Box plot as in (FIG. 17A), with outliers beyond whiskers not depicted.

FIGS. 18A-18I show that sequence features correlated with higher and lower inDelphi phi scores. FIG. 18A, Diagram of all unique alignment outcomes at an example 7-bp deletion accompanied with a table of their MH-less end-joining type, MH length, deletion length, and delta-position. FIG. 18B, Plot of function learned by the neural network modeling MH deletions (MH-NN) mapping MH length and % GC to a numeric score (psi). FIG. 18C, Plot of function learned by the neural network modeling MH-independent deletions (MHless-NN) mapping deletion length to a numeric score (psi). FIG. 18D, Histogram of MHless-NN phi scores by deletion length, normalized to sum to 1. FIG. 18E, Observed frequency of 1-bp insertion genotypes in 1,981 Lib-A target sequences with varying -4 nucleotides. The box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as fliers. FIG. 18F, Plot showing 1-bp insertion frequency in 1,996 Lib-A target sequences compared to their total phi score. Pearson correlation = -0.084 ($P=1.7 \times 10^{-4}$). FIG. 18G, Relationship between 1-bp insertion frequency in 1,996 Lib-A target sequences compared to the predicted deletion length precision score. Pearson correlation = 0.069 ($P=2.1 \times 10^{-3}$). FIG. 18H, Diagram of hypothesized repair mechanisms that give rise to the outcome categories used by inDelphi, based on known mechanisms of MMEJ, microhomology-mediated alt-NHEJ and c-NHEJ repair pathways. Microhomology-mediated repair begins with 5'-end resection, allowing overlap of 3'-overhangs. Microhomologous basepairing of the 3'-overhangs temporarily stabilizes the ssDNA ends. In microhomology deletion, non-paired 3'-overhangs are removed and polymerase and ligase fill in and connect the gaps to reconstitute a dsDNA strand. In microhomology-less deletions, one 3'-overhang is ligated to the dsDNA backbone and the opposing strand is removed entirely, giving rise to a unilateral deletion with loss of bases on one side of the cutsite only. DNA polymerase and ligation bridge the ssDNA to create a contiguous dsDNA strand. Microhomology-independent mutations occur as a combined result of exonu-

clease, polymerase, and ligase activity that results in the joining of modified ends at the double strand break cutsite, giving rise to microhomology-less deletions, insertions, and mixtures thereof. FIG. 18I shows the categories of Cas9-mediated genotypic outcomes in data from U2OS cells (n=1,958 lib-A target sites), which can be compared to the categories of Cas9-mediated genotypic outcomes shown in FIG. 17C with regard to data from endogenous contexts at VO target sites in K562 (n=88 target sites), HCT116 (n=92), HEK293 (n=89) cells.

FIGS. 19A-19F show performance of inDelphi at predicting Cas9-mediated indel length and repair genotypes. FIG. 19A, Box and swarm plot of the Pearson correlation at 189 held-out Lib-A target sequences comparing inDelphi predictions with observed mESC Lib-A genotype product frequencies. The box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range. FIG. 19B, Box and swarm plot of the Pearson correlation at 189 held-out Lib-A target sequences comparing inDelphi predictions with observed mESC Lib-A indel length frequencies for 1-bp insertions to 60-bp deletions. Box plot as in (FIG. 19A). FIG. 19C, Distribution of predicted frameshift frequencies among 1-60-bp deletions for SpCas9 gRNAs targeting exons, shuffled exons, and introns in the human genome. Dashed lines indicate means. ***P<10-100. FIG. 19D, Pie chart depicting the output of Delphi for specific outcome classes. MH deletions (58% of all products) and single-base insertions (9% of all products) are predicted at single-base resolution, and deletion length is predicted for MH-less deletions (25% of all products). FIGS. 19E and 19F show a comparison of two methods for frameshift predictions to observed values with Pearson's r in HCT116 cells (FIG. 19E, n=91 target sites) and K562 cells (FIG. 19F, n=82 target sites). The error band represents the 95% confidence intervals around the regression estimate with 1,000-fold bootstrapping.

FIGS. 20A-20K show target sequences with extremely high or low microhomology phi scores skew toward a single predictable Cas9-mediated edited product. FIG. 20A, Scatter plot of the frequency of microduplication repair in Lib-B target sequences with designed 7-25-bp regions of microduplication as a function of microduplication length in human U2OS (N=32) and HEK293T cells (N=39). The Error band represents the 95% C.I. around the regression estimate with 1,000-fold bootstrapping. FIG. 20B, Box plots displaying total deletion phi score, total precision scores, and 1-bp insertion frequencies for (blue) 312 Lib-B sequences in the low-microhomology cohort with four randomized bases flanking the cutsite (fourbp), (green) 89 VO sequences (VO), and (red) 71 Lib-B sequences in the high-microhomology cohort with microduplications ranging from 7-25 bp (longdup). Box displays median and first and third quartiles. Whiskers are at 1.5 times interquartile range (IQR). Either swarm plot or outlier fliers depicted for each box plot. *P=6.1x10-9; two-sided two-sample T-test, test statistic=-5.94, degrees of freedom=399, Hedges' g effect size=0.49. FIG. 20C, Scatterplot of 1-bp insertion frequency among all non-wild-type products when varying four bases surrounding the cutsite (positions -5 to -2 counted from the NGG-PAM at positions 0-2) with all x-tick labels depicted, contained within three target sequences (red, blue, green) from the low-microhomology cohort of Lib-B in mESCs (N=205). FIG. 20D, Distribution of the total frequency of all non-wild-type Cas9 editing products in the subset of target sequences from the low-microhomology cohort containing four randomized bases flanking the cutsite (fourbp) with >50% overall frequencies of 1-bp insertion (N=50), VO

sequences (N=89), and the high-microhomology cohort with microduplications ranging from 7-25 bp (longdup) in Lib-B editing in mESCs (N=56). FIG. 20E, Scatterplot displaying 1-bp insertion frequencies and Cas9 editing rate in 205 "fourbp" contexts with Pearson correlation of -0.35 (P=3.3e-07). FIG. 20F shows the frequency of 1-bp insertions in mESCs (n=1,981 lib-A target sites) and U2OS cells (n=1,918) with varying -4 nucleotides. FIGS. 20G and 20H show plots of 1-bp insertion frequency in mESCs (n=1,996 lib-A target sites) and U2OS cells (n=1,966) compared to their total phi score (FIG. 20G) and predicted deletion length precision score (FIG. 20H) with Pearson's r. FIG. 20I shows a comparison of 1-bp insertion frequencies among all edited products from 1,966 lib-A target sites in U2OS cells (combined data from n=2 independent biological replicates). FIG. 20J shows nucleotides and their effect on the frequency of 1-bp insertions in U2OS cells. Only bases with non-zero linear regression weights in 10,000-fold iterative cross-validation are shown. Total n=1,966 lib-A target sites. FIG. 20K shows the insertion frequency in mESCs (n=205) and U2OS cells (n=217) when varying four bases by the cleavage site (positions -5 to -2 counted from the NGG-PAM at positions 0-2) contained within three target sites designed with weak microhomology.

FIGS. 21A-21D show the precise repair of pathogenic microduplications. FIG. 21A, Observed frequencies of repair to wild-type genotype at 194 ClinVar pathogenic alleles vs. predicted frequencies in Lib-B in human HEK293T cells. FIG. 21B, Observed frequencies of repair to wild-type frame at 140 ClinVar pathogenic alleles vs. predicted frequencies in Lib-B in human HEK293T cells. FIG. 21C, Observed frequencies of repair to wild-type genotype at 49 Clinvar pathogenic alleles vs. predicted frequencies in Lib-B in human U2OS cells. FIG. 21D, Observed frequencies of repair to wild-type frame at 37 ClinVar pathogenic alleles vs. predicted frequencies in Lib-B in human U2OS cells.

FIGS. 22A-22E show altered distribution of Cas9-mediated genotypic products in Prkdc-/-Lig4-/- mESCs as compared to wild-type mESCs. FIG. 22A, Distribution of Cas9-mediated genotypic products by repair outcome class in Prkdc-/-Lig4-/- mESC for 1,446 target sequences.

FIG. 22B, Comparison of observed mean frequency of deletion products contributed by microhomology-less unilateral joining and medial joining deletions among all deletions comparing 1,995 Lib-A target sequences in wildtype mESC to 1,850 Lib-A target sequences in Prkdc-/-Lig4-/- mESC. *P<10-66; two-sided two-sample T-test, test statistic >17.7, degrees of freedom=3,843; Hedges' g effect size >0.58. FIG. 22C, Comparison of observed frequency of deletion products contributed by microhomology-less unilateral joining and medial joining deletions among all deletions, between 1,995 Lib-A target sequences in wildtype mESC to 1,850 Lib-A target sequences in Prkdc-/-Lig4-/- mESC. *and**as in (FIG. 22B). The box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as fliers. FIG. 22D, Observed mean frequency of 1-bp insertion genotypes at 1,055 target sequences with varying -4 nucleotides in Lib-A in Prkdc-/-Lig4-/- mESCs. The error bars show the 95% C.I. on the sample mean with 1,000-fold bootstrapping. FIG. 22E, Observed frequency of 1-bp insertion genotypes at 1,055 target sequences with varying -4 nucleotides in Lib-A in Prkdc-/-Lig4-/- mESCs. Box plot as in (b).

FIGS. 23A-23H show that template-free Cas9-nuclease editing of human cells containing pathogenic LDLR micro-

15

duplication alleles restores LDL uptake. FIG. 23A, Flow cytometric contour plots showing GFP fluorescence and LDL-Dylight550 uptake in HCT116 cells containing the denoted LDLR alleles and treated with SaCas9 and gRNA when denoted. FIG. 23B, Fluorescence microscopy of HCT116 cells containing the denoted LDLR alleles and treated with SaCas9 and gRNA when denoted. GFP fluorescence is shown in green, LDL-Dylight550 uptake in red, and Hoechst staining nuclei in blue. FIG. 23C, Fluorescence microscopy of U2OS cells containing the denoted LDLR alleles and treated with SaCas9 and gRNA when denoted. GFP fluorescence is shown in green, LDL-Dylight550 uptake in red, and Hoechst staining nuclei in blue. FIG. 23D, Flow cytometry gating strategy used for mESC+LDLRdup-P2A-GFP untreated. FIG. 23E, Flow cytometry gating strategy used for mESC+LDLRdup-P2A-GFP+SpCas9+gRNA. FIGS. 23F and 23G show the results of 12 pathogenic 1-bp deletion alleles selected by inDelphi for high 1-bp insertion frequency (combined data from n=2 independent biological replicates) compared to lib-A (f) and presented in a table (FIG. 23G). The box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as diamonds. *P=1.6×10⁻⁴, two-sided Welch's t-test. For detailed statistics, see Methods. In the table, the most frequent 1-bp insertion genotype predicted by inDelphi that does not correspond to the wild-type genotype is indicated by an asterisk. In fluorescence microscopy plots, GFP fluorescence is shown in green, LDL-Dylight550 uptake in red, and Hoechst staining nuclei in blue. FIG. 23H shows mESC-trained inDelphi genotype prediction accuracy as 40 library sites.

FIG. 24A is a schematic depicting an exemplary method of using a trained computational model (e.g., "inDelphi") in conjunction with a Cas-based genome editing system to edit a nucleotide sequence (e.g., a genome) to achieve a desired genetic outcome (e.g., a correction to a disease-causing mutation to treat a disease, or modification of a wildtype type gene to confer an improved trait or phenotype). For any given set of inputs (a context sequence and a selected cut site), the trained computational model computes the probability distribution of indel lengths and the probability distribution of genotype frequencies, enabling the user to select the optimal input (e.g., cut site) for conducting editing by a Cas-based genome editing system to achieve the highest frequency of desired genetic output. The computational method may be used to predict, for a given local sequence context, template-free repair genotypes and frequencies of occurrence thereof.

FIG. 24B is a schematic depicting an exemplary method of using a trained computational model (e.g., "inDelphi") in conjunction with a double-strand break (DSB)-inducing genome editing system to edit a nucleotide sequence (e.g., a genome) to achieve a desired genetic outcome (e.g., a correction to a disease-causing mutation to treat a disease, or modification of a wildtype type gene to confer an improved trait or phenotype). For any given set of inputs (a context sequence and a selected cut site), the trained computational model computes the probability distribution of indel lengths and the probability distribution of genotype frequencies, enabling the user to select the optimal input (e.g., cut site) for conducting editing by a DSB-inducing genome editing system to achieve the highest frequency of desired genetic output. The computational method may be used to predict, for a given local sequence context, template-free repair genotypes and frequencies of occurrence thereof.

FIG. 25A-25D provides a characterization of lib-B data including pathogenic microduplication repair in wild-type

16

mESCs, wild-type U2OS cells and mESCs treated with DPKi3, NU7026 and MLN4924. FIG. 25A shows box plots of the number of unique high-confidence editing outcomes called by simulating data subsampling in data at 2,000 lib-B target sites in mESCs (combined data from n=2 independent technical replicates) and U2OS cells (combined data from n=2 independent biological replicates). In 'all', the full non-subsampled data are presented (see Table 8 herein for read counts). Each box depicts data for 2,000 target sites. The box denotes the 25th, 50th, and 75th percentiles and whiskers show 1.5 times the interquartile range. Outliers are not depicted. FIG. 25B shows the frequencies of repair to wild-type genotype at 567 ClinVar pathogenic alleles versus predicted frequencies in lib-B in human U2OS cells with Pearson's r. FIG. 25C shows the frequencies of repair to wild-type frame at 437 ClinVar pathogenic alleles versus predicted frequencies in lib-B in human U2OS cells with Pearson's r. FIG. 25D shows the frequency of pathogenic microduplication repair in wild-type mESCs (n=1,480 target sites) compared to mESCs treated with MLN4924 (n=1,569), NU7041 (n=1,561) and DPKi3 (n=1,563).

FIG. 26A-26G shows the altered distributions of Cas9-mediated genotypic products in Prkdc-/-Lig4-/- mESCs and mESCs treated with DPKi3, NU7026, and MLN4924 compared to wild-type mESCs. FIG. 26A shows a comparison of MH deletions among all deletions at lib-B target sites in wild-type cells (n=1,909 target sites), cells treated with DPKi3 (n=1,999), MLN4924 (n=1,995) or NU7026 (n=1,999) and Prkdc-/-Lig4-/- cells (n=1,446). Statistical tests performed against wild-type population. *P=5.6×10⁻⁵, **P=3.5×10⁻¹³, ***P=5.0×10⁻⁴¹, two-sided Welch's t-test. FIG. 26B shows a comparison of the frequency of each class of MH-less deletions among all deletion products in wild-type (lib-A and lib-B target sites, n=3,829 target sites), DPKi3 (lib-B, n=1,990), MLN4924 (lib-B, n=1,980), NU7026 (lib-B, n=1,992) and Prkdc-/-Lig4-/- (lib-A and lib-B target sites, n=3,344). P values are compared to wild-type, two-sided Welch's t-test. FIG. 26C shows frequency of 1-bp insertions at 1,055 target sites in lib-A in Prkdc-/-Lig4-/- mESCs. FIG. 26D Frequencies of deletion repair to wild-type genotype in lib-B in wild-type mESCs (n=1,480 target sites, combined data from two technical replicates) compared to conditions, with combined data from two independent biological replicates for each of Prkdc-/-Lig4-/- (n=1,041 target sites), MLN4924 (n=1,569), NU7026 (n=1,561) and DPKi3 (n=1,563). FIG. 26E provides a table of Pearson's r of the change in disease correction frequency compared to wild-type at n=791 target sites for each pair of conditions. f, g, Annexin V-568 staining flow cytometry contour plots (FIG. 26F) and mean±standard deviation values (FIG. 26G) in wild-type and Prkdc-/-Lig4-/- lib-A mESCs following transfection with SpCas9-P2A-GFP (representative data for n=2 experiments). Box plots denote the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as diamonds.

DEFINITIONS

As used herein and in the claims, the singular forms "a," "an," and "the" include the singular and the plural reference unless the context clearly indicates otherwise. Thus, for example, a reference to "an agent" includes a single agent and a plurality of such agents. The term "Cas9" or "Cas9 nuclease" refers to an RNA-guided nuclease comprising a Cas9 protein, or a fragment thereof (e.g., a protein comprising an active or inactive DNA cleavage domain of Cas9,

and/or the gRNA binding domain of Cas9). A Cas9 nuclease is also referred to sometimes as a casn1 nuclease or a CRISPR (clustered regularly interspaced short palindromic repeat)-associated nuclease. CRISPR is an adaptive immune system that provides protection against mobile genetic elements (viruses, transposable elements and conjugative plasmids). CRISPR clusters contain spacers, sequences complementary to antecedent mobile elements, and target invading nucleic acids. CRISPR clusters are transcribed and processed into CRISPR RNA (crRNA). In type II CRISPR systems correct processing of pre-crRNA requires a trans-encoded small RNA (tracrRNA), endogenous ribonuclease 3 (mc) and a Cas9 protein. The tracrRNA serves as a guide for ribonuclease 3-aided processing of pre-crRNA. Subsequently, Cas9/crRNA/tracrRNA endonucleolytically cleaves linear or circular dsDNA target complementary to the spacer. The target strand not complementary to crRNA is first cut endonucleolytically, then trimmed 3'-5' exonucleolytically. In nature, DNA-binding and cleavage typically requires protein and both RNAs. However, single guide RNAs ("sgRNA", or simply "gRNA") can be engineered so as to incorporate aspects of both the crRNA and tracrRNA into a single RNA species. See, e.g., Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. *Science* 337:816-821(2012), the entire contents of which is hereby incorporated by reference. Cas9 recognizes a short motif in the CRISPR repeat sequences (the PAM or protospacer adjacent motif) to help distinguish self versus non-self. Cas9 nuclease sequences and structures are well known to those of skill in the art (see, e.g., "Complete genome sequence of an M1 strain of *Streptococcus pyogenes*." Ferretti et al., J. J., McShan W. M., Ajdic D. J., Savic D. J., Savic G., Lyon K., Primeaux C., Sezate S., Suvorov A. N., Kenton S., Lai H. S., Lin S. P., Qian Y., Jia H. G., Najjar F. Z., Ren Q., Zhu H., Song L., White J., Yuan X., Clifton S. W., Roe B. A., McLaughlin R. E., *Proc. Natl. Acad. Sci. U.S.A.* 98:4658-4663(2001); "CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III." Deltcheva E., Chylinski K., Sharma C. M., Gonzales K., Chao Y., Pirzada Z. A., Eckert M. R., Vogel J., Charpentier E., *Nature* 471:602-607(2011); and "A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity." Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. *Science* 337:816-821(2012), the entire contents of each of which are incorporated herein by reference). Cas9 orthologs have been described in various species, including, but not limited to, *S. pyogenes* and *S. thermophilus*. Additional suitable Cas9 nucleases and sequences will be apparent to those of skill in the art based on this disclosure, and such Cas9 nucleases and sequences include Cas9 sequences from the organisms and loci disclosed in Chylinski, Rhun, and Charpentier, "The tracrRNA and Cas9 families of type II CRISPR-Cas immunity systems" (2013) *RNA Biology* 10:5, 726-737; the entire contents of which are incorporated herein by reference. In some embodiments, a Cas9 nuclease has an inactive (e.g., an inactivated) DNA cleavage domain.

In some embodiments, Cas9 refers to Cas9 from: *Corynebacterium ulcerans* (NCBI Refs: NC_015683.1, NC_017317.1); *Corynebacterium diphtheria* (NCBI Refs: NC_016782.1, NC_016786.1); *Spiroplasma syphilicicola* (NCBI Ref: NC_021284.1); *Prevotella intermedia* (NCBI Ref: NC_017861.1); *Spiroplasma taiwanense* (NCBI Ref: NC_021846.1); *Streptococcus iniae* (NCBI Ref: NC_021314.1); *Belliella baltica* (NCBI Ref: NC_018010.1); *Psychroflexus torques* I (NCBI Ref: NC_018721.1); *Streptococcus thermophilus* (NCBI Ref: YP_820832.1); *Listeria innocua* (NCBI Ref:

NP_472073.1); *Campylobacter jejuni* (NCBI Ref: YP_002344900.1); or *Neisseria meningitidis* (NCBI Ref: YP_002342100.1).

The term "Cas-based genome editing system" refers to a system comprising any naturally occurring or variant Cas endonuclease (e.g., Cas9), or functional variant, homolog, or orthologue thereof, and a cognate guide RNA. The term "Cas-based genome editing system" may also refer to an expression vector having at least one expressible nucleotide sequence encoding a Cas protein (or homolog, variant, or orthologue thereof) and at least one other expressible nucleotide sequence encoding a guide RNA.

The term "DSB-based genome editing system" refers to a system comprising any naturally occurring or variant endonuclease which catalyzes the formation of a double strand break at a cut site (e.g., Cas9, Crf1, TALEN, or Zinc Finger), or functional variant, homolog, or orthologue thereof, and a cognate guide RNA if required (e.g., TALENs and Zinc Fingers do not require a guide RNA for targeting to a cut site). The term "DSB-based genome editing system" may also refer to an expression vector having at least one expressible nucleotide sequence encoding a DSB endonuclease protein (or homolog, variant, or orthologue thereof) and at least one other expressible nucleotide sequence encoding a guide RNA, if required (e.g., as required for Cas9 or Crf1).

The term "effective amount," as used herein, refers to an amount of a biologically active agent that is sufficient to elicit a desired biological response. For example, in some embodiments, an effective amount of a nuclease may refer to the amount of the nuclease that is sufficient to induce cleavage of a target site specifically bound and cleaved by the nuclease. In some embodiments, an effective amount of a recombinase may refer to the amount of the recombinase that is sufficient to induce recombination at a target site specifically bound and recombined by the recombinase. As will be appreciated by the skilled artisan, the effective amount of an agent, e.g., a nuclease, a recombinase, a hybrid protein, a fusion protein, a protein dimer, a complex of a protein (or protein dimer) and a polynucleotide, or a polynucleotide, may vary depending on various factors as, for example, on the desired biological response, the specific allele, genome, target site, cell, or tissue being targeted, and the agent being used.

The term "linker," as used herein, refers to a chemical group or a molecule linking two molecules or moieties, e.g., a binding domain and a cleavage domain of a nuclease. In some embodiments, a linker joins a gRNA binding domain of an RNA-programmable nuclease and the catalytic domain of a recombinase. In some embodiments, a linker joins a dCas9 and a recombinase. Typically, the linker is positioned between, or flanked by, two groups, molecules, or other moieties and connected to each one via a covalent bond, thus connecting the two. In some embodiments, the linker is an amino acid or a plurality of amino acids (e.g., a peptide or protein). In some embodiments, the linker is an organic molecule, group, polymer, or chemical moiety. In some embodiments, the linker is 5-100 amino acids in length, for example, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 30-35, 35-40, 40-45, 45-50, 50-60, 60-70, 70-80, 80-90, 90-100, 100-150, or 150-200 amino acids in length. Longer or shorter linkers are also contemplated.

The term "mutation," as used herein, refers to a substitution of a residue within a sequence, e.g., a nucleic acid or amino acid sequence, with another residue, or a deletion or insertion of one or more residues within a sequence. Muta-

tions are typically described herein by identifying the original residue followed by the position of the residue within the sequence and by the identity of the newly substituted residue. Various methods for making the amino acid substitutions (mutations) provided herein are well known in the art, and are provided by, for example, Green and Sambrook, *Molecular Cloning: A Laboratory Manual* (4th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2012)). Mutations can include a variety of categories, such as single base polymorphisms, microduplication regions, indel, and inversions, and is not meant to be limiting in any way.

The terms “nucleic acid” and “nucleic acid molecule,” as used herein, refer to a compound comprising a nucleobase and an acidic moiety, e.g., a nucleoside, a nucleotide, or a polymer of nucleotides. Typically, polymeric nucleic acids, e.g., nucleic acid molecules comprising three or more nucleotides are linear molecules, in which adjacent nucleotides are linked to each other via a phosphodiester linkage. In some embodiments, “nucleic acid” refers to individual nucleic acid residues (e.g. nucleotides and/or nucleosides). In some embodiments, “nucleic acid” refers to an oligonucleotide chain comprising three or more individual nucleotide residues. As used herein, the terms “oligonucleotide” and “polynucleotide” can be used interchangeably to refer to a polymer of nucleotides (e.g., a string of at least three nucleotides). In some embodiments, “nucleic acid” encompasses RNA as well as single and/or double-stranded DNA. Nucleic acids may be naturally occurring, for example, in the context of a genome, a transcript, an mRNA, tRNA, rRNA, siRNA, snRNA, a plasmid, cosmid, chromosome, chromatid, or other naturally occurring nucleic acid molecule. On the other hand, a nucleic acid molecule may be a non-naturally occurring molecule, e.g., a recombinant DNA or RNA, an artificial chromosome, an engineered genome, or fragment thereof, or a synthetic DNA, RNA, DNA/RNA hybrid, or including non-naturally occurring nucleotides or nucleosides. Furthermore, the terms “nucleic acid,” “DNA,” “RNA,” and/or similar terms include nucleic acid analogs, e.g., analogs having other than a phosphodiester backbone. Nucleic acids can be purified from natural sources, produced using recombinant expression systems and optionally purified, chemically synthesized, etc. Where appropriate, e.g., in the case of chemically synthesized molecules, nucleic acids can comprise nucleoside analogs such as analogs having chemically modified bases or sugars, and backbone modifications. A nucleic acid sequence is presented in the 5' to 3' direction unless otherwise indicated. In some embodiments, a nucleic acid is or comprises natural nucleosides (e.g. adenosine, thymidine, guanosine, cytidine, uridine, deoxyadenosine, deoxythymidine, deoxyguanosine, and deoxycytidine); nucleoside analogs (e.g., 2-aminoadenosine, 2-thiothymidine, inosine, pyrrolo-pyrimidine, 3-methyl adenosine, 5-methylcytidine, 2-aminoadenosine, C5-bromouridine, C5-fluorouridine, C5-iodouridine, C5-propynyl-uridine, C5-propynyl-cytidine, C5-methylcytidine, 2-aminoadenosine, 7-deazaadenosine, 7-deazaguanosine, 8-oxoadenosine, 8-oxoguanosine, O(6)-methylguanine, and 2-thiocytidine); chemically modified bases; biologically modified bases (e.g., methylated bases); intercalated bases; modified sugars (e.g., 2'-fluororibose, ribose, 2'-deoxyribose, arabinose, and hexose); and/or modified phosphate groups (e.g., phosphorothioates and 5'-N-phosphoramidite linkages).

The terms “protein,” “peptide,” and “polypeptide” are used interchangeably herein, and refer to a polymer of amino acid residues linked together by peptide (amide) bonds. The

terms refer to a protein, peptide, or polypeptide of any size, structure, or function. Typically, a protein, peptide, or polypeptide will be at least three amino acids long. A protein, peptide, or polypeptide may refer to an individual protein or a collection of proteins. One or more of the amino acids in a protein, peptide, or polypeptide may be modified, for example, by the addition of a chemical entity such as a carbohydrate group, a hydroxyl group, a phosphate group, a farnesyl group, an isofarnesyl group, a fatty acid group, a linker for conjugation, functionalization, or other modification, etc. A protein, peptide, or polypeptide may also be a single molecule or may be a multi-molecular complex. A protein, peptide, or polypeptide may be just a fragment of a naturally occurring protein or peptide. A protein, peptide, or polypeptide may be naturally occurring, recombinant, or synthetic, or any combination thereof. The term “fusion protein” as used herein refers to a hybrid polypeptide which comprises protein domains from at least two different proteins. One protein may be located at the amino-terminal (N-terminal) portion of the fusion protein or at the carboxy-terminal (C-terminal) protein thus forming an “amino-terminal fusion protein” or a “carboxy-terminal fusion protein,” respectively. A protein may comprise different domains, for example, a nucleic acid binding domain (e.g., the gRNA binding domain of Cas9 that directs the binding of the protein to a target site) and a nucleic acid cleavage domain or a catalytic domain of a recombinase. In some embodiments, a protein comprises a proteinaceous part, e.g., an amino acid sequence constituting a nucleic acid binding domain, and an organic compound, e.g., a compound that can act as a nucleic acid cleavage agent. In some embodiments, a protein is in a complex with, or is in association with, a nucleic acid, e.g., RNA. Any of the proteins provided herein may be produced by any method known in the art. For example, the proteins provided herein may be produced via recombinant protein expression and purification, which is especially suited for fusion proteins comprising a peptide linker. Methods for recombinant protein expression and purification are well known, and include those described by Green and Sambrook, *Molecular Cloning: A Laboratory Manual* (4th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2012)), the entire contents of which are incorporated herein by reference.

The term “RNA-programmable nuclease,” and “RNA-guided nuclease” are used interchangeably herein and refer to a nuclease that forms a complex with (e.g., binds or associates with) one or more RNA that is not a target for cleavage. In some embodiments, an RNA-programmable nuclease, when in a complex with an RNA, may be referred to as a nuclease:RNA complex. Typically, the bound RNA(s) is referred to as a guide RNA (gRNA). gRNAs can exist as a complex of two or more RNAs, or as a single RNA molecule. gRNAs that exist as a single RNA molecule may be referred to as single-guide RNAs (sgRNAs), though “gRNA” is used interchangeably to refer to guide RNAs that exist as either single molecules or as a complex of two or more molecules. Typically, gRNAs that exist as single RNA species comprise two domains: (1) a domain that shares homology to a target nucleic acid (e.g., and directs binding of a Cas9 complex to the target); and (2) a domain that binds a Cas9 protein. In some embodiments, domain (2) corresponds to a sequence known as a tracrRNA, and comprises a stem-loop structure. For example, in some embodiments, domain (2) is homologous to a tracrRNA as depicted in FIG. 1E of Jinek et al., *Science* 337:816-821 (2012), the entire contents of which is incorporated herein by reference. Other examples of gRNAs (e.g., those includ-

ing domain 2) can be found in U.S. Provisional Patent Application, U.S. Ser. No. 61/874,682, filed Sep. 6, 2013, entitled “Switchable Cas9 Nucleases And Uses Thereof,” and U.S. Provisional Patent Application, U.S. Ser. No. 61/874,746, filed Sep. 6, 2013, entitled “Delivery System For Functional Nucleases,” the entire contents of each are hereby incorporated by reference in their entirety. In some embodiments, a gRNA comprises two or more of domains (1) and (2), and may be referred to as an “extended gRNA.” For example, an extended gRNA will, e.g., bind two or more Cas9 proteins and bind a target nucleic acid at two or more distinct regions, as described herein. The gRNA comprises a nucleotide sequence that complements a target site, which mediates binding of the nuclease/RNA complex to said target site, providing the sequence specificity of the nuclease:RNA complex. In some embodiments, the RNA-programmable nuclease is the (CRISPR-associated system) Cas9 endonuclease, for example Cas9 (Csn1) from *Streptococcus pyogenes* (see, e.g., “Complete genome sequence of an M1 strain of *Streptococcus pyogenes*.” Ferretti J. J., McShan W. M., Ajdic D. J., Savic D. J., Savic G., Lyon K., Primeaux C., Sezate S., Suvorov A. N., Kenton S., Lai H. S., Lin S. P., Qian Y., Jia H. G., Najjar F. Z., Ren Q., Zhu H., Song L., White J., Yuan X., Clifton S. W., Roe B. A., McLaughlin R. E., Proc. Natl. Acad. Sci. U.S.A. 98:4658-4663(2001); “CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III.” Deltcheva E., Chylinski K., Sharma C. M., Gonzales K., Chao Y., Pirzada Z. A., Eckert M. R., Vogel J., Charpentier E., Nature 471:602-607(2011); and “A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity.” Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. Science 337:816-821(2012), the entire contents of each of which are incorporated herein by reference.

Because RNA-programmable nucleases (e.g., Cas9) use RNA:DNA hybridization to target DNA cleavage sites, these proteins are able to be targeted, in principle, to any sequence specified by the guide RNA. Methods of using RNA-programmable nucleases, such as Cas9, for site-specific cleavage (e.g., to modify a genome) are known in the art (see e.g., Cong, L. et al. Multiplex genome engineering using CRISPR/Cas systems. *Science* 339, 819-823 (2013); Mali, P. et al. RNA-guided human genome engineering via Cas9. *Science* 339, 823-826 (2013); Hwang, W. Y. et al. Efficient genome editing in zebrafish using a CRISPR-Cas system. *Nature biotechnology* 31, 227-229 (2013); Jinek, M. et al. RNA-programmed genome editing in human cells. *eLife* 2, e00471 (2013); Dicarlo, J. E. et al. Genome engineering in *Saccharomyces cerevisiae* using CRISPR-Cas systems. *Nucleic acids research* (2013); Jiang, W. et al. RNA-guided editing of bacterial genomes using CRISPR-Cas systems. *Nature biotechnology* 31, 233-239 (2013); the entire contents of each of which are incorporated herein by reference).

The term “subject,” as used herein, refers to an individual organism, for example, an individual mammal. In some embodiments, the subject is a human. In some embodiments, the subject is a non-human mammal. In some embodiments, the subject is a non-human primate. In some embodiments, the subject is a rodent. In some embodiments, the subject is a sheep, a goat, a cattle, a cat, or a dog. In some embodiments, the subject is a vertebrate, an amphibian, a reptile, a fish, an insect, a fly, or a nematode. In some embodiments, the subject is a research animal. In some embodiments, the subject is genetically engineered, e.g., a genetically engineered non-human subject. The subject may be of either sex and at any stage of development.

The terms “treatment,” “treat,” and “treating,” refer to a clinical intervention aimed to reverse, alleviate, delay the onset of, or inhibit the progress of a disease or disorder, or one or more symptoms thereof, as described herein. As used herein, the terms “treatment,” “treat,” and “treating” refer to a clinical intervention aimed to reverse, alleviate, delay the onset of, or inhibit the progress of a disease or disorder, or one or more symptoms thereof, as described herein. In some embodiments, treatment may be administered after one or more symptoms have developed and/or after a disease has been diagnosed. In other embodiments, treatment may be administered in the absence of symptoms, e.g., to prevent or delay onset of a symptom or inhibit onset or progression of a disease. For example, treatment may be administered to a susceptible individual prior to the onset of symptoms (e.g., in light of a history of symptoms and/or in light of genetic or other susceptibility factors). Treatment may also be continued after symptoms have resolved, for example, to prevent or delay their recurrence.

Use of ordinal terms such as “first,” “second,” “third,” etc. in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

Detailed Description of Certain Embodiments

Major research efforts focus on improving efficiency and specificity of genome editing systems, such as, CRISPR/Cas9, other Cas-based, TALEN-based, and Zinc Finger-based genome editing systems. For instance, with regard to CRISPR/Cas9 systems, efficiency may be improved by predicting optimal Cas9 guide RNA (gRNA) sequences, while specificity may be improved by modeling factors leading to off-target cutting, and by manipulating Cas9 enzymes. Variant Cas9 enzymes and fusion proteins may be developed to alter the protospacer adjacent motif (PAM) sequences acted on by Cas9, and to produce base-editing Cas9 constructs with high efficiency and specificity. For example, Cpf1 (also known as Cas12a) and other alternatives may be used in CRISPR genome editing in addition to, or instead of, Cas9.

The inventors have recognized and appreciated that less attention has been devoted to understanding and modulating repair outcomes. In that respect, nucleotide insertions and/or deletions resulting from template-free repair mechanisms (e.g., NHEJ, MMEJ, etc. and excluding homology-based repair (HDR)) are commonly thought to be random and therefore only suitable for gene knock-out applications. For gene knock-in or gain-of-function applications, a template-based repair mechanism such as HDR is typically used.

CRISPR/Cas with HDR allows arbitrarily designed DNA sequences to be incorporated at precise genomic locations. However, this technique suffers from low efficiency—HDR occurs rarely in typical biological conditions (e.g., around 10% frequency), because cells only permit HDR to occur after sister chromatids are synthesized in S phase but before M phase when mitosis splits the sister chromatids into

daughter cells. For many cell-types, the fraction of time spent in S-G2-M phases of a cell cycle is low. In sum, while outcomes are predictable when HDR does occur, HDR occurs infrequently, and therefore a desired DNA sequence will be incorporated into only a small percentage of cells. In addition, in post-mitotic cell-types of interest such as neurons, the HDR repair pathway is no longer used, further limiting HDR's utility for genetic engineering.

Some research has been done to improve efficiency of HDR, for example, through improved homology templates and small molecule modulation. Despite these efforts, template-based repair efficiency remains low, and proposed CRISPR/Cas gene knock-in or gain of function applications have thus far been limited to ex vivo applications where screening may be performed for cells with a desired repair genotype.

Unlike HDR, NHEJ is capable of occurring during any phase of a cell cycle and in post-mitotic cells. However, NHEJ, as discussed above, has been perceived as a random process that produces a large variety of repair genotypes with insertions and/or deletions, and has been used mainly to knock out genes. In short, NHEJ is efficient but unpredictable.

Recent work suggests that outcomes of some template-free repair mechanisms are actually non-random. For instance, it has been observed that MMEJ is involved in repair outcomes. Furthermore, repair outcomes have been analyzed to predict gRNAs that are more likely to produce frameshifts. However, there is still a need for accurate prediction of genotypic outcomes of CRISPR/Cas cutting and ensuing cellular DNA repair.

The present inventors have unexpectedly found through computational analyses that template-free DNA/genome editing systems, e.g., CRISPR/Cas9, Cas-based, Cpf1-based, or other DSB (double-strand break)-based genome editing systems, produce a predictable set of repair genotypes thereby enabling the use of such editing systems for applications involving or requiring precise manipulation of DNA, e.g., the correction of a disease-causing genetic mutation or modifying a wildtype sequence to confer a genetic advantage. This finding is contrary to the accepted view that DNA double-strand break repair (i.e., template-free, non-homology-dependent repair) following cleavage by genome editing systems produces stochastic and heterogeneous repair products and are therefore impractical for applications beyond gene disruption. Thus, the specification describes and discloses in various aspects and embodiments computational-based methods and systems for practically harnessing the innate efficiencies of template-free DNA repair systems for carrying out precise DNA and/or genomic editing without the reliance upon homology-based repair.

In accordance with some embodiments, techniques are provided for predicting genotypes of CRISPR/Cas editing outcomes. For instance, a high-throughput approach may be used for monitoring CRISPR/Cas cutting outcomes, and/or a computer-implemented method may be used to predict genotypic repair outcomes for NHEJ and/or MMEJ. The inventors have recognized and appreciated that accurate prediction of repair genotypes may allow development of CRISPR/Cas gene knock-in or gain-of-function applications based on one or more template-free repair mechanisms. This approach may simplify a genome editing process, by reducing or eliminating a need to introduce exogenous DNA into a cell as a template.

Additionally, or alternatively, using one or more template-free repair mechanisms for gene knock-in may provide improved efficiency. For instance, the inventors have rec-

ognized and appreciated that NHEJ and MMEJ may account for a large portion of CRISPR/Cas repair products. While template-free repair mechanisms may not always produce desired repair genotypes with sufficiently high frequencies, one or more desired repair genotypes may occur with sufficiently high frequencies in some specific local sequence contexts. For such a local sequence context, template-free repair mechanisms may outperform HDR with respect to simplicity and efficiency.

In some embodiments, one or more of the techniques provided herein may be used to predict, for a given local sequence context, template-free repair genotypes and frequencies of occurrence thereof, which may facilitate designs of gene knock-in or gain-of-function applications. For example, the inventors have recognized and appreciated that some disease-causing alleles, when cut at a selected location by CRISPR/Cas, may exhibit one or just a few repair outcomes that occur at a high frequency and transform the disease-causing allele into one or more healthy alleles. Disease-causing alleles may occur in genomic sequences that code for proteins or regulatory RNAs, or genomic sequences that regulate transcription or other genomic functions.

In some embodiments, one or more of the techniques provided herein may be used to predict, for a given local sequence context, template-free repair genotypes and frequencies of occurrence thereof, which may be used to select desirable one or more guide RNAs when HDR is employed to edit DNA. Since HDR does not occur 100% of the time, the template-free repair genotypes predicted by this method will be a natural byproduct of sites where HDR failed to occur. The one or more techniques provided herein allow these failed HDR byproducts to be predicted and one or more guide RNAs chosen that will produce the most desirable byproducts for HDR failures. For example, a disease-causing allele may be targeted for HDR repair, but if HDR does not occur at a specific site the template-free repair products can be chosen to transform a disease-causing allele into one or more healthy alleles or to not have deleterious effects. Deleterious effects could result from template-free repair that changed a weakly functional allele into a non-functional allele or into a dominant allele that negatively impacted health. In some embodiments, guide RNA selection consists of considering all guide RNAs that are compatible with HDR repair of a disease-causing allele, and for each guide RNA using one or more of the techniques provided herein to predict its template-free repair genotypes. One or more guide RNAs are then selected for use with the HDR template that have the template-free repair genotypes that are most advantageous for health. Alternatively in some embodiments, one or more guide RNAs are then selected for use with the HDR template that have the template-free repair genotypes that are most likely to disrupt gene function.

It should be appreciated that the techniques disclosed herein may be implemented in any of numerous ways, as the disclosed techniques are not limited to any particular manner of implementation. Examples of details of implementation are provided solely for illustrative purposes. For instance, while examples are given where CRISPR/Cas9 is used to perform genome editing, it should be appreciated that aspects of the present application are not so limited. In some embodiments, another genome editing technique, such as CRISPR/Cpf1, may be used. Furthermore, the disclosed techniques may be used individually or in any suitable combination, as aspects of the present disclosure are not limited to the use of any particular technique or combination of techniques.

25

FIG. 1 shows an illustrative DNA segment **100**, in accordance with some embodiments. For instance, the DNA segment **100** may be exon 43 of a dystrophin gene. About 4% of Duchenne's muscular dystrophy cases are caused by mutations in this exon. Therapeutic solutions showing success in clinical trials use antisense oligonucleotides to cause this exon to be skipped during translation, thereby restoring normal dystrophin function.

The inventors have recognized and appreciated that another therapeutic approach may be possible, using genome editing to make permanent changes to dystrophin exon 43. For instance, in some embodiments, CRISPR/Cas9 (or another suitable technique for cutting a DNA sequence, such as CRISPR/Cpf1) may be used to disrupt a donor splice site motif of dystrophin exon 43, and one or more template-free repair mechanisms may restore normal dystrophin function.

In one aspect, the specification discloses a computational model.

In certain embodiments, the computational model can predict and/or compute an optimized or preferred cut site for a DSB-based genome editing system for introducing a genetic change into a nucleotide sequence. In preferred embodiments, the repair does not require homology-based repair mechanisms.

In certain other embodiments, the computational model can predict and/or compute an optimized or preferred cut site for a Cas-based genome editing system for introducing a genetic change into a nucleotide sequence. In preferred embodiments, the repair does not require homology-based repair mechanisms.

In still other embodiments, the computation model provides for the selection of a optimized or preferred guide RNA for use with a Cas-based genome editing system for introducing a genetic change in a genome. In preferred embodiments, the repair does not require homology-based repair mechanisms.

In various embodiments, the computational model is a neural network model having one or more hidden layers.

In other embodiments, the computational model is a deep learning computational model.

In various embodiments, that the DSB-based genome editing system (e.g., a Cas-based genome editing system) edits the genome without relying on homology-based repair.

In various embodiments, that computational model is trained with experimental data to predict the probability of distribution of indel lengths for any given nucleotide sequence and cut site. In other embodiments, computational model is trained with experimental data to predict the probability of distribution of genotype frequencies for any given nucleotide sequence and cut site.

In embodiments, the computational model comprises one or more training modules for evaluating experimental data.

In an embodiment, the computational model comprises: a first training module (**305**) for computing a microhomology score matrix (**305**); a second training module (**310**) for computing a microhomology independent score matrix; and a third training module (**315**) for computing a probability distribution over 1-bp insertions, wherein once trained with experimental data the computational model computes a probability distribution over indel genotypes and a probability distribution over indel lengths for any given input nucleotide sequence and cut site.

In certain embodiments, the computational model predicts genomic repair outcomes for any given input nucleotide sequence (i.e., context sequence) and cut site.

26

In certain embodiments, the genomic repair outcomes comprise microhomology deletions, microhomology-less deletions, and 1-bp insertions.

In various embodiments, the one or more modules each comprising one more input features selected from the group consisting of: a target site nucleotide sequence; a cut site; a PAM-sequence; microhomology lengths relative at a cut site, % GC content at a cut site; and microhomology deletion lengths at a cut site.

In certain embodiments, the nucleotide sequence analyzed by the computational model is between about 25-100 nucleotides, 50-200 nucleotides, 100-400 nucleotides, 200-800 nucleotides, 400-1600 nucleotides, 800-3200 nucleotides, and 1600-6400 nucleotide, or more. In various embodiments, the computation model concerns predicting genetic repair outcomes at double-strand breaks cleaves induced by any DSB-based genomic editing system (e.g., CRISPR/Cas9, Cas-base, Cfr1-based, or others). FIG. 1 depicts the anatomy of a double strand break. In the example shown in FIG. 1, the DNA segment **100** includes a top strand **105A** and a bottom strand **105B**. These two strands are complementary and therefore encode the same information. In some embodiments, CRISPR/Cas9 may be used to create a double strand cut at a selected donor splice site motif, which may be a specific sequence of 6-10 nucleotides. In the example of FIG. 1, an NGG PAM may be used, as underlined and shown at **115**, so that a cut site **110** would occur within the selected donor splice site motif. Any suitable algorithm may be used to detect presence or absence of the splice site motif in repair products, thereby verifying if the splice site motif has been successfully eliminated.

FIGS. 2A-D show an illustrative matching of 3' ends of top and bottom strands of a DNA segment at a cut site and an illustrative repair product, in accordance with some embodiments. For instance, the strands may be the illustrative top strand **105A** and the illustrative bottom strand **105B** of FIG. 1, and the cut site may be the illustrative cut site **110** of FIG. 1. (To avoid clutter, the surrounding sequence context is omitted in FIGS. 2B-D.)

In some embodiments, a segment of double-stranded DNA may be represented such that the top strand runs 5' on the left to 3' on the right. Given a cut in this double stranded DNA, nucleotides and their complementary base-paired nucleotides that lie between the 5' end of the top strand and the cut site may be said to be located at the 5' side of the cut site. Likewise, nucleotides and their complementary base-paired nucleotides that lie between the cut site and the 3' end of the top strand may be said to be located at the 3' side of the cut site.

In the example shown in FIG. 2A, a deletion length of 5 base pairs is considered, for example, as a result of 5' end resection, where the top strand **105A** has an overhang **200A** of length 5 at the 5' side of the cut site **110**, and the bottom strand **105B** has an overhang **200B** of length 5 at the 3' side of the cut site **110**. As shown in FIG. 2B, there is no match between the overhangs **200A** and **200B** in the first three bases, but there is a match in each of the last two bases. Thus, in this example, a microhomology **205** is present, with a 2 base pair match.

FIG. 2C shows an illustrative result of flap removal, where the three mismatched bases in the overhang **200B** are removed. For instance, in some embodiments, given a microhomology, some or all nucleotides on the 3' side of the microhomology on the top strand, and/or some or all nucleotides on the 3' side of the microhomology on the bottom strand, may be resected. Pictorially, with the top strand running 5' to 3', nucleotides to the right of the microhomol-

27

ogy on the top strand may be resected, and nucleotides to the left of the microhomology on the bottom strand may be resected.

FIG. 2D shows an illustrative repair product resulting from polymerase fill-in and ligation, where three matching bases are added to the overhang **200B**.

FIG. 3A shows an illustrative machine learning model **300**, in accordance with some embodiments. The machine learning model **300** may be trained using experimental data to compute, given an input DNA sequence *seq* and a cut site location, a probability distribution over any suitable set of deletion and/or insertion genotypes, and/or a probability distribution over any suitable set of deletion and/or insertion lengths. For instance, in some embodiments, 1 base pair insertions and 1-60 base pair deletions may be considered. (These repair outcomes may also be referred to herein as +1 to -60 indels.) The inventors have observed empirically that indels outside of this range occur infrequently. However, it should be appreciated that aspects of the present disclosure are not limited to any particular set of repair outcomes. In some embodiments, only insertions (e.g., 1-2 base pair insertions), or only deletions (e.g., 1-28 base pair deletions), may be considered, for example, based on availability of training data.

The inventors have recognized and appreciated that accurate predictions of repair outcomes may be facilitated by separating the repair outcomes into three classes: microhomology (MH) deletions, microhomology-less (MH-less) deletions, and insertions. The inventors have further recognized and appreciated that different machine learning techniques may be applied to the different classes of repair outcomes. For instance, in the example of FIG. 3, the machine learning model **300** includes three modules: the MH deletion module **305**, the MH-less deletion module **310**, and the insertion module **315**. As discussed below, these modules may compute scores for various indel genotypes and/or indel lengths, which may in turn be used to compute a probability distribution over indel genotypes and/or a probability distribution over indel lengths. In some embodiments, one or more modules (e.g., the MH deletion module **305** and the MH-less deletion module **310**) may be trained jointly. In some embodiments, a module may be dependent upon one or more other modules. For instance, as discussed below, an input feature used in the insertion module **315** may be derived based on outputs of the MH deletion module **305** and/or the MH-less deletion module **310**.

In some embodiments, MH deletions may include deletions that are derivable analytically by simulating MMEJ. For instance, all microhomologies may be identified for deletion lengths of interest (e.g., deletion lengths 1-60). A genotypic outcome may be derived for each such microhomology by simulating polymerase fill-in, for example, as discussed in connection with FIGS. 2A-2D. (The inventors have recognized and appreciated that there is a one-to-one correspondence between the microhomologies and the genotypic outcomes.) A deletion that is derivable in this manner may be classified as a MH deletion, whereas a deletion that is not derivable in this manner may be classified as a MH-less deletion.

Techniques for identifying microhomologies for a given a deletion length *L* of interest (e.g., each deletion length between 1 and 60) are described below. However, it should be appreciated that aspects of the present disclosure are not limited to the use of any particular technique for identifying microhomologies.

In some embodiments, an input DNA sequence *seq* may be represented as a vector with integer indices, where each

28

element of the vector is a nucleotide from the set, {A, C, G, T}, and the cut site is between *seq*[-1] and *seq*[0], and *seq* is oriented 5' on the left to 3' on the right. A subsequence *seq*[*i*:*j*], *i* < *j*, may be a vector of length *j*-*i*, including elements *seq*[*i*] to *seq*[*j*-1]. For each deletion length *L* of interest (e.g., *L* between 1 and 60), *left*[*L*] may be used to denote *seq*[-*L*: 0], and *right*[*L*] may be used to denote *seq*[0, *L*]. Thus, with reference to the example shown in FIGS. 1, 2A, *left*[5] may be ACAAG, and *right*[5] may be GGTAG. Because the top strand **105A** and the bottom strand **105B** are complementary, a microhomology **205** may be identified by looking for exact matches between *left*[5] and *right*[5] (which may be equivalent to complementary matches between the overhang **200A** and the overhang **200B**). For instance, a match vector may be constructed for each deletion length *L* of interest (e.g., *L* between 1 and 60) as follows: *match*[*L*][*i*]='1' if *left*[*L*][*i*]=*right*[*L*][*i*], otherwise *match*[*L*][*i*]='0'. Such matching between *left*[5] and *right*[5] is illustrated below.

```
ACAAG
...||
GGTAG
```

In some embodiments, a microhomology may be identified by looking for *match*[*L*][*i*:*j*] such that *match*[*L*][*k*]='1' for all *i* ≤ *k* ≤ *j* and *match*[*L*][*i*] != '1' and *match*[*L*][*j*] != '1'. For instance, with reference to the example shown in FIG. 1, there may be no microhomology for deletion length 3, no microhomology for deletion length 4, one microhomology for deletion length 5, three microhomologies for deletion length 6, etc., as illustrated below.

```
AAG
...
GGT
...
CAAG
....
GGTA
....
ACAAG
...||
GGTAG
GACAAG
|..|..|
GGTAGG
```

In some embodiments, microhomologies identified for a suitable set of deletion lengths (e.g., 1-60) may be enumerated using indices *n*=1 . . . *N*, where *N* is the number of identified microhomologies. For each *n*, let *G*[*n*] denote the genotypic outcome corresponding to the microhomology *n*, let *ML* [*n*] denote the microhomology length of the microhomology *n*, let *C*[*n*] denote the GC fraction of the microhomology *n*, and let *DL* [*n*] denote the deletion length of the microhomology *n*.

Although examples of representations of DNA sequences and subsequences are discussed herein, it should be appreciated that aspects of the present disclosure are not limited to the use of any particular representation.

FIG. 3B shows an illustrative process **350** for building one or more machine learning models for predicting frequencies of deletion genotypes and/or deletion lengths, in accordance with some embodiments. For instance, the process **350** may be used to build the illustrative MH deletion module **305** and/or the illustrative MH-less deletion module **310** in the example of FIG. 3A. These modules may be used to compute, given an input DNA sequence *seq* and a cut site

29

location, a probability distribution over any suitable set of deletion genotypes and/or a probability distribution over any suitable set of deletion lengths.

In some embodiments, a probability distribution over deletion lengths from 1-60 may be computed. However, it should be appreciated that aspects of the present disclosure are not limited to any particular set of deletion lengths. In some embodiments, an upper limit of deletion lengths may be determined based on availability of training data and/or any other one or more suitable considerations.

Referring to FIG. 3B, act 355 of the process 350 may include, for each deletion length L of interest (e.g., each deletion length between 1-60), aligning subsequences of length L on the 5' and 3' sides of a cut site in an input DNA sequence to identify one or more microhomologies, as discussed in connection with FIG. 3A. This may be performed for an input DNA sequence and a cut site for which repair genotype data from an CRISPR/Cas9 experiment is available.

At act 360, one or more microhomologies identified at act 355 may be featurized. Any suitable one or more features may be used, as aspects of the present disclosure are not so limited. As one example, the inventors have recognized and appreciated that energetic stability of a microhomology may increase proportionately with a length of the microhomology. Accordingly, in some embodiments, a microhomology length j-i may be used as a feature for a microhomology match[L][i:j].

As another example, the inventors have recognized and appreciated that thermodynamic stability of a microhomology may depend on specific base pairings, and that G-C pairings have three hydrogen bonds and therefore have higher thermodynamic stability than A-T pairings, which have two hydrogen bonds. Accordingly, in some embodiments, a GC fraction, as shown below, may be used as a feature for a microhomology match[L][i:j], where indicator (boolean) equals 1 if boolean is true, and 0 otherwise.

$$\frac{\sum_{k=i}^{j-1} \text{indicator}(\text{top}[L][k] = 'G' \text{ or } 'C')}{j-i}$$

In some embodiments, a length N vector may be constructed for each feature (e.g., microhomology length, GC fraction, etc.), where N is the number of microhomologies identified at act 355 for a set of deletion lengths of interest (e.g., 1-60), as discussed in connection with FIG. 3A. As discussed above, the inventors have recognized and appreciated that there is a one-to-one correspondence between microhomologies and genotypic outcomes that are classified as MH deletions. Therefore, feature vectors for microhomologies may be viewed as feature vectors for MH deletions.

In some embodiments, acts 355 and 360 may be repeated for different input DNA sequences and/or cut sites for which repair genotype data from CRISPR/Cas9 experiments is available.

It should be appreciated that aspects of the present disclosure are not limited to any particular featurization technique. For instance, in some embodiments, two features may be used, such as microhomology length and GC fraction. However, that is not required, as in some embodiments one feature may be used (e.g., microhomology length, GC fraction, or some other suitable feature), or more than two features may be used (e.g., three, four, five, etc.). Examples of features that may be used for a microhomology match

30

[L][i:j] within a deletion of length L include, but are not limited to, a position of the microhomology within the deletion (e.g., as represented by

$$\frac{\sum_{k=i}^{j-1} k}{L * (j-i)}$$

and a ratio between a length of the microhomology (i.e., j-i) and the L*(j-i) deletion length L. As another example, the inventors have recognized and appreciated that deoxyribonuclease (DNase) hypersensitivity may be used to classify genomic sequences into open or closed chromatin, which may impact DNA repair outcomes. Accordingly, in some embodiments, open vs. closed chromatin may be used as a feature. Any one or more of these features, and/or other features, may be used in addition to, or instead of, microhomology length and GC fraction. Furthermore, in some embodiments, explicit featurization may be reduced or eliminated by automatically learning data representations (e.g., using one or more deep learning techniques). Returning to FIG. 3B, one or more machine learning models may be trained at act 365 to compute one or more target probability distributions. For instance, a neural network model may be built for the illustrative MH deletion module 305 in the example of FIG. 3A. This model may take as input a length N vector for each of one or more features, as constructed at act 360, and output a length N vector of MH scores, where N is the number of microhomologies identified at act 355 for a set of deletion lengths of interest (e.g., 1-60). Additionally, or alternatively, a neural network model may be built for the illustrative MH-less deletion module 310 in the example of FIG. 3A. This model may take as input a vector for each of one or more features, and output a vector of MH-less scores. Both of the input vector and the output vector may be indexed by the set of deletion lengths of interest (e.g., 1-60). These neural network models may then be trained jointly using repair genotype data collected from CRISPR/Cas9 experiments.

FIG. 4A shows an illustrative neural network 400A for computing MH scores, in accordance with some embodiments. For instance, the neural network 400A may be used in the illustrative MH deletion module 305 in the example of FIG. 3A, and may be trained at act 365 of the illustrative process 350 shown in FIG. 3B.

In some embodiments, the neural network 400A may have one input node for each microhomology feature being used. For instance, in the example shown in FIG. 4A, there are two input nodes, which are associated with microhomology length and GC fraction, respectively. Each input node may receive a length N vector, where N is the number of microhomologies identified for a set of deletion lengths of interest (e.g., 1-60), for example, as discussed in connection with act 355 in the example of FIG. 3B.

In some embodiments, the neural network 400A may include one or more hidden layers, each having one or more nodes. In the example shown in FIG. 4A, there are two hidden layers, each having 16 nodes. However, it should be appreciated that aspects of the present disclosure are not limited to the use of any particular number of hidden layers or any particular number of nodes in a hidden layer. Furthermore, different hidden layers may have different numbers of nodes.

In some embodiments, the neural network 400A may be fully connected. (To avoid clutter, the connections are not illustrated in FIG. 4A.) However, that is not required. For

31

instance, in some embodiments, a dropout technique may be used, where a parameter p may be selected, and during training each node's value is independently set to 0 with probability p . This may result in a neural network that is not fully connected.

In some embodiments, a leaky rectified linear unit (ReLU) nonlinearity sigma may be used in the neural network 400A. For instance, at hidden layer h and node i , an activation function may be provided as follows:

$$\text{unit}[h][i] = \text{sigma}(w[h][i] * \text{unit}[h-1] + b[h][i]), \text{ where} \\ \text{sigma}(x) = \max(0, x) + 0.001 * \min(0, x).$$

Thus, the neural network 400A may be parameterized by $w[h]$ and $b[h]$ for each hidden layer h . In some embodiments, these parameters may be initialized randomly, for example, from a spherical Gaussian distribution with some suitable center (e.g., 0) and some suitable variance (e.g., 0.1). These parameters may then be trained using repair genotype data collected from CRISPR/Cas9 experiments, for instance, as discussed below.

In some embodiments, the neural network 400A may have one output node, producing a length N vector Ψ_{MH} of scores, where N is the number of microhomologies identified for the set of deletion lengths of interest (e.g., 1-60). Thus, there may be one score for each identified microhomology.

In some embodiments, the neural network 400A may operate independently for each microhomology, taking as input the length of that microhomology (from the first input node) and the GC fraction of that microhomology (from the second input node), transforming those two values into 16 values (at the first hidden layer), then transforming those 16 values into 16 other values (at the second hidden layer), and finally outputting a single value (at the output node). In such an embodiment, parameters for the first hidden layer, $w[1][i]$ and $b[1][i]$, are vectors of length 2 for each node i from 1 to 16, whereas parameters for the second hidden layer, $w[2][i]$ and $b[2][i]$, are vectors of length 16 for each node i from 1 to 16, and parameters for the output layer, $w[3][1]$ and $b[3][1]$, are also vectors of length 16.

In some embodiments, the vector Ψ_{MH} of raw scores may be converted into a vector ϕ_{MH} of MH scores. The inventors have recognized and appreciated (e.g., from experimental data) that the strength of a microhomology decreases exponentially with deletion length. Accordingly, in some embodiments, an exponential linear model may be used to convert the raw scores into the MH scores. For instance, the following formula may be used:

$$\phi_{MH}[n] = \exp(\Psi_{MH}[n] - DL[n] * 0.25),$$

where n is an index for a microhomology (and thus a number between 1 and N), and $DL[n]$ is the deletion length of the microhomology n .

In some embodiments, 0.25 may be a hyperparameter value chosen to improve training speed by appropriate scaling. However, it should be appreciated that aspects of the present disclosure are not limited to the use of any particular hyperparameter value for exponential conversion, or any conversion at all. In some embodiments, the vector P of raw scores may be used directly as MH scores.

FIG. 4B shows an illustrative neural network 400B for computing MH-less scores, in accordance with some embodiments. For instance, the neural network 400B may be used in the illustrative MH-less deletion module 310 in the example of FIG. 3A, and may be trained at act 365 of the illustrative process 350 shown in FIG. 3B.

In some embodiments, deletion length may be modeled explicitly as an input to the neural network 400B. Thus, in

32

an example where the set of deletion lengths of interest is 1-60, an input node of the neural network 400B may receive a deletion length vector, $[1, 2, \dots, 60]$.

In some embodiments, the neural network 400B may include one or more hidden layers, each having one or more nodes. In the example shown in FIG. 4B, the neural network 400B has two hidden layers that are similarly constructed as the illustrative neural network 400A in the example of FIG. 4A. However, it should be appreciated that aspects of the present disclosure are not limited to the use of a similar construction between the neural network 400A and the neural network 400B.

In some embodiments, the neural network 400B may have an output node producing a vector $\Psi_{MH-less}$ of scores. There may be one score for each deletion length L of interest. Thus, in an example where the set of deletion lengths of interest is 1-60, the length of the vector $\Psi_{MH-less}$ may be 60.

In some embodiments, an exponential linear model may be used to convert the vector $\Psi_{MH-less}$ into a vector $\phi_{MH-less}$ of MH-less scores. For instance, the following formula may be used:

$$\phi_{MH-less}[L] = \exp(\Psi_{MH-less}[L] - L * 0.25),$$

where L is a deletion length of interest. However, it should be appreciated that aspects of the present disclosure are not limited to the use of any particular hyperparameter value for exponential conversion, or any conversion at all.

FIG. 4C shows an illustrative process 400C for training two neural networks jointly, in accordance with some embodiments. For instance, the process 400C may be used to jointly train the illustrative neural networks 400A and 400B of FIGS. 4A-4B.

In some embodiments, the MH score vector ϕ_{MH} and the MH-less score vector may be used to predict a probability distribution over MH deletion genotypes and/or a probability distribution over deletion lengths. For instance, given a microhomology n , a frequency may be predicted for the corresponding MH deletion genotype, out of all MH deletion genotypes. As discussed above, the inventors have recognized and appreciated that there is a one-to-one correspondence between microhomologies and genotypic outcomes that are classified as MH deletions. Thus, $n=1 \dots N$ may be used as an index both for microhomologies and for MH deletions.

In some embodiments, a frequency prediction for a microhomology n may depend on whether the microhomology n is full. A microhomology n is said to be full if the length of the microhomology n is the same as the deletion length associated with the microhomology n . For a microhomology n that is not full, a frequency may be predicted as follows, out of all MH deletion genotypes.

$$V_{MHG}[n] = \frac{\phi_{MH}[n]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{m=1}^N \phi_{MH-less}[L[m]] * \text{indicator (microhomology } m \text{ is full)}}$$

Here $DL[m]$ denotes the deletion length of the microhomology m , and $\text{indicator}(\text{boolean})$ equals 1 if boolean is true, and 0 otherwise.

The inventors have recognized and appreciated that, for a full microhomology, only a single deletion genotype is possible for the entire deletion length. Moreover, the single genotype may be generated via different pathways, such as MMEJ and MH-less end-joining. Therefore, full microhomologies may be modeled as receiving contributions from

MH-dependent and an MH-less mechanisms. Thus, for a microhomology n that is full, a frequency may be predicted as follows, out of all MH deletion genotypes.

$$V_{MHG}[n] = \frac{\phi_{MH}[n] + \phi_{MH-less}[DL[n]]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{m=1}^N \phi_{MH-less}[DL[m]] * \text{indicator (microhomology } m \text{ is full)}}$$

Because the predicted frequencies are normalized, V_{MHG} is a probability distribution over all microhomologies identified for the set of deletion lengths of interest, and hence also a probability distribution over all MH deletions.

In some embodiments, given a deletion length L , a frequency may be predicted as follows for the set of all deletions having the deletion length L , out of all deletions, taking into account contributions from MH-dependent and MH-less mechanisms.

$$V_{DL}[L] = \frac{\sum_{m=1}^N \phi_{MH}[m] * \text{indicator}(DL[m] == L) + \phi_{MH-less}[L]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l]}$$

Here $DL[m]$ denotes the deletion length of the microhomology m , and $\text{indicator}(\text{boolean})$ equals 1 if boolean is true, and 0 otherwise.

In some embodiments, the parameters $w[h]$ and $b[h]$ for each hidden layer h of the neural networks **400A** and **400B** may be trained using a gradient descent method with L2-loss:

$$\text{Loss} = \sum_{m=1}^N (V_{MHG}[m] - V_{MHG}^*[m])^2 + \sum_{l=1}^{60} (V_{DL}[l] - V_{DL}^*[l])^2,$$

where V_{MHG}^* is an observed probability distribution on MH deletion genotypes, and V_{DL}^* is an observed probability distribution on deletion lengths (e.g., based on repair genotype data collected from CRISPR/Cas9 experiments).

In some embodiments, multiple instantiations of the neural networks **400A** and **400B** may be trained with different loss functions. For instance, in addition to, or instead of L2-loss, a squared Pearson correlation function may be used.

$$\text{Loss} = -(\text{pearsonr}(V_{MHG}[m], V_{MHG}^*[m]))^2 - (\text{pearsonr}(V_{MHG}[m], V_{MHG}^*[m]))^2$$

The function $\text{pearsonr}(x, y)$ may be defined as follows for length N vectors x and y , where $\bar{x}$ and $\bar{y}$ denote the averages of x and y , respectively.

$$\text{pearsonr}(x, y) = \frac{\sum_{m=1}^N (x[m] - \bar{x})(y[m] - \bar{y})}{\sqrt{\sum_{m=1}^N (x[m] - \bar{x})^2} \sqrt{\sum_{m=1}^N (y[m] - \bar{y})^2}}$$

Although neural networks are used in the examples shown in FIGS. **4A-4C**, it should be appreciated that aspects of the present disclosure are not so limited. For instance, in some embodiments, one or more other types of machine learning techniques, such as linear regression, non-linear regression, random-forest regression, etc., may be used additionally or alternatively.

Furthermore, in some embodiments, one or more neural networks that are different from the neural networks **400A** and **400B** may be used additionally or alternatively. As one example, a different activation function may be used for one or more nodes, such as $\text{sigma}(x) = \max(0, x)$ (rectified linear

unit, or ReLU), $\text{sigma}(x) = \max(0.001x, x)$ (another example of leaky ReLU), $\text{sigma}(x) = 0.5 * (\tanh(x) + 1.0)$ or

$$\text{sigma}(x) = \frac{1}{1 + e^{-x}} \text{ (Sigmoid),}$$

$\text{sigma}(x) = \max(0, x) + \min(0, x) * 0.5 * (\tanh(x) + 1)$ (Swish), etc. As another example, batch normalization may be performed at one or more hidden layers. It should be appreciated that aspects of the disclosure are not limited to training the neural networks **400A** and **400B** jointly. For instance, given a microhomology n , a frequency may be predicted as follows for the corresponding MH deletion genotype, out of all MH deletion genotypes.

$$V'_{MHG}[n] = \frac{\phi_{MH}[n]}{\sum_{m=1}^N \phi_{MH}[m]}$$

Since this prediction does not depend on the $\phi_{MH-less}$ scores, the neural network **400A** may be trained independently.

In some embodiments, one or more other probability distributions may be predicted in addition to, or instead of V_{MHG} and V_{DL} . As one example, given a microhomology n , a frequency may be predicted as follows for the corresponding MH deletion genotype, out of all deletion genotypes (both MH and MH-less).

$$V''_{MHG}[n] = \frac{\phi_{MH}[n]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l]}$$

As another example, for a microhomology n that is not full, a frequency may be predicted as follows, out of all deletion genotypes (both MH and MH-less).

$$V'''_{MHG}[n] = \frac{\phi_{MH}[n]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l]}$$

For a microhomology n that is full, a frequency may be predicted as follows, out of all deletion genotypes (both MH and MH-less).

$$V'''_{MHG}[n] = \frac{\phi_{MH}[n] + \phi_{MH-less}[DL[n]]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l]}$$

Here $DL[n]$ denotes the deletion length of the microhomology n .

As another example, given a deletion length L , a frequency may be predicted as follows for the set of MH-less deletions having the deletion length L , out of all MH-less deletion genotypes.

$$V'_{DL}[L] = \frac{\phi_{MH-less}[L]}{\sum_{l=1}^{60} \phi_{MH-less}[l]}$$

As another example, given a deletion length L , a frequency may be predicted as follows for the set of MH-less

35

deletions having the deletion length L, out of all deletion genotypes (both MH and MH-less).

$$V_{DL}''[L] = \frac{\phi_{MH-less}[L]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l]}$$

Any one or more of the above predicted probability distributions may be used to train the neural networks **400A** and **400B**, with some suitable loss function.

FIG. **4D** shows an illustrative implementation of the insertion module **315** shown in FIG. **3A**, in accordance with some embodiments. In this example, the insertion module **315** includes two models. First, an insertion rate model **405** may be constructed to predict, given an input DNA sequence and a cut site, a frequency of 1 base pair insertions out of all +1 to -60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions). Second, an insertion base pair model **410** may be constructed to predict frequencies of 1 base pair insertion genotypes (i.e., A, C, G, T), again out of all +1 to -60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions). However, it should be appreciated that aspects of the present disclosure are not limited to any particular set of indels. In some embodiments, a small set of indels (e.g., 1 base pair insertions and 1-28 base pair deletions) may be considered, for instance, when less training data is available.

In some embodiments, the insertion rate model **405** may have one or more input features, which may be encoded as an M-dimensional vector of values for some suitable M. The insertion rate model **405** may have at least one output value. A set of training data for the insertion rate model **405** may include a plurality of M-dimensional training vectors and respective output values. Given an M-dimensional query vector, a k-nearest neighbor (k-NN) algorithm with weighting by inverse distance may be used to compute a predicted output value for the query vector. For instance, k=5 may be used, and five training vectors that are closest to the query vector may be identified, and a predicted output value for the query vector may be computed as a sum of the output values corresponding to the five closest training vectors, weighted by inverse distance, as follows.

$$y = \sum_{i=1}^5 \hat{y}[i] * \left(\frac{\sum_{j=1}^5 d(x, \hat{x}[j]) - d(x, \hat{x}[i])}{\sum_{j=1}^5 d(x, \hat{x}[j])} \right)$$

Here x is the query vector, d is a distance function for the M-dimensional vector space, $\hat{x}[1], \dots, \hat{x}[5]$ are the five closest training vectors, $\hat{y}[1], \dots, \hat{y}[5]$ are the output values corresponding respectively to $\hat{x}[1], \dots, \hat{x}[5]$, and y is the predicted output value for the query vector x.

It should be appreciated that aspects of the present disclosure are not limited to the use of any particular k, or to the use of any k-NN algorithm. For instance, any one or more of the following techniques, and/or Bayesian variants thereof, may be used in addition to, or instead of k-NN: gradient-boosted regression, linear regression, nonlinear regression, multilayer perceptron, deep neural network, etc. Also, any suitable distance metric d may be used, such as Euclidean distance.

In some embodiments, the insertion rate model **405** may have three input features: overall deletion score, precision score, and one or more cut site nucleotides. The overall deletion score may be computed based on outputs of the MH

36

deletion module **305** and the MH-less deletion module **310** in the example of FIG. **3A**, for instance, as follows.

$$\phi = \sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l]$$

Alternatively, log(ϕ) may be used as the overall deletion score.

In some embodiments, the precision score may be indicative of an amount of entropy in predicted frequencies of a suitable set of deletion lengths. The inventors have recognized and appreciated that it may be desirable to calculate precision based on a large set of deletion lengths, but in some instances a smaller set (e.g., 1-28) may be used due to one or more constraints associated with available data. As discussed above, given a deletion length L, a frequency may be predicted as follows for the set of all deletions having the deletion length L, out of all deletions, taking into account contributions from MH-dependent and MH-less mechanisms.

$$V_{DL}[L] = \frac{\sum_{m=1}^N \phi_{MH}[m] * \text{indicator}(DL[m] == L) + \phi_{MH-less}[L]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l]}$$

Here DL[m] denotes the deletion length of the microhomology m, and indicator(boolean) equals 1 if boolean is true, and 0 otherwise. The precision score may be computed as follows.

$$\text{precision} = 1 - \frac{-\sum_{L=1}^{28} V_{DL}[L] * \log(V_{DL}[L])}{\log(28)}$$

In some embodiments, the one or more cut site nucleotides may include nucleotides on either side of the cut site (i.e., seq[-1] and seq[0]). In the example shown in FIG. **1**, the cut site nucleotides are G and G, which are the third and fourth nucleotides to the left of the PAM sequence **115**. However, it should be appreciated that aspects of the present disclosure are not limited to the use of two cut side nucleotides as input features to the insertion rate model **405**. For instance, only one cut side nucleotide (e.g., seq[-1], which may be the fourth nucleotide to the left of the PAM sequence) may be used when less training data is available, whereas more than two cut side nucleotides (e.g., seq[-2], seq[-1], and seq[0], which may be the third, fourth, and fifth nucleotides to the left of the PAM sequence) may be used when more training data is available.

In some embodiments, one or more input features to the insertion rate model **405** may be encoded in some suitable manner. For instance, the one or more cut site nucleotides may be one-hot encoded, for example, as follows.

$$A=1000, C=0100, G=0010, T=0001$$

In some embodiments, encoded input features may be concatenated to form an input vector. In an example in which two cut side nucleotides are used, an input vector may

37

have a length of 10: four for each of the two cut side nucleotides, one for the precision score, and one for the overall deletion score.

In some embodiments, training data for a certain input DNA sequence may be organized into a matrix X. Each row in the matrix (X[i, -]) may correspond to a possible cut site, and may store a length M training vector for that cut site (e.g., M=10). In some embodiments, each column in the matrix (X[-j]) may be normalized to mean 0 and variance 1, as follows.

$$X[i, j] = \frac{X[i, j] - \text{mean}(X[-, j])}{\text{var}(X[-, j])}$$

In some embodiments, values in a query vector may be normalized in a like fashion. For instance, a jth value in a query vector x may be normalized as follows.

$$x[j] = \frac{x[j] - \text{mean}(X[-, j])}{\text{var}(X[-, j])}$$

In some embodiments, an output value may be computed for each row in the training matrix X. For instance, an output value Y[i], i corresponding to a possible cut site, may be a frequency of observed 1 base pair insertions, relative to all observed +1 to -60 indels, at that cut site.

In some embodiments, the insertion base pair model 410 may be constructed to predict frequencies of 1 base pair insertion genotypes (i.e., A, C, G, T). For instance, the insertion base pair model 410 may predict that the probability of a certain insertion genotype given one or more cut site nucleotides is the same as the frequency of that insertion genotype as observed in a subset of training data in which those one or more cut site nucleotides are observed. Thus, given an input DNA sequence seq and a cut site, the insertion base pair model 410 may determine one or more cut site nucleotides (e.g., seq[-1]="C"). The insertion base pair model 410 may then score the insertion genotypes A as follows.

$$\phi_{ins}["A"] = y * \text{ObsFreq}["A" | \text{seq}[-1] = "C"]$$

Here y is the frequency of 1 base pair insertions as predicted by the insertion rate model 405, and ObsFreq("A" | seq[-1]="C") is the observed frequency of insertion genotype A given that the nucleotide to the left of the cut site C. The other three insertion genotypes may be scored similarly.

In some embodiments, more than one cut site nucleotides may be considered. For instance, the insertion base pair model 410 may determine that seq[-2]="A", seq[-1]="C", and seq[0]="G". The insertion base pair model 410 may then score the insertion genotypes A as follows, and the other three insertion genotypes may be scored similarly.

$$\phi_{ins}["A"] = y * \text{ObsFreq}["A" | \text{seq}[-2] = "A" \& \text{seq}[-1] = "C" \& \text{seq}[0] = "G"]$$

In some embodiments, a frequency of 1 base pair insertion genotype A, out of all +1 to -60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions), may be predicted as follows. Frequencies for the other three insertion genotypes may be predicted similarly.

38

$$V_{ins}["A"] = \frac{\phi_{ins}["A"]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l] + \phi_{ins}["A"] + \phi_{ins}["C"] + \phi_{ins}["G"] + \phi_{ins}["T"]}$$

In some embodiments, a frequency of 1 base pair insertions, out of all +1 to -60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions), may be predicted as follows.

$$V_{ins} = \frac{\phi_{ins}["A"] + \phi_{ins}["C"] + \phi_{ins}["G"] + \phi_{ins}["T"]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l] + \phi_{ins}["A"] + \phi_{ins}["C"] + \phi_{ins}["G"] + \phi_{ins}["T"]}$$

In some embodiments, given a deletion length L, a frequency may be predicted as follows for the set of all deletions having the deletion length L, out of all +1 to -60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions), taking into account contributions from MH-dependent and MH-less mechanisms.

$$V_{DL+ins}[L] = \frac{\sum_{m=1}^N \phi_{MH}[m] * \text{indicator}(DL[m] == L) + \phi_{MH-less}[L]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l] + \phi_{ins}["A"] + \phi_{ins}["C"] + \phi_{ins}["G"] + \phi_{ins}["T"]}$$

In some embodiments, given a deletion length L, a frequency may be predicted as follows for the set of MH-less deletions having the deletion length L, out of all +1 to -60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions).

$$V_{DL+ins}''[L] = \frac{\phi_{MH-less}[L]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l] + \phi_{ins}["A"] + \phi_{ins}["C"] + \phi_{ins}["G"] + \phi_{ins}["T"]}$$

In some embodiments, given a microhomology n, a frequency may be predicted as follows for the corresponding MH deletion genotype, out of all +1 to -60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions).

$$V_{MHG+ins}''[n] = \frac{\phi_{MH}[n]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l] + \phi_{ins}["A"] + \phi_{ins}["C"] + \phi_{ins}["G"] + \phi_{ins}["T"]}$$

In some embodiments, for a microhomology n that is not full, a frequency may be predicted as follows, out of all +1 to -60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions).

39

$$V'''_{MHG+ins}[n] = \frac{\phi_{MH}[n]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l] + \phi_{ins}["A"] + \phi_{ins}["C"] + \phi_{ins}["G"] + \phi_{ins}["T"]}$$

For a microhomology n that is full, a frequency may be predicted as follows, out of all all +1 to -60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions).

$$V'''_{MHG+ins}[n] = \frac{\phi_{MH}[n] + \phi_{MH-less}[DL(n)]}{\sum_{m=1}^N \phi_{MH}[m] + \sum_{l=1}^{60} \phi_{MH-less}[l] + \phi_{ins}["A"] + \phi_{ins}["C"] + \phi_{ins}["G"] + \phi_{ins}["T"]}$$

Here $DL[n]$ denotes the deletion length of the microhomology n .

FIG. 5 shows an illustrative process 500 for processing data collected from CRISPR/Cas9 experiments, in accordance with some embodiments. For instance, the process 500 may be performed for each input DNA sequence and CRISPR/Cas9 cut site, and a resulting dataset may be used to train the illustrative computational models described in connection with FIGS. 4A-4D.

At act 505, repair genotypes observed from CRISPR/Cas9 experiments may be aligned with an original DNA sequence. Any suitable technique may be used to observe the repair genotypes, such as Illumina DNA sequencing. Any suitable alignment algorithm may be used for alignment, such as a Needleman-Wunsch algorithm with some suitable scoring parameters (e.g., +1 for match, -2 for mismatch, -4 for gap open, and -1 for gap extend, or +1 for match, -1 for mismatch, -5 for gap open, and -0 for gap extend).

At act 510, one or more filter criteria may be applied to alignment reads from act 505. For instance, in some embodiments, only those reads in which a deletion includes at least one base directly 5' or 3' of the CRISPR/Cas9 cut site are considered. This may filter out deletions that are unlikely to have resulted from CRISPR/Cas9.

At act 515, frequencies of indels of interest (e.g., from +1 to -60) may be normalized into a probability distribution.

FIG. 6 shows an illustrative process 600 for using a machine learning model to predict frequencies of indel genotypes and/or indel lengths, in accordance with some embodiments. Acts 605 and 610 may be similar to, respectively, acts 355 and 360 of the illustrative process 350 of FIG. 3B, except that acts 605 and 610 may be performed for an input DNA sequence seq and a cut site location for which repair genotype data from an CRISPR/Cas9 experiment may not be available. At act 615, one or more machine learning models, such as the machine learning models trained at act 365 of the illustrative process 350 of FIG. 3B, may be applied to an output of act 610 to compute a frequency distribution over deletion lengths of interest.

The inventors have recognized and appreciated that, while Cas9 is typically understood to induce a blunt-end double-strand break, some evidence suggests that Cas9 may generate a 1 base pair staggered end cut instead. FIG. 7 shows illustrative examples of a blunt-end cut and a staggered cut, in accordance with some embodiments.

FIG. 8A shows an illustrative plot 800A of predicted repair genotypes, in accordance with some embodiments. For instance, the plot 800A may be generated by applying one or more of the illustrative techniques described in

40

connection with FIGS. 2A-2D, 3A-3B, 4A-4D, 5-6 to the example shown in FIG. 1. Each vertical bar may correspond to a deletion length, and a height of the bar may correspond to a predicted frequency of that deletion length. The lighter color may indicate repair genotypes that successfully eliminate the donor splice site motif, whereas the darker color may indicate failure. In this example, about 90% of repair products in the 3-26 base pair deletion class are predicted to be successful for the illustrative local sequence context and cut site shown in FIG. 1.

The inventors have recognized and appreciated that the 3-26 base pair deletion class may occur as frequently as 50%, for example, when assaying selected sequences (e.g., patient genotypes underlying certain diseases) integrated into the genome of mouse embryonic stem cells, with a 14-day exposure to CRISPR/Cas9. Thus, in view of the 90% success rate predicted above for the 3-26 base pair deletion class, a genetic editing approach using CRISPR-Cas9 may be provided that achieves a desired result with a 45% rate. In contrast, genetic editing using HDR may achieve a success rate of 10% or lower, and may require a more complex experimental protocol.

FIG. 8B shows another illustrative plot 800B of predicted repair genotypes, in accordance with some embodiments. For instance, the plot 800B may be generated by applying one or more of the illustrative techniques described in connection with FIGS. 2A-2D, 3A-3B, 4A-4D, 5-6 to an illustrative DNA sequence 805B, which may be associated with spinal muscular atrophy (SMA).

In some patients, a specific single nucleotide polymorphism (SNP) in exon 7 of the SMA2 gene may induce exon skipping of exon 7, erroneously including exon 8 instead. Exon 8 includes a protein degradation signal (namely, EMLA-STOP, as shown in FIG. 8B), which causes degradation in the SMA2 gene product, thereby inducing spinal muscular atrophy. In this region, a disease genotype must have precisely EMLA-STOP. Nearly any other genotype is considered healthy.

In the example of FIG. 8B, each vertical bar corresponds to a deletion length, and a height of the bar corresponds to a predicted frequency of that deletion length. The lighter color may indicate repair genotypes that successfully disrupt the EMLA-STOP signal, whereas the darker color may indicate failure. In this example, over 90% of repair products in the 3-26 base pair deletion class are predicted to be healthy.

FIG. 8C shows another illustrative plot 800C of predicted repair genotypes, in accordance with some embodiments. For instance, the plot 800C may be generated by applying one or more of the illustrative techniques described in connection with FIGS. 2A-2D, 3A-3B, 4A-4D, 5-6 to an illustrative DNA sequence associated with breast-ovarian cancer.

In the example of FIG. 8C, a clinical observed patient genotype includes an abnormal duplication of 14 base pairs that a wild type sequence from a normal/health individual lacks. The patient genotype is incorporated into the genome of mouse embryonic stem cells, and then CRISPR/Cas9 is applied. It is observed that the 3-26 base pair deletion class occurs 65% out of all repair classes at this local sequence context. Moreover, as shown in FIG. 8C, repair to wild type is observed to occur at 89% rate among all 3-26 base pair deletions. Thus, an overall wild type repair rate is about 57%.

FIG. 8D shows a microhomology identified in the example of FIG. 8C.

41

As discussed above, the inventors have recognized and appreciated at least two tasks of interest: predicting frequencies of deletion lengths, as well as predicting frequencies of repair genotypes. In some embodiments, a single machine learning model may be provided that performs both tasks.

In some embodiments, repair genotypes corresponding to a deletion of length L may be labeled as follows: for every integer K ranging from 0 to L, a K-genotype associated with deletion length L may be obtained by concatenating left[L][−inf: K] with right[L][K: +inf]. A vector COLLECTION of length Q where each element is a tuple (K, L) may be constructed by enumerating each K-genotype for each deletion length L of interest and removing tuples that have the same repair genotype, e.g., (k', L) and (k, L) such that left[L][−inf: k'] concatenated with right[L][k': +inf] is equivalent to left[L][−inf: k] concatenated with right[L][k: +inf], for example, by retaining only the tuple with the larger K. A training data set may be constructed using observational data by constructing a vector X of length Q where X sums to 1 and X[q] represents an observed frequency of a repair genotype generated by COLLECTION[q].

In some embodiments, the vector COLLECTION may be featurized. This may be performed for a given tuple (k, l) by determining whether there is an index i such that match[1][i: k] is a microhomology. If no such i exists, then the tuple (k, l) may be considered to not partake in microhomology.

The inventors have recognized and appreciated that frequencies of repair products may be influenced by certain features of microhomologies such as microhomology length, fraction of G-C pairings, and/or deletion length. The inventors have also recognized and appreciated that some default values may be useful for repair genotypes that are considered to not partake in microhomology.

For example, the inventors have recognized and appreciated that energetic stability of a microhomology may increase proportionately with a length of the microhomology. Accordingly, in some embodiments, the microhomology length k−i may be used for a tuple (k, l), and a default value of 0 may be used if (k, l) does not partake in microhomology.

As another example, the inventors have recognized and appreciated that thermodynamic stability of a microhomology may depend on specific base pairings, and that G-C pairings have three hydrogen bonds and therefore have higher thermodynamic stability than A-T pairings, which have two hydrogen bonds. Accordingly, in some embodiments, a GC fraction, as shown below, may be used as a feature for (k, l), where indicator(boolean) equals 1 if boolean is true, and 0 otherwise. A default value of −1 may be used if (k, l) does not partake in microhomology.

$$\frac{\sum_{j=1}^{k-1} \text{indicator}(\text{left}[l][j] = 'G' \text{ or } 'C')}{k - i}$$

In some embodiments, a feature for deletion length may be considered, represented as 1 for the tuple (k, l).

The inventors have also recognized and appreciated (e.g., from experimental data) that 0-genotype and 1-genotype repair products may occur despite a lack of microhomology, and may occur through microhomology-free end-joining repair pathways. Accordingly, (k, l) may be featurized with a Boolean for 0-genotype that is equal to 1 if k=0 and (k, l) does not partake in microhomology, and 0 otherwise. A

42

Boolean feature for i-genotypes may also be used where it is equal to 1 if k=l and (k, l) does not partake in microhomology, and 0 otherwise.

FIG. 9 shows another illustrative neural network 900 for computing a frequency distribution over deletion lengths, in accordance with some embodiments.

In some embodiments, the neural network 900 may be parameterized by w[h] and b[h] for each hidden layer h. In some embodiments, these parameters may be initialized randomly, for example, from a spherical Gaussian distribution with some suitable center (e.g., 0) and some suitable variance (e.g., 0.1). These parameters may then be trained using repair genotype data collected from CRISPR/Cas9 experiments.

In some embodiments, the neural network 900 may operate independently for each microhomology, taking as input the length of that microhomology (from the first input node), the GC fraction of that microhomology (from the second input node), Boolean features for 0 and 1-genotypes (from the third and fourth input node, where N-flag corresponds to i-genotypes), and the length of the deletion (from the fifth input node), transforming those five values into 16 values (at the first hidden layer), then transforming those 16 values into 16 other values (at the second hidden layer), and finally outputting a single value (at the output node). In such an embodiment, parameters for the first hidden layer, w[1][i] and b[1][i], are vectors of length 5 for each node i from 1 to 16, whereas parameters for the second hidden layer, w[2][i] and b[2][i], are vectors of length 16 for each node i from 1 to 16, and parameters for the output layer, w[3][1] and b[3][1], are also vectors of length 16.

In some embodiments, the neural network 900 may be applied independently (e.g., as discussed above) to each featurized (k, l) in COLLECTIONS to produce a vector of Q microhomology scores called Z.

In some embodiments, Z may be normalized into a probability distribution over all unique repair genotypes of interest within all deletion lengths of interest (e.g., deletion lengths between 3 and 26). The inventors have recognized and appreciated (e.g., from experimental data) that frequency may decrease exponentially with deletion length. Accordingly, in some embodiments, an exponential linear model may be used to normalize the vector of repair genotype scores. For example, the following formula may be used:

$$Y[q] = \frac{\exp(Z[q] - \text{beta} * DL[q])}{\sum_{q'=1}^Q \exp(Z[q'] - \text{beta} * DL[q'])}$$

where DL[q]=1 for each q where COLLECTIONS[q]=(k, l), and beta is a parameter. In some embodiments, a probability distribution Y over all unique repair genotypes of interest within all deletion lengths of interest may be converted to a probability distribution Y' over all deletion lengths. The following formula may be used for this:

$$Y'[l] = \frac{\sum_{q=1}^Q Y[q] * \text{indicator}(DL[q] = l)}{\sum_{q=1}^Q Y[q]}$$

In some embodiments, the parameter beta may be initialized to -1. These parameters may then be trained using repair genotype data collected from CRISPR/Cas9 experiments.

In some embodiments, the parameters $w[h]$ and $b[h]$ for each hidden layer h and the parameters beta may be trained by using a gradient descent method with L2-loss on Y :

$$L(\text{pred}Y, \text{obs}Y) = \|\text{pred}Y - \text{obs}Y\|_2^2,$$

where $\text{pred}Y$ is a predicted probability distribution on deletion lengths (e.g., as computed by the neural network **900** using current parameter values), and $\text{obs}Y$ is an observed probability distribution on deletion lengths (e.g., based on repair genotype data collected from CRISPR/Cas9 experiments).

The inventors have recognized and appreciated that one or more of the techniques described herein may be used to identify therapeutic guide RNAs that are expected to produce a therapeutic outcome when used in combination with a genomic editing system without an HDR template. For instance, one or more of the techniques described herein may be used to identify a therapeutic guide RNA that is expected to result in a substantial fraction of genotypic consequences that cause a gain-of-function mutation in DNA in the absence of an HDR template. A therapeutic guide RNA may be used singly, or in combination with other therapeutic guide RNAs. An action of the therapeutic guide RNA may be independent of, or dependent on, one or more genomic consequences of the other therapeutic guide RNAs.

FIG. **10** shows, schematically, an illustrative computer **1000** on which any aspect of the present disclosure may be implemented. In the embodiment shown in FIG. **10**, the computer **1000** includes a processing unit **1001** having one or more processors and a non-transitory computer-readable storage medium **1002** that may include, for example, volatile and/or non-volatile memory. The memory **1002** may store one or more instructions to program the processing unit **1001** to perform any of the functions described herein. The computer **1000** may also include other types of non-transitory computer-readable medium, such as storage **1005** (e.g., one or more disk drives) in addition to the system memory **1002**. The storage **1005** may also store one or more application programs and/or external components used by application programs (e.g., software libraries), which may be loaded into the memory **1002**.

The computer **1000** may have one or more input devices and/or output devices, such as devices **1006** and **1007** illustrated in FIG. **10**. These devices can be used, among other things, to present a user interface. Examples of output devices that can be used to provide a user interface include printers or display screens for visual presentation of output and speakers or other sound generating devices for audible presentation of output. Examples of input devices that can be used for a user interface include keyboards and pointing devices, such as mice, touch pads, and digitizing tablets. As another example, the input devices **1007** may include a microphone for capturing audio signals, and the output devices **1006** may include a display screen for visually rendering, and/or a speaker for audibly rendering, recognized text.

As shown in FIG. **10**, the computer **1000** may also comprise one or more network interfaces (e.g., the network interface **1010**) to enable communication via various networks (e.g., the network **1020**). Examples of networks include a local area network or a wide area network, such as an enterprise network or the Internet. Such networks may be based on any suitable technology and may operate according

to any suitable protocol and may include wireless networks, wired networks or fiber optic networks.

Having thus described several aspects of at least one embodiment, it is to be appreciated that various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be within the spirit and scope of the present disclosure. Accordingly, the foregoing description and drawings are by way of example only.

The above-described embodiments of the present disclosure can be implemented in any of numerous ways. For example, the embodiments may be implemented using hardware, software or a combination thereof. When implemented in software, the software code can be executed on any suitable processor or collection of processors, whether provided in a single computer or distributed among multiple computers.

Also, the various methods or processes outlined herein may be coded as software that is executable on one or more processors that employ any one of a variety of operating systems or platforms. Additionally, such software may be written using any of a number of suitable programming languages and/or programming or scripting tools, and also may be compiled as executable machine language code or intermediate code that is executed on a framework or virtual machine.

In this respect, the concepts disclosed herein may be embodied as a non-transitory computer-readable medium (or multiple computer-readable media) (e.g., a computer memory, one or more floppy discs, compact discs, optical discs, magnetic tapes, flash memories, circuit configurations in Field Programmable Gate Arrays or other semiconductor devices, or other non-transitory, tangible computer storage medium) encoded with one or more programs that, when executed on one or more computers or other processors, perform methods that implement the various embodiments of the present disclosure discussed above. The computer-readable medium or media can be transportable, such that the program or programs stored thereon can be loaded onto one or more different computers or other processors to implement various aspects of the present disclosure as discussed above.

The terms “program” or “software” are used herein to refer to any type of computer code or set of computer-executable instructions that can be employed to program a computer or other processor to implement various aspects of the present disclosure as discussed above. Additionally, it should be appreciated that according to one aspect of this embodiment, one or more computer programs that when executed perform methods of the present disclosure need not reside on a single computer or processor, but may be distributed in a modular fashion amongst a number of different computers or processors to implement various aspects of the present disclosure.

Computer-executable instructions may be in many forms, such as program modules, executed by one or more computers or other devices. Generally, program modules include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types. Typically the functionality of the program modules may be combined or distributed as desired in various embodiments.

Also, data structures may be stored in computer-readable media in any suitable form. For simplicity of illustration, data structures may be shown to have fields that are related through location in the data structure. Such relationships may likewise be achieved by assigning storage for the fields

with locations in a computer-readable medium that conveys relationship between the fields. However, any suitable mechanism may be used to establish a relationship between information in fields of a data structure, including through the use of pointers, tags or other mechanisms that establish relationship between data elements.

Various features and aspects of the present disclosure may be used alone, in any combination of two or more, or in a variety of arrangements not specifically discussed in the embodiments described in the foregoing and is therefore not limited in its application to the details and arrangement of components set forth in the foregoing description or illustrated in the drawings. For example, aspects described in one embodiment may be combined in any manner with aspects described in other embodiments.

In an exemplary embodiment, a computational model described herein is trained with experimental data as outlined in Example 1. The method outlined in Example 1 for training a computational model with experimental data is meant to be non-limiting.

Accordingly, the specification discloses a method for training a computational model described herein, comprising: (i) preparing a library comprising a plurality of nucleic acid molecules each encoding a nucleotide target sequence and a cognate guide RNA, wherein each nucleotide target sequence comprises a cut site; (ii) introducing the library into a plurality of host cells; (iii) contacting the library in the host cells with a Cas-based genome editing system to produce a plurality of genomic repair products; (iv) determining the sequences of the genomic repair products; and (iv) training the computational model with input data that comprises at least the sequences of the nucleotide target sequence and/or the genomic repair products and the cut sites.

In another aspect, the specification discloses a method for training a computational model, comprising: (i) preparing a library comprising a plurality of nucleic acid molecules each encoding a nucleotide target sequence and a cut site; (ii) introducing the library into a plurality of host cells; (iii) contacting the library in the host cells with a DSB-based genome editing system to produce a plurality of genomic repair products; (iv) determining the sequences of the genomic repair products; and (iv) training the computational model with input data that comprises at least the sequences of the nucleotide target sequence and/or the genomic repair products and the cut sites.

Methods for preparing nucleic acid libraries, vectors, host cells, and sequencing methods are well known in the art. The instant description is not meant to be limiting in any way as to the construction and configuration of the libraries described herein for training the computational model.

Accordingly, the specification provides in one aspect a method of introducing a desired genetic change in a nucleotide sequence using a double-strand brake (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for introducing the desired genetic change into the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby introducing the desired genetic change in the nucleotide sequence at the cut site.

A cut site can be at any position in a nucleotide sequence and its position is not particularly limiting.

The nucleotide sequence into which a genetic change is desired is not intended to have any limitations as to

sequence, source, or length. The nucleotide sequence may comprise one or more mutations, which can include one or more disease-causing mutations.

In another aspect, the specification provides a method of treating a genetic disease by correcting a disease-causing mutation using a double-strand brake (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence comprising a disease-causing mutation; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for correcting the disease-causing mutation in the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby correcting the disease-causing mutation and treating the disease.

In yet another aspect, the specification provides a method of altering a genetic trait by introducing a genetic change in a nucleotide sequence using a double-strand brake (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for introducing the genetic change into the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby introducing the desired genetic change in the nucleotide sequence at the cut site and consequently altering the associated genetic trait.

In another aspect, the specification provides a method of selecting a guide RNA for use in a Cas-genome editing system capable of introducing a genetic change into a nucleotide sequence of a target genomic location, the method comprising: identifying in a nucleotide sequence of a target genomic location one or more available cut sites for a Cas-based genome editing system; and analyzing the nucleotide sequence and cut site with a computational model to identify a guide RNA capable of introducing the genetic change into the nucleotide sequence of the target genomic location.

In still another aspect, the specification provides a method of introducing a genetic change in the genome of a cell with a Cas-based genome editing system comprising: selecting a guide RNA for use in the Cas-based genome editing system in accordance with the method of the above aspect; and contacting the genome of the cell with the guide RNA and the Cas-based genome editing system, thereby introducing the genetic change.

In various embodiments, the cut sites available in the nucleotide sequence are a function of the particular DSB-inducing genome editing system in use, e.g., a Cas-based genome editing system.

In certain embodiments, the nucleotide sequence is a genome of a cell.

In certain other embodiments, the method for introducing the desired genetic change is done in vivo within a cell or an organism (e.g., a mammal), or ex vivo within a cell isolated or separated from an organism (e.g., an isolated mammalian cancer cell), or in vitro on an isolated nucleotide sequence outside the context of a cell.

In various embodiments, the DSB-inducing genome editing system can be a Cas-based genome editing system, e.g., a type II Cas-based genome editing system. In other embodiments, the DSB-inducing genome editing system can be a TALENS-based editing system or a Zinc-Finger-based genome editing system. In still other embodiments, the DSB-inducing genome editing system can be any such

endonuclease-based system which catalyzes the formation of a double-strand break at a specific one or more cut sites.

In embodiments involving a Cas-based genome editing system, the method can further comprise selecting a cognate guide RNA capable of directing a double-strand break at the optimal cut site by the Cas-based genome editing system.

In certain embodiments, the guide RNA is selected from the group consisting the guide RNA sequences listed in any of Tables 1-6. In various embodiments, the guide RNA can be known or can be newly designed.

In various embodiments, the double-strand brake (DSB)-inducing genome editing system is capable of editing the genome without homology-directed repair.

In other embodiments, the double-strand brake (DSB)-inducing genome editing system comprises a type I Cas RNA-guided endonuclease, or a variant or orthologue thereof.

In still other embodiments, the double-strand brake (DSB)-inducing genome editing system comprises a type II Cas RNA-guided endonuclease, or a functional variant or orthologue thereof.

The double-strand brake (DSB)-inducing genome editing system may comprise a Cas9 RNA-guided endonuclease, or a variant or orthologue thereof in certain embodiments.

In still other embodiments, the double-strand brake (DSB)-inducing genome editing system can comprise a Cpf1 RNA-guided endonuclease, or a variant or orthologue thereof.

In yet further embodiments, the double-strand brake (DSB)-inducing genome editing system can comprise a *Streptococcus pyogenes* Cas9 (SpCas9), *Staphylococcus pyogenes* Cas9 (SpCas9), *Staphylococcus aureus* Cas (Sa-Cas9), *Francisella novicida* Cas9 (FnCas9), or a functional variant or orthologue thereof.

In various embodiments, the desired genetic change to be introduced into the nucleotide sequence, e.g., a genome, is to a correction to a genetic mutation. In embodiments, the genetic mutation is a single-nucleotide polymorphism, a deletion mutation, an insertion mutation, or a microduplication error.

In still other embodiments, the genetic change can comprise a 2-60-bp deletion or a 1-bp insertion.

The genetic change in other embodiments can comprise a deletion of between 2-20, or 4-40, or 8-80, or 16-160, or 32-320, 64-640, or up to 1000, 2000, 3000, 4000, 5000, 6000, 7000, 8000, 9000, or 10,000 or more nucleotides. Preferably, the deletion can restore the function of a defective gene, e.g., a gain-of-function frameshift genetic change.

In other embodiments, the desired genetic change is a desired modification to a wildtype gene that confers and/or alters one or more traits, e.g., conferring increased resistance to a pathogen or altering a monogenic trait (e.g., eye color) or polygenic trait (e.g., height or weight).

In embodiments involving correcting a disease-causing mutation, the disease can be a monogenic disease. Such monogenic diseases can include, for example, sickle cell disease, cystic fibrosis, polycystic kidney disease, Tay-Sachs disease, achondroplasia, beta-thalassemia, Hurler syndrome, severe combined immunodeficiency, hemophilia, glycogen storage disease Ia, and Duchenne muscular dystrophy.

In any of the above aspects and embodiments, the step of identifying the available cut sites can involve identifying one or more PAM sequences in the case of a Cas-based genome editing system.

In various embodiments of the above methods, the computational model used to analyze the nucleotide sequence is a deep learning computational model, or a neural network

model having one or more hidden layers. In various embodiments, the computational model is trained with experimental data to predict the probability of distribution of indel lengths for any given nucleotide sequence and cut site. In still other embodiments, the computational model is trained with experimental data to predict the probability of distribution of genotype frequencies for any given nucleotide sequence and cut site.

In various embodiments, the computational model comprises one or more training modules for evaluating experimental data.

In various embodiments, the computational model can comprise: a first training module for computing a microhomology score matrix; a second training module for computing a microhomology independent score matrix; and a third training module for computing a probability distribution over 1-bp insertions, wherein once trained with experimental data the computational model computes a probability distribution over indel genotypes and a probability distribution over indel lengths for any given input nucleotide sequence and cut site.

In other embodiments, the computational model predicts genomic repair outcomes for any given input nucleotide sequence and cut site.

In various embodiments, the genomic repair outcomes can comprise microhomology deletions, microhomology-less deletions, and/or 1-bp insertions.

In still other embodiments, the computational model can comprise one or more modules each comprising one more input features selected from the group consisting of: a target site nucleotide sequence; a cut site; a PAM-sequence; microhomology lengths relative at a cut site, % GC content at a cut site; and microhomology deletion lengths at a cut site, and type of DSB-genome editing system.

In various embodiments, the nucleotide sequence analyzed by the computational model is between about 25-100 nucleotides, 50-200 nucleotides, 100-400 nucleotides, 200-800 nucleotides, 400-1600 nucleotides, 800-3200 nucleotides, and 1600-6400 nucleotide, or even up to 7K, 8K, 9K, 10K, 11K, 12K, 13K, 14K, 15K, 16K, 17K, 18K, 19K, 20K nucleotides, or more in length.

In another aspect, the specification relates to guide RNAs which are identified by various methods described herein. In certain embodiments, the guide RNAs can be any of those presented in Tables 1-6, the contents of which form part of this specification.

According to various embodiments, the RNA can be purely ribonucleic acid molecules. However, in other embodiments, the RNA guides can comprise one or more naturally-occurring or non-naturally occurring modifications. In various embodiments, the modifications can including, but are not limited to, nucleoside analogs, chemically modified bases, intercalated bases, modified sugars, and modified phosphate group linkers. In certain embodiments, the guide RNAs can comprise one or more phosphorothioate and/or 5'-N-phosphoramidite linkages.

In still other aspects, the specification discloses vectors comprising one or more nucleotide sequences disclosed herein, e.g., vectors encoding one or more guide RNAs, one or more target nucleotide sequences which are being edited, or a combination thereof. The vectors may comprise naturally occurring sequences, or non-naturally occurring sequences, or a combination thereof.

In still other aspects, the specification discloses host cells comprising the herein disclosed vectors encoding one more nucleotide sequences embodied herein, e.g., one or

more guide RNAs, one or more target nucleotide sequences which are being edited, or a combination thereof.

In other aspects, the specification discloses a Cas-based genome editing system comprising a Cas protein (or homolog, variant, or orthologue thereof) complexed with at least one guide RNA. In certain embodiments, the guide RNA can be any of those disclosed in Tables 1-6, or a functional variant thereof.

In still other aspects, the specification provides a Cas-based genome editing system comprising an expression vector having at least one expressible nucleotide sequence encoding a Cas protein (or homolog, variant, or orthologue thereof) and at least one other expressible nucleotide sequence encoding a guide RNA, wherein the guide RNA can be identified by the methods disclosed herein for selecting a guide RNA.

In yet another aspect, the specification provides a Cas-based genome editing system comprising an expression vector having at least one expressible nucleotide sequence encoding a Cas protein (or homolog, variant, or orthologue thereof) and at least one other expressible nucleotide sequence encoding a guide RNA, wherein the guide RNA can be identified by the methods disclosed herein for selecting a guide RNA.

In still a further aspect, the specification provides a library for training a computational model for selecting a guide RNA sequence for use with a Cas-based genome editing system capable of introducing a genetic change into a genome without homology-directed repair, wherein the library comprises a plurality of vectors each comprising a first nucleotide sequence of a target genomic location having a cut site and a second nucleotide sequence encoding a cognate guide RNA capable of directing a Cas-based genome editing system to carry out a double-strand break at the cut site of the first nucleotide sequence.

In another aspect, the specification provides a library and its use for training a computational model for selecting an optimized cut site for use with a DSB-based genome editing system (e.g., Cas-based system, TALAN-based system, or a Zinc-Finger-based system) that is capable of introducing a desired genetic change into a nucleotide sequence (e.g., a genome) at the selected cut site without homology-directed repair, wherein the library comprises a plurality of vectors each comprising a nucleotide sequence having a cut site, and optionally a second nucleotide sequence encoding a cognate guide RNA (in embodiments involving a Cas-based genome editing system).

Also, the concepts disclosed herein may be embodied as a method, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

EXAMPLES

In order that the invention described herein may be more fully understood, the following examples are set forth. It should be understood that these examples are for illustrative purposes only and are not to be construed as limiting this invention in any manner.

Example 1: Demonstration of Predictable and Precise Template-Free CRISPR Editing of Pathogenic Variants

SUMMARY

DNA double-strand break repair following cleavage by Cas9 is generally considered stochastic, heterogeneous, and impractical for applications beyond gene disruption. Here, it is shown that template-free Cas9 nuclease-mediated DNA repair is predictable in human and mouse cells and is capable of precise repair to a predicted genotype in certain sequence contexts, enabling correction of human disease-associated mutations. A genomically integrated library of guide RNAs (gRNAs) was constructed, each paired with its corresponding DNA target sequence, and trained a machine learning model, inDelphi, on the end-joining repair products of 1,095 sequences cleaved by Cas9 nuclease in mammalian cells. The resulting model accurately predicted frequencies of 1- to 60-bp deletions and 1-bp insertions (median $r=0.87$) with single-base resolution at 194 held-out library sites and -90 held-out endogenous sequence contexts in four human and mouse cell lines. The inDelphi model predicts that 26% of all *Streptococcus pyogenes* Cas9 (SpCas9) gRNAs targeting the human genome result in outcomes in which a single predictable product accounts for $\geq 30\%$ of all edited products, while 5% of gRNAs are “high-precision guides” that result in repair outcomes in which one product accounts for $\geq 50\%$ of all edited products. It was experimentally confirmed that 183 human disease-associated microduplication alleles can each be corrected to their wild-type genotypes with $\geq 50\%$ frequency among edited products following Cas9 cleavage in mammalian cells. Using these insights, genotypic and functional rescue of pathogenic LDLR microduplication alleles was achieved in human and mouse cells, and restored to wild-type an endogenous genomic Herman-sky-Pudlak syndrome (HPS1) pathogenic allele in primary patient-derived fibroblasts. This study establishes that template-free Cas9 nuclease activity can be harnessed for precise genome editing applications.

More in particular, this study developed a high-throughput *Streptococcus pyogenes* Cas9 (SpCas9)-mediated repair outcome assay to characterize end-joining repair products at Cas9-induced double-stranded breaks using 1,872 target sites based on sequence characteristics of the human genome. The study used the resulting rich set of repair product data to train the herein disclosed machine-learning algorithm (i.e., inDelphi), which accurately predicts the frequencies of the substantial majority of template-free Cas9-induced insertion and deletion events at single-base resolution (which is further described in M. Shen et al., “Predictable and precise template-free CRISPR editing of pathogenic variants,” *Nature*, vol. 563, Nov. 29, 2018, pp. 646-651, and including Extended Data). This study finds that in contrast to the notion that end-joining repair is heterogeneous, inDelphi identifies that 5-11% of SpCas9 gRNAs in the human genome induce a single predictable repair genotype in $\geq 50\%$ of editing products.

Building on this idea of precision gRNAs, this study further uses inDelphi to design 14 gRNAs for high-precision template-free editing yielding predictable 1-bp insertion genotypes in endogenous human disease-relevant loci and experimentally confirmed highly precise editing (median 61% among edited products) in two human cell lines. As described herein, inDelphi was used to reveal human pathogenic alleles that are candidates for efficient and precise template-free gain-of-function genotypic correction and

achieved template-free correction of 183 pathogenic human microduplication alleles to the wild-type genotype in $\geq 50\%$ of all editing products. Finally, these developments were integrated to achieve high-precision correction of five pathogenic low-density lipoprotein receptor (LDLR) microduplication alleles in human and mouse cells, as well as correction of endogenous pathogenic microduplication alleles for Hermansky-Pudlak syndrome (HPS1) and Menkes disease (ATP7A) to the wild-type sequence in primary patient-derived fibroblasts.

Results

Cas9-Mediated DNA Repair Products are Predictable

To capture Cas9-mediated end-joining repair products across a wide variety of target sequences, a genome-integrated gRNA and target library screen was designed in which many unique gRNAs are paired with corresponding 55-bp target sequences containing a single canonical “NGG” SpCas9 protospacer-adjacent motif (PAM) that directs cleavage to the center of each target sequence (FIG. 11A). To explore repair products among sequences representative of the human genome, 1,872 target sequences were computationally designed that collectively span the human genome’s distributions of % GC, number of nucleotides participating in microhomology, predicted Cas9 on-target cutting efficiency⁴, and estimated precision of deletion products²⁴ (FIGS. 16A-16C). Through a multi-step process (FIGS. 16A-16C), the library (Lib-A—see Table 4) was cloned into a plasmid backbone allowing Tol2 transposon-based integration into the genome²⁵, gRNA expression, and hygromycin selection for cells with genomically integrated library members.

Lib-A was stably integrated into the genomes of mouse embryonic stem cells (mESCs). Next, these cells were targeted with a Tol2 transposon-based SpCas9 expression plasmid containing a blasticidin expression cassette and selected for cells with stable Cas9 expression. Sufficient numbers of cells were maintained throughout the experiment to ensure $>2,000$ -fold coverage of the library. After one week, genomic DNA was collected from three independent replicate experiments from these cells and performed paired-end high-throughput DNA sequencing (HTS) using primers flanking the gRNA and the target site to reveal the spectrum of repair products at each target site. Using a sequence alignment procedure, the resulting 96,838,690 sequence reads were tabulated into observed frequencies of, on average, 1,262 unique repair genotypes for each target site.

To test the correspondence between library repair products and endogenous repair products, Lib-A included the 55-bp sequences surrounding 90 endogenous genomic loci for which the products of Cas9-mediated repair were previously characterized by HTS²⁴. Previously reported repair products from this endogenous dataset (VO) in three human cell lines (HCT116, K562, and HEK293) reveal that 94% of endogenous Cas9-mediated deletions are 30 bp or shorter (FIGS. 16A-16C), suggesting that the Lib-A analysis method is capable of assessing the vast majority of Cas9-mediated editing products. It was found that repair outcomes for these Lib-A members corresponding to the VO sites in mESCs are consistent with previously reported endogenous repair products in human cells (median $r=0.76$, FIGS. 17A-17D). Lib-A repair genotype frequencies are also consistent between experimental replicates (median $r=0.89$, FIGS. 17A-17D), confirming that Cas9-mediated editing products of the target library reflect previously reported endogenous target locus editing products in human cells.

In Lib-A data from mESCs and in the three VO datasets from endogenous HEK293, K562, and HCT116 cells, end-

joining repair of Cas9-mediated double-strand breaks primarily causes deletions (73-87% of all products) and insertions (13-25% of all products) (FIGS. 11B, 11C, FIGS. 17A-17D). Rarer Cas9-mediated repair products were also detected such as combination insertion/deletions (0.5-2% of all products) and deletions and insertions distal to the cutsite (3-5% of all products), which occur more often on the PAM-distal side of the double-strand break (FIGS. 17A-17D). The majority of products are deletions containing microhomology consistent with MMEJ (53-58% of all products, and 70-75% of deletions) (FIGS. 11B, 11C, FIGS. 17A-17D for a definition of microhomology-containing deletions).

Using the wealth of Cas9 outcome data provided by Lib-A, a novel machine learning model, inDelphi, was trained to predict the spectrum of Cas9-mediated editing products at a given target site. This model consists of three interconnected modules aimed at predicting the three major classes of repair outcomes: microhomology deletions (MH deletions), microhomology-less deletions (MH-less deletions), and single-base insertions (1-bp insertions, FIG. 12A). These three repair classes are defined as constituting all major editing outcomes and note that they comprise 80-95% of all observed editing products (FIGS. 11B, 11C). Motivated by the abundance of MH deletion products in Lib-A and VO data, a deep neural network was designed to predict MH deletions as one module of inDelphi. This module simulates MH deletions using the MMEJ repair mechanism, where 5'-to-3' end resection at a double-strand break reveals two 3' ssDNA overhangs that can anneal through sequence microhomology. Extraneous ssDNA overhangs are eliminated, and DNA synthesis and ligation generates a dsDNA repair product²⁶ (FIG. 12B). Through this mechanism, each potential microhomology results in a distinct deletion genotype, allowing a 1:1 mapping between possible microhomologies at a target site and available MH deletion outcome genotypes (FIG. 12B). inDelphi models MH deletions as a competition between different MH-mediated hybridization possibilities. Using the input features of MH length, MH % GC, and deletion length, inDelphi outputs a score (ϕ) reflecting the predicted strength of each microhomology (FIG. 12A). From training data, inDelphi learned that strong microhomologies tend to be long and have high GC content (FIGS. 18A-18H).

To account for all deletions that cannot be simulated through the MMEJ mechanism, inDelphi also contains a second neural network module that predicts the distribution of MH-less deletion lengths using the minimum required resection length as the only input feature (FIG. 12A). Because there are many MH-less genotypes for each deletion length with frequencies that do not fit a simple pattern, inDelphi predicts the frequencies of deletion lengths but not of genotypic outcomes for MH-less deletions. This module learned from training data that the frequency of MH-less deletions decays rapidly with increasing length (FIGS. 18A-18H). It is hypothesized that MH-less deletions arise primarily from the activity of the classical and alternative NHEJ pathways²⁷. The two neural networks were jointly trained using observed distributions of deletion genotypes from 1,095 Lib-A target sites (FIG. 12A).

The inDelphi model contains a third module to predict 1-bp insertions (FIG. 12A). In VO and Lib-A data, insertions represent a major class of DNA repair at Cas9-mediated double-strand breaks (13-25% of all products, FIGS. 11B, 11C, FIGS. 17A-17D). Among insertions, 1-bp insertions are dominant (9-21% of all products, FIGS. 11B, 11C, FIGS. 17A-17D). Surprisingly, it was found that the frequency of

1-bp insertions and their resultant genotypes depend strongly on local sequence context. In endogenous and library settings, 1-bp insertions predominantly comprise duplications of the -4 nucleotide (counting the NGG PAM as nucleotides 0-2, FIG. 12A), with higher precision when the -4 nucleotide is an A or T and with lower precision when it is a C or G (FIG. 12C). While 1-bp insertions were observed occurring in 9% of products on average in Lib-A, this frequency varies significantly depending on the nucleotide at position -4, falling to less than 4% on average when the -4 nucleotide is G (FIG. 12D, $P < 10^{-34}$). While position -4 is most strongly associated with 1-bp insertion frequency, other surrounding bases also contribute to insertion frequency (FIG. 12E). In addition, it was found that target sites with poor microhomology (low total phi score) and target sites with imprecise deletion product distributions are more likely to contain insertions at the expense of deletions (FIGS. 18A-18H). Based on these empirical observations, inDelphi models insertions and deletions as competitive processes in which the total deletion phi score (overall microhomology strength) and predicted deletion precision influence the relative frequency of 1-bp insertions, and the local sequence context influences the relative frequency and genotypic outcomes of 1-bp insertions (FIG. 12A). inDelphi integrates these factors into predictions of 1-bp insertion genotype frequencies using a k-nearest neighbor approach. Collectively, from sequence context alone, inDelphi predicts the indel lengths of 80-95% of Cas9-mediated editing products and the single-base resolution genotypes of 65-80% of all products (FIG. 13A, FIGS. 19A-19D).

Trained on data from 1,095 Lib-A sequence contexts in mESCs, inDelphi demonstrates highly accurate genotypic prediction of 1-bp insertions and 1-60-bp deletions at 87-90 VO target sequences previously characterized experimentally in endogenous K562, HCT116, and HEK293 cells (median $r=0.87$, FIG. 13B). It is noted that the Lib-A versions of these target sites were held out of inDelphi training. The inDelphi model also performs well when predicting indel length distributions from 1-bp insertions to 60-bp deletions at the endogenous VO sites in three human cell lines (median $r=0.84$, FIG. 13C). Additionally, inDelphi accurately predicts relative frequencies of genotypic outcomes (median $r=0.94$) and indel length distributions (median $r=0.91$) of 189 held-out Lib-A targets in mESCs (FIGS. 19A-19D). As a control that the features used in training inDelphi are crucial for its performance, the MH length feature was deleted from the inDelphi MH deletion module and found that inDelphi's performance predicting genotype frequency was reduced to the performance of a model with random weights. A second control in which the deep neural networks were replaced with linear models showed 10-24% reduced performance on the genotype frequency and indel length prediction tasks. Together, these controls indicate that inDelphi's computational structure is important for its accuracy. An online implementation of inDelphi is provided to predict the spectrum of Cas9-mediated products at any target site (crisprindelphi.design). Taken together, these results establish that in data from human and mouse cells, the relative frequencies of most Cas9 nuclease-mediated editing outcomes are highly predictable.

The ability of Cas9-mediated end-joining repair to induce frameshifts enables efficient gene knockout²⁸. It was reasoned that inDelphi's accurate prediction of the indel length distribution of 80-95% of template-free Cas9-mediated editing products should also enable accurate prediction of Cas9-induced frameshifts. This task was simulated in 86 endogenous VO target sequences in HEK293 by tabulating

the observed frequency of indels resulting in +0, +1, and +2 frameshifts. The observed frequency of indels in each frame predicted by inDelphi (median $r=0.81$) compare favorably to those generated by Microhomology Predictor (median $r=0.37$), a previously published method²⁹ (FIG. 13D). Thus, it is expected that inDelphi facilitates Cas9-mediated gene knockout approaches by allowing a priori selection of gRNAs that induce high or low knockout frequencies. To this end, an online tool is provided to predict frameshift frequencies for any SpCas9 gRNA targeting the coding human and mouse genome (crisprindelphi.design). It is noted that human exons have a significant tendency ($p < 10^{-100}$, FIGS. 19A-19D) to favor frame-preserving deletion repair compared to shuffled exon sequences or non-coding human DNA. Taken together, the results show that inDelphi provides accurate single-base resolution predictions for the relative frequencies of most Cas9 nuclease-mediated end-joining repair outcomes, including frameshifts.

Designing High-Precision Template-Free Cas9 Nuclease-Mediated Editing

While end-joining repair is highly efficient at inducing mutations after Cas9 treatment, its tendency to induce a heterogeneous mixture of repair genotypes has limited its application primarily to gene disruption and removal of intervening sequences between two double-stranded breaks³⁰⁻³³. Motivated by inDelphi's ability to predict Cas9-mediated repair outcomes from target sequences alone, it was sought to identify target sites for which the repair profile is highly skewed toward a single outcome. In principle, the ability of inDelphi to identify such sites may enable efficient, template-free, nuclease-mediated precision gene editing.

It was reasoned that a single strong microhomology hybridization possibility with a high phi score would out-compete a background of weaker alternative microhomologies to yield efficient and precise repair to a single deletion genotype. Microduplications, in which a stretch of DNA is repeated in tandem, contain stretches of exact microhomology and thus are predicted by inDelphi to collapse precisely through deletion upon MMEJ repair (FIG. 14A).

To test this prediction, a second high-throughput Cas9 substrate library (Lib-B—see Table 5) was designed and constructed that contains three families of target sequences with microduplications of each length from 7-25 bp. Cas9-mediated double-strand break repair products were analyzed in Lib-B in mESCs and in human U2OS and HEK293T cells using the same procedure as for Lib-A evaluation. Highly precise repair was consistently observed in which 40-80% of all repair events correspond to a single repair genotype (FIG. 14B), substantially higher than the 21% median frequency of the most abundant deletion genotype in 90 VO sites that were not pre-selected for microhomology. The fraction of microduplication repair to a single collapsed product as compared to other outcomes increased with microduplication length in mESCs, U2OS, and HEK293T cells ($r=0.35$, $p < 7 \times 10^{-5}$, FIG. 14B, FIGS. 20A-20E). It is noted that these sites have significantly higher phi scores and precision scores compared to VO sites, and significantly fewer 1-bp insertions (FIGS. 20A-20E). Thus, sites with strong MH deletion candidates are enriched in that specific deletion outcome at the expense of MH-less deletion and 1-bp insertion outcomes.

It is also hypothesized that sequence contexts with no strong MHs (low total phi scores) could enable precise 1-bp insertion repair. To test this possibility, three target sequence frameworks with low total phi scores were included in Lib-B (FIGS. 20A-20E) containing randomization at the four posi-

tions surrounding the Cas9 cleavage site (positions -5 to -2 with respect to the PAM at positions 0-2; see FIG. 12A). Cas9 nuclease treatment of 205 such sequences in Lib-B resulted in highly precise (up to 90% of all repair events) and reproducible ($r=0.90$ between mESC replicates) 1-bp insertions (FIG. 14C, FIGS. 20A-20E). Strikingly, the efficiency of 1-bp insertions is strongly influenced by the nucleotide identities in positions -5 to -2 (FIG. 14C). Similar to the findings from Lib-A (FIG. 12E, FIGS. 20A-20E), -4T and -3G correlate with higher relative frequencies of 1-bp insertion among all products while -4G correlates with lower frequencies of insertion (FIG. 14D). Among these three fixed sequence contexts in Lib-B with low total MH, 1-bp insertions comprise a median of 29% of all repair products, which is significantly higher than in VO sites (FIGS. 20A-20E). Moreover, a median of 61% of all products are 1-bp insertions at sites with TG at the -4 and -3 positions (FIG. 14D), revealing that precise 1-bp insertion can be obtained through Cas9-mediated end-joining at specific, predictable sequence contexts.

It is noted that sequences that support higher insertion efficiencies (>50%) have on average 33% lower total efficiencies of Cas9-mediated indels than sequences that yield lower insertion efficiencies ($r=-0.35$, $p=3.3 \times 10^{-7}$, FIGS. 20A-20E), possibly because the lower efficiency of MMEJ at such sites decreases the likelihood of mutagenic repair of the Cas9-induced double-strand break. These observations collectively establish that Cas9-mediated repair of target sites with predictable sequence features can lead to precise editing favoring one particular outcome.

Based on these findings, inDelphi was used to predict gRNAs that lead to such precise outcomes. A metric was defined using information entropy to measure the precision of a repair outcome spectrum as a score ranging from zero (highest entropy, lowest precision) to one (lowest entropy, highest precision) and demonstrated that inDelphi is capable of predicting the precision of Cas9-induced deletions in 86 VO target sequences in HEK293 cells (median $r=0.64$, FIG. 14E). inDelphi was then used to discover SpCas9 gRNAs that support precise end-joining repair in the human genome.

It was found that substantial fractions of all genome-targeting SpCas9 gRNAs are predicted to produce relatively precise outcomes (Table 2). Indeed, inDelphi predicts that 26% of SpCas9 gRNAs that target human exons and introns are "precision gRNAs" (FIG. 14F), which are defined as gRNAs predicted to produce a single genotypic outcome in >30% of all major editing products, with 20% of gRNAs predicted to be produce a single deletion genotype at $\geq 30\%$ efficiency and 6.2% predicted to produce a single 1-bp insertion genotype at >30% efficiency. Moreover, inDelphi predicts that 4.8% of SpCas9 gRNAs targeting human exons and introns are "high-precision gRNAs," which are defined as gRNAs that produce a single genotype in $\geq 50\%$ of all major editing products (FIG. 14F, 3.8% producing high-precision deletion, 0.94% producing high-precision 1-bp insertion). These findings suggest that Cas9-mediated end-joining outcomes at many target sites are both predictable and precise, and that precision and high-precision gRNAs offer new opportunities for precision deletion and insertion by Cas9 nuclease-mediated editing. An online tool is provided to predict the precision of a given gRNA and to identify precision and high-precision SpCas9 gRNAs targeting the human and mouse genomes (crisprindelphi.design).

Efficient Template-Free Repair of Pathogenic Alleles to Wild-Type Genotypes

Next, inDelphi-classified high-precision gRNAs were used to identify new targets for therapeutic genome editing. Starting with 23,018 insertion, short deletion, and microdu-

plication disease genotypes from the ClinVar and HGMD databases^{16,17}, inDelphi was tasked with identifying pathogenic alleles that are suitable for template-free Cas9-mediated editing to effect precise gain-of-function repair of the pathogenic genotype. Two genetic disease allele categories that have not been previously identified as targets for Cas9-mediated repair are predicted by inDelphi to be candidates for high-precision repair. The first category is a selected subset of pathogenic frameshifts in which, because of high-precision repair, inDelphi predicts that 50-90% of Cas9-mediated deletion products will correct the reading frame compared to the average frequency of 34% among all disease-associated frameshift mutations. The second category is pathogenic microduplication alleles in which a short sequence duplication leads to a frameshift or loss-of-function protein sequence changes (FIG. 15A).

To test the accuracy of inDelphi at predicting repair genotypes of therapeutically relevant alleles, 1,592 pathogenic human loci that inDelphi identified to have the highest predicted rates of frameshift or microduplication repair to the wild-type sequence, were included in Lib-B. Cas9-mediated repair of genome-integrated Lib-B in mESCs and human U2OS and HEK293T cells confirmed highly efficient and precise gain-of-function editing. It was observed that 183 human disease microduplication alleles included in Lib-B were repaired to wild-type in $\geq 50\%$ of all products (FIG. 15B), and 508 pathogenic human frameshift alleles were restored into proper reading frame in $\geq 50\%$ of all products in mESCs (FIG. 15C), in agreement with inDelphi's predictions ($r=0.64$ for frame restoration, $r=0.64$ for wildtype repair). Similar results were observed in HEK293T and U2OS cells (FIGS. 21A-21D). While microduplication repair to the wild-type genotype unambiguously restores wild-type protein function, it is noted that frameshift restoration that alters coding sequence requires case-by-case analysis to validate rescue of protein function.

To determine if the efficiency of microduplication repair can be increased by manipulation of DNA repair pathways, Cas9 cleavage of Lib-B was performed in $Prkdc^{-/-}Lig4^{-/-}$ mESCs³⁴, which are deficient for two proteins involved in NHEJ repair³⁵. As expected, the frequency of MH-less deletion repair in cells with impaired NHEJ is decreased (25% to 16%) (FIGS. 22A-22E). It was also observed that the precision of 1-bp insertions that result in duplication of the -4 nucleotide is increased in $Prkdc^{-/-}Lig4^{-/-}$ mESCs (FIGS. 22A-22E). Importantly, the frequency of MH-dependent deletion repair is substantially increased (58% to 72%) in $Prkdc^{-/-}Lig4^{-/-}$ mESCs, enabling a subset of pathogenic alleles to be repaired to wild-type with strikingly high precision. In wild-type mESCs, 183 pathogenic alleles are repaired to wild-type in $\geq 50\%$ of all edited products and 11 pathogenic alleles are repaired to wild-type in $\geq 70\%$ of all edited products, while in $Prkdc^{-/-}Lig4^{-/-}$ mESCs, 286 pathogenic alleles are repaired to wildtype in $\geq 50\%$ of all edited products and 153 pathogenic alleles are repaired to wild-type in $\geq 70\%$ of products (FIG. 15D, Table 6). Thus, impairing NHEJ can further increase the precise repair of pathogenic microduplications to wild-type ($p=7.8 \times 10^{-12}$, FIG. 15D). These data support the model that competing end-joining repair mechanisms determine the relative frequencies of specific editing outcome types and demonstrate that template-free genotypic correction of hundreds of pathogenic microduplication alleles in genes such as PKD1 (corrected in 92% of edited $Prkdc^{-/-}Lig4^{-/-}$ mESC alleles), GJB2 (91%), MSH2 (88%), LDLR (87%), and BRCA1 (82%) can be optimized to occur with strikingly high efficiency by manipulation of repair pathways. inDelphi's prediction of highly efficient wild-type repair was further tested on pathogenic LDLR microduplication alleles, which cause dominantly inherited familial hypercholesterolemia³⁶. Five pathogenic LDLR microduplication alleles were separately introduced within a full-length LDLR coding

57

sequence upstream of a P2A-GFP cassette into the genome of human and mouse cells, such that Cas9-mediated repair to the wild-type LDLR sequence should induce phenotypic gain of LDL uptake and restore the reading frame of GFP. Cas9 and a gRNA that is specific to each pathogenic allele and does not target the wild-type repaired sequence were then deleivered. Robust restoration of LDL uptake was observed as well as restoration of GFP fluorescence in mESCs, U2OS cells, and HCT116 cells in up to 79% of cells following transfection with Cas9 and inDelphi gRNAs (FIGS. 15E, 15F, FIGS. 23A-23E). HTS confirms efficient genotypic repair to wild-type of these five LDLR microduplication alleles in human and mouse cells as well as of three other pathogenic microduplication alleles in the GAA, GLB1, and PORCN genes introduced to cells using the same method (Table 1, Table 3). Importantly, in these experiments, high-frequency LDLR phenotypic correction was observed when cutting with either SpCas9 or *Streptococcus aureus* Cas9 (SaCas9)³⁷ (Table 3), suggesting that microdu-

58

plication repair is a feature of cellular repair after a Cas9-mediated double-strand break that does not require a specific nuclease.

Finally, precise template-free Cas9-mediated MMEJ was used to repair an endogenous pathogenic 16-bp microduplication in primary fibroblasts from a Hermansky-Pudlak syndrome (HPS1) patient. HPS1 causes blood clotting deficiency and albinism in patients and is particularly common in Puerto Ricans³⁸. Simultaneous delivery of Cas9 and gRNA specific to the pathogenic microduplication allele induced high-efficiency correction to the wild-type sequence (mean frequency=71% of edited alleles, N=3, Table 1). These findings suggest the potential of template-free, precise Cas9 nuclease-mediated repair of microduplication alleles to achieve efficient repair to the wild-type sequence for therapeutic gain-of-function genome editing.

The following tables are referenced in this specification.

TABLE 1

Repair of microduplication pathogenic alleles through template-free Cas9-nuclease treatment.				
Pathogenic microduplication genotype	Cell type	Predicted frequency of repair to wild-type genotype in all major editing products (%)	Observed frequency of repair to wild-type genotype in all edited products in Lib-B, mESCs (%)	Observed frequency of repair to wild-type genotype in all edited outcomes in endogenous data (%)
HPS1:c.1472_1487 dup16	Primary patient fibroblasts	88	53	71
LDLR:c.1662_1689 dupGCTGGTGA	mESCs	85	61	65
LDLR:c.1662_1669 dupGCTGGTGA	HCT116	85	61	89
LDLR:c.1662_1669 USOS dupGCTGGTGA	USOS	85	51	77

TABLE 2

Frequency of gRNAs in the human genome with denoted Cas9-mediated outcome precision.				
Fraction of 1,003,524 SpCas9 gRNAs in human exons and introns for which the most-common repair genotype comprises >XX % of all major editing products				
	Precision gRNA frequency (%)	Precise product is a deletion (% of gRNAs)	Precise product is a 1-bp insertion (% of gRNAs)	Any precise product of gRNAs)
50	10	86	35	96
	15	63	23	78
	30	43	14	55
	35	29	10	39
55	30	20	6.2	26
	35	13	4.1	17
	40	8.6	2.6	11
	45	5.8	1.3	7.1
60	50	3.8	0.94	4.8
	55	2.4	0.52	3.0
	60	1.5	0.39	1.9
	65	0.98	0.086	1.1
65	70	0.58	0.026	0.61
	75	0.29	0.024	0.32
	80	0.12	0	0.12
	85	0.040	0	0.040
	90	0.0049	0	0.0049

TABLE 3

Repair of eight pathogenic microduplication alleles in individual cellular experiments.													
Pathogenic allele	LDLRdup1	LDLRdup2	LDLRdup2	LDLRdup2	LDLRdup3	LDLRdup4	LDLRdup5	PORCndup	GAAdup	GLB1dup	HPS1dup	ATP7Adup	
#AlleleID	245617	245706	245706	245706	245709	245715	246266	25739	354180	354180	98805	ND	ND
Predicted frequency of deletions restoring frame	79	98	96	96	95	86	94	90	76	93	95	ND	ND
Flow cytometric frameshift frequency (%)	57	95	57	57	90	72	87	ND	79	74	85	ND	ND
Predicted frequency of repair to wild-type genotype among all major editing products (%)	72	90	83	83	94	85	86	89	74	91	79	88	43
Flow cytometric phenotypic repair frequency, mESC (%)	36	69	30	30	53	33	78	ND	ND	ND	ND	ND	ND
Observed frequency of repair to wild-type genotype among all edited products in HTS, mESC (%)	ND	67	39	39	25	15	65	48	76	59	42	ND	ND
Observed frequency of repair to wild-type genotype among all edited products in HTS, U2OS (%)	100	88	ND	ND	ND	ND	77	ND	ND	ND	ND	ND	ND
Observed frequency of repair to wild-type genotype among all edited products in HTS, HCT116 (%)	ND	ND	ND	ND	24	ND	89	ND	ND	ND	ND	ND	ND

TABLE 3-continued

Repair of eight pathogenic microduplication alleles in individual cellular experiments.												
Pathogenic allele	LDLRdup1	LDLRdup2	LDLRdup2	LDLRdup3	LDLRdup4	LDLRdup5	PORCNDup	GAAdup	GAADup	GLB1dup	HPS1dup	ATP7Adup
Observed frequency of repair to wild-type genotype among all edited products in Lib-B, mESCs (%)	ND	ND	ND	ND	ND	58	42	ND	63	41	ND	ND
Observed frequency of repair to wild-type genotype among all edited products in primary patient fibroblasts (%)	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	88 ± 14*	98
gRNA sequence (SEQ ID Nos: 11577-11588) from left to right	TGCGA AGATG GCTCG GAGGC	GCAAG GATAA ATTG ACAGG	TCTC GTCAG ATTG TCTG	CTGCA AGGAC AAATC TGAGG	TTTCC TCGTC AGATT TGTCG	ACATC TACTC GCTGG TGAGC	CTGTC CCTGG CTTTT ATCCC	AGCTG CAGAA GGTGA CTGCA	CTGCA GAGG TGA GCAGA	TGTGA ACTAT GGTGC ATATA	CAGCA GGGGA GGCC CCAGC	TTTTT CCATA TAAGA TAAGA
Cas9 Type	KKH SaCas9	KKH SaCas9	KKH SaCas9	KKH SaCas9	KKH SaCas9	SaCas9	SaCas9	KKH SaCas9	SaCas9	SaCas9	SaCas9	SaCas9

ND, not determined.

LDLRdup1, LDLR:c.526_533dupGGCTCGGA. LDLRdup2, LDLR:c.668_681dupAGCAGAAATCTGAC. (SEQ ID NO: 1)

LDLRdup3, LDLR:c.669_680dupGGAGCAAAATCTGA. LDLRdup4, LDLR:c.672_683dupCAAAATCTGAGCA. LDLRdup5, LDLR:c.1662_1669dupGCTGGTGA. (SEQ ID Nos: 2 and 3)

PORCNDup, PORC:c.1059_1071dupCCTGGCTTTTATC. GAAdup, GAA:c.2704_2716dupCAGAGGTGACTG. (SEQ ID Nos: 4 and 5)

GLB1dup, GLB1:c.1456_1466dupGGTGCATATAT (SEQ ID NO: 6)

Table 4: Lib-A sequences (presented below between the end of this specification and Table 5).

Table 5: Lib-B sequences (presented below between Table 4 and Table 6).

Table 6: inDelphi predictions and observed results. Table 6 comprises Table 6A: inDelphi predictions and observed results for Lib-B, showing all sequences with replicate-consistent mESC results; Table 6B (continued from 6A); Table 6C (continued from 6B); Table 6D (continued from 6C); and Table 6E (continued from 6D) (presented below between Table 5 and the claims).

TABLE 7

Frequency of gRNAs in the human genome with denoted Cas9-mediated outcome precision						
Precision-X threshold (%)	inDelphi trained on Lib-A data from mESCs for 1-bp ins. module			inDelphi trained on Lib-A data from U2OS cells for 1-bp ins. module		
	Precise product is a deletion (% of gRNAs)	Precise product is a 1-bp insertion (% of gRNAs)	Total % of gRNAs that are precise-X	Precise product is a deletion (% of gRNAs)	Precise product is a 1-bp insertion (% of gRNAs)	Total % of gRNAs that are precise-X
10	82	38	93	70	78	97
15	61	23	75	44	64	87
20	43	15	55	27	53	72
25	30	10	39	17	44	58
30	21	6.6	28	11	36	46
35	15	4.4	19	6.9	28	34
40	10	2.9	13	4.1	21	25
45	6.5	1.9	6.4	2.4	15	18
50	4.3	1.3	5.6	1.4	10	12
55	2.8	0.8	3.6	0.6	6.7	7.5
60	1.8	0.5	2.3	0.5	4.0	4.4
65	1.1	0.3	1.5	0.2	2.2	2.4
70	0.7	0.2	0.9	0.1	1.1	1.2
75	0.4	0.1	0.5	0.04	0.5	0.5
80	0.2	0.06	0.3	0.01	0.2	0.2
85	0.06	0.04	0.1	0.003	0.07	0.06
90	0.03	0.02	0.05	0.0007	0.03	0.03

TABLE 8

Endogenous repair of 24 designed high-precision gRNAs in human cell lines				
Gene, exon/chr, outside (hg19)	Observed frequency among all edited products from deep sequencing at endogenous loci (%)			
	Frameshift, U2OS	Most frequent genotype, U2OS	Frameshift HEK293T	Most frequent genotype, HEK293T
VEGFA exon1: 456	91, 67	36, 34*	90, 90	43, 40*
VEGFR2 exon5: 2	91, 91	50, 53*	91, 91	50, 24*
PDCD1 exon5: 208	90, 90	20, 21*	91, 90	29, 13*
APOB exon25: 147	83, 83	22, 21*	87, 85	35, 18*
VEGFA exon3: 127	85, 89	27, 29*	93, 91	55, 32*
CCR5 exon1: 1941	82, 81	20, 21*	86, 84	43, 27*
CD274 exon2: 271	85, 86	9, 10*	84, 82	31, 14*
APOB exon26: 5590	91, 89	26, 25*	88	37*
VEGFR2 exon26: 19	82, 82	35, 33*	82, 82	40, 24*
CXCR4 exon1: 825	86, 86	32, 33*	91	54*
PCSK9 exon11: 15	81, 78	28, 25†	78	27†
CCR5 exon1: 885	84, 85	55, 52†	67	46†

TABLE 8-continued

Endogenous repair of 24 designed high-precision gRNAs in human cell lines				
Gene, exon/chr, outside (hg19)	Observed frequency among all edited products from deep sequencing at endogenous loci (%)			
	Frameshift, U2OS	Most frequent genotype, U2OS	Frameshift HEK293T	Most frequent genotype, HEK293T
CCR5 exon1: 1027	92, 94	61, 60†	91, 92	49, 58†
APOB exon26: 5573	93, 93	75, 74†	93, 95	69, 81†
CCR5 exon1: 61	94, 94	37, 25†	83, 89	29, 38†
CCR5 exon1:1577	61, 61	28, 29†	80, 83	29, 43†
APOB exon22: 100	89, 89	25, 27†	91, 89	23, 38†
APOBEC3B exon3: 202	83, 84	50, 52†	75, 88	51, 60†
MACCHC chr1: 45973892	97, 95	80, 77†‡	97, 98	78, 85†‡
PROK2 chr3: 71821967	93, 94	44, 41†‡	93, 93	45, 53†‡
IDS chrX: 148564700	95, 95	72, 74†‡	93, 95	64, 80†‡
ECM1 chr1: 150484936	87, 89	44, 47†‡	89, 89	32, 35†‡

TABLE 8-continued

Endogenous repair of 24 designed high-precision gRNAs in human cell lines				
Observed frequency among all edited products from deep sequencing at endogenous loci (%)				
Gene, exon/chr, outsite (hg19)	Frameshift, U2OS	Most frequent genotype, U2OS	Frameshift HEK293T	Most frequent genotype, HEK293T
KCNH2	40	25†‡	65, 85	35, 14†‡
chr7: 150644566				
LDLR	90, 91	76, 77†‡	90, 96	77, 83†‡
chr19: 11222303				

Discussion

The Cas9-mediated end-joining repair products of thousands of target DNA loci integrated into mammalian cells were used to train a machine learning model, inDelphi, that accurately predicts the spectrum of genotypic products resulting from double-strand break repair at a target DNA site of interest. The ability to predict Cas9-mediated products enables new precision genome editing research applications and facilitates existing applications.

The inDelphi model identifies target loci in which a substantial fraction of all repair products consists of a single genotype. The findings suggest that 26% of SpCas9 gRNAs targeting the human genome are precision gRNAs, yielding a single genotypic outcome in $\geq 30\%$ of all major repair products, and 5% are high-precision gRNAs in which $\geq 50\%$ of all major repair products are of a single genotype. Such precision and high-precision gRNAs enable uses of Cas9 nuclease in which the major genotypic products can be predicted a priori. Indeed, it was experimentally shown that high-precision, template-free Cas9-mediated editing can mediate efficient gain-of-function repair at hundreds of pathogenic alleles including microduplications (FIGS. 15B, 15E, 15F) in cell lines and in patient-derived primary cells (Table 1).

Moreover, evidence is presented that manipulation of available DNA repair pathways can further increase the precision of template-free repair outcomes. Suppressing NHEJ augments repair of pathogenic microduplication alleles, suggesting that temporary manipulation of DNA repair pathways could be combined with Cas9-mediated editing to favor specific editing genotypes with high precision. Genome editing currently lacks flexible strategies to correct indels in post-mitotic cells because of the limited efficiency of HDR in non-dividing cells³⁹. As MMEJ is thought to occur throughout the cell cycle⁴⁰, inDelphi may provide access to predictable and precise post-mitotic genome editing in a wider range of cell states. It is also anticipated that, given appropriate training data, inDelphi will also be able to accurately predict repair genotypes from other double-strand break creation methods, including other Cas9 homologs, Cpf1, transcription activator-like effector nucleases (TALENs), and zinc-finger nucleases (ZFNs)^{37,41-43}. This work establishes that the prediction and judicious application of template-free Cas9 nuclease-mediated genome editing offers new capabilities for the study and potential treatment of genetic diseases.

Cellular Repair of Double-Stranded DNA Breaks and inDelphi

DNA double-strand breaks are detrimental to genomic stability, and as such the detection and faithful repair of genomic lesions is crucial to cellular integrity. A large number of genes have evolved to respond to and repair DNA double-strand breaks, and these genes can be broadly grouped into a set of DNA repair pathways²⁶, each of which differs in the biochemical steps it takes to repair DNA double-strand breaks. Accordingly, these pathways tend to produce characteristically distinguishable non-wildtype genotypic outcomes.

The goal of the machine learning algorithm, inDelphi, is to accurately predict the identities and relative frequencies of non-wildtype genotypic outcomes produced following a CRISPR/Cas9-mediated DNA double-strand break. To accomplish this goal, parameters were developed to classify three distinct categories of genotypic outcomes, microhomology deletions, microhomology-less deletions, and insertions, informed by the biochemical mechanisms underlying the DNA repair pathways that typically give rise to them.

Double strand breaks are thought to be repaired via four major pathways: classical non-homologous end-joining (c-NHEJ), alternative-NHEJ (alt-NHEJ), microhomology-mediated end-joining (MMEJ), and homology-directed repair (HDR)¹. To create inDelphi, three machine learning modules were developed to model genotypic outcomes assuming characteristic of the c-NHEJ, microhomology mediated alt-NHEJ, and MMEJ pathways. While template-free CRISPR/Cas9 DNA double-strand break may lead to HDR repair via endogenous homology templates that exist in trans⁴⁵, HDR-characteristic outcomes are not explicitly modeled using the algorithm.

Before proceeding, it is important to note that while specific DNA repair pathways are characteristically associated with distinct genotypic outcomes, the proteins involved in the various pathways and the resulting repair products may at times overlap. This fact has several implications. First, conclusive statements cannot be made about the role of specific proteins or pathways in specific genotypic outcomes without perturbation experiments (e.g. the comparison of wildtype and *Prkdc*^{-/-}*Lig4*^{-/-} mESCs can illuminate the roles of these proteins, specifically). Second, because assigning genotypic outcomes to biochemical mechanisms is likely imperfect, machine learning methods were used to identify trends and patterns in genotype frequencies that refine this crude binning process.

In the first step of the inDelphi method, genotypic outcomes were separated into three classes: microhomology deletions (MH deletions), microhomology-less deletions (MH-less deletions), and single-base insertions (1-bp insertions) (FIG. 12A). Below the algorithmic definitions of each genotypic outcome class are outlined, the pathways associated with each class, and the DNA sequence parameters included in inDelphi training of each class. For more detailed technical algorithmic definitions of the genotypic outcome classes.

MH Deletions are Predicted from MH Length, MH GC Content, and Deletion Length

The majority of Cas9-mediated double-strand break repair genotypes observed in the datasets are what are classified as MH deletions (53-58% in mESC, K562, HCT116, and HEK293). It is hypothesized that these deletions occur through MMEJ-like processes and use known features of this pathways to inform a machine learning module to predict MH deletion outcomes. Following 5'-end resection as occurs in MMEJ, alt-NHEJ, and HDR²⁶, micro-

67

homologous basepairing of single-stranded DNA (ssDNA) sequences occurs across the border of the double strand breakpoint^{46, 47}. To restore a contiguous double-strand DNA chain, the 5'-overhangs not participating in the microhomology are removed up until the paired microhomology region, and the unpaired ssDNA sequences are extended by DNA polymerase using the opposing strand as a template (FIG. 12B, FIGS. 18A-18H).

Assuming these same processes, inDelphi calculates the set of all MH deletions available given a specific sequence context and cleavage site.

As an example workflow, given the following sequence and its cleavage site:

```

                                (SEQ ID NO: 7)
ACGTG|CATGA

                                (SEQ ID NO: 11589)
TGCAC|GTACT

```

for every possible deletion length from 1-bp to 60-bp deletions, the 3'-overhang is overlapped downstream of the cut site under the upstream 3'-overhang and it is determined if there is any microhomologous basepairing. As an example, given the 4-bp deletion length:

```

ACGTG
 |  ||
GTACT

```

it is seen that there are three microhomologous basepairing events.

Then a particular microhomology is chosen (here, the italicized C:G):

```

ACGTG
 |  ||
GTACT

```

then generate its unique repair genotype by following left-to-right along the top strand and jumping down to the complement of the bottom strand to simulate DNA polymerase fill-in.

Here, this yields:

```

ACATGA
TGTACT

```

This can also be displayed as an alignment. It is noted that by "jumping down" after the first base in the top strand, this outcome can also be described using the delta-position 1. (See section on delta-positions). A deletion at delta-position 0 yields the same genotype.

```

Deletion b:      ACGTG---A
Wt:              ACGTGCATGA (SEQ ID NO: 7)

```

Thus, there may be multiple MH deletion outcome genotypes for a given deletion length, and there is always a 1:1 mapping between the microhomologous basepairing used in that MH deletion and the resultant genotypic outcome. The set of MH deletions thus includes all 1-bp to 60-bp deletions that can be derived from the steps above that simulate the MMEJ mechanism.

68

MMEJ efficiency has been reported to depend on the thermodynamic favorability and stability of a candidate microhomology^{46, 47}. To parameterize MH deletions using the biochemical sequence features that influence this form of DNA repair, inDelphi calculates the MH length, MH GC content, and resulting deletion length for each possible MH deletion. These features are input into a machine learning module as the microhomology neural network (MH-NN) to learn the relationship between these features and the frequency of an MH deletion outcome in a training CRISPR/Cas9 genotypic outcome dataset. While it was predicted and empirically found that favored MH deletions have long MH lengths relative to total deletion length and high MH GC-contents, any explicit direction or comparative weighting to these parameters are not provided at the outset. inDelphi then outputs a phi-score for any MH deletion genotype (whether it was in the training data or not) that represents the favorability of that outcome as predicted by MH-NN.

It is important to emphasize that the phi-score of a particular MH deletion does not itself represent the likelihood of that MH deletion occurring in the context of all MH deletions at a given site. Some CRISPR/Cas9 target sites may have many possible favorable MH deletion outcomes while other sites have few, and thus phi-score must be normalized for a given target site to generate the fractional likelihood of that genotypic outcome at that site. Total unnormalized MH deletion phi-score is one factor that is further used to predict the relative frequency of the different repair classes: MH deletions, MH-less deletions, and insertions.

MH-Less Deletions are Predicted from their Length

MH-less deletions are defined as all possible deletions that have not been accounted for by the workflow described above for MH deletions. Mechanistically, the data analysis suggests that MH deletions are associated with repair genotypes produced by c-NHEJ and microhomology-mediated alt-NHEJ pathways.

Following a double-strand break, c-NHEJ-associated proteins rapidly bind the DNA strands flanking the double-strand DNA breakpoint and recruit ligases, exonucleases, and polymerases to process and re-anneal the breakpoint in the absence of 5'-end resection (FIGS. 18A-18H)^{26, 35}. Commonly, c-NHEJ repair is error-free; however, in the context of Cas9-mediated cutting, faithful repair leads to repeated cutting, thereby increasing the eventual likelihood of mutagenic repair. Erroneous c-NHEJ repair products are mainly thought to consist of small insertions or deletions or combinations thereof that most frequently occur in the direct vicinity of the DNA break point^{35, 48, 49}. The resulting deletions, which are referred to as medial end-joining MH-less deletions, have often lost bases both upstream and downstream of the cleavage site.

Microhomology-mediated alt-NHEJ is a distinct pathway that produces MH-less deletion products. In contrast to c-NHEJ, which is microhomology independent, this form of alt-NHEJ repair occurs following 5'-end resection and is mediated by microhomology in the sequence surrounding the double-strand break-point1. Microhomologous basepairing stabilizes the 3'-ssDNA overhangs following 5'-end resection, similarly to in MMEJ, allowing DNA ligases to join the break across one of the strands of this temporarily configured complex. The opposing un-annealed flap is then removed, and newly synthesized DNA templated off of the remaining strand is annealed to repair the lesion (FIGS. 18A-18H).

While alt-NHEJ uses microhomology, the repair products it produces do not follow the predictable genotypic patterns

induced by MMEJ and are thus grouped into MH-less deletion genotypes. MH deletions are a direct merger of both annealed strands, in which the outcome genotype switches from top to bottom strand at the exact end-point of a microhomology. In contrast, while alt-NHEJ employs microhomology in its repair mechanism, the deletion outcomes it generates comprise bases exclusively derived from either the top or bottom strand. Mechanistically, this occurs because ligation of a 3'-overhang to its downstream ligation partner results in removal of the entire opposing ssDNA overhang up until the point of ligation. This process prevents any deletion from occurring in the 3'-overhang strand that is first attached to the DNA backbone, while inducing loss of an indeterminate length of sequence on the opposing strand. The resulting deletion genotypes, which are referred to as unilateral end-joining MH-less deletions, do not retain information on the exact microhomology causal to their occurrence, and are thus also referred to as MH-less.

Consequently, the various mechanisms that give rise to MH-less deletions are capable of generating a vast number of genotypic outcomes for any given deletion length. Having less information on the biochemical mechanisms that impact the relative frequency of NHEJ deletion products, inDelphi models these deletions without assuming any particular mechanism. inDelphi detects MH-less deletions from training data as the set of all deletions that are not MH deletions and parameterizes them solely by the length of the resulting deletion. This is based on the simple assumption that c-NHEJ and alt-NHEJ processes are most likely to produce short deletions, supported by the empirical observation. As with MH deletions, this assumption is not explicitly coded into the inDelphi MH-less deletion prediction module, instead allowing it to be "learned" by a neural network called MHless-NN.

MHless-NN optimizes a phi-score for a given MH-less deletion length, grounded in the frequency of MH-less deletion outcomes of that length observed in the training data. It was observed that MHless-NN learns a near-exponential decaying phi-score for increasing deletion length, that reflects the sum total frequency of all MH-less deletion genotypes. The total unnormalized MH-less deletion phi-score for a given target and cut site is also employed to inform the relative frequency of different repair classes.

1-Bp Insertions are Predicted from Sequence Context and Deletion Phi-Scores

Lastly, inDelphi predicts 1-bp insertions from both the broader sequence context and the immediate vicinity of the cleavage site. It was empirically found that 1-bp insertions are far more common than longer insertions, so the focus is on their prediction. It is classically assumed that short sequence insertions are the result of c-NHEJ^{48,49}, however, little else is known about their biochemical mechanism as it pertains to local sequence context to help inform prediction. Nonetheless, powerful correlations were found between the identities of the bases surrounding the Cas9 cleavage site and the frequency and identity of the inserted base (see main text). Motivated by these empirical observations, inDelphi is fed with training data on 1-bp insertion frequencies and identities at each training site parameterized with the identities of the -3, -4, and -5 bases upstream of the NGG PAM-sequence (when the training set is sufficiently large, and the -4 base alone when training data is limited) as features. Also added as features are the precision score of the deletion length distribution and the total deletion phi-score at that site. These features are combined into a k-nearest neighbor algorithm that predicts the relative frequencies and identities of 1-bp insertion products at any target site.

The Combination of the MH, MH-Less, and Insertion Model Predict Genotype Fractions

Altogether, informed by known paradigms of DNA repair, 2 neural networks and a k-nearest neighbor model were built to predict genotypic outcomes following Cas9 cutting. These models compete and collaborate in inDelphi to generate predictions of the relative frequencies of these products. This competition within inDelphi among repair types reflects empirical evidence from Lib-A and Lib-B that sequence contexts do influence classes of repair outcomes. Sequence contexts with high phi scores (high microhomology) have higher efficiencies of MH deletions among all editing outcomes (FIG. 14B, FIGS. 20A-20E), and sequence contexts with low phi scores (low microhomology) have higher efficiencies of 1-bp insertions among all editing outcomes (FIGS. 14C, 14D, FIGS. 18A-18H, FIGS. 20A-20E). While it is tempting to generalize that the competition and collaboration among outcome classes modeled by inDelphi reflects interactions among components of distinct DNA repair pathways, the classes of outcomes considered by inDelphi do not necessarily arise from distinct DNA repair pathways as they are described above. inDelphi is trained on the repair outcomes only and cannot distinguish between the nature of genotypes when they may occur through MH-mediated and MH-less mechanisms, and it is imaginable that some repair products result through more than one repair pathway.

As an additional note, while NHEJ is generally assumed to dominate double-strand break repair from environmentally induced damage³⁵, it was found in the context of Cas9 cutting that MH deletion genotypes are more common than MH-less deletions and insertions. It is possible that error-free c-NHEJ is occurring frequently in response to Cas9 cutting but that its perfect repair allows for recurring Cas9 cutting that goes undetected by the workflow, thus skewing the observed relative frequency profile of mutagenic outcomes toward MMEJ-type repair.

Prkdc^{-/-}Lig4^{-/-} Mutants have Distinct and Predictable DNA Repair Product Distributions

While it is generally true that the work cannot establish roles for specific DNA repair pathways in specific types of Cas9-mediated outcomes, an experiment has been performed in which Cas9-mediated genotypic outcomes were measured from mESCs that are lacking Prkdc and Lig4, two proteins known to be key in c-NHEJ5. An increase in relative frequency of MH deletions was found as compared to MH-less deletions in Prkdc^{-/-}Lig4^{-/-} mESCs as compared to wild-type mESCs (see main text), which is suggestive of an increase in MMEJ outcomes at the expense of NHEJ outcomes.

Intriguingly, it was also found that Prkdc^{-/-}Lig4^{-/-} mESCs are impaired in unilateral deletions, where only bases from one side of the cutsite are removed, but not medial MH-less deletion outcomes that have loss of bases on both sides of the breakpoint. (FIGS. 22A-22E). As discussed earlier, microhomology-mediated alt-NHEJ, which it was hypothesized may give rise to unilateral MH-less deletions, proceeds through a mechanism in which DNA repair intermediates that mimic MMEJ-mediated repair are formed initially (FIGS. 18A-18H), as microhomology basepairing temporarily stabilizes 3'-overhangs following 5'-end resection. Subsequently, ligation joins one 3' overhang with the sequence on the other side of the DNA double-strand break, giving rise to a unilateral deletion. If the unilateral joining products observed in the experiments indeed arise through similar mechanisms as those described by this form of alt-NHEJ, it is conceivable that the MMEJ pathway may

overtake 3'-end ligation at this microhomology-containing intermediate step when ligation is impaired through loss of Lig4. Thus, cross-talk of microhomology-mediated repair pathways could account for loss of unilateral end-joining MH-less outcomes and concomitant increase in MH deletion outcomes. Medial joining outcomes are not hypothesized to originate from intermediates that overlap with microhomology-mediated deletion products (FIGS. 18A-18H). Therefore, the repair genotypes generated via this orthogonal pathway may be afforded more time to be completed by ligases other than Lig4, thus explaining why these outcomes appear unaffected by NHEJ impairment.

While DNA repair products in *Prkdc*^{-/-}*Lig4*^{-/-} mESCs differ substantially from those in wild-type cells, it was found that these DNA repair products are also highly predictable. In particular, inDelphi performed well on held-out *Prkdc*^{-/-}*Lig4*^{-/-} data when trained on *Prkdc*^{-/-}*Lig4*^{-/-} data (indel genotype prediction median Pearson correlation=0.84, indel length frequency prediction Pearson correlation=0.80), showing that the modeling approach is robustly capable of learning accurate predictions for Cas9 editing data in not just wild-type experimental settings but also settings with significant biochemical perturbation. As such, it is suggested here that inDelphi's modeling approach can be useful on additional tasks unexplored here provided that inDelphi is supplied with appropriate training data.

Methods

Library Cloning.

Specified pools of 2000 oligos were synthesized by Twist Bioscience and amplified with NEBNext polymerase (New England Biolabs) using primers OligoLib_Fw and OligoLib_Rv (see below), to extend the sequences with overhangs complementary to the donor template used for circular assembly. To avoid over-amplification in the library cloning process, qPCR was first performed by addition of SybrGreen Dye (Thermo Fisher) to determine the number of cycles required to complete the exponential phase of amplification. The PCR reaction was run for half of the determined number of cycles at this stage. Extension time for all PCR reactions was extended to 1 minute per cycle to prevent skewing towards GC-rich sequences. The 246-bp fragment was purified using a PCR purification kit (Qiagen).

Separately, the donor template for circular assembly was amplified with NEBNext polymerase (New England Biolabs) for 20 cycles from an SpCas9 sgRNA expression plasmid (Addgene 71485)³⁴ using primers CircDonor_Fw and CircDonor_Rv (see below) to amplify the sgRNA hairpin and terminator, and extended further with a linker region meant to separate the gRNA expression cassette from the target sequence in the final library. The 146-bp amplicon was gel-purified (Qiagen) from a 2.5% agarose gel.

The amplified synthetic library and donor templates were ligated by Gibson Assembly (New England Biolabs) in a 1:3 molar ratio for 1 hour at 50° C., and unligated fragments were digested with Plasmid Safe ATP-Dependent DNase (Lucigen) for 1 hour at 37° C. Assembled circularized sequences were purified using a PCR purification kit (Qiagen), linearized by digestion with SspI for ≥3 hours at 37° C., and the 237-bp product was gel purified (Qiagen) from a 2.5% agarose gel.

The linearized fragment was further amplified with NEBNext polymerase (New England Biolabs) using primers PlasmidIns_Fw and PlasmidIns_Rv (see below) for the addition of overhangs complementary to the 5'- and 3'-regions of a Tol2-transposon containing gRNA expression

plasmid (Addgene 71485)³⁴ previously digested with BbsI and XbaI (New England Biolabs), to facilitate gRNA expression and integration of the library into the genome of mammalian cells. To avoid over-amplification, qPCR was performed by addition of SybrGreen Dye (Thermo Fisher) to determine the number of cycles required to complete the exponential phase of amplification, and then ran the PCR reaction for the determined number of cycles. The 375-bp amplicon was gel-purified (Qiagen) from a 2.5% agarose gel.

The 375-bp amplicon and double-digested Tol2-transposon containing gRNA expression plasmid were ligated by Gibson Assembly (New England Biolabs) in a 3:1 ratio for 1 hour at 50° C. Assembled plasmids were purified by isopropanol precipitation with GlycoBlue Coprecipitant (Thermo Fisher) and reconstituted in milliQ water and transformed into NEB10beta (New England Biolabs) electrocompetent cells. Following recovery, a small dilution series was plated to assess transformation efficiency and the remainder was grown in liquid culture in DRM medium overnight at 37° C. A detailed step-by-step library cloning protocol is provided below.

The plasmid library was isolated by Midiprep plasmid purification (Qiagen). Library integrity was verified by restriction digest with SapI (New England Biolabs) for 1 hour at 37° C., and sequence diversity was validated by high-throughput sequencing (HTS) as described below.

Library Cloning Primers

```
OligoLib_Fw                                     (SEQ ID NO: 8)
TTTTTGTGTTTCTGTGTTCCGTTGTCCGTGCTGTAACGAAAGGATGGGTGC
GACGCGTCAT

OligoLib_Rv                                     (SEQ ID NO: 9)
GTTGATAACGGACTAGCCTTATTTAACTTGCTATGCTGTTTCCAGCATA
GCTCTTAAAC

CircDonor_Fw                                     (SEQ ID NO: 10)
GTTTAAGAGCTATGCTGGAAACAGC

CircDonor_Rv                                     (SEQ ID NO: 11)
ATGACGCGTCGACCCATCCTTTTCGTTACAGCAGCGACAACGGAACACAG
AAAACAAAAAGCACCGACTC

PlasmidIns_Fw                                    (SEQ ID NO: 12)
GTAAGTGAAGTATTTTCGATTTCTTGGCTTTATATATCTTGTGAAAGG
ACGAAACACC

PlasmidIns_Rv                                    (SEQ ID NO: 13)
TTGTGGTTTGTCCAACTCATCAATGTATCTTATCATGTCTGCTCGAAGC
GGCCGTACCTCTAGATTCAGACGTGTGCTCTTCCGATCT
```

Cloning.

A base plasmid was constructed starting from a Tol2-transposon containing plasmid (Addgene 71485)³⁴. The sequence between Tol2 sites was replaced with a CAGGS promoter, multi-cloning site, P2A peptide sequence followed by eGFP sequence, and Puromycin resistance cassette

to produce p2T-CAG-MCS-P2A-GFP-PuroR. The full sequence of this plasmid is appended in the Sequences section below, and this plasmid has been submitted to Addgene.

Plasmids with this backbone and containing wildtype and micro-duplication mutation versions of LDLR and three other genes, GAA, GLB1, and PORCN, were constructed. Information on cloning these genes is provided below, and the gene sequences are appended below.

LDLR: To generate p2T-CAGGS-LDLRwt-P2A-GFP-PuroR, LDLR (NCBI Gene ID #3949, transcript variant 1 CDS) was PCR amplified from a base plasmid ordered from the Harvard PlasmID resource core and cloned between the BamHI and NheI sites of the base plasmid.

The following mutants were generated through InFusion (Clontech) cloning. Sequences are provided below, and the internal allele nomenclature is in parentheses:

LDLR:c.526_533dupGGCTCGGA (LDLRdup252)
(SEQ ID NO: 14)
LDLR:c.668_681dupAGGACAAATCTGAC (LDLRdup254/255)
(SEQ ID NO: 15)
LDLR:c.669_680dupGGACAAATCTGA (LDLRdup258)
(SEQ ID NO: 16)
LDLR:c.672_683dupCAAATCTGACGA (LDLRdup261)
LDLR:c.1662_1669dupGCTGGTGA (LDLRdup264)

PORCN: NCBI Gene ID #64840, transcript variant C CDS was PCR amplified from HCT116 cDNA and cloned between the BamHI and NheI sites of the base plasmid. PORCN:c.1059_1071dupCCTGGCTTTTATC (SEQ ID NO: 17) (PORCNdup20) was generated through InFusion cloning.

GLB1: NCBI Gene ID #2720, transcript variant 1 CDS was PCR amplified from HCT116 cDNA and cloned between the BamHI and NheI sites of the base plasmid. GLB1:c.1456_1466dupGGTGCATATAT (SEQ ID NO: 18) (GLB1dup84) was generated through InFusion cloning.

GAA: NCBI Gene ID #2548, transcript variant 1 CDS was PCR amplified from a base plasmid ordered from the Harvard PlasmID resource core and cloned between the BamHI and NheI sites of the base plasmid. GAA:c.2704_2716dupCAGAAGGTGACTG (SEQ ID NO: 19) (GAAdup327/328) was generated through InFusion cloning.

SpCas9¹ and KKH SaCas9⁹ were constructed starting from a Tol2-transposon containing plasmid (Addgene 71485)³⁴. The sequence between Tol2 sites was replaced with a CAGGS promoter, Cas9 sequence, and blasticidin resistance cassette to produce p2T-CAG-SpCas9-BlastR and p2T-CAG-KKHSaCas9-BlastR. These plasmids have been submitted to Addgene.

SpCas9 guide RNAs were cloned as a pool into a Tol2-transposon containing gRNA expression plasmid (Addgene 71485)³⁴ using BbsI plasmid digest and Gibson Assembly (NEB). SaCas9 guide RNAs were cloned into a similar Tol2-transposon containing SaCas9 gRNA expression plasmid (p2T-U6-sgsaCas2xBbsI-HygR) which has been submitted to Addgene using BbsI plasmid digest and Gibson Assembly. Protospacer sequences used are listed below, using the internal nomenclature which matches the duplication alleles.

LDLR gRNAs
sgsaLDLRdup252: (SEQ ID NO: 20)
GCTGCGAAGATGGCTCGGAGGC
sgsaLDLRdup254: (SEQ ID NO: 21)
GTGCAAGGACAAATCTGACAGG
sgsaLDLRdup255: (SEQ ID NO: 22)
GTTCTCTCGTCAGATTTGCTCTG
sgsaLDLRdup258: (SEQ ID NO: 23)
GACTGCAAGGACAAATCTGAGG
sgsaLDLRdup261: (SEQ ID NO: 24)
GTTTTCTCTCGTCAGATTTGTCTG
sgspLDLRdup264: (SEQ ID NO: 25)
GACATCTACTCGCTGGTGAGC
PORCN gRNAs
sgspPORCNdup20: (SEQ ID NO: 26)
GCTGTCCCTGGCTTTTATCCC
GLB1 gRNAs
sgspGLB1dup84: (SEQ ID NO: 27)
GTGTGAACTATGGTGCATATA
GAA gRNAs
sgsaGAAdup327: (SEQ ID NO: 28)
GCAGCTGCAGAAGGTGACTGCA
sgspGAAdup328: (SEQ ID NO: 29)
GCTGCAGAAGGTGACTGCAGA

Cell Culture.

Mouse embryonic stem cell lines used have been described previously and were cultured as described previously⁴⁴. HEK293T, HCT116, and U2OS cells were purchased from ATCC and cultured as recommended by ATCC. For stable Tol2 transposon plasmid integration, cells were transfected using Lipofectamine 3000 (Thermo Fisher) using standard protocols with equimolar amounts of Tol2 transposase plasmid²⁵ (a gift from Koichi Kawakami) and transposon-containing plasmid. For library applications, 15-cm plates with >10⁷ initial cells were used, and for single gRNA targeting, 6-well plates with >10⁶ initial cells were used. To generate lines with stable Tol2-mediated genomic integration, selection with the appropriate selection agent at an empirically defined concentration (blasticidin, hygromycin, or puromycin) was performed starting 24 hours after transfection and continuing for >1 week. In cases where sequential plasmid integration was performed such as integrating gRNA/target library and then Cas9 or micro-duplication plasmid and then Cas9 plus gRNA, the same Lipofectamine 3000 transfection protocol with Tol2 transposase plasmid was performed each time, and >1 week of appropriate drug selection was performed after each transfection. Deep Sequencing.

Genomic DNA was collected from cells after >1 week of selection. For library samples, 16 µg gDNA was used for each sample; for individual locus samples, 2 µg gDNA was used; for plasmid library verification, 0.5 µg purified plasmid DNA was used.

For individual locus samples, the locus surrounding CRISPR/Cas9 mutation was PCR amplified in two steps using primers >50-bp from the Cas9 target site. PCR1 was

performed using the primers specified below. PCR2 was performed to add full-length Illumina sequencing adapters using the NEBNext Index Primer Sets 1 and 2 (NEB) or internally ordered primers with equivalent sequences. All PCR2s were performed using NEBNext polymerase (New England Bioscience). Extension time for all PCR reactions was extended to 1 min per cycle to prevent skewing towards GC-rich sequences. The pooled samples were sequenced using NextSeq (Illumina) at the Harvard Medical School Biopolymers Facility, the MIT BioMicro Center, or the Broad Institute Sequencing Facility.

Library Prep Primers:

For LDLRDup252, 254, 255, 258, 261:
120417_LDLRDup254_r1_seq_A (SEQ ID NO: 30)
CTTTCCTACACGACGCTCTTCCGATCT NNN
ACTCCAGCTGGCGCTGTGAT
120417_LDLR254_r2seq_A (SEQ ID NO: 31)
GGAGTTCAGACGTGTGCTCTTCCGATCT
CAACTTCATCGCTCATGTCCTTG
For LDLRDup264:
120817_LDLR264_r1seq_B (SEQ ID NO: 32)
CTTTCCTACACGACGCTCTTCCGATCT
NNNAACTCCCGCCAAGATCAAGAAAG
120817_LDLR264_r2seq_B (SEQ ID NO: 33)
GGAGTTCAGACGTGTGCTCTTCCGATCT
CAGCCTCTTTTCATCCTCCAAGA
For PORCDup20:
120517_PORCN20_r1_seq (SEQ ID NO: 34)
CTTTCCTACACGACGCTCTTCCGATCT NNN
CCTCTACATGGCTTCAGTTTCC
120517_PORCN20_r2seq (SEQ ID NO: 35)
GGAGTTCAGACGTGTGCTCTTCCGATCT
CCAGAGCTCCAAAGAGCAAGTTT
For GLB1Dup84:
120517_GLB_184_r1seq (SEQ ID NO: 36)
CTTTCCTACACGACGCTCTTCCGATCT NNN
AGCCACTCTGGACCTTCTGGTA
120517_GLB_184_r2seq (SEQ ID NO: 37)
GGAGTTCAGACGTGTGCTCTTCCGATCT
CCAGTCCGTGAGGATATTGGAAC
For GAADup327/328:
120517_GAA327_r1seq (SEQ ID NO: 38)
CTTTCCTACACGACGCTCTTCCGATCT NNN
GATCGTGAATGAGCTGGTACGTG
120517_GAA327_r2seq (SEQ ID NO: 39)
GGAGTTCAGACGTGTGCTCTTCCGATCT
AACAGCGAGACACAGATGTCCAG

Data Availability.

High-throughput sequencing data have been deposited in the NCBI Sequence Read Archive database under accession codes SRP141261 and SRP141144.

5 Code Availability.

All data processing, analysis, and modeling code is available at github.com/maxwshen/inDelphi-dataprocessing-analysis. The inDelphi model is available online at the URLcrisprindelfi.design.

10 Library Cloning Protocol

Synthesized Oligo Library Sequence

(SEQ ID NO: 40)

GATGGGTGCGACGCGTCAT[55bpTarget]AGATCGGAAGAGCACA

CGTCTGAATATTGTGGA AAGGACGAAACACCG[19/20-nt

PROTOSPACER depending on whether it naturally

starts with a G] GTTTAAGAGCTATGCTGGAAACAGC

25 Linker Region/Oligo Library Amplification Primer
Anneal Region

Read 2 Sequencing Primer Stub

SspI Restriction Site

U6-Promoter Stub

30 sgRNA-Hairpin Stub

1. Oligo library QPCR to determine number of amplification cycles for Oligo Library PCR

Notes: Amplification of oligos with relatively low GC-content is less efficient than GC-rich sequences. It was found that NEBNext polymerase was the least biased in amplification of the library. Increasing the elongation time to 1 min per cycle for all cloning and sequencing library prep PCR2s eliminates GC-skewing of library sequences and reduces the rate of PCR-recombination.

40 Set up the following reaction:

0.4 ng	Synthesized Oligo Library
10 ul	NEBNext 2x Master Mix
0.5 ul	20 uM OligoLib_Fw
0.5 ul	20 uM OligoLib_Rv
0.2 ul	SybrGreen Dye (100x)
to 20 ul	H ₂ O

50 67° C. annealing temperature

Check 246 bp amplicon size on 2.5% agarose gel.

Determine the point that signal amplification has plateaued.

2. Oligo Library PCR Amplification

55 Set up the following reaction:

4 ng	Synthesized Oligo Library
50 ul	NEBNext 2x Master Mix
2.5 ul	20 uM OligoLib_Fw
2.5 ul	20 uM OligoLib_Rv
to 100 ul	H ₂ O

67° C. annealing temperature, 1 minute extension time.

65 Cycle number is half the number of cycles needed to reach signal amplification plateau in the QPCR in step 1, reduced by 1 cycle to scale for DNA input.

77

PCR purify amplified sequence. 3. Donor template amplification

Set up the following reaction:

5 ng	spCas9 sgRNA plasmid (71485)
50 ul	NEBNext 2x Master Mix
2.5 ul	20 uM CircDonor_Fw
2.5 ul	20 uM CircDonor_Rv
to 100 ul	H ₂ O

62° C. annealing temperature

20 cycles

Gel purify 167 bp band from 2.5% agarose gel.

4. Circular assembly and restriction digest linearization

Note: A molar ratio of donor template to amplified oligo library of 3:1 was used. An increase in amplified oligo library compounds cross-over within library members resulting in mismatch of protospacer and target sequences.

Set up the following reaction:

429 ng	Donor template
239 ul	Amplified Oligo Library
30 ul	Gibson Assembly 2x Master Mix
to 60 ul	H ₂ O

50° C. incubation for 1 hour.

Exonuclease treatment

60 ul	Circular assembly reaction
9 ul	ATP (25 mM)
9 ul	10x Plasmid Safe Buffer
3 ul	Plasmid Safe Nuclease
9 ul	H ₂ O

37° C. incubation for 1 hour.

PCR purify and elute in 50 ul.

Digest to linearize library

50 ul	Purified assemblies
10 ul	10x CutSmart Buffer
3 ul	SspI-HF
37 ul	H ₂ O

37° C. incubation for ≥3 hours.

Gel purify 273 bp band from 2.5% agarose gel.

Note: Band is sometimes fuzzy and poorly visible. If not clearly discernible, proceed with gel isolation between 200-300 bp.

5. Linearized Library QPCR to Determine Number of Amplification Cycles for PCR Amplification

Set up the following reaction:

0.5%	Purified linearized library
10 ul	NEBNext 2x Master Mix
0.5 ul	20 uM PlasmidIns_Fw
0.5 ul	20 uM PlasmidIns_Rv
0.2 ul	SybrGreen Dye (100x)
to 20 ul	H ₂ O

65° C. annealing temperature

Determine the point that signal amplification has plateaued.

78

6. Linearized Library PCR amplification

Set up the following reaction:

50%	Purified linearized library
50 ul	NEBNext 2x Master Mix
2.5 ul	20 uM PlasmidIns_Fw
2.5 ul	20 uM PlasmidIns_Rv
to 100 ul	H ₂ O

65° C. annealing temperature, 1 minute extension time.

Cycle number is number of cycles needed to reach signal amplification plateau in the QPCR in step 5, reduced by 4 cycles to scale for increased DNA input.

15 Gel purify 375 bp band from 2.5% agarose gel.

7. Vector backbone digest

Set up the following reaction:

2 ug	spCas9 sgRNA plasmid (71485)
10	10x Buffer 2.1
3	BbsI
2	XbaI
to 100 ul	H ₂ O

25 37° C. incubation for ≥3 hours.

Gel purify 5.9 kb band from 1% agarose gel.

8. Vector assembly and cleanup

Note: Include a ligation with water for insert as a control.

30 Set up the following reaction:

300 ng	Digested vector backbone
42 ng	Amplified Oligo Library
30 ul	Gibson Assembly 2x Master Mix
to 60 ul	H ₂ O

50° C. incubation for 1 hour.

Isopropanol precipitation

40 ul	Vector assembly reaction
0.4 ul	GlycoBlue Coprecipitant
0.8 ul	50 mM NaCl
38.8 ul	Isopropanol

45 Vortex and incubate at room temperature for 15 minutes. Spin down at ≥15,000 g for 15 minutes, and carefully remove supernatant.

50 Wash pellet with 300 ul 80% EtOH and repeat spin at ≥15,000 g for 5 minutes.

Carefully remove all liquid without disturbing pellet, and let air dry for 1-3 minutes.—Dissolve dried pellet in 10 ul H₂O at 55° C. for 10 minutes.

55 9. Transformation

Note: Electroporation competent cells give a higher transformation efficiency than chemically competent cells. NEB10beta electro-competent cells were used, however these can be substituted for other lines and transformed according to the manufacturer's instructions.

60 Note: DRM was used as recovery and culture medium to enhance yield. If substituting for a less rich medium such as LB, it is recommended scaling up the culture volume to obtain similar plasmid DNA quantities.

65 Note: Antibiotic-free recovery time should be limited to 15 minutes to prevent shedding of transformed plasmids from replicating bacteria.

Note: Also transform water ligation as control.

Pre-warm 3.5 mL recovery medium per electroporation reaction, at 37° C. for 1 hour.

Pre-warm LB-agar plates containing appropriate antibiotic.

Per reaction, add 1 ul purified vector assembly to 25 ul competent cells on ice.

Perform 8 replicate reactions.

Electroporate according to the manufacturer's instructions.

Immediately add 100 ul pre-warmed recovery media per cuvette and pool all replicates into culture flask.

Add 1 mL recovery media per replicate reaction to culture flask and shake at 200 rpm 37° C. for 10-15 minutes.

Plate a dilution series from 1:104-1:106 on LB-agar plates containing antibiotic and grow overnight at 37° C.

Add 2 mL media per replicate reaction and admix appropriate antibiotic.

Grow overnight in shaking incubator at 200 rpm 37° C.

Assess transformation efficiency from serial dilution LB-agar plates. Expect ~10⁶ clones.

The development of this cloning protocol was guided by work described in Videgal et al. 2015.

Sequence Alignment and Data Processing

For library data, each sequenced pair of gRNA fragment and target was associated with a set of designed sequence contexts G by finding the designed sequence contexts for all gRNAs whose beginning section perfectly matches the gRNA fragment (read 1 in general does not fully sequence the gRNA), and by using locality sensitive hashing (LSH) with 7-mers on the sequenced target to search for similar designed targets. An LSH score on 7-mers between a reference and a sequenced context reflects the number of shared 7-mers between the two. If the best reference candidate scored, through LSH, greater than 5 higher than the best LSH score of the reference candidates obtained from the gRNA-fragment, the LSH candidate is also added to G. LSH was used due to extensive (~33% rate) PCR recombination between read1 and read2 which in sequenced data appears as mismatched read1 and read2 pairs. The sequenced target was aligned to each candidate in G and the alignment with the highest number of matches is kept. Sequence alignment was performed using the Needleman-Wunsch algorithm using the parameters: +1 match, -1 mismatch, -5 gap open, -0 gap extend. For library data, starting gaps cost 0. For all other data, starting and ending gaps cost 0. For VO data, sequence alignments were derived from SAM files from SRA.

Alignments with low-accuracy or short matching sections flanked by long (10 bp+) insertions and deletions were filtered out as PCR recombination products (observed frequency of ~5%). These PCR recombination products are different than that occurring between read1 and read2; these occur strictly in read2. Alignments with low matching rates were removed. Deletions and insertions were shifted towards the expected cleavage site while preserving total alignment score. CRISPR-associated DNA repair events were defined as any alignment with deletions or insertions occurring within a 4 bp window centered at the expected cut site and any alignment with both deletions and insertions (combination indel) occurring with a 10 bp window centered at the expected cut site. All CRISPR-associated DNA repair events observed in control data had their frequencies subtracted from treatment data to a minimum of 0.

Replicate experiments were carried out for library data in each cell type. For each cell-type, each sequence context not fulfilling the following data quality criteria was filtered: data

at this sequence context in the two replicates with the highest read-counts must have at least 1000 reads of CRISPR editing outcomes in both replicates, and a Pearson correlation of at least 0.85 in the frequency of microhomology-based deletion events. The class of microhomology-based deletion events was used for this criterion since it is a major repair class with the highest average replicability across experiments. For disease library data in U2OS and HEK, a less stringent read count threshold of 500 was used instead.

Details on Alignment Processing

All alignments with gaps were shifted as much as possible towards the cleavage site while preserving the overall alignment score. Then, the following criteria were used to categorize the alignments into noise, not-noise but not CRISPR-associated (for example, wildtype); as well as primary and secondary CRISPR activity. All data used in modeling and analysis derive solely from outcomes binned into primary CRISPR activity.

The following criteria was used to filter library alignments into "noise" categories. Homopolymer: Entire read is homopolymer of a single nucleotide. Not considered a CRISPR repair product.

Has N: Read contains at least one N. Discarded as noise, not considered a CRISPR repair product.

PCR Recombination: Contains recombination alignment signature: (1) if a long indel (10 bp+) followed by chance overlap followed by long indel (10 bp+) of the opposite type, e.g., insertion-randommatch-deletion and deletion-randommatch-insertion. OR, if one of these two indels is 30 bp+, the other can be arbitrarily short. If either criteria is true, and if the chance overlap is length 5 or less, or any length with less than 80% match rate, then it satisfies the recombination signature. In addition, if both indels are 30 bp+, regardless of the middle match region, it satisfies the recombination signature. Finally, if randommatch is length 0, then indel is allowed to be any length. Not considered a CRISPR repair product.

Poor-Matches: 55 bp designed sequence context has less than 5 bp representation (could occur from 50 bp+ deletions or severe recombination) or less than 80% match rate. Not considered a CRISPR repair product.

Cutsite-Not-Sequenced: The read does not contain the expected cleavage site.

Other: An alignment with multiple indels where at least one non-gap region has lower than an 80% match rate. Or generally, any alignment not matching any defined category above or below. In practice, can include near-homopolymers. Not considered a CRISPR repair product.

The following criteria was used to filter library alignments into "main" categories.

Wildtype: No indels in all of alignment. Not considered a CRISPR repair product.

Deletion: An alignment with only a single deletion event. Subdivided into: Deletion—Not CRISPR: Single deletion occurs outside of 2 bp window around cleavage site. Not considered a CRISPR repair product.

Deletion—Not at cut: Single deletion occurring within 2 bp window around cleavage site, but not immediately at cleavage site. Considered a CRISPR repair product.

Deletion: Single deletion occurring immediately at cleavage site. Considered a CRISPR repair product.

Insertion: An alignment with only a single insertion event. Subdivided into:

Insertion—Not CRISPR: Single insertion occurs outside of 10 bp window around cleavage site. Not considered a CRISPR repair product.

Insertion—Not at cut: Single insertion occurring within 2 bp window around cleavage site, but not immediately at cleavage site. Considered a CRISPR repair product.

Insertion: Single insertion occurring immediately at cleavage site. Considered a CRISPR repair product.

Combination indel: An alignment with multiple indels where all non-gap regions have at least 80% match rate. Subdivided into:

Combination Indel: All indels are within a 10 bp window around the cleavage site. Considered a primary CRISPR repair product.

Forgiven Combination Indel: At least two indels, but not all, are within a 10 bp window around the cleavage site. Considered a rarer secondary CRISPR repair product, ignored.

Forgiven Single Indel: Exactly one indel is within a 10 bp window around the cleavage site. Considered a rarer secondary CRISPR repair product, ignored.

Combination Indel—Not CRISPR: No indels are within a 10 bp window around the cleavage site. Not considered a CRISPR repair product.

It is noted that deletion and insertion events, even those spanning many bases, are defined to occur at a single location between bases. As such, events occurring up to 5 bp away from the cleavage site are defined as events where there are five or fewer matched/mismatched alignment positions between the event and the cleavage site, irrespective of the number of gap dashes in the alignment.

Selection of Variants from Disease Databases

Disease variants were selected from the NCBI ClinVar database (downloaded Sep. 9, 2017)¹⁶ and the Human Gene Mutation Database (publicly available variant data from before 2014.3)¹⁷ for computational screening and subsequent experimental correction.

A total of 4,935 unique variants were selected from Clinvar submissions where the functional consequence is described as complete insertions, deletions, or duplications where the reference or alternate allele is of length less than or equal to 30 nucleotides. Variants were included where at least one submitting lab designated the clinical significance as ‘pathogenic’ or ‘likely pathogenic’ and no submitting lab had designated the variant as ‘benign’ or ‘likely benign’, including variants with all disease associations. More complex indels and somatic variants were included. A total of 18,083 unique insertion variants were selected from HGMD which were between 2 to 30 nucleotides in length. Variants were included with any disease association with the HGMD classification of ‘DM’ or disease-causing mutation.

SpCas9 gRNAs and their cleavage sites were enumerated for each disease allele. Using a previous version of inDelphi, genotype frequency and indel length distributions were predicted for each tuple of disease variant and unique cleavage site. Among each unique disease, the single best gRNA was identified as the gRNA inducing the highest predicted frequency of repair to wildtype genotype, and if this was impossible (due to, for example, a disease allele with 2+ bp deletion), then the single best gRNA was identified as the gRNA inducing the highest predicted frame-shift repair rate. 1327 sequence contexts were designed in this manner for Lib-B. An additional 265 sequence contexts were designed by taking the 265 sequence contexts in any disease in decreasing order of predicted wildtype repair rate, starting with Clinvar, stopping at 45% wildtype repair rate, then continuing with HGMD. This yielded 1592 total sequences derived from Clinvar and HGMD.

Definition of Delta-Positions

Using the MMEJ mechanism, deletion events can be predicted at single-base resolution. For computational convenience, the tuple (deletion length, delta-position) was used to construct a unique identifier for deletion genotypes. A delta-position associated with a deletion length N is an integer between 0 and N inclusive (FIGS. 19A-19D). In a sequence alignment, a delta-position describes the starting position of the deletion gap in the read w.r.t. the reference sequence relative to the cleavage site. For a deletion length N and a cleavage site at position C such that seq[:C] and seq[C:] yield the expected DSB products where the vector slicing operation vector[index1:index2] is inclusive on the first index and exclusive on the second index (python style), a delta-position of 0 corresponds to a deletion gap at seq[C-N+0: C+0], and generally with a delta-position of D, the deletion gap occurs at seq[C-N+D: C+D]. Microhomologies can be described with multiple delta-positions. To uniquely identify microhomology-based deletion genotypes, the single maximum delta-position in the redundant set is used. Microhomology-less deletion genotypes are associated with only a single delta position and deletion length tuple; this was used as its unique identifier.

Another way to define delta-positions can be motivated by the example workflow shown above on MH deletions describing how each microhomology is associated with a deletion genotype. In that workflow, the delta-position is the number of bases included on the top strand before “jumping down” to the bottom strand.

MH-less medial end-joining products correspond to all MH-less genotypes with delta-position between 1 and N-1 where N is the deletion length. MH-less unilateral end-joining products correspond to MH-less genotypes with delta-position 0 or N. It is noted that a deletion genotype with delta position N does not immediately imply that it is a microhomology-less unilateral end-joining product since it may contain microhomology (it’s possible that delta-positions N-j, N-j+1, . . . , N all correspond to the same MH deletion.)

Definition of Precision Score

For a distribution X, where |X| indicates its cardinality (or length when represented as a vector):

$$PrecisionScore(X) = 1 - \frac{-\sum_{i=1}^n P(x_i) \log(P(x_i))}{\log(|X|)}$$

This precision score ranges between zero (minimally precise, or highest entropy) to one (maximally precise, or lowest entropy).

inDelphi Deletion Modeling: Neural Network Input and Output

inDelphi receives as input a sequence context and a cleavage site location, and outputs two objects: a frequency distribution on deletion genotypes, and a frequency distribution on deletion lengths.

To model deletions, inDelphi trains two neural networks: MH-NN and MHless-NN.

MH-NN receives as input a microhomology that is described by two features: microhomology length and GC fraction in the microhomology. Using these features, MH-NN outputs a number (psi). MHless-NN receives as input the deletion length. Using this feature, MHless-NN outputs a number (psi).

A phi score is obtained from a psi score using: phi_i = exp(psi_i - 0.25 * deletion_length), where 0.25 is a “redundant”

hyperparameter that serves to reduce training speed by helpful scaling. This relationship between psi and phi is differentiable and encodes the assumption that the frequency of an event exponentially increases with neural network output psi (which empirically appears to reflect MH strength) and exponentially decreases with its minimum necessary resection length (deletion length).

The architecture of the MH-NN and MHless-NN networks are input-dimension $\rightarrow 16 \rightarrow 16 \rightarrow 1$ for a total of two hidden layers where all nodes are fully connected. Sigmoidal activations are used in all layers except the output layer. All neural network parameters are initialized with Gaussian noise centered around 0.

inDelphi Deletion Modeling: Making Predictions

Given a sequence context and cleavage site, inDelphi enumerates all unique deletion genotypes as a tuple of its deletion length and its delta-position for deletion lengths from 1 bp to 60 bp. For each microhomology enumerated, an MH-phi score is obtained using MH-NN. In addition, for each deletion length from 1 bp to 60 bp, an MHindep-phi score is obtained using MHless-NN.

inDelphi combines all MH-phi and MHindep-phi scores for a particular sequence context into two objects—a frequency distribution on deletion genotypes, and a frequency distribution on deletion lengths—which are both compared to observations for training. The model is designed to output two separate objects because both are of biological interest, and separate but intertwined modeling approaches are useful for generating both. By learning to generate both objects, inDelphi jointly learns about microhomology-based deletion repair and microhomology-less deletion repair.

To generate a frequency distribution on deletion genotypes, inDelphi assigns a score for each microhomology. Score assignment considers the concept of “full” microhomology and treats full and not full MHs differently.

A microhomology is “full” if the length of the microhomology is equal to its deletion length. The biological significance of full microhomologies is that there is only a single deletion genotype possible for the entire deletion length, while in general, a single deletion length is consistent with multiple genotypes. In addition, this single genotype can be generated through not just the MH-dependent MMEJ mechanism but also through MH-less end-joining, for example as mediated by Lig4. Therefore, full microhomologies were modelled as receiving contributions from both MH-containing and MH-less mechanisms by scoring full microhomologies as $MH\text{-}\phi[i] + MH\text{indep-}\phi[j]$ for deletion length j and microhomology index i . Microhomologies that are not “full” are assigned a score of $MH\text{-}\phi[i]$ for MH index i .

Scores for all deletion genotypes assigned this way are normalized to sum to 1 to produce a predicted frequency distribution on deletion genotypes.

To generate a frequency distribution on deletion lengths, inDelphi assigns a score for each deletion length. Score assignment integrates contributions from both MH-dependent and MH-independent mechanisms via the following procedure: For each deletion length j , its score is assigned as $MH\text{indep-}\phi[j]$ plus the sum of $MH\text{-}\phi$ for each microhomology with that deletion length. Scores for all deletion lengths are normalized to sum to 1 to produce a frequency distribution.

inDelphi trains its parameters using a single sequence context by producing both a predicted frequency distribution on deletion genotypes and deletion lengths and minimizing the negative of the sum of squared Pearson correlations for both objects to their observed versions. In practice, deletion

genotype frequency distributions are formed from observations for deletion lengths 1-60, and deletion length frequency distributions are formed from observations for deletion lengths 1-28. Both neural networks are trained simultaneously on both tasks. inDelphi is trained with stochastic gradient descent with batched training sets. inDelphi is implemented in Python using the autograd library. A batch size of 200, an initial weight scaling factor of 0.10, an initial step size of 0.10, and an exponential decaying factor for the step size of 0.999 per step were used.

inDelphi Deletion Modeling: Summary and Revisiting Assumptions

In summary, inDelphi trains MH-NN, which uses as input (microhomology length, microhomology GC content) to output a psi score which is translated into a phi score using deletion length. This phi score represents the “strength” of the microhomology corresponding to a particular MH deletion genotype. It also trains MHless-NN which uses as input (deletion length) to directly output a phi score representing the “total strength” of all MH-independent activity for a particular deletion length.

While the model assumes that microhomology and microhomology-less repair can overlap in contributions to a single repair genotype, this assumption is made conservatively by assuming that their contributions overlap only when there is no alternative. Specifically, in the context of a deletion length with full microhomology, the model assumes that they must overlap, while in the context of a deletion length without full microhomology, inDelphi allows MHindep-phi to represent all MH-less repair genotypes and none of the MH-dependent repair genotypes which are represented solely using their MH-phi scores. This can be seen by noting that at a deletion length j without full microhomology, MH genotypes are scored using their MH-phi scores, while the length j is scored by $MH\text{indep-}\phi[j]$ plus the sum of MH-phi for each microhomology. Therefore, the subset of MH-less genotypes at this deletion length have a score $MH\text{indep-}\phi[j]$.

When the subset of MH-less genotypes includes only one MH-less genotype, this single genotype’s score is equal to $MH\text{indep-}\phi[j]$. In general, multiple MH-less genotypes are possible, in which case the total score of all of the MH-less genotypes is equal to $MH\text{indep-}\phi[j]$.

The relative frequency of MH deletions and MH-less deletions is learned implicitly by the balancing between the sum of all MH-phi and MHindep-phi. Since MHindep-phi does not vary by sequence context while MH-phi does, the model assumes that variation in the fraction of deletions that use MH can at least partially be explained by varying sequence microhomology as represented by MH-NN.

inDelphi Insertion Modeling

Once inDelphi is trained on both deletion tasks, it predicts insertions from a sequence context and cleavage site by using the precision score of the predicted deletion length distribution and total deletion phi (from all MH-phi and MHindep-phi). inDelphi also uses one-hot-encoded binary vectors encoding nucleotides -4 and -3 . In a training set, these features are collected and normalized to zero mean and unit variance, and the fraction of 1-bp insertions over the sum counts of 1-bp insertions and all deletions are tabulated as the prediction goal. A k-nearest neighbor model is built using the training data. inDelphi uses the default parameter $k=5$.

On test data, the above procedure is used to predict the frequency of 1-bp insertions out of 1-bp insertions and all deletions for a particular sequence context. Once this frequency is predicted, it is used to make frequency predictions

for each of the 4 possible insertion genotypes, which are predicted by deriving from the training set the average insertion frequency for each base given its local sequence context. When the training set is small, only the -4 nucleotide is used. When the training set is relatively large, nucleotides -5, -4, and -3 are used.

To produce a frequency distribution on 1-bp insertions and 1-60 bp deletion genotypes, scores for all deletion genotypes and all 1-bp insertions are normalized to sum to 1. To produce a frequency distribution on indel lengths (+1 to -60), scores for all deletion lengths and 1-bp insertions are normalized to sum to 1.

inDelphi: Repair Classes Predicted at Varying Resolution

inDelphi predicts MH-deletions and 1-bp insertions at single base resolution. Measuring performance on the task of genotype frequency prediction considers this subset of repair outcomes only (about 60-70% of all outcomes). inDelphi predicts MH-less deletions to the resolution of deletion length. That is, inDelphi predicts a single frequency corresponding to the sum total frequency of all unique MH-less deletion genotypes possible for a particular deletion length. This modeling choice was made because genotype frequency replicability among MH-less deletions is substantially lower than among MH deletions.

Measuring performance on the task of indel length frequency considers MH deletions, MH-less deletions, and 1-bp insertions (90% of all outcomes).

In practice, if end-users desire, they can extend inDelphi predictions to frequency predictions for specific MH-less deletion genotypes by noting that MH-less deletions are distributed uniformly between 0 delta-position genotypes, medial genotypes, and N delta-position genotypes.

Comparison with a Linear Baseline Model

inDelphi was compared to a baseline model with the same model structure but replacing the deep neural networks with linear models. The comparison was done using Lib-A mESC data. While inDelphi achieves a mean held-out Pearson correlation of 0.851 on deletion genotype frequency prediction and 0.837 on deletion length frequency prediction, the linear baseline model achieves a mean held-out Pearson correlation of 0.816 on deletion genotype frequency prediction and 0.796 on deletion length frequency prediction. When including the third model component for 1-bp insertion modeling and testing on genotype frequency prediction for 1-bp insertions and all deletions, inDelphi achieves a median held-out Pearson correlation of 0.937 and 0.910 on the task of indel length frequency prediction. The linear baseline model achieves a median held-out Pearson correlation of 0.919 and 0.900 on the two tasks respectively.

From these results, it is shown that much of the model's power is derived from its designed structure which is independent of the choice of linear or non-linear modeling. While the baseline does not significantly cripple the model, the use of deep nonlinear neural networks offers a substantial performance improvement (10-24%) above linear modeling. In addition, the strong performance of the linear baseline model highlights that the prediction task, given the model structure, is relatively straightforward. This suggests that the model should be able to generalize well to unseen data.

The deep neural network version of MH-NN learns that microhomology length is more important than % GC (FIGS. 18A-18H). The linear version learns the same concept, with a weight of 1.1585 for MH length and 0.332 for % GC. Comparison with a Baseline Model Lacking Microhomology Length as a Feature

Microhomology length is an important feature for MH-NN (FIGS. 18A-18H). A model was trained that uses only %

GC as input to MH-NN while keeping the rest of the model structure identical. On held-out data, this baseline model at convergence achieves to a mean Pearson correlation of 0.59 on the task of predicting deletion genotype frequencies, and a mean Pearson correlation of 0.58 on the task of predicting deletion length frequencies. Notably, a model at iteration 0 with randomly initialized weights achieves mean Pearson correlations of 0.55 and 0.54 on the two respective tasks on held-out data. This basal Pearson correlation is relatively high due to the model structure, in particular, the exponential penalty on deletion length. In sum, removing MH length as a feature severely impacts model performance, restricting it to predictive performance not appreciably better than random chance.

inDelphi Training and Testing on Data from Varying Cell-Types

For predicting genotype and indel length frequencies in any particular cell-type C where data D is available, inDelphi's deletion component was first trained on a subset of Lib-A mESC data. Then, k-fold cross-validation was applied on D where D is iteratively split into training and test datasets. For each cross-validation iteration, the training set is used to train the insertion frequency model (k-nearest neighbors) and insertion genotype model (matrix of observed probabilities of each inserted base given local sequence context, which is just the -4 nucleotide when the training dataset is small, and -5, -4 and -3 nucleotides when the training dataset is large). For each cross-validation iteration, predictions are made at each sequence context in the test set which are compared to observations for each sequence context to yield a Pearson correlation. For any particular sequence context, the median test-time Pearson correlation across all cross-validation iterations is used as a single number summary of the overall performance of inDelphi. For all reported results, 100-fold cross-validation was used with 80%/20% training and testing splits. Empirically, small variance in test-time Pearson correlation was observed, highlighting the stability of inDelphi's modeling approach.

inDelphi Testing on Endogenous VO Data

On this task, the deletion component of inDelphi was trained on a subset of the Lib-A mESC data. For each cell type in HCT116, K562, and HEK293T, all VO sequence contexts (about 100) were randomly split into training and test datasets 100 times. During each split, the training set was used for k-nearest neighbor modeling of 1-bp insertion frequencies. Feature normalization to zero mean and unit variance was not performed. The average frequency of each 1-bp insertion genotype was derived from the training set as well. For each of the ~100 sequence contexts, the median test-time Pearson correlation was used for plotting in FIGS. 13A-13D. Due to the small size of the training set, only the -4 nucleotide was used for modeling both the insertion frequency and insertion genotype frequencies.

inDelphi Testing on Library Data

On this task, the deletion component of inDelphi was trained on a subset of the Lib-A mESC data. The remaining test set was used for measuring test-time prediction performance on Lib-A. Nucleotides -5, -4, and -3 were used for the insertion genotype model. For testing on Lib-B, Lib-B was split into training and test datasets in the same manner as with VO data. Nucleotide -4 was used for the insertion genotype model. The median test-time Pearson correlation is used as a single number summary of the overall performance of inDelphi on any particular sequence context. For reporting predictive results in FIGS. 15A-15F, sequence contexts

with low replicability (less than 0.85 Pearson correlation) in observed editing outcome frequencies were first removed. inDelphi Training and Testing on Prkdc^{-/-}Lig4^{-/-} Data

inDelphi was trained on data from 946 Lib-A sequence contexts and tested on 168 held-out Lib-A sequence contexts. Nucleotide -4 was used for insertion rate modeling, all other modeling choices were standard as described above. On held-out data, this version of inDelphi achieved a median Pearson correlation of 0.84 on predicting indel genotype frequencies, and 0.80 on predicting indel length frequencies. Training the Online Public Version of inDelphi and its Expected Properties

For general-use on arbitrary cell types, a version of inDelphi was trained using additional data from diverse types of cells. Deletion modeling was trained using data from 2,464 sequence contexts from high-replicability Lib-A and Lib-B data (including clinical variants and microduplications, fourbp, and longdup) in mES and data from VO sequence contexts in HEK293 and K562. Insertion frequency modeling is implemented as above. Insertion genotype modeling uses nucleotides -5, -4, and -3. The insertion frequency model and insertion genotype model are trained on VO endogenous data in K562 and HEK293T, Lib-A data in mESC, and Lib-B data (including clinical variants and microduplications, fourbp, and longdup) in mESC and U2OS.

Though MHless-NN, as trained on library data, never receives information on deletion lengths beyond 28, it was allowed to generalize its learned function and make predictions on deletion lengths up to 60 bp to match the supported range of MH-NN.

inDelphi makes predictions on 1-bp insertions and 1–60-bp deletions, which were empirically shown to consist of higher than 90% of all Cas9 editing outcomes in data from multiple human and mouse cell lines. Nevertheless, there is a subset of repair (about 8% on average) that inDelphi does not attempt to predict. It is suggested that end-users, depending on what predictive quantities are of interest, take this into account when using inDelphi. For example, if inDelphi predicts that 60% of 1-bp insertions and 1–60-bp deletions at a disease allele correspond to repair to wildtype genotype, a quantity of interest may be the rate of wildtype repair in all Cas9 editing outcomes (including the 8% not predicted by inDelphi). In such a situation, this quantity can be calculated as $(92\% \cdot 60\%) = 55.2\%$.

By the design of 1872 sequence contexts in Lib-A, the training dataset has rich and uniform representation across all quintiles of several major axes of variation including GC content, precision, and number of bases participating in microhomology as measured empirically in the human genome. This design strategy enables inDelphi to generalize well to arbitrary sequence contexts from the human genome.

These training data further include data in the outlier range of statistics of interest, including extremely high and low precision repair distributions, and extremely weak and strong microhomology (minimal microhomology and extensive microduplication microhomology sequences). The availability of such sequences in the training data enables inDelphi to generalize well to sequence contexts of clinical interest and sequence contexts supporting unusually high frequencies of precision repair. In particular, by training on more than 1000 examples of repair at clinical microduplications, inDelphi has received strong preparation for accurate prediction on other clinical microduplications.

By training on data from many cell-types, inDelphi was enabled to make predictions that are generally applicable to many human cell-types. It is noted that the HCT116 human

colon cancer cell line experiences a markedly higher frequency of single base insertions compared to all other cell lines that were studied, possibly due to the MLH1 deficiency of this cell line leading to impaired DNA mismatch repair. For this reason, HCT116 data was excluded from the training dataset. For best results, it is suggested that end-users keep in mind that repair class frequencies can be cell type-dependent, and this issue has not been well-characterized thus far.

It is noted that inDelphi's main error tendency is on the side of overestimating rather than underestimating the precision of repair (FIGS. 14A-14F, FIGS. 15A-15F). In general, this tendency can be explained by noting that inDelphi only considers sequence microhomology as a factor, while it's plausible and likely in biological experimental settings that even sequence contexts with very strong sequence microhomology may not yield precise results due to noise factors that are not considered by inDelphi. For best results, it is recommended that end-users take this tendency into account when using inDelphi predictions for further experiments. In particular, if gRNAs are designed by using a minimum precision threshold, end-users should recognize that observed repair outcomes may have empirical precision under this threshold. However, conversely, it is unlikely that a gRNA will have precision higher than what inDelphi predicts.

Lib-A Design (See Table 4)

All designed sequence contexts were 55 bp in length with cutting between the 27th and 28th base.

1872 sequence contexts were designed by empirically determining the distribution of four statistics in sequence contexts from the human genome. These four statistics are GC content, total sum of bases participating in microhomology for 3–27-bp deletions, Azimuth predicted on-target efficiency score, and the statistical entropy of the predicted 3–27-bp deletion length distribution from a previous version of inDelphi. For each of these statistics, empirical quintiles were derived by calculating these statistics in a large number of sequence contexts from the human genome. For the library, sequence contexts were designed by randomly generated DNA that categorized into each combination of quintiles across each of the four statistics. For example, a sequence context falling into the 1st quintile in GC, 2nd quintile of total MH, 1st quintile of Azimuth score, and 5th quintile of entropy, was found by random search. With four statistics and five bins each (due to quintiles), there are $5^4 = 625$ possible combinations. For each combination, it was attempted to design three sequence contexts for a total of 1875; 3 sequences could not be designed (for a total of 1872) though each bin was filled. 90 sequence contexts were designed from VO sequence contexts. Other sequence contexts were also designed for a total of 2000 sequence contexts in Lib-A. Lib-A sequence names, gRNAs, and sequence contexts are listed in Table 4 (appended, forming part of the instant specification).

Lib-B Design (See Table 5)

All designed sequence contexts were 55 bp in length with cutting between the 27th and 28th base.

1592 sequence contexts were designed from Clinvar and HGMD (see section on Selection of variants from disease databases). Some disease sequence contexts were designed that such that the corrected wildtype or frameshift allele supports further cutting by the original gRNA; data from such sequence contexts were ignored during analysis. "longdup" sequence contexts were designed by repeating the following procedure three times: for N=7 to 25, an N-mer was randomly generated, then duplicating and surrounded

by randomly generated sequences, while ensuring that SpCas9 NGG was included and appropriately positioned for cutting between positions 27 and 28. 90 sequence contexts were designed from VO sequence contexts. 228 “fourbp” sequence contexts were designed at 3 contexts with random sequences (with total phi score on average lower than VO sequence contexts) while varying positions -5 to -2; for each of the 3 “low-microhomology” contexts, 76 four bases were randomly designed while ensuring representation from all possible 2 bp microhomology patterns including no microhomology, one base of microhomology at either position, and full two bases of microhomology. Other sequence contexts were also designed for a total of 2000 sequence contexts in Lib-B. Lib-B sequence names, gRNAs, and sequence contexts are listed in Table 5.

Generating a DNA Motif for 1-Bp Insertion Frequencies

Nucleotides from positions -7 to 0 were one-hot-encoded and used in ridge regression to predict the observed frequency of 1-bp insertions out of all Cas9 editing events in 1996 sequence contexts from Lib-A mESC data. The data were split into training and testing sets (80/20 split) 10,000 times to calculate a bootstrapped estimate of linear regression weights and test-set predictive Pearson correlation. The median test-set Pearson correlation was found to be 0.62. To generate a DNA motif, any features that included 0 within the bootstrapped weight range were excluded (probability that the weight is zero >1e-4). The average bootstrapped weight estimate was used as the “logo height” for all remaining features. Each feature is independent; vertical stacking of features follows the published tradition of DNA motifs.

Plasmid and insert sequences
P2T-CAG-MCS-P2A-GFP-PuroR complete plasmid sequence (SEQ ID NO: 41)
CCACCTAAATTGTAAGCGTTAATATTTTGTAAAATTCGCGTTAAATTTTGTAAAT
CAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCCTTATAAATCAAAAGA
ATAGACCGAGATAGGGTTGAGTGTGTTCAGTTTGAACAAGAGTCCACTATTAA
AGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTATCAGGGCGATGGCCC
ACTACGTGAACCATCACCTAATCAAGTTTTTTGGGGTCGAGGTGCGGTAAAGCAC
TAAATCGGAACCCCTAAAGGGAGCCCCGATTAGAGCTTGACGGGGAAGCCGGC
GAACGTGGCGAGAAAGGAAGGAAGAAAGCGAAAGGAGCGGGCGCTAGGGCGC
TGGCAAGTGTAGCGGTACGCTGCGCGTAACCACCACACCCGCCGCTTAATGC
GCCGCTACAGGGCGCGTCCCATTCGCCATTAGGCTGCGCAACTGTTGGGAAGG
GCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGATGTGC
TGCAAGGCGATTAGTTGGGTAACGCCAGGGTTTTCCAGTCACGACGTTGTAAAA
CGACGGCCAGTGAGCGCGCTAATACGACTCACTATAGGGCGAATTGGGTACCG
GCATATGGTCTTGACAGAGGTGTAAAAAGTACTCAAAAATTTTACTCAAGTGAAAG
TACAAGTACTTAGGGAAAAATTTACTCAATTAAAGTAAAGTATCTGGCTAGAATC
TTACTTGAGTAAAGTAAAAAGTACTCCATTAAAAATGTACTTGAGTATTAAGGAA
GTAAAGTAAAGCAAGAAAGATCGATCTCGAAGGATCTGGAGGCCACCATGGTG
TCGATAACTTCGTATAGCATACATTATACGAAGTTATCGTGCTCGACATTGATTATT
GACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATATATGGA
GTTCCGCGTTACATAACTTACGGTAAATGGCCCGCTGGCTGACCGCCCAACGAC
CCCCGCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGGAC
TTTCCATTGACGTCAATGGGTGGAGTATTACGGTAAACTGCCCACTTGGCAGTAC
ATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGG
CCCCCTGGCATTATGCCAGTACATGACCTTATGGGACTTTCTACTTGGCAGTA
CATCTACGTATTAGTCATCGCTATTACCATGGTCGAGGTGAGCCCCACGTTCTGCT
TCACTCTCCCCATCTCCCCCCCCCTCCCAACCCCAATTTGTATTATTTATTTTAA
ATTATTTTGTGCAGCGATGGGGCGGGGGGGGGGGGGGGGGCGCGCCAGGCG
GGGCGGGCGGGCGAGGGCGGGCGGGCGGGCGAGGCGAGAGGTGCGGCGG
CAGCCAATCAGAGCGGCGGCTCCGAAAGTTTCTTTTATGGCGAGGCGGCGG
GGCGGCGGCCCTATAAAAAGCGAAGCGCGCGGGCGGGGAGTCGCTGCGAC

- continued

GCTGCCTTCGCCCCGTGCCCCGCTCCGCCGCCGCTCGCGCCGCCGCCCGG
CTCTGACTGACCGCTTACTCCACAGGTGAGCGGGCGGGACGGCCCTTCTCCTC
CGGGCTGTAATTAGCGCTTGGTTTAAATGACGGCTTGTTCCTTTCTGTGGCTGCGT
GAAAGCCTTGAGGGGCTCCGGGAGGGCCCTTGTGCGGGGGAGCGGCTCGGG
GGGTGCGTGCGTGTGTGTGCGTGGGGAGCGCCGCGTGC GGCTCCGCGCTGC
CCGGCGGCTGTGAGCGCTGCGGGCGCGGCGCGGGCTTGTGCGCTCCGAGT
GTGCGCGAGGGGAGCGCGGCCGGGGCGGTGCCCCGCGGTGCGGGGGGGCT
GCGAGGGGAACAAAGGCTGCGTGC GGGGTGTGTGCGTGGGGGGTGAAGAGG
GGTGTGGGCGCTCGGTGCGGCTGCAACCCCCCTGCACCCCCCTCCCGAGTT
GCTGAGCACGGCCCCGCTTCGGGTGCGGGCTCCGTACGGGCGTGGCGCGGG
GCTCGCCGTGCCGGGCGGGGGTGGCGGCAGGTGGGGGTGCCGGGCGGGCG
GGCCCGCTCGGGCCGGGAGGGCTCGGGGAGGGGCGCGCGGCCCGGA
GCGCCGGCGGCTGTGAGGCGCGGCGAGCCGAGCCATTGCCTTTTATGTAAT
CTGTGCGAGAGGGCGCAGGACTTCCTTTGTCCAAATCTGTGCGAGCCGAAATC
TGGAGGCGCCCGCCGACCCCTCTAGCGGGCGCGGGCGAAGCGGTGCGGCG
CCGGCAGGAAGGAAATGGGCGGGGAGGGCTTCGTGCGTCCCGCGCCCGCT
CCCCCTCTCCCTCTCCAGCCTCGGGGCTGTCCGCGGGGGACGGCTGCCCTTCGG
GGGGACGGGGCAGGCGGGGTTGCGCTTCTGGCGTGTGACCGGCGGCTCTAG
AGCCTCTGCTAACCATGTTTCATGCTTCTTCTTTTCTACAGCTCCTGGGCAACGT
GCTGGTTATTGTGCTGTCTCATCTTTTGGCAAAGAATTCCTCGAGCGGCCCCAG
TGTGATGGATATCGGATCCGCTAGCGCTACTAATTCAGCCTGCTGAAGCAGGCT
GGAGACGTGGAGGAGAACCCTGGACCTGGACCGGTGCGCCACCATGGTGAAG
GGCGAGGAGCTGTTACCGGGTGGTGCCATCCTGGTCGAGCTGGACGGCGAC
GTAAACGGCCACAAGTTCAGCGTGTCCGGCGAGGGCGAGGGCGATGCCACCTAC
GGCAAGCTGACCTGAAGTTCATCTGCACCACCGCAAGCTGCCCGTGCCCTGG
CCCACCTCGTGACCACCTGACCTACGGCGTGAGTGCTTCAGCCGCTACCCCG
ACCACATGAAGCAGCAGACTTCTTCAAGTCCGCCATGCCCGAAGGCTACGTCCA
GGAGCGCACCATCTTCTTCAAGGACGACGGCAACTACAAGACCGCGCCGAGGT
GAAGTTCGAGGGCGACACCCTGGTGAACCGCATCGAGCTGAAGGGCATCGACTT
CAAGGAGGACGGCAACATCCTGGGCGACAAGCTGGAGTACAACATAACAGCCAC
AACGCTCTATATCATGGCCGACAAGCAGAAGAACGGCATCAAGGTGAATTCAGAT
CCGCCACAACATCGAGGACGGCAGCGTGCAGCTCGCCGACCACTACCAGCAGAA
CACCCCATCGGCGACGGCCCCGTGCTGCTGCCCCGACAACCACTACCTGAGCAC
CCAGTCCGCCCTGAGCAAAGACCCAACGAGAAGCGCGATCACATGGTCCTGCTG
GAGTTCGTGACCGCCGCGGGATCACTCTCGGCATGGACGAGCTGTACAAGTAAA
GCGGCCCGCACCGCGTGGAGCTCGAATTAATTCATCGATGATGATCCAGACATG
ATAAGATACATTGATGAGTTTGGACAAACCAACTAGAAATGCAGTAAAAAATG
CTTTATTTGTGAAATTTGTGATGCTATTGCTTTATTTGTAACCATTATAAGCTGCAAT
AAACAAGTTAACAACAACAATTGCATTCAATTTATGTTTCAGGTTCAGGGGAGGTG
TGGGAGGTTTTTTAAAGCAAGTAAACCTCTACAAATGTGGTATGGCTGATTATGAT

-continued

CCTCTAGAGTCGGTGGGCCTCGGGGCGGGTGCGGGTCGGCGGGCCGCCCC
GGGTGGCTTCGGTCGGAGCCATGGGGTCGTGCGCTCCTTTCGGTCGGGCGCTGC
GGGTCGTGGGGCGGGCGTCAGGCACCGGCTTGCGGGTCATGCACCAGGTGCG
CGGTCCCTTCGGGCACCTCGACGTCGGCGGTGACGGTGAAGCCGAGCCGCTCGTA
GAAGGGGAGGTTGCGGGGCGCGAGGTCTCCAGGAAGCGGGCACCCGGCGC
GTCGCGCCGCTCCACTCCGGGGAGCACGACGGCGCTGCCAGACCTTGCCCT
GGTGGTCGGGCGAGACGCCGACGGTGGCCAGGAACCACGCGGGCTCCTTGGGC
CGGTGCGGCGCAGGAGGCCTTCCATCTGTTGCTGCGCGGCCAGCCGGGAACCG
CTCAACTCGGCCATGCGCGGGCCGATCTCGGCGAACACGCCCCGCTTCGACG
CTCTCCGGCGTGGTCCAGACCGCCACCGCGCGCGTCGTCCGCGACCCACACC
TTGCCGATGTCGAGCCGACGCGCTGAGGAAGAGTTCTTGACGCTCGGTGACC
CGCTCGATGTGGCGGTCCGGGTCGACGGTGTGGCGCGTGGCGGGTAGTCGGC
GAACGCGGCGGCGAGGGTGCGTACGGCCCGGGGACGTCGTCGCGGTGGCGA
GGCGCACCGTGGGCTTGTAAGTCTCGGTGAGGAGGTGCTCTCTTGAGGGGTCA
GGGCGTGGGTGAGGGATGGTGGCGGCACCGGTGTCGCGGCCGACCTGCAG
GCATGCAAGCTTTTGGAAAAGCCTAGGCCCTCAAAAAGCCTCCTCACTACTTCT
GGAATAGCTCAGAGCCGAGGCGGCTCGGCCCTGTCATAAATAAAAAATTAG
TCAGCCATGGGCGGAGAATGGGCGGAAGTGGGCGGAGTTAGGGCGGGATGG
GCGGAGTTAGGGGCGGACTATGGTTGCTGACTAATTGAGATGCATGCTTTGCAT
ACTTCTGCCTGCTGGGAGCCTGGGGACTTTCCACACCTGTTGCTGACTAATTG
AGATGCATGCTTTGCATACTTCTGCCTGCTGGGAGCCTGGGGACTTTCCACACC
CTAACTGACACACATTCCACAGAATTCAAGTGATCTCAAAAAATAAGTACTTTTGG
ACTGTAAATAAAATTGTAAGGAGTAAAAGTACTTTTTTTCTAAAAAATGTAATT
AAGTAAAAGTAAAAGTATTGATTTTAATTGTAAGTAAAGTAAAAATCCCAA
AAATAAATACTTAAGTACAGTAATCAAGTAAATTACTCAAGTACTTTACACCTCTGG
TTCTTGACCCCTACCTTCAGCAAGCCAGCAGATCCGAGCTCCAGCTTTTGTTCCT
TTAGTGAGGGTTAATTGCGCGCTTGGCGTAATCATGGTCATAGCTGTTTCTGTGTG
AAATTGTTATCCGCTCACAATTCACACAACATACGAGCCGAAGCATAAAGTGTA
AAGCCTGGGTGCTAATGAGTGAGCTAACTCACATTAATTGCGTTGCGCTCACTG
CCCGCTTTCCAGTCGGGAAACCTGTCGTGCCAGCTGCATTAATGAATCGGCCAAC
GCGCGGGGAGAGCGGTTTGGCTATTGGGCGCTCTTCCGCTTCCTCGCTCACTGA
CTCGCTGCGCTCGGTGCTTGGCTGCGGCGAGCGGTATCAGCTCACTCAAAGGC
GGTAATACGTTATCCACAGAATCAGGGGATAACGCAGGAAGAATGTCGAGCA
AAAGGCCAGCAAAAGGCCAGGAACCGTAAAAAGGCCGCTTGCTGGCGTTTTTCC
ATAGGCTCCGCCCCCTGACGAGCATCAAAAAATCGACGCTCAAGTCAGAGGTG
GCGAAACCCGACAGGACTATAAAGATACCAGGCGTTTCCCCTGGAAGCTCCCTC
GTGCGCTCTCCTGTTCCGACCTGCCGCTTACCGGATACCTGTCCGCTTTCTCC
CTTCGGGAAGCGTGGCGCTTTCTCATAGCTCACGCTGTAGGTATCTCAGTTCGGT
GTAGTTCGTTGCTCCAAGCTGGGCTGTGTGCACGAACCCCGTTACGCCCAGC
CGCTGCGCTTATCCGTAACATCGTCTTGAGTCCAACCCGGTAAGACACGACTT
ATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAGCGAGGTATGTAGGC

- continued

GGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACTACGGCTACACTAGAAAGGACAG
TATTTGGTATCTGCGCTCTGCTGAAGCCAGTTACCTTCGGAAAAAGAGTTGGTAGC
TCTTGATCCGGCAAACAAACCACCGCTGGTAGCGGTGGTTTTTTTGTTCAGCA
GCAGATTACGCGCAGAAAAAAGGATCTCAAGAAGATCCTTGATCTTTCTACGG
GGTCTGACGCTCAGTGGAACGAAAACTCACGTTAAGGGATTTTGGTCATGAGATTA
TCAAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTAAATCAATC
TAAAGTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATCAGTGAGGCA
CCTATCTCAGCGATCTGTCTATTTTCGTTTCATCCATAGTTGCCTGACTCCCCGTCGTGT
AGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTCTGCAATGATACC
GCGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACAGCCAGCCGGA
AGGGCCGAGCGCAGAAGTGGTCTTGCAACTTTATCCGCCTCCATCCAGTCTATTA
ATTGTTGCCGGAAGCTAGAGTAAGTAGTTTCGCCAGTTAATAGTTTGCACAACGTT
GTTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTCGTTGGTATGGCTTCATT
CAGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATCCCCATGTTGTGCAAAA
AAGCGGTTAGCTCCTTCGGTCTCCGATCGTTGTCAGAAGTAAGTTGGCCGAGT
GTTATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGT
AAGATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTAT
GCGGCGACCGAGTTGCTCTTGCCCGCGTCAATACGGGATAATACCGGCCACAT
AGCAGAACTTTAAAAGTGCTCATCATTGGAACAGTTCTTCGGGGCGAAAACTCTC
AAGGATCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCACT
GATCTTCAGCATCTTTTACTTTTACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGG
CAAAATGCCGCAAAAAGGAATAAGGGCGACACGGAATGTTGAATACTCATACT
CTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGAGCGGATAC
ATATTGAATGTATTAGAAAAATAACAATAGGGGTCCGCGCACATTTCCCCGA
AAAGTG

LDLRwt

(SEQ ID NO: 42)

ATGGGGCCCTGGGGCTGGAAATTGCGCTGGACCGTCGCCTTGCTCCTCGCCGCG
GCGGGGACTGCAGTGGGCGACAGATGCGAAGAAACGAGTTCCAGTGCCAAGAC
GGGAAATGCATCTCTACAAGTGGGTCTGCGATGGCAGCGCTGAGTGCCAGGATG
GCTCTGATGAGTCCCAGGAGACGTGCTTGTCTGTACCTGCAATCCGGGGACTT
CAGCTGTGGGGCCGTGTCAACCGCTGCATTCTCAGTTCTGGAGGTGCGATGGC
CAAGTGGA CTGCGACAACGGCTCAGACGAGCAAGGCTGTCCCCCAAGACGTGC
TCCCAGGACGAGTTTCGCTGCCACGATGGGAAGTGCATCTCTCGGCAGTTCGTCT
GTGACTCAGACCGGGACTGCTTGGACGGCTCAGACGAGGCCTCCTGCCCGGTGC
TCACCTGTGGTCCCGCCAGCTTCCAGTGCAACAGCTCCACCTGCATCCCCCAGCT
GTGGGCCTGCGACAACGACCCCGACTGCGAAGATGGCTCGGATGAGTGCCGCA
GCGCTGTAGGGGTCTTTACGTGTTCCAAGGGACAGTAGCCCTGCTCGGCCTTC
GAGTTCCACTGCCTAAGTGGCGAGTGCATCCACTCCAGCTGGCGCTGTGATGGTG
GCCCCGACTGCAAGGACAAATCTGACGAGGAAAAGTGCCTGTGGCCACCTGTCTG
CCCTGACGAATTCCAGTGCTCTGATGGAACTGCATCCATGGCAGCCGGCAGTGT

- continued

GACCCGGGAATATGACTGCAAGGACATGAGCGATGAAGTTGGCTGCGTTAATGTGA
CACTCTGCGAGGGACCCAAACAAGTTCAAGTGTACAGCGGCGAATGCATCACCCCT
GGACAAAGTCTGCAACATGGCTAGAGACTGCCGGGACTGGTCAGATGAACCCATC
AAAGAGTGCGGGACCAACGAATGCTTGGACAACAACGGCGGCTGTTCCACGTCT
GCAATGACCTTAAGATCGGCTACGAGTGCCTGTGCCCCGACGGCTTCAGCTGGT
GGCCAGCGAAGATGCGAAGATATCGATGAGTGTGAGGATCCCGACACCTGCAGC
CAGCTCTGCGTGAACCTGGAGGGTGGCTACAAGTGCCAGTGTGAGGAAGGCTTC
CAGCTGGACCCCCACAGGAAGGCTGCAAGGCTGTGGGCTCCATCGCCTACCTCT
TCTTCACCAACCGGCACGAGGTCAGGAAGATGACGCTGGACCGGAGCGAGTACA
CCAGCCTCATCCCCAACCTGAGGAACGTGGTCGCTCTGGACACGGAGGTGGCCA
GCAATAGAATCTACTGGTCTGACCTGTCCCAGAGAATGATCTGCAGCACCAGCTT
GACAGAGCCCCACGGCGTCTCTTCCTATGACACCGTCATCAGCAGAGACATCCAGG
CCCCCGACGGGCTGGCTGTGGACTGGATCCACAGCAACATCTACTGGACCGACTC
TGTCTGGGCACTGTCTCTGTTGCGGATACCAAGGGCGTGAAGAGGAAAACGTTA
TTCAGGGAGAACGGCTCCAAGCCAAGGGCCATCGTGGTGGATCCTGTTTCATGGCT
TCATGTACTGGACTGACTGGGGAACCTCCCGCCAAGATCAAGAAAGGGGGCCTGAA
TGGTGTGGACATCTACTCGCTGGTGAAGAACATTGAGTGGCCCAATGGCATCA
CCCTAGATCTCCTCAGTGGCCGCTCTACTGGGTTGACTCCAACTTCACTCCATC
TCAAGCATCGATGTCAATGGGGGCAACCGGAAGACCATCTTGGAGGATGAAAAGA
GGCTGGCCACCCCTTCTCCTTGGCCGTCTTTGAGGACAAAGTATTTTGGACAGAT
ATCATCAACGAAGCCATTTTCAGTGCCAACCGCCTCACAGGTTCCGATGTCAACTT
GTTGGCTGAAAACCTACTGTCCCCAGAGGATATGGTCTCTTCCACAACCTCACCC
AGCCAAGAGGAGTGAACCTGGTGTGAGAGGACCACCTGAGCAATGGCGGCTGCC
AGTATCTGTGCTCCTCGCCCGCAGATCAACCCCACTCGCCCAAGTTTACCTG
CGCTGCCCCGACGGCATGCTGCTGGCCAGGGACATGAGGAGCTGCCTCACAGA
GGCTGAGGCTGCAGTGGCCACCCAGGAGACATCCACCGTCAGGCTAAAGGTCAG
CTCCACAGCCGTAAGGACACAGCACACAACCAACCGGCTGTTCCCGACACCTCC
CGGCTGCCTGGGGCCACCCCTGGGCTACCCACGGTGGAGATAGTGACAATGTCT
CACCAAGCTCTGGGCGACGTTGCTGGCAGAGGAAATGAGAAGAAGCCAGTAGC
GTGAGGGCTCTGTCCATTGTCCTCCCATCGTGTCTCGTCTTCCTTTGCCTGGG
GGTCTTCCTTCTATGGAAGAACTGGCGGCTTAAGAACATCAACAGCATCAACTTTG
ACAACCCCGTCTATCAGAAGACCACAGAGGATGAGGTCCACATTTGCCACAACCA
GGACGGCTACAGCTACCCCTCGAGACAGATGGTCAGTCTGGAGGATGACGTGGCG

LDLRDup252 with surrounding region

(SEQ ID NO: 43)

CCCCAAGACGTGCTCCAGGACGAGTTTCGCTGCCACGATGGGAAGTGACATCTC
TCGGCAGTTCGTCTGTGACTCAGACCGGACTGCTTGGACGGCTCAGACGAGGC
CTCTTGCCCGGTGCTACCTGTGGTCCCGCAGCTTCCAGTGCAACAGCTCCACC
TGCATCCCCAGCTGTGGGCTGCGACAACGACCCCGACTGCGAAGATGGCTCG
GAGGCTCGGATGAGTGGCCGACGCTGTAGGGGTCTTTACGTGTTCCAAGGGG

-continued

ACAGTAGCCCCCTGCTCGGCCTTCGAGTTCCACTGCCTAAGTGGCGAGTGCATCCA
CTCCAGCTGGCGCTGTGATGGTGGCCCCGACTGCAAGGACAAATCTGACGAGGA
AAACTGCG

LDLRDup254/255 with surrounding region

(SEQ ID NO: 44)

CCCCCAAGACGTGCTCCCAGGACGAGTTTCGCTGCCACGATGGGAAGTGCATCTC
TCGGCAGTTCGTCTGTGACTCAGACCGGACTGCTTGGACGGCTCAGACGAGGC
CTCCTGCCCCGGTGCTCACCTGTGGTCCCGCCAGCTTCCAGTGCAACAGCTCCACC
TGCATCCCCCAGCTGTGGGCTGCGACAACGACCCCGACTGCGAAGATGGCTCG
GATGAGTGGCCGACGCTGTAGGGGTCTTTACGTGTTCCAAGGGGACAGTAGC
CCCTGCTCGGCCTTCGAGTTCCACTGCCTAAGTGGCGAGTGCATCCACTCCAGCT
GGCGCTGTGATGGTGGCCCCGACTGCAAGGACAAATCTGACAGGACAAATCTGAC
GAGGAAAACCTGCGCTGTGGCCACCTGTCGCCCTGACGAATTCAGTGCTCTGATG
GAAACTGCATCCATG

LDLRDup258 with surrounding region

(SEQ ID NO: 45)

CCCCCAAGACGTGCTCCCAGGACGAGTTTCGCTGCCACGATGGGAAGTGCATCTC
TCGGCAGTTCGTCTGTGACTCAGACCGGACTGCTTGGACGGCTCAGACGAGGC
CTCCTGCCCCGGTGCTCACCTGTGGTCCCGCCAGCTTCCAGTGCAACAGCTCCACC
TGCATCCCCCAGCTGTGGGCTGCGACAACGACCCCGACTGCGAAGATGGCTCG
GATGAGTGGCCGACGCTGTAGGGGTCTTTACGTGTTCCAAGGGGACAGTAGC
CCCTGCTCGGCCTTCGAGTTCCACTGCCTAAGTGGCGAGTGCATCCACTCCAGCT
GGCGCTGTGATGGTGGCCCCGACTGCAAGGACAAATCTGAGGACAAATCTGACGA
GGAAAACCTGCGCTGTGGCCACCTGTCGCCCTGACGAATTCAGTGCTCTGATGGA
AACTGCATCCATG

LDLRDup261 with surrounding region

(SEQ ID NO: 46)

CCCCCAAGACGTGCTCCCAGGACGAGTTTCGCTGCCACGATGGGAAGTGCATCTC
TCGGCAGTTCGTCTGTGACTCAGACCGGACTGCTTGGACGGCTCAGACGAGGC
CTCCTGCCCCGGTGCTCACCTGTGGTCCCGCCAGCTTCCAGTGCAACAGCTCCACC
TGCATCCCCCAGCTGTGGGCTGCGACAACGACCCCGACTGCGAAGATGGCTCG
GATGAGTGGCCGACGCTGTAGGGGTCTTTACGTGTTCCAAGGGGACAGTAGC
CCCTGCTCGGCCTTCGAGTTCCACTGCCTAAGTGGCGAGTGCATCCACTCCAGCT
GGCGCTGTGATGGTGGCCCCGACTGCAAGGACAAATCTGACGACAAATCTGACGA
GGAAAACCTGCGCTGTGGCCACCTGTCGCCCTGACGAATTCAGTGCTCTGATGGA
AACTGCATCCATG

LDLRDup264 with surrounding region

(SEQ ID NO: 47)

CTTCATGTACTGGAAGTGGGGAAGTCCCGCAAGATCAAGAAAGGGGCTG
AATGGTGTGGACATCTACTCGTGGTGAGCTGGTGAAGTGAACATTCAAGTGGCC
CAATGGCATCACCCTAG

GAAwt

(SEQ ID NO: 48)

ATGGGAGTGAGGCACCCGCCCTGCTCCACCGGCTCCTGGCCGTCTGCGCCCTC
GTGTCCTTGGCAACCGCTGCACTCCTGGGGCACATCCTACTCCATGATTTCTTGCT
GGTTCCTCCGAGAGCTGAGTGGCTCCTCCCGAGTCTGGAGGAGACTCACCCAGCT

- continued

CACCAGCAGGGAGCCAGCAGACCAGGGCCCCGGGATGCCAGGCACACCCCGG
CCGTCCCAGAGCAGTGCCACACAGTGCGACGTCCCCC AACAGCCGCTTCGA
TTGCGCCCCGTGACAAGGCCATCACCAGGAACAGTGCAGGCCCCGCGCTGTTG
CTACATCCCTGCAAAGCAGGGGCTGCAGGGAGCCAGATGGGGCAGCCCTGGTG
CTTCTTCCACCCAGCTACCCAGCTACAAGCTGGAGAACCTGAGCTCCTCTGAAA
TGGGTACACGGCCACCCTGACCCGTACCACCCCACTTCTTCCCCAAGGACAT
CCTGACCCTGCGGCTGGACGTGATGATGGAGACTGAGAACC GCCTCCACTTCACG
ATCAAAGATCCAGCTAACAGGCGCTACGAGGTGCCCTTGAGACCCCGCATGTCC
ACAGCCGGGCACCGTCCCCACTCTACAGCGTGGAGTTCTCCGAGGAGCCCTTCG
GGGTGATCGTGCGCCGGCAGCTGGACGGCCGCGTGCTGCTGAACACGACGGTG
GCGCCCCGTGTTCTTTGCGGACCAGTTCCTTCAGCTGTCCACCTCGCTGCCCTCGC
AGTATATCACAGGCCTCGCCGAGCACCTCAGTCCCCTGATGCTCAGCACCAGCTG
GACCAGGATCACCCCTGTGGAACCGGGACCTTGCGCCACGCCCGGTGCGAACCT
CTACGGGTCTCACCCCTTCTACCTGGCGCTGGAGACGGCGGGTCGGCACACGG
GGTGTTCCTGCTAAACAGCAATGCCATGGATGTGGTCTTGCAGCCGAGCCCTGCC
CTTAGCTGGAGGTCGACAGGTGGGATCCTGGATGTCTACATCTTCTGGGCCAG
AGCCCAAGAGCGTGGTGACGAGTACCTGGACGTTGTGGGATACCCGTTTCATGCC
GCCATACTGGGGCTGGGCTTCCACCTGTCCGCTGGGGCTACTCCTCCACCGCT
ATCACCCGCCAGGTGGTGGAGAACATGACCAGGGCCCACTTCCCCCTGGACGTC
CAGTGGAAACGACCTGGACTACATGGACTCCCGGAGGGACTTCACGTTCAACAAGG
ATGGCTTCCGGGACTTCCCGCCATGGTGCAGGAGCTGCACCAGGGCGGCCGGC
GCTACATGATGATCGTGGATCCTGCCATCAGCAGCTCGGGCCCTGCCGGGAGCTA
CAGGCCCTACGACGAGGGTCTGCGGAGGGGGTTTTTCATCACCAACGAGACCGG
CCAGCCGCTGATTGGGAAGGTATGGCCCGGTCCACTGCCTTCCCCGACTTCACC
AACCCACAGCCCTGGCCTGGTGGGAGGACATGGTGGCTGAGTTCCATGACCAG
GTGCCCTTCGACGGCATGTGGATTGACATGAACGAGCCTTCCAATTCATCAGGG
GCTCTGAGGACGGCTGCCCCAACAATGAGCTGGAGAACCACCTACGTGCCTG
GGGTGGTTGGGGGACCCCTCAGGCGGCCACCATCTGTGCCTCCAGCCACCAGT
TTCTCTCCACACACTACAACCTGCACAACCTCTACGGCCTGACCGAAGCCATCGCC
TCCCACAGGGCGCTGGTGAAGGCTCGGGGACACGCCCATTTGTGATCTCCCGC
TCGACCTTTGCTGGCCACGGCCGATACGCCGGCCACTGGACGGGGACGTGTGG
AGCTCCTGGGAGCAGCTCGCCTCCTCCGTGCCAGAAATCCTGCAGTTTAACCTGC
TGGGGGTGCCTCTGGTCGGGGCCGACGTCTGCGGCTTCTGGGCAACACCTCAG
AGGAGCTGTGTGTCGCTGGACCCAGCTGGGGGCCTTCTACCCCTTCATGCGGAA
CCACAACAGCCTGCTCAGTCTGCCCCAGGAGCCGTACAGCTTCAGCGAGCCGGC
CCAGCAGGCCATGAGGAAGGCCCTCACCTGCGCTACGCACTCCTCCCCACCT
CTACACACTGTTCCACCAGGCCACGTGCGGGGAGACCGTGGCCCGGCCCT
CTTCTGGAGTTCCCCAAGGACTCTAGCACCTGGACTGTGGACCACCAGCTCCTG
TGGGGGAGGCCCTGCTCATCCCCAGTGCTCCAGGCCGGAAGGCCGAAGTG
ACTGGCTACTTCCCCTTGGGCACATGGTACGACCTGCAGACGGTGCCAGTAGAGG

- continued

CCCTTGGCAGCCTCCACCCCCACCTGCAGCTCCCGTGAGCCAGCCATCCACAG
CGAGGGGCAGTGGGTGACGCTGCCGGCCCCCTGGACACCATCAACGTCCACCT
CCGGGCTGGGTACATCATCCCCCTGCAGGGCCCTGGCCTCACAACCACAGAGTC
CCGCCAGCAGCCCATGGCCCTGGCTGTGGCCCTGACCAAGGGTGGGGAGGCC
GAGGGGAGCTGTTCTGGGACGATGGAGAGAGCCTGGAAGTGCTGGAGCGAGGG
GCCTACACACAGGTCACTTCTCGCCAGGAATAACACGATCGTGAATGAGCTGG
TACGTGTGACCAGTGAGGGAGCTGGCCTGCAGCTGCAGAAGGTGACTGTCTTGG
GCGTGGCCACGGCGCCCCAGCAGGTCTCTCCAACGGTGTCCCTGTCTCCAACTT
CACCTACAGCCCCGACACCAAGGTCCTGGACATCTGTGTCTCGCTGTTGATGGGA
GAGCAGTTTCTCGTCAGCTGGTGT

GAADup327/328

(SEQ ID NO: 49)

ATGGGAGTGAGGCACCCGCCCTGCTCCACCGGCTCCTGGCCGTCTGCGCCCTC
GTGTCTTGGCAACCGCTGCACTCCTGGGGCACATCCTACTCCATGATTTCTTGCT
GGTTCCTCCGAGAGCTGAGTGGCTCCTCCCACTCCTGGAGGAGACTCACCAGCT
CACCAGCAGGGAGCCAGCAGACCAGGGCCCCGGGATGCCAGGCACACCCCGG
CCGTCCAGAGCAGTGCCACACAGTGCGACGTCCCCC AACAGCCGCTTCGA
TTGCGCCCCTGACAAGGCCATCACCAGGAACAGTGCGAGGCCCGCGCTGTTG
CTACATCCCTGCAAAGCAGGGGCTGCAGGGAGCCAGATGGGGCAGCCCTGGTG
CTTCTTCCACCCAGCTACCCAGCTACAAGCTGGAGAACCTGAGCTCCTCTGAAA
TGGGTACACGGCCACCTGACCCGTACACCCCACTTCTTCCCCAAGGACAT
CCTGACCTTGCGGCTGGACGTGATGATGGAGACTGAGAACCGCTCCACTTCACG
ATCAAAGATCCAGCTAACAGGCGCTACGAGGTGCCCTTGGAGACCCGCATGTCC
ACAGCCGGGCACCGTCCCACTCTACAGCGTGGAGTTCTCCGAGGAGCCCTTCG
GGGTGATCGTGCGCCGGCAGCTGGACGGCCGCGTGCTGCTGAACACGACGGTG
GCGCCCTGTCTTTTGGGACCAGTTCTTTCAGCTGTCCACCTCGCTGCCCTCGC
AGTATATCACAGGCCTCGCCGAGCACCTCAGTCCCTGATGCTCAGCACCAGCTG
GACCAGGATCACCTGTGGAACCGGGACCTTGCGCCACGCCCGGTGCGAACCT
CTACGGGTCTCACCTTTCTACCTGGCGCTGGAGACGGCGGTCGGCACACGG
GGTGTCTCTGCTAAACAGCAATGCCATGGATGTGGTCTTGACCCGAGCCCTGCC
CTTAGCTGGAGGTCGACAGGTGGGATCCTGGATGTCTACATCTTCTGGGCCAG
AGCCCAAGAGCGTGGTGACGAGTACCTGGACGTTGTGGGATACCCGTTTCATGCC
GCCATACTGGGGCTGGGCTTCCACCTGTGCCGCTGGGGCTACTCCTCCACCGCT
ATCACCCGCCAGGTGGTGGAGAACATGACCAGGGCCCACTTCCCCCTGGACGTC
CAGTGGAAACGACCTGGACTACATGGACTCCCGAGGGACTTCACGTTCAACAAGG
ATGGCTTCGGGACTTCCCGCCATGGTGACAGGAGCTGCACCAGGGCGGCCGGC
GCTACATGATGATCGTGGATCCTGCCATCAGCAGCTCGGGCCCTGCCGGGAGCTA
CAGGCCCTACGACGAGGGTCTGCGGAGGGGGTTTTCATACCAACGAGACCGG
CCAGCCGCTGATTGGGAAGGTATGGCCCGGTCCACTGCCTTCCCGACTTCACC
AACCCACAGCCCTGGCTGGTGGGAGGACATGGTGGCTGAGTTCCATGACCAG
GTGCCCTTCGACGGCATGTGGATTGACATGAACGAGCCTTCCAATTTCATCAGGG
GCTCTGAGGACGGCTGCCCCAACAATGAGCTGGAGAACCCACCTACGTGCCTG

- continued

GGGTGGTTGGGGGACCCTCCAGGCGGCCACCATCTGTGCCTCCAGCCACCAGT
TTCTCTCCACACTACAACCTGCACAACCTCTACGGCCTGACCGAAGCCATCGCC
TCCCACAGGGCGCTGGTGAAGGCTCGGGGACACGCCCATTTGTGATCTCCCGC
TCGACCTTTGCTGGCCACGGCCGATACGCCGGCCACTGGACGGGGGACGTGTGG
AGCTCCTGGGAGCAGCTCGCCTCCTCCGTGCCAGAAATCCTGCAGTTTAACCTGC
TGGGGGTGCCTCTGGTGGGGCCGACGTCTGCGGCTTCTGGGCAACACCTCAG
AGGAGCTGTGTGTGCGCTGGACCCAGCTGGGGGCCTTCTACCCCTTCATGCGGAA
CCACAACAGCCTGCTCAGTCTGCCCCAGGAGCCGTACAGCTTCAGCGAGCCGGC
CCAGCAGGCCATGAGGAAGGCCCTCACCTGCGCTACGCACTCCTCCCCACCT
CTACACACTGTTCCACCAGGCCACGTGCGGGGGAGACCGTGGCCCCGCCCT
CTTCCTGGAGTTCCCCAAGGACTCTAGCACCTGGACTGTGGACCACCAGCTCCTG
TGGGGGAGGCCCTGCTCATCACCCAGTGCTCCAGGCCGGAAGGCCGAAGTG
ACTGGCTACTTCCCCCTGGGCACATGGTACGACCTGCAGACGGTGCCAGTAGAGG
CCCTTGGCAGCCTCCACCCCCACCTGCAGCTCCCCGTGAGCCAGCCATCCACAG
CGAGGGGCAGTGGGTGACGCTGCCGCCCCCCCTGGACACCATCAACGTCCACCT
CCGGGCTGGGTACATCATCCCCCTGCAGGGCCCTGGCCTCACAACCACAGAGTC
CCGCCAGCAGCCCATGGCCCTGGCTGTGGCCCTGACCAAGGGTGGGGAGGCC
GAGGGGAGCTGTTCTGGGACGATGGAGAGAGCCTGGAAGTGCTGGAGCGAGGG
GCCTACACACAGGTCACTTCTGGCCAGGAATAACACGATCGTGAATGAGCTGG
TACGTGTGACCAGTGAGGGAGCTGGCCTGCAGCTGCAGAAGTGACTGCAGAAG
GTGACTGTCTGGGCGTGGCCACGGCGCCCCAGCAGGTCTCTCCAACGGTGTC
CCTGTCTCCAACCTTCACTACAGCCCCGACCAAGGTCTGGACATCTGTGTCTC
GCTGTTGATGGGAGAGCAGTTTCTCGTCAGCTGGTGT
GLB1wt
(SEQ ID NO: 50)
ATGCCGGGTTCTCGTTCGCATCCTCCTCTGTTGCTGGTTCTGCTGCTTCTGG
GCCCTACGCGCGGCTTGCGCAATGCCACCCAGAGGATGTTTGAAATTGACTATAG
CCGGGACTCCTTCTCAAGGATGGCCAGCCATTTTCGTACATCTCAGGAAGCATTC
ACTACTCCCGTGTGCCCCGCTTCTACTGGAAGGACCGGCTGCTGAAGATGAAGAT
GGCTGGGCTGAACGCCATCCAGACGTATGTGCCCTGGAACTTTCATGAGCCCTGG
CCAGGACAGTACCAGTTTCTGAGGACCATGATGTGGAATATTTTCTTCGGCTGGC
TCATGAGCTGGGACTGCTGGTTATCCTGAGGCCCGGGCCCTACATCTGTGCAGAG
TGGGAAATGGGAGGATTACCTGCTTGGCTGCTAGAGAAAGAGTCTATTCTTCTCCG
CTCCTCCGACCCAGATTACCTGGCAGCTGTGGACAAGTGGTTGGGAGTCCTTCTG
CCCAAGATGAAGCCTCCTCTATCAGAATGGAGGGCCAGTTATAACAGTGCAGG
TTGAAATGAATATGGCAGCTACTTTGCCTGTGATTTTGACTACCTGCGCTTCTGTC
AGAAGCGCTTTCGCCACCATCTGGGGGATGATGTGGTTCTGTTTACCACTGATGGA
GCACATAAAACATTCTGAAATGTGGGGCCCTGCAGGGCCTTACACCACGGTGG
ACTTTGGAACAGGCAGCAACATCACAGATGCTTTCCTAAGCCAGAGGAAGTGTGA
GCCCCAAGGACCCCTTGATCAATTCTGAATTCTATACTGGCTGGCTAGATCACTGGG
GCCAACCTCACTCCACAATCAAGACCGAAGCAGTGGCTTCTCCTCCTATGATATA

- continued

CTTGCCCGTGGGGCGAGTGTGAAC TTGTACATGTTTATAGGTGGGACCAATTTTGC
CTATTGGAATGGGGCCAACTCACCCTATGCAGCACAGCCACCAGCTACGACTAT
GATGCCCCACTGAGTGAGGCTGGGGACCTCACTGAGAAGTATTTTGCTCTGCGAA
ACATCATCCAGAAGTTTGAAAAAGTACCAGAAGGTCCTATCCCTCCATCTACACCA
AAGTTTGCAATATGGAAAGGTCAC TTGGAAAAGTTAAAGACAGTGGGAGCAGCTCT
GGACATTCTGTGTCCCTCTGGGCCCATCAAAGCCTTTATCCCTTGACATTATCCA
GGTGAAACAGCATTATGGGTTTGTGCTGTACCGACAACACTTCCTCAAGATTGCA
GCAACCCAGCACCTCTCTCTTCACCCCTCAATGGAGTCCACGATCGAGCATATGTT
GCTGTGGATGGGATCCCCAGGGAGTCCTTGAGCGAAACAATGTGATCACTCTGA
ACATAACAGGGAAAGCTGGAGCCACTCTGGACCTTCTGGTAGAGAACATGGGACG
TGTGAACTATGGTGCATATATCAACGATTTTAAGGGTTTGGTTTCTAACCTGACTCT
CAGTTCCAATATCCTCACGGACTGGACGATCTTCCACTGGACACTGAGGATGCAG
TGTGCAGCCACCTGGGGGGCTGGGGACACCGTGACAGTGGCCACCATGATGAAG
CCTGGGCCCACTCATCCAAC TACACGCTCCCGGCCTTTTATATGGGGAAC TTC
TCCATTCCAGTGGGATCCCAGACTTGCCCCAGGACACCTTATCCAGTTTCTTGG
ATGGACCAAGGGCCAGGTCTGGATTAATGGCTTTAACCTTGGCCGCTATTGGCCA
GCCCCGGGCCCTCAGTTGACCTTGT TGTGCCCCAGCACATCCTGATGACCTCGG
CCCCAAACACCATCACCGTGCTGGAAC TGAGTGGGCACCCCTGCAGCAGTGATGA
TCCAGAACTATGTGCTGTGACGTTCTGTGGACAGGCCAGTTATTGGCTCATCTGTGA
CCTACGATCATCCCTCCAAACCTGTTGAAAAAGACTCATGCCCCACCCCGCAA
AAAAACAAAGATT CATGGCTGGACCATGTA

GLB1Dup84

(SEQ ID NO: 51)

ATGCCGGGGTTCTCGTTCGCATCCTCCCTCTGTTGCTGGTTCTGCTGCTTCTGG
GCCCTACGCGCGGCTTGCGCAATGCCACCCAGAGGATGTTTGAAATTGACTATAG
CCGGGACTCCTTCTCAAGGATGGCCAGCCATTTCTGCTACATCTCAGGAAGCATTC
ACTACTCCCGTGTGCCCCGCTTCTACTGGAAGGACCGGCTGTGAAGATGAAGAT
GGCTGGGCTGAACGCCATCCAGACGTATGTGCCCTGGAAC TTT CATGAGCCCTGG
CCAGGACAGTACCAGTTTCTGAGGACCATGATGTGGAATATTTTCTTCGGCTGGC
TCATGAGCTGGGACTGCTGGTTATCCTGAGGCCCGGCCCTACATCTGTGCAGAG
TGGGAAATGGGAGGATTACCTGCTTGGCTGCTAGAGAAAGAGTCTATTCTTCTCCG
CTCCTCCGACCCAGATTACCTGGCAGCTGTGGACAAGTGGTTGGGAGTCCTTCTG
CCCAAGATGAAGCCTCTCCTCTATCAGAATGGAGGGCCAGTTATAACAGTGCAGG
TTGAAATGAATATGGCAGCTACTTTGCCTGTGATTTTGACTACCTGCGCTTCCTGC
AGAAGCGCTTTCGCCACCATCTGGGGGATGATGTGGTTCTGTTTACCACTGATGGA
GCACATAAAACATTCTTGAAATGTGGGGCCCTGCAGGGCCTCTACACCACGGTGG
ACTTTGGAACAGGCAGCAACATCACAGATGCTTTCCTAAGCCAGAGGAAGTGTGA
GCCCCAAGGACCC TTGATCAATTCTGAATTCTATACTGGCTGGCTAGATCACTGGG
GCCAACCTCACTCCACAATCAAGACCGAAGCAGTGGCTTCCTCCCTCTATGATATA
CTTGCCCGTGGGGCGAGTGTGAAC TTGTACATGTTTATAGGTGGGACCAATTTTGC
CTATTGGAATGGGGCCAACTCACCCTATGCAGCACAGCCACCAGCTACGACTAT
GATGCCCCACTGAGTGAGGCTGGGGACCTCACTGAGAAGTATTTTGCTCTGCGAA

- continued

ACATCATCCAGAAGTTTGAAAAAGTACCAGAAGGTCCTATCCCTCCATCTACACCA
AAGTTTGCATATGGAAAGGTCACCTTGGAAAAGTTAAAGACAGTGGGAGCAGCTCT
GGACATTCTGTGTCCCTCTGGGCCCATCAAAGCCTTTATCCCTTGACATTTATCCA
GGTGAAACAGCATTATGGGTTTGTGCTGTACCGGACAACACTTCCTCAAGATTGCA
GCAACCCAGCACCTCTCTTTCACCCCTCAATGGAGTCCACGATCGAGCATATGTT
GCTGTGGATGGGATCCCCAGGGAGTCCTTGAGCGAAACAATGTGATCACTCTGA
ACATAACAGGGAAAGCTGGAGCCACTCTGGACCTTCTGGTAGAGAACATGGGACG
TGTGAACTATGGTGCATATATGGTGCATATATCAACGATTTTAAGGGTTTGGTTTCT
AACCTGACTCTCAGTTCCAATATCCTCACGGACTGGACGATCTTCCACTGGACAC
TGAGGATGCAAGTGTGCAGCCACTGGGGGGCTGGGGACACCGTGACAGTGGCCA
CCATGATGAAGCCTGGGCCCAACTCATCCAACACACGCTCCCGGCCTTTTATA
TGGGGAACCTTCCATTCCAGTGGGATCCCAGACTTGCCCCAGGACACCTTTATC
CAGTTTCTGGATGGACCAAGGGCCAGGTCCTGGATTAATGGCTTTAACCTTGGCC
GCTATTGGCCAGCCCGGGCCCTCAGTTGACCTTGTTTGTGCCCCAGCACATCCT
GATGACCTCGGCCCAACACCATCACCGTGTGGAAGTGGAGTGGCACCCCTG
CAGCAGTGATGATCCAGAAGTATGTGCTGTGACGTTCTGTGGACAGGCCAGTTATT
GGCTCATCTGTGACCTACGATCATCCCTCCAAACCTGTTGAAAAAAGACTCATGCC
CCCACCCCGCAAAAAACAAAGATTTCATGGCTGGACCATGTA

PORCNwt

(SEQ ID NO: 52)

ATGGCCACCTTTAGCCGCCAGGAATTTTCCAGCAGCTACTGCAAGGCTGTCTCCT
GCCTACTGCCCAGCAGGGCCTTGACCAGATCTGGCTGCTCCTTGCCATCTGCCTC
GCCTGCCGCCCTCCTCTGGAGGCTCGGGTTGCCATCCTACCTGAAGCATGCAAGCA
CCGTGGCAGGCGGGTTCTTCAGCCTCTACCACTTCTTCCAGCTGCACATGGTTTG
GGTCGTGCTGCTCAGCCTCCTGTGCTACCTCGTGTCTTCTCTGCCGACATTCCT
CCCATCGAGGCGTCTTCTATCCGTACCATCCTCATCTACCTACTCATGGGTGAG
ATGCACATGGTAGACACCGTGACATGGCACAAGATGCGAGGGGCACAGATGATTG
TGCCCATGAAGGCAGTGTCTCTGGGCTTCGACCTGGACCGGGGCAGGTGGGTA
CGGTGCCCTCGCCAGTGGAGTTCATGGGCTACCTCTACTTCTGTGGGCACCATCGT
CTTCGGGCCCTGGATATCCTTCCACAGCTACCTACAAGCTGTCCAAGGCCGCCCA
CTGAGCTGCCGGTGGCTGCAGAAGGTGGCCCGGAGCCTGGCACTGGCCCTGCTG
TGCTTGTGCTGTCCACTTGCGTGGGCCCTACCTCTTCCCGTACTTCATCCCCCT
CAACGGTGACCGCTCCTTCGCAAGGGCACCATGGTAAGGTGGCTGCGAGCCTA
CGAGAGTGTGTCTCCTTCCACTTCAGCAACTATTTTGTGGGCTTTCTTCCGAGG
CCACGGCCACGTTGGCGGGGGCTGGCTTTACCGAGGAGAAGGATCACCTGGAAT
GGGACCTGACGGTGTCCAAGCCACTGAATGTGGAGCTGCCTCGGTCAATGGTGG
AAGTTGTCACAAGCTGGAACCTGCCCATGTCTTATTGGCTAAATAACTATGTTTCA
AGAATGCTCTCCGCCTGGGACCTTCTCGGCTGTGCTGGTCACCTATGCAGCCAG
CGCCCTCTACATGGCTTCAGTTTCCACCTGGCTGCGGTCCTGTGTCCCTGGCT
TTTATCACTTACGTGGAGCATGTCTCCGGAAGCGCCTGGCTCGGATCCTCAGTG
CCTGTGTCTTGTCAAAGCGGTGCCCGCCAGACTGTTGCAACCAGCATCGCTTGGG

-continued

CCTGGGGGTGCGAGCCTTAAACTTGCTCTTTGGAGCTCTGGCCATCTTCCACCTG
 GCCTACCTGGGCTCCCTGTTTATGTCGATGTGGATGACACCACAGAGGAGCAGG
 GCTACGGCATGGCATACTGTCCACAAGTGGTCAGAGCTCAGCTGGGCCAGTCA
 CTGGGTCACTTTTGGATGCTGGATCTTCTACCGTCTCATAGGC
 PORCNDup20 (SEQ ID NO: 53)
 ATGGCCACCTTTAGCCGCCAGGAATTTTCCAGCAGCTACTGCAAGGCTGTCTCCT
 GCCTACTGCCCAGCAGGGCCTTGACCAGATCTGGCTGCTCCTTGCCATCTGCCTC
 GCCTGCCGCTCTCTGGAGGCTCGGGTTGCCATCCTACCTGAAGCATGCAAGCA
 CCGTGGCAGGCGGGTTCTTCAGCCTCTACCATTCTTCCAGCTGCACATGGTTTG
 GGTCTGTCTGCTCAGCCTCCTGTGTACCTCGTGTCTTCTCTGCCGACATTCCT
 CCCATCGAGGCGTCTTCTATCCGTCACCATCCTCATCTACCTACTCATGGGTGAG
 ATGCACATGGTAGACACCGTGACATGGCACAAGATGCGAGGGGCACAGATGATTG
 TGGCCATGAAGGCAGTGTCTCTGGGCTTCGACCTGGACCGGGCGAGGTGGGTA
 CGGTGCCCTCGCCAGTGGAGTTCATGGGCTACCTCTACTTCGTGGGCACCATCGT
 CTTCCGGGCCCTGGATATCCTTCCACAGCTACCTACAAGCTGTCCAAGGCCGCCCA
 CTGAGCTGCCGGTGGCTGCAGAAGGTGGCCCGAGCCTGGCACTGGCCCTGCTG
 TGCCTTGTCTGTCCACTTGCGTGGGCCCTACCTCTTCCCGTACTTCATCCCCCT
 CAACGGTGACCGCTCCTTCGCAAGGGCACCATGGTAAGGTGGCTGCGAGCCTA
 CGAGAGTGCTGTCTCCTTCCACTTCAGCAACTATTTTGTGGGCTTTCTTTCCGAGG
 CCACGGCCACGTGGCGGGGCTGGCTTTACCGAGGAGAAGGATCACCTGGAAT
 GGGACCTGACGGTGTCCAAGCCACTGAATGTGGAGCTGCCTCGGTCAATGGTGG
 AAGTTGTCACAAGCTGGAACCTGCCCATGTCTTATTGGCTAAATAACTATGTTTCA
 AGAATGCTCTCCGCTGGGGACCTTCTCGGCTGTCTGGTCACCTATGCAGCCAG
 CGCCCTCTACATGGCTTCAGTTTCCACCTGGCTGCGGTCCTGTGTCCCTGGCT
 TTTATCCCTGGCTTTTATCACTTACGTGGAGCATGTCCTCCGGAAGCGCCTGGCTC
 GGATCCTCAGTGCCGTGTCTTGTCAAAGCGGTGCCCGCCAGACTGTTTCGCACCA
 GCATCGCTTGGGCCTGGGGGTGCGAGCCTTAAACTTGCTCTTTGGAGCTCTGGCC
 ATCTTCCACCTGGCCTACCTGGGCTCCCTGTTTGATGTCGATGTGGATGACACCAC
 AGAGGAGCAGGGCTACGGCATGGCATACTGTCCACAAGTGGTCAGAGCTCAG
 CTGGGCCAGTCACTGGGTCACTTTTGGATGCTGGATCTTCTACCGTCTCATAGGC

EXECUTIVE SUMMARY

It was found that template-free DNA repair of Cas9-
 cleaved and Cpf1-cleaved DNA produces a predictable set
 of repair genotypes that can result in the gain-of-function
 repair of human disease mutations. Contrary to the assump-
 tion that end-joining following double-strand breaks is ran-
 dom and difficult to harness for applications beyond gene
 disruption, here it is shown that template-free end-joining
 repair of DNA cleaved by CRISPR-associated nucleases
 produces a predictable set of repair genotypes. A library of
 2000 guide RNAs paired with target DNA sites was con-
 structed, and they were integrated into mouse and human
 genomes, applied Cas9, and performed high-throughput
 sequencing of repair genotypes. Data from this assay are
 consistent with results from 98 endogenous loci. Building
 upon prior work, it is shown that the majority of repair

50

genotypes in cells with saturated exposure to both CRISPR-
 Cas9 and Cpf1 are deletions associated with sequence
 microhomology. Using 1,588 sequence contexts from the
 data, CRISPR-Texture, a machine learning method that
 accurately predicted the frequencies of template-free Cas9-
 mediated microhomology-associated deletions as well as 1
 bp insertions, was trained. On 282 held-out sequence con-
 texts, CRISPR-Texture predicted frameshift rates more
 accurately than published methods and accurately predicted
 the statistical entropy of repair product distributions.
 Applied to the human genome, CRISPR-Texture identified
 an appreciable fraction of Cas9 target sites supporting high-
 precision repair distributions that are dominated by a single
 genotype. Further, it was found that a class of human
 disease-associated micro-duplication mutations can be
 repaired to wildtype at high frequency by template-free Cas9
 nuclease editing and used the assay to validate hundreds of

60

65

such alleles. Template-free Cas9 nuclease-mediated rescue of pathogenic LDLR alleles to wildtype phenotype in cellular models was also validated. This work establishes a strategy for predicting the outcomes of template-free end-joining and demonstrates that CRISPR editing can also mediate efficient gain-of-function editing at certain disease alleles without homology-directed repair.

REFERENCES

1. Cong, L. et al. Multiplex Genome Engineering Using CRISPR/Cas Systems. *Science* 339, 819 (2013).
2. Mali, P. et al. RNA-Guided Human Genome Engineering via Cas9. *Science* 339, 823–826 (2013).
3. Jinek, M. et al. RNA-programmed genome editing in human cells. *eLife* 2, e00471 (2013).
4. Doench, J. G. et al. Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. *Nat. Biotechnol.* 34, 1–12 (2016).
5. Kleinstiver, B. P. et al. High-fidelity CRISPR-Cas9 nucleases with no detectable genome-wide off-target effects. *Nature* 529, 490–495 (2016).
6. Slaymaker, I. M. et al. Rationally engineered Cas9 nucleases with improved specificity. *Science* 351, 84–88 (2016).
7. Fu, Y., Sander, J. D., Reyon, D., Cascio, V. M. & Joung, J. K. Improving CRISPR-Cas nuclease specificity using truncated guide RNAs. *Nat. Biotechnol.* 32, 279–284 (2014).
8. Kleinstiver, B. P. et al. Engineered CRISPR-Cas9 nucleases with altered PAM specificities. *Nature* 523, 481–485 (2015).
9. Kleinstiver, B. P. et al. Broadening the targeting range of *Staphylococcus aureus* CRISPR-Cas9 by modifying PAM recognition. *Nat. Biotechnol.* 33, 1293–1298 (2015).
10. Hu, J. H. et al. Evolved Cas9 variants with broad PAM compatibility and high DNA specificity. *Nature* 1–24 (2018). doi:10.1038/nature26155
11. Komor, A. C., Kim, Y. B., Packer, M. S., Zuris, J. A. & Liu, D. R. Programmable editing of a target base in genomic DNA without double-stranded DNA cleavage. *Nature* 533, 420–424 (2016).
12. Gaudelli, N. M. et al. Programmable base editing of A•T to G•C in genomic DNA without DNA cleavage. *Nature* 1–27 (2017). doi:10.1038/nature24644
13. Chu, V. T. et al. Increasing the efficiency of homology-directed repair for CRISPR-Cas9-induced precise gene editing in mammalian cells. *Nat. Biotechnol.* 33, 543–548 (2015).
14. Richardson, C. D., Ray, G. J., DeWitt, M. A., Curie, G. L. & Corn, J. E. Enhancing homology-directed genome editing by catalytically active and inactive CRISPR-Cas9 using asymmetric donor DNA. *Nat. Biotechnol.* 34, 339–344 (2016).
15. Paquet, D. et al. Efficient introduction of specific homozygous and heterozygous mutations using CRISPR/Cas9. *Nature* 533, 1–18 (2016).
16. Landrum, M. J. et al. ClinVar: Public archive of interpretations of clinically relevant variants. *Nucleic Acids Res.* 44, D862–D868 (2016).
17. Stenson, P. D. et al. Human Gene Mutation Database: towards a comprehensive central mutation database. *J. Med. Genet.* 45, 124 (2008).
18. Shin, H. Y. et al. CRISPR/Cas9 targeting events cause complex deletions and insertions at 17 sites in the mouse genome. *Nat. Commun.* 8, 1–10 (2017).
19. Sakuma, T., Nakade, S., Sakane, Y., Suzuki, K.-I. T. & Yamamoto, T. MMEJ-assisted gene knock-in using TALENs and CRISPR-Cas9 with the PITCh systems. *Nat. Protoc.* 11, 118–133 (2015).
20. Suzuki, K. et al. In vivo genome editing via CRISPR/Cas9 mediated homology-independent targeted integration. *Nature* 540, 144–149 (2016).
21. Nakade, S. et al. Microhomology-mediated end-joining-dependent integration of donor DNA in cells and animals using TALENs and CRISPR/Cas9. *Nat. Commun.* 5, 5560–5560 (2014).
22. Kraft, K. et al. Deletions, Inversions, Duplications: Engineering of Structural Variants using CRISPR/Cas in Mice. *Cell Rep.* 10, 833–839.
23. Koike-Yusa, H., Li, Y., Tan, E.-P., Velasco-Herrera, M. D. C. & Yusa, K. Genome-wide recessive genetic screening in mammalian cells with a lentiviral CRISPR-guide RNA library. *Nat. Biotechnol.* 32, 267–273 (2013).
24. van Overbeek, M. et al. DNA Repair Profiling Reveals Nonrandom Outcomes at Cas9-Mediated Breaks. *Mol. Cell* 63, 633–646 (2016).
25. Urasaki, A., Morvan, G. & Kawakami, K. Functional dissection of the Tol2 transposable element identified the minimal cis-sequence and a highly repetitive sequence in the subterminal region essential for transposition. *Genetics* 174, 639–649 (2006).
26. Ceccaldi, R., Rondinelli, B. & D'Andrea, A. D. Repair Pathway Choices and Consequences at the Double-Strand Break. *Spec. Issue Qual. Control* 26, 52–64 (2016).
27. Deriano, L. & Roth, D. B. Modernizing the Nonhomologous End-Joining Repertoire: Alternative and Classical NHEJ Share the Stage. *Annu. Rev. Genet.* 47, 433–455 (2013).
28. Evers, B. et al. CRISPR knockout screening outperforms shRNA and CRISPRi in identifying essential genes. *Nat. Biotechnol.* 34, 631–633 (2016).
29. Bae, S., Kweon, J., Kim, H. S. & Kim, J.-S. Microhomology-based choice of Cas9 nuclease target sites. *Nat. Methods* 11, 705–706 (2014).
30. Cornu, T. I., Mussolino, C. & Cathomen, T. Refining strategies to translate genome editing to the clinic. *Nat. Med.* 23, 415 (2017).
31. Shalem, O., Sanjana, N. E. & Zhang, F. High-throughput functional genomics using CRISPR-Cas9. *Nat. Rev. Genet.* 16, 299 (2015).
32. Mandal, P. K. et al. Efficient Ablation of Genes in Human Hematopoietic Stem and Effector Cells using CRISPR/Cas9. *Cell Stem Cell* 15, 643–652 (2014).
33. Tabebordbar, M. et al. In vivo gene editing in dystrophic mouse muscle and muscle stem cells. *Science* 351, 407 (2016).
34. Arbab, M., Srinivasan, S., Hashimoto, T., Geijsen, N. & Sherwood, R. I. Cloning-free CRISPR. *Stem Cell Rep.* 5, 908–917 (2015).
35. Davis, A. J. & Chen, D. J. DNA double strand break repair via non-homologous end-joining. *Transl. Cancer Res.* 2, 130–143 (2013).
36. Bourbon, M., Alves, A. C. & Sijbrands, E. J. Low-density lipoprotein receptor mutational analysis in diagnosis of familial hypercholesterolemia. *Curr. Opin. Lipidol.* 28, 120–129 (2017).
37. Ran, F. A. et al. In vivo genome editing using *Staphylococcus aureus* Cas9. *Nature* 520, 186 (2015).
38. Oh, J. et al. Positional cloning of a gene for Hermansky-Pudlak syndrome, a disorder of cytoplasmic organelles. *Nat. Genet.* 14, 300–306 (1996).

39. Orthwein, A. et al. A mechanism for the suppression of homologous recombination in G1 cells. *Nature* 528, 422 (2015).
40. Biehs, R. et al. DNA Double-Strand Break Resection Occurs during Non-homologous End Joining in G1 but Is Distinct from Resection during Homologous Recombination. *Mol. Cell* 671–684 (2017). doi:10.1016/j.molcel.2016.12.016
41. Zetsche, B. et al. Cpf1 Is a Single RNA-Guided Endonuclease of a Class 2 CRISPR-Cas System. *Cell* 163, 759–771 (2015).
42. Christian, M. et al. Targeting DNA Double-Strand Breaks with TAL Effector Nucleases. *Genetics* 186, 757 (2010).
43. Kim, Y. G., Cha, J. & Chandrasegaran, S. Hybrid restriction enzymes: zinc finger fusions to Fok I cleavage domain. *Proc. Natl. Acad. Sci.* 93, 1156 (1996).
44. Sherwood, R. I. et al. Discovery of directional and nondirectional pioneer transcription factors by modeling DNase profile magnitude and shape. *Nat. Biotechnol.* 32, 171–178 (2014).
45. DiCarlo, J. E., Chavez, A., Dietz, S. L., Esvelt, K. M. & Church, G. M. Safeguarding CRISPR-Cas9 gene drives in yeast. *Nat. Biotechnol.* 33, 1250 (2015).
46. McVey, M. & Lee, S. E. MMEJ repair of double-strand breaks (director's cut): deleted sequences and alternative endings. *Trends Genet.* 24, 529–538 (2008).
47. Yu, A. M. & McVey, M. Synthesis-dependent microhomology-mediated end joining accounts for multiple types of repair junctions. *Nucleic Acids Res.* 38, 5706–5717 (2010).
48. Heidenreich, E., Novotny, R., Kneidinger, B., Holzmann, V. & Wintersberger, U. Non-homologous end joining as an important mutagenic process in cell cycle; arrested cells. *EMBO J.* 22, 2274 (2003).
49. Pfeiffer, P., Goedecke, W. & Obe, G. Mechanisms of DNA double-strand break repair and their potential to induce chromosomal aberrations. *Mutagenesis* 15, 289–302 (2000).

Example 2: Workflow Description of Using inDelphi to Design Cas9 gRNAs for Efficient Genome Editing to Induce Exon Skipping

One application of CRISPR-Cas9 for therapeutic purposes is to alter the genome to cause RNA editing to skip a pathogenic exon by changing the splicing regulatory sites controlling the inclusion of the pathogenic exon. In some cases, a single exon in a gene may contain a pathogenic variant, and the entire exon can be skipped to produce an alternative isoform with normal function. For example, certain cases of Duchenne muscular dystrophy (DMD) are caused by a deleterious mutation in exon 23 of the *Dmd* gene that results in a premature stop codon that produces a dysfunctional *Dmd* protein. To restore function the deleterious exon 23 can be skipped by editing the DNA proximal to the diseased exon with a single gRNA (Long, 2016).

This Example tests a new method for selecting gRNAs for CRISPR editing to disrupt splice site sequences to cause the skipping of pathogenic exons to restore wild-type cellular function. Splice site acceptor DNA motifs occur at the boundary between introns and exons, depicted as 5'-intron-AG-exon-3', where the AG is a highly-conserved element of the splice site acceptor motif and is considered to reside in the intron. The splice site acceptor DNA motif is considered to be as long as 23 bp in length including the AG. Algorithms such as MaxEntScan (Yeo, 2004) receive as input a DNA

sequence and output a numerical score representing how strongly the splice site acceptor motif is present at the DNA sequence.

CRISPR-Cas9 induces a DNA double-strand break at a specific location specified by the gRNA. DNA repair fixes the double-strand break, inducing insertions and deletions through non-homologous end-joining (NHEJ) and microhomology-mediated end-joining (MMEJ) when a homology template is not present for repair through homology-directed repair (HDR). InDelphi, as disclosed herein, predicts the frequency distribution of NHEJ/MMEJ-mediated repair genotypes following Cas9 cutting. In this Example, inDelphi's ability to predict the spectrum of repair genotypes is utilized to identify gRNAs that ablate the splice site acceptor motif at a high frequency out of all non-wild-type repair outcomes.

Other computational methods have focused on predicting on-target efficiency of CRISPR-Cas9 cutting. This Example aims to identify gRNAs that will efficiently cut and induce non-wild-type repair outcomes (otherwise described as high on-target activity). A relevant published method (Doench et al., 2016, *Nature*, aka "Azimuth") uses the DNA sequence surrounding the gRNA, as well as the position of the cutsite in the protein, to predict the gRNA's ability to knock out the protein as observed in gRNA enrichment in screens of cell-essential genes, where a higher score indicates higher gRNA cutting efficacy. However, Doench et al. does not directly predict the frequency of non-wild-type repair frequency. It is reasoned here that the frequency of protein knockdown depends on the rate of non-wild-type repair, and the rate of frameshift repair out of all non-wild-type repair outcomes. Despite this concern, gRNAs are filtered based on a minimum threshold Azimuth score in order to maximize the chances that selected gRNAs have high on-target activity (increase true positives), at the risk of filtering away some gRNAs that also have high on-target activity (increase false negatives). In alternative embodiments, this Azimuth filtering step may be skipped to decrease the rate of false negatives at the risk of decreasing the rate of true positives.

To more directly address the question of on-target efficiency, this Example developed an algorithm referred to as the Basic On-Target Model (BOTM) which directly predicts the frequency of non-wild-type observations from DNA sequence features. BOTM uses DNA sequence as input and outputs a predicted frequency of non-wild-type repair, where non-wild-type repair is defined as the sum frequency of CRISPR-associated deletions and insertions (defined as reads aligning to the reference with exactly one gap which resides within 1-bp of the cutsite, using alignment scores +1 match, -1 mismatch, -5 gap open, -0 gap extend), over the denominator of sum frequency of non-noise outcomes consisting of CRISPR-associated indels, wildtype repair, and reads with multiple indels with at least one occurring near the cutsite, and reads with exactly one indel occurring anywhere outside the cutsite). BOTM is implemented as an ensemble of 100 gradient boosted regression trees, each with maximum depth 3, that are fitted in consecutive stages on the negative gradient of the least squares loss function. BOTM uses the following input features: one-hot encoded nucleotides at positions -7 to 0 (such that "NGG" occupies positions 0 to 2), the GC fraction of the 40-bp window around the cutsite, and the following features from inDelphi: log phi score (microhomology score), precision score (ranging from 0 to 1, with 1 being more precise), expected value of the indel length distribution, the frequency of 1-bp insertions, microhomology deletions, and microhomology-less deletions, the highest frequency of any single 1-bp

insertion outcome, the highest frequency of any single deletion outcome, and the highest frequency of any single outcome. Trained on deep sequencing data at 3,600 target sites from our genome-integrated library construct in mES also used to train inDelphi, BOTM achieves a Pearson correlation of 0.42 at predicting the observed frequency of non-wild-type repair on 400 held-out target sites from our genome-integrated library construct in mES. On held-out data, it was manually determined a BOTM predicted frequency of 0.65 or greater for gRNAs that have a high frequency of non-wildtype repair.

One computational workflow for identifying Cas9 gRNAs with clinical relevance for the correction of genetic diseases by inducing exon skipping consists of four steps: identify relevant exons, select gRNAs for these exons with effective targeting, determine the genotypic products of each gRNA using inDelphi, and select gRNAs with genotypic products that are predicted to disrupt the relevant splicing motif. Using this approach, we have identified 4000 gRNAs that target splice sites to correct genetic diseases (Appendix attached).

First, 6805 exons with the following characteristics were determined: the exon length is evenly divisible by 3 so that skipping them preserves frame; the exon contains at least one HGMD pathogenic indel, which are likely to disrupt normal protein function (basal frameshift rate ~66%, column "hgmd_indel_count" in Appendix spreadsheet); the exon is not constitutive, measured by <100% presence in Ensembl transcripts (Ensembl); and the exon does not contain an annotated protein domain in Pfam (Pfam). The last two criteria are used to identify exons that may not be essential for wild-type protein function. The resulting 6,805 exons were candidates for disease correction by exon skipping.

Then, SpCas9 gRNAs (NGG PAM) with cutsites in a 6 bp window surrounding and including the AG motif were selected, resulting in an average of 2.2 SpCas9 gRNAs per exon. We then ensured high predicted on-target editing efficiency by removing all gRNAs with Azimuth score below 0.20. (threshold set manually) or BOTM score below 0.65 which is chosen to separate gRNAs with high versus low frequencies of non-wild-type repair).

Each gRNA and exon target site for splice site motif disruption were scored. We obtained this prediction by first using inDelphi to predict the frequency distribution of 1-bp insertion and deletion (1–60 bp) genotypes resulting from template-free DNA repair of a CRISPR gRNA induced cut at the target exon site.

Finally, for each genotype predicted by inDelphi, we classified a genotype as "motif disrupting" when its MaxEntScan score is <0.9 of its unedited MaxEntScan score; otherwise we classified the genotype as "no effect". This classification ruleset was provided by (Tang, 2016) and validated on experimental splicing data to achieve a sensitivity of 83.6% and specificity of 79.2% (Tang, 2016). The total frequency of all motif-disruption repair genotypes was used to predict the splice site motif disruption frequency out of all inDelphi predicted genotypes.

The top 4000 gRNA and target site pairs were selected based on this predicted frequency of splice site disruptions. Long, 2016 identified several SpCas9 gRNAs that, in a mouse model of muscular dystrophy, restored some degree of dystrophin protein expression and improved skeletal muscle function by inducing exon skipping of exon 23 (containing a non-sense mutation) via NHEJ-mediated DNA repair of a Cas9-induced cut. Without considering the results of their experiments and focusing solely on the DNA

sequence context and background biological knowledge, our computational workflow recognizes that exon 23 of DMD is a good candidate for disease correction via exon skipping: the exon has a length evenly divisible by 3, is associated with a pathogenic non-sense variant that destroys normal protein function, and is not constitutive or required for normal protein function. Long 2016 reports results for only one SpCas9 gRNA targeting the 5' end of exon 23 called sgRNA-L8 ATAATTTCTATTATATTACA (SEQ ID NO: 11590) with PAM GGG. In their experiments with sgRNA-L8, they observe 9/18 pups with exon 23 skipping. This gRNA targets mm10 chrX: 83,803,134–83,803,156 (minus strand), while the exon 23 boundary is 149-bp downstream at mm10 chrX: 83,803,305. This Example's computational workflow for now only identifies gRNAs cutting within a 6-bp window of the AG motif at the exon 5' boundary, so as described our workflow does not identify Long's sgRNA-L8. Other methods of selecting exons for splice site acceptor removal include selecting exons with mutant splice regulatory sites that result in the inappropriate inclusion of exons in RNA transcripts (Sterne-Weiler 2014). Alternatively, subsequent expressed exons can be skipped to restore reading frame. In this case, reading frame can be restored by skipping a subsequent expressed exon where the length of the subsequent skipped exon and the length of the indel sum to 0 mod 3. In addition, constitutive exons and/or exons known to contain annotated protein domains in Pfam can be selected for exon skipping as an alternative method for knocking out a gene.

Correcting Genetic Disorders Using Predictable CRISPR/Cas9-Induced Exon Skipping

Exon skipping has emerged as a powerful method to restore gene function in a number of genetic disorders. These therapies force the splicing machinery to bypass exons that contain deleterious point mutations or frameshifts. The FDA has recently approved an antisense oligonucleotide therapy that induces exon skipping in Duchenne muscular dystrophy to restore dystrophin function, and several other related strategies have shown pre-clinical promise. Yet, oligonucleotide therapies are transient treatments that require frequent dosing.

CRISPR/Cas9 instead promises to alleviate genetic disease permanently, through genome alteration. Using a high-throughput experimental-computational pipeline, as described herein, the inventors have developed an algorithm capable of highly accurate prediction of CRISPR/Cas9 genotypic alterations. At a predictable subset of genomic target sites, CRISPR/Cas9 induces precise sequence deletions. These modifications are highly specific and have excellent potential for therapeutic genome editing through controlled deletion of splice-acceptor sites.

The inventors will systematically evaluate this new approach to treat genetic disorders using CRISPR/Cas9 deletions. At intron-exon junctions, we will induce small deletions to bypass exons containing deleterious variants that affect protein function or alter the reading frame. While not every splice site can be successfully deleted through CRISPR/Cas9 modification, and not every exon can be skipped without compromising gene function, the inventors expect that this approach will succeed in enough genes to have broad therapeutic implications.

To measure the applicability of this approach to treat disease throughout the genome, the inventors will establish a principled computational approach to identify exons known to harbor disease-causing mutations where omission is unlikely to impact gene function. The inventors will then apply a novel, high-throughput CRISPR/Cas9 assay that

quantifies the impact of high-precision genome editing on splicing at thousands of these intron-exon boundaries. After determining a set of candidate exons that can be skipped efficiently, the inventors will measure the impact of CRISPR/Cas9-mediated exon skipping on transcript structure and gene function for dozens of human disease exons. This exhaustive approach promises to chart a systematic path toward classifying disease genes that would be most amenable for future pre-clinical evaluation of permanent therapeutic exon skipping.

Induce Exon Skipping Using CRISPR/Cas9 at Thousands of Exons that Harbor Disease Variants.

By mapping coding variants known to be associated with genetic disorders, the inventors will develop a set of exons whose skipping could feasibly provide clinical benefit. The inventors will use measures of selective constraint from large-scale population data and alternative splicing data to prioritize exons whose skipping is least likely to compromise protein function. Using the herein described algorithm which predicts the genotypic consequence of targeting with CRISPR/Cas9, the inventors will refine a list of up to 10,000 intron-exon boundaries where modification is predicted to induce exon skipping at high rates. To test these predictions in high-throughput, the inventors will adapt our CRISPR/Cas9 cutting assay to read out context-specific splicing outcomes in human cells *in vitro*. This novel assay will allow paired evaluation of CRISPR/Cas9 cutting genotype and splicing phenotype for hundreds of distinct replicates in each of 10,000 human exons. The inventors will perform this assay in several human cell lines and will computationally identify exons that can be skipped at high frequency for further study. The inventors will also explore Cas9 base-editing in the same high-throughput system to determine if splicing can be altered by single base alterations. Using these data, the inventors will derive computational rules for which sequence alterations do and do not lead to exon skipping.

Evaluate the Consequences of CRISPR/Cas9 Exon Skipping on Transcript Structure and Function.

Using results from the high-throughput assay, the inventors will select up to 100 exon-skipping guide RNAs to pursue in greater depth. The inventors will prioritize exons that are natively excluded from at least one experimentally validated splice isoform, that lack characterized protein domains, and that are under relaxed selective constraint. For these exons, the inventors will edit native genomes in an appropriate cell line given the gene function and disease process. By performing transcript-specific RNA deep sequencing, the inventors will determine the rate of exon skipping and the transcript structure after the exon is skipped, monitoring for the appearance of aberrant splice acceptors. The inventors will also assay the function of these genes with skipped exons, using appropriate cellular and biochemical assays for each gene. This analysis will identify a set of disease genes that are promising candidates for further study in mutated cell lines and animal models. Overall, this systematic study will elucidate which disease genes are compelling candidates for pre-clinical evaluation of CRISPR/Cas9-mediated exon skipping therapy.

REFERENCES

Long C, Amoasii L, Mireault A, McAnally J, Li H, Sanchez-Ortiz E, Bhattacharyya S, Shelton J, Bassel-Duby R, Olson E. Postnatal genome editing partially restores dystrophin expression in a mouse model of muscular dystrophy. *Science* 2016; 351 (6271): 400–403.

Yeo G, Burge CB. Maximum entropy modeling of short sequence motifs with applications to RNA splicing signals. *J Comput Biol.* 2004; 11(2–3):377–94.

John G. Doench*, Nicolo Fusi*, Meagan Sullender*, Mudra Hegde*, Emma W. Vaimberg*, Katherine F. Donovan, Ian Smith, Zuzana Tothova, Craig Wilen, Robert Orchard, Herbert W. Virgin, Jennifer Listgarten*, David E. Root. Optimized sgRNA design to maximize activity and minimize off-target effects for genetic screens with CRISPR-Cas9. *Nature Biotechnology* January 2016.

Rongying Tang, Debra O. Prosser, and Donald R. Love, “Evaluation of Bioinformatic Programmes for the Analysis of Variants within Splice Site Consensus Regions,” *Advances in Bioinformatics*, vol. 2016, Article ID 5614058, 10 pages, 2016.

Sterne-Weiler T, Sanford J. Exon identity crisis: disease-causing mutations that disrupt the splicing code. *Genome Biology* 2014 15:201.

OTHER EMBODIMENTS

The foregoing has been a description of certain non-limiting embodiments of the invention. Those of ordinary skill in the art will appreciate that various changes and modifications to this description may be made without departing from the spirit or scope of the present invention, as defined in the following claims.

EQUIVALENTS AND SCOPE

In the claims articles such as “a,” “an,” and “the” may mean one or more than one unless indicated to the contrary or otherwise evident from the context. Claims or descriptions that include “or” between one or more members of a group are considered satisfied if one, more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process unless indicated to the contrary or otherwise evident from the context. The invention includes embodiments in which exactly one member of the group is present in, employed in, or otherwise relevant to a given product or process. The invention includes embodiments in which more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process.

Furthermore, the invention encompasses all variations, combinations, and permutations in which one or more limitations, elements, clauses, and descriptive terms from one or more of the listed claims is introduced into another claim. For example, any claim that is dependent on another claim can be modified to include one or more limitations found in any other claim that is dependent on the same base claim. Where elements are presented as lists, e.g., in Markush group format, each subgroup of the elements is also disclosed, and any element(s) can be removed from the group. It should be understood that, in general, where the invention, or aspects of the invention, is/are referred to as comprising particular elements and/or features, certain embodiments of the invention or aspects of the invention consist, or consist essentially of, such elements and/or features. For purposes of simplicity, those embodiments have not been specifically set forth in *haec verba* herein. It is also noted that the terms “comprising” and “containing” are intended to be open and permits the inclusion of additional elements or steps. Where ranges are given, endpoints are included. Furthermore, unless otherwise indicated or otherwise evident from the context and understanding of one of ordinary skill in the art, values that are expressed as

121

ranges can assume any specific value or sub-range within the stated ranges in different embodiments of the invention, to the tenth of the unit of the lower limit of the range, unless the context clearly dictates otherwise.

This application refers to various issued patents, published patent applications, journal articles, and other publications, all of which are incorporated herein by reference. If there is a conflict between any of the incorporated references and the instant specification, the specification shall control. In addition, any particular embodiment of the present invention that falls within the prior art may be explicitly excluded from any one or more of the claims. Because such embodiments are deemed to be known to one of ordinary skill in the art, they may be excluded even if the exclusion is not set forth explicitly herein. Any particular embodiment of the invention can be excluded from any claim, for any reason, whether or not related to the existence of prior art.

Those skilled in the art will recognize or be able to ascertain using no more than routine experimentation many equivalents to the specific embodiments described herein. The scope of the present embodiments described herein is not intended to be limited to the above Description, but rather is as set forth in the appended claims. Those of ordinary skill in the art will appreciate that various changes and modifications to this description may be made without departing from the spirit or scope of the present invention, as defined in the following claims.

Lengthy table referenced here

US12406749-20250902-T00001

Please refer to the end of the specification for access instructions.

Lengthy table referenced here

US12406749-20250902-T00002

Please refer to the end of the specification for access instructions.

122

Lengthy table referenced here

US12406749-20250902-T00003

Please refer to the end of the specification for access instructions.

Lengthy table referenced here

US12406749-20250902-T00004

Please refer to the end of the specification for access instructions.

Lengthy table referenced here

US12406749-20250902-T00005

Please refer to the end of the specification for access instructions.

Lengthy table referenced here

US12406749-20250902-T00006

Please refer to the end of the specification for access instructions.

Lengthy table referenced here

US12406749-20250902-T00007

Please refer to the end of the specification for access instructions.

LENGTHY TABLES

The patent contains a lengthy table section. A copy of the table is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12406749B2>). An electronic copy of the table will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12406749B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

123

What is claimed is:

1. A method of introducing a genetic change into a target genomic location encoding a pathogenic allele to modify the pathogenic allele to become a non-pathogenic allele using a Cas-based double strand break genome editing system, the method comprising:

using a computer hardware processor to perform:

selecting a guide RNA for use in introducing the genetic change into the target genomic location by analyzing inputs indicating a nucleotide sequence of the target genomic location and one or more available cut sites for the Cas-based double strand break genome editing system, the selecting comprising:

(a) determining a microhomology score matrix using a first neural network, the determining comprising:

determining a plurality of pairs of overhang sequences using the inputs;

determining a microhomology length vector and/or a microhomology GC fraction vector using the inputs; and

applying the first neural network to the plurality of pairs of overhang sequences and the microhomology length vector and/or the microhomology GC fraction vector to obtain the microhomology score matrix;

(b) determining a microhomology-independent score matrix using a second neural network, the determining comprising:

determining a deletion length vector using the inputs and the plurality of pairs of overhang sequences; and

applying the second neural network to the deletion length vector to obtain the microhomology-independent score matrix;

(c) determining a probability distribution over 1-bp insertions;

(d) determining, using the microhomology score matrix, the microhomology-independent score matrix and the probability distribution over 1-bp insertion, a probability distribution over indel genotypes and a probability distribution over indel lengths for the nucleotide sequence of the target genomic location and the one or more available cut sites;

(e) determining, using the probability distribution over indel genotypes and the probability distribution over indel lengths, for each guide RNA of a plurality of guide RNAs, a predicted frequency of introducing the genetic change into the target genomic location using the Cas-based double strand break genome editing system and the guide RNA;

(f) selecting, using the predicted frequencies of (e), a guide RNA of the plurality of guide RNAs for use in introducing the genetic change into the target genomic location using the Cas-based double strand break genome editing system; and

introducing the genetic change into the target genomic location using the guide RNA selected at (f) and the Cas-based double strand break genome editing system, wherein the genetic change is a 1 base pair insertion or a 1–60 base pair deletion that modifies the pathogenic allele to become the non-pathogenic allele.

2. The method of claim 1, wherein analyzing inputs comprises identifying, for a cut site of the one or more available cut sites, a plurality of pairs of overhang sequences, each pair of the plurality of pairs of overhang sequences comprising:

(i) a 3'-overhang sequence; and

(ii) a 5'-overhang sequence.

124

3. The method of claim 1, wherein determining the probability distribution over 1-bp insertion comprises:

determining, using the inputs, an overall deletion score, a precision score, and/or cut site nucleotides; and

applying a third neural network to the overall deletion score, the precision score, and/or the cut site nucleotides to obtain the probability distribution over 1-bp insertion.

4. The method of claim 1, comprising:

contacting a plurality of cells with the guide RNA and the Cas-based double strand break genome editing system; and

introducing, using the selected guide RNA and the Cas-based double strand break genome editing system, the genetic change into the target genomic location of at least 30% of cells of the plurality of cells.

5. The method of claim 1, comprising:

contacting a plurality of cells with the guide RNA and the Cas-based double strand break genome editing system; and

introducing, using the selected guide RNA and the Cas-based double strand break genome editing system, the genetic change into the target genomic location of at least 50% of cells of the plurality of cells.

6. The method of claim 1, wherein the Cas-based double strand break genome editing system comprises a type II Cas RNA-guided endonuclease, or a functional variant or ortholog thereof.

7. The method of claim 1, wherein the Cas-based double strand break genome editing system comprises a Cas9 RNA-guided endonuclease, or a variant or orthologue thereof.

8. The method of claim 7, wherein the Cas9 RNA-guided endonuclease is *Streptococcus pyogenes* Cas9 (SpCas9), *Staphylococcus pyogenes* Cas9 (SpCas9), *Staphylococcus aureus* Cas (SaCas9), *Francisella novicida* Cas9 (FnCas9), or a functional variant or orthologue thereof.

9. The method of claim 1, further comprising:

treating a genetic disease in a subject caused by a pathogenic allele in a genome of a cell of the subject by contacting the genome of the cell of the subject with the selected guide RNA and the Cas-based double strand break genome editing system, in order to correct the pathogenic allele into a non-pathogenic allele in the genome of the cell.

10. The method of claim 1, wherein selecting the guide RNA for use in introducing the genetic change into the target genomic location using the Cas-based double strand break genome editing system comprises selecting a guide RNA that corresponds to a cut site of the target genomic location.

11. The method of claim 1, wherein selecting the guide RNA comprises selecting a guide RNA that has a predicted frequency of introducing the genetic change into the target genomic location of at least 30%.

12. The method of claim 1, wherein selecting the guide RNA comprises selecting a guide RNA that has a predicted frequency of introducing the genetic change into the target genomic location of at least 50%.

13. The method of claim 1, wherein the genetic change is a 1 base pair insertion or a 1–60 base pair deletion that modifies the pathogenic allele to become a wildtype allele.

* * * * *